NOT MEASUREMENT SENSITIVE

MIL-HDBK-844B(AS)
13 January 2014
SUPERSEDING
MIL-HDBK-844A(AS)
30 December 2003

DEPARTMENT OF DEFENSE HANDBOOK

AIRCRAFT REFUELING HANDBOOK FOR NAVY/MARINE CORPS AIRCRAFT

This handbook is for guidance only. Do not cite this document as a requirement.

AMSC N/A FSC 9130

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This handbook is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

2. For technical guidance on the contents of this document, contact the Naval Air Systems Command, Attention: Air 4.4.5.

Telephone:
DSN 757-3406/3410,
Commercial 301-757-3406/3410
Telefax numbers:
DSN 757-3614,
Commercial 301-757-3614
Message address:
COMNAVAIRSYSCOM PATUXENT RIVER MD//4.4.5//

3. Comments, suggestions, or questions should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 4L8000B120-3, Highway 547, Lakehurst, NJ 08733-5100, or email to michael.sikora@navy.mil. Since contact information can change you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
1.	SCOPE	1
1.1	Scope	1
1.2	Purpose	1
1.3	Sources of information	1
2.	APPLICABLE DOCUMENTS	2
2.1	General	2
2.2	Government documents	2
2.2.1	Specifications and Standards	2
2.2.2	Other Government documents, drawings, and publications	2
2.3	Non-Government publications	3
2.4	Order of precedence	4
3.	DEFINITIONS	5
3.1	Definitions	5
4.	ORGANIZATION AND TRAINING	6
4.1	Organization	6
4.1.1	General	6
4.1.1.1	Organizational flexibility	6
4.1.1.2	Recommended organizational structure	6
4.1.2	Responsibilities and duties	6
4.1.3	Shore activity fuels organization	6
4.1.3.1	Fuel Management Officer (FMO)	6
4.1.3.2	Assistant Fuel Management Officer (AFMO)	7
4.1.3.3	Fuel delivery branch	7
4.1.3.4	Storage and transfer duties	7
4.1.3.5	Quality Surveillance (QS) branch	8
4.1.3.6	Inventory branch	8
4.1.3.7	Training branch	8
4.2	Training	9
5.	CHARACTERISTICS OF AVIATION FUELS	10
5.1	Introduction	10
5.1.1	General	10
5.1.2	Manufacturing aviation fuels	10
5.1.3	Compatibility with other materials	10
5.1.4	Aviation fuel use	10
5.2	Turbine engine fuels	10
5.2.1	Comparison of volatilities	11
5.2.1.1	JP-5 (NATO Code F-44)	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
5.2.1.2	JP-4 (NATO Code F-40)	12
5.2.1.3	JP-8 (NATO Code F-34)	12
5.2.1.4	JP-8+100 (NATO Code F-37)	12
5.2.1.5	JET A (NATO Code F-35)	12
5.2.1.6	F-24 (NATO Code F-24)	12
5.2.1.7	F-27 (NATO Code F-27)	12
5.2.2	Turbine fuel additives.	12
5.2.2.1	Fuel System Icing Inhibitor (FSII)	13
5.2.2.1.1	Icing protection	13
5.2.2.1.1.1	Consequence of lack of FSII	13
5.2.2.1.2	Biostat	13
5.2.2.1.3	FSII materials	13
5.2.2.2	Lubricity additive	14
5.2.2.3	Antioxidant additives	14
5.2.2.4	Static Dissipater Additive (SDA)	15
5.2.2.4.1	SDA and JP-5	15
5.2.2.4.2	Conductivity	15
5.2.2.4.2.1	Testing conductivity at Navy/Marine Corps air activities	15
5.2.2.4.2.2	When SDA should be added to JP-5	15
5.2.2.5	+100 additive	15
5.3	Aviation gasoline (AVGAS)	16
5.3.1	General	16
5.3.2	AVGAS grades	16
5.3.3	Interchangeability	16
5.3.4	AVGAS and fuel contaminants	16
5.3.5	Tetraethyl lead	16
6.	CONTAMINATION OF AIRCRAFT FUELS	17
6.1	General	17
6.1.1	Requirement for quality surveillance of aviation fuel	17
6.1.2	Objective	17
6.1.3	Engine problems due to fuel contamination or incorrect fuel	17
6.1.4	“Clear” and “Bright”	17
6.2	Types and sources of contamination	17
6.2.1	General	17
6.2.2	Particulate matter	18
6.2.2.1	Coarse matter	18
6.2.2.2	Fine matter	18
6.2.2.3	Most prevalent particulate contamination	19
6.2.2.4	Experience reducing the level of particulate contamination	19
6.2.2.5	Hose talc	19
6.2.3	Water	19
6.2.3.1	Dissolved water	19

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
6.2.3.2	Free water	20
6.2.3.2.1	Detrimental effect of free water	20
6.2.3.2.2	Free water allowable limit	20
6.2.4	Chemical contamination	20
6.2.4.1	Determining chemical contamination	20
6.2.4.2	Sources of chemical contamination	21
6.2.4.3	Mixing of different grades of fuel	21
6.2.5	Microbiological growth	21
6.2.5.1	Location of microbiological growth	21
6.2.5.2	Prevention of microbiological growth	22
6.3	Common sources of contamination	22
6.4	Procedures for preventing contamination	22
6.4.1	General	22
6.4.2	Problem prevention and detection	22
6.4.3	Vigilance in fuel delivery operations	22
6.5	Deterioration of aircraft fuels	23
6.5.1	Aviation turbine fuel	23
6.5.2	Aviation gasoline	23
6.6	Sampling of aviation fuel	23
7.	FLEET QUALITY SURVEILLANCE TESTS	24
7.1	General	24
7.2	Particulate contamination	24
7.2.1	Combined contaminated fuel detector (CCFD)	24
7.2.1.1	Capability of CCFD	24
7.2.1.2	Accuracy of CCFD	24
7.2.1.3	Calibration of CCFD and correlation with gravimetric method	24
7.2.2	CCFD test technique	25
7.2.3	CCFD operating procedure	25
7.2.4	Alternate methods	26
7.3	Water contamination	26
7.3.1	General	26
7.3.2	FWD test technique	27
7.3.3	FWD operating procedure	27
7.3.4	Alternate method	28
7.4	Fuel System Icing Inhibitor (FSII) content	28
7.4.1	General	28
7.4.2	B/2 test technique	28
7.4.3	Standard B/2 operating procedures	28
7.4.3.1	Digital B/2 operating procedures	29
7.5	Conductivity	29
7.5.1	General	29
7.5.2	Fuel conductivity meter test procedure	29

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
8.	SAFETY IN FUEL HANDLING OPERATIONS	30
8.1	General	30
8.1.1	Continuous development of safe procedures	30
8.2	Abnormal fuel operations	30
8.3	Fire and explosion	30
8.3.1	General	30
8.3.2	Flammable fuel-air mixtures	31
8.3.2.1	Low vapor pressure products	31
8.3.2.2	Intermediate vapor pressure products	31
8.3.2.3	High vapor pressure products	31
8.3.2.4	Mixing different grades of aviation turbine fuel	32
8.3.3	Flame spread rates	33
8.3.3.1	Relationship between flashpoint, temperature and extinguishment	34
8.4	Sources of ignition	34
8.4.1	Static electricity	35
8.4.1.1	Internal static	35
8.4.1.1.1	Charge generation	35
8.4.1.1.2	Charge accumulation	36
8.4.1.1.3	Static discharge or ignition	36
8.4.1.1.4	Control measures	36
8.4.1.1.4.1	USAF approach to handling internal static charges	37
8.4.1.1.4.2	Procedures for reducing internal static charges	37
8.4.1.2	External static	37
8.4.1.2.1	Charge generation	37
8.4.1.2.2	Charge accumulation and dissipation	37
8.4.1.2.3	Control measures for external static	38
8.4.2	Operating engines	38
8.4.3	Arcing of electrical circuits	38
8.4.4	Open flames	38
8.4.5	Electromagnetic energy	38
8.4.6	Hot surfaces or environment	39
8.5	Extinguishment	39
8.5.1	General	39
8.5.2	Fire chemistry	39
8.5.3	Classification of fires	39
8.5.3.1	Class A fires	39
8.5.3.2	Class B fires	39
8.5.3.3	Class C fires	39
8.5.3.4	Class D fires	40
8.5.4	Fire extinguisher types, agents, and methods of application.	40

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH PAGE

8.5.4.1 Halon 1211 (Bromochlorodifluoro-methane) portable and wheeled unit extinguishers 40
8.5.4.1.1 Extinguishment mechanism 41
8.5.4.1.2 Application 41
8.5.4.2 Carbon Dioxide (CO_2) 15-pound portable units and 50-pound wheeled extinguisher units 41
8.5.4.2.1 Application 42
8.5.4.3 Purple-K-Powder (PKP) dry chemical powder extinguishers 42
8.5.4.3.1 Application 43
8.5.5 Fire extinguisher requirements 43
8.6 Health hazards 45
8.6.1 General 45
8.6.2 Principal health hazards 46
8.6.2.1 Toxic vapor effect 46
8.6.2.1.1 Symptoms and remedial action 46
8.6.2.2 Lead poisoning 46
8.6.2.3 Injury to skin and eyes 46
8.6.2.4 Swallowing aviation fuels 47
8.6.2.5 Fuel tank and filter/separator water bottoms 47
8.6.2.6 Specific procedures for avoiding the health hazards of aircraft fuels 47

9. NOTES 48
9.1 Intended use 48
9.2 Subject term (key word) listing 48
9.3 Changes from previous issue 48

APPENDIX
A. Visual contamination 49
B. Petroleum testing laboratories 52
C. Aircraft information summaries 58
D. Glossary 395

CONCLUDING MATERIAL 400

TABLES
Table I Grades of turbine engine fuels 11
Table II Grades of aviation gasoline 16
Table III Allowable contamination with other products 21
Table IV Vapor pressures of aviation fuels 31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

FIGURES

Figure 1 Enlargement of small particles and comparison to human hair 18
Figure 2 Relationship between temperature, rvp, and flammable limits of petroleum products at sea level 32
Figure 3 Flame spread rate versus fuel temperature 33
Figure 4 Flash point of turbine fuel mixtures 34
Figure 5 Airfield fire protection requirements 44

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. The contents of this handbook are limited to general, technical, and operational information. Specific operating procedures and equipment requirements are contained in the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109. Accounting and stock control procedures are not included in this handbook or in the NATOPS manual. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Purpose. This handbook provides basic information on the properties and characteristics of aviation fuels along with general information on the standards, equipment, and operating principles related to the handling of these fuels at Navy and Marine Corps activities. This handbook is designed to supplement the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, by providing background information and guidance on the requirements and procedures contained in the NATOPS manual.

1.3 Sources of information. The information contained herein has been derived from a number of diverse sources including the fueling experience of the Navy, Marine Corps, and commercial companies; the recommended practices of the American Petroleum Institute (API) and the American Society for Testing and Materials (ASTM); and the published findings of research activities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-25017	-	Inhibitor, Corrosion Inhibitor/Lubricity Improver, Fuel Soluble (NATO S-1747)
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-DTL-85470	-	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3004	-	Quality Assurance/Surveillance for Fuels, Lubricants, and Related Products

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

DEPARTMENT OF THE AIR FORCE

Technical Order (T.O.) 42B-1-1	-	Quality Control of Fuels and Lubricants

(Copies of this document are available through the AF Corrosion Prevention & Control Office website at http://afcpo.com.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF THE NAVY

NAVAIR 00-80R-14	-	U.S. Navy Aircraft Firefighting and Rescue Manual
NAVAIR 00-80T-109	-	Aircraft Refueling NATOPS Manual

(Copies of these documents are available from the Naval Air Systems Command, AIR-4.4.5, Bldg. 2360, 22229 Elmer Rd., Unit 4, Patuxent River, MD 20670-1906; or email Douglas.Mearns@navy.mil.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM D910	-	Standard Specification for Aviation Gasolines
ASTM D1655	-	Standard Specification for Aviation Turbine Fuels
ASTM D2276	-	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM D2624	-	Standard Test Methods for Electrical Conductivity of Aviation and Distillate Fuels
ASTM D3240	-	Standard Test Method for Undissolved Water in Aviation Turbine Fuels
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products

(Copies of these documents are available from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959; (610) 832-9585 or through their website at http://www.astm.org.)

AMERICAN PETROLEUM INSTITUTE (API)

EI-1529	-	Aviation Fueling Hose
EI-1581	-	Specification and Qualification Procedures for Aviation Jet Fuel Filter/Separators

(Copies of these documents are available from American Petroleum Institute, 1220 L Street, NW, Washington, DC 20005-4070; (202) 682-8000 or through their website at http://www.api.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA)

Dangerous Goods Regulations

(Copies of these documents are available from IATA, 703 Waterford Way, NW 62nd Avenue, Suite 600, Miami, FL 33126; (305) 264-7772; or through their website at http://www.iata.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 407 - Standard for Aircraft Fuel Servicing

(Copies of this document are available through the National Fire Protection Association website at http://nfpa.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the NATOPS manuals or STANAGs cited herein, the text of the NATOPS manuals or STANAGs takes precedence. In the event of a conflict between the text of this document and any other cited references, the text of this document takes precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Definitions. See Appendix D, Glossary.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

4. ORGANIZATION AND TRAINING

4.1 Organization.

4.1.1 General. The allowances for a Fuel Division should include an adequate number of personnel who possess sufficient grade structure, training, and seniority to ensure responsible operation of facilities and equipment in response to any operational demand.

4.1.1.1 Organizational flexibility. The organization should be flexible enough to efficiently handle increased workload on short notice, which can best be accomplished by cross training and cross manning. Leave schedules and school attendance can be adjusted to accommodate workload peaks. Scheduled leave can be deferred if unexpected peaks are encountered. Lengthening working shifts and nonstandard duty section, including stand-by duty section assignments, should be last-resort measures.

4.1.1.2 Recommended organizational structure. Chapter 8 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, outlines a recommended standard organizational structure for Navy and Marine Corps fuel operations divisions. Shipboard organizational structures are established by applicable instructions, regulations and/or standard operating procedures.

4.1.2 Responsibilities and duties. The Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, delineates fuels responsibilities at shore activities. For shipboard operations, consult Shipboard Operation Regulation Manual 3120.32. For tactical refueling operations, refer to the appropriate instructions, Standard Operating Procedures (SOPs), and NAVAIR 00-80T-109.

4.1.3 Shore activity fuels organization. The following paragraphs list the normal duties assigned to various personnel within a typical shore activity fuels organization.

4.1.3.1 Fuel Management Officer (FMO). The FMO directs and supervises integrated fuel operations. An FMO typically:

1. Estimates quantities of fuel products to be consumed and fuel service requirements.
2. Develops proposed fuel budget.
3. Performs contract administration services.
4. Prepares and revises the activity fuel instruction in accordance with the Aircraft Refueling NATOPS Manual and other applicable documents.
5. Prepares an oil spill prevention and countermeasure plan.
6. Prepares environmental impact statements for the U.S. Coast Guard on any procedural change and modification to plant facilities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

7. Plans and initiates military construction, repair, and improvement fuel projects.

8. Performs liaison with fuel service customers, activity departments, other Governmental agencies, community officials, and commercial concerns.

9. Represents fuel interests on official boards and committees.

4.1.3.2 Assistant Fuel Management Officer (AFMO). The AFMO assists the FMO in the supervision of the integrated fuel operations and performs the following special duties:

1. Directs the quality assurance program for fuel products.

2. Manages the petroleum laboratory.

3. Directs entire fuel training program.

4. Supervises inspections.

5. Maintains inventory control.

4.1.3.3 Fuel delivery branch. The employee placed in charge of the fuel delivery section or branch is normally delegated the following specific duties:

1. Delivery of aviation Petroleum, Oil, Lubricants (POL) products alongside aircraft.

2. Operation of hydrants to fuel aircraft with engines idling (hot refueling).

3. Operation of aircraft defuelers.

4. Dispatching of personnel and/or equipment and maintenance of dispatch log.

5. Delivery of ground products on automatic fill basis or as requested.

6. Pickup of waste oil.

7. Operational maintenance of facilities and equipment.

4.1.3.4 Storage and transfer duties. The branch or section head in charge of fuel storage and delivery usually is assigned the following duties:

1. Receipt of POL products by pipeline, tanker, barge, tank car or tank truck.

2. Storage of products.

3. Operation of the distribution systems and transferring of products.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

4. Operation of vehicle service stations.

5. Grass cutting in hazardous areas.

6. Receipt and storage of packaged POL products.

7. Operational maintenance of facilities and equipment.

4.1.3.5 Quality Surveillance (QS) branch. The QS branch normally performs the following:

1. Sampling POL products at point of receipt and in aircraft tanks.

2. Surveillance of fuel handling operations.

3. Surveillance of POL filtration, water removal, and monitoring equipment including the maintenance of pressure differential graphs.

4. Operation of the POL laboratory.

5. Inspection and surveillance of facilities and equipment including contractor-owned equipment.

4.1.3.6 Inventory branch. The inventory branch usually performs the following functions:

1. Estimates of POL requirements.

2. Scheduling of product deliveries.

3. Preparation of fuel requisitions for replenishments.

4. Maintenance of daily inventory records.

5. Processing of receipt and issue documents.

6. Monitoring of contract refueling fuel deliveries.

4.1.3.7 Training branch. The training branch normally performs the following duties:

1. Preparation of training guide.

2. Conducting classroom and on-the-job training.

3. Certifying qualifications and issuing certificates.

4. Maintenance of training and qualification records.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5. Reviewing and supplementing contractors' training program.

4.2 Training. The importance of proper training cannot be over-emphasized. It is essential to the safety of fuel handling operations that the personnel involved be properly trained. Historical records disclose that the instinctive reactions of experienced fuel operators during emergency situations have minimized personnel injuries, fuel losses, and the destruction of government property including aircraft. Conversely, the records show that the reactions of relatively untrained personnel, under similar situations, cannot be relied upon.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5. CHARACTERISTICS OF AVIATION FUELS

5.1 Introduction.

5.1.1 General. The basic type of aircraft fuel in use at Navy and Marine Corps air activities is aviation turbine fuel. Aviation gasoline (AVGAS) is only available in limited quantities at certain air facilities. Knowledge of the basic properties and characteristics of aviation fuels is necessary to understand the importance of delivering the proper fuel to the aircraft. Such knowledge is also valuable in understanding the need for safety and caution in handling aviation fuels.

5.1.2 Manufacturing aviation fuels. Turbine engine fuels and aviation gasolines are petroleum products manufactured from crude oil by refineries. Both are classified as flammable liquids and will burn when ignited. Under the right conditions, they will explode with a force similar to that of dynamite. Death can result if the vapors of either type of fuel are inhaled in sufficient quantities and serious skin irritation can result from contact with the fuels in the liquid form. In liquid form, aircraft fuels are lighter than water, and in vapor form they are heavier than air. Consequently, any water present in the fuels will usually settle to the bottom of the container. On the other hand, vapors of these fuels, when released in the air, tend to remain close to the ground, thus increasing the danger to personnel and property. For safety and health considerations, both aviation gasolines and turbine engine fuels should be handled with equal caution. Additional information on the properties and characteristics of aviation fuels and their effects on the safe handling of these materials is contained in Section 6 of this handbook.

5.1.3 Compatibility with other materials. All aviation fuels are extremely good solvents. For example, AVGAS will dissolve common lubricants (such as oils and greases used in pumps, valve, packing, and other equipment) and can cause serious deterioration of many rubber materials. It is therefore extremely important that only materials specially designed, tested, and approved for use with aviation fuels be allowed to come into contact with fuels. Never use substitute greases, lubricants, packing, etc., on or with fuel handling equipment without first obtaining agreement from the cognizant technical authority for the piece of equipment in question.

5.1.4 Aviation fuel use. Aviation gasolines are used for piston-type (reciprocating) engines, which are similar to automotive engines in terms of their basic operating principle. Turbine engine fuels, on the other hand, are intended for use in an entirely different type of engine. In place of the pistons found in reciprocating engines, turbine engines comprise an air compressor, a combustor, and a turbine that is used to turn fuel into usable propulsion energy. Since these are completely different types of engines, they require different types of fuels for proper operation. The following paragraphs provide some specific information on each type of fuel.

5.2 Turbine engine fuels. While aircraft piston engines are sensitive to the fuel used and will operate safely and satisfactorily only on one grade of aviation gasoline, most aircraft turbine engines can use a variety of grades of aviation turbine fuels. The difference between the grades of turbine fuels is their volatility. Table I lists the more common grades of military and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

commercial aviation turbine fuels and provides a rough comparison of their volatilities through their flash points (appendix D.1.33) and freeze points (appendix D.1.36). Detailed information on the properties and requirements of each grade of fuel is contained in the applicable specification.

5.2.1 Comparison of volatilities. Although kerosene type turbine fuels, JP-5 and JP-8, are much less volatile than JP-4 and AVGAS, under the right conditions, such as severe agitation, mists that are as flammable and explosive as AVGAS can form. *All aviation fuels should be handled carefully*.

TABLE I. Grades of turbine engine fuels.

	Grade	Specification	Fuel Type	NATO CODE	Freeze Point	Flash Point	Density (°API Gravity)
Military	JP-5	MIL-DTL-5624	Kerosene (High Flash)	F-44	–51 °F –46 °C	140 °F 60 °C	36.0 – 48.0
	JP-4	MIL-DTL-5624	Wide-Cut	F-40	–72 °F –58 °C	Below –4 °F –20 °C	45.0 – 57.0
	JP-8	MIL-DTL-83133	Kerosene	F-34	–53 °F –47 °C	100 °F 38 °C	37.0 – 51.0
	JP-8+100	MIL-DTL-83133	Kerosene	F-37	–53 °F –47 °C	100 °F 38 °C	37.0 - 51.0
	F-24	STANAG 3747	Kerosene	F-24	–40 °F –40 °C	100 °F 38 °C	37.0 – 51.0
	F-27	STANAG 3747	Kerosene	F-27	–40 °F –40 °C	100 °F 38 °C	37.0 – 51.0
	TS-1	GOST 10227-86	Kerosene	NONE	–58 °F –50 °C	82 °F 28 °C	51.0 (max)
	F-76	MIL-DTL-16884	Diesel	F-76	–1 °C 30 °F	140 °F 60 °C	N/A
Commercial	Jet A	ASTM D1655	Kerosene	none	–40 °F –40 °C	100 °F 38 °C	37.0 – 51.0
	Jet A-1	ASTM D1655	Kerosene	F-35	–53 °F –47 °C	100 °F 38 °C	37.0 – 51.0
	Jet B	ASTM D6615	Wide-Cut	none	–72 °F –58 °C	Below –4 °F –20 °C	45.0 – 57.0
	100LL	ASTM D910	Aviation Gasoline	F-18	–72 °F –58 °C	Below –40 °F –40 °C	N/A

5.2.1.1 JP-5 (NATO Code F-44). JP-5 is a kerosene fuel with an especially high flash point facilitating safety in shipboard handling. It is the only fuel that can be used for turbine engine aircraft aboard ships and is used widely at USN and USMC air stations. Because it has the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

highest density of all the aviation fuels, JP-5 has the greatest affinity for dirt, rust, and water contaminants.

5.2.1.2 JP-4 (NATO Code F-40). JP-4 is no longer widely used as a primary aviation turbine fuel by the U.S. military. It is used in a limited area (Alaska and some northern tier USAF bases) and may be encountered in other regions of the world (particularly Saudi Arabia and Canada). Jet B is the commercial equivalent of JP-4. JP-4 is a wide boiling range petroleum product including both gasoline and kerosene boiling range components. JP-4 exhibits better low temperature starting than JP-5. JP-4 fuel is intermediate between AVGAS and JP-5 with respect to its tendencies to acquire and hold dirt, rust, and water contaminants. It is an alternate fuel to JP-5 for USN and USMC jet aircraft. Both JP-4 and JP-5 fuels are procured under MIL-DTL-5624.

5.2.1.3 JP-8 (NATO Code F-34). JP-8, procured under MIL-DTL-83133, is a kerosene fuel similar to commercial jet fuel, ASTM Jet A-l, except that JP-8 contains the military additive package (see 5.2.2). It is also similar to JP-5 with respect to most fuel properties except flash point and freeze point. Since its flash point is not as high as JP-5's, it cannot be used for shipboard operations. JP-8 is the primary aviation turbine fuel used at all Air Force and Army installations and by NATO.

5.2.1.4 JP-8 +100 (NATO Code F-37). JP-8+100 is a version of JP-8 with an additive that increases its thermal stability by 56 °C (a difference of 100 °F). The additive is a combination of a surfactant, metal deactivator, and an antioxidant, and was introduced to reduce coking and fouling in USAF engine fuel systems.

Note

The use of fuel containing the +100 additive is restricted to "emergency only". If fuel containing the +100 additive is used, the air crews are to note the use of the fuel containing the +100 additive and need to inform ground refueling personnel if defueling is required in the following three refueling cycles. Disposition of the fuel has to be in accordance with NATOPS 00-80T-109.

5.2.1.5 JET A (NATO Code F-35). Jet A is a kerosene-type fuel similar to JP-8. The primary physical difference between the two is freeze point (the temperature at which wax crystals disappear in a laboratory test). Jet A, which is mainly used in the United States, is to have a freeze point of -40 ºC or below, while JP-8 is to have a freeze point of -47 ºC or below.

5.2.1.6 F-24 (NATO Code F-24). F-24 is JET A with the military additives listed in section 5.2.2.

5.2.1.7 F-27 (NATO Code F-27). F-27 is F-24 with the +100 additive used in JP-8 +100.

5.2.2 Turbine fuel additives. Although JP-5 and JP-8 are quite similar to commercial turbine fuels Jet A and Jet A-1, and JP-4 is basically the same fuel as Jet B, there are some very

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

important differences. In addition to small but significant differences in volatility, all three military fuels contain the following additives, which commercial jet fuels normally do not:

1. Fuel System Icing Inhibitor (FSII).
2. Lubricity Additive (corrosion inhibitor).
3. Antioxidants (storage stability additives).
4. Static Dissipater Additive (JP-4 and JP-8 fuels only).
5. +100 Additive (Air Force JP-8 +100 and F-27 fuel only).

5.2.2.1 Fuel System Icing Inhibitor (FSII). FSII is added to the fuel for two reasons: it provides icing protection and also acts as a biostat.

5.2.2.1.1 Icing protection. Even when the free-water content of the fuel is maintained below the 5 ppm level, FSII is essential because, in addition to free water, aviation fuel contains a significant amount of dissolved water. In general, the amount of dissolved water a fuel will hold in parts per million (ppm) is approximately equal to the temperature of the fuel in degrees Fahrenheit. For example, a fuel that is 70 °F (21 °C) contains approximately 70 ppm dissolved water. If this fuel were cooled to 20 °F (-7 °C) it would then contain 50 ppm free water and 20 ppm dissolved water. When an aircraft is exposed to cold temperatures such as high altitudes or very cold weather conditions at sea level, the fuel contained in its tanks will drop in temperature allowing the water that is dissolved in the fuel to condense out into tiny droplets of free water. If the temperature of the fuel drops low enough (below 32 °F (0 °C)) these droplets will form ice crystals that will collect on screens or filters in the fuel system quickly blocking them. In addition, the ice crystals can cause fuel system valves to stick or malfunction preventing the aircraft pilot from using or distributing his fuel load.

5.2.2.1.1.1 Consequence of lack of FSII. The result of using fuel without sufficient FSII can be the loss of an aircraft. Certain aircraft are more susceptible to these problems than others because of differences in their fuel system designs as well as possible flight profiles. These aircraft are the S-3 and SH-60. Thus, these aircraft require a minimum FSII level of 0.03 percent by volume in their fuel. All other USN and USMC aircraft do not require FSII and may use JP-5 or other approved fuel even if it does not contain any FSII.

5.2.2.1.2 Biostat. FSII prevents the growth of fungi and other microorganisms that can develop at the interface between the fuel and any water that collects at the bottom of the tank (aircraft as well as fuel storage). Since these microbiological growths can form rapidly, clog filters, and degrade the fuel, every effort should be made to maintain FSII levels as high as possible in order to maximize their biostatic effect. The best way of maintaining these levels is to minimize contact of the fuel with water, which tends to leach out the FSII from the fuel. JP-5 (NATO Code F-44) tanks should, therefore, be stripped of water on a daily basis if any significant amount of water accumulates within a 24-hour period.

5.2.2.1.3 FSII materials. Diethylene Glycol Monomethyl Ether (DiEGME) (NATO Code S-1745) is the FSII used in U.S. military and NATO aviation turbine fuels. A similar commercial

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

product, known by the tradename Prist, is approved for use in commercial aviation turbine fuels. The U.S. military specification that defines the properties of DiEGME and governs its procurement is MIL-DTL-85470. DiEGME is considered mutagenic in its neat state but is considered safe once blended into fuel. As a general rule, DiEGME is injected at either the refinery or Defense Fuel Supply Point (DFSP). Personnel involved in the handling and injection of FSII materials are advised to follow all instructions, wear appropriate personal protective equipment (gloves, aprons, and goggles at a minimum), and minimize their exposure as much as possible. Since FSII tends to become concentrated in the water that collects in the bottoms of fuel tanks (aircraft and storage) and filter/separator vessels, personnel handling these water bottoms are advised to follow similar precautions.

Note

Shipboard addition of FSII is not authorized.

Note

The addition of concentrated Fuel System Icing Inhibitor (e.g., DiEthylene Glycol Monomethyl Ether (DiEGME) defined by MIL-DTL-85470) directly into an aircraft's fuel tank is not authorized. The FSII additive does not readily dissolve into aviation fuels, and consequently, the direct addition of concentrated FSII into an aircraft fuel tank during a refueling cycle may not achieve sufficient mixing to fully dissolve the additive into the fuel. This may result in some of the FSII settling to the bottom of the aircraft fuel tank in a concentrated form. In the neat (concentrated) form, the additive is not compatible with fuel system materials. Incidents (non-military) of aircraft fuel bladder and filter damage have been reported due to FSII addition using aerosol can products (such as Prist, or Dice). In addition, since not all of the FSII is mixed with the fuel, the resultant fuel has inadequate icing protection.

5.2.2.2 Lubricity additive. A combination lubricity improver and corrosion inhibitor additive, procured under MIL-PRF-25017, is injected in all military turbine fuels at the refinery in order to improve the lubricating characteristics of the fuel. A series of several contiguous (one immediately following another) flights with fuel that does not contain one of these additives may cause abnormal wear or malfunctions of aircraft and/or engine fuel system components. A few flights (one or two) will not contribute to such problems since the additives tend to leave a protective coating on the components.

5.2.2.3 Antioxidant additives. These materials, which are injected into the fuel at the refinery, are particularly important for fuels that have been processed at the refinery with hydrogen. They ensure that the fuel will be stable when placed in long-term storage (a few months to several years). Commercial fuels do not need these additives since they are usually consumed within a few weeks to a couple of months. The first fuel property to drop below specification minimums in an unstable fuel is usually its thermal stability. Other properties such as total acid number, copper strip corrosion, and existent gums can also fall below acceptable minimums.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5.2.2.4 Static dissipater additive (SDA). JP-4, F-24, and JP-8 fuels are injected with a special additive that increases the fuel's conductivity and helps relax static electric charges, which are produced during fuel handling operations (filtration, pipeline movement, etc.). SDA, as this additive is now called, was originally added to these fuels to prevent small static initiated explosions that were occurring during refueling of polyester foam-filled USAF aircraft tanks. The USN and USMC have never experienced similar problems with foam-filled tanks (probably due to aircraft tank and refueling equipment design differences).

5.2.2.4.1 SDA and JP-5. Because SDA additives adversely affect the performance of shipboard centrifugal purifiers, SDA is not added to JP-5 (see 5.2.2.4.2.2 for the exception to this policy).

5.2.2.4.2 Conductivity. When USAF or foreign government aircraft are refueled with JP-4, F-24, or JP-8, the fuel should exhibit conductivity above 100 pS/m as measured by a portable conductivity meter, NSN 6630-01-115-2398, to protect the accuracy of certain sensitive aircraft fuel quantity gauging systems. Since SDA is depleted in the supply distribution system, USAF policy has been to add the additive as close as possible to the using activity or base. SDA is therefore most often added at Defense Fuel Support Points (DFSPs).

5.2.2.4.2.1 Testing conductivity at Navy/Marine Corps air activities. It is not necessary for Navy or Marine Corps activities to frequently test the conductivity levels of their stocks of JP-8 fuel when they are only refueling USN or USMC aircraft. However, occasional testing of the conductivity of these fuels is recommended in order to ensure that SDA is being injected at the proper level, especially since excessively high levels can affect the accuracy of certain aircraft gauging systems. If on such checks the fuel is found to be out of the specification conductivity range, the injecting facility (DFSP) should be informed immediately so that the injection rate can be adjusted accordingly. In some instances where the refueling of USAF aircraft is involved, manual addition of SDA at base level may be necessary in order to prevent aircraft damage. More information and assistance on this subject may be obtained from the Air Force Petroleum Office Technical Support Team located at Wright-Patterson Air Force Base, OH (DSN 785-4311 or commercial 937-255-4311).

5.2.2.4.2.2 When SDA should be added to JP-5. In a couple of special situations where very large numbers of USAF aircraft are frequently refueled, it may be necessary to have the JP-5 supplied to a Navy/Marine Corps air activity injected with SDA. The above deterioration use limits are then applicable to JP-5.

5.2.2.5 +100 additive. This additive is a thermal stability improver that increases the thermal stability of a fuel by about 56 °C (100 °F). The additive is a combination of a surfactant, metal deactivator, and an antioxidant, and was introduced to reduce coking and fouling in USAF engine fuel systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5.3 Aviation gasoline (AVGAS).

5.3.1 General. Very little AVGAS is currently being used by the U.S. Military services. For this reason the Military AVGAS Specification, MIL-G-5572, was canceled in 1988 and military needs are being satisfied via the commercial specification ASTM D910.

5.3.2 AVGAS grades. AVGAS is graded according to its performance in a similar manner to the "octane" ratings used for automotive gasolines. "Performance Numbers," as they are called, are also based on the performance of the fuel in preventing engine knock, an extremely serious problem in aircraft engines due to the continuous high power demands placed upon them. The grades of AVGAS were formerly designated by 2 numbers, e.g., 100/130. The first number indicated the knock rating with a lean fuel-air mixture, while the second number indicated the knock rating with a rich fuel-air mixture. The new ASTM designations for these fuels now refer only to the lean knock rating. The different grades of AVGAS are dyed various colors so they can be easily distinguished. Table II shows the various grades of AVGAS and their colors.

5.3.3 Interchangeability. Grade 100 (high lead) and Grade 100LL (low lead) have exactly the same performance characteristics and can be used interchangeably. Grade 100 and Grade 100LL may also be commingled in storage tanks at air stations. Note that if these two fuels are mixed an unusual color may result.

5.3.4 AVGAS and fuel contaminants. Since AVGAS is the lightest (lowest density) naval aviation fuel, it has the least tendency to acquire and hold in solution dirt, rust, and water contaminants making it the easiest aviation fuel to keep clean.

5.3.5 Tetraethyl lead. All AVGAS contains some tetraethyl lead for the improvement of its anti-knock performance. The presence of this fuel additive makes it very deleterious to turbine engines so it is extremely important that turbine fuels not be contaminated with even small amounts of AVGAS.

TABLE II. Grades of aviation gasoline.

ASTM D910	MIL-G-5572	Color	NATO Symbol
Grade 100	100/130	Green	-----
Grade 100LL	100/130 Low Lead	Blue	F-18

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

6. CONTAMINATION OF AIRCRAFT FUELS

6.1 General.

6.1.1 Requirement for quality surveillance of aviation fuel. Basic information and minimum requirements pertaining to the quality surveillance of aviation fuels at Navy and Marine Corps air activities are contained in Chapters 3 and 9 of the NATOPS Aircraft Refueling Manual, NAVAIR 00-80T-109. This chapter provides a brief review of these requirements with amplifying information.

Note

At activities where the fuel is a Defense Working Capital Funds (DWCF) product, the bulk fuel storage quality surveillance has to be in accordance with the latest version of MIL-STD-3004.

6.1.2 Objective. The major objective of fuel handling personnel is to deliver clean, water-free, and correct fuel to aircraft. The fuel systems of modern aircraft are complex and will not function properly if fuel is contaminated with dirt, water, or other foreign matter. Foreign matter in fuel can plug or restrict fuel pumping and metering equipment and accelerate the clogging of fuel filters. Fuel contaminated with water is harmful because ice may be formed at high altitudes also clogging aircraft components. The presence of water also permits the growth of microorganisms in aircraft fuel tanks, which hinder the operation of components and cause corrosion in fuel systems.

6.1.3 Engine problems due to fuel contamination or incorrect fuel. Aircraft engine failure or poor performance may also be caused by incorrect fuel or by contamination of the proper fuel with other petroleum products. For example, a small amount of turbine engine fuel in aviation gasoline can significantly reduce its anti-knock quality. Similarly diesel fuel, lubricating oils, and hydraulic fluids are harmful to the quality of both types of aviation fuels. Any contamination of aviation fuels is to be avoided.

6.1.4 “Clear” and “Bright”. As a general rule, for aviation fuel to be acceptable to aircraft it should be clear, bright, and contain no free water. The terms “clear” and “bright” are independent of the natural color of the fuel. The various grades of AVGAS have dyes added. Turbine fuels are not dyed and may be any color from water-white to straw yellow. “Clear” means the absence of any cloud, emulsion, readily visible particulate matter, or entrained water. “Bright” refers to the shiny appearance of clean, dry fuels. A cloud, haze, specks of particulate matter, or entrained water indicates that the fuel is unsuitable and points to a breakdown in fuel handling equipment or procedures.

6.2 Types and sources of contamination.

6.2.1 General. Aircraft fuel can be contaminated with particulate matter, free water, foreign chemicals, microorganisms, or any combination of the four. In addition to the following paragraphs that discuss these various types of contamination, Appendix A is a guide to help in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the detection and understanding of the consequences of the various types of contamination. In practice, any significant (deleterious) amount of coarse material contaminant can usually be detected visually.

6.2.2 Particulate matter. Particulate matter appears as dust, powder, grains, flakes, fibers, or stain. Particulate matter, or solid contamination, can be separated into two categories — coarse matter and fine matter.

6.2.2.1 Coarse matter. Coarse matter can be seen and easily settles out of fuel or can be removed by adequate filtration. Ordinarily, particles 10 microns in size and larger are regarded as coarse matter (a micron is 1/1,000,000th of a meter or approximately 1/25,000th of an inch). Figure 1 illustrates just how small these particles can be. Coarse particles clog orifices and wedge in sliding valve clearances and shoulders, causing malfunctions of fuel controls and metering equipment. They are also effective in clogging nozzle screens and other fine screens throughout the aircraft fuel system.

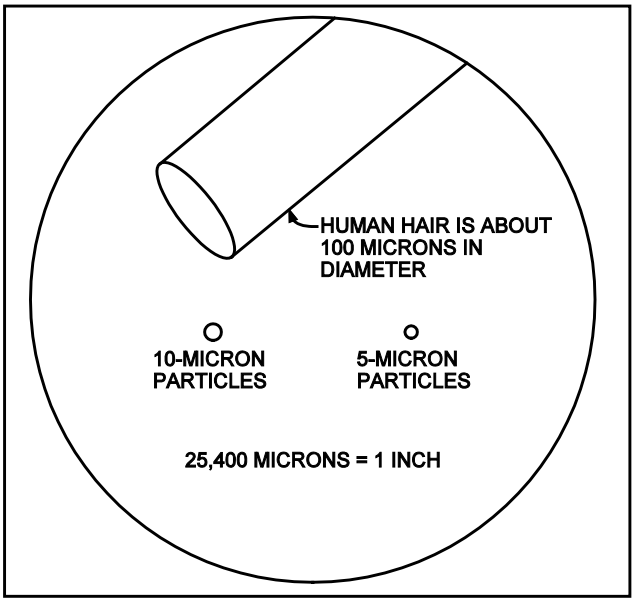


FIGURE 1. Enlargement of small particles and comparison to human hair.

6.2.2.2 Fine matter. Fine matter is defined as particles smaller than 10 microns. To a limited degree, fine matter can be removed by settling, filtration, and the use of centrifugal purifiers. Particles in this range accumulate throughout fuel controls, appearing as a dark shellac-like surface on sliding valves, and may also be settled out in rotating chambers as sludge-like matter, causing sluggish operation of fuel metering equipment. Fine particles are not visible to the naked eye as distinct or separate particles; they will, however, scatter light and may appear as point flashes of light or a slight haze in fuel. Occasionally, a fuel contaminated with gross amounts of fine particulate matter that does not respond to normal filtration and cleanup procedures may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

encountered. For shipboard applications notify the type commander (TYCOM) or higher authority for guidance.

6.2.2.3 Most prevalent particulate contamination. The most prevalent types of particulate matter contamination in aircraft fuels are iron rust and scale, sand, and airborne dirt. The principal source of iron rust and scale is the corrosion in pipelines, storage tanks, and other fuel containers.

6.2.2.4 Experience reducing the level of particulate contamination. Experience has shown that solid contaminants (rust and dirt) can be held well below a level of 1 milligram per liter (mg/liter) in a properly functioning fuel distribution system. If solid contaminants in fuel at aircraft dispensing points exceed 1 mg/liter when measured by the Combined Contaminated Fuel Detector (CCFD), or by laboratory analysis using ASTM D2276, investigative and corrective action should be taken to improve fuel quality. If solid contaminants exceed 2 mg/liter, delivery of fuel to aircraft should be stopped and corrective measures completed prior to resumption of fueling operations.

6.2.2.5 Hose talc. Hose talc can be a source of fuel contamination with some older types of hoses (especially collapsible hose). Talc (soapstone) material, which is applied to the interior of the hose by the manufacturer to aid the curing process, can be dislodged during normal handling and reeling operations. Prolonged soaking in fuel also tends to loosen talc. Fueling stations subjected to a reduced level of activity are particularly prone to talc contamination. Although hose pickling is designed to prevent this type of contamination it is not always successful in doing so. Hoses manufactured to EI-1529 or NFPA 407 do not need to be pickled.

6.2.3 Water. Free water (undissolved water) is a common contaminant of fuels and refueling personnel should be concerned with it in two forms: 1) entrained in the fuel, and 2) as a separate phase (liquid water). Entrained water is found in fuels in the form of very small droplets, fog, or mist and it may or may not be visible. When large quantities of entrained water are present, the fuel will have a hazy or milky appearance. Water usually becomes entrained in the fuel when it is broken up into small droplets and thoroughly mixed with the fuel in equipment such as pumps or meters. Given sufficient time and the proper conditions, entrained water will settle and separate from aviation gasoline; however, since they are fairly dense compared to AVGAS or motor gasoline, turbine engine fuels will hold entrained water in suspension for long periods of time. Once separated and settled from the fuel, water will collect at the bottoms of tanks, pipes, and other fuel system components.

6.2.3.1 Dissolved water. Fuel will actually dissolve a small amount of water. Dissolved water is absorbed into the fuel and is not visible. The amount a fuel will hold in a dissolved state is dependent upon the fuel’s temperature. A rough correlation can be made between a fuel’s temperature in degrees Fahrenheit and the amount of water that can be dissolved in it. For example, a fuel at 60 °F will hold approximately 60 parts per million (ppm) dissolved water, while at 30 °F it will hold only 30 ppm. It is important to note that as a fuel cools down; the water dissolved in it at the higher temperature will come out of solution and become free water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

6.2.3.2 Free water. Free water may be fresh or saline. Free water may be in the form of a cloud, emulsion, entrained droplets, or in gross amounts in the bottom of a tank or container. Any form of free water can cause icing in the aircraft fuel system components. A fuel system icing inhibitor (FSII) is added to JP-4, JP-5, and JP-8 to prevent the formation of ice in aircraft fuel systems when temperatures fall below the freezing point of water at high altitudes. Because FSII is preferentially soluble in water, prevention and elimination of water from fuel transportation and storage systems are essential. Failure to eliminate water could result in the loss of FSII below an acceptable use limit.

6.2.3.2.1 Detrimental effect of free water. Free water (water dispersed as a haze, cloud, or droplets) in fuel can be disastrous in aircraft fuel systems — particularly sea water. It can cause filter and fuel control icing, fuel quantity probe fouling, and corrosion of fuel system components. Water is also the one item essential for microbiological growth to develop in aircraft tanks.

6.2.3.2.2 Free water allowable limit. The maximum allowable limit of free water in fuel at aircraft dispensing points is 10 ppm when tested by the free water detector. A satisfactorily performing filter/separator will provide fuel containing less than 10 ppm of free water. Should the level of free water in fuel at an aircraft dispensing point exceed 10 ppm, a second sample will be taken immediately to ascertain if the second sample confirms that the free water exceeds 10 ppm. If so, fueling will be stopped until changes in procedure and equipment reduce the free water to 10 ppm or below.

Note

EI-1581 allows 15 ppm of free water in the effluent fuel of a filter-separator vessel being tested using the EI-1581 test procedure. This high free water limit is necessary because of the large quantities of free water introduced during EI-1581 qualification (and acceptance) testing of filter-separator vessels. Facilities with fuel distribution systems which contain filter-separator vessels certified to EI-1581 should be aware that the 15 ppm free water limit is not applicable to in-service EI-1581 filter-separator vessels. All properly functioning filter-separator vessels are capable of providing aviation jet fuel having less than 10 ppm of free water (as measured by the free water detector or Aqua-Glo instruments).

6.2.4 Chemical contamination. Chemical contamination usually results from the inadvertent mixing of petroleum products. This type of contamination affects the chemical and physical properties of the fuel and can generally be detected only by specific laboratory tests. These tests are conducted at refineries, bulk terminals, and petroleum testing laboratories. Chemical contamination is prevented by isolating fuels and providing separate handling systems. Pilots and personnel servicing aircraft will seldom be confronted with chemical or petroleum contamination and will be able to detect contamination only by an unusual color, appearance, or odor.

6.2.4.1 Determining chemical contamination. When doubts as to fuel quality cannot be resolved by application of the standard fleet test methods, fuel samples should be drawn and shipped immediately to at least one of the petroleum testing laboratories listed in Appendix B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

6.2.4.2 Sources of chemical contamination. During transportation by truck, railroad, barge, tanker, or fleet oiler and during terminal or station storage there are frequent opportunities for contamination with other bulk petroleum products. In some instances it is not feasible to completely eliminate the possibility of some contamination. The use limits for fuel chemical and physical properties specified in Appendix B of the Aircraft NATOPS Refueling Manual should not be used to determine the acceptability of fuels from a commercial contractor. The procurement specification should be adhered to in determining acceptability of products from contractors. If the product no longer meets the procurement specification but conforms to the chemical and physical property limits specified in Aircraft Refueling NATOPS Manual, Appendix B, a request for waiver will be submitted to the Naval Supply (NAVSUP) Energy Office (see table B-I for address) together with pertinent facts concerning the circumstances and nature of the contamination. It is important to stress that the prescribed use limits are designed to serve as parameters in determining the quality of fuel in storage and not as procurement criteria. Known contamination with other products is limited to the percentages shown in table III.

TABLE III. Allowable contamination with other products.

Product Being Handled	Contaminating Product		
	Aviation Gasoline	Jet Fuel	Diesel
Aviation Gasoline	N/A	0.5%	0.5%
Jet Fuel[1/]	0.0%	N/A	0.5%

1/ Jet fuel refers to JP-5, JP-8, JP-8 +100, Jet A, F-24, F-27, TS-1

6.2.4.3 Mixing of different grades of fuel. Extreme vigilance needs to be used to ensure that different grades of fuel are not mixed. All personnel should know and remember that small quantities of one fuel can seriously contaminate and render unusable another aircraft fuel. For example, the alternate use of a truck-mounted refueler for turbine fuel and aviation gasoline can contaminate aviation gasoline sufficiently to cause failure of an engine designed for high-octane aviation gasoline.

6.2.5 Microbiological growth. Microbiological growth consists of living organisms that grow at the fuel-water interface. Fungus is the major source responsible for problems associated with microbiological contamination of jet fuels. Fungus is a form of plant life; it holds rust and water in suspension and acts as a stabilizing agent for fuel-water-sediment emulsion. It clings to glass and metal surfaces and can cause erroneous readings in fuel quantity gauging systems, sluggish fuel control operation, and sticking of flow dividers.

6.2.5.1 Location of microbiological growth. Microbiological growth is generally found wherever pockets of water exist in fuel tanks. Microorganism contamination appears as a brown slime-like deposit that adheres to the inner surfaces of fuel tanks. Although bacteria and fungi are present in most turbine fuels, the conditions necessary for their growth include water, fuel, and trace minerals. Water remains the key ingredient. Without free water there is no growth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

6.2.5.2 Prevention of microbiological growth. The presence of microbiological growth in fuel being delivered to aircraft is a reliable indication of the presence of free water and the failure of fuel cleanup equipment. FSII in sufficient concentrations in the water bottoms of aircraft fuel tanks prevents the growth of microorganisms; however, this does not alleviate the requirement for the daily removal of all water from the low point drains since this action is necessary to prevent corrosion and the deterioration of tank coatings. The growth of microorganisms and their resultant contamination is usually most severe in tropical climates where temperatures and humidity are high.

6.3 Common sources of contamination. Some of the most common sources of contamination of fuel supplies are:

1. Fuel storage tanks containing water bottoms that cannot be completely drained.
2. Floating roof tanks that allow the entry of rainwater and airborne dust.
3. Water introduced by ballasting or leaks during transport in tanks, tankers, or barges.
4. Previously contaminated fuel being defueled from aircraft into storage tanks.

6.4 Procedures for preventing contamination.

6.4.1 General. Contamination of aircraft fuel can be prevented only by the use of proper equipment and by following proper operating procedures. “Retail” or “Ready Issue” fuel handling systems are to be used at all shore station aircraft refueling activities to contain and process the fuel immediately prior to issue to aircraft. These systems include an approximate 10-day supply (based on normal base issues to aircraft) of storage capacity. Storage tanks used in this system should have sloping bottoms, bottom suction (pick-up), and continuous recirculation through a filter/separator that removes both water and particulates. Additional filter/separators further clean and dry the fuel as it is loaded onto trucks at truck fill stands or as it enters and/or exits direct refueling or hydrant systems. Proper care and operation of these systems will help ensure that only clean dry fuel enters aircraft.

6.4.2 Problem prevention and detection. Even though retail fuel delivery systems are designed with multiple filtration steps, the success of these systems is dependent upon the manner in which they are operated and maintained. The pressure drops across filter/separators should be routinely observed and recorded in order to detect failures or problems. In addition, every precaution should be taken to prevent the introduction of any particulate matter (dirt) into the fuel. All openings and connections, including refueling nozzles, should have dust-tight caps or covers that remain in place at all times except when in use.

6.4.3 Vigilance in fuel delivery operations. The mixing of fuels or the delivery of the wrong fuel can be avoided only by alert and careful personnel who know and follow the proper procedures. Accidents caused by fuel mixing are solely the responsibility of fuel handling personnel. Fortunately most Navy and Marine Corps air facilities handle only one aviation fuel, aviation turbine fuel, JP-5 (F-44) or JP-8 (F-34). While this situation helps to minimize the problem, it does not reduce the need for vigilance. Where more than one type or grade of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

aviation fuels are handled, complete separate handling facilities and equipment for each type and grade of fuel are essential in preventing contamination.

6.5 Deterioration of aircraft fuels. The following are some of the most frequently encountered situations in which some form of contamination leads to deterioration of an aircraft fuel.

6.5.1 Aviation turbine fuel.

1. *Reduction in Flash Point (JP-5, F-24, and JP-8).* The flash points of these fuels will be reduced when contaminated with other fuels having lower flash points.

2. *Contamination with Dirt, Rust, and Water*. This is due to normal handling procedures. JP-5, F-24, and JP-8 have a greater affinity for these contaminants than AVGAS or JP-4; therefore contaminant removal is more difficult. If adequate surveillance is not practiced, contamination is almost certain to result.

3. *Contamination with Naval Distillate Fuel (F-76).* This is a shipboard handling problem. Naval distillate in amounts of 0.5 percent or more may effect fuel properties such as freeze point and thermal stability.

6.5.2 Aviation gasoline.

1. *Lowering of Vapor Pressure*. The change in this property is usually due to prolonged storage in vented tanks in warm climates.

2. *Loss of a Fuel's Performance Rating*. The performance rating of a fuel will usually be degraded as a result of contamination with another petroleum product.

3. *Increase in Copper Corrosion*. The corrosive properties of AVGAS often increase as a result of storage in warm climates in tanks with water bottoms or sludge accumulations. Bacteria growing in the water bottoms and/or sludge generate hydrogen sulfide, which dissolves in the fuel.

4. *Contamination with Dirt, Rust, and Water*. These types of contamination are due to normal handling and are not difficult to remove, utilizing settling or filtration methods.

6.6 Sampling of aviation fuel. Detailed information on sampling practices and techniques is contained in ASTM D4057. All activities that handle aviation fuel should have a copy of this document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

7. FLEET QUALITY SURVEILLANCE TESTS

7.1 General. Fleet activities should monitor the fuel they issue to aircraft for particulate and free water contamination and FSII content. In some special situations it may also be necessary for activities to monitor the conductivity of their fuel. This chapter describes the test equipment and general procedures used to monitor these fuel properties.

7.2 Particulate contamination.

7.2.1 Combined contaminated fuel detector (CCFD). The CCFD, procured under specification MIL-D-22612, is a portable unit for use in the field and aboard ship to determine the solid contamination and free water level existent in aviation fuels. The instrument has a range of 0 to 10 mg/liter for solids and 0 to 20 ppm for free water. The CCFD (NSN 6640-01-013-5279), includes a built-in Free Water Detector (FWD) Viewer Kit.

7.2.1.1 Capability of CCFD. This unit provides operating activities with a capability to determine the solids content and free water level existent in aviation fuels. While the unit is comparatively simple to use, it is a precision instrument and should be treated accordingly. The maximum value of this unit can best be realized by placing it in the hands of the person responsible for quality control and inspection of aviation fuels for the activity.

7.2.1.2 Accuracy of CCFD. The accuracy and value of a unit of this nature will depend upon the personnel operating it. If the results are to be valid, the fuel samples should be truly representative and the whole operation conducted so that nothing extraneous is introduced. Normally it will be easier to bring the fuel sample to the instrument than the instrument to the fuel.

7.2.1.3 Calibration of CCFD and correlation with gravimetric method. It should be recognized that this instrument is only a secondary standard that supplements laboratory analysis and does not replace the requirements for periodic laboratory gravimetric analysis. Extensive field tests have demonstrated that the calibration curve furnished with this unit is valid for the majority of fuel samples, but there are occasional samples that do not fit the normal pattern. It may become necessary to establish a new or modified calibration curve in a few unusual cases where the contaminants in a particular system do not follow normal patterns. Duplicate samples sent to the laboratory for gravimetric analysis will give a cross check on the instrument and quickly pinpoint these unusual systems. In addition, CCFD operators are requested to visually inspect the Millipore filter pads for any large particles or unusual spots and stains that may cause erratic or erroneous CCFD results. Any such situations that frequently reoccur should be reported to the Naval Air Systems Command (AIR-4.4.5) at the following address:

Commander
Naval Air Systems Command
AIR-4.4.5, Bldg. 2360
22229 Elmer Rd., Unit 4
Patuxent River, MD 20670-1906

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Telephone Numbers:
DSN — 757-3406/3410
Commercial — 301-757-3406/3410

Telefax Numbers:
DSN — 757-3614
Commercial — 301-757-3614

Message Address:
COMNAVAIRSYSCOM PATUXENT RIVER MD//4.4.5//

7.2.2 CCFD test technique. A sample of fuel to be tested is obtained in the sample bottle provided. This fuel is filtered through two membrane filters (NSN 1H 6630-00-877-3157) placed in series. The solid contaminants will be collected on the top filter. A light is shown through each filter and a meter measures the decrease in transparency of the filters due to the trapped solids. Use of two filters eliminates errors due to variations in color of different fuels. A calibration chart is provided to convert the meter readings to contamination levels in mg/liter. The special filters used to calibrate the unit are available from the Navy Ships Parts Control Center, Mechanicsburg, PA, under NSN 1H 6630-00-849-5288.

7.2.3 CCFD operating procedure. Complete details on the operation of the detector are contained in the technical manual supplied with each detector. The following procedure applies to all CCFDs in general, and may be used in the event the manual is unavailable.

1. Turn the detector on and allow it to warm up for 2 to 3 minutes.

2. Ensure the vacuum-receiving flask is empty and the drain cock closed.

3. Place two membrane filters, one on top of the other, in the bottle receiver.

Note

The membrane filters are packaged with blue paper between the filters. Dispose of the blue paper dividers. Do not use them for filtration.

4. Fill the sample bottle provided with 800 milliliters of fuel to be tested. Fit the bottle receiver onto the top of the sample bottle. Plug the grounding wire from the bottle receiver into the receptacle provided on the CCFD.

5. Turn the vacuum pump on. Holding the bottle receiver snug against the sample bottle, invert the assembly and fit the bottle receiver into the top of the vacuum flask.

6. Gently swirl the fuel in the sample bottle while the fuel is being filtered to ensure any contaminants are washed down with the fuel through the filters. The movement of the bottle should also be sufficient to vent the bottle, allowing air bubbles into the bottle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

7. When all of the fuel has passed through the filters, turn the pump off, and remove the sample bottle from the bottle receiver.

8. Drain the fuel from the vacuum receiving flask into an appropriate container for disposal.

9. With no filter in the tray, and the tray fully inserted, adjust the panel meter to a reading of 600 milliamperes.

10. Place a small amount of pre-filtered fuel in the wetting dish depression provided on the top of the detector. Open the bottle receiver and remove the top membrane filter with forceps. Gently lay the filter in the wetting dish so that the entire filter is wetted with fuel.

11. Place the wetted filter into the filter tray, insert the tray into the photocell housing, and record the milliammeter reading. Remove the filter from the tray.

12. Confirm the meter still reads 600 milliamperes with no filter in place. Repeat the wetting process with the second membrane filter, and place it in the filter tray. Insert the tray in the photocell housing and record the milliammeter reading. Remove the filter from the tray and dispose of both filters.

13. Subtract the lower of the milliammeter readings from the higher allocated set. Locate the change in reading value on the vertical left axis of the calibration chart, and read the corresponding value of contamination (where the change in reading intersects the curve) in milligrams per liter from the horizontal bottom axis of the chart. Report the contamination in milligrams per liter.

Note

Detailed instructions or assistance in operating or calibrating a specific model detector are available from Naval Air Systems Command AIR-4.4.5 (see 7.2.1.3).

7.2.4 Alternate methods. An alternate system for the field detection of particulate contaminants in aviation fuel, which is approved for use at shore facilities only, is the USAF's system of in-line sampling coupled with a visual assessment technique. Details on this method and the necessary equipment can be obtained from USAF Technical Order 42B-1-1, Quality Control of Fuels and Lubricants, 5-7. Copies of this document are available from the Naval Air Systems Command, AIR-4.4.5 (see 7.2.1.3). Another version of this visual particulate testing technique that is also approved for use at shore activities is contained in the U.S. Army's Aviation Fuel Test Kit.

7.3 Water contamination.

7.3.1 General. The viewer kit, free water detector procured under MIL-D-81227, NSN 6640-00-999-2786, is a simple, small unit for use in the field or the laboratory to determine the free

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

water content of aviation fuels. It was designed for use in conjunction with the CCFD and will accurately measure undissolved water in jet fuels. The current version of this device is sometimes referred to as “AEL Mk II,” while an earlier version was designated “AEL Mk I.” CCFDs now in use include free-water detector capability and may be used instead of a separate FWD.

7.3.2 FWD test technique. A sample of fuel to be tested is passed through a chemically treated filter pad (NSN 9L 6640-00-999-2785) by using the filter holder and vacuum pump of the CCFD. The chemical on the pad is sensitive to any free water in the fuel, producing a fluorescent pattern readily visible under ultra-violet light. After filtration, the pad is examined under the ultra-violet light contained in the CCFD. The amount of free water in the fuel sample is determined by the intensity of fluorescence on the test pad. Visual comparison is made with a series of standards representing known quantities of water. Standards, which are available under NSN 9L 6640-00-999-2784, tend to deteriorate with time and exposure to ultra-violet light. They should, therefore, be replaced every 6 months.

7.3.3 FWD operating procedure.

1. Turn the detector on and allow it to warm up for 2 to 3 minutes.

2. Ensure the vacuum receiving flask is empty and the drain cock closed.

3. Place the 47 mm free water pad into the bottle receiver.

4. Fill the sample bottle provided with 500 ml of fuel to be tested. Shake the sample vigorously for approximately 30 seconds. Fit the bottle receiver onto the top of the sample bottle. Plug the grounding wire from the bottle receiver into the receptacle provided.

5. Turn the vacuum pump on. Holding the bottle receiver snug against the sample bottle, invert the assembly and fit the bottle receiver into the top of the vacuum flask.

6. Gently swirl the fuel in the sample bottle while the fuel is being filtered to ensure any contaminants are washed down with the fuel through the filters. The movement of the bottle should also be sufficient to vent the bottle, allowing air bubbles into the bottle.

7. When all of the fuel has passed through the filters, turn the pump off, and remove the sample bottle from the bottle receiver.

8. Drain the fuel from the vacuum receiving flask into an appropriate container for disposal.

9. Remove the free water pad from the bottle receiver and place in the free-water detector slide. Insert the slide into the free-water detector and turn on the ultra-violet light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

10. Visually compare the fluorescence on the free water pad with that on the free-water standards inside the detector. The free water standards provided show the fluorescence at 0, 5, 10, and 20 ppm of free water. Estimate the corresponding free-water content by fluorescence intensity and droplet pattern on the free water pad.

Note

If the result is greater than 20 ppm, take a new sample half the volume of the original (250 ml), read the fluorescence, and double the value.

11. Report the free water results in parts per million.

Note

The free-water test utilizing the FWD should be executed as soon as possible following sampling since the results are directly affected by any temperature change in the fuel sample. This test, conducted by the sampling activity, is the only free-water determination now required and is more accurate than can be obtained from a sample sent to a laboratory.

7.3.4 Alternate method. ASTM D3240 is an approved alternate method for determining the free-water content of aviation fuels using 25 mm pads. Instruments included in this method are the Aqua Glo Water Detector and the Jet Fuel Water Analyzer.

7.4 Fuel System Icing Inhibitor (FSII) content.

7.4.1 General. The B/2 Anti-Icing Additive Refractometer and Test Kit (NSN 6630-01-165-7133) provides a simple and accurate means of determining the FSII content of aviation fuels.

7.4.2 B/2 test technique. A sample of fuel to be tested is placed in a separatory funnel along with a small quantity of tap water. After agitation, a few drops of the water layer are placed on the cell of the refractometer and a reading is taken directly from the appropriate scale. During this process the FSII is “washed” from the fuel and collects in the water layer. Since the FSII content of the water changes its refractive index, the extent of this change is used to determine the concentration of FSII in the fuel.

7.4.3 Standard B/2 operating procedures. Detailed instructions are provided with the test kit. The following procedure summarizes the operation of the kit.

1. Procure 1 pint (473 ml) of fuel to be tested in a clean and dry container.

2. Assemble the ring stand. Fill an aluminum dish one half full of water. Tap water can be used.

3. Pre-treat the graduated cylinder and separatory funnel with the test fuel. Place a small amount of fuel in the cylinder, swirl to wet the sides of the cylinder, then pour out the fuel. With the drain cock closed, place a small amount of fuel in the separatory funnel and swirl to wet the sides of the funnel. Pour the fuel out of the top of the funnel; do not use the drain cock for this step.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

4. Transfer exactly 160 ml of the fuel from step 1 to the separatory funnel. Some kits may have a separatory funnel with a line marking the 160 ml capacity instead of a graduated cylinder. Fill to that line if the kit is so equipped.

5. Using one of the piston pipettes contained in the set, add exactly 2 ml of water to the separatory funnel from the aluminum dish supply. Cap the funnel and shake vigorously for 3 minutes. Swirl funnel and place in ring stand.

6. Open the cover of the refractometer's window and make sure that it is clean. Apply several drops of water to it from the aluminum dish supply. Close the cover and observe through the eyepiece the location of the shadow in the viewer. Remove the plastic rod from the instrument's base and adjust the setscrew in the base if necessary, so that the shadow line intersects the zero line of the scale. Clean and cover the refractometer's window.

7. Carefully rotate the separatory funnel's drain cock so that a trickle of fluid can be taken in a clean, dry aluminum dish. Use two to three drops.

8. Using the same technique as step 6, transfer the fluid from the aluminum dish to the refractometer's window. Close the cover and observe the position of the shadow line. If testing JP-5 with DiEGME, read the scale on the left. If testing JP-4 with EGME, read the scale on the right. Record results.

9. Properly dispose of the fluids. Wash the apparatus in soap and water and properly dry all items. Treat the refractometer as an optical instrument and avoid damage to the lens and window elements.

7.4.3.1 Digital B/2 operating procedures. Follow manufacturer's operating instructions.

7.5 Conductivity.

7.5.1 General. The EMCEE Fuel Conductivity Meter (NSN 6630-01-115-2398) provides a simple method of measuring the electrical conductivity of aviation turbine fuels with SDA injected in them.

7.5.2 Fuel conductivity meter test procedure. A sample of the fuel to be tested is extracted into a sampling bottle or can. The meter's probe is inserted into the fuel sample and its conductivity is read directly from the meter. Detailed instructions for the calibration and use of this meter are available in ASTM D2624 and USAF Technical Order 42B-1-1, Quality Control of Fuels and Lubricants, paragraph 5-14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8. SAFETY IN FUEL HANDLING OPERATIONS

8.1 General. This chapter identifies and explains the more hazardous elements of aviation fuel handling with particular emphasis on electrostatic phenomena and their effects on fuel operations. The better these hazards are known and understood by refueling personnel, the better refueling personnel will be at avoiding or correcting unsafe situations.

8.1.1 Continuous development of safe procedures. The development of safe and efficient fuel handling and aircraft refueling procedures is a continuously evolving process. Scientific investigations are coupled with actual field experience in order to establish the safest and simplest procedures possible. One of our most important sources of information in this process often is the investigation of field accidents or problems. It is therefore extremely important that knowledgeable personnel are involved in accident investigations especially whenever explosions or fires have occurred. This section is designed to provide a basic education on the subject of the hazards of fuel handling; however, it is advisable to request the assistance and participation of experts whenever major fuels accidents are being investigated to ensure that correct conclusions are drawn. NAVAIR (AIR-4.4.5) and Naval Supply (NAVSUP) Energy Office will assist in the identification of appropriate experts for any investigation.

8.2 Abnormal fuel operations. Do not depart from the requirements and operating procedures contained in the NATOPS Aircraft Refueling Manual, NAVAIR 00-80T-109, or of the activity fuel instruction without the full cognizance of the FMO. When in doubt, operators should contact the FMO. Furthermore, fuel operators should discontinue any fuel operation that does not appear to be progressing in a normal fashion (such as appearing to be taking much longer than would normally be expected, pressures are too high, etc.) and immediately notify the FMO of their apprehensions. This is not to say that a special or one-time operation should not be conducted, but rather that it should be closely monitored and performed under the surveillance of the most knowledgeable fuel personnel available.

8.3 Fire and explosion.

8.3.1 General. Three factors are necessary for combustion of fuel — fuel in the form of vapor, oxygen from the air, and sufficient heat to raise a material to the ignition temperature. These three factors should all be present to produce a fire. The removal of any one of the factors will prevent combustion. All refueling or defueling operations contain two essential factors, fuel and air, therefore the elimination of sources of ignition is the most effective way of preventing fire. Reducing or controlling the generation of fuel vapor is extremely important in preventing fires and explosions. See the NATOPS manual for specific procedures that reduce the generation of fuel vapor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.3.2 Flammable fuel-air mixtures. The probability of a fuel vapor-air mixture being flammable is dependent on the vapor pressure and flash point of the product. Table IV provides Reid vapor pressures for aviation fuels. These properties may be used to classify refined products into low, intermediate and high vapor pressure categories.

8.3.2.1 Low vapor pressure products. Low vapor pressure products such as, JP-5, JP-8, commercial Jet A, diesel fuel, kerosene, furnace oil, safety solvents, etc., usually have flash points above 38 °C (100 °F). Since these products are normally handled at temperatures well below their flash points, no hazard is involved because no flammable vapors will develop. However, conditions for ignition may exist if these products are handled at temperatures above their flash points, if they are mixed with intermediate or high vapor pressure products, are loaded into tanks where flammable vapor may be present from previous usage (switch loading), or are splash loaded.

TABLE IV. Vapor pressures of aviation fuels.

VAPOR PRESSURE CATEGORY	FUEL TYPE	FUEL GRADES	NATO CODE	VAPOR PRESSURE, PSI (kPa)	
				MINIMUM	MAXIMUM
High	AVGAS	100LL (100, 100/130, 115/145)	F-18	5.5 (38.5)	7.0 (49.0)
Intermediate	Wide-Cut Turbine Fuel	JP-4 (Jet B)	F-40	2.0 (14)	3.0 (21)
Low	Kerosene Turbine Fuel	JP-8 (Jet A-1, Jet A, F-24, F-27)	F-34	0.0	0.1 (0.7)[1/]
Low	High Flash Kerosene Turbine Fuel	JP-5	F-44	0.0	0.035 (0.245)[1/]
Low	Kerosene Turbine Fuel	TS-1	N/A	0.0	0.342 (2.35)[1/2/]

1/ These are approximate values. The vapor pressure of kerosene fuels is indirectly limited by the flash point.
2/ This value is representative of TS-1 at 40 °C, 12 °C above the flash point.

8.3.2.2 Intermediate vapor pressure products. These materials may create flammable mixtures in the vapor space at some normal handling temperatures. Examples of these products are JP-4, commercial Jet B, and solvents such as xylene, benzene, and toluene.

8.3.2.3 High vapor pressure products. These products are so volatile that under equilibrium conditions at normal handling temperatures, between 2 °C and 38 °C (35 °F and 100 °F), they produce a “too rich” mixture to be flammable in a restricted vapor space. When high vapor pressure products are loaded into gas-free tanks, the vapor space will pass through the flammable range, but vapor just above the surface becomes over-rich almost immediately. In this and other tank filling operations, it is possible that a stratified layer of flammable vapor will be raised to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the top of the tank by the filling process. Products that do not create flammable vapor mixtures in the normal handling temperature range could do so under extreme temperature conditions. Figure 2, which shows the approximate correlation of Reid Vapor Pressure (RVP) and product temperature to the flammable range, may be useful in determining whether a flammable vapor-air mixture is likely to exist. The above information is important to understand from the safety standpoint, but in actual practice individual flammability determinations will not be made by operators. All fuel handling operations, regardless of the product, are conducted in accordance with established procedures that are designed for the most hazardous product or combination thereof.

8.3.2.4 Mixing different grades of aviation turbine fuel. The mixing of different grades of aviation turbine fuels in ground handling operations is held to the minimum level practicable; however, all fuel defueled (removed) from aircraft is almost invariably a mixture of grades. For this reason defueled fuel should be handled as a separate grade of fuel designated “JP” or “jet fuel” and not “JP-5,” “JP-8,” “JP-4,” or the equivalent analogous NATO Code Number until laboratory testing determines actual grade.

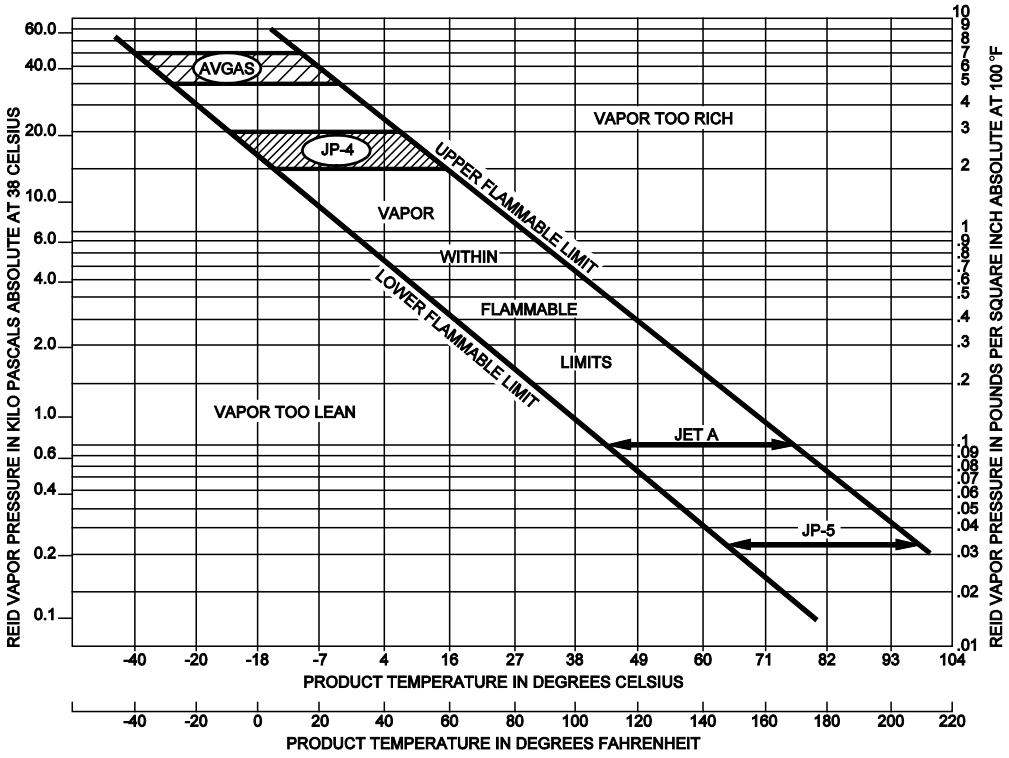


FIGURE 2. Relationship between temperature, RVP and flammable limits of petroleum products at sea level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.3.3 Flame spread rates. The Naval Research Laboratory (NRL) has investigated the relationship among a fuel's flash point, temperature, and the speed with which a flame will spread across its surface. The results of this investigation are summarized on figure 3. The data clearly discloses a dramatic change in the behavior of a fuel. Once the fuel has been heated 20 to 30 °F above 140 °F (11 °C to 16.7 °C above 60 °C) above its flash point, the flame spread rate greatly increases. This is an important factor to keep in mind when handling the various types of fuels or aircraft containing them. If fuel spills onto a hot surface such as a flight deck, it will quickly assume the temperature of that surface and behave accordingly. Avoid introducing fuels into environments where the ambient or surface temperature exceeds the flash point of the fuel by greater than 20 °F (11 °C). Figure 4 illustrates the effects on flash point of mixing the different types of turbine fuels.

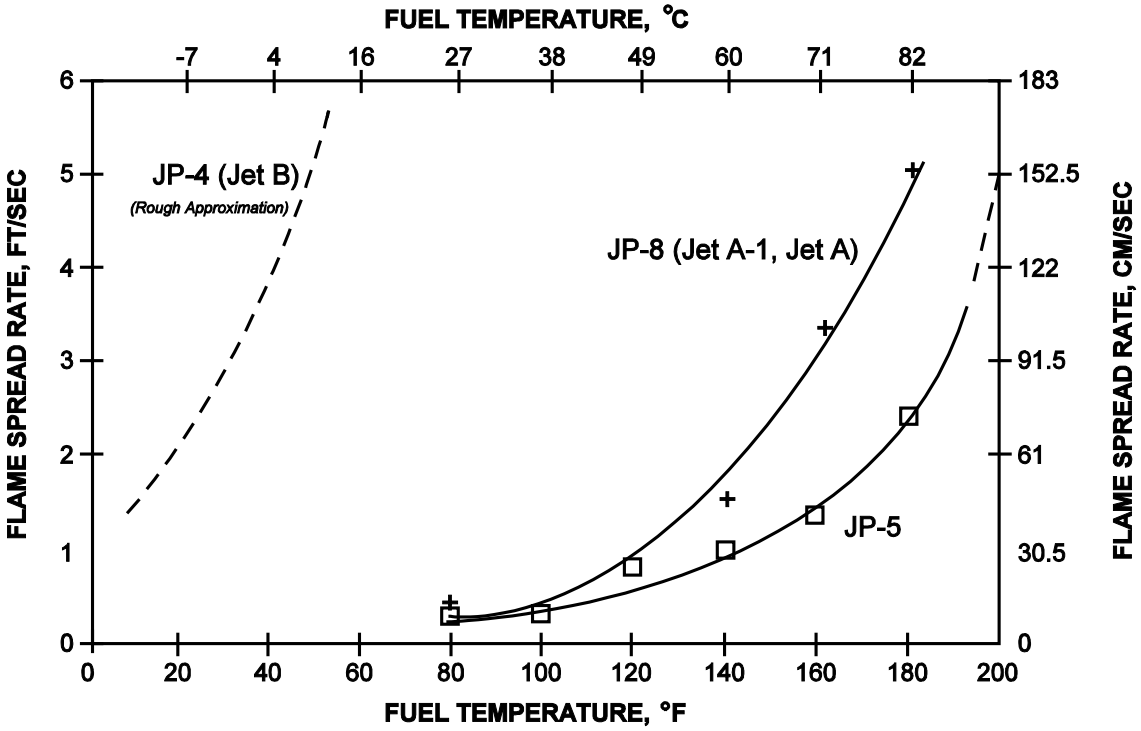


FIGURE 3. Flame spread rate versus temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

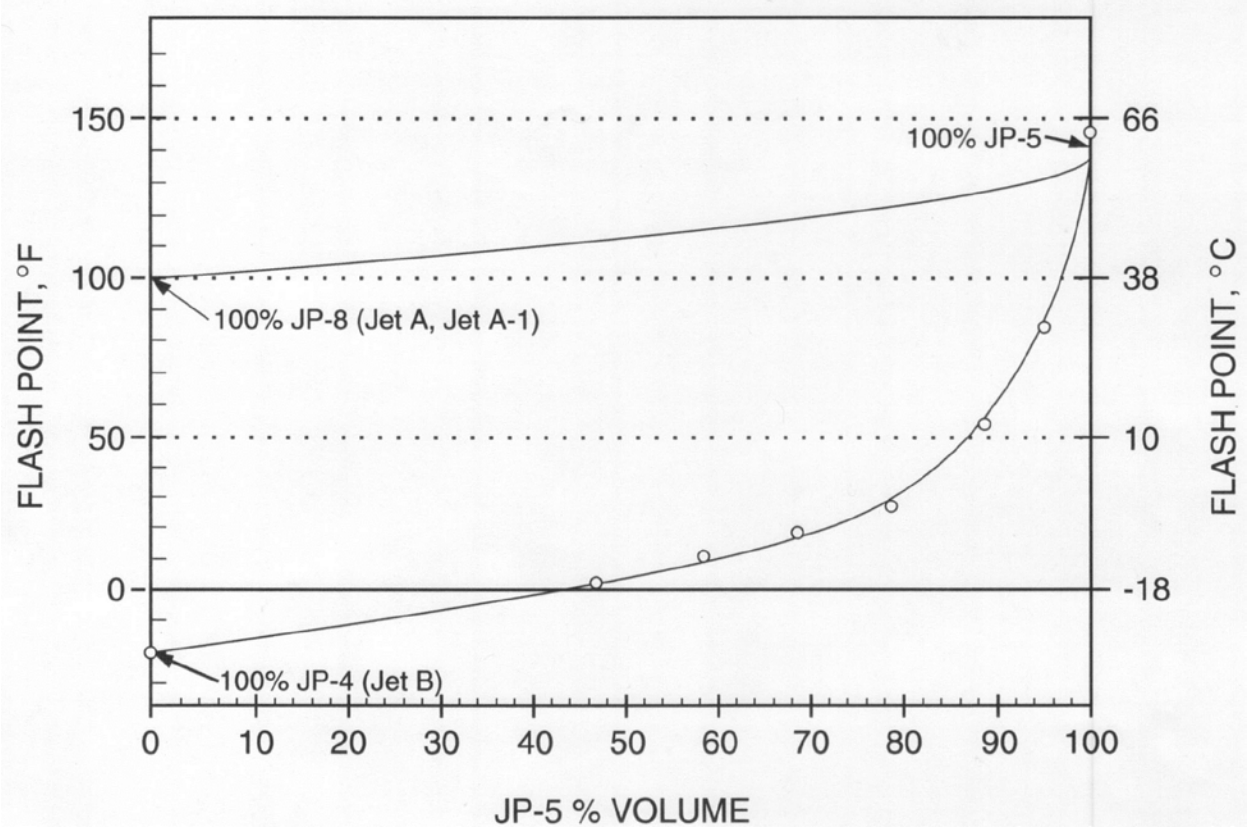


FIGURE 4. Flash point of turbine fuel mixtures.

8.3.3.1 Relationship between flashpoint, temperature and extinguishment. NRL has also demonstrated a relationship among a fuel's flash point, temperature, and the degree of difficulty in extinguishing a fire. Similar to flame spread rates, the investigators found that the fire was significantly more difficult to extinguish if the temperature of the fuel was greater than or equal to 20 °F (11 °C) above the fuel's flash point.

8.4 Sources of ignition. There are many potential ignition sources, but the ones most likely to be present during the refueling and defueling operations are:

1. Static electricity spark.
2. Operating engines.
3. Arcing of electrical circuits.
4. Open flames.
5. Electromagnetic energy.
6. Hot surfaces or environment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.4.1 Static electricity. Static electricity is the term applied to the accumulation of electrical charges on materials and objects and the subsequent recombination (relaxation or discharge) of these charges. Static charges are created when two materials (or objects of differing composition) are rubbed or passed across each other. Negative charges (electrons) migrate to one material while positive charges accumulate in the other. For significant charges to be developed, the bodies should become and remain insulated from one another so that the electrons, which have passed from one surface to the other, become trapped when separation occurs. Insulation may occur through complete physical separation of the bodies or because at least one of the bodies is a poor conductor. The development of electrical charges may not of itself be a potential fire or explosion hazard. There should be a discharge or sudden recombination of separated positive and negative charges (sparking). In order for static to be a source of ignition four conditions should be fulfilled:

1. There should be an effective means of static generation.
2. There should be a means of accumulating the separated charges and maintaining a suitable difference of electrical potential.
3. There should be a spark discharge of adequate energy.
4. The spark should occur in an ignitable mixture.

8.4.1.1 Internal static. Since the mid-1950s, nearly 200 fires and explosions that occurred during aircraft fuel handling operations have been attributed to static discharges. Of all possible sources of static build-up that could occur during fueling and defueling operations, the real danger lies in the electrostatic charges generated by the movement of the fuel itself.

8.4.1.1.1 Charge generation. Internal static charges can be generated by the following:

1. Filtration. Whenever a hydrocarbon liquid, such as jet fuel, flows past any surface, a potential difference (electrical charge) is generated between the liquid and the surface. Filtration media provide tremendous amounts of surface area upon which such charge separation can take place as the fuel moves through them. For this reason filter/separators and fuel monitors are prolific charge generators.

2. Splash filling. Charges may be generated by splashing or spraying of a stream of fuel as it enters a tank, by the disturbance of the water bottom by the incoming stream, or by pumping substantial amounts of entrained air with the fuel. The amount of charge generated by splash filling depends on the configuration of the tank, the fuel inlet, the conductivity, foaming tendency and velocity of the fuel, and the existence in the fuel of pro-static agents. Pro-static agents are materials that greatly increase the charging tendency of the fuel without a corresponding increase in conductivity. Water, for example, has been found to be an efficient pro-static agent.

3. Foam filled aircraft tanks. During the 1970s the USAF and Navy began placing polyurethane foam into aircraft tanks for fire protection purposes. The foam acts like a three-dimensional flame arrestor and suppresses ignition of fuel vapor by incendiary projectiles. Unfortunately the polyurethane foam did not prove durable and was

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

replaced with a polyester foam, which has an approximately 10 times greater tendency to create a static charge when fuel flows across it. The designs of several USAF aircraft incorporate splash filling of their tanks through this foam medium. As a result, several of these aircraft have experienced small flash fires/explosions during refueling operations. While the damage to the aircraft is limited, repairs are usually expensive. Fortunately U.S. Navy and Marine Corps aircraft do not incorporate splash filling in their tanks and have subsequently not had similar problems.

8.4.1.1.2 Charge accumulation. Since hydrocarbon fuels are relatively poor conductors of electricity, the charges built up in them relax (dissipate) slowly. It is therefore possible to pump highly charged fuel into a tank or aircraft. Once the fuel is in the tank, the unit charges of similar sign within the liquid are repelled from each other toward the outer surfaces of the liquid, including not only the surfaces in contact with the tank walls, but also the top surface adjacent to the air space. It is this “surface charge” that is of concern in fuel operations. If fuel were a perfect insulator the charge would remain indefinitely. However, there are no perfect insulators and the isolated charges in the fuel continually leak away to recombine with their counterparts on the tank shell which have migrated there from the filter/separator on the dispenser via the bonding cable. This leakage characteristic is called relaxation. Under the continuous influence of a charge generating mechanism, the accumulated voltage on the fuel (an insulated body) continues to increase. As the voltage becomes greater, the rate at which the charge will leak through the insulation (the fuel) to the tank shell will increase. At some voltage that is lower than the sparking potential, the leakage of the charge will be equal to the rate at which charge is being placed on the fuel and a stabilized condition will exist.

8.4.1.1.3 Static discharge or ignition. When sparking potential is reached before stabilization, a spark discharges from a portion of the fuel surface, the voltage immediately drops, and the entire process is resumed from the beginning. The fact that a spark occurs does not mean that a flammable mixture will ignite. For ignition, there should be sufficient energy transferred from the spark to the surrounding fuel vapor. Many factors such as material and shape of electrodes, the space of the gap, temperature, and pressure could decrease the availability of stored energy to the flammable mixture. The worst condition is a spark discharge from a floating, suspended, or attached but unbonded object in the tank since the entire amount of stored charge is released in a single discharge.

8.4.1.1.4 Control measures. The sizable static charges built up in fuel as it passes through filtration media can be effectively reduced to safe levels by passing the fuel through a relaxation chamber. Such devices slow down the velocity of the fuel and bring it into contact with metal surfaces bonded to the filter housing, thus giving the fuel both the time (a minimum of 30 seconds after it has passed through the filter) and the means to have its static charges recombine with those produced on the filter medium. This is the control approach used by the U.S. Navy and Marine Corps in fuel handling equipment. Relaxation chambers can be either actual tanks or large diameter pipes; the exact dimensions of the chamber should be determined from the maximum flow rate of the system. Tanks used for relaxation chambers should be baffled to avoid a tunneling effect as fuel enters and leaves. The relaxation chamber in a system does not negate

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

the need for bonding the refueling equipment to the aircraft. The chamber only reduces the static charge on the fuel to safer levels it does not completely eliminate the charge.

8.4.1.1.4.1 USAF approach to handling internal static charges. The USAF has taken a different approach to handling static charges generated in fuels. SDA is injected into JP-4 and JP-8 fuels to increase the fuel's conductivity and therefore the speed with which the static charges are relaxed. If the fuel's conductivity is high enough, charges will be sufficiently reduced prior to the fuel exiting the filter housing. SDAs tend to be lost in the supply system and should be periodically reinjected, thus, requiring frequent monitoring of the fuel's conductivity level as well as facilities for reinjection.

8.4.1.1.4.2 Procedures for reducing internal static charges. The NATOPS Aircraft Refueling Manual contains a detailed list of procedures and actions that can be followed to help ensure the safe relaxation of internal static charges.

8.4.1.2 External static. Accident records indicate that the fire and explosion hazard from exterior static discharges during fuel operations is minimal; however refueling personnel should be aware of the possible sources of external static and take proper precautions to further minimize and eliminate its impact.

8.4.1.2.1 Charge generation. There are five possible sources of exterior electrostatic charges on aircraft:

1. Movement of airborne charged particles, such as, snow, ice crystals, dust, or smoke.

2. Aircraft engines turning or other systems operating on the aircraft.

3. Cloud induced fields. The heavily charged clouds of an electrical storm set up an electrostatic field over a large area of the earth's surface below them. The "field" induces static on the earth, aircraft, refueler, etc.

4. Movement of a charged object onto aircraft. Fitting in this category is static from over-the-wing nozzles, probably transferred from personnel while the nozzle was insulated from both the refueler and aircraft prior to the initiation of fuel flow.

5. Lightning. Electrostatic currents resulting from both distant as well as direct lighting strikes are more severe by orders of magnitude than those from any other source.

8.4.1.2.2 Charge accumulation and dissipation. The amount of electrostatic charge that can accumulate on an insulated body, such as a rubber-tired aircraft or refueler, depends upon the rate at which the static is generated and the resistance of the paths by which the charge leaks off. On concrete parking aprons the tire-surface contact-resistance paths will provide sufficient discharge capability to prevent the accumulation of static charges. Asphalt, especially when dry, is a much better insulator than concrete and therefore dissipates static charges much more slowly. The extremely large static charges that have accumulated on aircraft during flight are dissipated in short periods of time after landing; but fueling operations should not commence until adequate time has elapsed to ensure the dissipation of accumulated charges. Turning

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

engines and helicopter rotors are also strong static generators and sufficient time should be allowed after they have been turned off for the charges created to dissipate. Since most refueling operations are conducted on concrete, less than a minute is needed to completely relax these charges; however, waiting 3 minutes will ensure safety. The refueling equipment should always be electrically bonded to the aircraft or truck into which the fuel is being loaded. Any refueling operations conducted with aircraft engines or other equipment operating (with the exception of fuel system switches) should be considered abnormal and also subject to special procedures including earthing.

8.4.1.2.3 Control measures for external static. During normal fueling and defueling operations on concrete surfaces, tire-surface contact can be relied upon to prevent the accumulation of static charge. Specific procedures to be followed are contained in the NATOPS Aircraft Refueling Manual.

8.4.2 Operating engines. The operation of aircraft engines, automobile engines, or other internal combustion engines can provide sources of ignition. Ignition of vapors may occur through the arcing of distributor points, arcing at spark plugs, loose spark plug wires, hot engine exhaust piping, and backfiring. The starting of engines with their susceptibility to false starts, malfunctions, and fires during the cranking phase, are likely sources of fuel vapor ignition. Stopping an engine, as opposed to leaving it running, is more apt to provide an ignition source because of possible dieseling, backfiring, or malfunctioning when the engine's ignition is turned off.

8.4.3 Arcing of electrical circuits. Arcing of electrical circuits is another common source of ignition in fuel handling operations. Sparks may occur when battery terminals are connected or when an electrical switch is operated. Other examples of sparks from electrical circuits are arcing of generator brushes, arcing of welding machine brushes, arcing of brushes on electrical motors and tools, and the arcing that occurs in short circuits. Defective aircraft electrical circuits have been known to melt the plastic coatings on bonding and earthing cables during refueling operations; cables have even turned cherry-red, burned, and disintegrated, moments after the earth cable was attached to the earth point. Static earthing cables are not adequate protection against such stray currents.

8.4.4 Open flames. Open flames and lights are obvious sources of ignition. These include cigarettes, cigars, pipes, exposed flame heaters, welding torches, blow torches, flare pots, gasoline or kerosene lanterns, matches, cigarette lighters, and others. Similar to this hazard of open flames is that of standard electric light bulbs. Should any of these bulbs break, the filament will become hot enough to ignite a vapor-air mixture and cause a fire or explosion.

8.4.5 Electromagnetic energy. High-frequency transmitting equipment, including radar, mounted in aircraft or mobile units, automotive equipment, or on the ground can provide sufficient electromagnetic energy to ignite fuel vapors. For this reason, refueling or defueling operations should not be conducted within 30 meters (100 feet) of operating airborne radar equipment, or within 91 meters (300 feet) of operating ground radar equipment installations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.4.6 Hot surfaces or environment. Fuel or fuel vapors can come into contact with surfaces sufficiently hot to cause auto ignition of the fuel or vapor. Examples are a high-pressure fuel leak hitting a hot brake or metal particles from a welding, filing, or cutting operation contacting the fuel vapor. Some areas of the aircraft, such as the bomb bay, may provide a hot enough environment for auto ignition of any fuel that leaks into the area. Although actual auto ignition temperatures may vary a great deal depending on the specific fuel involved and its environment, the minimum temperatures can safely be assumed to be above 700 °F for turbine fuels.

8.5 Extinguishment.

8.5.1 General. Although the Air Station's Crash Crew (Crash, Fire, and Rescue) or Ship's Crash Crew has primary responsibility for firefighting, all fuel handling personnel should be aware of the basic principles involved in extinguishing fires as well as the equipment used. They should also make certain that appropriate firefighting equipment, in good condition, is readily available whenever and wherever fuel handling operations are being conducted. For maximum effectiveness and safety, fire extinguishers should be operated in accordance with the specific procedures developed for each individual type. The following information has been extracted from the U.S. Navy Crash, Fire and Rescue NATOPS Manual, NAVAIR 00-80R-14, and is presented here to provide refueling personnel with general information on fire chemistry, classification, and methods of extinguishment.

8.5.2 Fire chemistry. For many years, fire was considered to be the product of a combination of three elements — fuel, an oxidizing agent, and temperature. Research in the past 30 years has indicated the presence of a fourth critical element. The fourth element is the chemical chain reaction that takes place in a fire and allows the fire to both sustain itself and grow. For example, in a fuel fire, as the fire burns, fuel molecules are reduced within the flame to simpler molecules. As the combustion process continues, the increase in the temperature causes additional oxygen to be drawn into the flame area. Then, more fuel molecules will break down, enter into the chain reaction, reach their ignition point, begin to burn, cause a temperature increase, draw additional oxygen, and continue the chain reaction. As long as there is fuel and oxygen and as long as the temperature is sustained, the chain reaction will cause the fire to grow.

8.5.3 Classification of fires.

8.5.3.1 Class A fires. Class A fires, which include burning wood and wood products, cloth, textiles and fibrous materials, paper and paper products, are extinguished with water in straight or fog pattern. If fire is deep seated, aqueous film forming foam (AFFF) can be used as a wetting agent.

8.5.3.2 Class B fires. Class B fires, which include gasoline, jet fuels, oil, and other flammable/combustible liquids, may be extinguished with AFFF, Halon 1211, purple-K-powder (PKP), or carbon dioxide (CO_2).

8.5.3.3 Class C fires. A Class C fire involves energized electrical equipment. Extinguishment tactics are: de-energize and treat as a Class A, B, or D fire; attack with application of non-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

conductive agents (CO_2, Halon, PKP); or attack with application of fresh or salt water in fog patterns maintaining nozzle at least 4 feet from the energized object.

8.5.3.4 Class D fires. Class D fires, which include combustible metals such as magnesium and titanium, are extinguished with water in large quantities such as high-velocity fog. When water is applied to burning Class D material, there may be small explosions. The firefighter should apply water from a safe distance or from behind shelter.

8.5.4 Fire extinguisher types, agents, and methods of application. The best application technique varies with the type of extinguishing agent and associated hardware. Some fire extinguishers deliver their entire quantity of extinguishing agent within 10 seconds, while others are designed to be operated for 30 seconds or longer. The agent should be applied correctly at the outset since there is seldom time to experiment. Using a portable extinguisher at too close a range may scatter the fire; using it at a distance beyond the effective range will simply waste the extinguishing agent.

WARNING

- Firefighters should use caution in fighting fuel fires and be prepared to back out well before the extinguisher contents are exhausted.

- Halon, PKP, and CO_2 are all rapidly dissipated and no vapor sealing property is developed, so the fuel is always subject to reignition. Discharge should be continued for a short time after the flames are extinguished to prevent possible reflash and to cool any ignition sources in or near the fire.

WARNING

Portable and wheeled Halon, PKP, and CO_2 extinguishers should be discharged in an upright position. If the extinguisher is on its side or inverted, the siphon tube will not reach the agent and an unsatisfactory discharge will result.

8.5.4.1 Halon 1211 (Bromochlorodifluoro-methane) portable and wheeled unit extinguishers. These extinguishers are intended primarily for use on Class B and C fires; however, Halon 1211 is effective on Class A fires. Halon 1211 is a colorless, faintly sweet smelling, electrically nonconductive gas that leaves no residue to clean up.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.5.4.1.1 Extinguishment mechanism. Halon 1211 extinguishes fires by inhibiting the chemical chain reaction of the combustion process. Halon 1211 is virtually non-corrosive, nonabrasive, and is at least twice as effective as CO_2 on Class B fires when compared on a weight-of-agent basis. Although the agent is retained under pressure in a liquid state and is self-expelling, a booster charge of nitrogen is added to ensure proper operation. Upon actuation, the vapor pressure causes the agent to expand so that the discharge stream consists of a mixture of liquid droplets and vapor. Halon 1211 extinguishers are marked with a reflective silver band around the tank.

WARNING

Do not use Halon 1211 on Class D fires. It has no blanketing effect and, if it reaches a Class D fire in the liquid state, the possibility of an explosive reaction exists.

WARNING

The discharge of Halon 1211 to extinguish a fire creates a hazard to personnel, such as dizziness and impaired coordination, from the natural Halon 1211 product and from the products of decomposition that result through exposure of the agent to the fire. Exposure to the agent is of less concern than is the exposure to the products of decomposition. In using extinguishers of this type in unventilated spaces or confined areas, operators should use positive-pressure, self-contained breathing apparatus.

8.5.4.1.2 Application. Initial application should start close to the fire. On all fires, the discharge should be directed at the base of the flames. Sweep the agent stream back and forth across the leading edge of the fire, overshooting on both sides, and continue to push the leading edge of the fire back until the fire is extinguished. These units have an effective discharge range of 10 to 30 feet, depending on ambient conditions, and a discharge time of 15 to 40 seconds, depending on the extinguisher size and application rate.

8.5.4.2 Carbon Dioxide (CO_2) 15-pound portable units and 50-pound wheeled extinguisher units. These units are intended primarily for use on Class B and C fires. CO_2 is a colorless, odorless gas that is approximately 1-1/2 times heavier than air. It is stored in rechargeable containers designed to hold pressurized carbon dioxide in liquid form at atmospheric temperatures. Fire suppression is accomplished by the displacement of oxygen in the atmospheric to a level below the percent required to support combustion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

WARNING

Exposure to CO_2 in high concentrations for extended periods of time can be fatal.

WARNING

The use of portable CO_2 extinguishers to inert flammable atmospheres is prohibited. When a portable CO_2 extinguisher is discharged, the liquid CO_2 expanding through the nozzle and cone becomes solid (commonly called “snow”). This “snow” contacting and separating from the extinguisher cone becomes electrically charged, as does the extinguisher itself. If the charged “snow” contacts an insulated metal object, it will cause the object to become charged. Tests indicate that voltages greater than 15 kilovolts can be developed on insulated metal objects from a 1- to 2-second application of CO_2 from an extinguisher. This voltage is sufficient to cause a spark.

8.5.4.2.1 Application. Agent application should commence at the upwind edge and be directed slowly in a side-to-side sweeping motion, gradually moving toward the back of the fire. These extinguishers have a limited discharge range of 3 to 8 feet and a discharge time of 8 to 44 seconds, depending on the extinguisher size and application rate.

8.5.4.3 Purple-K-Powder (PKP) dry chemical powder extinguishers. These extinguishers are intended primarily for use on Class B fires. The principal base chemical used in the production of PKP dry chemical agent is potassium bicarbonate. Various additives are mixed with the base material to improve its stowage, flow, and water repellency characteristics. The ingredients used in PKP are nontoxic; however, the discharge of large quantities may cause temporary breathing difficulty, may seriously interfere with visibility, and may cause disorientation. Dry chemical agent does not produce a lasting inert atmosphere above the surface of flammable liquid; consequently, its use will not result in permanent extinguishment if reignition sources are present. These extinguishers are marked with a purple band around the tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION

- Chemical agents may harden after being exposed to moisture. It is therefore important to avoid exposing them to any moisture during stowage, handling, and recharging evolutions.

- When PKP is used as the fire suppression agent on an aircraft fire and the agent is directed or ingested into an engine or accessory section, the fire chief, on-site personnel using the extinguisher, or senior fire official should notify the maintenance officer of the unit involved or, in the case of a transient aircraft, the supporting facility. PKP injected into a jet engine cannot be completely removed without disassembly of the engine to remove deposits that penalize engine performance and restrict internal cooling air passages.

8.5.4.3.1 Application. These extinguishers have a discharge range of approximately 3 to 30 meters (10 to 40 feet), depending on extinguisher size. Discharge time varies from 8 to 60 seconds. When used on flammable liquid fires, the stream should be directed at the base of the flame, gradually moving toward the back of the fire while sweeping the nozzle rapidly from side to side.

8.5.5 Fire extinguisher requirements. Figure 5 lists the requirements of fire extinguishers around different types of operations. This figure has been extracted from NAVAIR 00-80R-14, which is the controlling document and should be consulted for the latest requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TYPE OF OPERATION	FIRE PROTECTION REQUIREMENT
Note All references to Halon 1211 extinguishers are for the 150-pound wheeled unit. Current in-service 150-pound PKP extinguishers are acceptable until Halon units can be placed in service.	
FLIGHT LINE PARKING AREA	
A. SMALL OR MEDIUM TYPE AIRCRAFT C-12, T-2, T-34, T-38, T-44, T-45, TA-4, EA-6B, AV-8, F-4, F-14, F-15, F-16, F-117, F-22, F/A-18, JSF, T-39, S-3, E-2, C-2, UH-1, AH-1, SH-2, H-3, TH-57, H-46, H-60	1 Halon 1211 extinguisher per 3 aircraft.
B. LARGE TYPE AIRCRAFT P-3, C-9, C-20, DC-10, C-5 (See Note), C-17, C-130, C-131, C-141, E-6A, KC-135, 707, 727, 737, 747, L-1011, KC-10, H-53, V-22	1 Halon 1211 extinguisher per 2 aircraft.
Note • Two Halon 1211 extinguishers are required for each C-5 aircraft. • If any substitutions are required, the substituted unit should be approved by the fire department of CFR unit. As a general rule, any substitute portable extinguishing unit should have the same extinguishing capability as the Halon unit replaced.	
C. HOT REFUELING — Transfer of fuel into aircraft tanks with one or more aircraft engines operating.	The minimum fire protection shall be two 150-pound Halon 1211 wheeled units positioned at each lane/island.
Note A crash fire vehicle (TAU) shall be on standby alert during multiple aircraft hot refueling operations. The location and response criteria will be determined by the Fire Chief/CFR Officer.	
D. CONCURRENT FUELING AND SERVICING OR AIRCRAFT WITHOUT PASSENGERS	1 Halon 1211 extinguisher. Additionally, one major crash fire vehicle shall be capable or responding to the scene within 3 minutes.
E. FUELING OF AIRCRAFT WITH PASSENGERS	1 Halon 1211 extinguisher. Additionally a TAU or major crash fire vehicle shall be positioned at the aircraft.
Note Fire department/CFR unit shall be alerted at least 15 minutes prior to commencement of fueling operations. The number of passengers on board the aircraft shall be included in the notification.	

FIGURE 5. Airfield fire protection requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TYPE OF OPERATION	FIRE PROTECTION REQUIREMENT
F. FUELING OR SERVICING OF MEDICAL EVACUATION FLIGHTS WITH PASSENGERS/PATIENTS ON BOARD	1 150-pound Halon 1211 wheeled extinguisher. Additionally, one major CFR vehicle shall be positioned at the aircraft for optimum response. Turrets shall be manned and agent pumping equipment/systems ready for instant activation.
Note Fire department/CFR unit shall be alerted at least 15 minutes prior to commencement of fueling operations. The number of passengers on board the aircraft shall be included in the notification.	
G. HIGH POWER AND NEW ENGINE TURN-UP	2 Halon 1211 extinguishers located in immediate vicinity. One major CFR vehicle shall be capable of responding to the site within 3 minutes.
Note Fire department/CFR unit shall be notified at least 15 minutes prior to commencement of new engine turn-up.	
H. COMBAT AIRCRAFT ORDNANCE LOADING AREA	1 Halon 1211 extinguisher per 2 aircraft.
Note Fire department/CFR unit shall be notified of daily ordnance loading/unloading schedule.	

FIGURE 5. Airfield fire protection requirements – Continued.

8.6 Health hazards.

8.6.1 General. Aviation fuels should be handled with caution because of the obvious dangers associated with possible fires, explosions or both. These materials also present a danger to the health of fuel handling personnel. These dangers are equally important as those of fires and explosions even though they are not so well known. In an attempt to comply with a recently enacted California law, Exxon Company U.S.A., has issued the following warning statement to inform people of the potential health hazards associated with all petroleum products including fuels:

> Detectable amounts of chemicals known to the State of California to cause cancer, birth defects or other reproductive harm may be found in petroleum products, intermediate products, by-products, waste products, and chemical products, and their vapors or result from their use. Read and follow label directions and other

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

product safety and health information that you have been provided when handling or using these materials.

8.6.2 Principal health hazards. The following paragraphs discuss the principal health dangers from aviation fuels. Once these dangers are known and understood, they can be easily avoided by controlling or minimizing exposure of personnel to both the liquid and vapor forms of the fuels.

8.6.2.1 Toxic vapor effect. One of the greatest health hazards to fuel handling personnel is the toxic effect of the fuel vapors. Only a very small amount of fuel vapor in the air is sufficient to cause harmful effects. Since most fuel handling operations are usually conducted in the open, fuel vapors are sufficiently diluted by the ambient air so that there is no hazard to personnel. In the event of a fuel spill or a leak in an enclosed area, personnel should be very careful to avoid inhalation of the fuel vapors.

8.6.2.1.1 Symptoms and remedial action. The first symptoms of the toxic effect of breathing fuel vapors are nausea, dizziness, and headaches. If any of these symptoms are noted while conducting fuel handling operations, personnel should immediately stop the operation and move to a fresh air location. If personnel are overcome by vapors, they should receive prompt medical attention. First aid procedures for personnel overcome by vapors include removal to fresh air, treatment for shock, and administering of artificial respiration if breathing has stopped.

8.6.2.2 Lead poisoning. The tetraethyl lead that is added to all grades of AVGAS is very poisonous. It is harmful if the vapors are inhaled or if the compound enters the body through the mouth by skin contact. The principal danger of lead poisoning occurs when it is necessary to enter or repair containers that have been used for leaded gasolines. Refueling and defueling personnel normally will not have to do such work, but it is important that they be aware of the danger, and that they know never to enter a tank or vessel which has contained leaded gasoline until the necessary safety precautions have been followed. The toxic effect of tetraethyl lead may also occur from prolonged exposure to gasoline vapors or liquids and, therefore, such exposure should be avoided. Another danger from leaded gasoline is from the fumes given off by stoves or lanterns burning a gasoline containing tetraethyl lead. For this reason, leaded gasoline should never be used for such purposes.

8.6.2.3 Injury to skin and eyes. AVGAS and jet engine fuels may cause irritation if brought in contact with the skin. For this reason, direct skin contact with fuel should be avoided. If fuel is accidentally spilled on personnel, affected clothing should be removed immediately. Any skin areas exposed to fuel should be promptly washed with soap and water.

Note

If a person accidentally gets gasoline or jet fuel in the eyes, the fuel should be immediately removed by rinsing them with plenty of water. Medical attention should be administered as soon as possible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

8.6.2.4 Swallowing aviation fuels. Swallowing even small amounts of aviation fuels is very harmful and medical attention should be obtained immediately.

8.6.2.5 Fuel tank and filter/separator water bottoms. The water that collects in the bottom of fuel tanks and filter/separator often contains a high percentage (40 to 60 percent) of fuel system icing inhibitor (FSII). Since FSII (DiEGME) is considered toxic in a pure or concentrated state, exposure of personnel to water bottoms of filter separators and tanks should be kept to the absolute minimum. If exposure occurs, wash with soap and water followed by a thorough fresh water rinse. Disposal of water bottoms should be accomplished in accordance with local environmental regulations.

8.6.2.6 Specific procedures for avoiding the health hazards of aircraft fuels. The Aircraft Refueling NATOPS Manual (NAVAIR 00-80T-109) contains a list of procedures which will minimize the dangers to the health of fuel handling personnel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

9. NOTES

9.1 Intended use. This handbook is intended for use by Navy and Marine Corps aircraft refueling organizations afloat and ashore.

9.2 Subject term (key word) listing. The following subject terms (key words) are applicable to this handbook:

AVGAS
Aviation turbine fuel
B/2 Test Kit
Biostat
CCFD
Conductivity
Fire extinguisher
Free water
FSII
FWD
Icing inhibitor
JP-4
JP-5
JP-8
Microbiological growth
Quality surveillance
SDA
Static electricity
Tetraethyl lead
Water bottoms

9.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

VISUAL CONTAMINATION

A.1 SCOPE

A.1.1 Scope. This appendix describes, in tabular format, the most commonly seen types of aviation fuel contamination along with their visual appearance, effect on aircraft, and acceptable limits.

A.2 CONTAMINATION

A.2.1 Visual contamination table. See Table A-I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE A-I. Visual contamination.

CONTAMINANT	APPEARANCE	CHARACTERISTICS	EFFECTS OF AIRCRAFT	ACCEPTABLE LIMITS
A. Water				
1. Dissolved Water	Not Visible	Fresh water only. Precipitates out as cloud when fuel is cooled.	None unless precipitated out by cooling of fuel. Can then cause ice to form on low pressure fuel filter if fuel temperature is below freezing.	Any amount up to saturation.
2. Free Water	Light cloud. Heavy Cloud. Droplets adhering to sides of bottle. Gross amounts settled in bottom.	Free water may be saline or fresh. Cloud usually indicates water-in-fuel emulsion.	Icing of fuel system — usually low pressure fuel filters. Erratic fuel gauge readings. Gross amounts of water can cause flame-outs. Sea water will cause corrosion of fuel system components.	Zero — fuel should contain no visually detectable free water.
B. Particulate Matter				
1. Rust	Red or black powder, rouge, or grains. May appear as dye-like material in fuel.	Red rust (Fe_2O_3) — nonmagnetic Black rust (Fe_3O_4) — magnetic Rust generally comprises major constituent of particulate matter.	Will cause sticking and sluggish or general malfunction of fuel controls, flow dividers, pumps, nozzles, etc.	Fuel should contain less than 2 mg/liter.
2. Sand or Dust	Crystalline, granular, or glass-like.	Usually present and occasionally constitutes major constituent.	Will cause sticking, and sluggish or general malfunction of fuel controls, flow dividers, pumps, nozzles, etc.	Fuel should contain less than 2 mg/liter.
3. Aluminum	White or grey powder or paste.	Sometimes very sticky or gelatinous when wet with water. Often present and occasionally represents major constituent.	Will cause sticking, and sluggish or general malfunction of fuel controls, flow dividers, pumps, nozzles, etc.	Fuel should contain less than 2 mg/liter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE A-I. Visual contamination – Continued.

CONTAMINANT	APPEARANCE	CHARACTERISTICS	EFFECTS OF AIRCRAFT	ACCEPTABLE LIMITS
C. Microbiological Growth	Brown, gray, or black. Stringy or fibrous.	Usually found with other contaminants in the fuel. Very light weight; floats or "swims" in fuel longer than water droplets or solid particles. Develops only when free water is present.	Fouls fuel quantity probes, sticks, flow dividers, makes fuel controls sluggish.	Zero.
D. Emulsions				
1. Water-in-fuel Emulsions	Light cloud. Heavy cloud.	Finely divided drops of water in fuel. Same as free water could. Will settle to bottom in minutes, hours, or weeks depending upon nature of emulsion.	Same as free water.	Zero — Fuel should contain no visually detectable free water.
2. Fuel and water or "stabilized"	Reddish, brownish, grayish or blackish. Sticky material variously described as gelatinous, gummy, or like catsup or mayonnaise.	Finely divided drops of fuel in water. Contains rust or microbiological growth that stabilizes or "firms" the emulsion. Will adhere to many materials normally in contact with the fuels. Usually present as "globules" or stringy, fibrous-like material in clear or cloudy fuel. Will stand from days to months without separating. This material contains half to three-fourths water, a small amount of fine rust or microbiological growth and is one-third to one-half fuel.	Same as free water and sediment, only more drastic. Will quickly cause filter plugging and erratic readings in fuel quantity probes.	Zero.
E. Miscellaneous				
1. Interface Material	Lacy bubbles or scum at interface between fuel and water. Sometimes resembles jellyfish.	Extremely complicated chemically. Occurs only when emulsion and free water are present.	Same as microbiological growth.	Zero — There should be no free water.
2. Air Bubbles	Cloud in fuel.	Disperses upward within a few seconds.	None.	Any amount.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PETROLEUM TESTING LABORATORIES

B.1 SCOPE

B.1.1 Scope. This appendix provides guidance on the two types of aviation fuel samples and provides the addresses of military petroleum testing laboratories.

B.2 INFORMATION

B.2.1 General Information. Aviation fuels require quality surveillance from the point of initial acceptance until they are actually used in the aircraft. Every activity and individual in the supply system that transports, stores, distributes, or issues these products is responsible for some phase of quality surveillance. Two types of samples are required to ensure quality fuel for aircraft: Special and Routine. Definitions for each of these types of samples and procedures for field personnel to follow in submitting them are contained in the following paragraphs.

B.3 SAMPLES

B.3.1 Special samples. These are samples that are submitted for testing because the quality of the fuel is suspect either as a result of aircraft malfunctions or for other reasons.

B.3.1.1Shipping Destinations. Special samples should be taken in duplicate and shipped to both the Naval Air Systems Command (AIR-4.4.5) and to the nearest regional laboratory contained in table B-I (regional laboratories are indicated by an asterisk (*) next to their location). AIR-4.4.5 is available to assist field personnel worldwide in the identification and resolution of quality problems and is responsible for determining whether off-specification fuel can be issued to aircraft. AIR-4.4.5's address and telephone numbers are listed in table B-II. NAVSUP Energy Office (see table B-III for address) should be informed of all special samples sent when off specification fuel is suspected.

B.3.1.2 Sample size. Each laboratory will require a minimum quantity of 1 gallon (3.8 liters). This means two 1-gallon samples of fuel should be extracted from each problem location. Ship both samples by the fastest traceable means — the first gallon to AIR-4.4.5 and the second to the nearest regional laboratory.

B.3.1.3 Shipping container. Use the standard MIL-K-23714 specification red shipping container or other International Air Transport Association (IATA) approved packaging.

B.3.1.4 Documentation. When a fuel quality problem is encountered the activity should contact AIR-4.4.5 via telephone immediately and follow up with a naval message, telefax, or letter that describes the problem and identifies a point of contact. All samples should be properly labeled with all pertinent information. (See sections 3.3.3, 9.3.3 or 15.3.3 in the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

B.3.2 Routine samples. Routine samples are samples taken when no fuel problems or aircraft problems attributable to the fuel are known or suspected. These samples and their test results serve two purposes:

1. They assist the activity in monitoring the performance of their local fuel testing equipment and methods.

2. They provide TYCOM and SYSCOM cognizant offices with information on the general quality of the fuel delivered to aircraft and the performance of the fleet's quality control equipment.

Routine sample size and submission frequency are contained in the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, sections 3.3, 9.3, and 15.3. Routine fuel samples should be shipped to the nearest or most convenient laboratory contained in table B-I.

B.4 PETROLEUM TESTING LABORATORIES

B.4.1 Petroleum laboratories listing. The laboratories listed in table B-I are equipped to perform all tests required by aviation fuel specifications. All samples sent for testing should be properly labeled with all pertinent information as specified in NAVAIR 00-80T-109 Aircraft Refueling NATOPS Manual sections 3.3.3, 9.3.3, and 15.3.3. Special samples (see B.3.1) should be shipped to the closest laboratory annotated with an asterisk (*) next to its location in table B-I.

TABLE B-I. Petroleum testing laboratories.

Location	Laboratory Shipping Address	Lab Mailing Address
East Coast		
*Norfolk, VA	Mid-Atlantic Fuels Testing Laboratory 9673 Virginia Ave. Building W-388, Code 134.14 Norfolk, VA 23511-3323 Telephone: DSN — 564-4364 Commercial — 757-444-2761(Lab)	Same as shipping address.
Jacksonville, FL	NAVSUP FLC Jacksonville Fuel Depot Lab 8808 Somers Road, Bldg. 56 Jacksonville, FL 32226-2600 Telephone: DSN — 942-4907 Commercial — 904-696-6556	Same as shipping address.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE B-I. Petroleum testing laboratories – Continued.

Location	Laboratory Shipping Address	Lab Mailing Address
*Dayton, OH	Director, Aerospace Fuels Laboratory HQ AFPET/PTPLA 2430 C Bldg. 70, Area B Wright-Patterson AFB, OH 45433-7632 Telephone: DSN — 785-0739 Commercial — 937-255-0739	Same as shipping address.
West Coast		
*San Diego, CA	NAVSUP FLC San Diego Bldg. 70A 199 Rosecrans San Diego, CA 92106 Telephone: DSN — 553-1326 Commercial — 619-553-1326	Director, Fuel Department FISC San Diego Code 700 937 North Harbor Dr., Suite 480 San Diego, CA 92132-0480
Seattle, WA	NAVSUP FLC Puget Sound 7501 Beach Drive East Port Orchard, WA 98366 Telephone: DSN — 439-2135 Commercial — 360-476-3724	Director, Fuel Department (Code 700) FISC Puget Sound P.O. Box 8 Manchester, WA 98353-0008
Pacific		
*Pearl Harbor, HI	NAVSUP FLC Pearl Harbor POL Lab Director, Fuel Department 194 Gaffney Street Suite 100 Bldg 1685 Pearl Harbor, HI 96860-5300 Telephone: DSN — 471-9344 Commercial — 808-471-9344	Same as shipping address.
Note: NSC Pearl Harbor facilities should be used by units in the Central Pacific.		
Guam, Marianas Island	Commander (Code 701) FLC Yokosuka Guam Area Fuel Division Laboratory PSC 455, Box 190 FPO AP 96540-1500 Telephone: DSN — 339-7106 Commercial — 671-343-7106	Same as shipping address.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE B-I. Petroleum testing laboratories – Continued.

Location	Laboratory Shipping Address	Lab Mailing Address
Sasebo, Japan	Officer in Charge (Code 804) U.S. FISC Yokosuka, Det Sasebo PSC 476, Box 7 FPO AP 96322-1504 Telephone: DSN — 252-4136/4126 Commercial — 87-956-24-4136/4126	Same as shipping address.
Okinawa, Japan	Director, Aerospace Fuels Laboratory AFPA/PTPLG Unit 5161, Bldg 854 Kadena AB Japan APO AP 96368-5161 Telephone: DSN — 315-634-1602 Commercial — 81-611-734-1602	Same as shipping address.
Yokosuka, Japan	Fuel Department Director FISC Code 700 PSC 473, Box 11 FPO AP 96349-1500 Telephone: DSN — 244-2733 Commercial — 87-311-744-2733	Same as shipping address.
North Atlantic		
*Keflavik, Iceland	Commanding Officer NAS Keflavik, IC Fuels Officer (Code 403) PSC 1003, Box 32 FPO AE 09728-0332 Telephone: DSN — 450-6299 Commercial — 611-354-425-6299	Same as shipping address.
*Suffolk, England UK	Director, Aerospace Fuels Laboratory OL SA-ALC/SFTLF Bldg. 725 West Row Gate #6 Mildenhall AB, UK APO AE 09459-5000 Telephone: DSN — 314-238-2043/2797 Commercial — 011-44-163-854-2043/2797	OL SA-ALC/SFTLF Unit 5025 APO AE 09459-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE B-I. Petroleum testing laboratories – Continued.

Location	Laboratory Shipping Address	Lab Mailing Address
Mediterranean and Middle East		
*Rota, Spain	US Naval Station Rota Fuels Division Base Naval De Rota Aptd 33 – Edificio 600 Rota, Cadiz 11520, Spain Telephone: DSN — 727-2569 Commercial — 34-956-82-4101/4010	Same as shipping address
Indian Ocean		
*Diego Garcia	Commanding Officer U.S. Naval Support Facility Diego Garcia Attn: Petroleum Testing Lab Code 407 PSC 466, Box 24 FPO AP 96595-0004 Telephone: Operator Assisted — 246-370-8445	Same as shipping address.

Additions, corrections, or changes to the table should be forwarded to:

Commander
Naval Air Systems Command
AIR-4.4.5, Bldg. 2360
22229 Elmer Rd., Unit 4
Patuxent River, MD 20670-1906

Telephone number:
DSN — 757-3614
Commercial — 301-757-3614

Telefax number:
DSN — 757-3410/3409
Commercial — 301-757-3410/3409

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLE B-II. AIR-4.4.5 petroleum testing laboratory.

Laboratory Shipping Address	Lab Mailing Address
Naval Air Station Fuel Sample (AIR-4.4.5) HAZMART, Bldg 2385 22680 Hammond Rd. Patuxent River, MD 20670 Telephone: DSN — 757-3417/3410/3409/3405 Commercial — 301-757-3417/3410/3409/3405 Telefax: DSN — 757-3614 Commercial — 301-757-3614 UIC: N00421 Message Address: COMNAVAIRSYSCOM PATUXENT RIVER MD//4.4.5//	Commander Naval Air Systems Command AIR-4.4.5, Bldg 2360 22229 Elmer Rd., Unit 4 Patuxent River, MD 20670-1906

TABLE B-III. Naval Supply Energy Office address.

Director NAVSUP ENERGY OFFICE 8725 John J. Kingman Rd., Suite 3719 Ft. Belvoir, VA 22060-6224 Telephone: DSN — 427-7341/7328/7333 Commercial — 703-767-7341/7328/7333 Telefax: DSN — 427-7389 Commercial — 703-767-7389

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

APPENDIX C
AIRCRAFT: AV-8B

AIRCRAFT INFORMATION SUMMARIES

C.1 SCOPE

C.1.1 SCOPE. This appendix provides general information about Navy and Marine Corps aircraft configurations, characteristics, and requirements that can impact aircraft refueling operations. This appendix has been assembled for use by fuel handling personnel for general planning and other purposes and is current as of the effective date of this handbook. The information has been extracted from specific aircraft NATOPS and maintenance manuals. Users of this handbook are advised to consult the specific aircraft NATOPS and maintenance manuals for the most accurate, up-to-date information.

Type of Aircraft	Aircraft Model	Page
Fixed Wing	AV-8B	60
	C-9	66
	C-12	73
	C-20	79
	C-26	92
	C-37	99
	C-40	110
	C/KC-130 series (except model J)	123
	C/KC-130J	132
	E-2/C-2	143
	E-2C 2000	150
	E-6	164
	EA-6	178
	F-5	185
	F/A-18A/B/C/D	193
	F/A-18E/F	200
	P-3	208
	P-8A	214
	S-3	227
	T-2	234
	T-6	239
	T-34	245
	T-39	250
	T-44	258
	T-45	264
	TA-4	271
	UC-35	281
Rotary Wing	AH-1	288
	H-2	294
	H-3	301

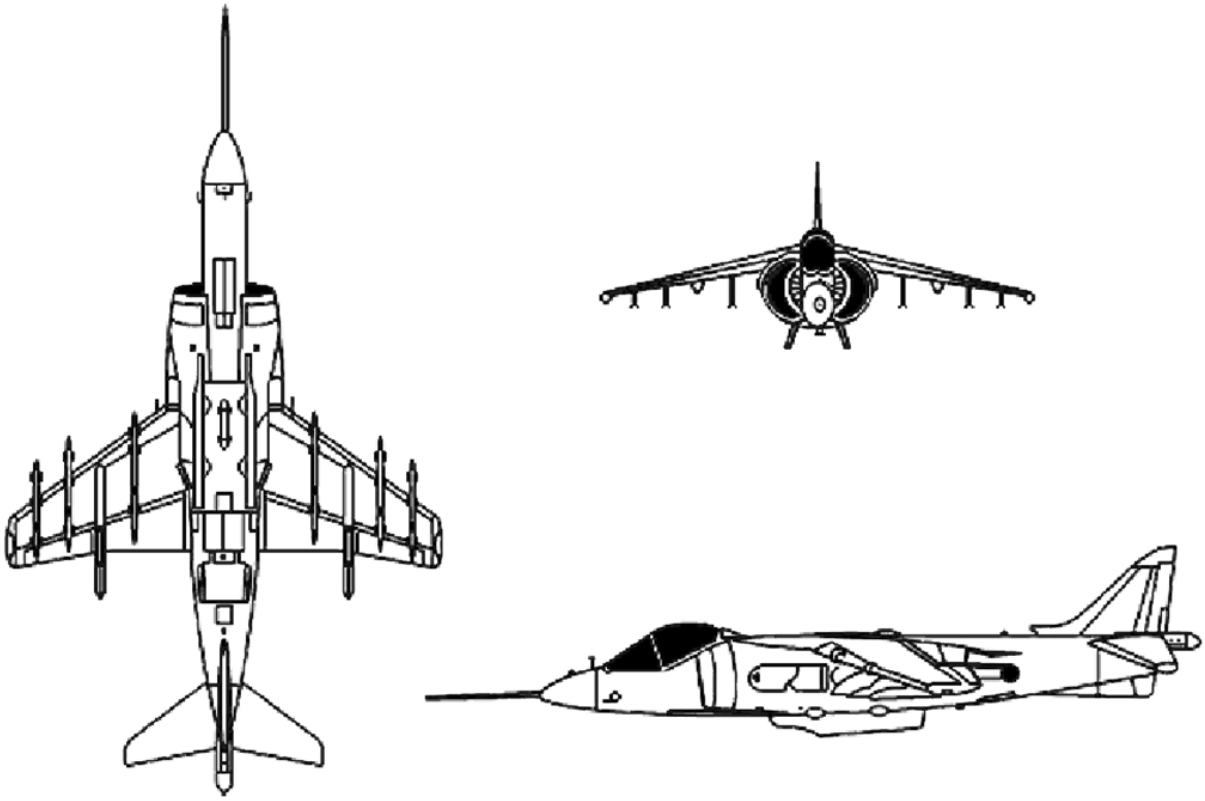
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-46 307
H-53 314
H-60 322
H/UH-1 329
TH-57 337
MQ-8B 345
RQ-4 358
MV-22B 376

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AV-8B

The AV-8B is a transonic, single cockpit, single engine, and jet propelled day/night tactical fighter built by McDonnell Douglas Aerospace. The aircraft is powered by a Rolls Royce axial flow, twin spool turbo fan engine.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: AV-8B

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread	30 ft 4 in	Maximum Gross Weight — 32,000 lbs
Length	46 ft 4 in	Maximum Footprint — 115 psi
Height	11 ft 9 in	

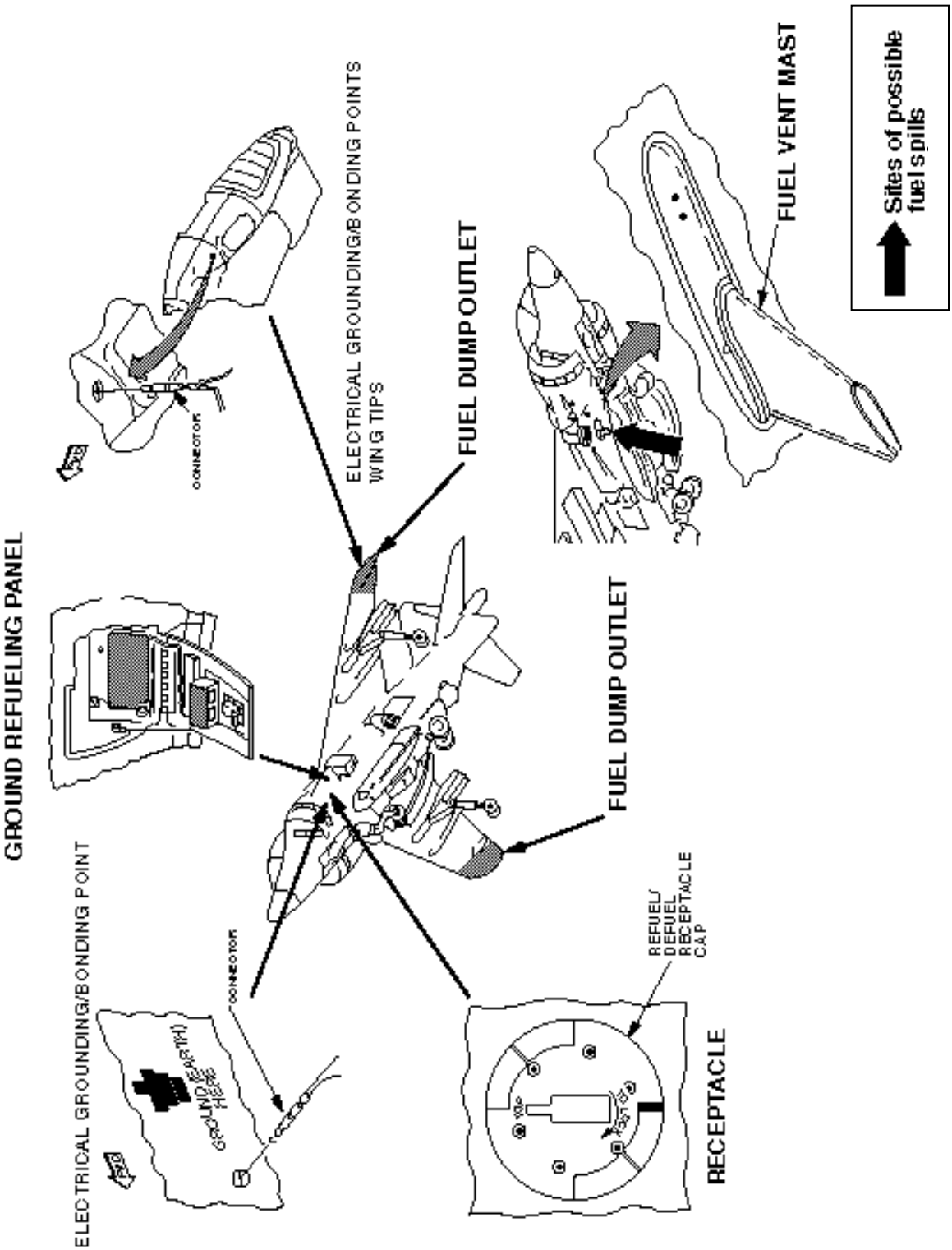
TABLE OF FUEL CAPACITIES

Tank			Gallons	Pounds (JP-5)
Internal	Fuselage	Left Tank	47	320
		Left Front	80	545
		Right Front	80	545
		Right Center	47	320
		Rear	162	1100
	Wing	Left	362	2450
		Right	362	2450
Total Internal			1140	7750
External Tanks (Each)			290	1970
Totals	With 2 External Tanks		1720	11700
	With 4 External Tanks		2328	15830

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

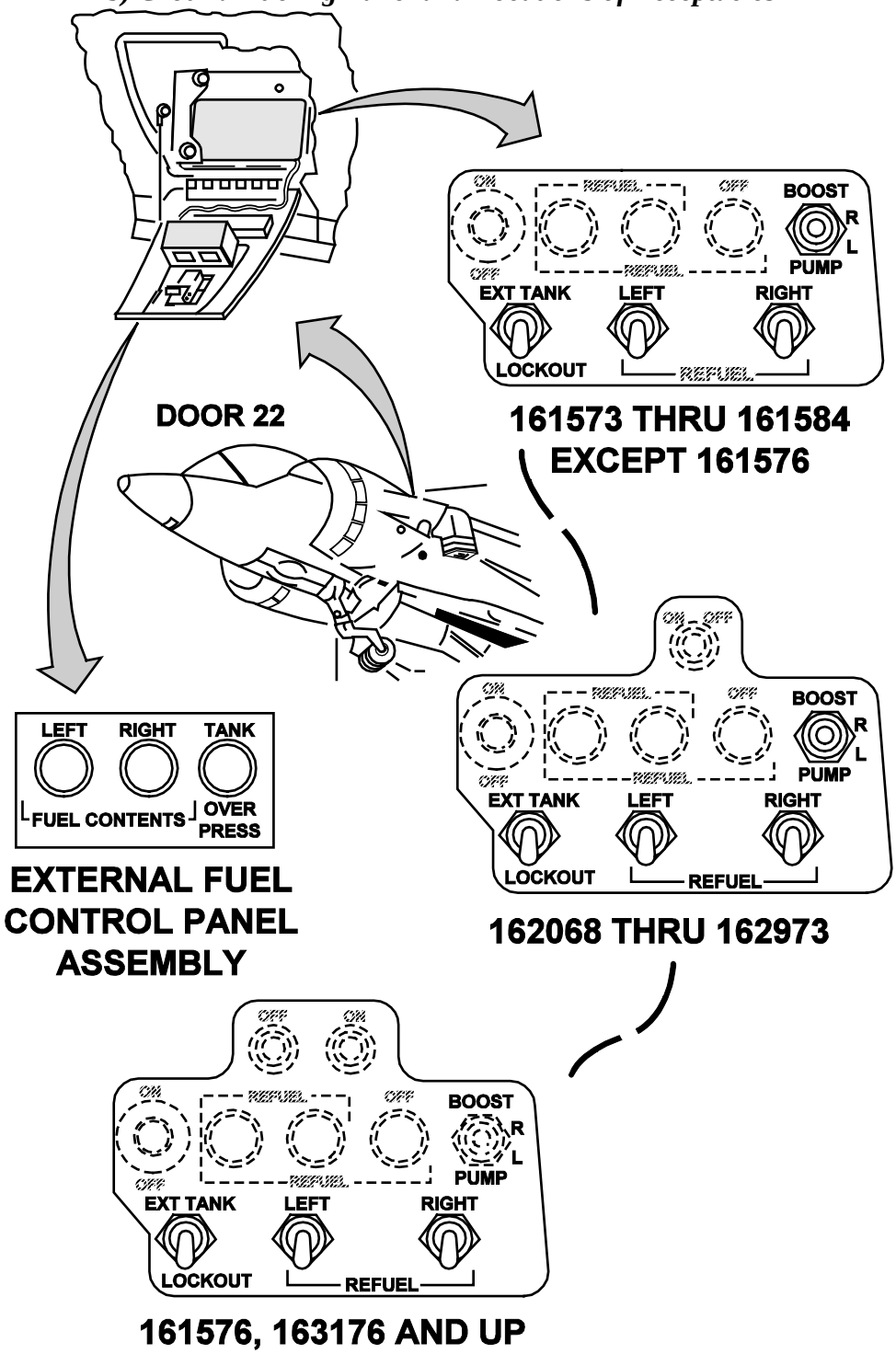
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Location of Fuel Vents, Dump Ports, and Low Point Drains



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

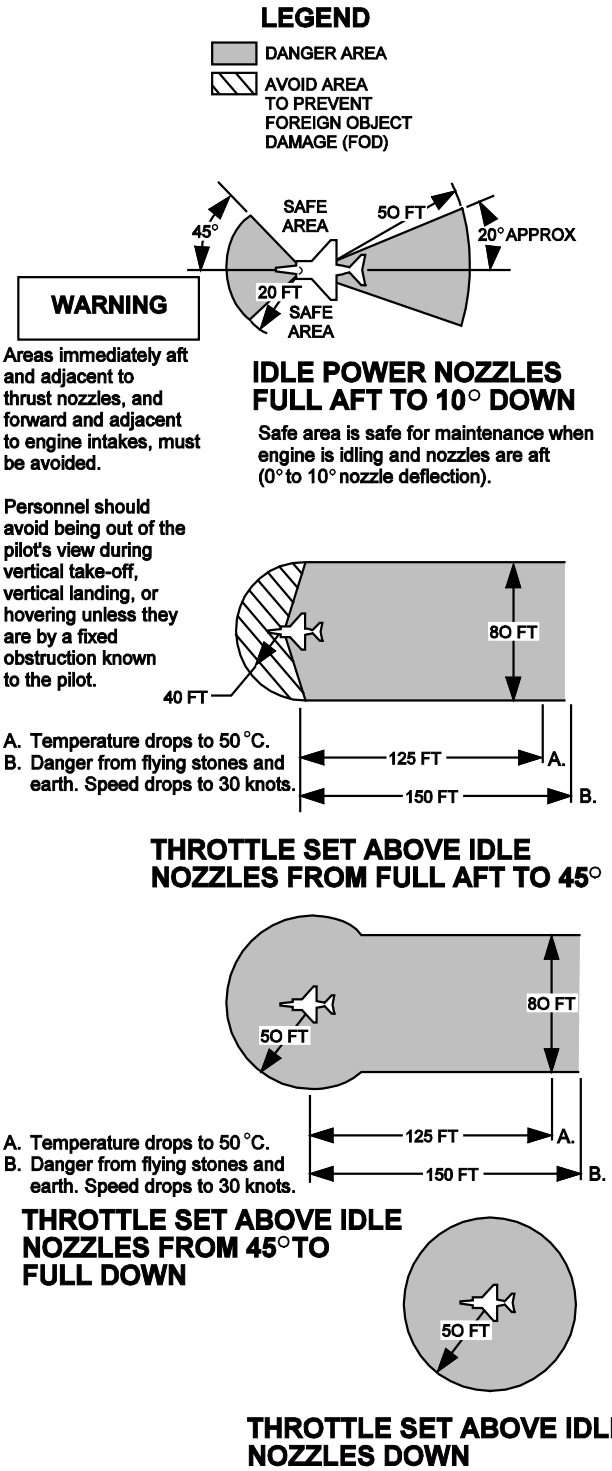
b) Ground Fueling Panel and Locations of Receptacles



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Engine Inlet/Exhaust/Blast Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the AV-8B aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Remove receptacle cap and attach refueling nozzle.
2. Open door 22.
3. Press-to-test "TANK OVER PRESS" warning light on ground refueling panel for lamp operation.
4. Set "LEFT" and "RIGHT REFUEL" switches on the ground refueling panel (door 22) to the "REFUEL" position (toggles down) and commence refueling.
5. Make sure "LEFT" and "RIGHT FUEL CONTENTS" lights are on.

CAUTION

If any of the three lamps fails to operate in steps 3 and 5 above, discontinue refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

6. Fuel can be stopped from entering the external tanks by setting "EXT TANK LOCKOUT" switch to "LOCKOUT" position.
7. Monitor the "TANK OVER PRESS" warning light.

CAUTION

Immediately stop fuel flow if "TANK OVER PRESS" indicator illuminates. "Hot refueling" cannot be performed on the aircraft until problem is resolved.

8. After 60 to 120 gallons of fuel have entered the aircraft's tanks, make sure air is venting from FUEL VENT MAST.
9. Watch the "LEFT" and "RIGHT FUEL CONTENTS" lights. When the lights go off, the aircraft is full. Immediately stop fuel flow and close and lock the nozzle's flow control handle.
10. Set "LEFT" and "RIGHT REFUEL" switches to flight position (toggles up).

Note

If any of the high level shut off valves fails to operate correctly, fuel will spill from the vent mast. Turn off fuel flow immediately!

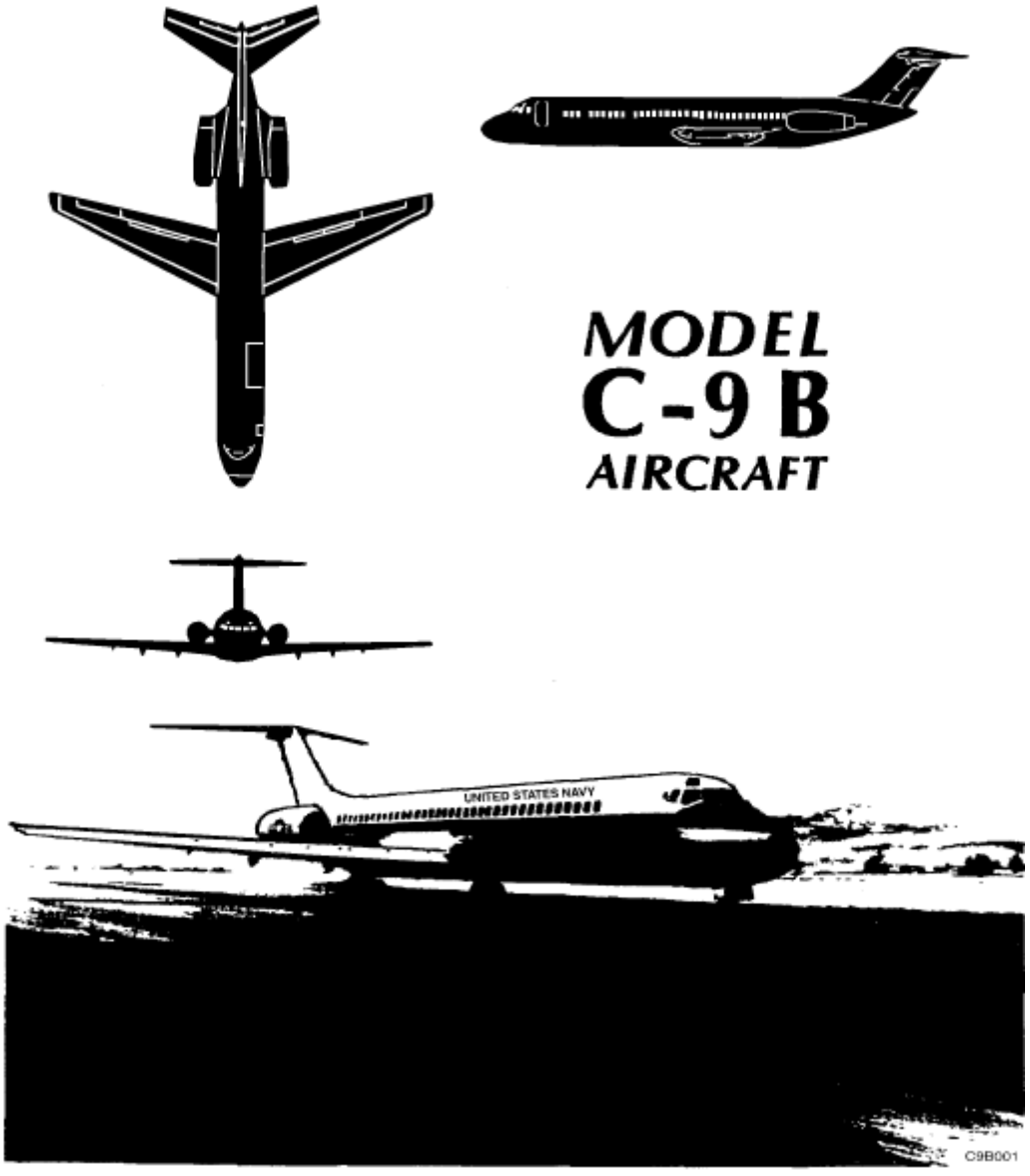
SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. This aircraft is NOT equipped with a precheck system.
2. Under normal conditions, all air being displaced by fuel in the tanks, including external tanks, exits the aircraft through the common "Fuel Vent Mast."
3. If any high level shut-off valves fails to operate correctly, fuel may spill from the "Fuel Vent Mast."
4. A malfunction within an external fuel tank may cause fuel to spill from the bottom center of tank (pressure relief vent).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-9

The Navy Model C-9B Skytrain II aircraft is manufactured by the McDonnell Douglas Corporation and is designed for the transportation of personnel and cargo. The aircraft is a low-wing monoplane equipped with a fully retractable landing gear that is powered by aft-mounted Pratt & Whitney JT8D-9A turbofan engines flat rated at 14,500 pounds (each) static takeoff thrust at sea level up to 84˚F.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

AIRCRAFT DIMENSIONS		Weight
Length	119.3 ft	Maximum Gross
Height (top of vertical tail)	27.5 ft	111,000 lbs
Wing Span	93.3 ft	

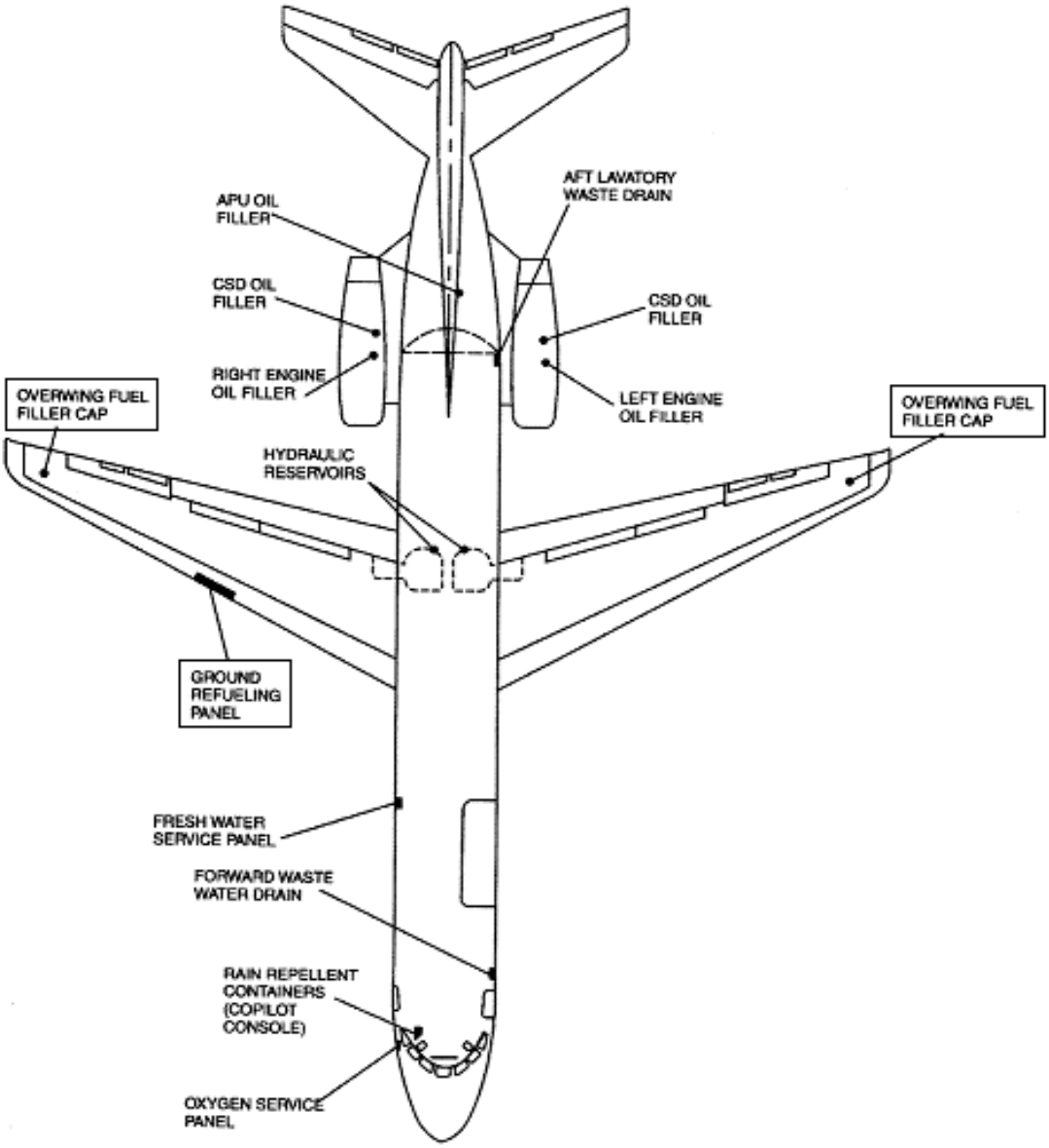
TABLE OF FUEL CAPACITIES

Fuel Tank	Gallons
Left Main	1,386
Right Main	1,386
Center	907
FWD Fuselage	1,250
Aft Fuselage	1,000
TOTAL	5,929

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

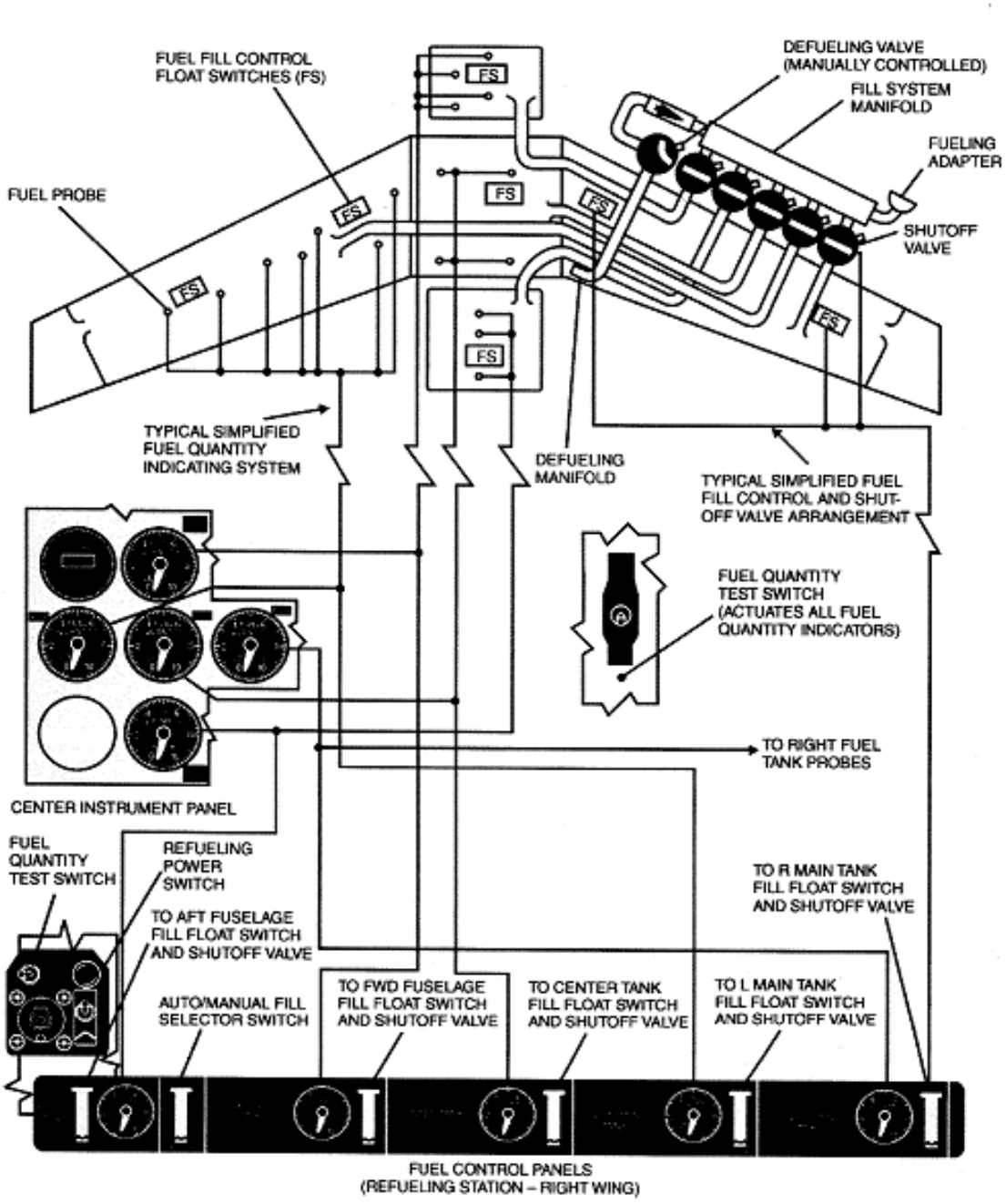
AIRCRAFT CONFIGURATION

Locations of Filler Caps, Service Panels, and Drains



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

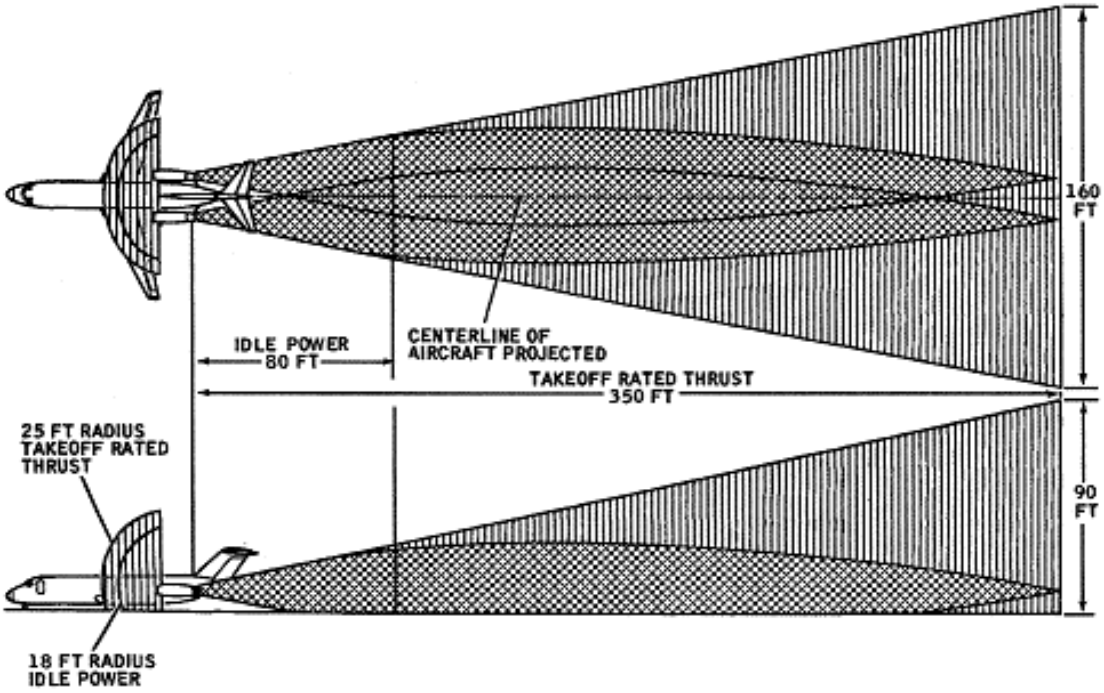
Ground Fueling Panel and Location of Various Valves and Switches



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Engine Exhaust/Blast Area

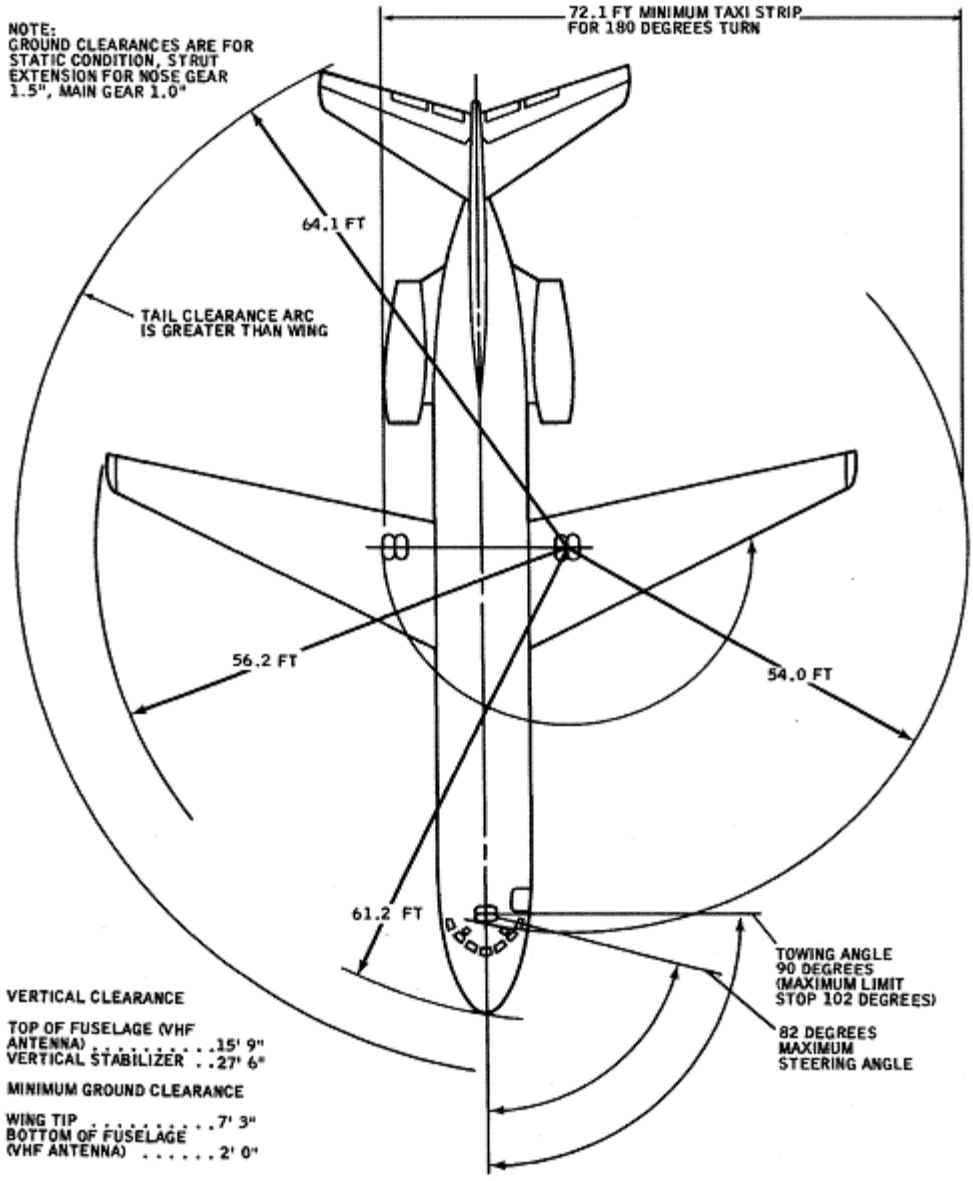


DISTANCE BEHIND EXHAUST EXIT		75 FT	100 FT	200 FT	350 FT
AT START OF TAXI ROLL	BLAST PRESSURE-PSF	18	8	3	2
	VELOCITY-KNOTS	75	50	30	20
	TEMPERATURE RISE (1)	50° F	30° F	5° F	
		28° C	17° C	3° C	
AT IDLE POWER	VELOCITY-KNOTS	40	25	20	10
	TEMPERATURE RISE (1)	30° F	10° F	5° F	
		17° C	6° C	3° C	
AT TAKEOFF RATED THRUST	VELOCITY-KNOTS	165	125	60	36
	TEMPERATURE RISE (1)	90° F	70° F	20° F	20° F
		50° C	39° C	11° C	11° C

NOTE:
(1) TEMPERATURES ARE ABOVE AMBIENT CONDITIONS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Turning Radius



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures are extracted from the C-9 NATOPS manual. In addition to these procedures, the applicable, basic refueling procedures contained in Chapter 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. **Pressure Fueling.** The aircraft fueling control panel is located in the right midwing leading edge, approximately 5-1/2 feet from ramp level. Access to the fuel control panel is through a forward swinging hinged access door located in the lower surface of the wing leading edge.

   Pressure fueling can be accomplished with or without electrical power. Fuel flow, through the pressure-fueling adapter (manifold), is directed to independently operated left and right main, center, forward, and aft fuselage tank fuel fill valves. The fuel fill valves can be operated electrically or manually to control fuel flow through the individual tank fill lines.

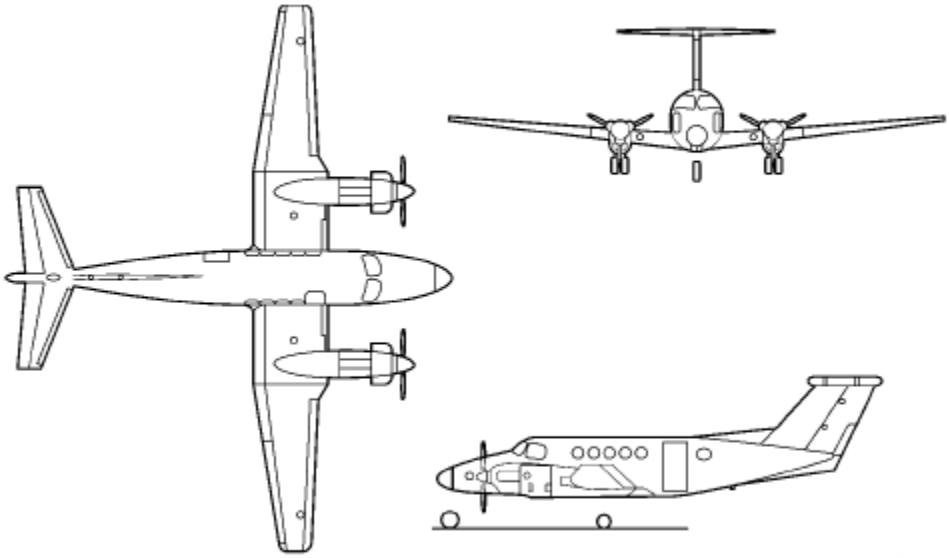
   Fuel level control float switches, provided in each tank, automatically close the respective fuel fill valves when the tanks are full, provided electrical control is used. If nose gear strut is near the fully extended position, the oleo switch will be actuated and electrical power to fuel control panel will be off. When the tanks are being fueled to less than maximum capacity, the fuel fill valves should be closed individually, either electrically or manually, when the desired fuel load in each tank is attained. Fuel tank capacities and acceptable fuel specifications are shown on the fuel quantity data and fuel grade and limits charts.

2. **Gravity Fueling.** The left and right main fuel tanks can be gravity fueled through the overwing fill adapters located approximately 10 feet inboard of each wingtip. Procedures for gravity fueling the left and right main tanks are identical. The center, forward, and aft fuselage tank fueling is accomplished by using the right main tank boost pumps to pump fuel through the defueling line and fuel fill valve, and into the applicable tank fill line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-12

The TC-12B aircraft, manufactured by Beech Aircraft Corporation, is an all metal, low-wing, twin-turboprop, T-tail monoplane with an axial engine mounted on each wing. The TC-12B is powered by two 850-shaft horsepower PT6A-41 turboprop engines manufactured by Pratt & Whitney of Canada Limited. The primary mission of the TC-12B is to train student military aviators to fly multiengine turboprop aircraft.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	54 ft 6 in	Maximum Gross Weight — 13,500 lbs
Length	43 ft 10 in	Maximum Landing Weight – 12,500 lbs
Height	14 ft 6 in	Maximum Footprint — 64 psi

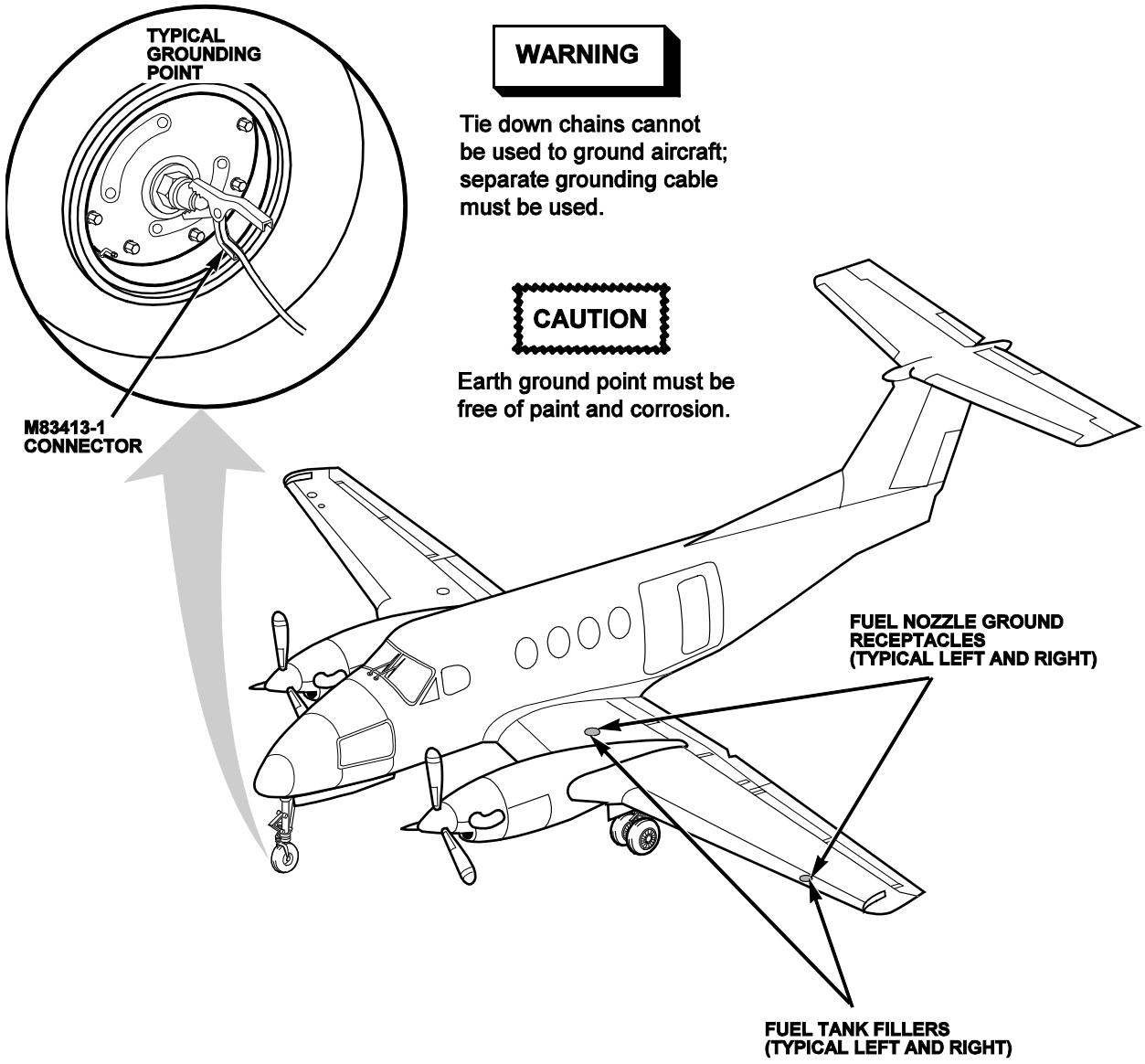
TABLE OF FUEL CAPACITIES

Tank	Gallons
Wing & Nacelle	195.0
Auxiliary	79.5
Total Useable	544.0
Total Capacity	549.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

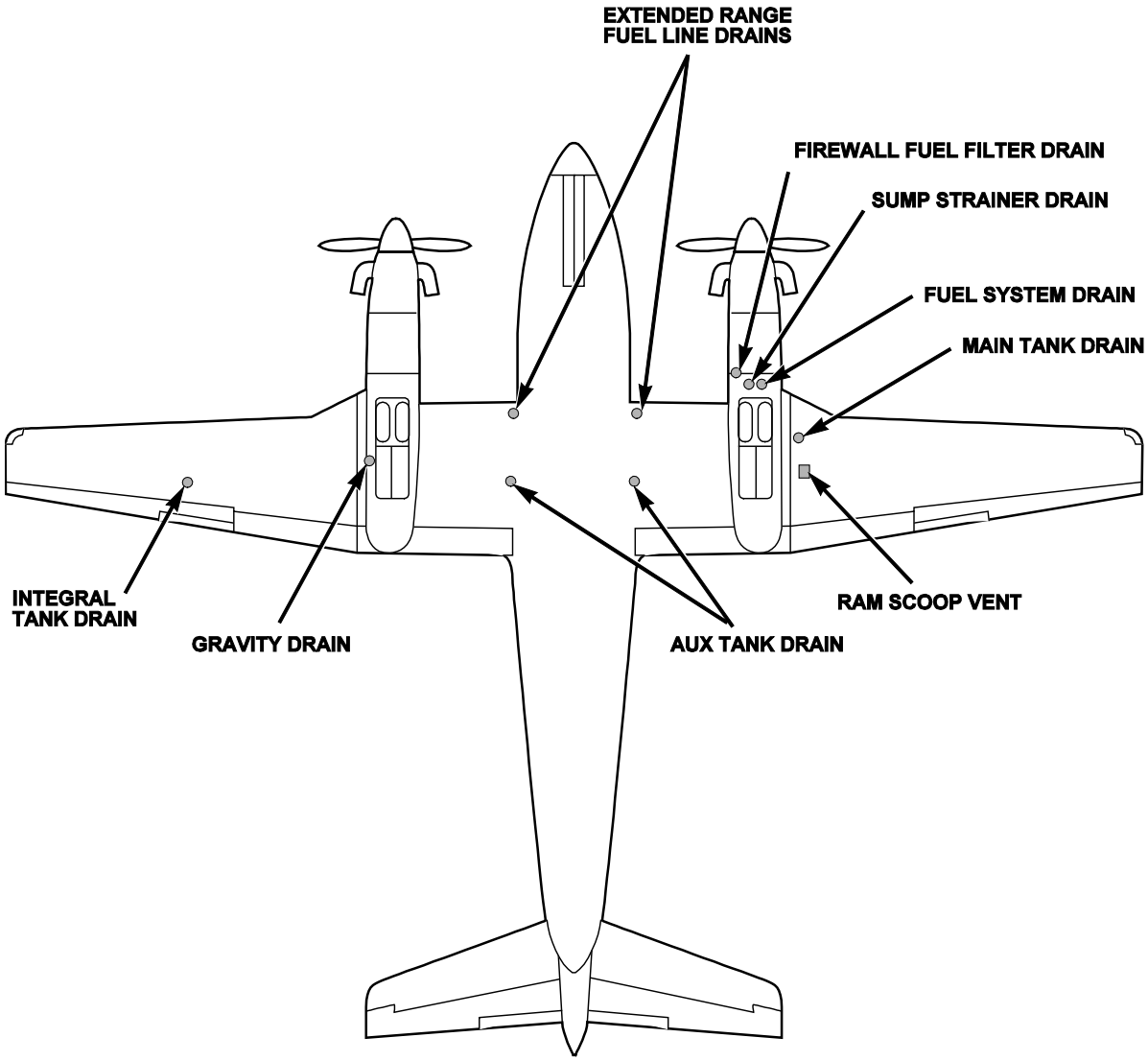
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

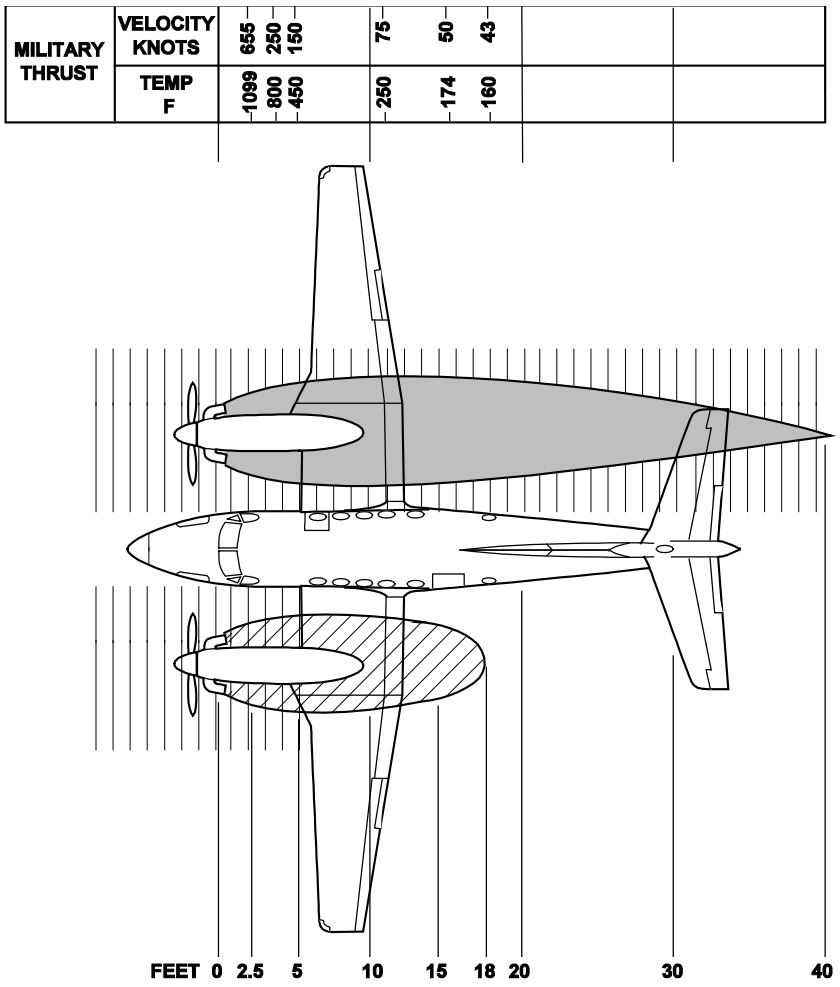
b) Location of Fuel Vents, Dump Ports, and Low Point Drains



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Exhaust/Blast Area



LEGEND

EXHAUST DANGER AREAS (GROUND IDLE)
EXHAUST DANGER AREAS (MAX POWER)
PROPELLER DANGER AREA

NOTE

- The exhaust danger area does not include propeller wake, which increases velocity and significantly reduces temperature.
- Exhaust gas temperature and velocity at ground idle are very low, however, the immediate area of exhaust discharge should be avoided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The C-12 aircraft does not have a single point refueling adapter and, therefore, can only be gravity refueled.

GRAVITY FUELING

1. Attach bonding cables to aircraft.
2. Attach bonding cable from hose nozzle to ground.
3. Open applicable fuel tank filler cap.



Do not insert fuel nozzle completely into fuel cell because of possible damage to bottom of fuel cell.

4. Fill fuel tank with fuel.
5. Secure applicable fuel tank filler cap.



Make sure latch-tab on cap is pointed aft.

6. Disconnect bonding cables from aircraft.



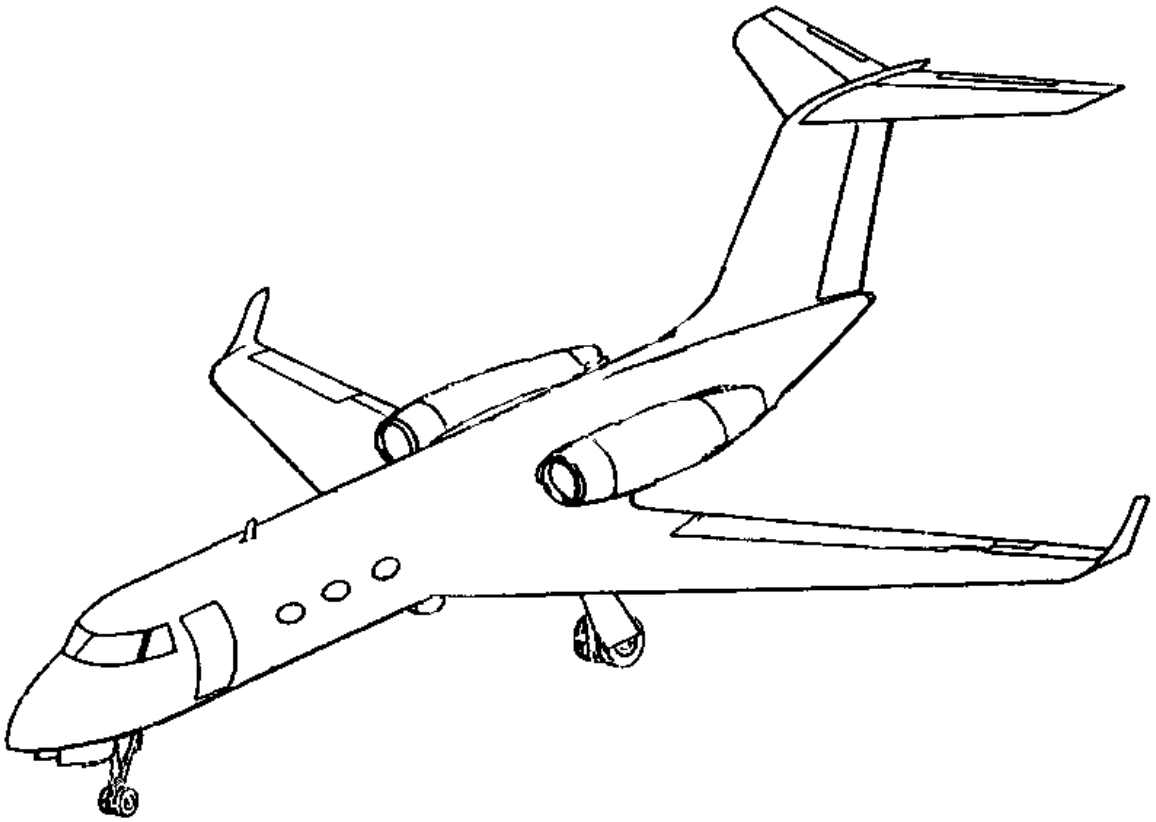
Hot Gravity refueling of the C-12 is strictly prohibited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-20

The C-20, designed and manufactured by Gulfstream Aerospace Corporation, is a two engine, swept wing, long range, high altitude, high speed aircraft used as a special mission aircraft. The aircraft is powered by two turbofan Spey Mark 511-8 Rolls Royce engines.

There are two versions of the C-20 in Navy service: the C-20D and the C-20G. Both aircraft are outwardly similar and have similar fuel systems. Differences are noted where appropriate.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Length (overall)		Maximum Gross Takeoff:
C-20D	83 ft 2 in	C-20D – 69,700 lbs
C-20G	88 ft 4 in	C-20G – 73,200 lbs
Height (to top of tail)		Maximum Gross Ramp:
C-20D	24 ft 6 in	C-20D – 70,200 lbs
C-20G	24 ft 5-1/8 in	C-20G – 73,600 lbs
Wingspan	77 ft 10 in	

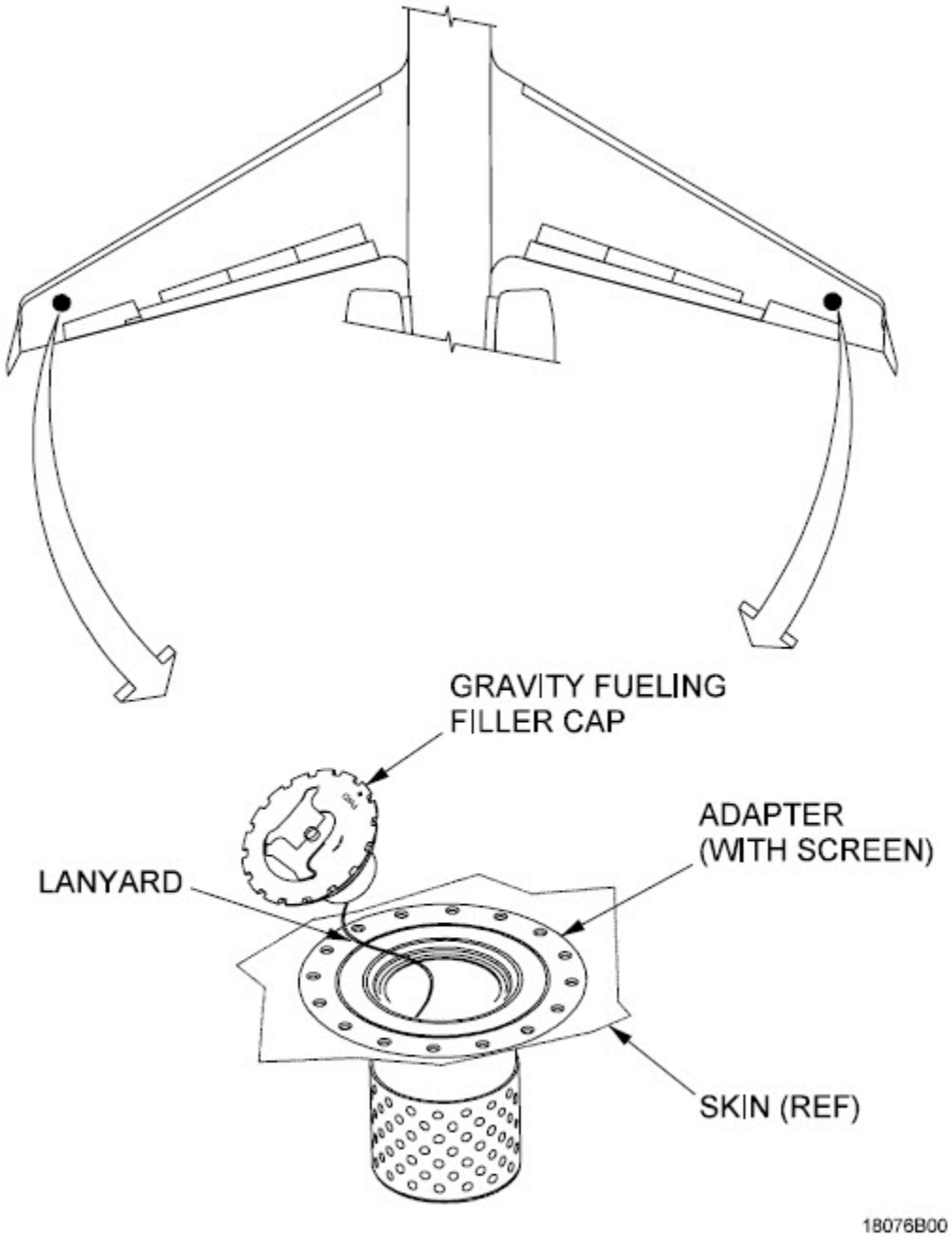
TABLE OF FUEL CAPACITIES

Tank	Gallons
C-20D: Left	2,096
Right	2,096
Total	4,192
C-20G: Left	2,185
Right	2,185
Total	4,370

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

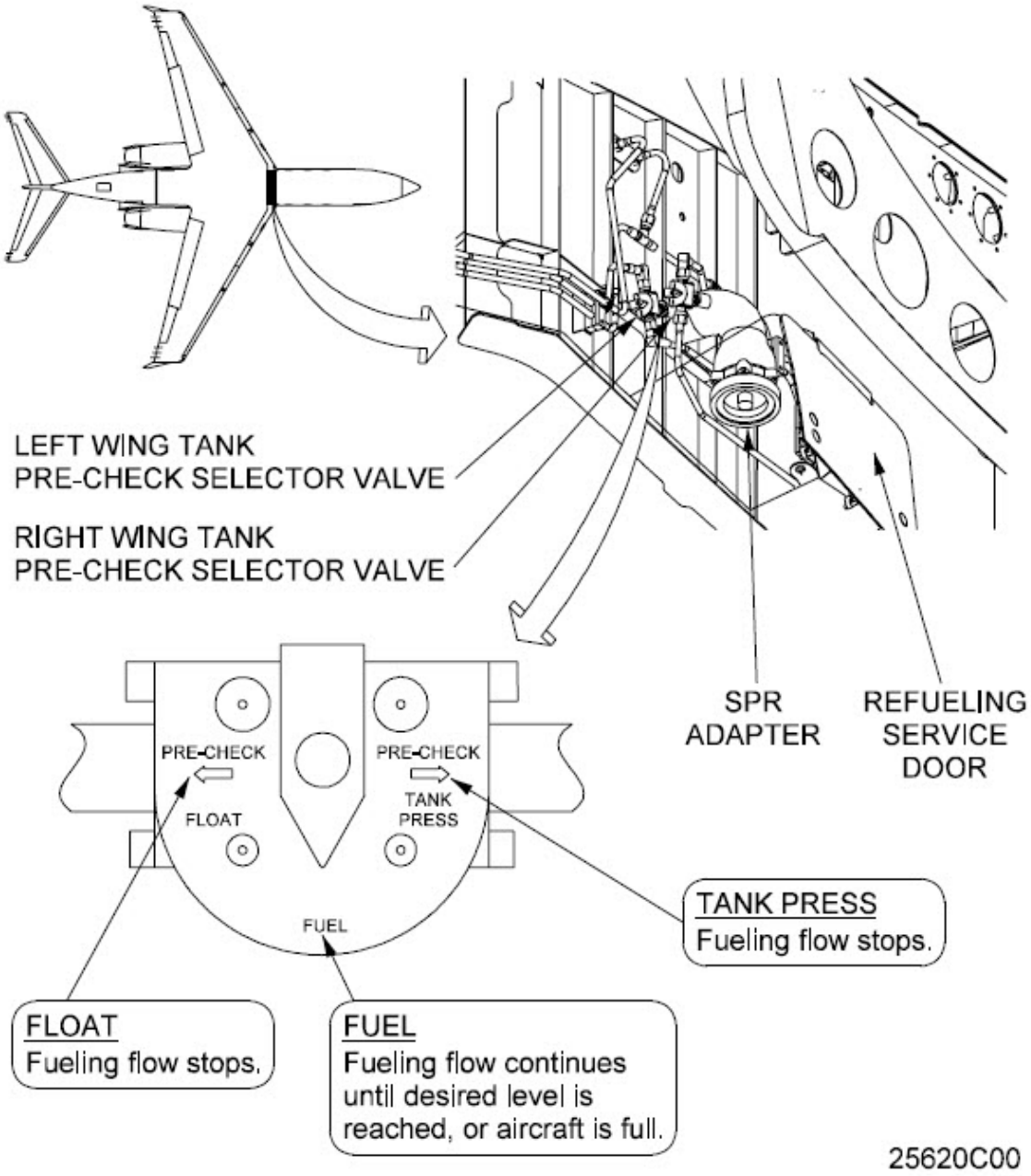
a) Gravity refueling locations



Over-The-Wing Refueling Adapters

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Single-point refueling location



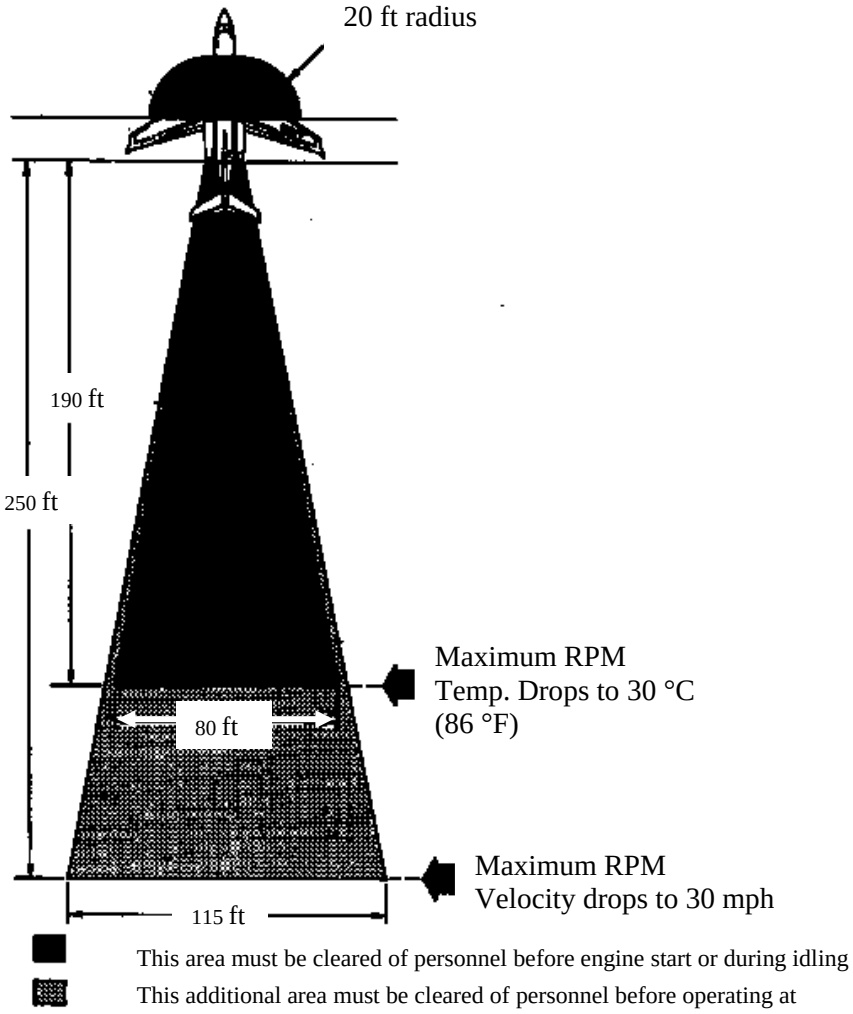
Single Point Refueling Adapter / PRE-CHECK Valves

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

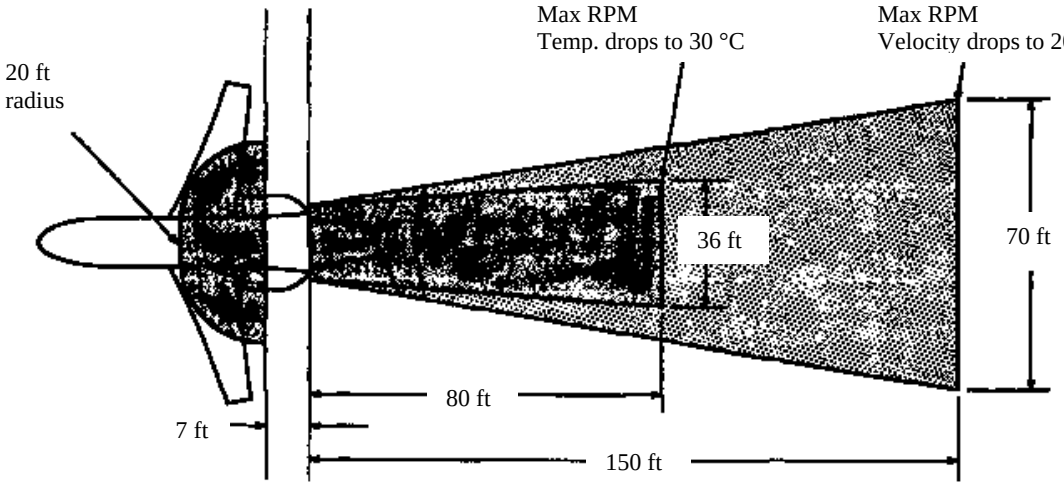
a) Engine/APU Inlet/Exhaust/Blast Area

1. C-20D Engine

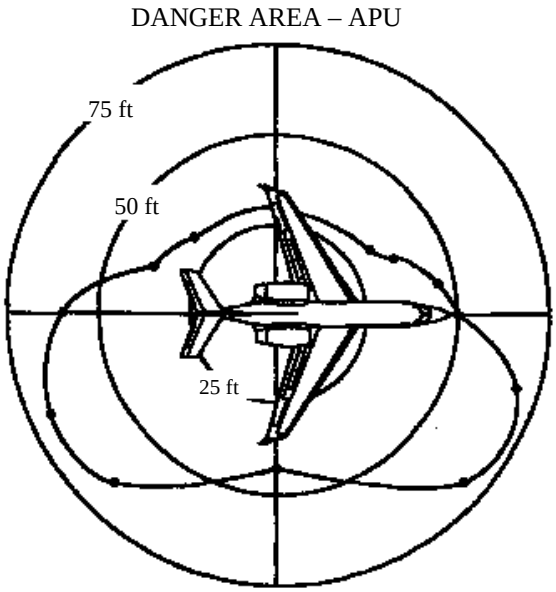


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

2. C-20G Engine



APU



TWO HOUR NOISE EXPOSURE BOUNDARY (92 DB(A)) – NO HEARING PROTECTION. APU OPERATING ONLY. NOTE – NO RESTRICTTION IF EAR PLUGS ARE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

Location of Refueling Adapters:

Pressure refueling adapter: Right Hand Wing Fillet Leading Edge
Gravity refueling adapters: Upper Skin of Each Outer Wing

Location of Bonding/Grounding Jack: Leading Edge of Each Wing

The following procedures are extracted from the C-20 D and G NATOPS manuals. In addition to these procedures, the applicable, basic refueling procedures contained in Chapter 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Gravity Fueling

a. C-20D

A fuel filler cap and adapter assembly is located in the upper skin of each outer wing. The adapter has a standpipe filler neck that limits a wing tank capacity to approximately 2,067 gallons.

A grounding jack, located in the wing leading edge, is provided so that the fueling nozzle may be grounded during fueling.

CAUTION

It is recommended that fueling be accomplished over the leading edge of the wing, rather than the trailing edge in order to prevent damage to the aileron.

If full fuel capacity is required when gravity fueling, it will be necessary to remove the inboard vent cover which is located on the lower skin, BL 26.

During gravity fueling, the fuel is allowed to run inboard, through the ribs, to the centerline. The tank is then filled from the centerline rib, including the hopper, outboard.

b. C-20G

A fuel filler cap and adapter assembly is located in the upper skin of each outer wing. The adapter has a standpipe filler neck that limits a wing tank capacity to approximately 2,185 gallons.

A grounding jack, located in the wing leading edge, is provided so that the fueling nozzle may be grounded during fueling.

CAUTION

It is recommended that fueling be accomplished over the leading edge of the wing, rather than the trailing edge to prevent damage to control surfaces.

During gravity fueling, the fuel is allowed to run inboard through the ribs to the centerline. The tank is then filled from the centerline rib, including the hopper, outboard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Perform overwing refueling as follows:

1. Check quantity in tanks using cockpit standby gauges.

2. Check that main vent on lower surface of wing is open.

3. Remove gravity filler cap from adapter at outboard location, upper skin surface.

4. Ground fuel nozzle to aircraft structure through grounding jack located in the upper leading edge.

5. Fuel as required.

6. When fueling has been completed, replace filler cap and check for security.

2. Pressure Fueling

c. C-20D

The aircraft can be refueled from a single point manifold located in the RH wing/fuselage fillet leading edge. The system is a hydromechanical system, with capability for cockpit or filler station intermediate shut off. During normal single-point refueling conditions, fuel enters the Pressure Fueling Adapter (figure 2-21). The fuel proceeds through the main fill line to a cross fitting. A directional check valve located downstream of the straight-through portion of the cross fitting stops flow into the RH hopper while fuel continues to flow through the arms of the cross fitting and through the LH and RH fill lines to the shutoff valves. The shutoff, when fueling to capacity, is controlled by a fluid force differential on the shutoff poppet. As fuel tills the cavity, the air is vented and expelled overboard through the vent lines. As the fuel level nears capacity, the float in the hi-level pilot rises and, by means of mechanical linkages, closes the shutoff poppet, Fuel coming into this line from the downstream side of the shutoff valve poppet increases in pressure until the valve closes. Two two-position rocker switches labeled REMOTE FUELING SHUTOFF switches (figure 2-22) are located on the Overhead Control Panel which allow shutting off of fuel to a particular tank when selected. When the switches are selected to the CLOSED position, the left and right fueling valves are closed and fueling is shut off. Fuel can also be shutoff by manually operating the pre-check selector valves. Power for operation of the Remote Fueling Shutoff Valves and switches is from Essential 28 VDC Bus, through two circuit breakers labeled L and R FUELING S/O (Hl, H2) on the Pilot's Circuit Breaker Panel.

When refueling on a significant side slope, the fuel gaging system is inaccurate and considerable fuel unbalance may occur. To preclude fuel unbalance, determine the proper amount of gallons/liters/etc to upload for each tank. Then, refuel one tank at a time using the truck gages to determine proper upload.



When Remote Fueling Shutoff switches are used to terminate fueling, the fueling nozzle should be removed prior to either placing the Remote Fueling Shutoff Switches to OPEN or removing power. Pressure from the refueling vehicle should be 35-55 psi for proper operation of the shutoff valves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION

Empennage damage may occur if aft tail compartment ladder is down and pinned during fuel servicing.

Pressure fueling pre-check valves. The pressure fueling pre-check valves (figure 2-21) are located on the pressure fueling adapter. The purpose of these valves, one for each tank, is to check the shutoff operation due to actuation of the tank pressure sense valve and the hi-level pilot. Operation of the pre-check valves is identical for the left hand and right hand sides. As fuel enters the pressure fueling adapter, the position of the pre-check selector should be on fuel. Selecting press allows some fuel to flow through the selector valve to the tank pressure sense valve entering the valve on the tank side of the diaphragm. This operation simulates a buildup of tank pressure acting on the diaphragm against the ambient pressure and results in a fuel shut-off to the affected tank. If the float position is selected, fuel is diverted to the hi-level pilot. The fuel enters the float cage at a much faster rate than it is allowed to bleed off, causing the float to rise and simulating a full tank capacity which closes the poppet. This simulates the normal operation of the fuel shutoff valve to the particular tank. The procedure is shown in figure 2-23.

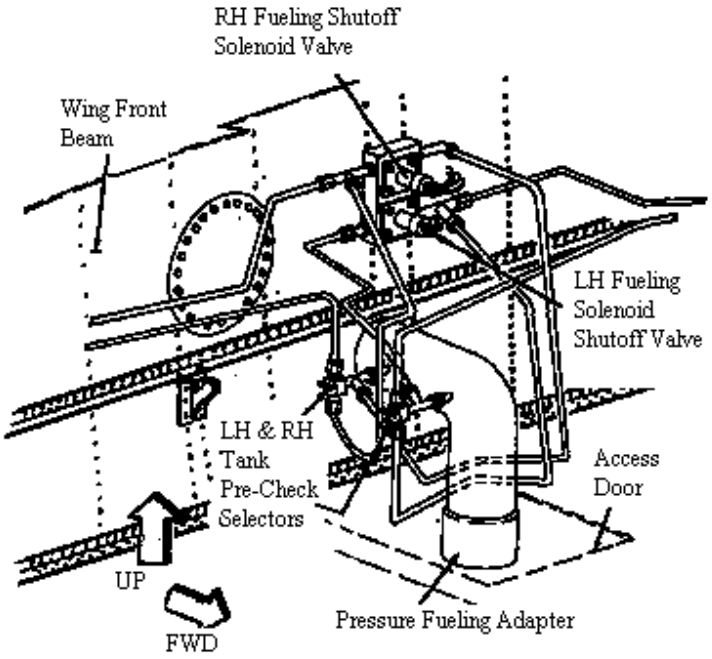
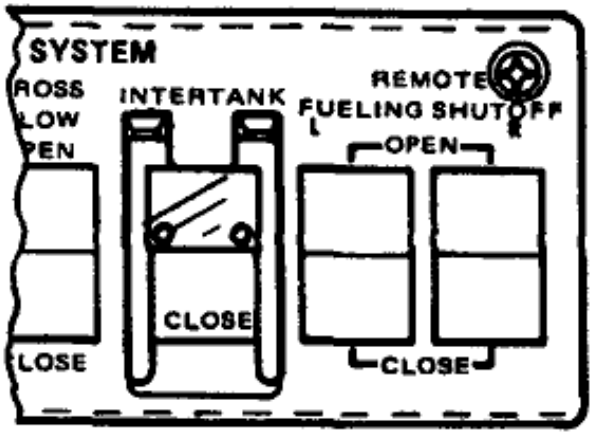


Figure 2-21: Pressure Fueling Adapter



Located on Overhead Control Panel

Figure 2-22: Remote Fueling Shutoff Switches

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Procedure	PRESSURE FUELING PRECHECK VALVES	
	LH TANK	RH TANK
a) Check Precheck selectors are in FUEL and fuel is flowing	FUEL	FUEL
b) Position both LH Tank & RH Tank selectors to FLOAT. Fueling should stop.	FLOAT	FLOAT
c) Position RH Tank selector to PRESS. Fuel may start to flow as the selector passes FUEL and should stop again.	FLOAT	PRESS
d) Reposition RH Tank selector to FLOAT. Fuel may start to flow as the selector passes FUEL and should stop	FLOAT	FLOAT
e) Position LH Tank selector to PRESS. Fuel may start to flow as the selector passes FUEL and should stop again.	PRESS	FLOAT
f) Return the LH Tank & RH Tank selectors to FUEL and continue fueling.	FUEL	FUEL

CAUTION **IF DURING THIS PROCEDURE FUEL FLOW INTO THE TANK DOES NOT STOP WHEN INDICATED, TERMINATE FUELING AT THE NOZZLE AND TROUBLESHOOT.**

Figure 2-23: Pressure Fueling Precheck Valve Check Procedure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d. C-20G

The single-point pressure fueling system is a hydromechanical system, with capability for cockpit (through the solenoid valve) or filler station (manual) immediate shutoff.

During normal pressure fueling conditions, fuel enters the pressure fueling adapter, which is located in the leading edge of the right wing/ fuselage fillet. The fuel proceeds through the main till line to a cross fitting. A directional check valve located downstream of the straight-through portion of the cross fitting stops flow into the right hopper while fuel continues to flow through the arms of the cross fitting and through the left and right till lines, the shutoff valves, and into the tanks.

When the fuel level rises enough to close the outboard high-level pilot valves, pressure-sensing lines divert fuel pressure to the downstream side of the shutoff valve poppets, causing the shutoff valves to close and fuel to stop entering the wings.

Pressure fueling precheck valves are located on the pressure fueling adapter. The purpose of these valves, one for each tank, is to check the shutoff operation due to actuation of the tank pressure sensing valve and the high-level pilot valve. The valves can be used to drive the shutoff valves to the closed position, thereby shutting off fuel to the respective tank.

Pressure fueling may also be terminated from the cockpit by using the L and/or R REMOTE FUELING SHUTOFF switches (Figure 3-2) on the overhead panel.
These switches, when positioned to CLOSED, energize the fuel shutoff solenoid valves located above the pressure fueling adapter, terminating the pressure fueling operation.

Note

Pressure from the refueling vehicle should be 35-55 psi for proper operation of the shutoff valves.

Full Capacity Pressure Fueling

1. Gain access to pressure fueling adapter through access door located in the fillet between the fuselage and right wing.
2. Remove dust cover from adapter.
3. Ground pressure fueling nozzle.
4. Connect pressure fueling nozzle to pressure fueling adapter.
5. Check that precheck selectors, mounted to pressure fueling adapter, are in FUEL position.
6. Open pressure fueling nozzle.
7. Perform prechecks during refueling operation as follows:
    a. Check that precheck selectors are in FUEL and fuel is flowing.
    b. Position both LH tank and RH tank selectors to FLOAT. Fueling should stop.
    c. Position RH tank selector to PRESS. Fuel may start to flow as the selector passes FUEL and should again stop.
    d. Reposition RH tank selector to FLOAT. Fuel may start to flow as the selector passes FUEL and should again stop.
    e. Position LH tank selector to PRESS. Fuel may start to flow as the selector passes FUEL and should again stop.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

f. Return the LH tank and RH tank selectors to FUEL and continue fueling.

8. After completion of the precheck procedures, perform a functional check of the pressure fueling shutoff from the cockpit as follows:

   a. Make sure that the L and R REMOTE FUELING SHUTOFF switches are in the OPEN position.
   b. Energize the essential 28-vdc bus.
   c. Place the L SHUTOFF switch to the CLOSED position. Fueling to the left tank should terminate.
   d. Place the L SHUTOFF switch to the OPEN position. Fuel to the left tank should start to flow.
   e. Place the R SHUTOFF switch to the CLOSED position. Fueling to the right tank should terminate.
   f. Place the R SHUTOFF switch to the OPEN position. Fuel to the right tank should start to flow.
   g. De-energize the essential 28-vdc bus.

9. Fueling will terminate automatically by operation of the high-level shutoff pilot valve.

10. Close pressure fueling nozzle.

11. Remove pressure fueling nozzle from pressure fueling adapter.

12. Remove ground.

13. Replace dust cover on pressure fueling adapter and check for security.

14. Close access door.

Partial Loading Pressure Fueling

1. Gain access to pressure fueling adapter through access door located in the fillet between the fuselage and right wing.

2. Remove dust cover from adapter.

3. Ground pressure fueling nozzle.

4. Connect pressure fueling nozzle to the pressure fueling adapter.

5. Check that precheck selectors, mounted to pressure fueling adapter, are in FUEL position.

6. Make sure that the L and R REMOTE FUELING SHUTOFF switches (located in the overhead panel) are in the OPEN position.

7. Open pressure fueling nozzle.

8. Perform prechecks during refueling operation.

Note

- Steps 9 through 11 cover partial loading using the precheck selector at the pressure fueling adapter.

- Steps 12 through 17 cover partial loading using the remote fueling shutoff switches.

9. When the required amount of fuel has been uplifted to one side, place the precheck selector for that tank to the FLOAT position.

10. When the required amount of fuel has been uplifted into the other side, close the pressure fueling nozzle.
11. Reposition selector valve from FLOAT (step 9) to FUEL.

12. Energize the essential 28-vdc bus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

13. When the required amount of fuel has been uplifted into one side, place the REMOTE FUELING SHUTOFF switch for that tank to the CLOSED position.

14. When the required amount of fuel has been uplifted into the other side, place the other REMOTE FUELING SHUTOFF switch to the CLOSED position.

15. Close the pressure fueling nozzle.

16. Place the REMOTE FUELING SHUTOFF switches to the OPEN position.

17. De-energize the essential 28-vdc bus.

18. Remove pressure fueling nozzle from pressure fueling adapter.

19. Remove ground.

20. Replace dust cover from pressure fueling adapter and check for security.

21. Check that precheck selectors are in FUEL position.

22. Close access door.

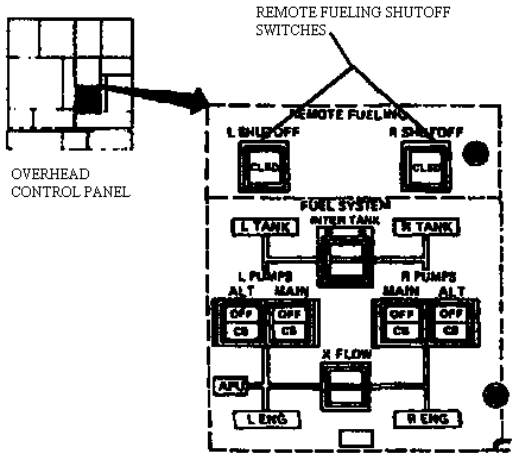


FIGURE 3-2. Remote Fueling Shutoff Switches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-26

The C-26 is the military variant of the commercial Model SA-227-DC (Metro 23) aircraft series manufactured by Fairchild Aircraft Corporation. It is an all-metal, low wing, cabin-class aircraft with two Allied Signal TPE331-12 turboprop engines. The flight deck and cabin are pressurized for high-altitude flight. This aircraft incorporates a cargo door with an integral airstair door to permit easy entry and egress. With a maximum payload capacity of 5,020 pounds, the cabin can be configured to accommodate passengers, cargo or both.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	57 ft	Maximum Ramp Weight — 16,600 lbs
Length	59 ft 4 in	Maximum Landing Weight – 15,675 lbs
Height	16 ft 8 in	

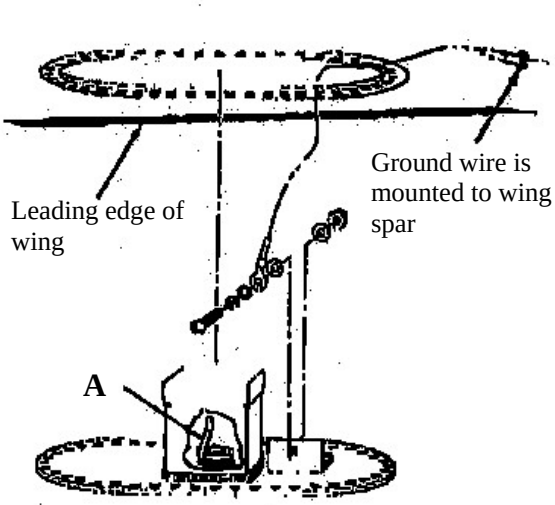
TABLE OF FUEL CAPACITIES

Tank	Gallons
Total Useable	648

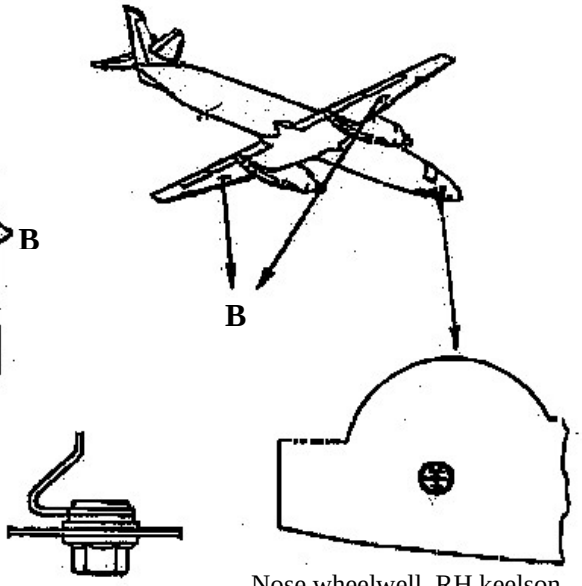
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Electrical Grounding



View looking up at underside of RH wing



DETAIL A
View looking outboard

Nose wheelwell, RH keelson

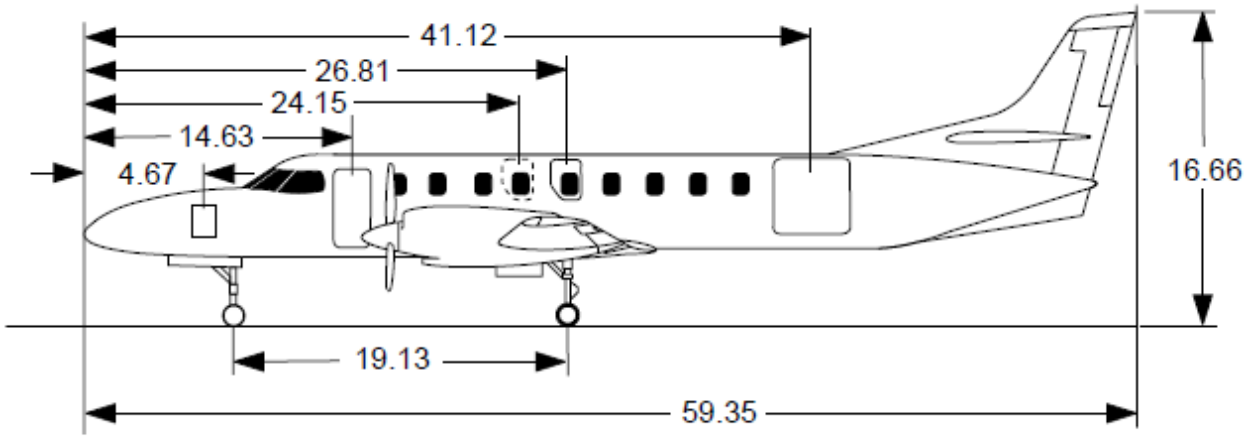
NOTE

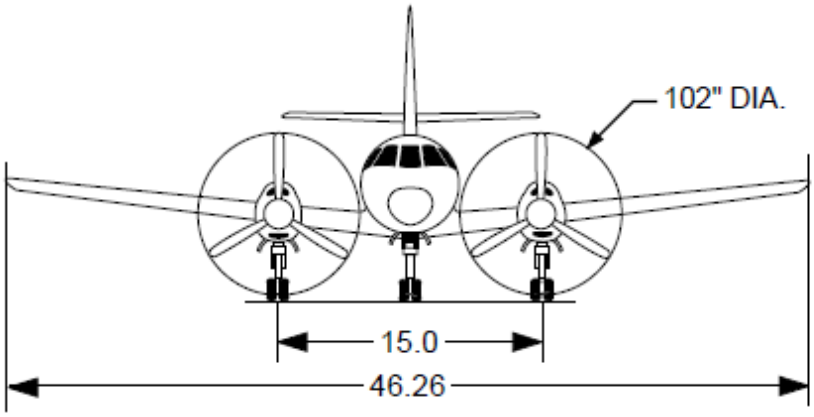
Connect overwing refueling nozzle bonding plug to underwing grounding receptacle prior to initiating refueling

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Propeller Location





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

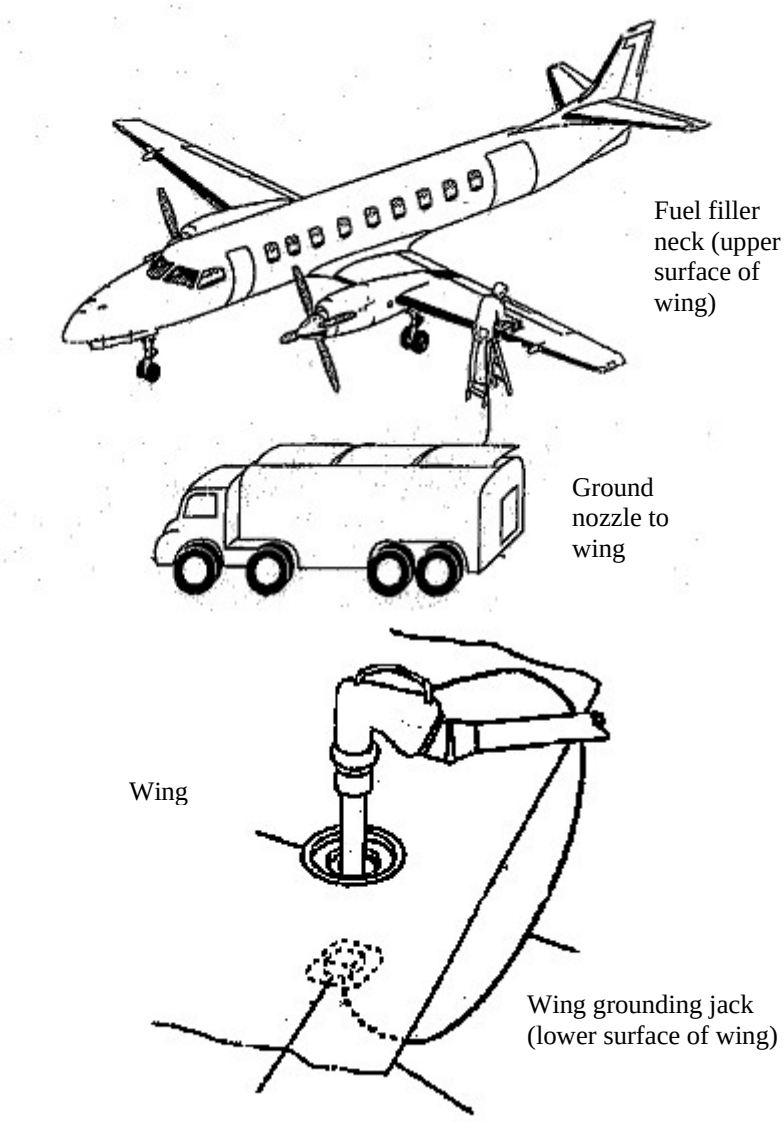
PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The C-26 aircraft is designed for overwing (gravity) refueling with one fuel cap on the upper surface of each wing. No provisions for single-point (pressure) refueling are provided. Use a small stepladder to access the refueling receptacle. Bring the refueling hose over the leading edge of the wing being careful to protect the wing deicing boot on the wing leading edge.

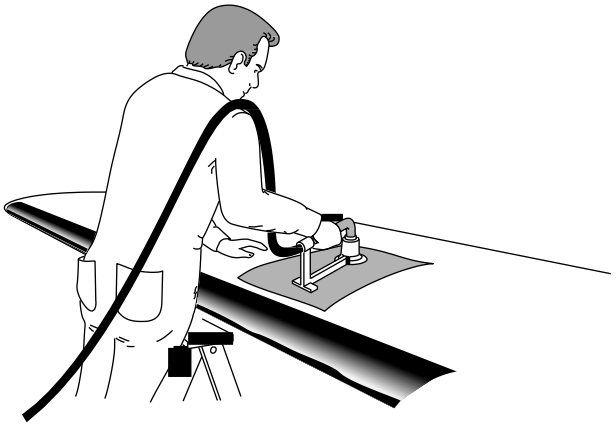
WARNING

Hot Gravity refueling of the C-26 is strictly prohibited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



CAUTION

DO NOT ALLOW AFT PORTION OF FUEL NOZZLE OR HOSE TO REST ON DEICE BOOT. THE DEICE BOOTS ARE MADE OF SOFT, FLEXIBLE, SYNTHETIC RUBBER WHICH MAY BE DAMAGED IF HOSES ARE DRAGGED OVER THE SURFACE OR IF LADDERS AND PLATFORMS ARE RESTED AGAINST THEM.

SPECIAL NOTES

Usable fuel capacity is 324 US gallons per wing (646 gallons total). The fuel quantity gauge compensates for fuel density and reads tank quantity in pounds.

NOTE

On a sloping ramp, the uphill tank may not accept a full fuel load if the downhill wing tank is filled first. When refueling on a sloping ramp, the uphill tank can be filled to maximum capacity by refilling both tanks simultaneously, by refilling the uphill wing tank first, or by adding fuel to the wings alternately in approximately 125 gallon increments. When less than maximum capacity is required, this special fueling procedure is not necessary.

When filling the wing tanks to maximum capacity under all conditions, expect the tanks to accept the last 20 or 30 gallons at a slower rate because the last of the fuel requires extra time to travel to all of the fuel bays.

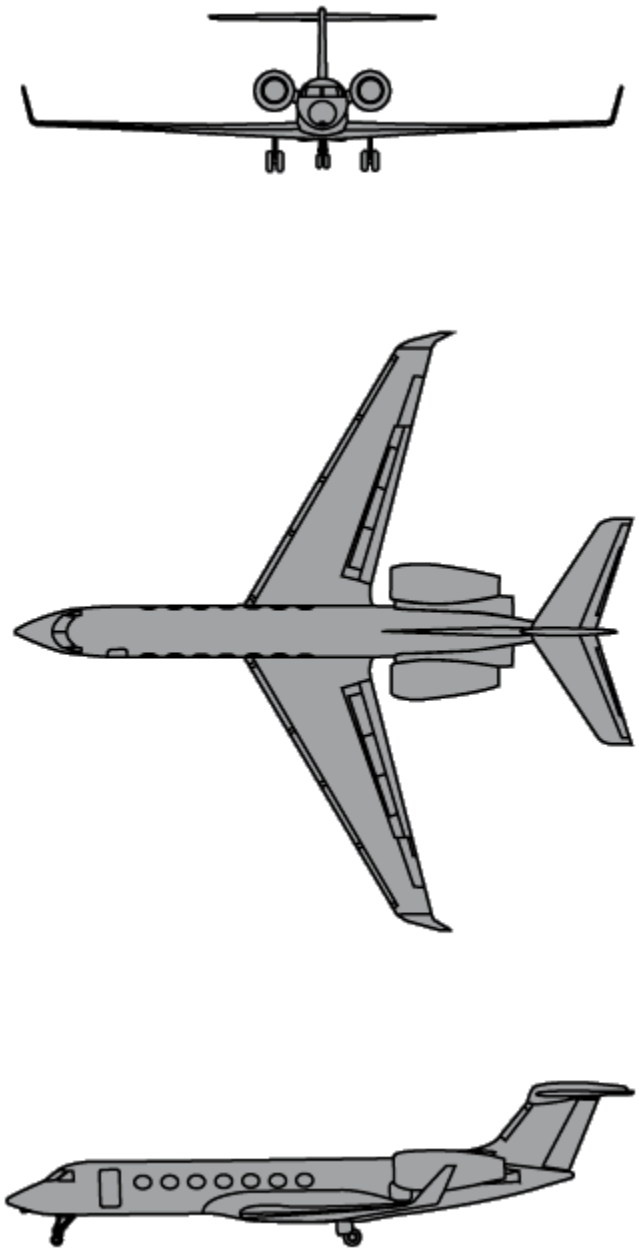
After fuel tanks are serviced, remove fuel nozzle before disconnecting ground.

Ensure tank caps are securely seated and locked (latch facing aft).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-37

The C-37 is the Navy's version of the Gulfstream G V corporate jet aircraft.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	93 ft 6 in	Maximum Takeoff Weight — 91,000 lbs
Length	96 ft 5 in	
Height	25 ft 10 in	

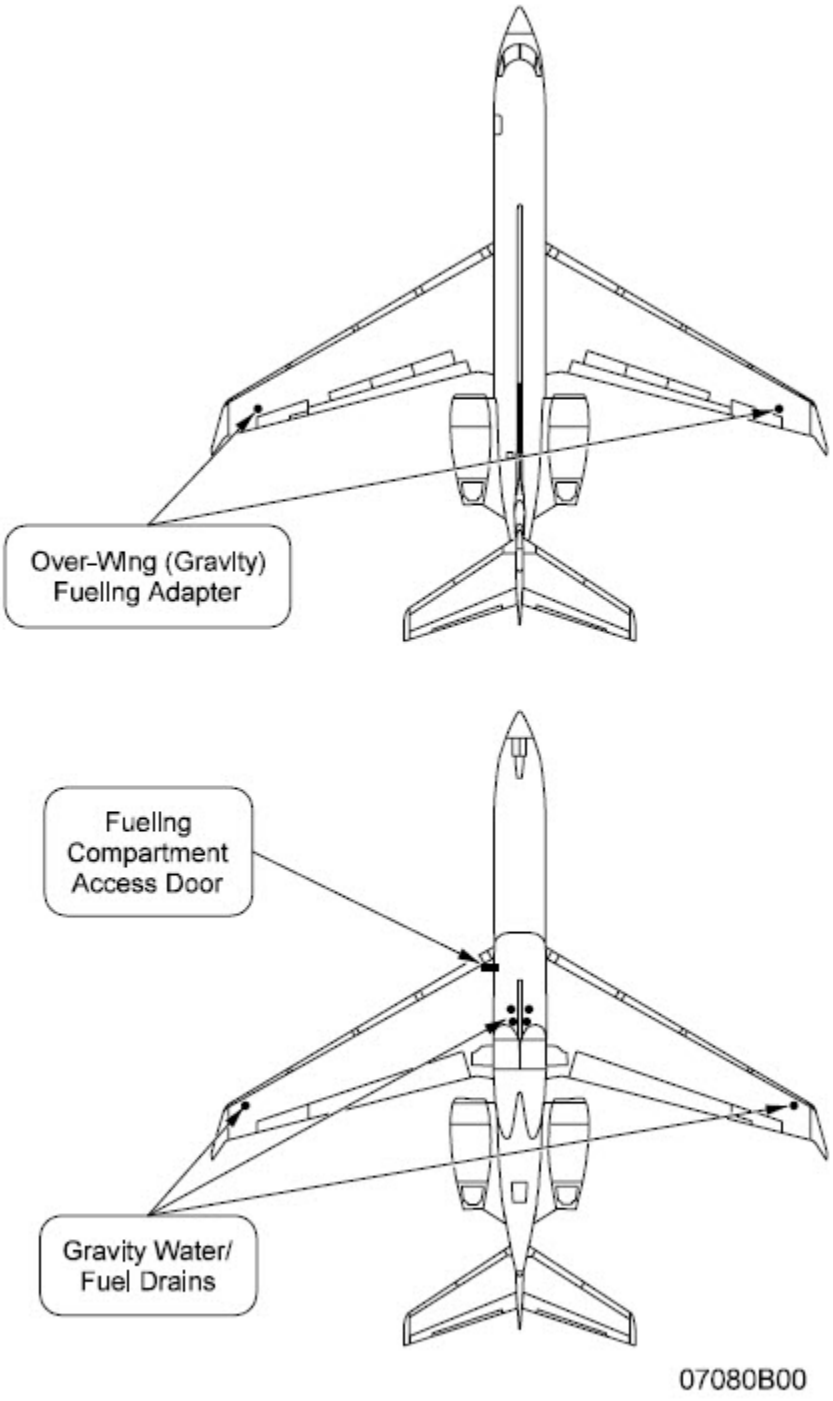
TABLE OF FUEL CAPACITIES

Tank	Gallons
Total Useable	6120

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

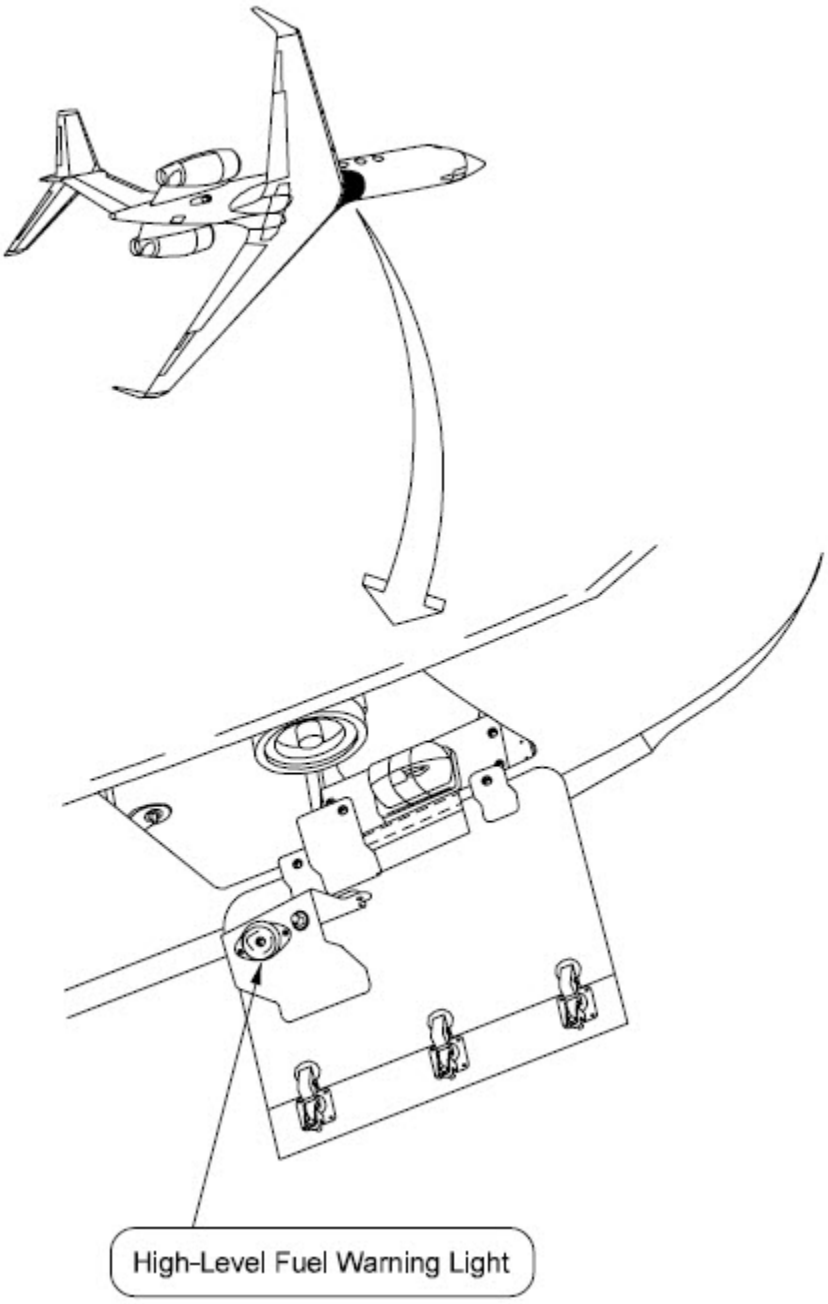
a) Gravity Refueling Locations and Single-Point Refueling Access Door



Fuel Storage System External Controls

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

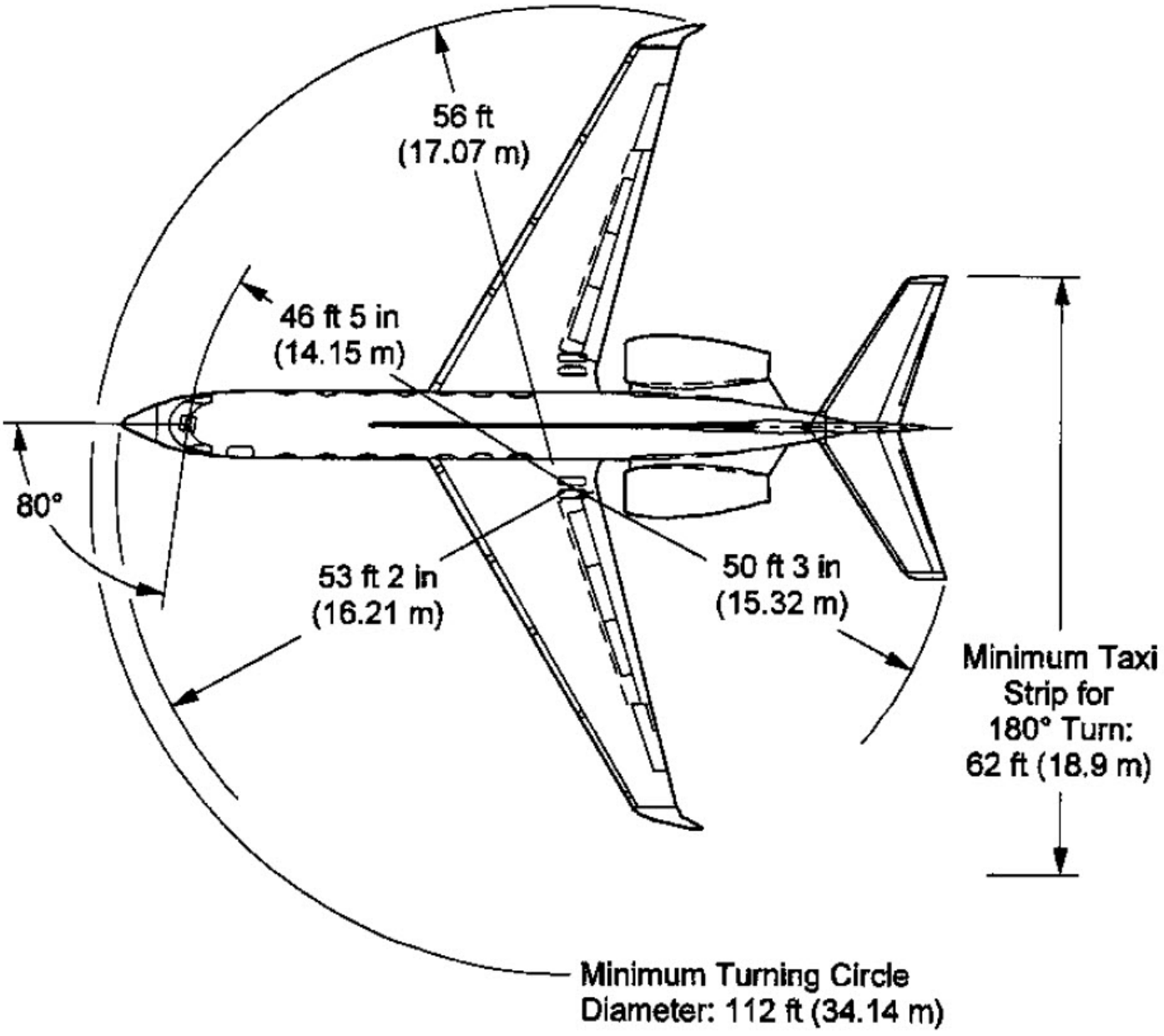
b) Fueling Compartment Access Door



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

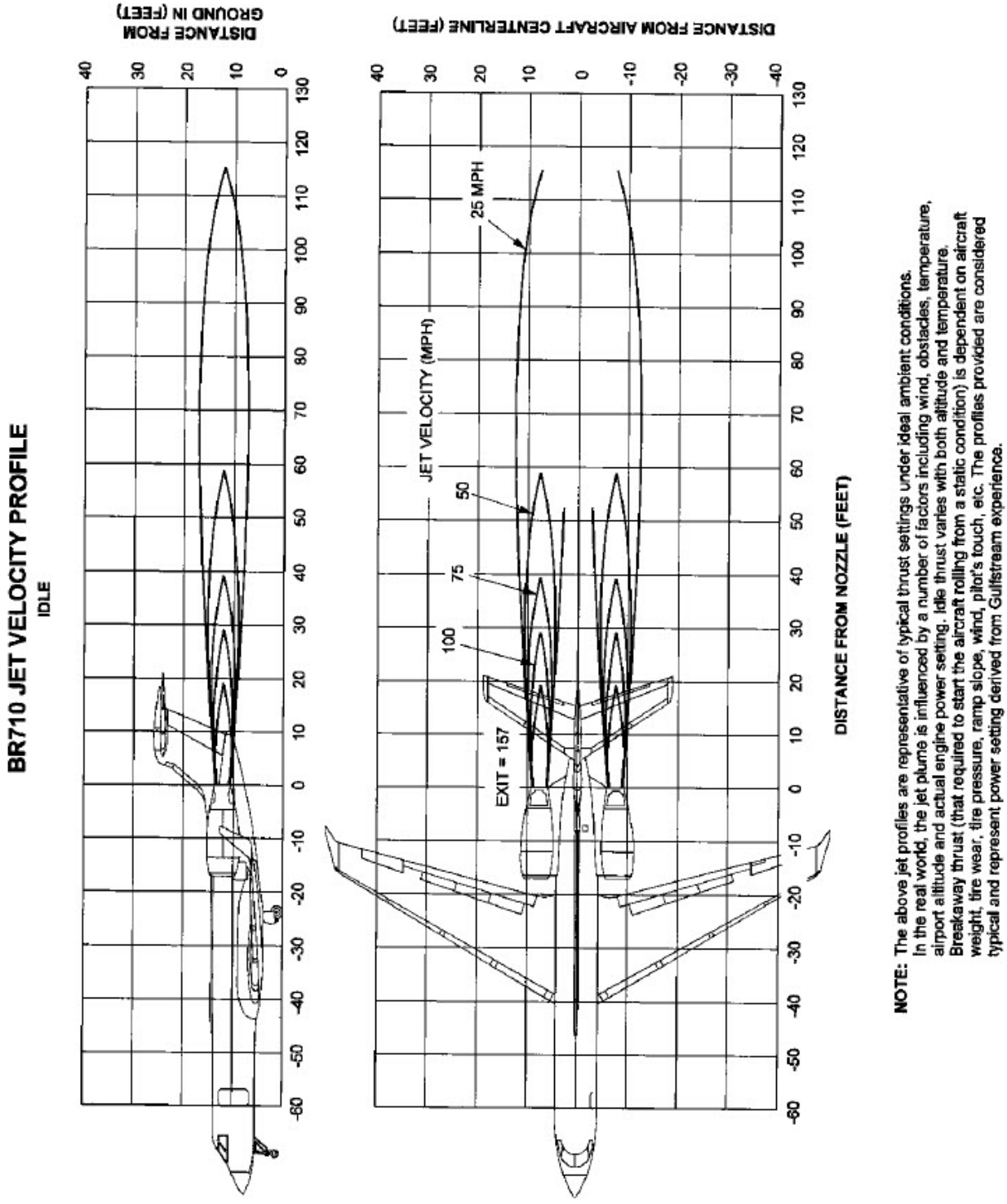
PERSONNEL DANGER ZONES

a) Turning Radius



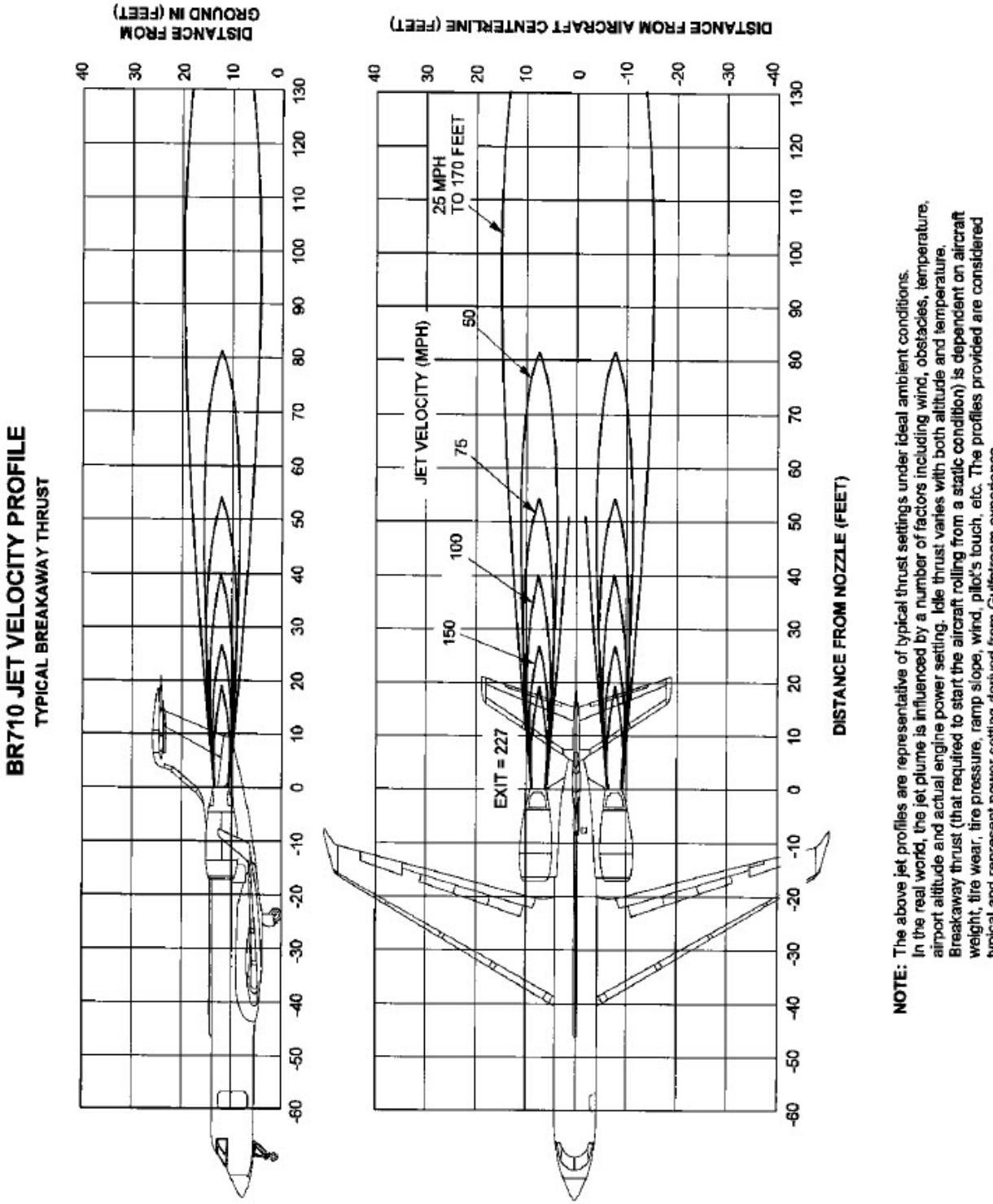
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Idle Thrust: Jet Velocity Profile



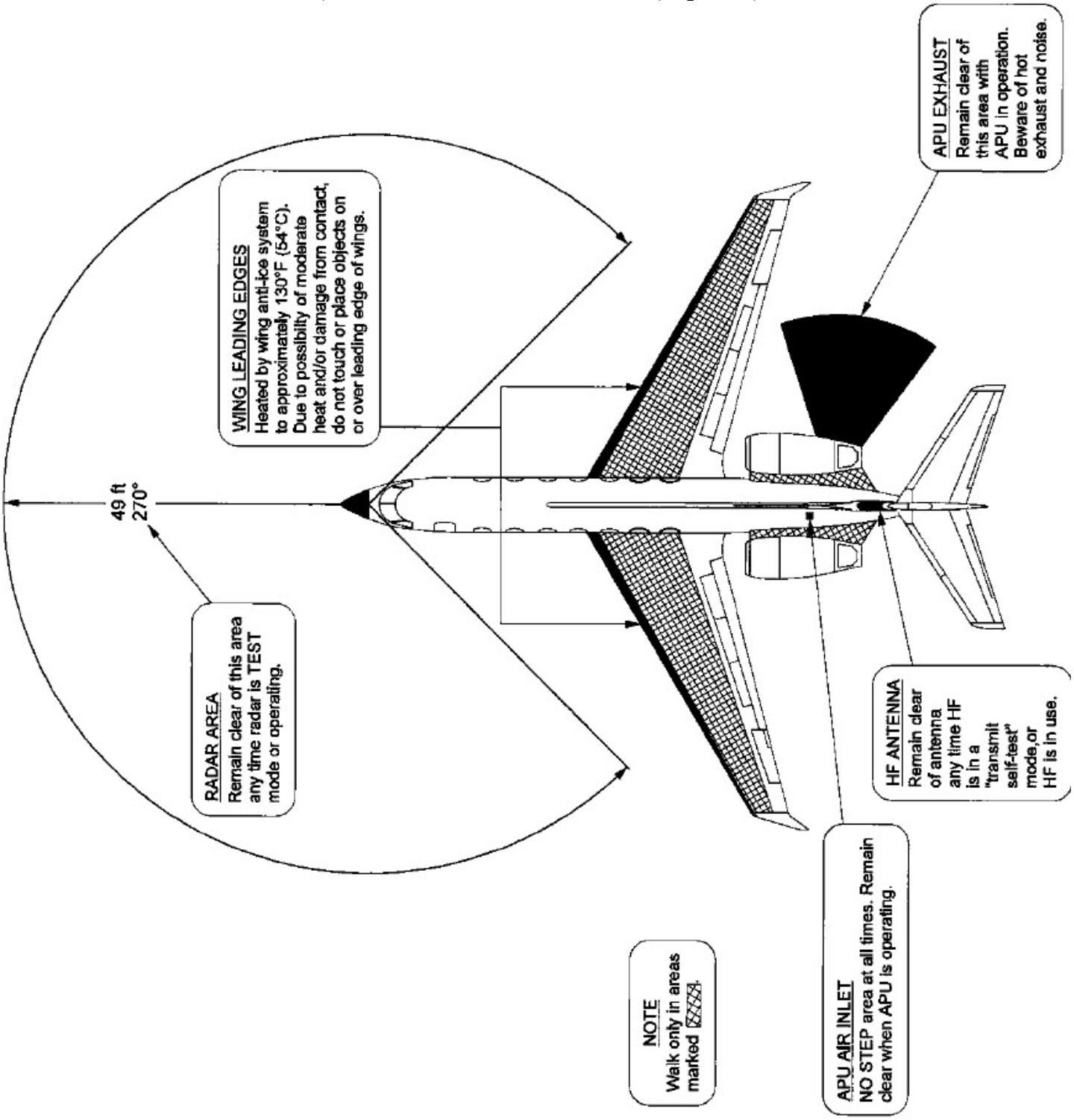
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Breakaway Thrust: Jet Velocity Profile



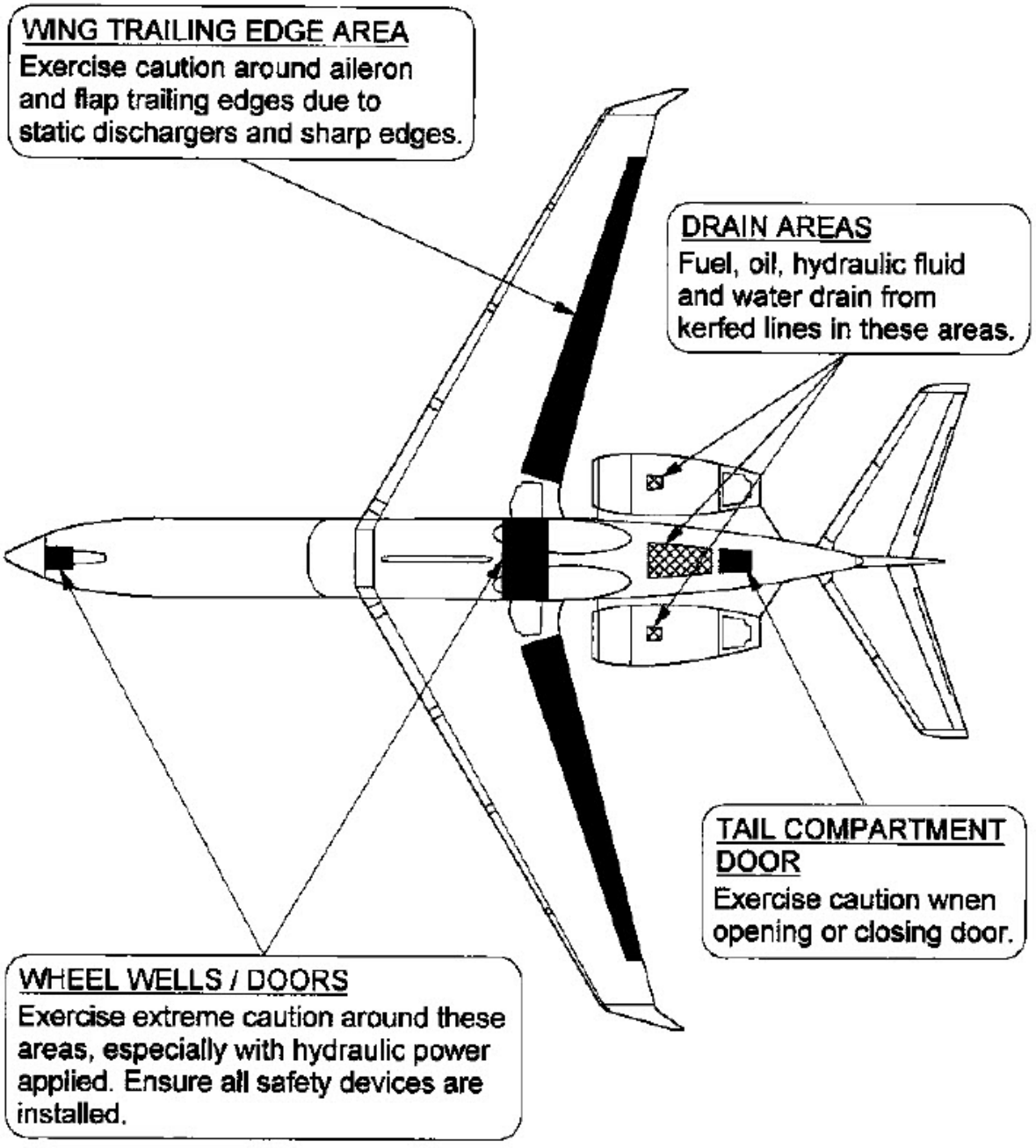
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Radar Area and APU Exhaust (Top View)



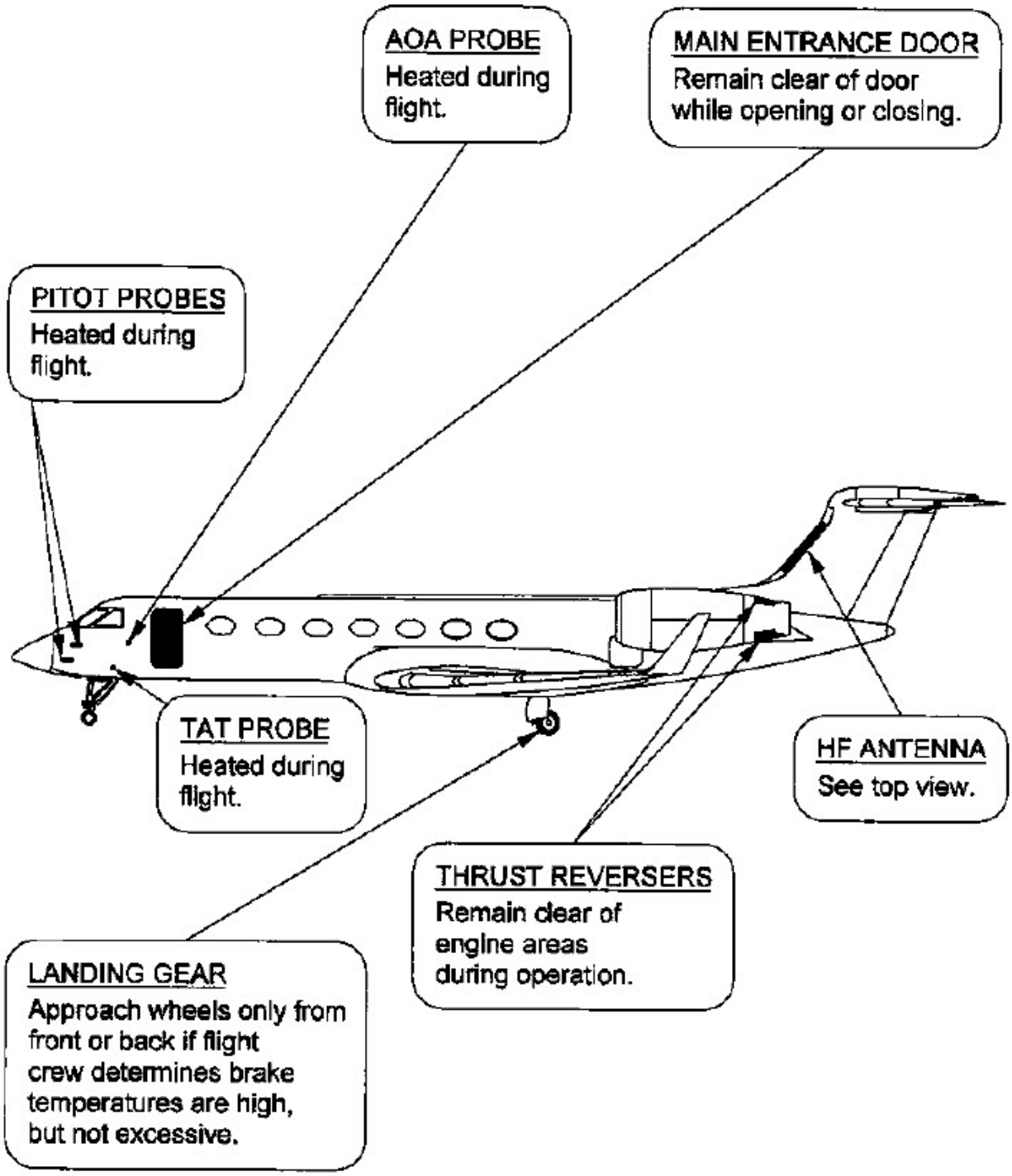
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

e) Danger and Caution Areas (Bottom View)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

f) Danger and Caution Areas (Left Fuselage; Right Fuselage does not contain main entrance door



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

For refueling procedures, refer to the C-37 (G V) flight manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C-40

The C-40 Clipper is the Navy version of the Boeing B737-700C.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	112 ft 7 in	Maximum Takeoff Weight — 171,000 lbs
Length	110 ft 4 in	
Height	41 ft 2 in	

TABLE OF FUEL CAPACITIES

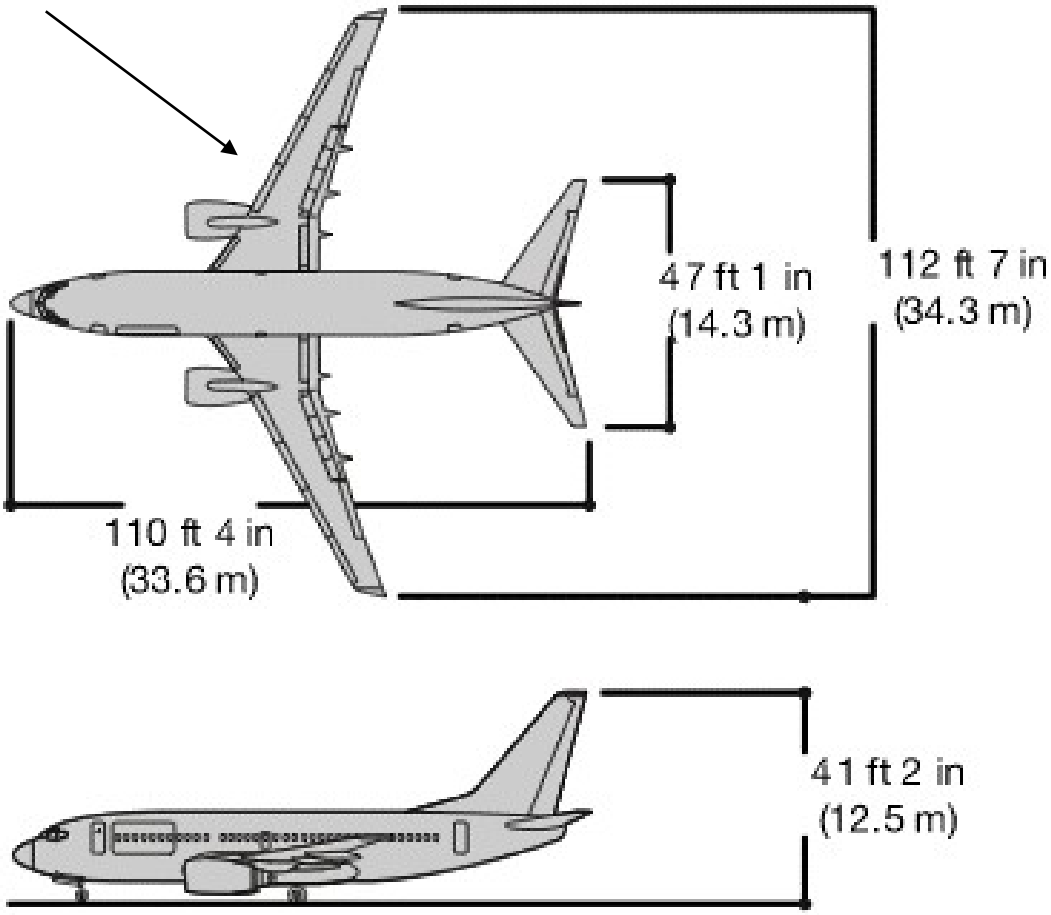
Tank		Gallons	Pounds (Approx) JP-5 @ 6.7lbs/gal
Main and Center wing tanks	M1 (Left)	1,288	8,630
	M2 (Right)	1,288	8,630
	Center	4,299	28,803
Total Main and Center		6,875	46,063
Surge Tanks	Each	35	235

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Ground Fueling Panel and Location

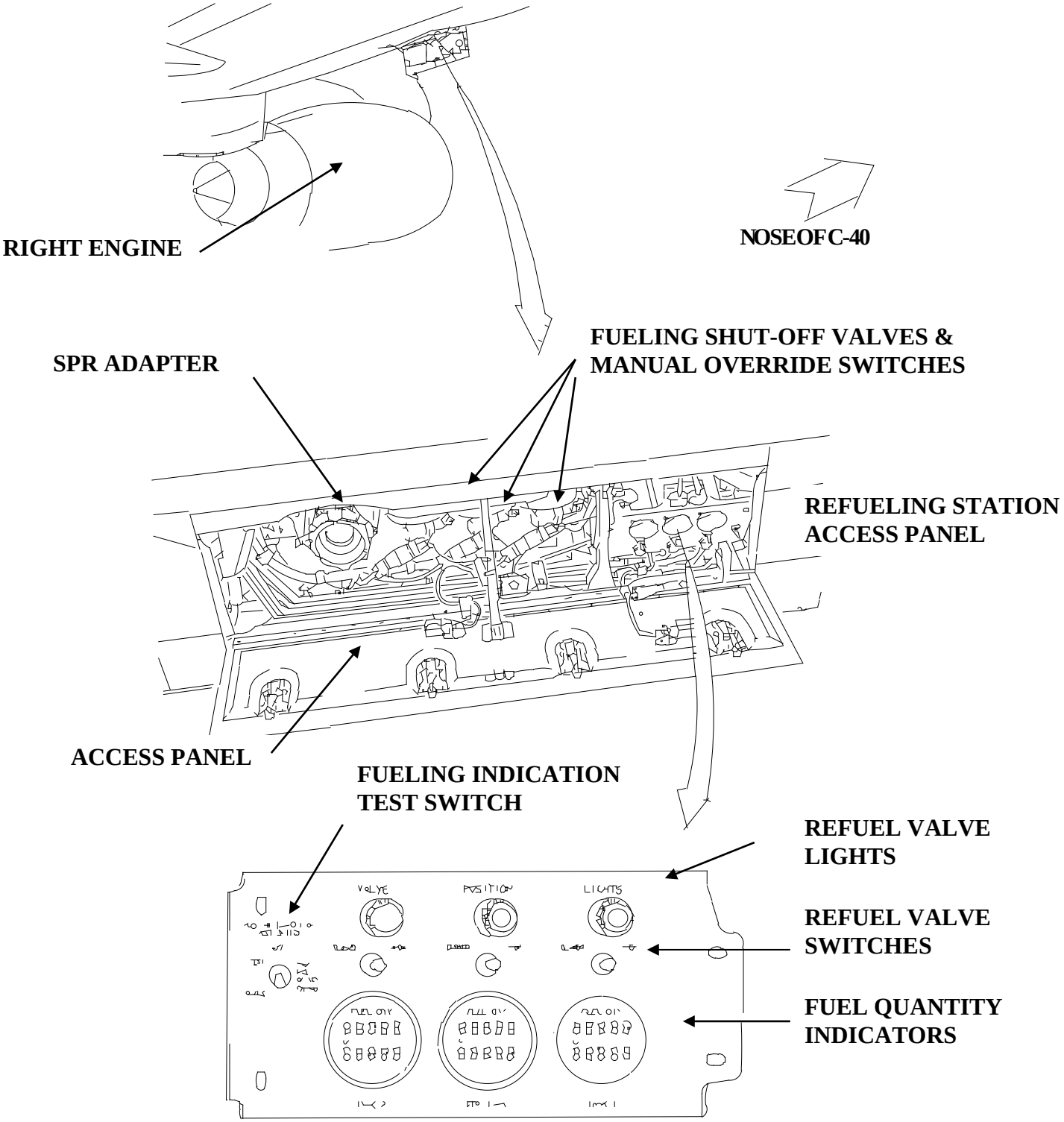
REFUELING PANEL LOCATION



The C-40 refueling panel is located just behind the leading edge of the starboard wing outboard of the engine nacelle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

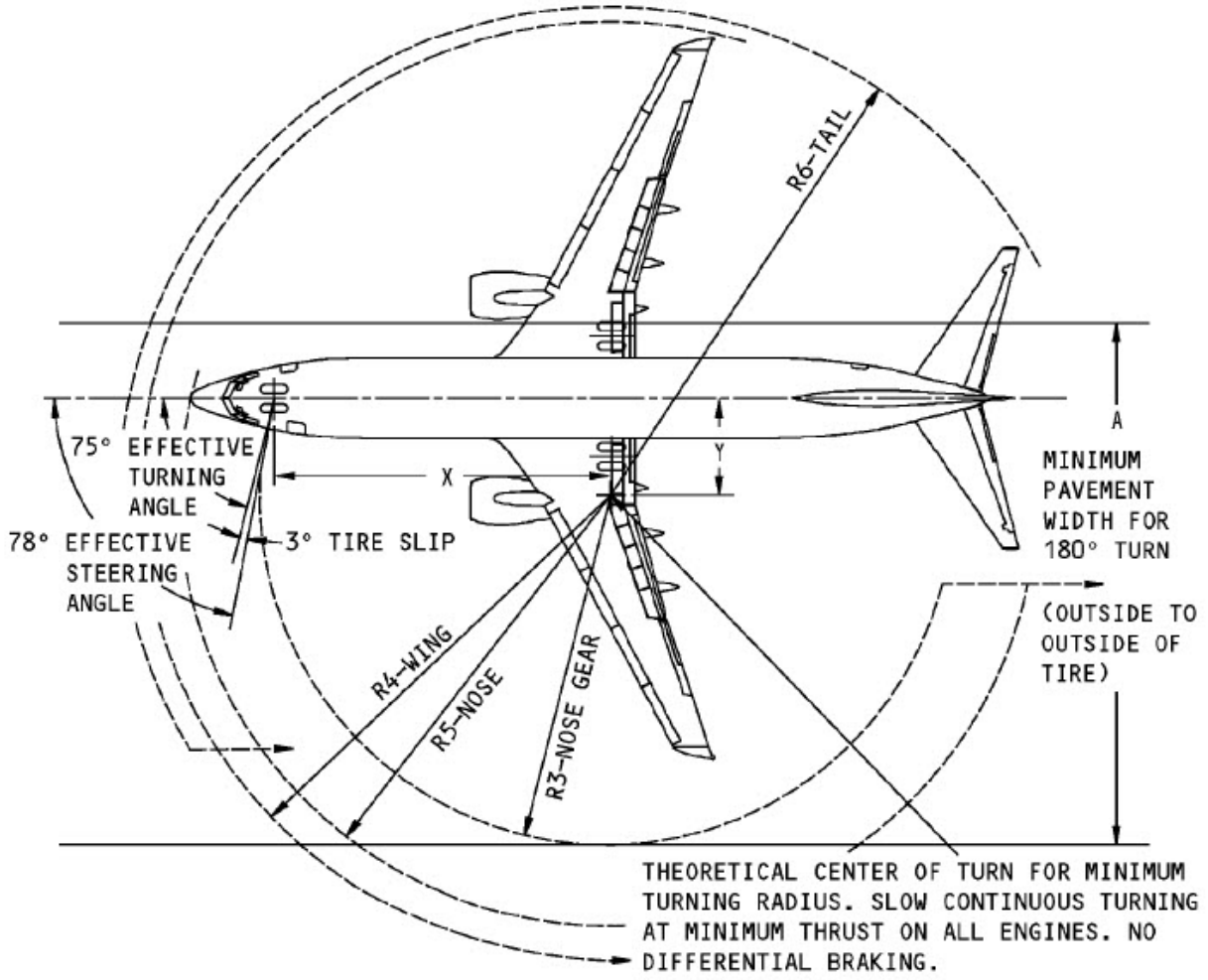
b) Indepth View of Ground Fueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Minimum Towing Turning Radius

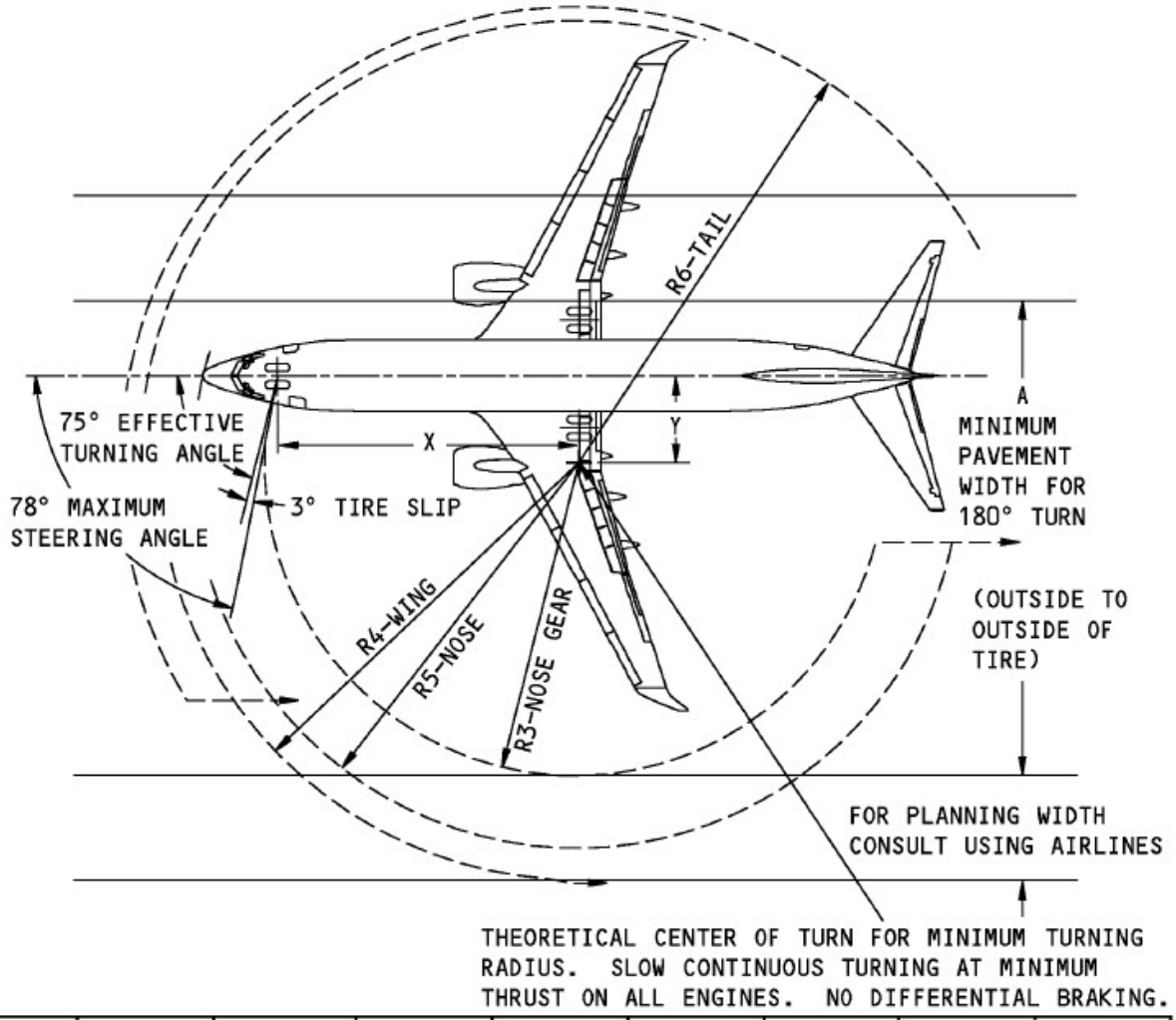


AIRPLANE MODEL	EFFECTIVE TURNING ANGLE (°)	X		Y		A		R3 NOSE GEAR		R4 WING		R5 NOSE		R6 TAIL	
		FT	M	FT	M	FT	M	FT	M	FT	M	FT	M	FT	M
737-700	75	41.3	12.6	11.1	3.4	66.4	20.3	43.8	13.3	69.6	21.2	55.9	17.0	65.5	20.0

NOTE:
- 3° TIRE SLIP ANGLE APPROXIMATE ONLY FOR 78° STEERING ANGLE.
- DIMENSIONS ROUNDED TO NEAREST 0.1 FOOT AND 0.1 METER.
- EFFECTIVE TURNING ANGLE IS 75°.
- NO WINGLET.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Minimum Towing Turning Radius with Winglets

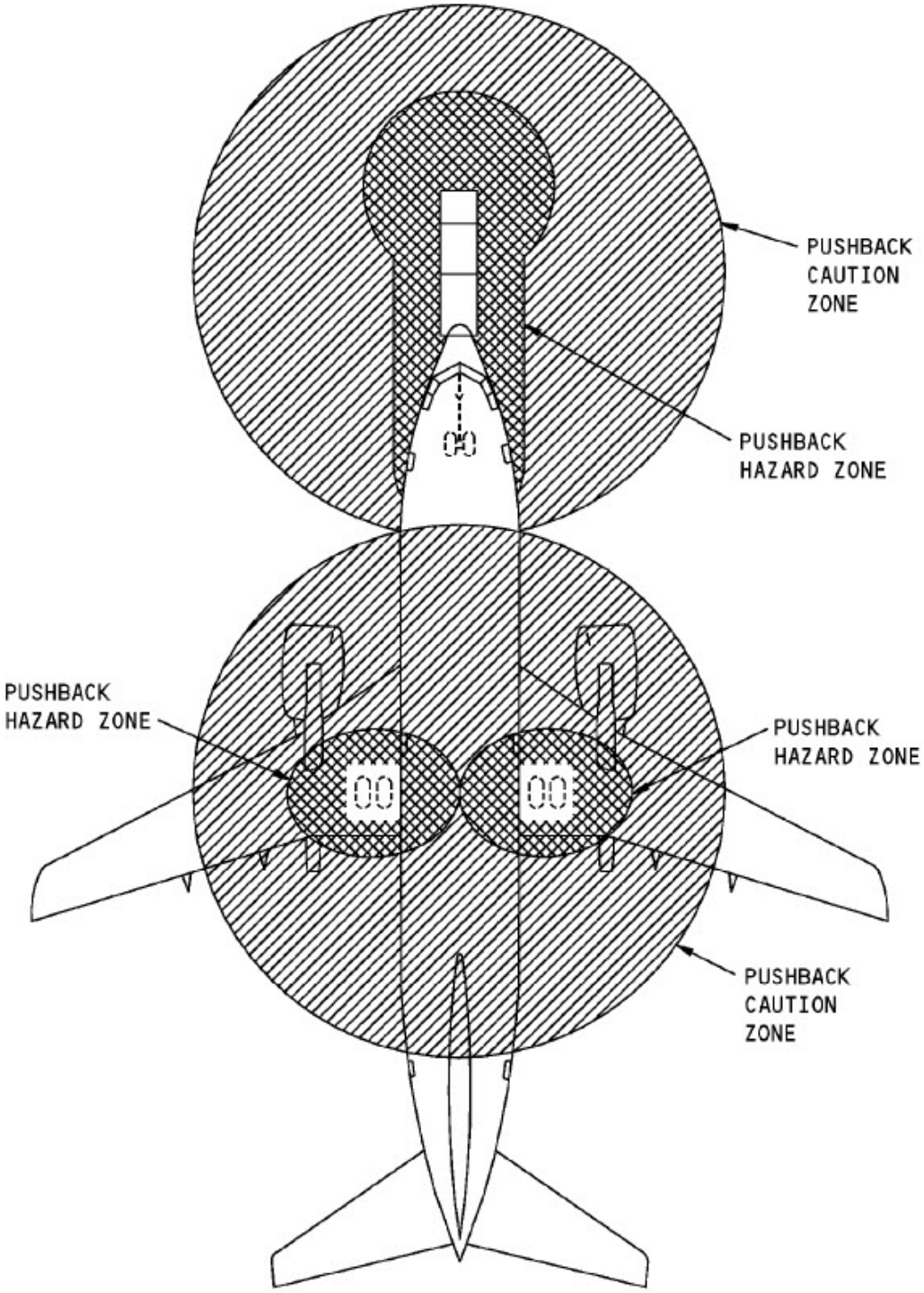


AIRPLANE MODEL	EFFECTIVE TURNING ANGLE (°)	X		Y		A		R3 NOSE GEAR		R4 WING		R5 NOSE		R6 TAIL	
		FT	M	FT	M	FT	M	FT	M	FT	M	FT	M	FT	M
737-700 AND 737-700 IGW/BBJ	75	41.3	12.6	11.1	3.4	66.4	20.3	43.8	13.3	72.6	22.1	55.9	17.0	65.5	20.0

NOTE:
- 3° TIRE SLIP ANGLE APPROXIMATE ONLY FOR 78° STEERING ANGLE.
- DIMENSIONS ROUNDED TO NEAREST 0.1 FOOT AND 0.1 METER.
- EFFECTIVE TURNING ANGLE IS 75°.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

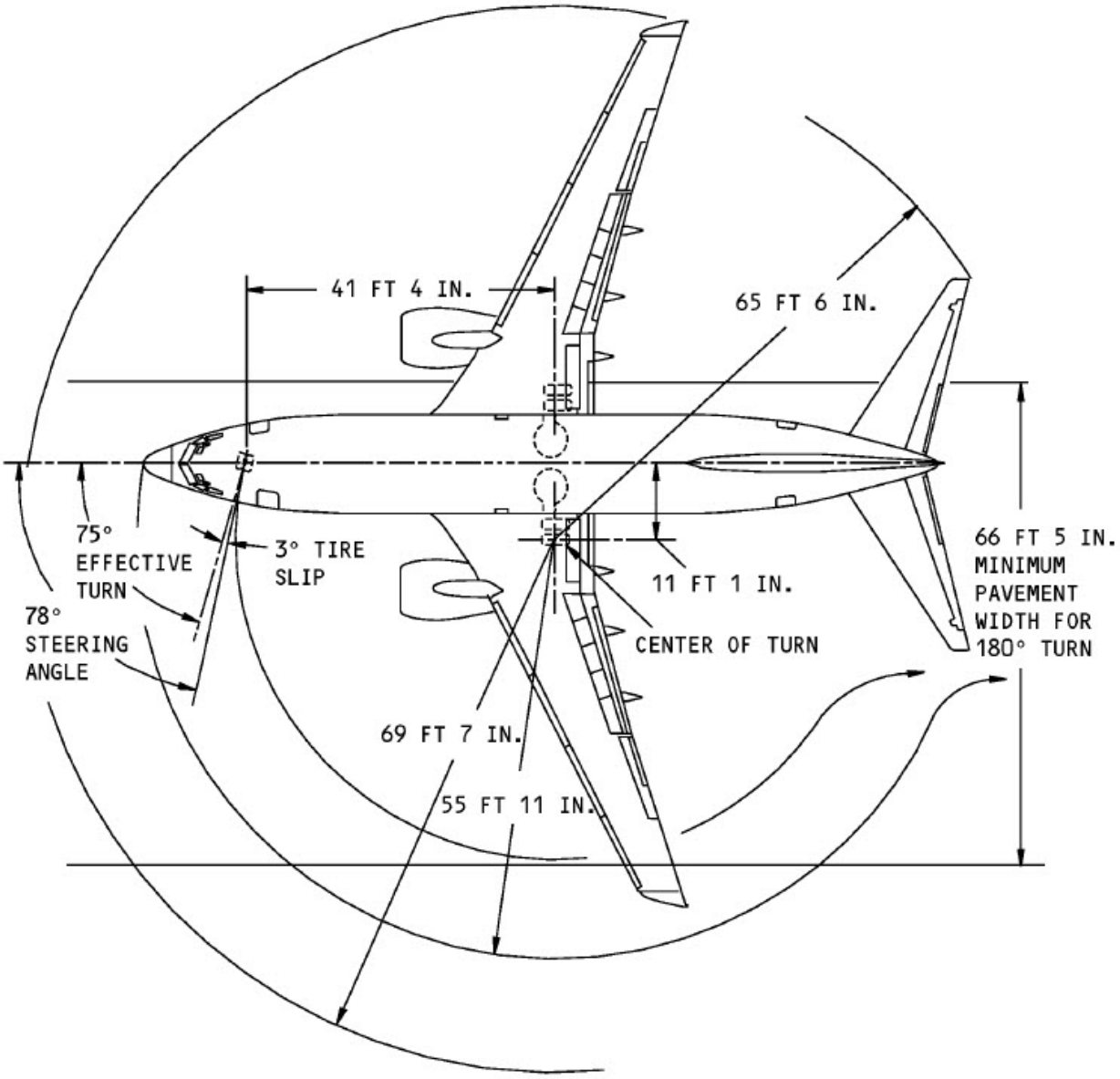
c) Towing Hazard Zones



WARNING: MAINTAIN A MINIMUM OF TEN FEET (3 METERS) SEPARATION BETWEEN PERSONS ON THE GROUND, AND THE NOSE WHEELS, THE TOW BAR AND TOW VEHICLE, AND THE MAIN WHEELS WHILE THE AIRPLANE IS MOVING.

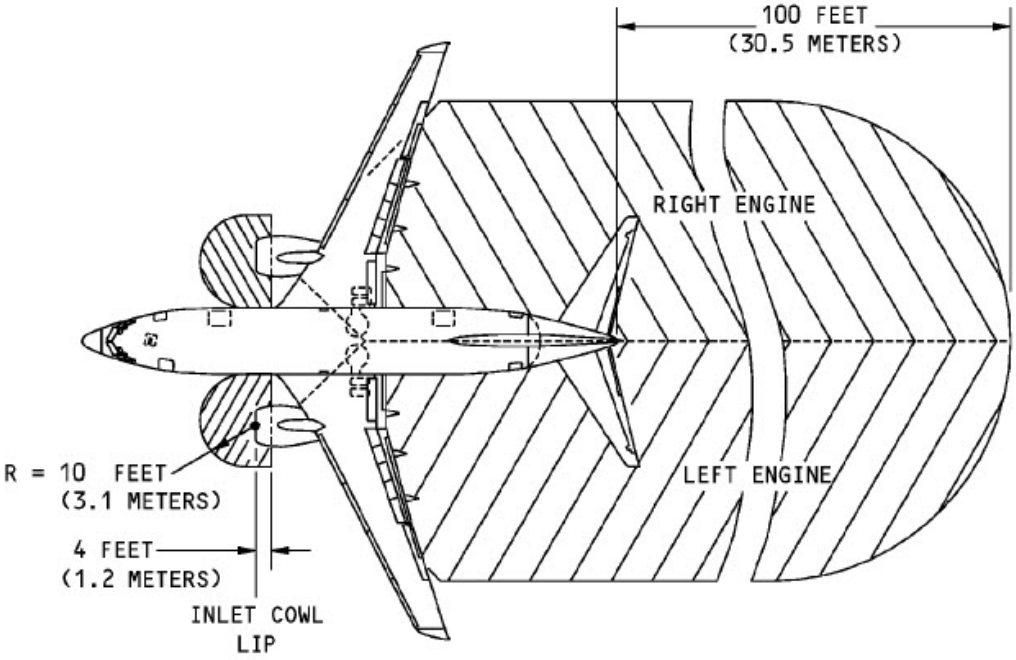
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Taxi Turning Radius



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

e) Idle Power Exhaust Danger Zones



WARNING: KEEP ALL PERSONS OUT OF THE DANGEROUS AREA DURING ENGINE OPERATION. IF THE SURFACE WIND IS MORE THAN 25 KNOTS, INCREASE THE DANGEROUS AREA AT THE ENGINE INLET BY 20 PERCENT.

CLEAN THE RAMP IF THERE IS SNOW, ICE, WATER, OIL OR OTHER CONTAMINATION ON THE RAMP. YOU CAN ALSO MOVE AIRPLANE TO A LOCATION THAT IS CLEAN.

MAKE SURE THAT ALL PERSONS ARE SAFE BEFORE YOU START THE ENGINE.

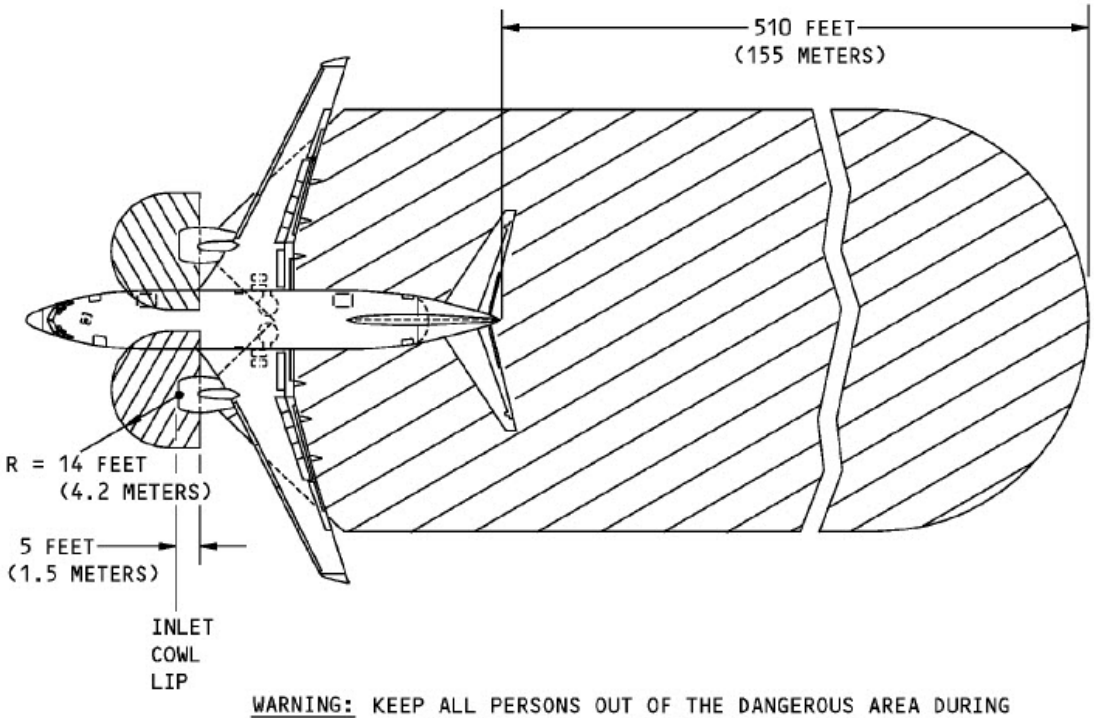
MAKE SURE THE PERSONS IN THE FLIGHT COMPARTMENT CAN SPEAK TO ALL PERSONS NEAR THE DANGEROUS AREA DURING ENGINE OPERATION.

OBEY ALL OF THE GROUND SAFETY PRECAUTIONS FOR THE ENGINES.

THE ENGINES CAN PULL PERSONS OR UNWANTED MATERIALS INTO THEM AND CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

f) Breakaway Power Exhaust Danger Zones



WARNING: KEEP ALL PERSONS OUT OF THE DANGEROUS AREA DURING ENGINE OPERATION. IF THE SURFACE WIND IS MORE THAN 25 KNOTS, INCREASE THE DANGEROUS AREA AT THE ENGINE INLET BY 20 PERCENT.

CLEAN THE RAMP IF THERE IS SNOW, ICE, WATER, OIL OR OTHER CONTAMINATION ON THE RAMP. YOU CAN ALSO MOVE THE AIRPLANE TO A LOCATION THAT IS CLEAN.

MAKE SURE THAT ALL PERSONS ARE SAFE BEFORE YOU START THE ENGINE.

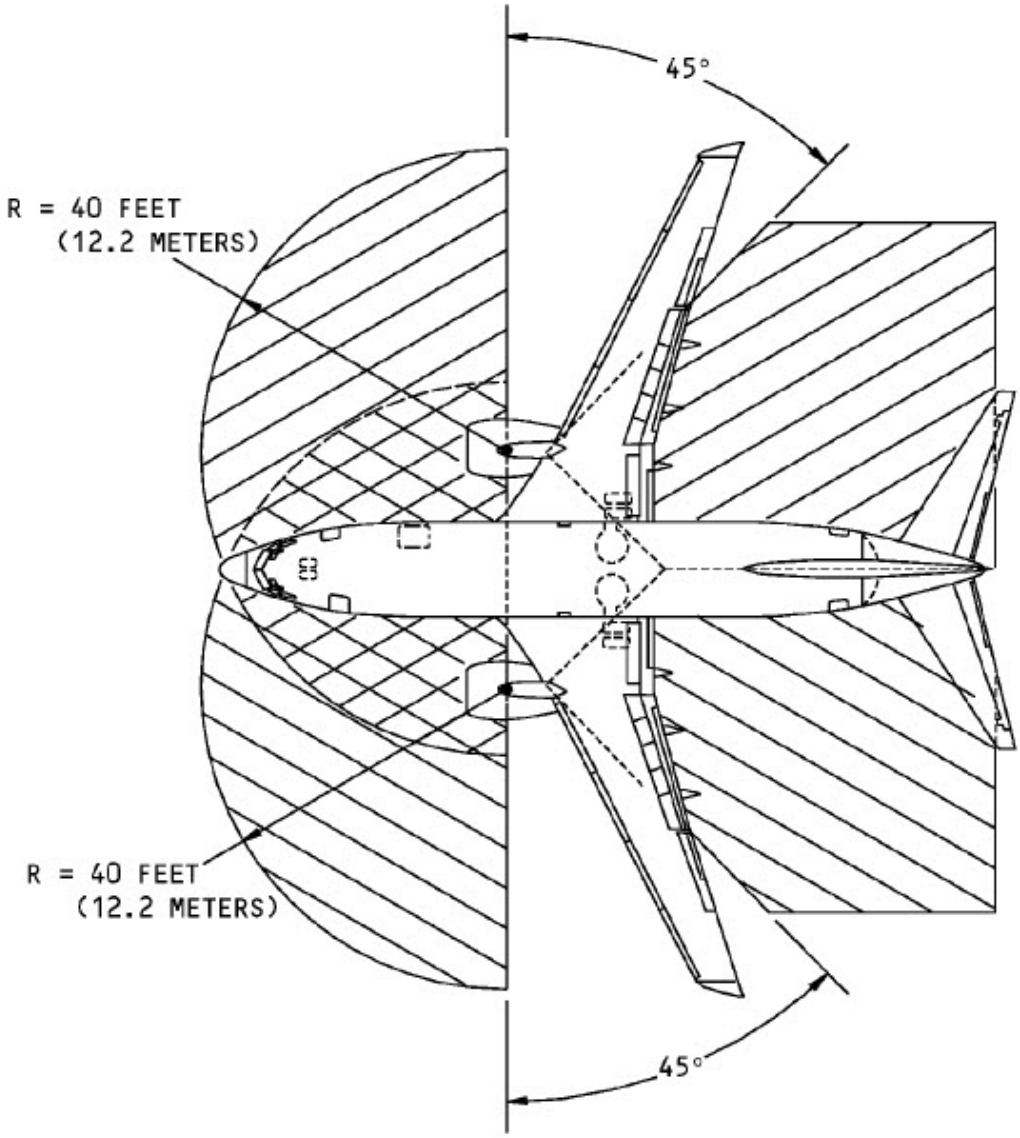
MAKE SURE THE PERSONS IN THE FLIGHT COMPARTMENT CAN SPEAK TO ALL PERSONS NEAR THE DANGEROUS AREA DURING ENGINE OPERATION.

OBEY ALL OF THE GROUND SAFETY PRECAUTIONS FOR THE ENGINES.

THE ENGINES CAN PULL PERSONS OR UNWANTED MATERIALS INTO THEM AND CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

g) Idle Power Exhaust Danger Zones (Reverse)



WARNING: MAKE SURE THAT ALL PERSONS ARE SAFE BEFORE YOU START THE ENGINE.

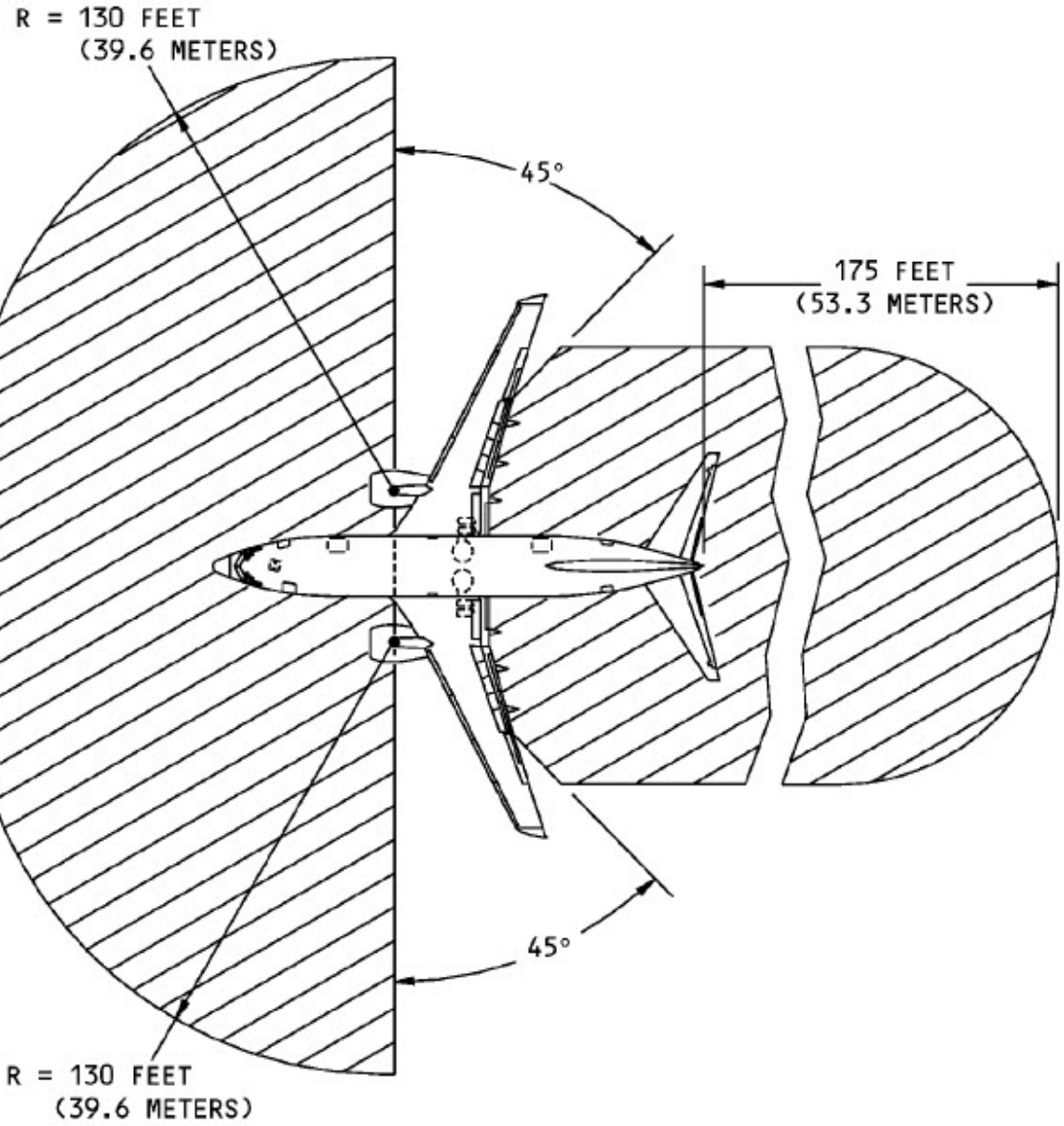
MAKE SURE THE PERSONS IN THE FLIGHT COMPARTMENT CAN SPEAK TO ALL PERSONS NEAR THE DANGEROUS AREA DURING ENGINE OPERATION.

OBEY ALL OF THE GROUND SAFETY PRECAUTIONS FOR THE ENGINES.

THE ENGINES CAN PULL PERSONS OR UNWANTED MATERIALS INTO THEM AND CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

h) Breakaway Power Exhaust Danger Zones (Reverse)



WARNING: MAKE SURE THAT ALL PERSONS ARE SAFE BEFORE YOU START THE ENGINE.

MAKE SURE THE PERSONS IN THE FLIGHT COMPARTMENT CAN SPEAK TO ALL PERSONS NEAR THE DANGEROUS AREA DURING ENGINE OPERATION.

OBEY ALL OF THE GROUND SAFETY PRECAUTIONS FOR THE ENGINES.

THE ENGINES CAN PULL PERSONS OR UNWANTED MATERIALS INTO THEM AND CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the C-40 aircraft. They have been extracted from standard commercial refueling procedures for the B737-700 series aircraft. For more detailed information on refueling the C-40, consult the C-40 operations manual. In addition to the following procedures, the applicable basic refueling procedures contained in Chapter 12 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Connect the bonding cable between the refueling equipment and the aircraft.

2. Open the aircraft REFUELING STATION ACCESS PANEL.

3. Connect pressure refueling nozzle to the spr adapter.

4. Hold the FUELING INDICATION TEST SWITCH to ensure indicators operate correctly.

a. The fueling indicators will blank for 2 seconds
b. The led segments will then go on for 2 seconds
c. The above sequence will continue while the TEST SWITCH is held for 20 seconds
d. Release the TEST SWITCH
e. If an internal error is found, the indicator will show fail

5. Push and release the FUELING VALVE LIGHTS to test the bulbs. Make sure each FUELING VALVE LIGHT comes on and then goes off.

6. Set REFUEL VALVE SWITCHES to OPEN for tanks to be filled (if full fuel load is required, set all REFUEL VALVE SWITCHES to OPEN).

7. Activate deadman to begin refueling aircraft.

8. Monitor FUEL QUANTITY INDICATORS to ensure that they do not start to FLASH.

CAUTION

If INDICATORS flash, STOP fueling as tank can be overfilled.

NOTE

Manual returning of a fueling valve to the OFF position during fueling will STOP the fuel flow to THAT TANK.

9. When the required amount of fuel has been loaded onto the aircraft, set each FUELING VALVE SWITCH to the CLOSED position.

NOTE

Fueling valves will close automatically as each tank reaches FULL.

10. Release the DEADMAN, disconnect pressure refueling nozzle from the SPR adapter, and re-install the SPR adapter cap.

11. Verify that all switches are in the closed position.

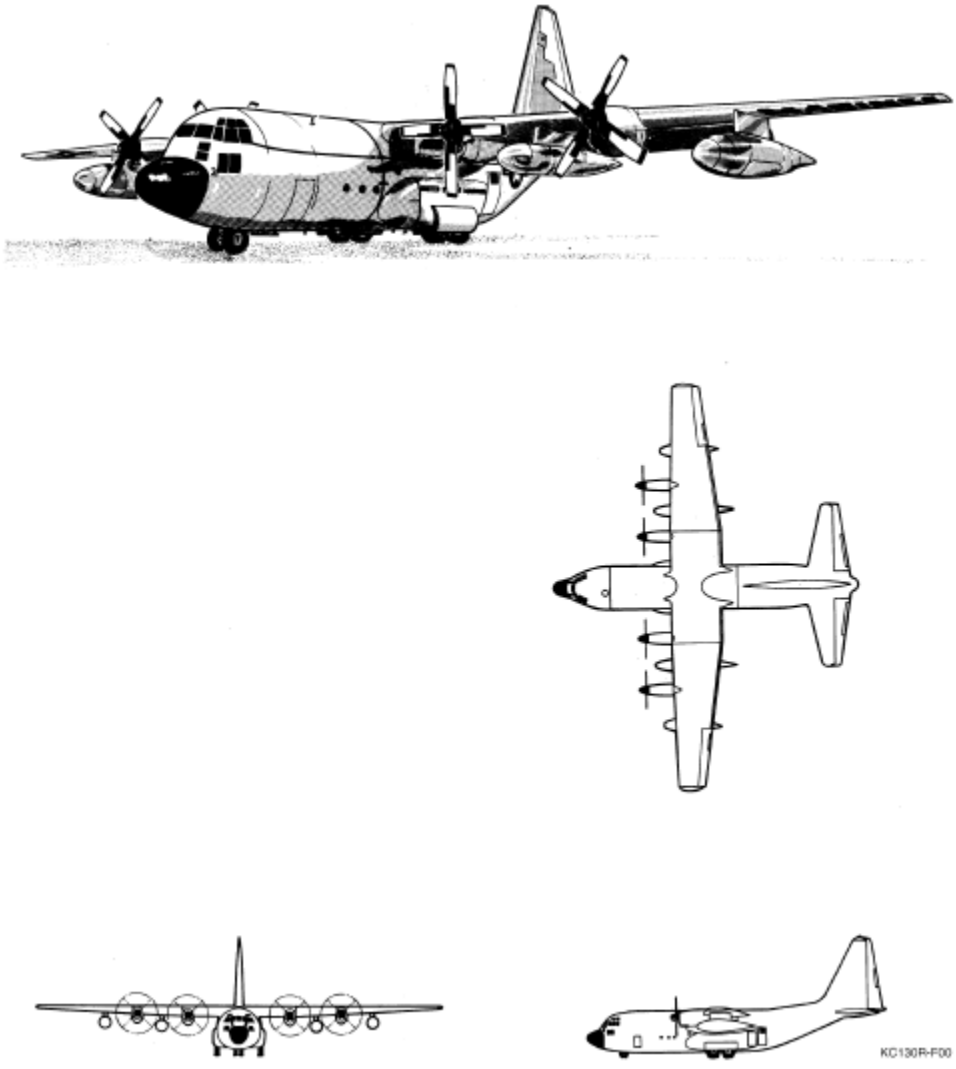
12. Close and securely latch the REFUELING STATION ACCESS PANEL.

13. Disconnect the bonding cable from the aircraft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C/KC-130 SERIES (EXCEPT MODEL J)

The Lockheed C/KC-130 series are high-wing, all-metal, long-range land-based monoplanes. Power is supplied by four Allison T56-A-16, turboprop, constant-speed engines. The mission of these aircraft is to provide in-flight refueling or rapid transportation of personnel or cargo for delivery by parachute or landing.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: C/KC-130 SERIES (EXCEPT MODEL J)

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	132 ft 7 in	Maximum Gross Weight — 175,000 lbs
Length	97 ft 5 in	Maximum Footprint — 123 psi
Height	39 ft 2 in	

TABLE OF FUEL CAPACITIES

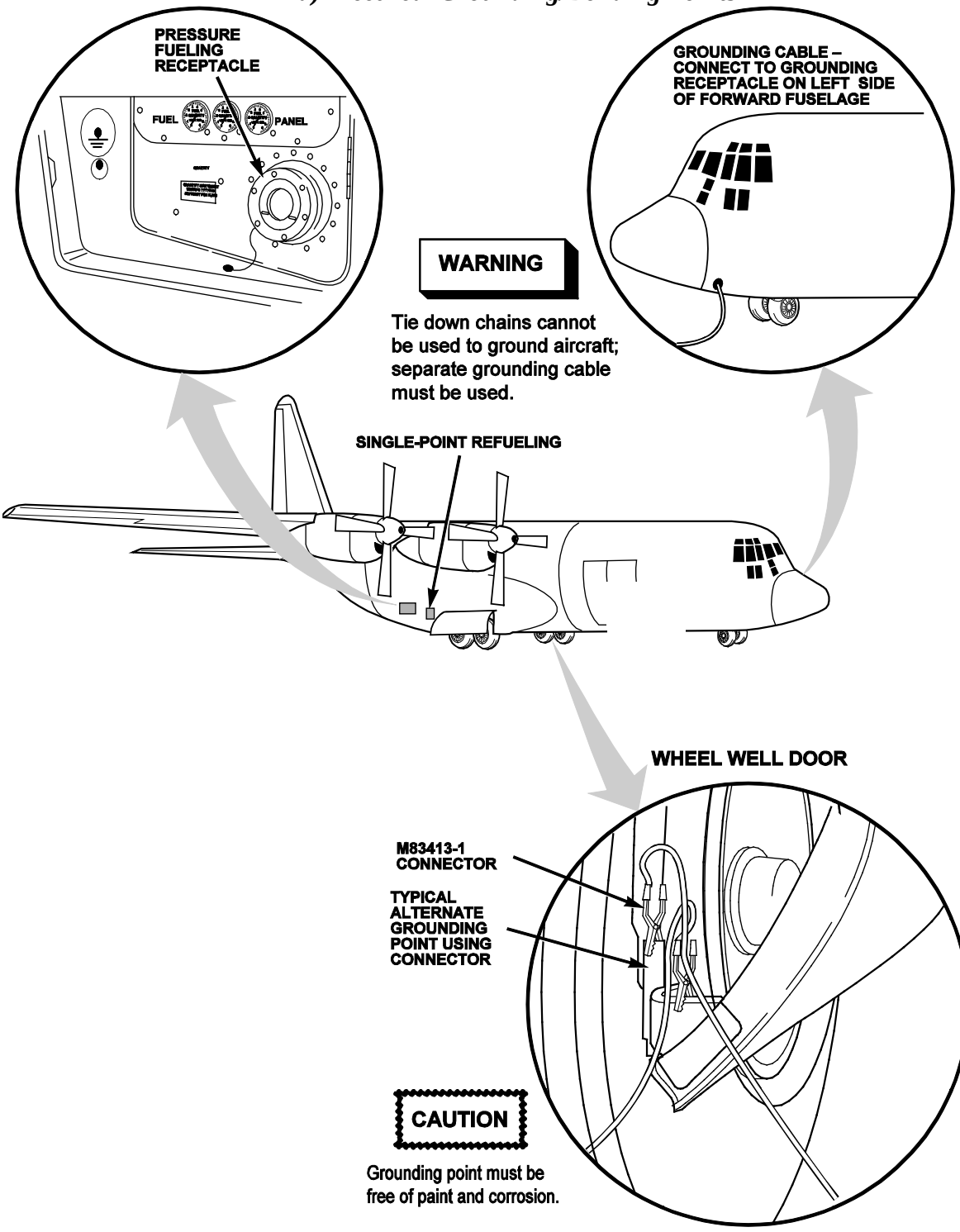
NOTE: Each series has varying fuel tank sizes that can be +/- 100 Gallons.

Tank		Gallons	Pounds
Internal	Tank No. 1	1200	8160
	Tank No. 2	1100	7480
	Tank No. 3	1100	7480
	Tank No. 4	1200	8160
	Left Auxiliary	860	5810
	Right Auxiliary	860	5810
Internal Total		6320	42900
External	Left	1310	8900
	Right	1310	8900
External Total		2620	17800
Total	Internal + External	8940	60700

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

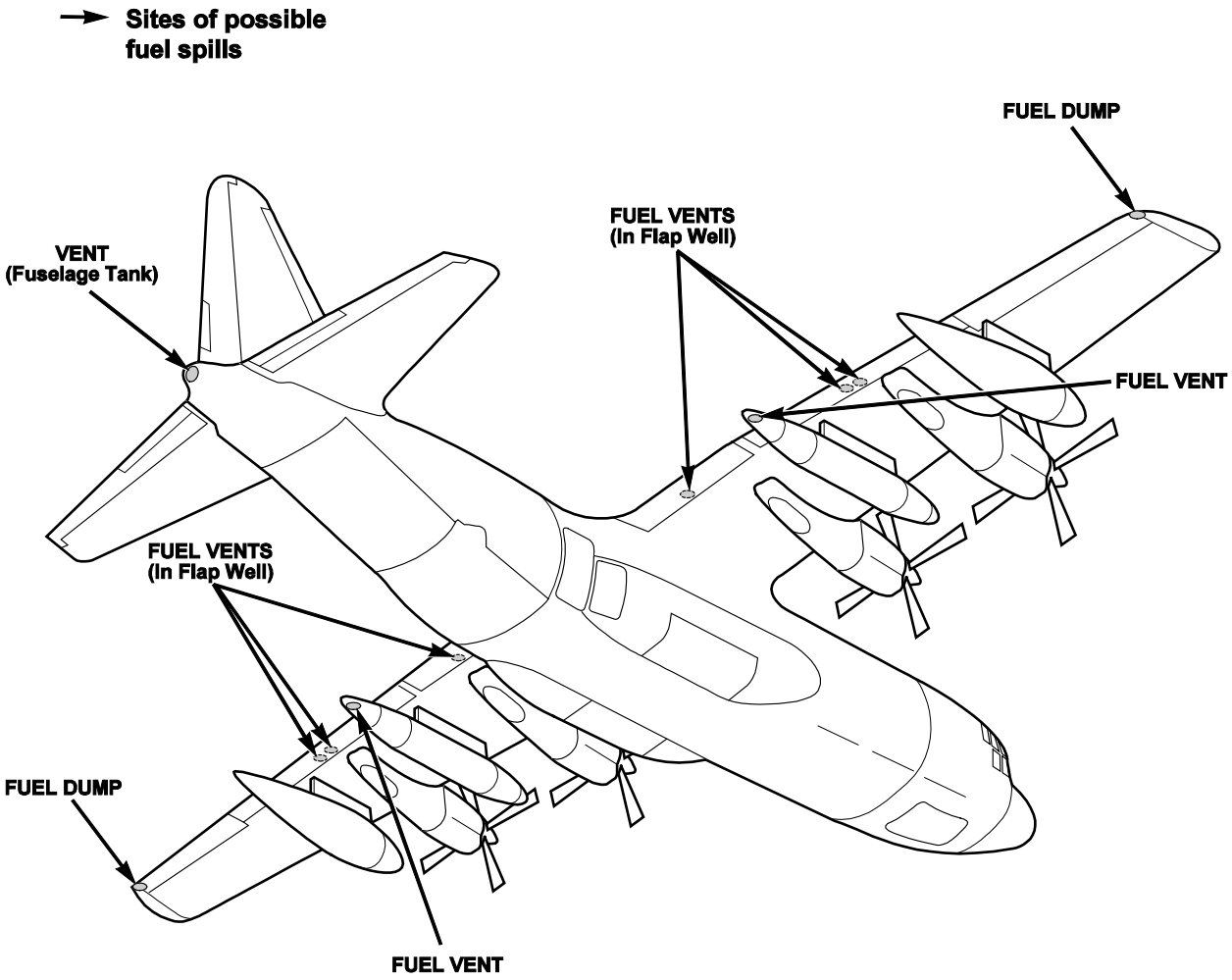
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points



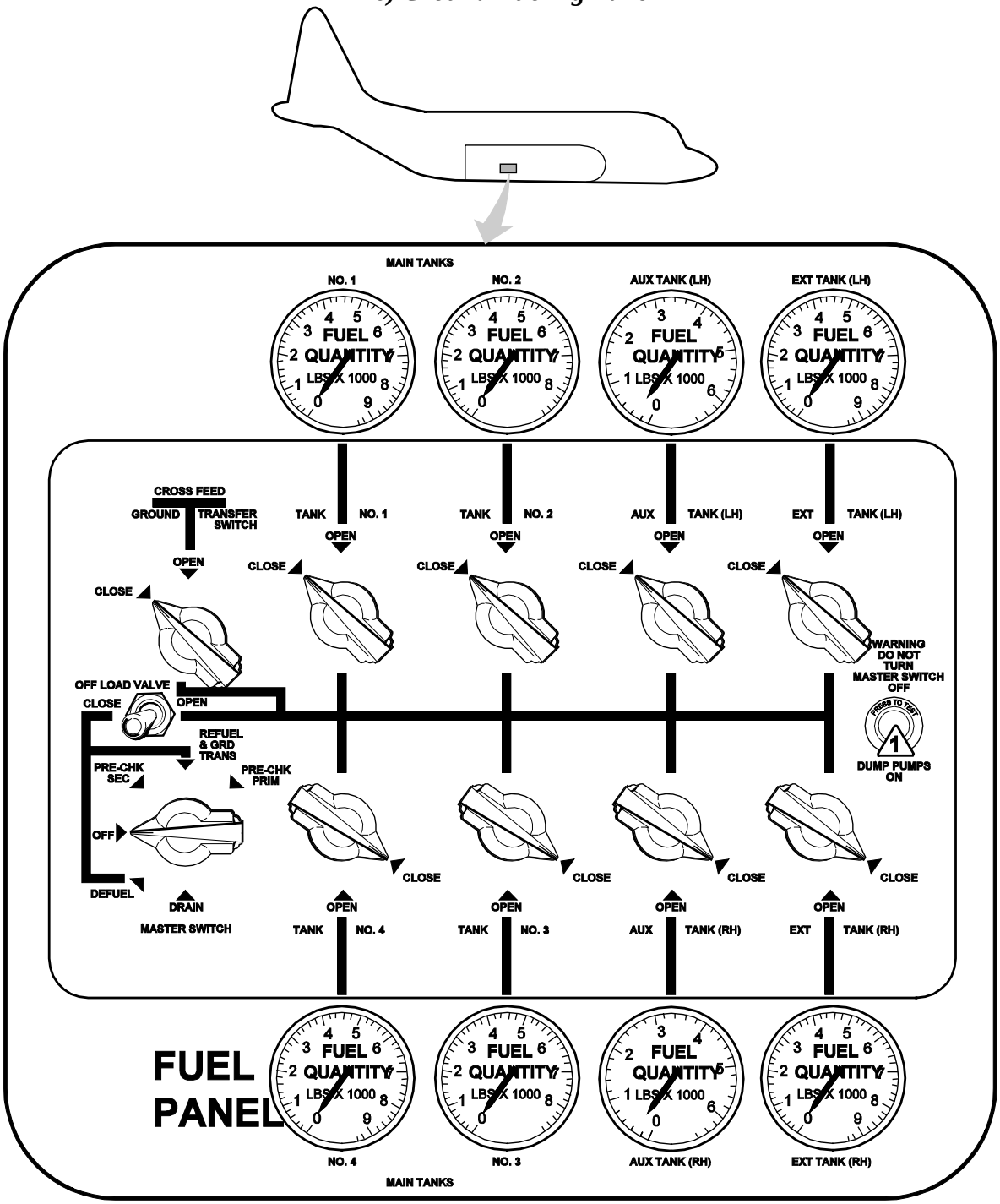
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Location of Fuel Vents and Dump Ports



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Ground Fueling Panel



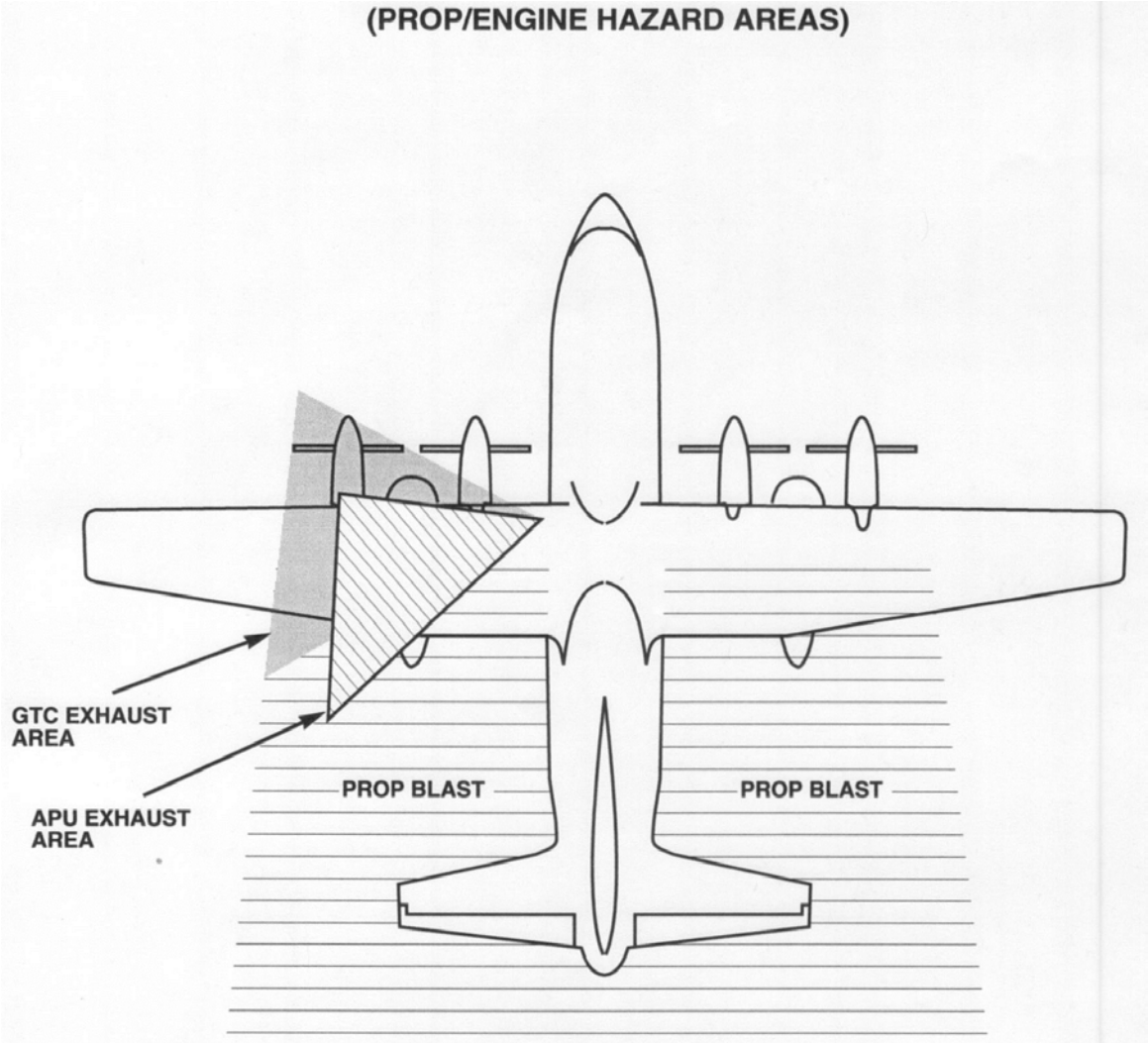
NOTE

1 AIRCRAFT 151888 THROUGH 159469.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Engine/APU Exhaust/Blast Area



MAXIMUM POWER, NO WIND

DISTANCE AFT OF PROPS – FEET	100	200	300	400	500
WAKE VELOCITY – KNOTS	128	107	92	80	69

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only the refueling procedures that are unique to the C-130 aircraft, primarily the operation of the "pre-check" system.

1. Prepare aircraft and refueling system in accordance with the applicable procedure in Chapter 6, 12, or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109.

2. Place the master switch in the REFUEL & GRD TRANS position; place the tank selector switches for the tanks to be refueled in the OPEN position, and place the offload valve switch in the OPEN position.

WARNING

Maintain balance weight of fuel within 1,000 pounds between each pair of symmetrical tanks and 1,500 pounds between wings, except for auxiliary tanks. It is permissible for one auxiliary tank to be empty when the opposite tank is full.

3. Start the fuel truck pump or fuel pit pump, and establish fuel flow.

Refuel wing tanks as follows:

1. If the tanks are not to be filled to capacity, control the servicing with the tank selector switches. Turn the switches to the CLOSE position when the desired fuel level is reached.

2. If the tanks are to be filled to their single-point capacity, make a check of the float-operated shutoff valves as follows:

Note

Each tank can be checked individually by directing refueling flow to one tank at a time, or all tanks can be checked simultaneously.

a. Place the master switch in the PRE-CHK PRI position, and observe the fuel quantity gauges. Fuel flow should stop within 15 seconds.

b. Place the master switch in die REFUEL & GRD TRANS position, and reestablish fuel flow.

c. Place the master switch in the PRE-CHK SEC position, and observe the fuel quantity gauges and servicing truck flow meter. Fuel flow should stop within 15 seconds.

d. If either step a or c stops fuel flow into the wing tanks, automatic shutoff is functioning and will stop fuel flow when the single-point capacity is reached. If fuel flow into the tanks does not stop during at least one of the checks, both float valves for that tank have failed and will not shut off fuel flow when the single-point capacity is reached. Therefore, refueling should be accomplished through individual filter ports to prevent overfilling the tank with the inoperative float valves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Note

There are no wing filler ports for the auxiliary fuel tanks in the center wing.

3. Return the master switch to the REFUEL & GRD TRANS position.

4. After wing tanks refueling is complete, place all tank selector switches in the CLOSE position.

5. Place the master switch to the OFF position and the offload valve switch to the CLOSE position.

6. Perform after refueling steps.

Refuel fuselage tank as follows:

Note

An additional man is required to position controls at the flight deck since the fuselage tank is refueled using both the SPR panel and the auxiliary fuel control panel.

1. Place the master switch to the REFUEL & GRD TRANS position and the offload valve switch to the OPEN position.

2. Place the fuselage tank crossfeed switch on the auxiliary fuel control panel to the no-flow position.

3. Place the fuselage tank fill valve switch on the auxiliary fuel control panel to the OPEN position.
4. If the fuselage tank is not to be filled to capacity, control the servicing with the fuselage tank fill valve switch. Turn the fuselage tank fill valve switch to the CL position when the desired fuel capacity on the fuel quantity indicator on the auxiliary fuel control panel is reached.

5. If the fuselage tank is to be filled to the single-point capacity, check the float-operated shutoff valve as follows:

   a. Place the fuselage tank fill valve switch on the auxiliary fuel control panel to the PRETEST PRIM position, and observe the servicing truck flowmeter. Fuel flow should stop within 15 seconds.

   b. Place the fuselage tank fill valve switch in the OPEN position and reestablish fuel flow into the fuselage tank.

   c. Place the fuselage tank fill valve switch on the auxiliary fuel control panel to the PRETEST SEC position and observe the servicing truck flowmeter. Fuel flow should stop within 15 seconds.

   d. Place the fuselage tank fill valve switch in the OPEN position and reestablish the fuel flow to the fuselage tank.

   e. If either step a or c stops fuel flow into the fuselage tank, automatic shutoff is functioning and will stop fuel flow when the single-point capacity is reached. If fuel flow into the fuselage tank does not stop during at least one of the checks, both float valves for the fuselage tank have failed and will

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

not shut off fuel flow when the single-point capacity is reached. Therefore, fuselage tank fueling should be stopped at least 1,000 pounds less than full in accordance with step 4.



If the fuselage tank is filled to capacity, filling the wing tanks to capacity will exceed the gross weight limit.

6. Place the offload valve switch on the SPR panel to the CLOSE position.

7. Place the fuselage tank fill valve switch on the auxiliary fuel control panel to the CL position.

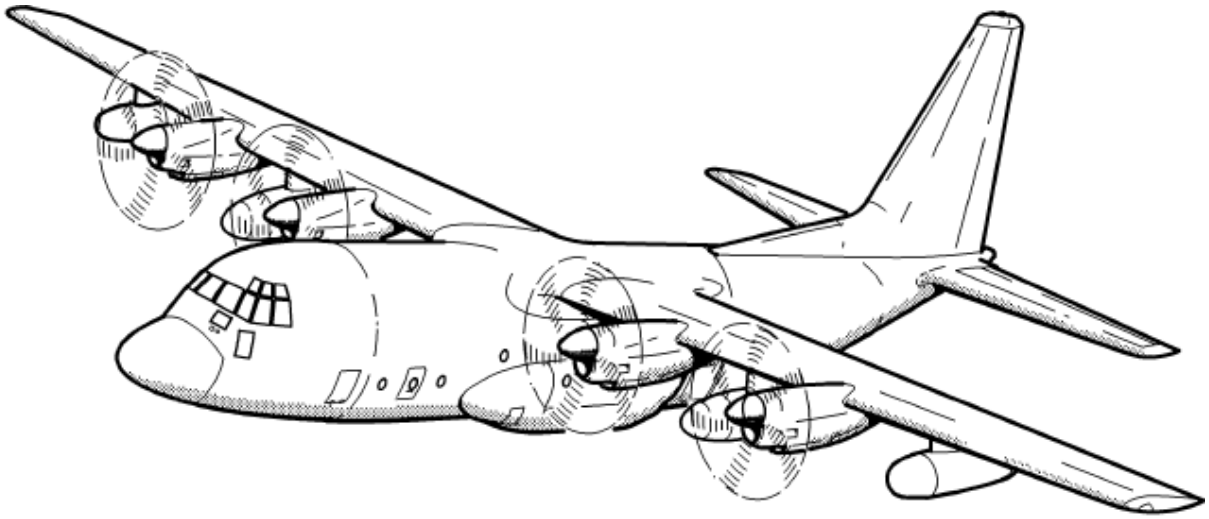
Note

Refuel the aircraft through the single-point refueling system by using battery power only when neither external power nor over-the-wing equipment is available. Follow special procedures contained in the applicable NATOPS Manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

C/KC-130J

Externally, the KC-130J aircraft looks similar to earlier models of the C/KC-130 family. The noticeable difference is the 6-bladed propellers in place of the older models' 4-bladed propellers.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: C/KC-130J

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	132 ft 7 in	Maximum Gross Weight — 175,000 lbs
Length	99 ft 5 in	Maximum Footprint — 123 psi
Height	38 ft 4 in	

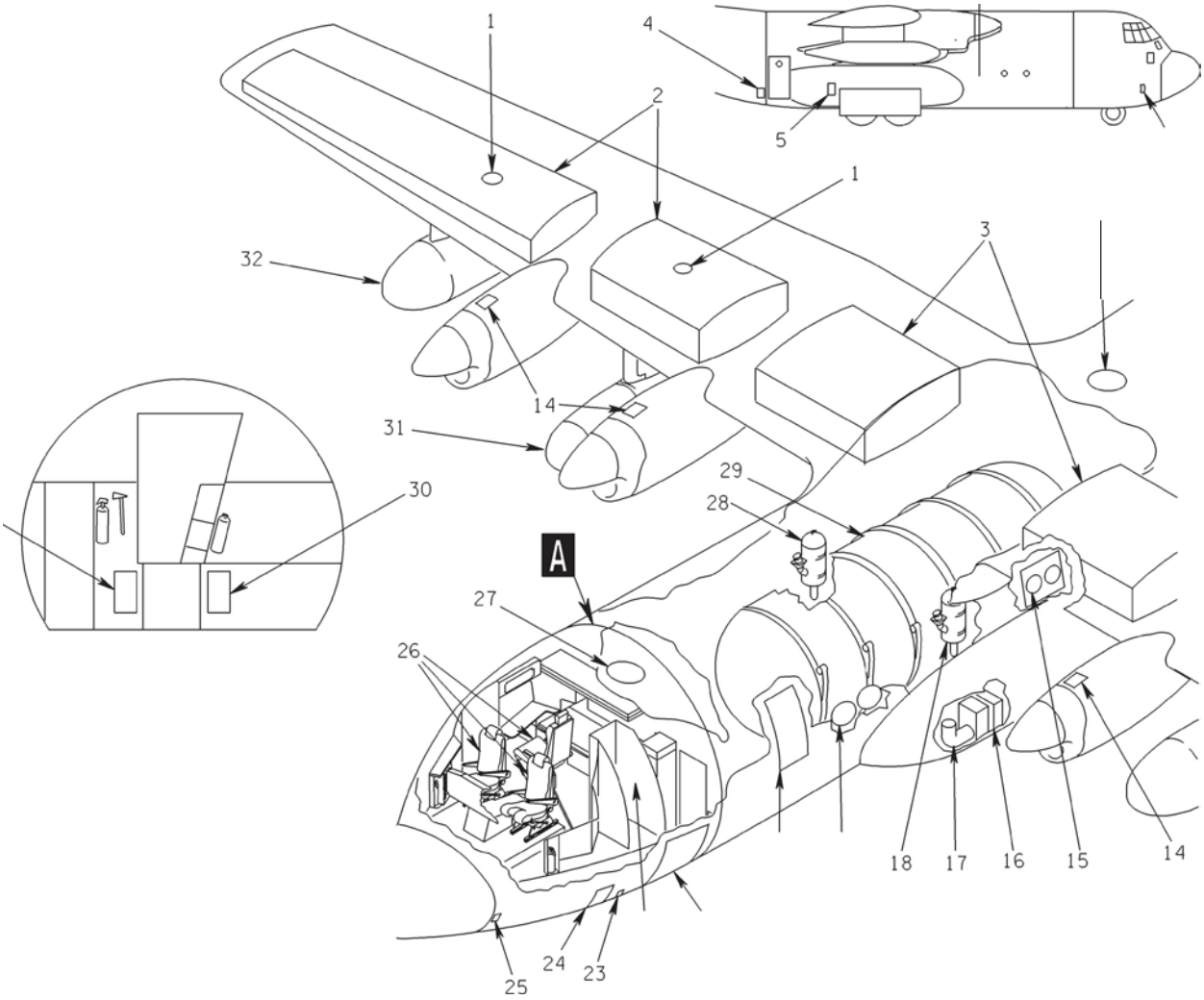
TABLE OF FUEL CAPACITIES

Tank		Gallons	Pounds
Internal	Tank No. 1	1220	8310
	Tank No. 2	1120	7660
	Tank No. 3	1120	7660
	Tank No. 4	1220	8310
	Left Auxiliary	860	5810
	Right Auxiliary	860	5810
Internal Total		6400	43560
External	Left	1310	8900
	Right	1310	8900
External Total		2620	17800
Total	Internal + External	9020	61360

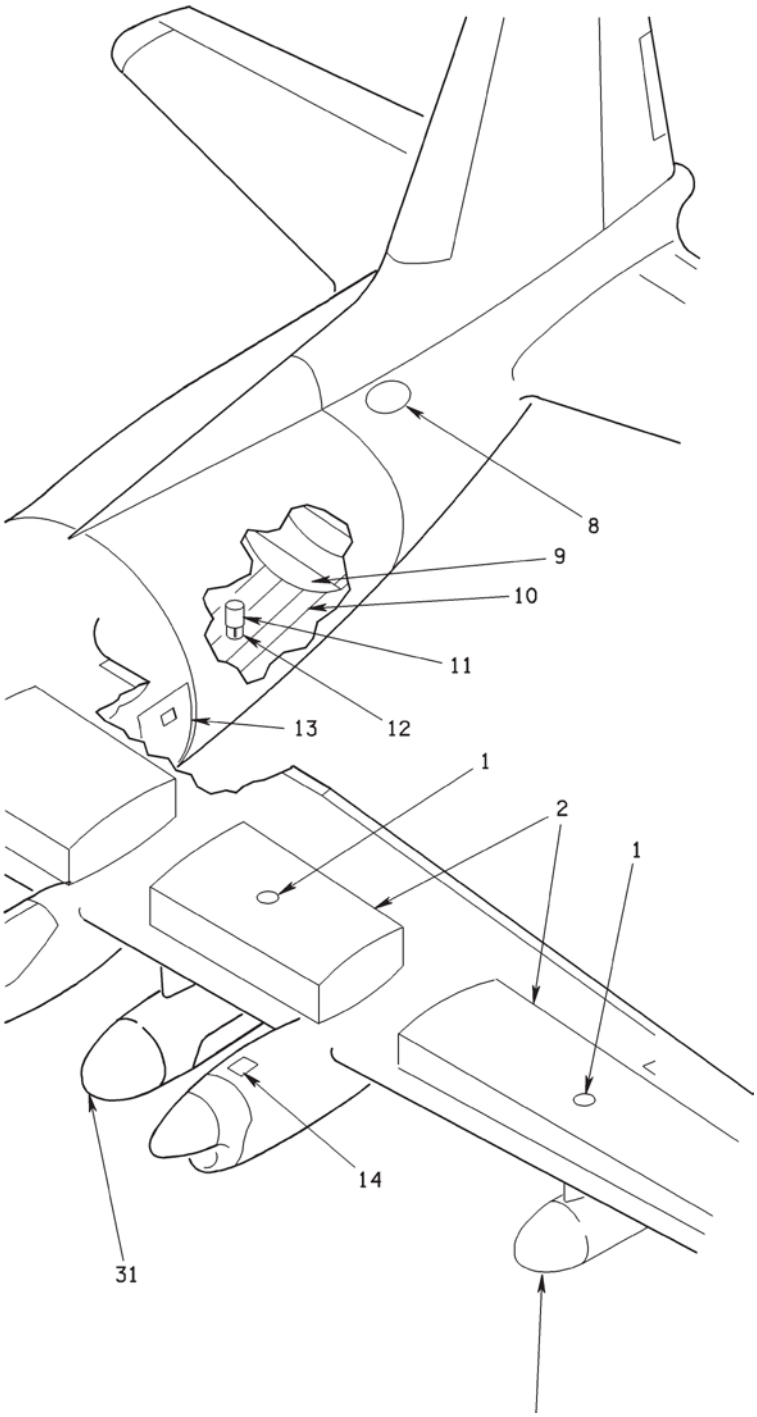
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

Servicing (Figure 3-1 from 01-75-GAJ-01)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



1. FUEL FILLER POINTS (4 PLACES)
2. MAIN FUEL TANKS (4 PLACES)
3. AUXILIARY FUEL TANKS (2 PLACES)
4. TOILET SERVICING ADAPTER
5. SINGLE-POINT REFUELING ADAPTER
6. LIQUID OXYGEN FILLER
7. CENTER ESCAPE HATCH
8. AFT ESCAPE HATCH
9. CARGO DOOR
10. RAMP
11. AUXILIARY HYDRAULIC SYSTEM RESERVOIR
12. AUXILIARY HYDRAULIC PUMP
13. PARATROOP DOORS <LH SHOWN – RH OPPOSITE)
14. ENGINE OIL TANKS (4 PLACES)
15. FIRE EXTINGUISHER AGENT BOTTLES
16. APU OIL RESERVOIR
17. APU
18. UTILITY HYDRAULIC SYSTEM RESERVOIR
19. ENGINE AIR INTAKE SHIELDS
20. SIDE EMERGENCY EXIT <LEFT SHOWN- RH OPPOSITE>
21. CREW ENTRANCE DOOR
22. GALLEY
23. EXTERNAL ELECTRICAL POWER RECEPTACLE
24. BATTERY COMPARTMENT
25. EXTERNAL INTERPHONE CONNECTION
26. CREW SEATS
27. FORWARD ESCAPE HATCH
28. BOOSTER HYDRAULIC SYSTEM RESERVOIR
29. FUSELAGE FUEL TANK
30. URINALS (2 PLACES)
31. EXTERNAL FUEL TANK
32. REFUELING POD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

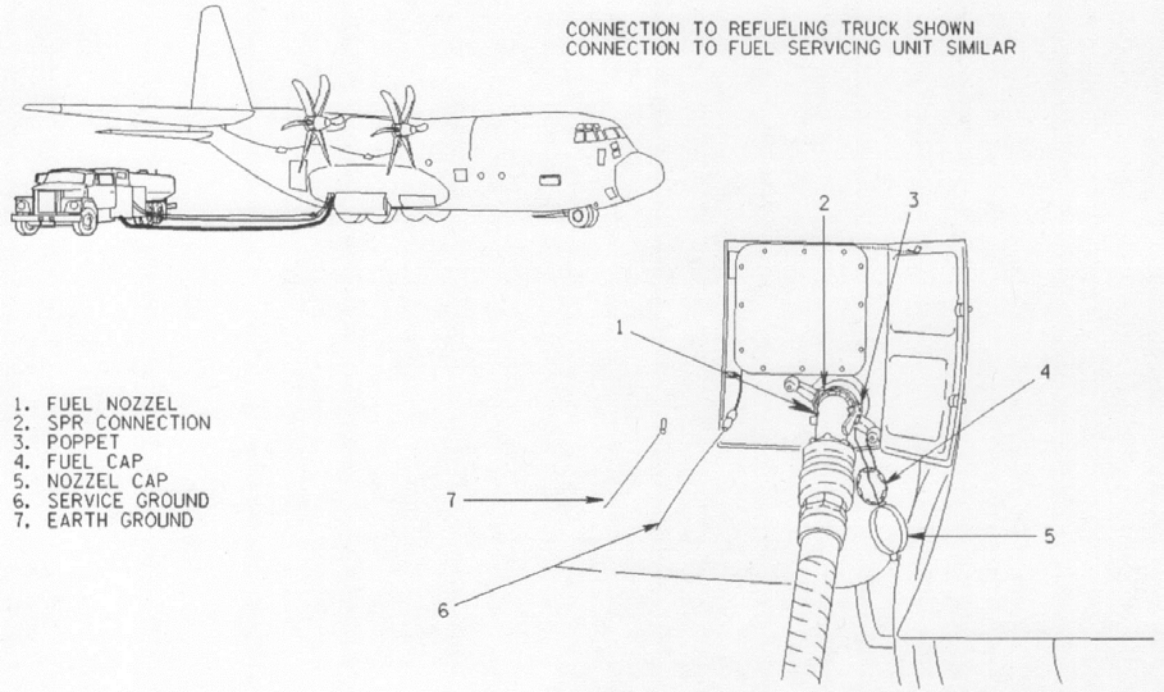


Figure 3-4 from 01-75-GAJ-01 (SPR Connection)

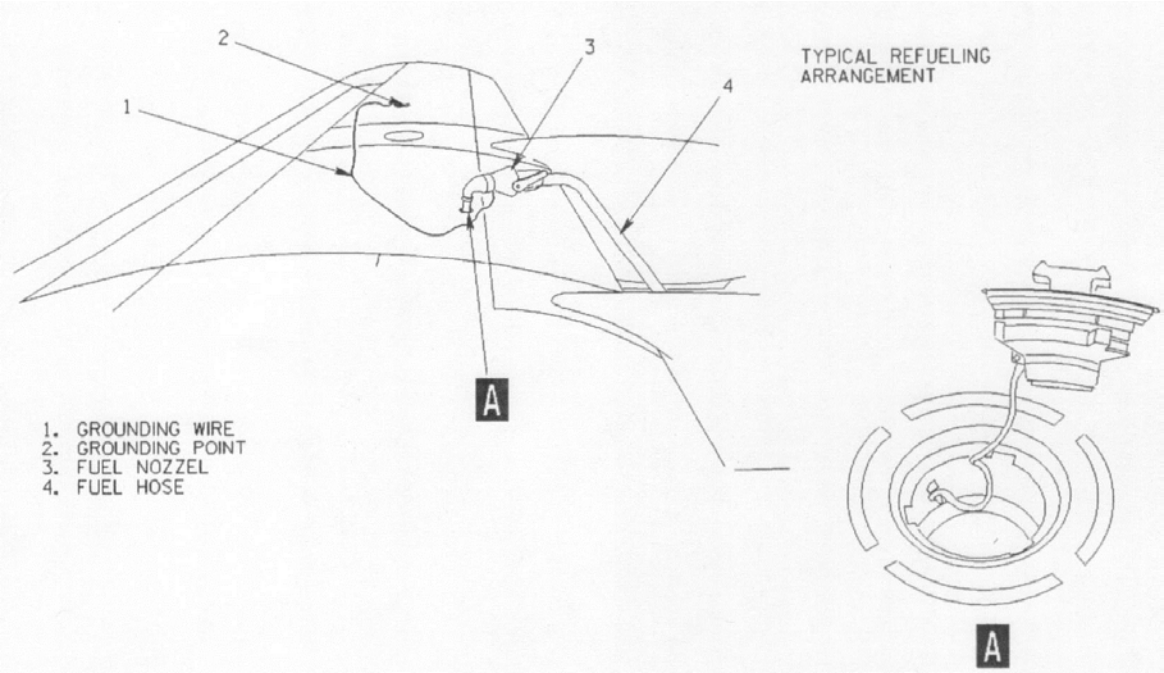
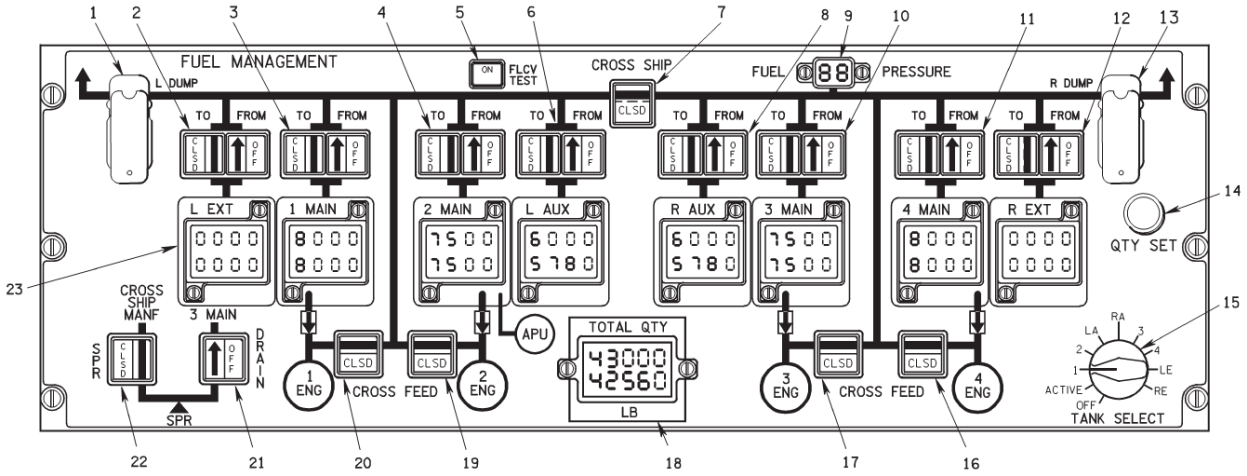


Figure 3-5 from 01-75-GAJ-01 Wing Refueling Filler Point (Typical)

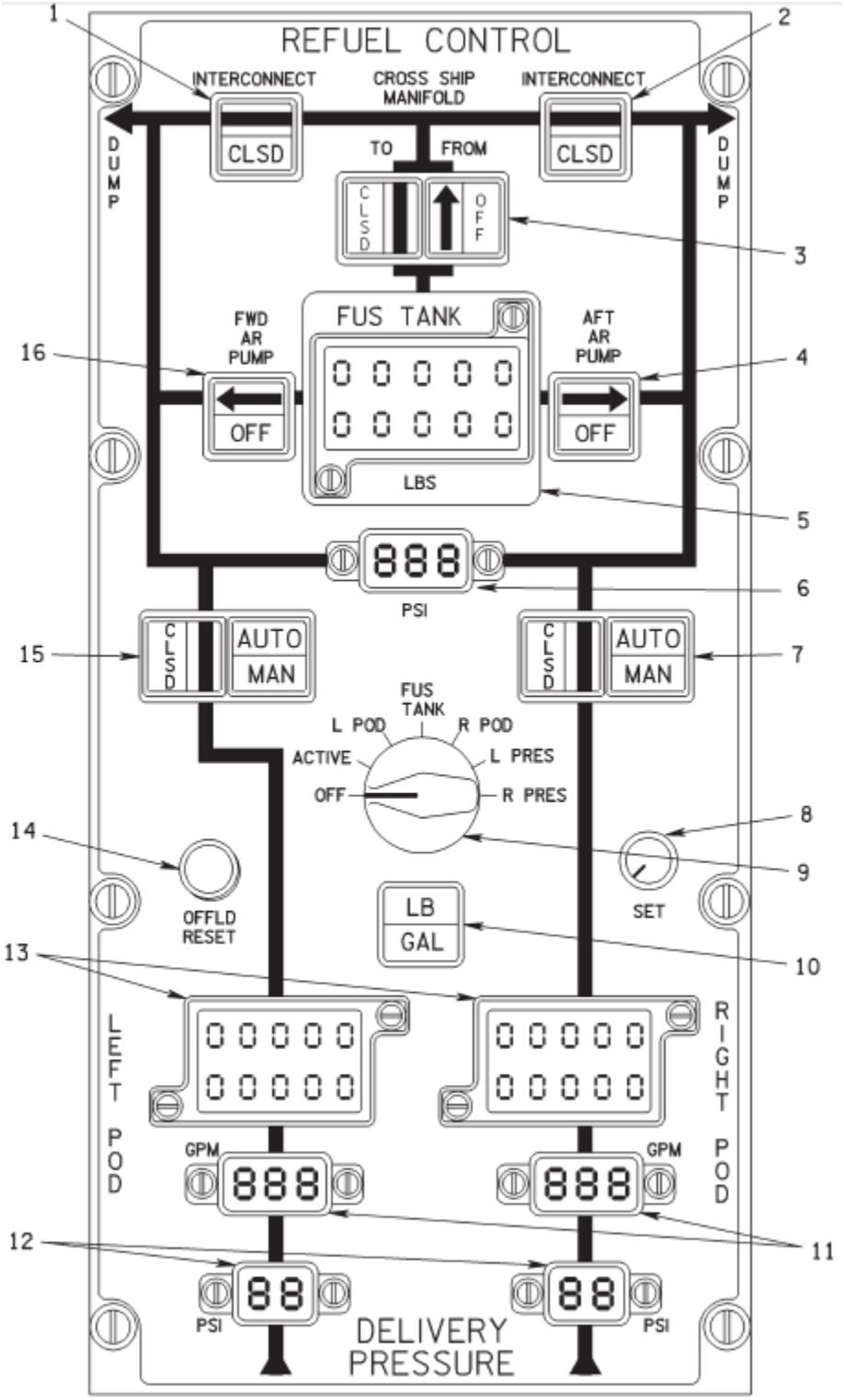
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



1. LEFT DUMP SWITCH
2. LEFT EXTERNAL TRANSFER SWITCH
3. NO. 1 MAIN TRANSFER SWITCH
4. NO. 2 MAIN TRANSFER SWITCH
5. FLCV TEST SWITCH
6. LEFT AUXILIARY TRANSFER SWITCH
7. CROSS-SHIP SEPARATION VALVE SWITCH
8. RIGHT AUXILIARY TRANSFER SWITCH
9. FUEL PRESSURE INDICATOR
10. NO. 3 MAIN TRANSFER SWITCH
11. NO. 4 MAIN TRANSFER SWITCH
12. RIGHT EXTERNAL TRANSFER SWITCH
13. RIGHT DUMP SWITCH
14. QUANTITY SET SWITCH
15. TANK SELECT SWITCH
16. NO. 4 ENGINE CROSSFEED SWITCH
17. NO. 3 ENGINE CROSSFEED SWITCH
18. TOTAL QUANTITY INDICATOR
19. NO. 2 ENGINE CROSSFEED SWITCH
20. NO. 1 ENGINE CROSSFEED SWITCH
21. SPR DRAIN SWITCH
22. SPR TRANSFER VALVE SWITCH
23. FUEL QUANTITY INDICATOR (1 PER TANK)

Figure 2-12 from 01-75-GAJ-01 (Fuel Management Panel)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



1. LEFT INTERCONNECT VALVE SWITCH
2. RIGHT INTERCONNECT VALVE SWITCH
3. FUSELAGE TANK TRANSFER SWITCH
4. AFT AR PUMP SWITCH

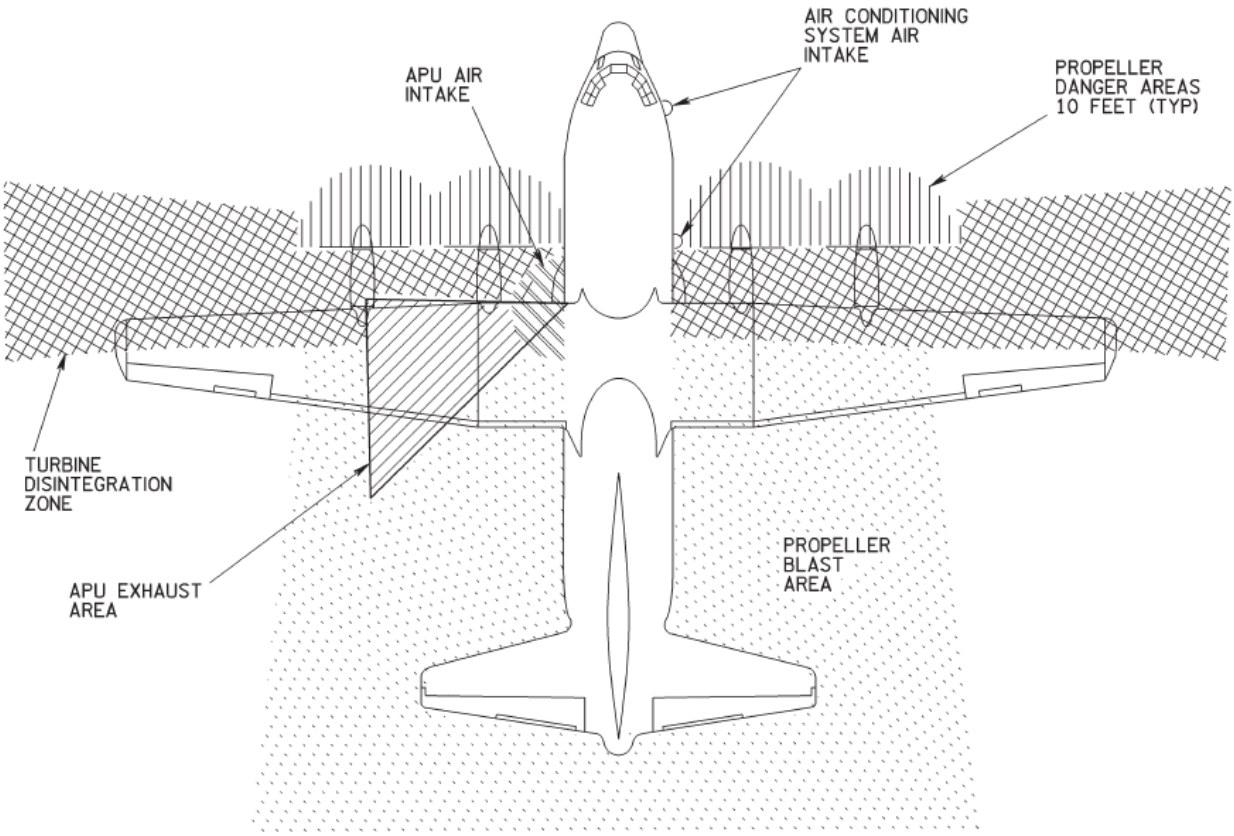
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

5. FUSELAGE TANK QUANTITY INDICATOR
6. AR MANIFOLD PRESSURE
7. RIGHT POD SUPPLY SWITCH
8. QUANTITY /PRESSURE SET SWITCH
9. DISPLAY SELECT SWITCH
10. LB/GAL SELECT SWITCH
11. FLOW RATE INDICATORS
12. FUEL PRESSURE INDICATORS
13. REFUEL QUANTITY SET /DELIVERED INDICATOR
14. OFFLOAD RESET SWITCH
15. LEFT POD SUPPLY SWITCH
16. FORWARD AR PUMP SWITCH

(Figure 15-15 From 01-75-GAJ-01 Refuel Control Panel)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE



MAXIMUM TAKEOFF POWER, NO WIND

DISTANCE AFT OF ENGINE TAILPIPE - FEET	5	100	200	300	400	500
WAKE VELOCITY - KNOTS	178	157	135	120	108	97

WARNING

BECAUSE OF ENGINE EXHAUST HEAT, WHILE OPERATING IN HOTEL MODE WITH FLAPS EXTENDED, DO NOT ENTER OR EXIT THE AIRCRAFT THROUGH THE LEFT AND RIGHT PARATROOP DOORS. EXTREME CARE SHOULD BE TAKEN WHEN USING THESE DOORS IN OTHER ENGINE/FLAP CONFIGURATIONS.

(01-75-GAJ-01 Figure 3-6 Danger Areas)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the KC-130J aircraft. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

The KC-130J refueling panel is located in the same area of the aircraft as earlier model C/KC-130 aircraft (on the right wheel well bulge near the right side troop door). The KC-130J refueling panel is not the same as the earlier models' ground refueling panels; major differences are:

1. The KC-130J refueling panel has only one SPR adapter.
2. The KC-130J refueling panel does not have all of the switches and gauges found on the earlier models' ground refueling panels (all controls/indicators are located in the cockpit)

Refer to NAVAIR 01-75GAJ-12JG-10-1 Fuel Servicing Manual for additional info.

CAUTION

To simplify the refueling/defueling process, one tank may be empty while the symmetrical tank is full, provided that all other pairs of symmetrical tanks are either balanced or weighted toward the light wing. Upon completion of refueling/defueling, the lateral fuel limits of no more than 1,000 pounds difference between symmetrical tanks and 1,500 pounds difference between wings (excluding auxiliary tanks) needs to be established before moving the airplane. Failure to comply can cause structural damage.

NOTE

If fuel does not flow into tank(s), stop refueling operation in affected and symmetrical tanks. Unaffected tanks may continue to be refueled by SPR. Affected and symmetrical tanks have to be filled by over-the-wing method (12-11-11).

NOTE

Foam-filled fuel tanks may vent fuel overboard during refueling at normal pressure. This is considered normal and can be minimized by reducing refueling pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION

Fuel tank(s) may be filled to capacity by SPR provided fuel flow to tank(s) stops during fuel level control valve test. Failure to comply could cause damage to tank due to overfilling.

CAUTION

Take care when moving refueling hose during over wing gravity fueling to avoid damaging wing leading edge with hose. The wing leading edge is of double skin construction to conduct hot air for anti-icing. Dents in wing leading edge can cause hot spots, which cause material failure.

NOTE

When moving fuel nozzle to another tank, it may also be necessary to move the refueling source. Ensure all grounds are properly installed after refueling source is moved.

CAUTION

All obstructions need to be removed from tank vents prior to defueling. Failure to remove obstructions will cause excessive suction inside tanks resulting in wing damage.

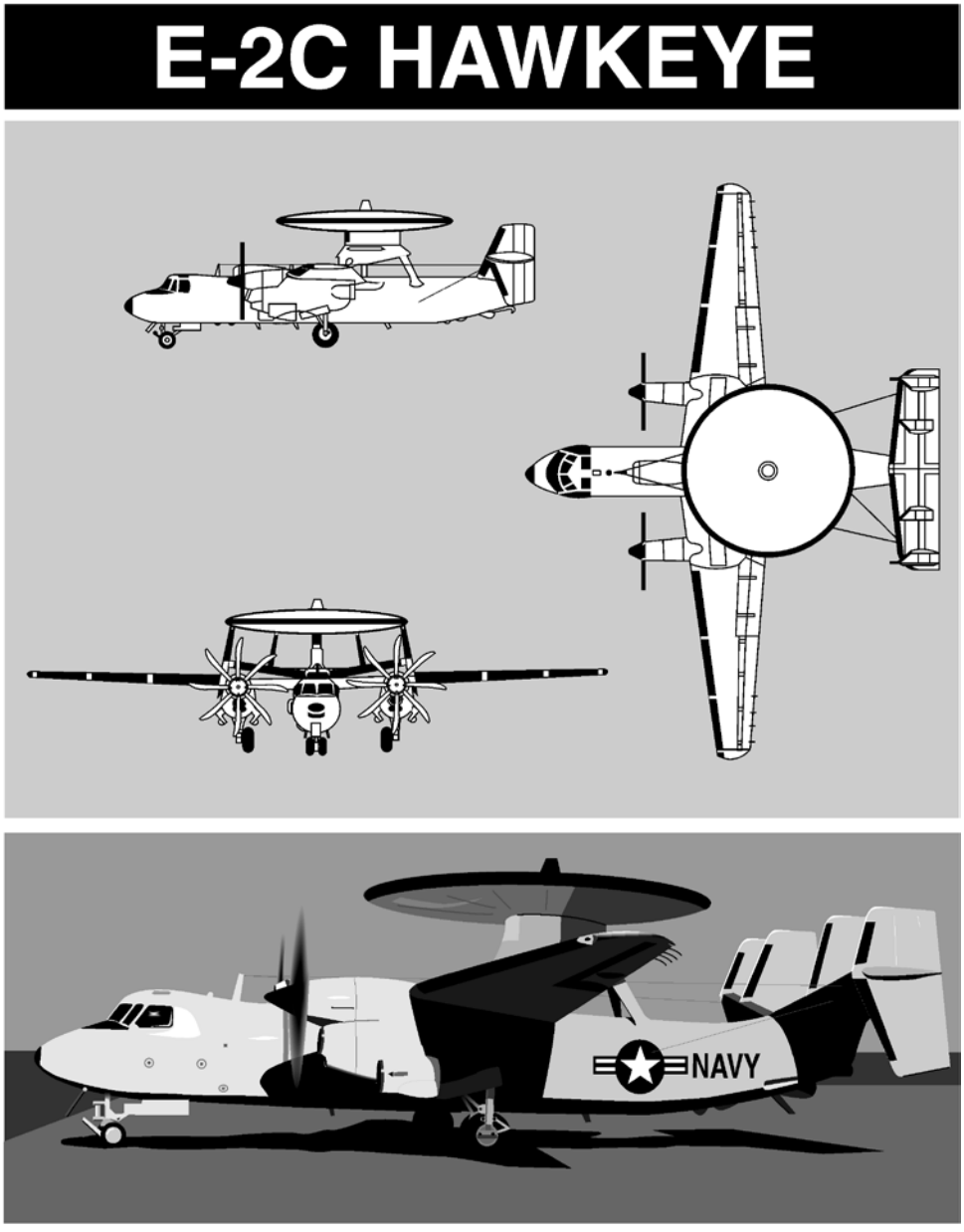
ADDITIONAL INFORMATION

The auxiliary tanks and fuselage fuel tank have no over-the-wing or external refueling capability. These tanks are to be filled by transferring fuel from other tanks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

E-2/C-2

The E-2C Hawkeye is a high-wing, all-weather, carrier-and shore-based airborne early warning and control (AEW&C) aircraft that patrols task for defense perimeters. It provides early warning of approaching enemy aircraft and vectors interceptors into attack position. In addition to this primary AEW function, the Hawkeye can also provide strike and traffic control, area surveillance, search and rescue guidance, navigational assistance, and communications relay.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: E-2/C-2

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread Folded	 80 ft 7 in 29 ft 4 in	Maximum Gross Weight — 55,000 lbs
Length	57 ft 7 in	Maximum Footprint — 260 psi
Height	20 ft 3 in	

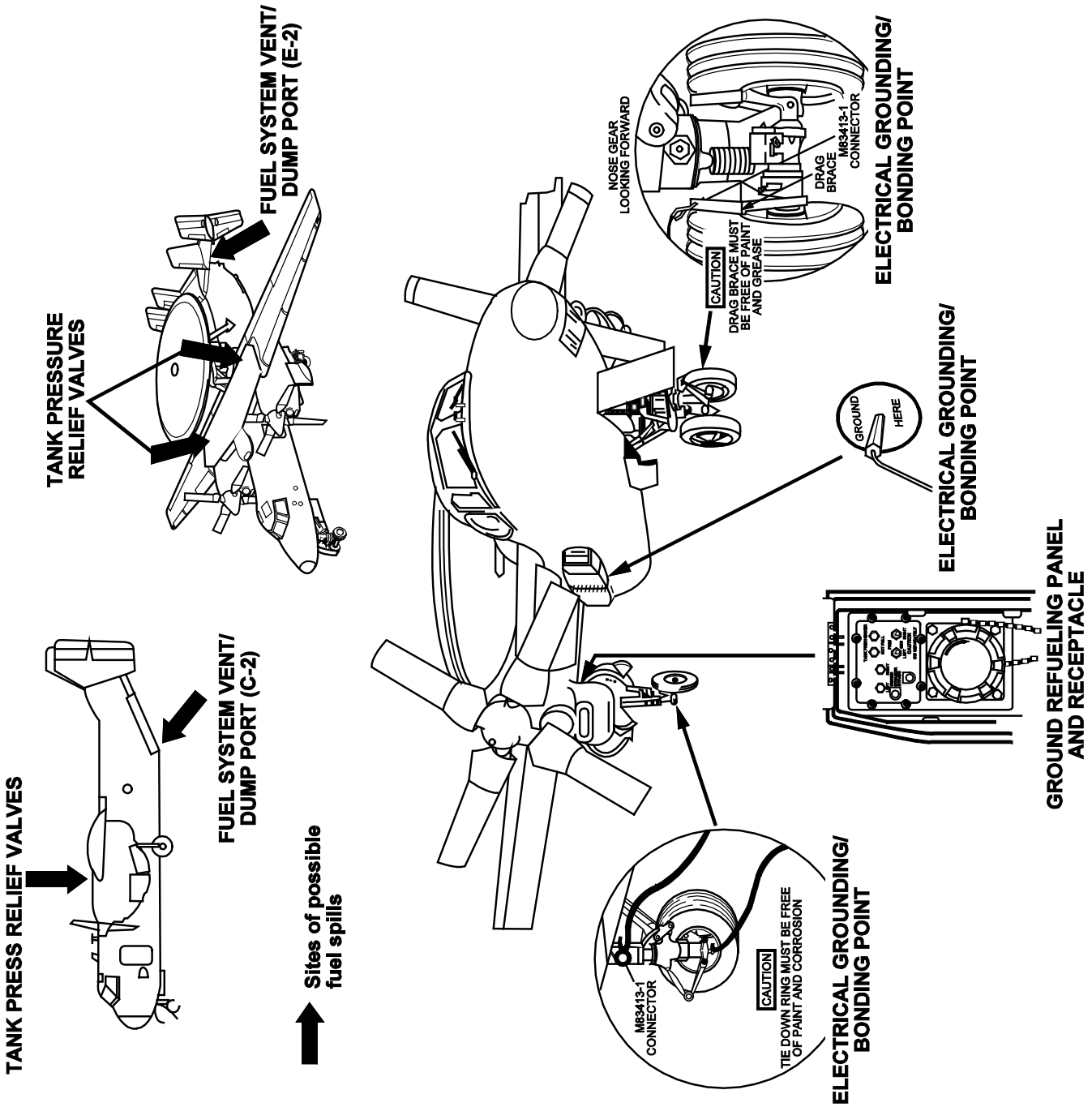
TABLE OF FUEL CAPACITIES

Tank		Gallons	Pounds
Wings	Left	912	6,200
	Right	912	6,200
	Total Internal	1,824	12,400

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

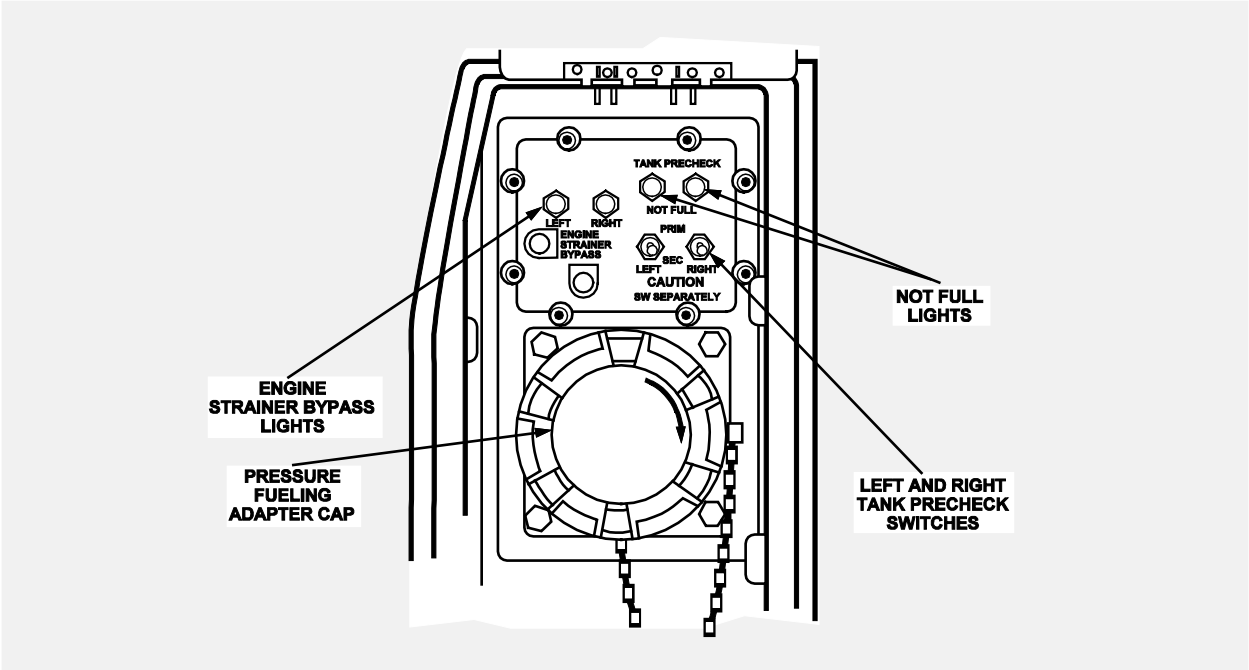
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Location of Fuel Vents, Dump Ports, and Relief Valves



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

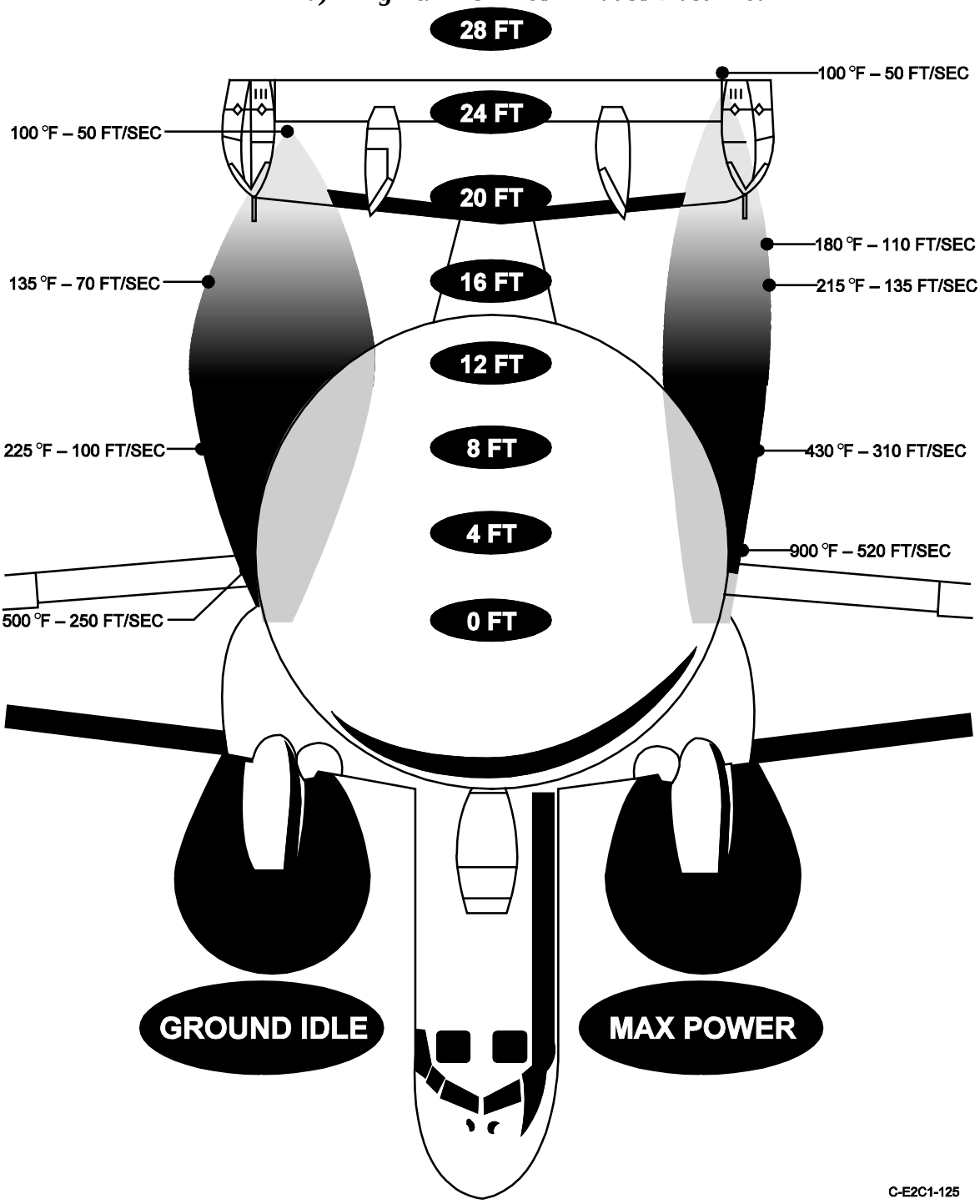
b) Ground Refueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Inlet/Exhaust/Blast Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the E-2/C-2 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

CAUTION

- E-2 aircraft are normally hot refueled with both engines operating. DO NOT ASSUME RIGHT ENGINE TURNED OFF until confirmation obtained.

1. Open "PRESS FUEL STA" access door.

2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position and initiate fuel flow. "NOT FULL" lights on pressure fueling station should illuminate.

CAUTION

- Discontinue Hot Refueling operation immediately if "NOT FULL" lights fail to illuminate.

WARNING

- Ensure that vent at tail of aircraft is clear and that tanks are venting during fueling. If no venting is indicated, cease fueling operation immediately.

3. Hold left "TANK PRECHECK" switch to "PRIM." After the left "NOT FULL" light goes off (approximately 3 seconds later) hold the right "TANK PRECHECK" switch to "SEC." After the right "NOT FULL" light goes off fuel flow into the aircraft should stop.

4. Release switches and fuel flow should resume.

5. Repeat the precheck process in step 3 above, this time holding the left "TANK PRE-CHECK" switch to "SEC" and the right "TANK PRECHECK" switch to "PRIM."

CAUTION

- If fuel flow does not stop when "NOT FULL" lights go off in either step 3 or 5 above, discontinue hot refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

6. Release switches and refuel aircraft until pilot gives hand signal indicating aircraft is full or he wishes refueling terminated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

WARNING

- Aircraft should not be completely filled during hot refueling operations, since failure of a high-level shut-off valve will result in fuel flowing out of the pressure relief valve and onto an engine exhaust pipe.

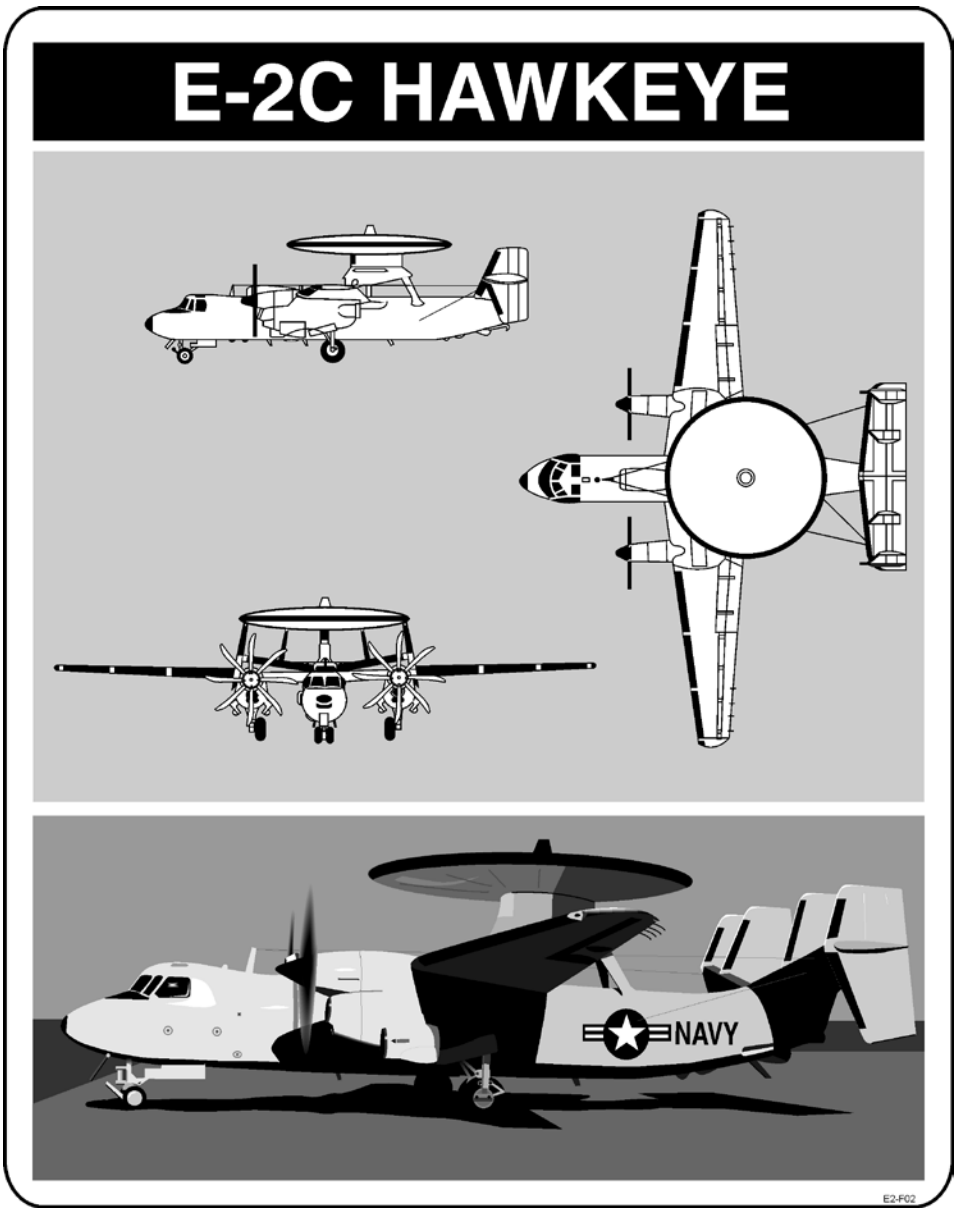
SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common "Fuel System Vent/Dump Port" (in the tail on the E-2 and on the fuselage on the C-2).
2. If any high-level shut-off valve fails to operate correctly, fuel may spill from special pressure relief valves on the tops of the wings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

E-2C 2000

The E-2C Hawkeye is a high-wing, all-weather, carrier- and shore-based airborne early warning and control aircraft that patrols task force defense perimeters. It provides early warning of approaching enemy aircraft and vectors interceptors into attack position. The E-2C 2000 includes a larger variety of system upgrades compared to the E-2C+ and E-2C+ AFC-399 variants.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	20 ft 3 in	Maximum Gross	55,00 lbs
Length	57 ft 7 in		
Wingspan Spread Folded	 80 ft 7 in 29 ft 4 in	Maximum Footprint	260 psi

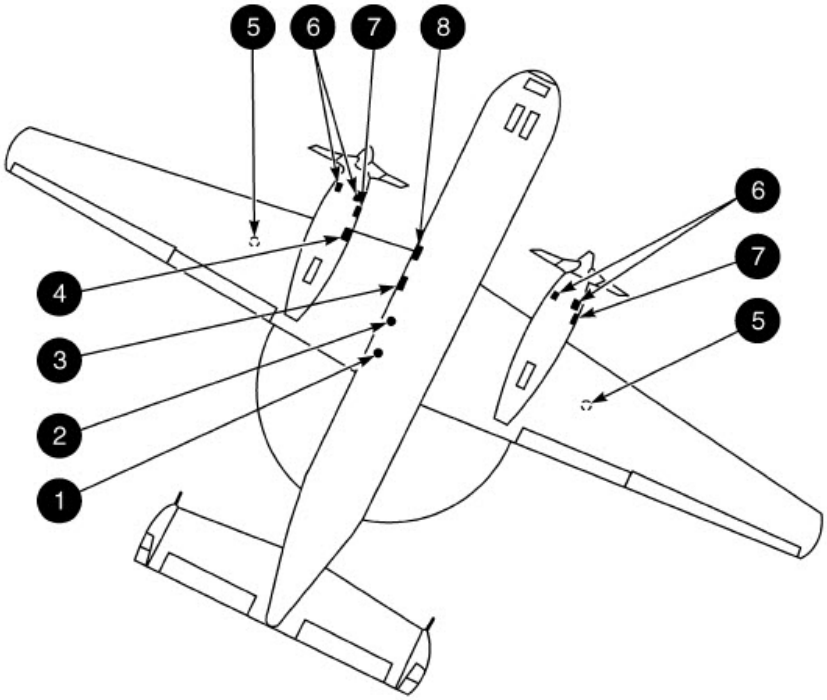
FUEL CAPACITIES

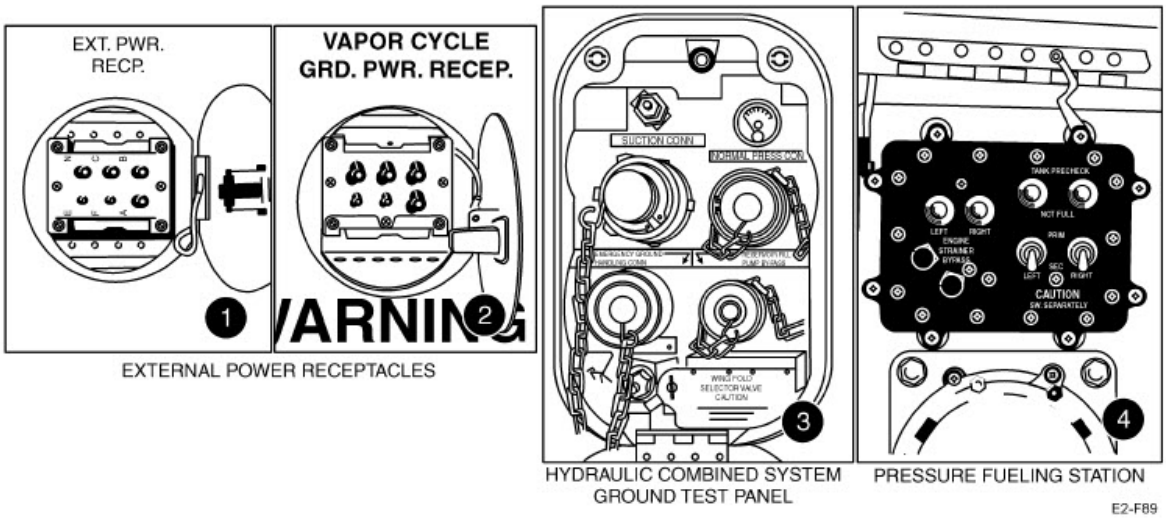
Tank		Gallons	Pounds (JP-5)
Wings	Left	912	6,200
	Right	912	6,200
	Total Internal	1,824	12,400

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

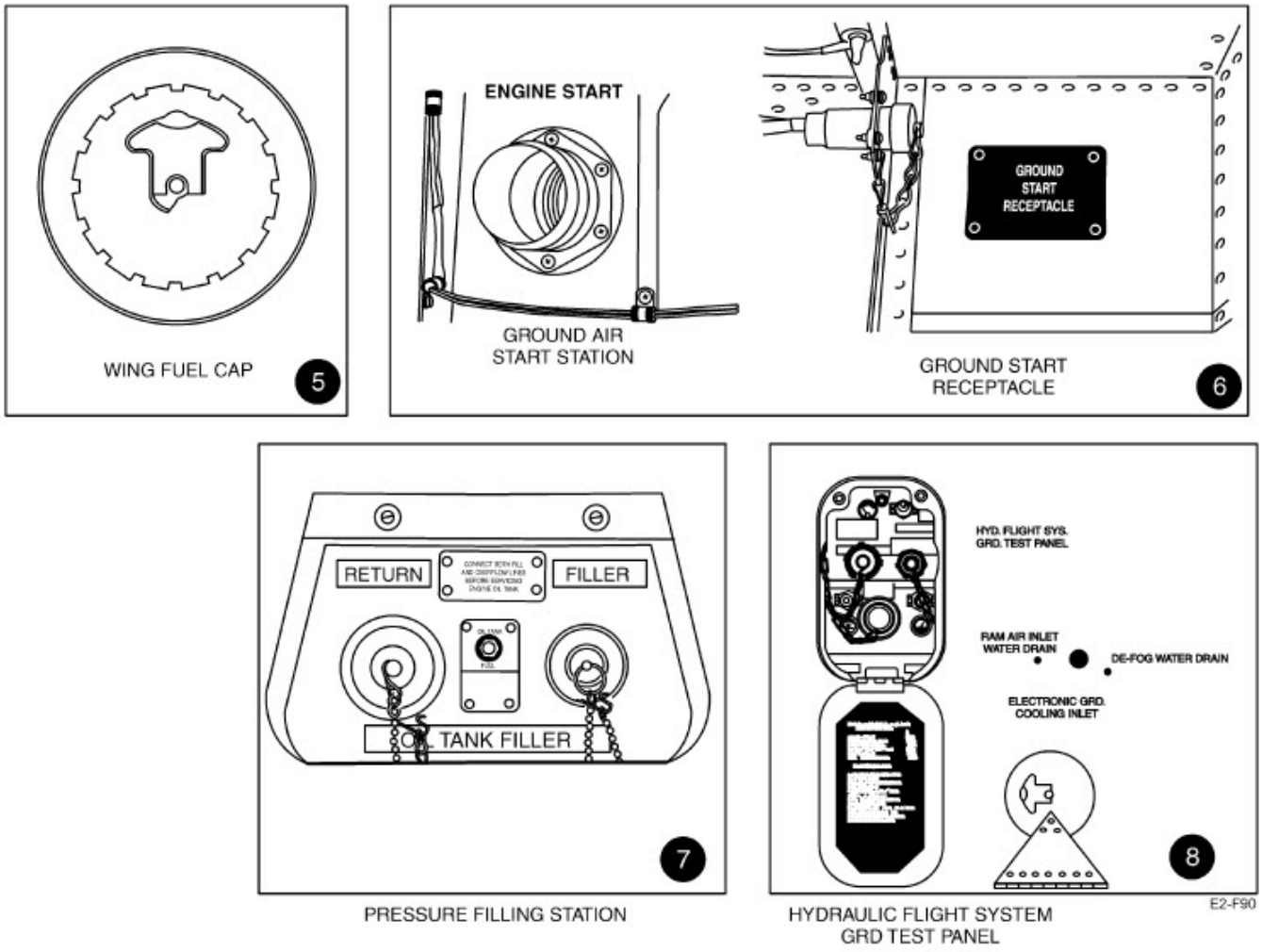
a) Location of External Power Receptacles, Hydraulic Combined System Ground Test Panel, and Pressure Refueling Station





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

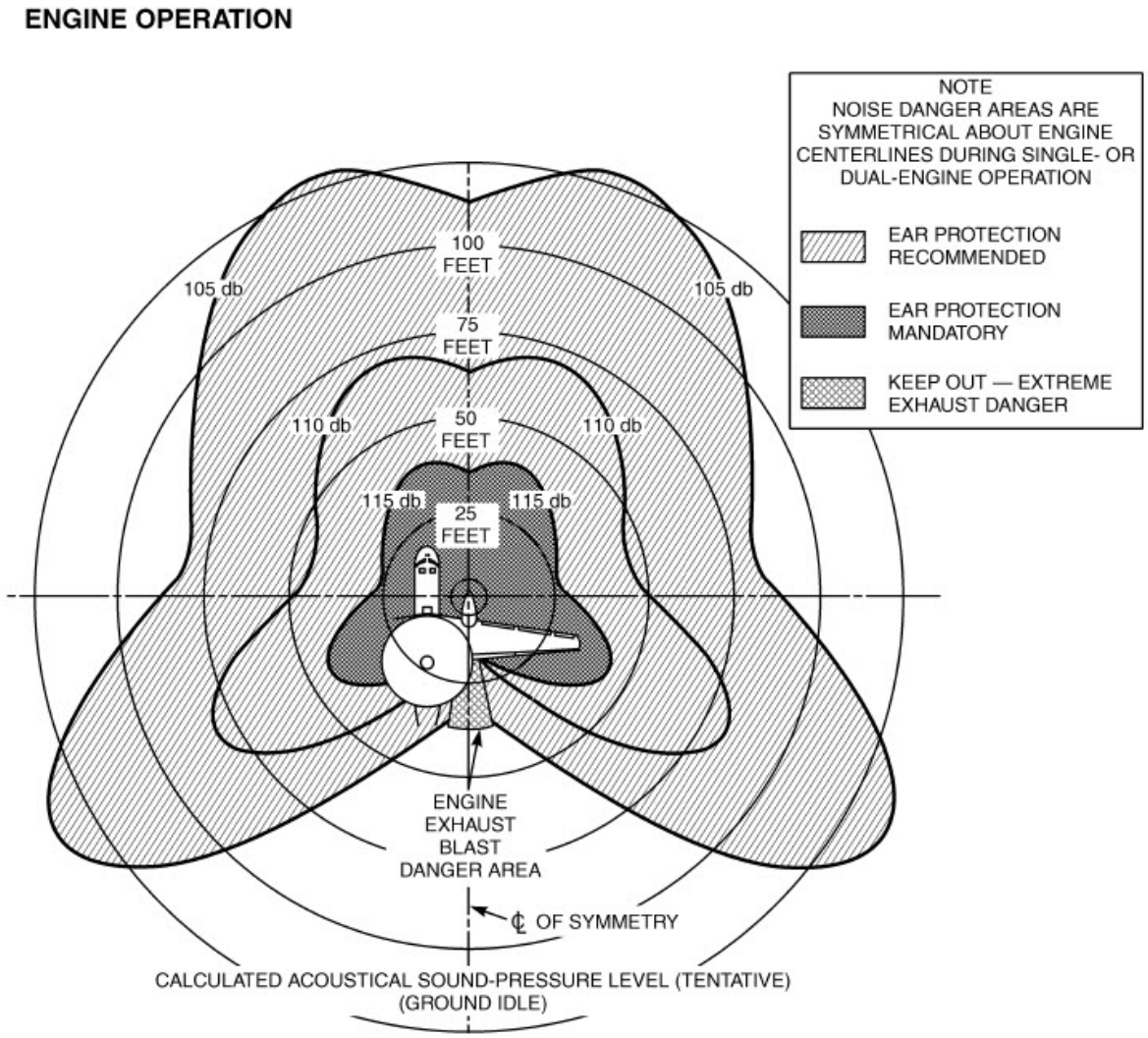
b) Location of Servicing Panels Continued: Wing Fuel Cap, Ground Start Receptacle, Pressure Filling Station, Hydraulic Flight System Ground Test Panel

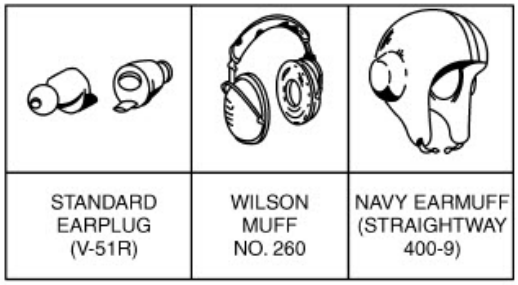


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Exhaust Blast Area, Acoustic Sound Pressure Levels





STANDARD EARPLUG (V-51R)	WILSON MUFF NO. 260	NAVY EARMUFF (STRAIGHTWAY 400-9)

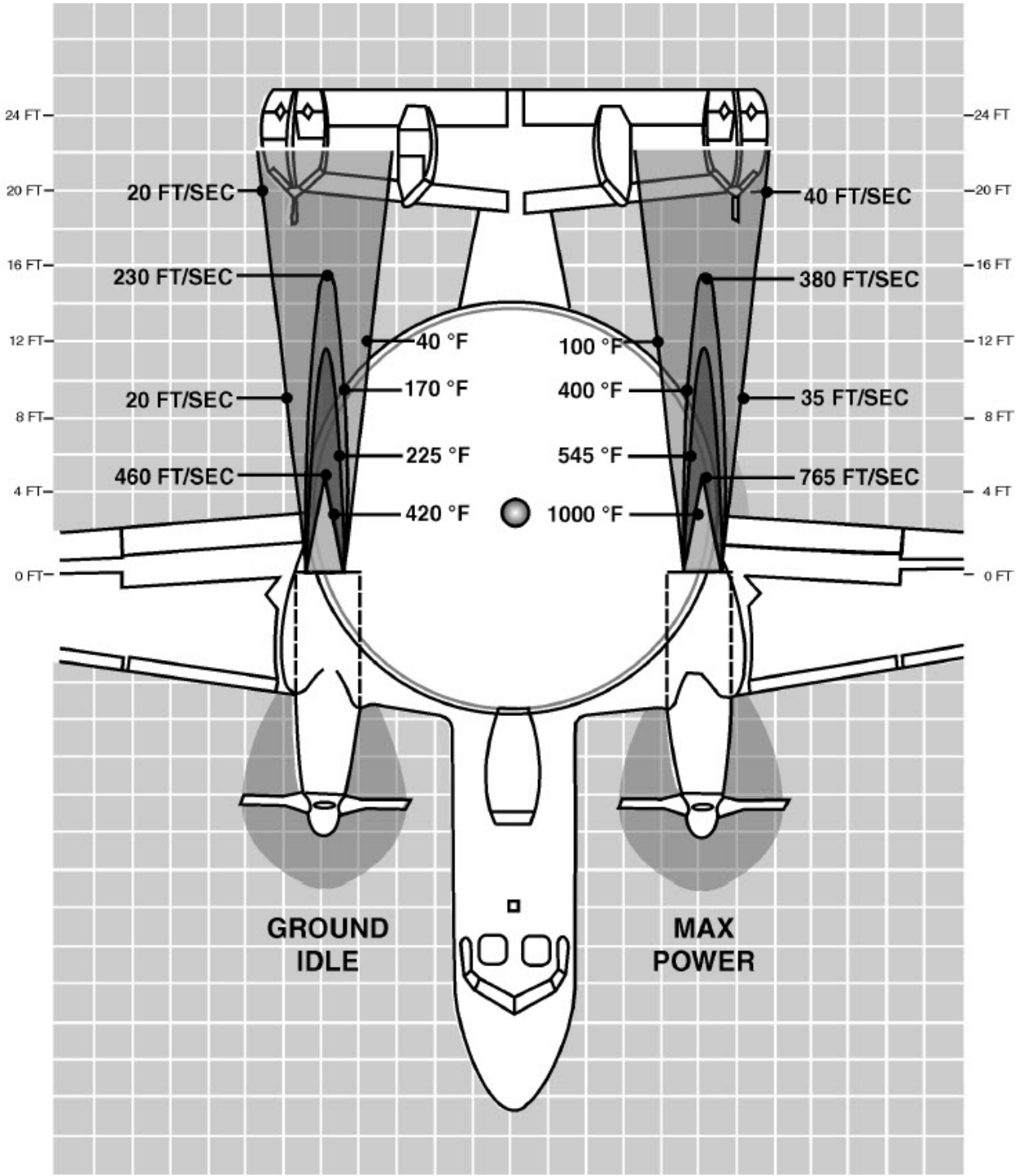
DAMAGE RISK CRITERIA

EAR PROTECTION NECESSARY	EXPOSURE TIME DURATION PER DAY				
	5 MINUTES	15 MINUTES	30 MINUTES	1 HOUR	2 HOURS
NO PROTECTION	99 db	95 db	92 db	89 db	86 db
EARPLUGS	129 db	124 db	121 db	118 db	115 db
EARPLUGS AND EARMUFFS	139 db	134 db	131 db	128 db	125 db

REF. WADC TN 55-355 TO DAMAGE RISK CRITERIA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

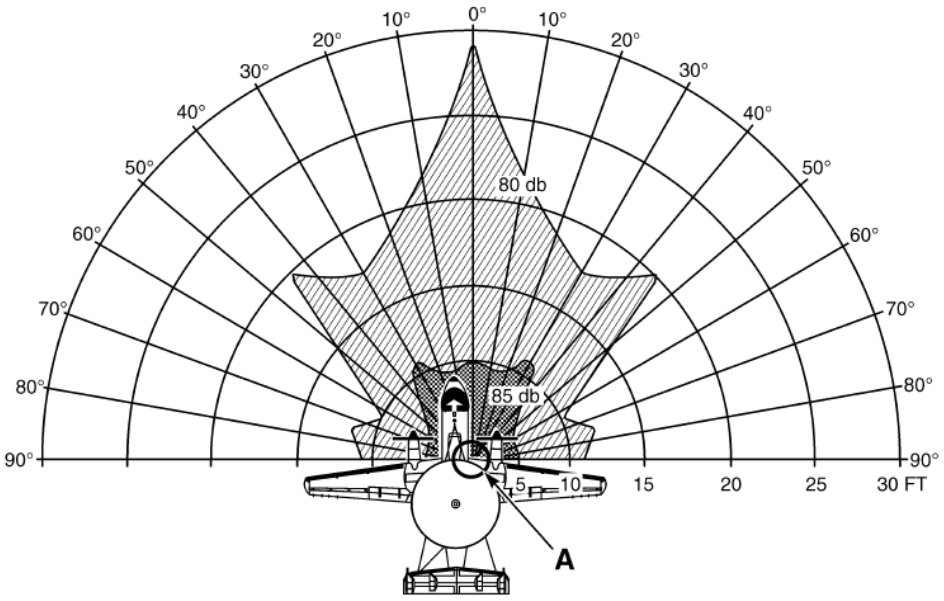
b) Local Engine Blast Area

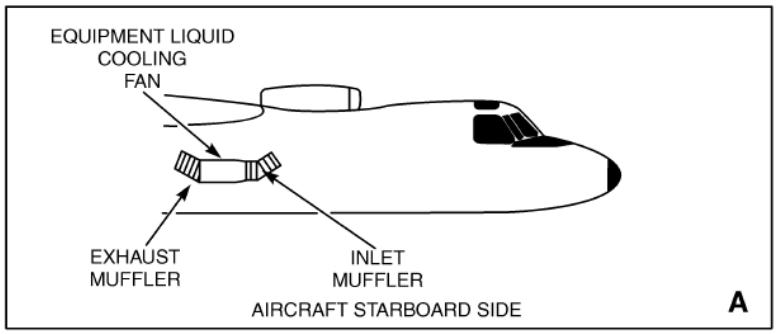


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Equipment Cooling Fan Forward Noise Levels

EQUIPMENT LIQUID COOLING FAN
FORWARD QUADRANT





TOTAL DURATION OF NOISE EXPOSURE ALLOWABLE DURING AN 8-HOUR DAY (BUMEDINST 6260.5 26 APRIL 1984). EXPOSURE SHALL NOT EXCEED THAT SHOWN BELOW.

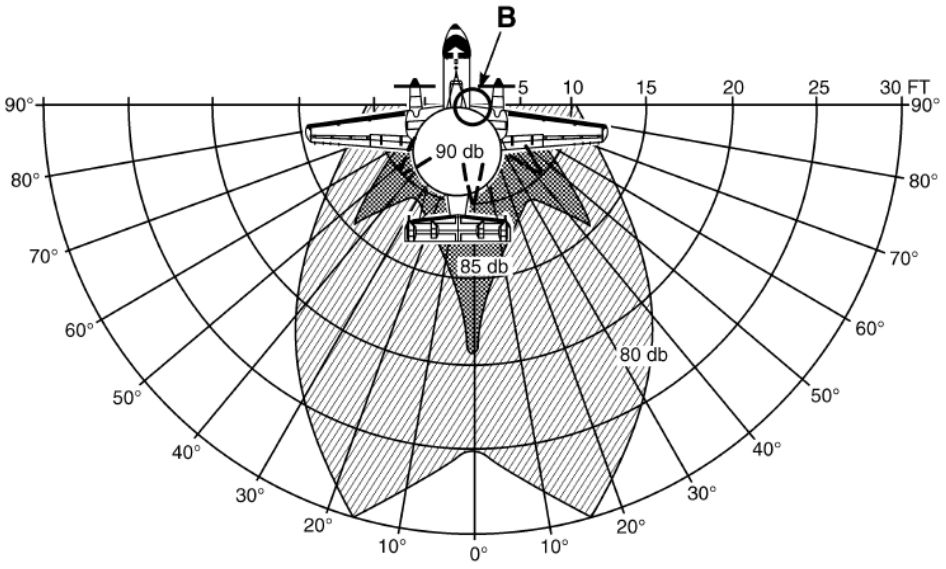
DURATION PER 8-HOUR DAY	SOUND LEVEL db
8 HR	90
6 HR	92
4 HR	95
3 HR	97
2 HR	100
1 HR 30 MIN	102
1 HR	105
30 MIN	110
15 MIN	115

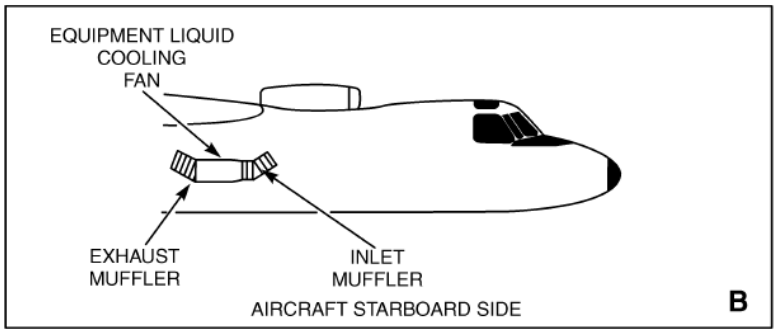
HE2K-F093

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Equipment Cooling Fan Aft Noise Levels

EQUIPMENT LIQUID COOLING FAN
AFT QUADRANT





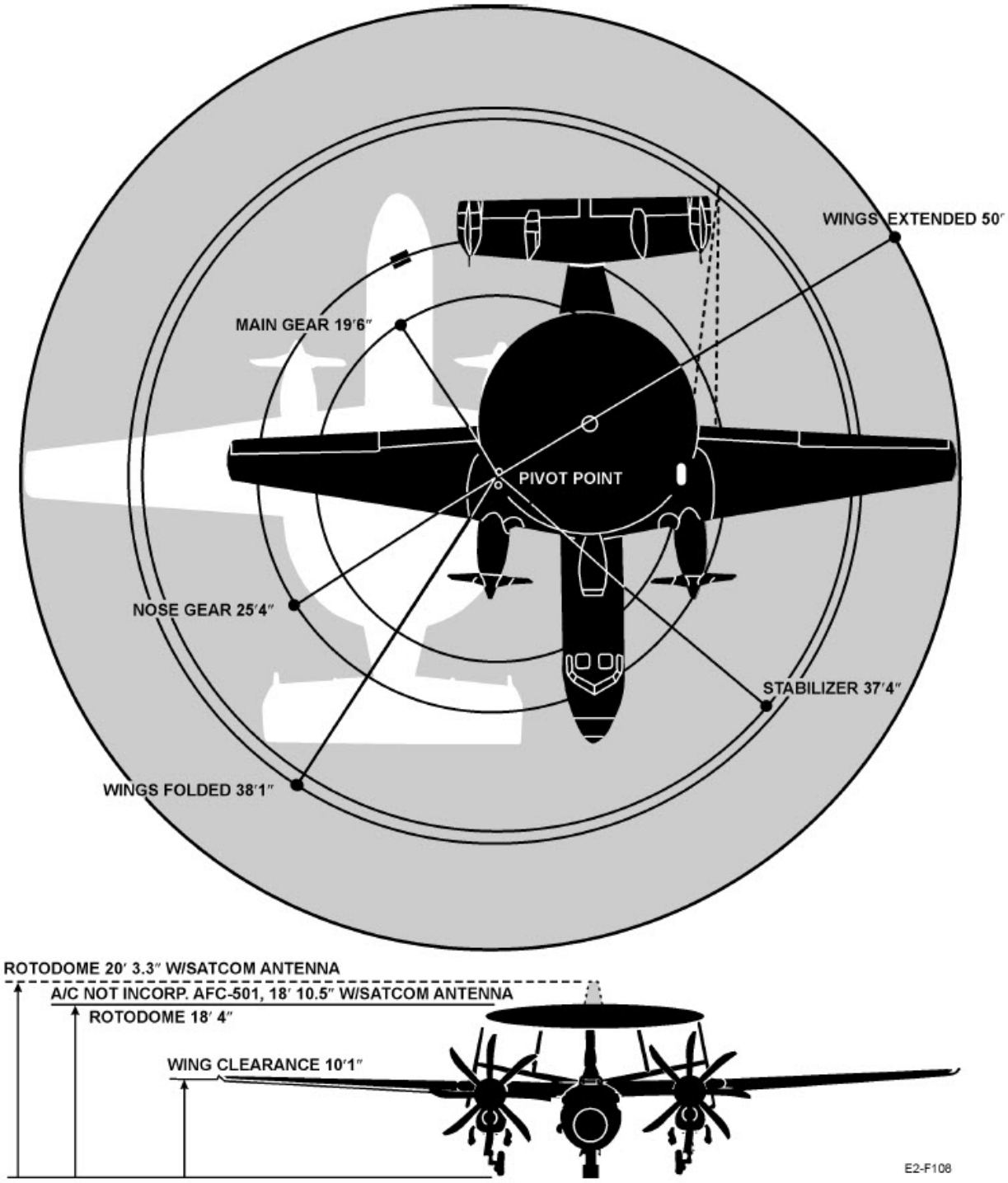
TOTAL DURATION OF NOISE EXPOSURE ALLOWABLE DURING AN 8-HOUR DAY (BUMEDINST 6260.5 26 APRIL 1984). EXPOSURE SHALL NOT EXCEED THAT SHOWN BELOW.

DURATION PER 8-HOUR DAY	SOUND LEVEL db
8 HR	90
6 HR	92
4 HR	95
3 HR	97
2 HR	100
1 HR 30 MIN	102
1 HR	105
30 MIN	110
15 MIN	115

HE2K-F094

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

e) Turning radius and ground clearance



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

Pressure Fueling System. Fuel is admitted to both tanks simultaneously by a pressure fueling system that provides high-speed fueling and defueling. The single-point fueling station is on the inboard side of the right nacelle. This station has the fueling adapter and the tank precheck panel on which the switches and lights necessary for checking the system are mounted. Though the system is hydromechanical, external electrical power should be available to check that all components are functioning properly.

Fueling is initiated by pressure (from the fueling hydrant or truck) acting upon the fueling valves in the tanks forcing the valves open. When the valves are open, two green lights (NOT FULL) on the TANK PRECHECK panel go on to indicate that fuel is entering each tank. Fueling is automatically terminated in each tank by float-operated, high-level control valves. Each of these valves is, in effect, two valves because it consists of two independently operating floats: one primary float and one secondary float. In addition to the dual protection of the two high-level control valves, each tank has a relief valve that prevents the tanks from rupturing if the control valves malfunction. When one of these floats is actuated by the rising fuel level, it causes the fueling shutoff valve to close and the light on the tank precheck panel goes off to indicate that the tank is full and that fueling has stopped. The relief valves start to open when tank pressure reaches 11.5 psig. They are fully open at 12.5 psig and permit fuel to flow overboard.

Precheck Switches. Two tank precheck toggle switches are on the TANK PRECHECK panel at the pressure fueling station. They are spring loaded to the off (center) position and have positions PRIM and SEC. Setting the switch to SEC actuates a solenoid in the high-level control valve. This permits a calibrated spring to lift the secondary float, thereby simulating a full tank condition. The fueling shutoff valve then closes and the NOT FULL warning light goes off to indicate that the tank has been shut down. Setting the switch to PRIM repeats the cycle for the primary float.

CAUTION

Sequential actuation of the left and right tank high level control valve solenoid is necessary to ensure that fueling operation automatically terminates.

Fuel Tank Not Full Warning Lights. The NOT FULL warning lights are green press-to-test lights on the tank precheck panel. They go on to indicate that the fueling shutoff valve is open to permit fuel to enter the tank.

Preparation for Fueling. Only authorized and qualified personnel should be permitted to operate fueling equipment. During the fueling process, loose pyrotechnics, smoking, striking matches, working on aircraft, LOX servicing, or producing flame within 50 feet of the aircraft or truck is strictly prohibited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Position of Aircraft and Truck. The aircraft should not be located in the vicinity of possible sources of ignition such as blasting, drilling, or welding operations. A minimum of 50 feet should be maintained from other aircraft and 1,000 feet from any operating radar set.

Grounding. Prior to fueling, grounding devices on the aircraft and drag chains on the truck will be inspected by fueling personnel for proper ground.

Electrical Hazards. Turn off all radio and electrical switches in aircraft prior to fueling. Check that no electrical apparatus supplies by outside power (electrical cords, droplights, floodlights, etc.) is in or near the aircraft. For night fueling, safety flashlights should be used.

Attaching Ground Wire. Before removing the pressure fueling adapter cap, the hose nozzle grounding attachment should be connected to the aircraft ground connection.

Fire Extinguisher and Attendant. During fueling, a secondary operator or assistant plane captain should man a properly serviced CO2 hand extinguisher, with a properly serviced second extinguisher readily available.

Fueling Procedure

1. Install ground locks.

2. Close FUELING and LEFT TR BUS FDR circuit breakers on main power distribution box circuit breaker panel.

3. Connect external electrical power to aircraft.

4. Turn the EXT PWR NO. 1 AC PWR switch to RESET then ON.

5. Open PRESS FUEL STA access door (right nacelle, inboard side only) and check LEFT and RIGHT ENGINE STRAINER BYPASS lights. If either light is on, respective engine low-pressure fuel filters are clogged and corrective maintenance should be performed. Notify supervisor.

6. Before removing pressure-fueling adapter cap, the hose nozzle grounding attachment should be connected to the aircraft ground connection.

7. Remove cap from pressure-fueling adapter, insert pressure-fueling nozzle, and lock nozzle in adapter.

8. Commence fueling. NOT FULL lights on pressure-fueling station go on, indicating that fuel is entering tanks. Ensure that vent at tail of aircraft is clear and that tanks are venting during fueling. If no venting is indicated, cease fueling operation immediately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION

Maximum fueling pressure should be limited to 60 psi and a flow rate of 250 GPM.

NOTE

When only partial fuel load is required in each tank, close FUEL QTY LEFT TANK and RIGHT TANK circuit breakers, and station observer in cockpit to monitor fuel quantity indicators. If aircraft is to be fully fueled, cockpit monitor is not required.

9. Hold left TANK PRECHECK switch to PRIM. After the left NOT FULL light goes off (approximately 3 seconds later) hold the right TANK PRECHECK switch to SEC. After the right NOT FULL light goes off, fueling should terminate; release switches and continue fueling. Before aircraft fueling is complete, repeat the above procedure, this time holding the left TANK PRECHECK switch to SEC and the right TANK PRECHECK to PRIM. If fuel flow does not stop when NOT FULL lights go off, turn off fueling source immediately and disconnect pressure fueling nozzle. Notify supervisor.

10. When fuel tanks are full, NOT FULL lights go off and fueling stops automatically.

11. When partial fuel load is required, cockpit monitor should signal to stop fueling when 100 pounds of fuel less than fuel load required for respective tank is indicated on fuel quantity indicator. When signal to stop fueling is given by monitor, fuel flow to each tank is terminated by holding respective TANK PRECHECK switch to SEC.

12. Turn off fueling source and remove pressure fueling nozzle from pressure-fueling adapter. Install cap.

13. Remove hose nozzle grounding cable from aircraft grounding connection.

14. Close and secure PRESS FUEL STA access door.

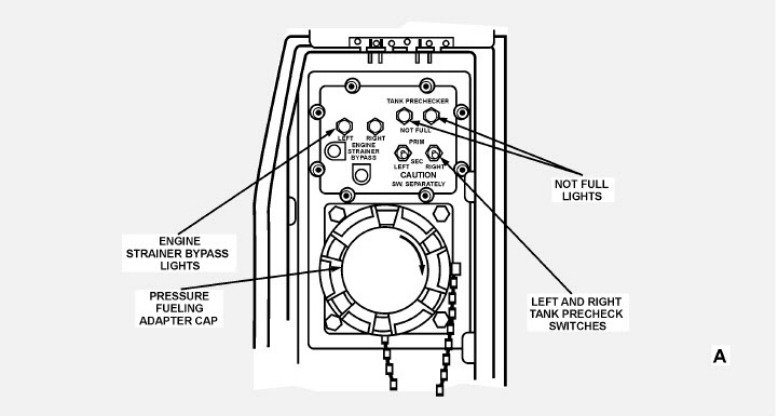
Fueling without Electrical Power. If electrical power is not available, the aircraft may still be pressure refueled. The tanks may be completely filled, but without electrical power, the operator cannot check the left and right tank not full lights. Without a proper check, the operator will not be certain that fueling will automatically terminate when the tank is full. During refueling without electrical power tank prechecks cannot be performed. Consequently there is no assurance that pressure-fueling will automatically terminate. Personnel and support equipment

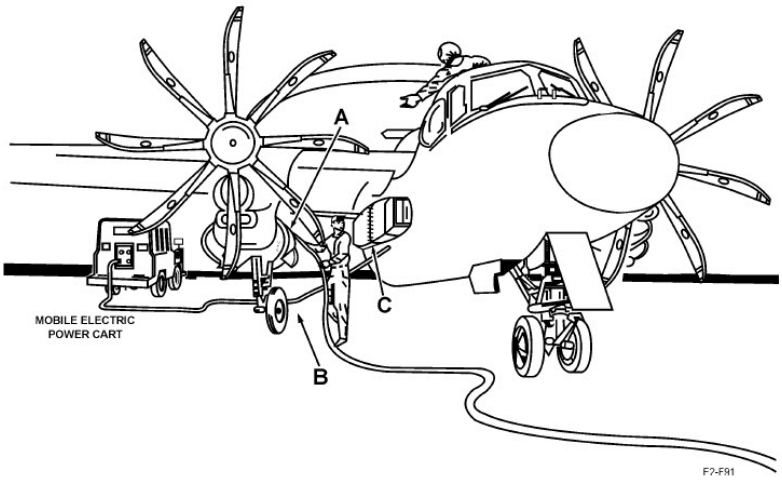
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

should be kept clear of the areas aft of the overwing and tail fuel vents.

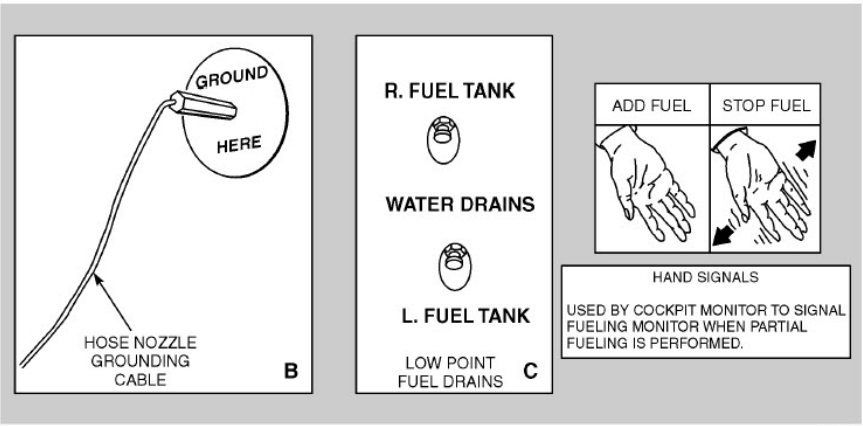
Emergency Fueling. Normally, the aircraft is fueled through the pressure-fueling panel on the inboard side of the right engine nacelle. However, if a pressure-fueling nozzle or truck is not available, or if it is otherwise impossible to fuel the aircraft in the normal manner, the tanks can be gravity-filled through tank caps on top of each wing. A ground wire attachment is located at each wing leading edge for proper grounding.

FUEL SERVICING PANEL





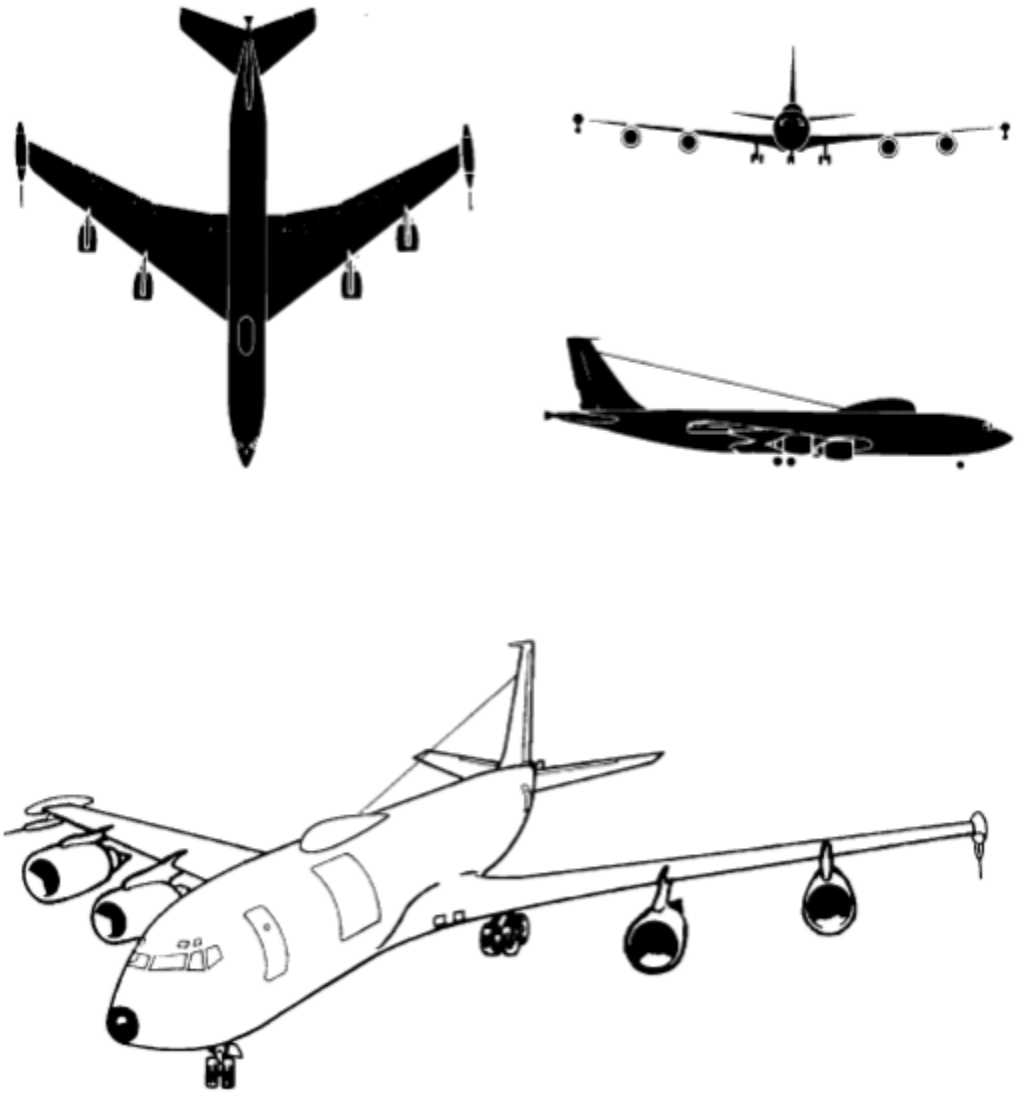
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

E-6

The Boeing E-6B is an airborne command post and communication relay aircraft based on the Boeing 707-320 airframe with modifications and special equipment. The aircraft is a low-wing, all metal, long-range, land-based four-engine jet monoplane.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	148 ft 4 in	Maximum Gross Weight - 342,000 lbs
Length	150 ft 4 in	
Height	42 ft 5 in	

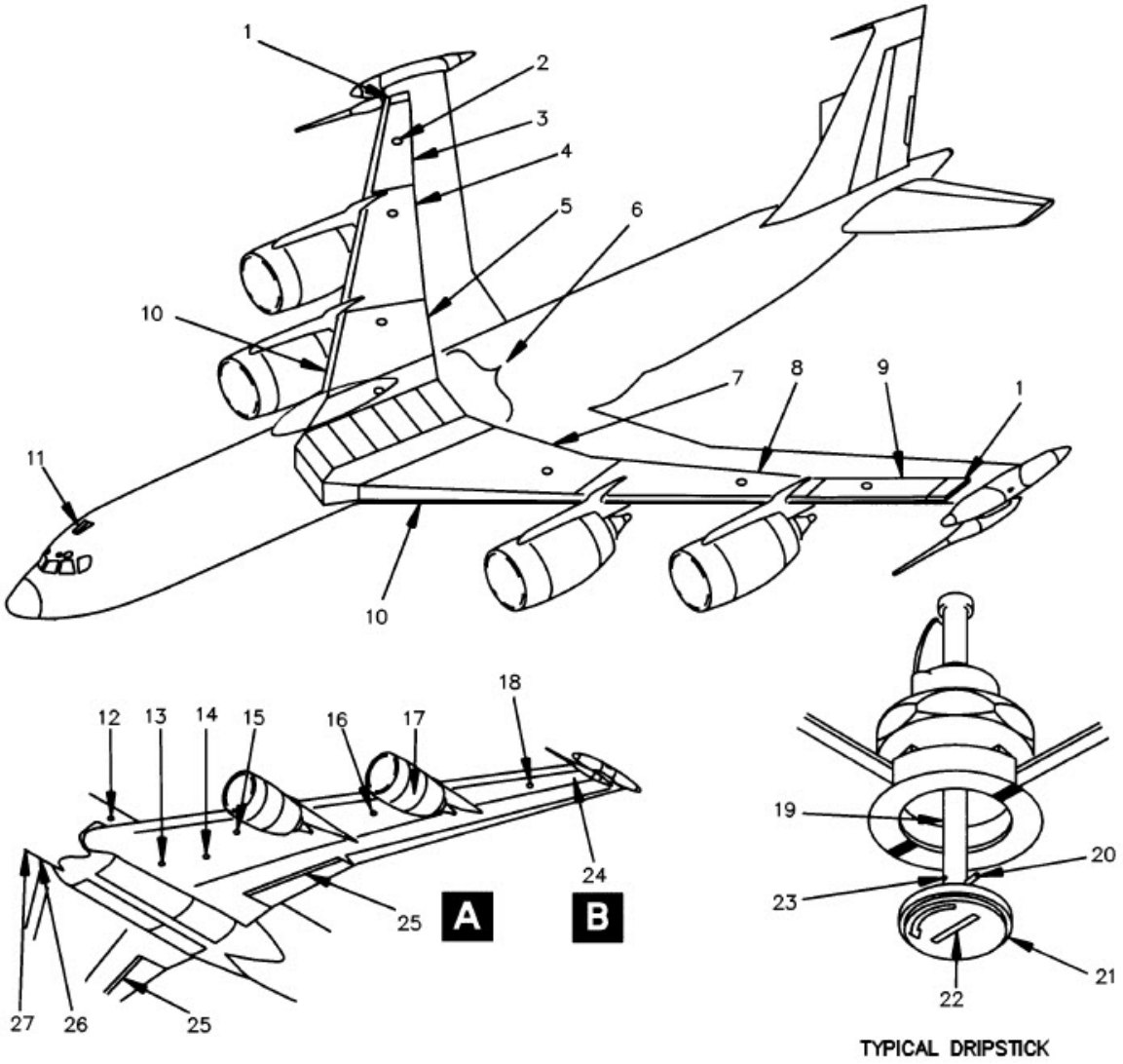
TABLE OF FUEL CAPACITIES

Tank		Gallons	Pounds
Internal	No. 1 Main	2353	15854
	No. 2 Main	4069	27324
	No. 3 Main	4069	27324
	No. 4 Main	2353	15854
	Center	10193	68502
Reserve	Left	439	2946
	Right	439	2946
Total		23915	160750

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Fuel System Component Locations

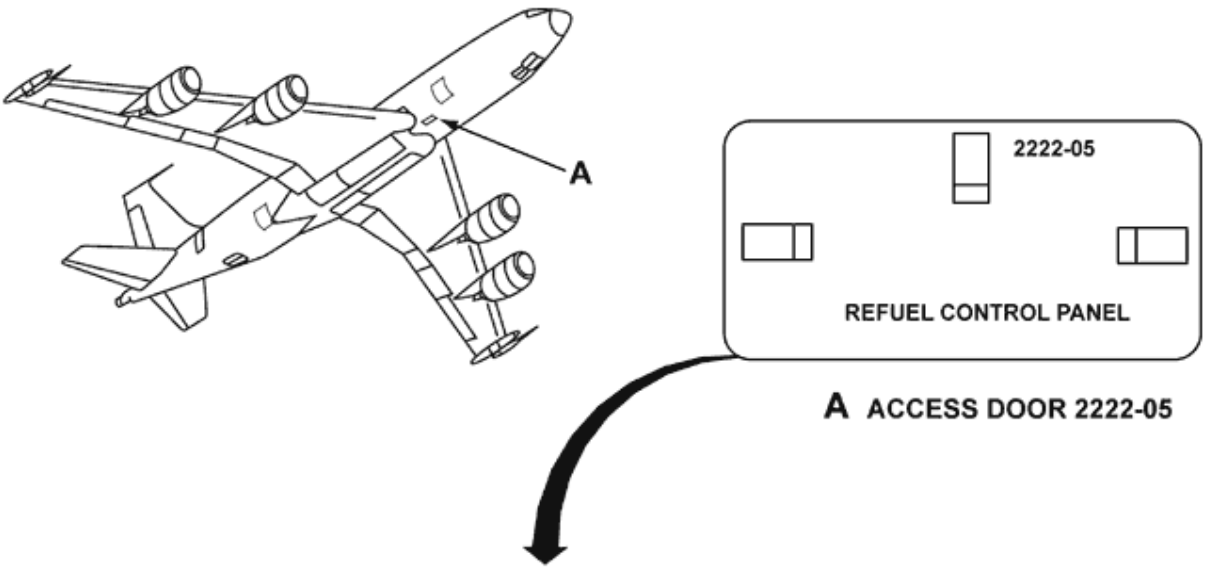


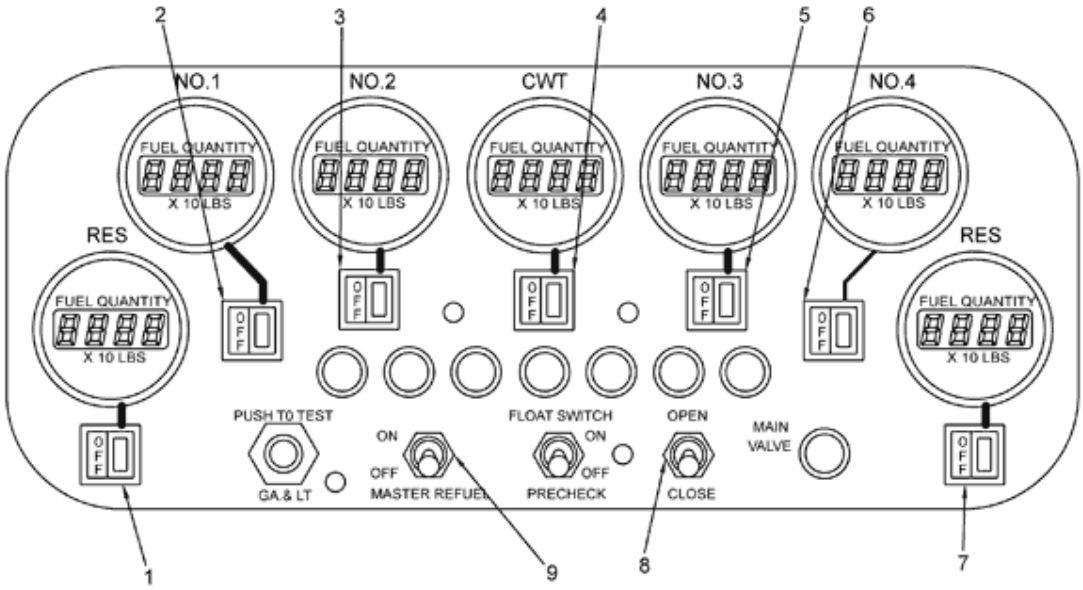
TYPICAL DRIPSTICK

1. VENT SURGE TANK
2. OVERWING FILLER PORT (7 PLACES) (TYPICAL)
3. RESERVE TANK NO. 4
4. MAIN TANK NO. 4.
5. MAIN TANK NO. 3
6. CENTER TANK
7. MAIN TANK NO. 2
8. MAIN TANK NO. 1
9. RESERVE TANK NO. 1
10. DEFUEL VALVE ACCESS
11. AIR REFUELING SLIPWAY DOORS
12. CENTER WING CAVITY VENT
13. CENTER WING TANK DRIPSTICK
14. MAIN TANK NO. 2 INBOARD DRIPSTICK
15. MAIN TANK NO. 2 OUTBOARD DRIPSTICK
16. MAIN TANK NO. 1 INBOARD DRIPSTICK
17. MAIN TANK NO. 1 OUTBOARD DRIPSTICK (NOT SHOWN)
18. RESERVE TANK NO. 1 DRIPSTICK
19. CALIBRATED SCALE
20. LOCKING PIN
21. CAP
22. SCREW DRIVER SLOT
23. DRIPHOLE
24. FUEL VENT SCOOP
25. FUEL DUMP CHUTES
26. GROUND REFUELING PANEL (RIGHT SIDE OF FUSELAGE)
27. SINGLE–POINT REFUELING RECEPTACLE (RIGHT SIDE OF FUSELAGE)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Ground Refueling Panel



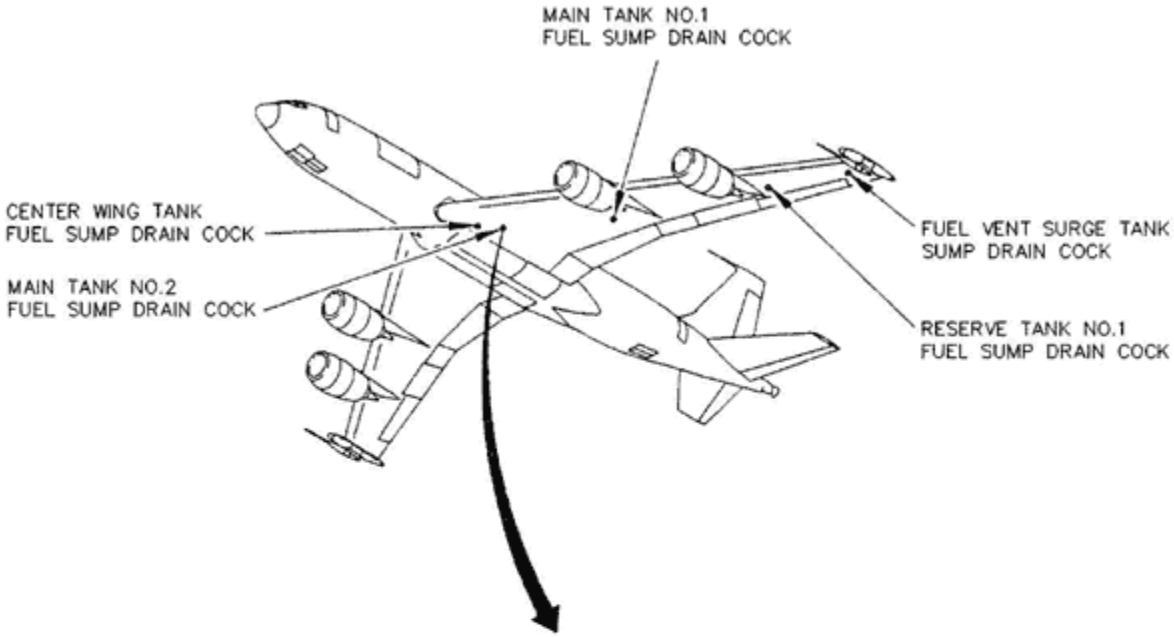


P21 REFUEL CONTROL PANEL

1. RES 1 SWITCHLIGHT
2. NO 1 SWITCHLIGHT
3. NO. 2 SWITCHLIGHT
4. CWT SWITCHLIGHT
5. NO. 3 SWITCHLIGHT
6. NO 4 SWITCHLIGHT
7. RES 4 SWITCHLIGHT
8. MAIN VALVE SWITCH
9. MASTER REFUEL SWITCH

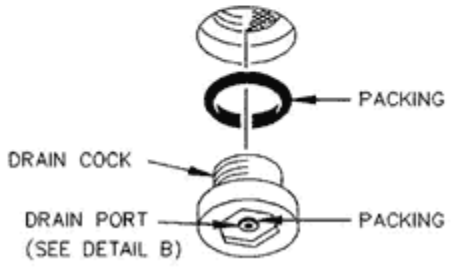
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Location of Sump Drains



NOTE

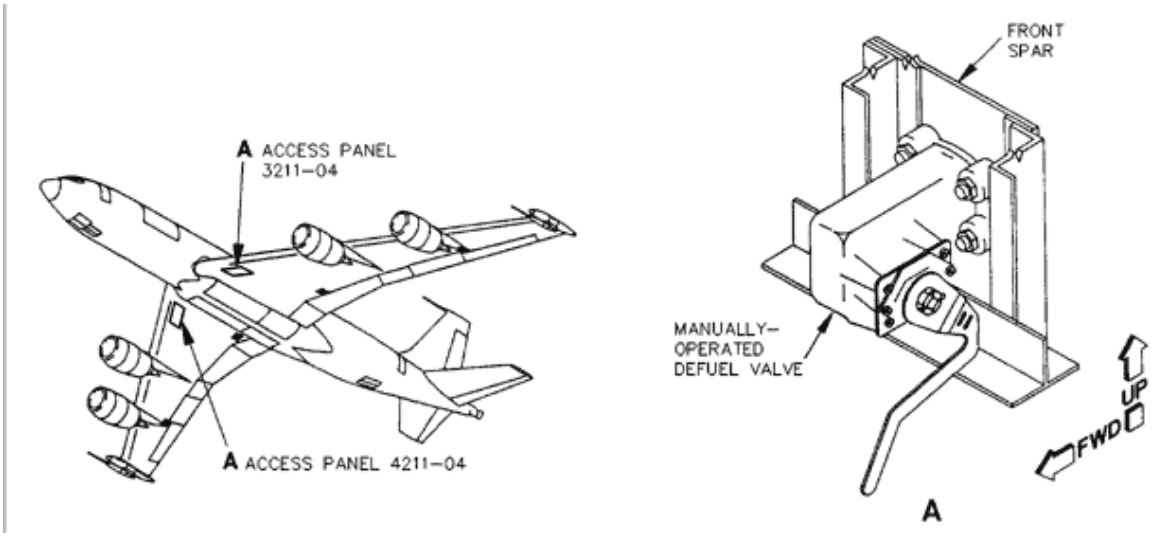
RESERVE TANK NO.1, MAIN TANK NO.1, MAIN TANK NO.2, AND CENTER WING TANK SUMP DRAIN LOCATIONS SHOWN; RESERVE TANK NO.4, MAIN TANK NO.3, AND MAIN TANK NO.4 OPPOSITE.



FUEL SUMP DRAIN COCK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

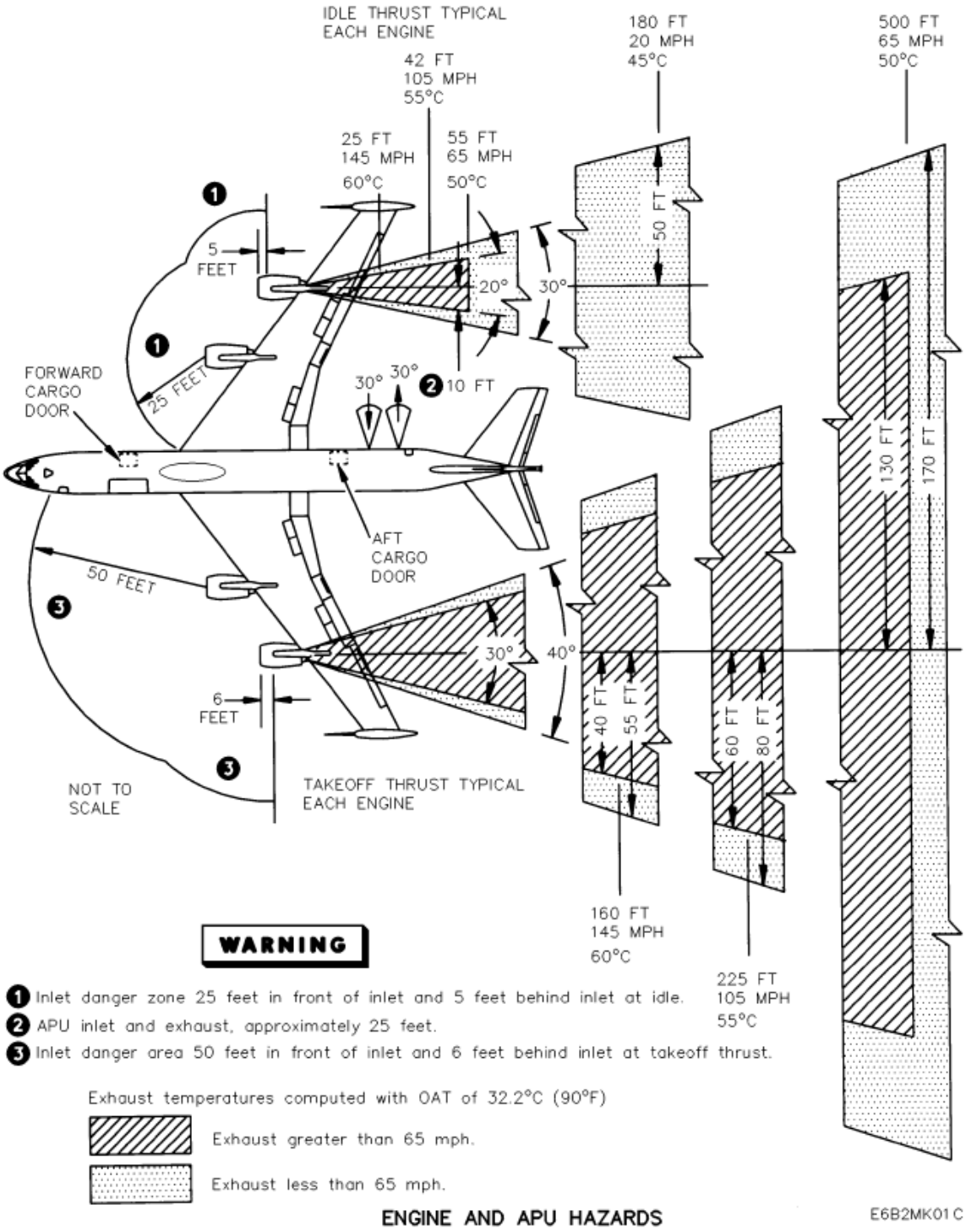
d) Aircraft Defueling Valve



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Inlet/Exhaust/Blast Area

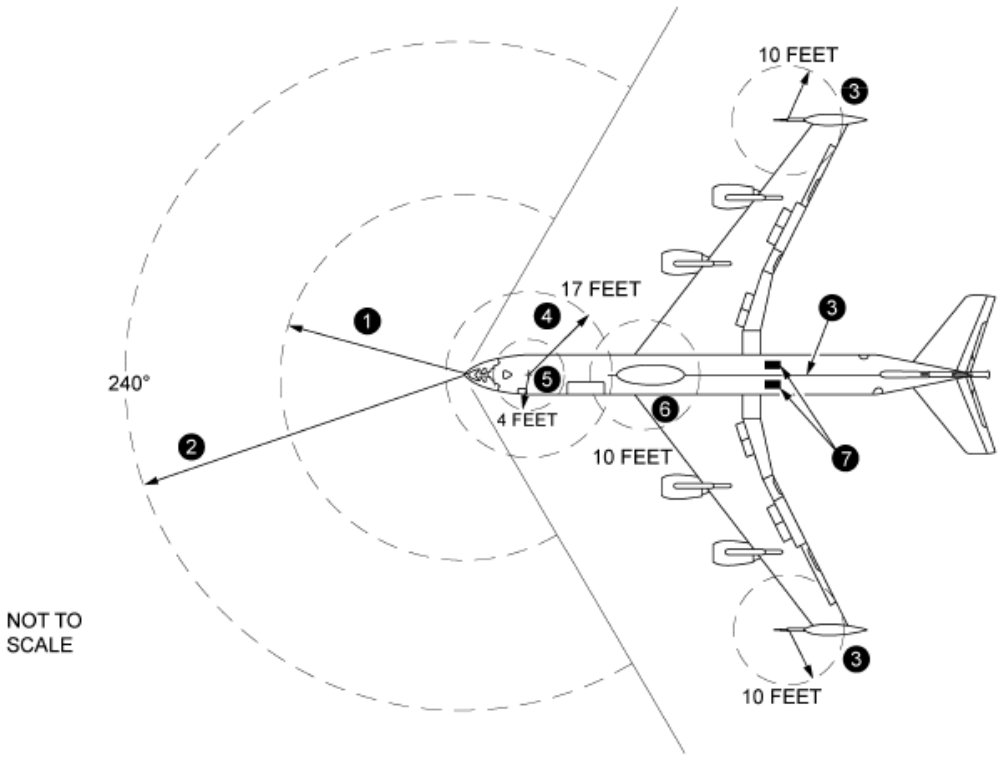


ENGINE AND APU HAZARDS

E6B2MK01C

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Radiation Hazards



WARNING

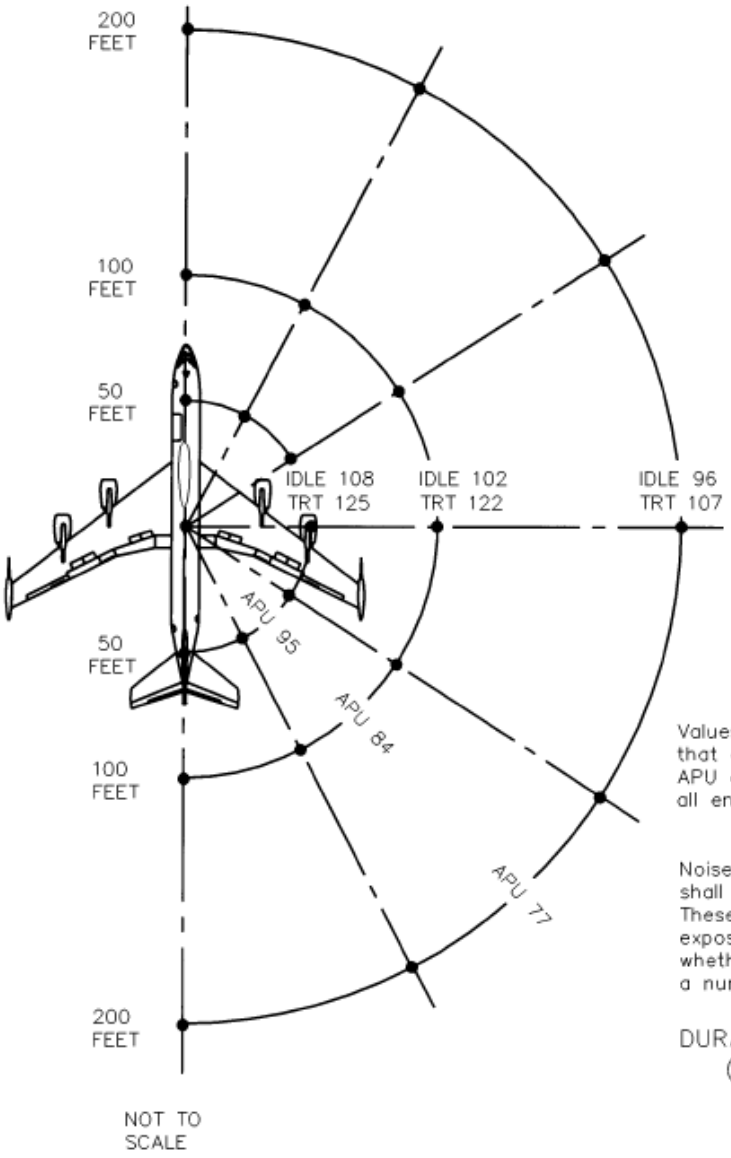
1. Radar area hazardous to personnel (60-foot radius).
2. Radar area of possible fuel ignition (300-foot radius) and electro-explosive device detonation (100-foot radius).
3. HF radio area hazardous to:
   1. Personnel (contact with external aircraft skin surface or headsets connected to external jacks or 10 feet from any point or pod or antenna wire).
   2. Possible fuel ignition (200 feet from any point on pod or antenna wire).
   3. Possible electro-explosive device detonation (150 feet from any point on pod or antenna wire).
4. UHF C3 TX antennas (1, 2, and 3) hazardous to personnel (17-foot radius) when transmitting with the HPA on. UHF C3 TX antennas (1, 2, and 3) hazardous to personnel (1-foot radius) when transmitting with the HPA off.
5. The DNS TX antenna 2 hazardous to personnel (4-foot radius) when transmitting with system set to LOW power.
6. The DNS TX antenna 1 hazardous to personnel (10-foot radius) when transmitting.
7. INMARSAT SATCOM antennas (2) hazardous to:
   1. Personnel (8-foot radius) when transmitting through the HGA.
   2. Possible electro-explosive device detonation (150 feet from antenna) when transmitting through the HGA.

RADIATION HAZARDS

TPL5660

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Exterior Noise



Note

Values shown are the maximum dB recorded at that distance. Recordings conducted with the APU operating, with all engines at idle, then with all engines at takeoff rated thrust (TRT).

Noise exposure (exterior and interior) shall not exceed times shown below. These values apply to total time of exposure per working day regardless of whether this is one continuous exposure or a number of short term exposures.

DURATION PER DAY (IN HOURS)	SOUND LEVEL DB
8	90
4	95
2	100
1	105
1/2	110
1/4	115

WARNING

Ear protection required within 60–foot radius of any engine at idle including the APU and 150–foot radius of any engine at takeoff rated thrust.

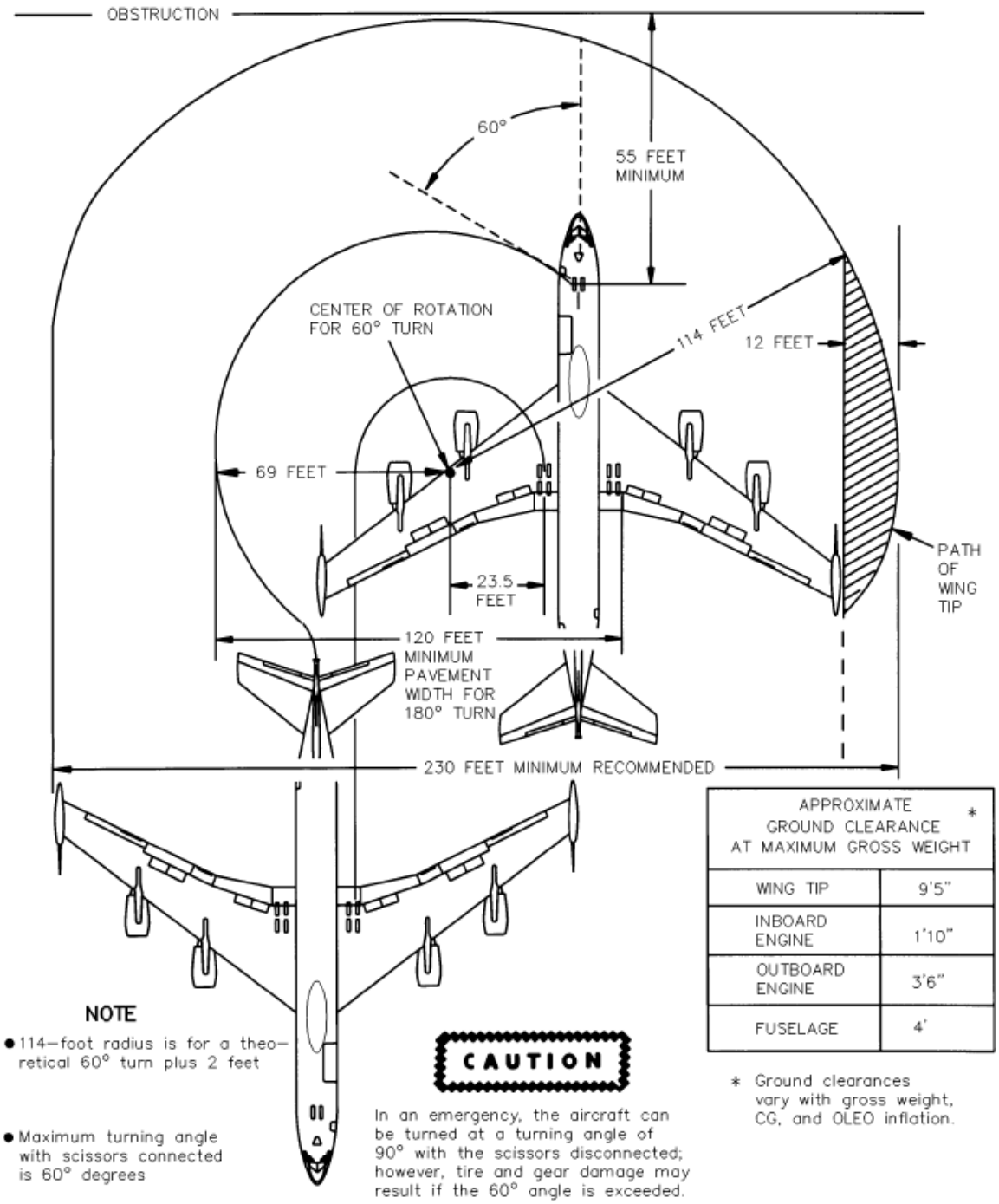
Ear plugs can reduce the noise by 25 to 30 dB. Sound attenuators (mickey mouse ears) can reduce noise by an additional 10 to 15 dB, if used with ear plugs. Sound attenuators alone reduce noise by 30 dB.

EXTERIOR NOISE

E6B2MK04

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Turning Radius



APPROXIMATE GROUND CLEARANCE AT MAXIMUM GROSS WEIGHT *	
WING TIP	9'5"
INBOARD ENGINE	1'10"
OUTBOARD ENGINE	3'6"
FUSELAGE	4'

* Ground clearances vary with gross weight, CG, and OLEO inflation.

NOTE

- 114–foot radius is for a theoretical 60° turn plus 2 feet
- Maximum turning angle with scissors connected is 60° degrees
- Drawing not to scale

CAUTION

In an emergency, the aircraft can be turned at a turning angle of 90° with the scissors disconnected; however, tire and gear damage may result if the 60° angle is exceeded.

E6B2MM01

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The aircraft can be refueled using either the single-point refueling receptacles or overwing filler ports.

REFUELING OPERATIONS PREPARATION AND PRECAUTIONS
CAUTION: The E-6B has a critical center of gravity and care needs to be taken in fuel distribution to ensure safe fuel loading.

CAUTION: Allow only electrical power necessary for the fueling operation on the aircraft. Do not operate any switches in the aircraft except those necessary for the fueling operation. Do not open or close any circuit breakers on the aircraft during the actual fueling operation.

- Ensure all OXYGEN REGULATOR switches are set to OFF.
- Ensure thrust levers 1, 2, 3, and 4 on aisle control stand aft electronics panel, are at the aft stop position.
- Ensure ENGINE-START LEVERS 1, 2, 3, and 4 are at the CUTOFF position (to ensure engine fuel shutoff valves are closed).
- Ensure all interior lighting switches are in OFF position.
- On P67-1 panel, ensure AISLE lighting circuit breaker is closed.
- Ensure defuel valve control handles are in the CLOSED position and DEFUEL VALVE access panels, 3211-04 and 4211-04, are installed.
- Ensure B-2 maintenance platform is positioned by forward entry door for emergency egress.
- Ensure B-2 maintenance platform will not contact aircraft when aircraft settles during fueling.
- Remove all support equipment not associated with aircraft fueling operations.

NOTE: During inclement weather, forward entry door may be partially closed with upper and lower gates left open.

- Open forward entry door and secure.
- Ensure ELECTRONICS ACCESS/BATTERY ACCESS door and forward cargo door are closed and secured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

- Ensure fuel tank sump drains are closed.

- Inspect fuel hoses for breaks or worn spots. Replace damaged hoses before refueling.

- Ensure aircraft single-point refueling receptacle and single-point fuel servicing nozzle locking devices are satisfactory for use before connecting to aircraft.

- The refuel team needs to be constantly alert during refueling operations for fuel spills, such as fuel flow from aircraft fuel overflow vents or shroud drains. Fuel overflow vents have to be monitored by refuel team members during the fueling operations. If fuel is noted leaking from shroud drains, stop fueling and check ARR system.

SINGLE-POINT REFUELING

WARNING: No portion of the aircraft electrical system, including the radio, is to be activated during single-point refueling operations except that portion required for refueling. Before beginning the refueling operation, review emergency shutdown procedures.

WARNING: Circuit breakers and switches are to be set in position required prior to aircraft fueling and are to remain in that position until fueling operations are complete.

NOTE: The aircraft can be refueled using any one of three different sources of electrical power: battery power, external electrical power, and APU power. NAVAIR A1-E6AAB-GAI-000, WP026, Table 1 provides switch/control positioning when using any one of these three different sources of electrical power.

WARNING: When refueling using APU power, a member of the refuel service team will be positioned so in the event of fire in the APU, he can activate the APU fire extinguishers.

NOTE: OFF indicator lights on P21 refuel control panel will go off upon flow into the system.

NOTE: The refueling operation can be controlled from either P11 flight engineer FUEL panel or P21 refuel control panel. The preferred refueling method is by using P21 refuel control panel.

WARNING: Ensure refueling hose nozzle ground strap is bonded in positive contact with aircraft and maintained until fuel cap is replaced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

NOTE: Two hoses may be used for maximum fuel flow if both hoses are connected to the same pressure source.

WARNING: Personnel handling the nozzle need to discharge static electricity from their person frequently during the fueling operation by grasping an approved grounding strap or line before handling or touching metal in the vapor area.

NOTE: If more than one fuel truck is utilized during a fueling period, unpressurized and pressurized test need only be accomplished for first fuel truck.

CAUTION: Refueling equipment pressure is not to exceed 55 psig (except for momentary surges) during refueling. Momentary surges are not to exceed 120 psig.

NOTE: If any tank is fueled to automatic shutoff, expansion of fuel can cause activation of the secondary shutoff valves. This will prevent the addition of more fuel to the aircraft by closing the main refueling valve. In case of unintentional closing of main refueling valve because of secondary float switch action, open main refueling valve.

CAUTION: During refueling, an inoperative dual pilot valve will allow the tank to take fuel even when related refuel valve is in the OFF position. If any tank does not close (indicated by fuel quantity indicator reading 300 pounds or more above quantity showing when switch was pressed), discontinue single-point refueling. Refueling may be completed using overwing refueling.

NOTE: When tank refueling is stopped by pressing the switchlight on P21 refuel control panel, watch the applicable tank fuel quantity indicator for a few seconds to ensure tank does not take on additional fuel.

OVERWING REFUELING

WARNING: No portion of the aircraft electrical system, including the radio and radar, is to be activated during servicing operations except that portion required for servicing. Do not provide electrical power to the aircraft when the overwing filler port is opened. Prior to providing electrical power to aircraft to read fuel quantity gages: shut down refueling source, remove hose from filler port, disconnect hose ground wire, and replace fuel cap. After reading gages and removing electrical power from aircraft: attach refueling hose ground wire, remove fuel cap, insert the hose, and continue the refueling operation.

NOTE: External filler ports are located on the upper surface of the wing for each wing tank. A grounding receptacle is located near each filler port. Filler caps are opened and closed with a screwdriver.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION: Observe wing walk areas.

NOTE: Fill one tank at a time on each wing to maintain proper fuel load distribution.

WARNING: When refueling over the wing, ensure fuel hose nozzle ground is bonded in positive contact with the aircraft before fuel tank filler cap is removed and that the bond is maintained throughout the fueling operations until filler cap is replaced.

WARNING: Open and close fuel hose nozzle valve slowly to prevent kickback of nozzle and high-pressure surge. Personnel handling the nozzle need to discharge static electricity from their person frequently during the fueling operation by grasping an approved grounding strap or line before handling or touching metal in the vapor area.

WARNING: Fuel hose nozzles have to be manually operated at all times and not be blocked in an open position.

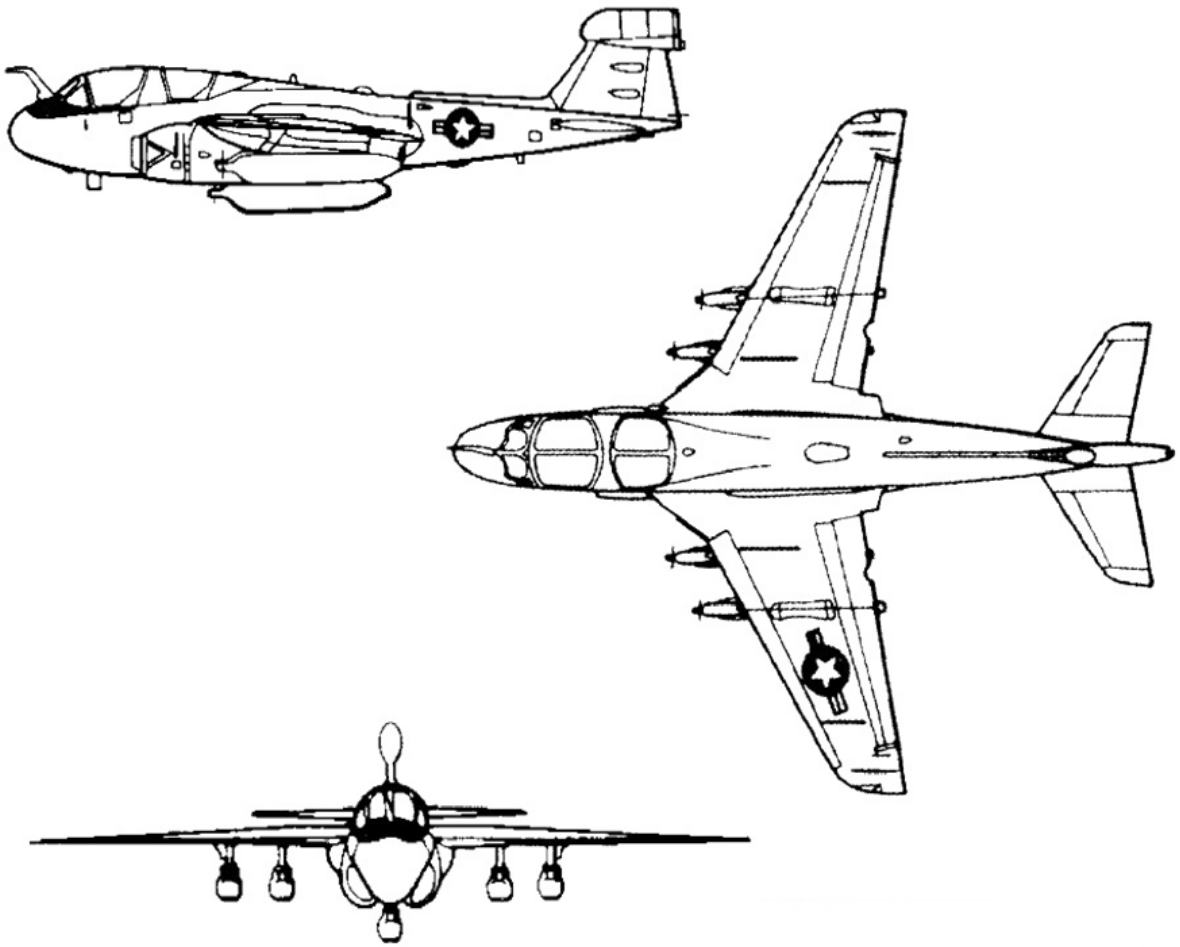
NOTE: To maintain proper center of gravity during overwing fueling, fill tanks in sequence called for.

CAUTION: Whenever total fuel load exceeds 80,000 pounds, both reserve tanks have to be filled to capacity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

EA-6

The EA-6B is a four-place, twin-engine, mid-wing monoplane manufactured by the Grumman Aerospace Corporation, Bethpage, Long Island, New York. The aircraft is designed for carrier and advanced base operation, is a modification of the basic two place A-6 airframe.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread Folded	 53 ft 25 ft 10 in	Maximum Gross Weight —61,500 lbs
Length	59 ft 10 in	Maximum Footprint — 230 psi
Height	16 ft 8 in	

TABLE OF FUEL CAPACITIES

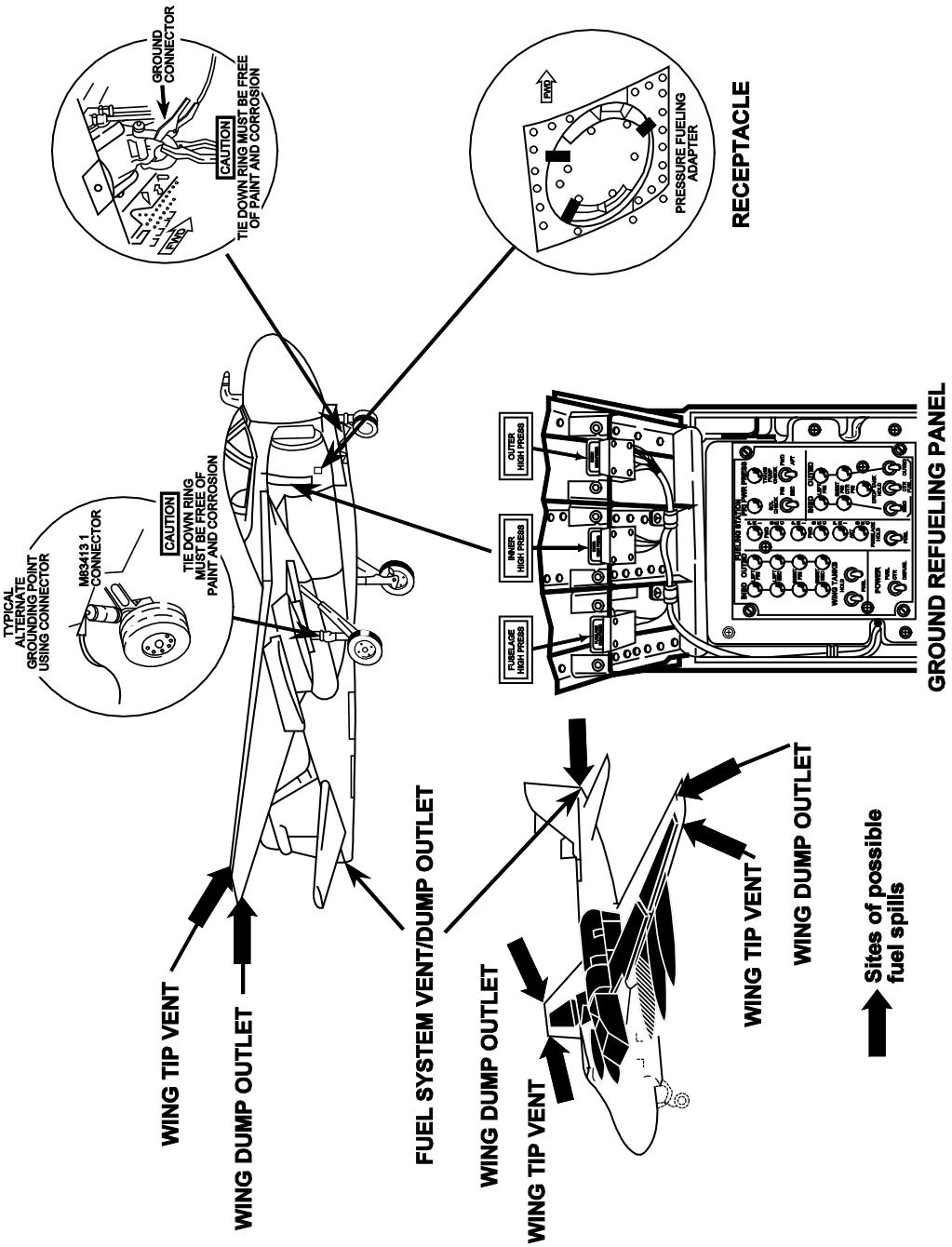
Tank		Gallons
Internal	FWD Cell	400
	MID Cell	260
	AFT Cell	640
	Wing	700
	Total Internal	2,000
External	Drop Tank (Each)*	300
		400
	Air Refueling Store	300

*Two sizes of tanks are available

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

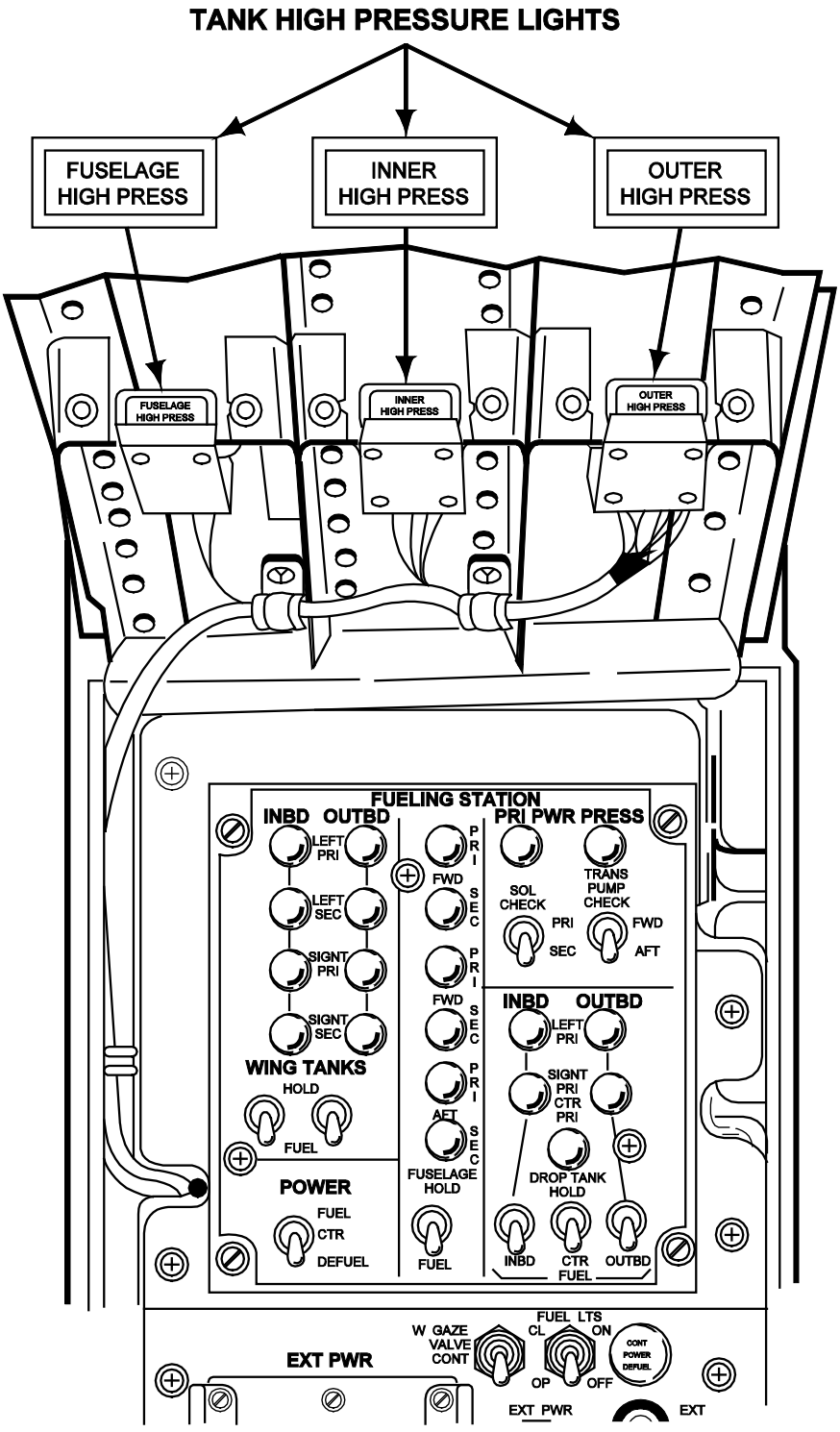
AIRCRAFT CONFIGURATION

a) Locations of Filler Caps, Service Panels, and Vents



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

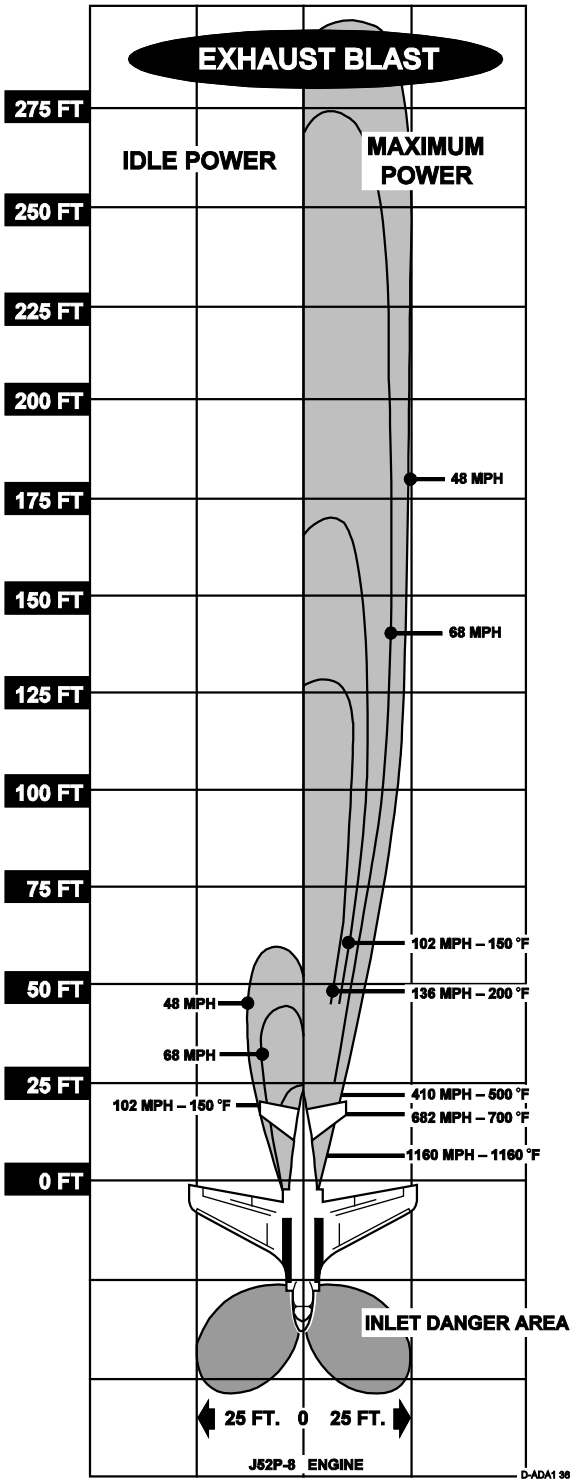
b) Ground Refueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Inlet/Exhaust/Blast Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the EA-6 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

WARNING

EA-6 aircraft are normally hot refueled with both engines operating. DO NOT ASSUME RIGHT ENGINE TURNED OFF until confirmation obtained.

1. Set "POWER" switch on ground refueling panel to "FUEL," and set "WING TANKS," "FUS TANKS," and "DROP TANKS" switches to "FUEL."

2. On ground refueling panel access door, press "OUTER HIGH PRESS" indicator. All three "High Pressure" indicator lights on the door should illuminate.

3. On ground refueling panel, momentarily press each "WING TANKS," "FUS TANKS," and "DROP TANKS" indicator that is not lighted; each indicator will illuminate.

   If **CAUTION**

   indicators checked in steps 2 and 3 malfunction, do not fuel aircraft.

4. Attach refueling nozzle to aircraft receptacle and place nozzle flow control handle in the open position.

5. Initiate fuel flow and monitor tank "High Pressure" lights on door above ground refueling panel.

   CAUTION

   Immediately stop fuel flow if any of the three "High Pressure" indicators illuminates during the refueling operation. Cause of vent system overpressure **should** be corrected before fueling is resumed.

6. Exercise precheck system. Set and hold "SOL CHECK" switch to "PRI." All "WING TANKS PRI," "FUS TANKS PRI," and "DROP TANKS PRI" indicator lights will go out and fuel flow into the aircraft will stop prior to aircraft receiving 45 gallons of fuel.

7. When fuel flow stops, check flow rate. If flow rate exceeds 3 gallons per minute, discontinue hot refueling operation.

8. On ground refueling panel, set "DROP TANKS" switches to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

"HOLD" and release "SOL CHECK" switch.

9. When fuel flow starts, immediately set and hold "SOL CHECK" switch to "SEC." All "WING TANKS SEC" and FUS TANKS SEC" indicator lights will go out and fuel flow automatically stop prior to aircraft receiving 30 gallons of fuel.

10. When fueling stops, check fuel flow rate. If flow rate exceeds 3 gallons or 20 pounds per minute, shut off fueling unit and disconnect fuel nozzle.

11. On ground refueling panel, set "WING TANKS" switches to "HOLD" and release "SOL CHECK" switch.

12. Fuel fuselage tanks until "FUS TANK" indicators on ground refueling panel go out.

13. On ground refueling panel, set "WING TANKS" switches to "FUEL"; on aircraft with external tanks, set "DROP TANKS" switches to "FUEL." When "WING TANKS," "FUS TANK," and "DROP TANKS" indicators are all out, aircraft is fueled.

14. Set "POWER" switch to "OFF" on ground refueling panel.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks, including external tanks, exits the aircraft through the common "Fuel System Vent/Dump Outlet."

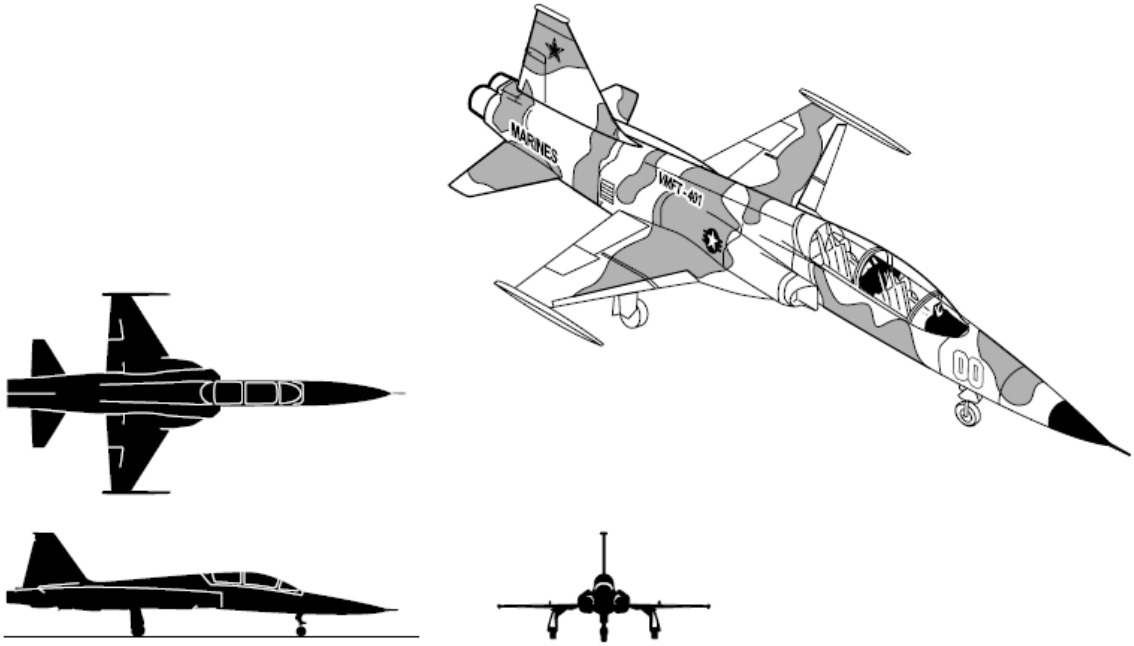
2. If any high level shut-off valves fail to operate correctly, fuel may spill from the "Fuel System Vent/Dump Outlet" or the "Wing Tip Vents."

3. A malfunction within an external fuel tank may cause fuel to spill from the bottom center of tank (pressure relief vent).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

F-5

The F-5 is part of a family of widely-used light supersonic fighter aircraft, designed and built by Northrop. The F-5N/F variants remain in service as an adversary trainer.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wingspan	26 ft 8 in	Maximum Gross Weight — 15,650 lbs
Length	48 ft 2 in	
Height	13 ft 4 in	

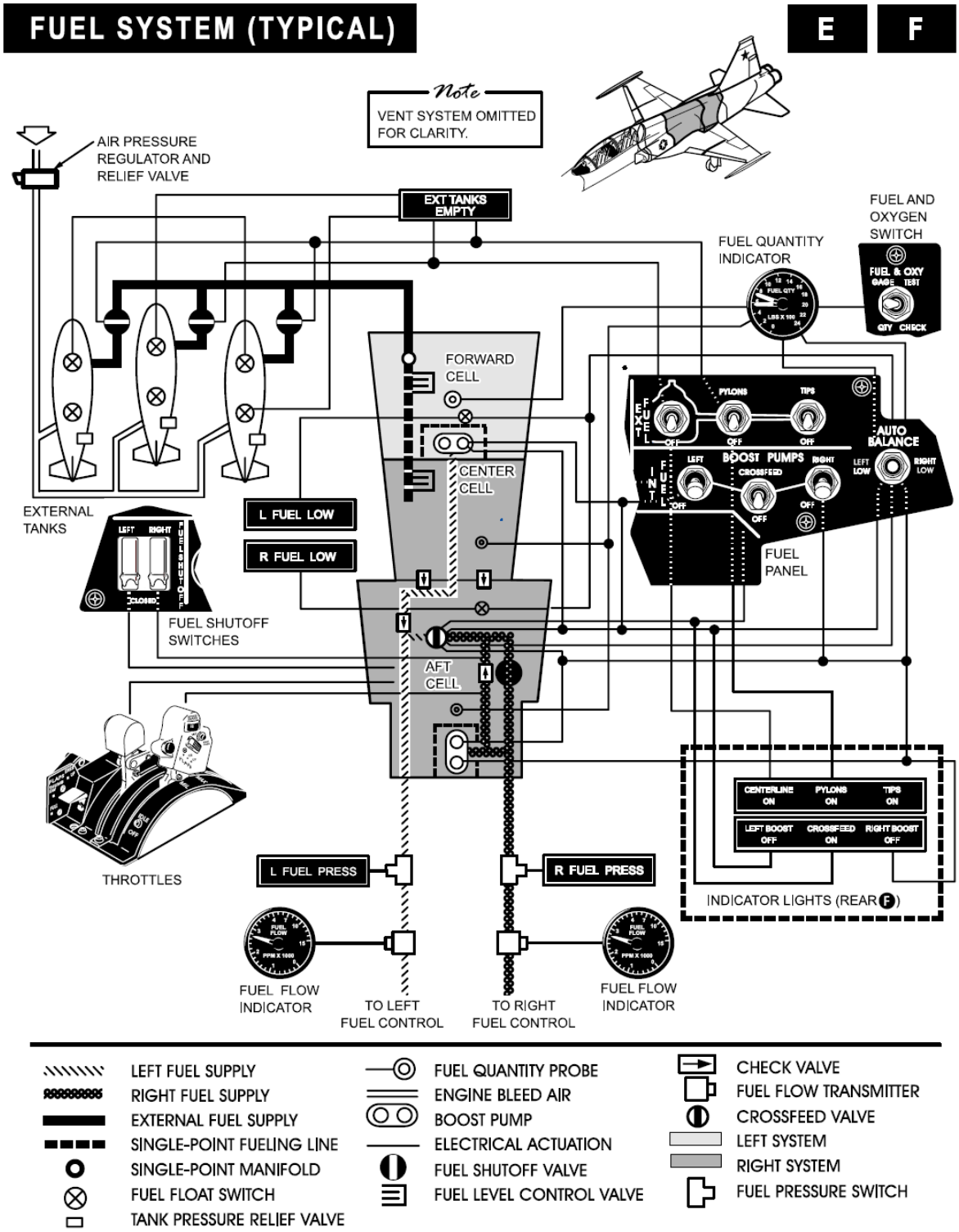
TABLE OF FUEL CAPACITIES

Tank	Gallons	Weight (JP-5)
Left	306	2081
Right	392	2666
Total internal	698	4746

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

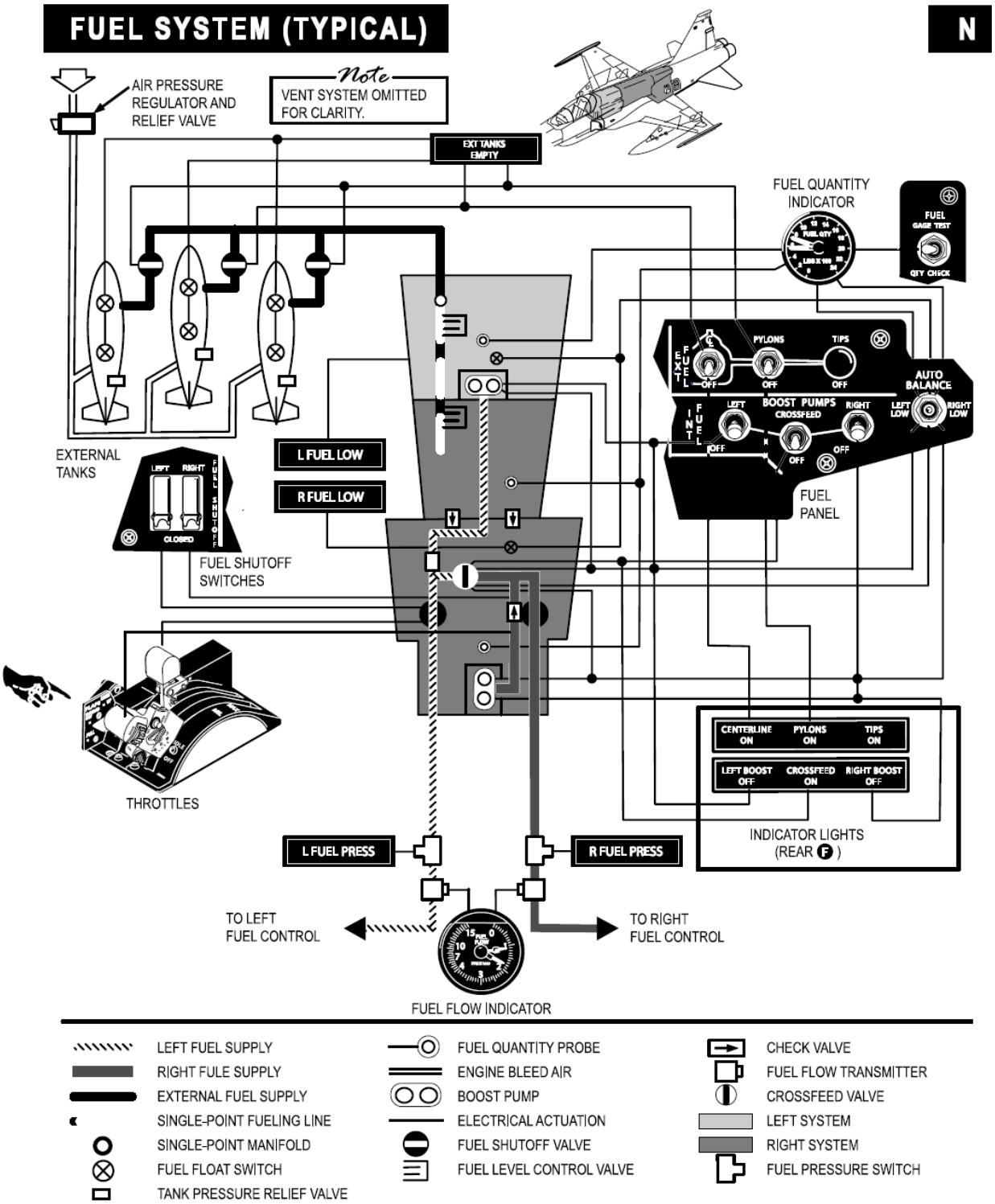
AIRCRAFT CONFIGURATION

a) Fuel System (E/F Series)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

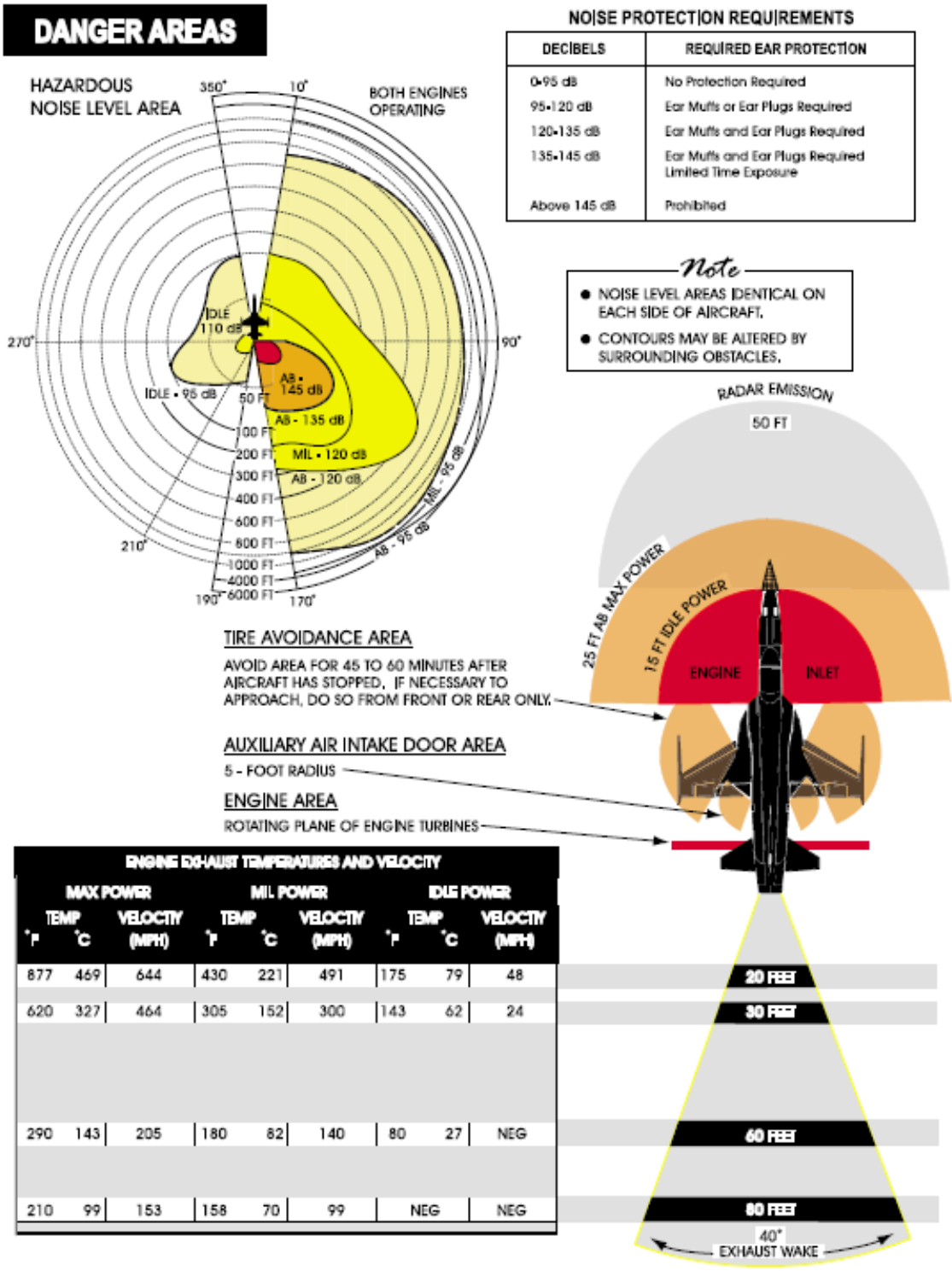
b) Fuel System (N Series)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Hazardous Noise Level, Engine Inlet and Exhaust Areas

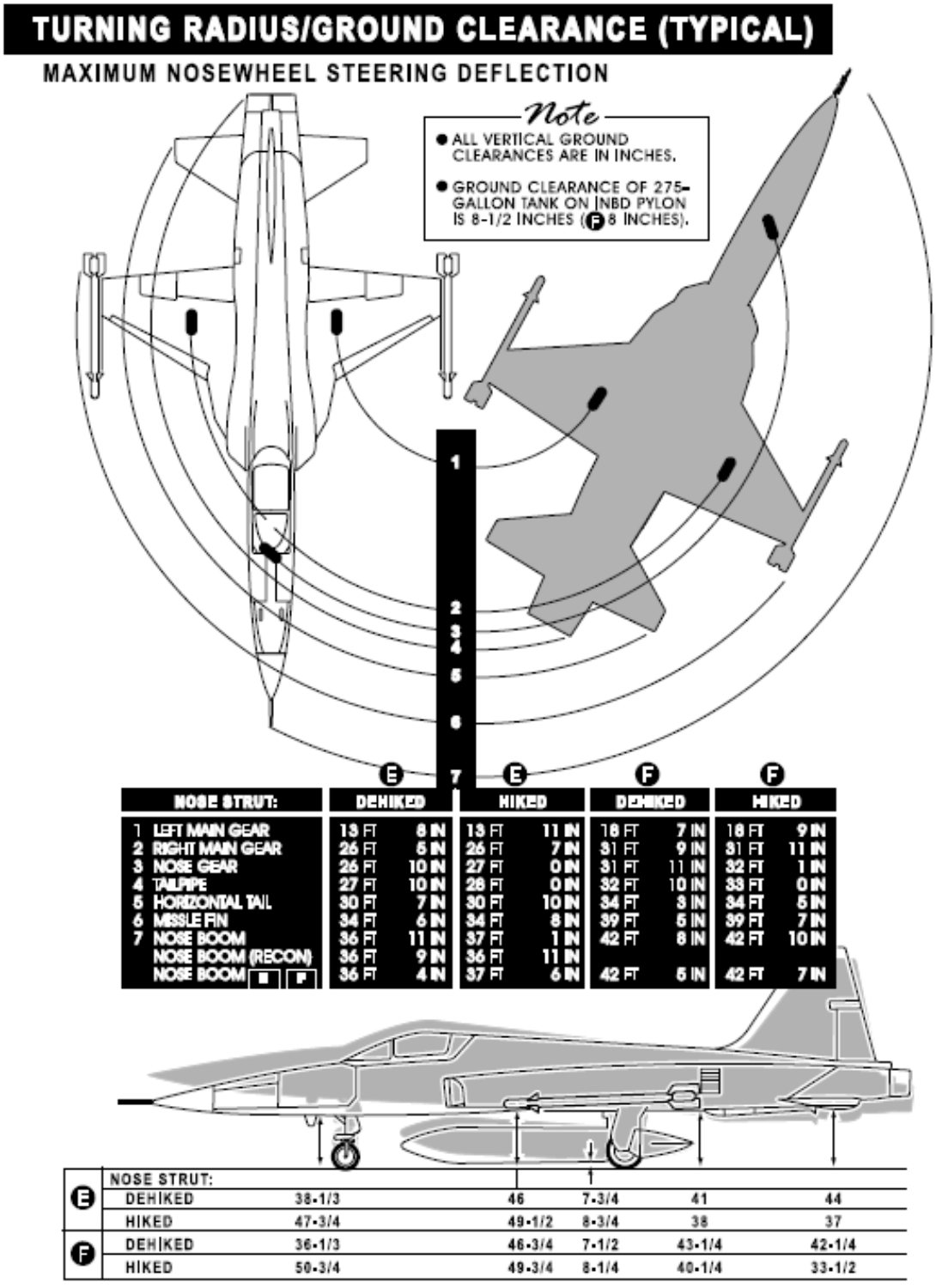


ENGINE EXHAUST TEMPERATURES AND VELOCITY

MAX POWER			MIL POWER			IDLE POWER		
TEMP °F	TEMP °C	VELOCITY (MPH)	TEMP °F	TEMP °C	VELOCITY (MPH)	TEMP °F	TEMP °C	VELOCITY (MPH)
877	469	644	430	221	491	175	79	48
620	327	464	305	152	300	143	62	24
290	143	205	180	82	140	80	27	NEG
210	99	153	158	70	99	NEG		NEG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Turning Radius



	NOSE STRUT:					
E	DEHIKED	38-1/3	46	7-3/4	41	44
	HIKED	47-3/4	49-1/2	8-3/4	38	37
F	DEHIKED	36-1/3	46-3/4	7-1/2	43-1/4	42-1/4
	HIKED	50-3/4	49-3/4	8-1/4	40-1/4	33-1/2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

SINGLE POINT-PRECHECK

WARNING

Before completing steps 1 through 6, request fuel vehicle operator set dial for static pressure between 45 and 55 psi (15 and 25 psi if using a fuel grade other than was previously contained in the tanks) to preclude excessive electrostatic discharge when refueling with a grade other than previously contained in tanks.

1. Precheck Valve Button PRI (after fuel flow starts) – Press. Fuel flow stops within 10 seconds.
2. Precheck Valve Button – Release.
3. Precheck Valve Button SEC (after fuel flow starts) – Press. Fuel flow stops within 10 seconds.
4. Precheck Valve Button – Release.
5. Precheck Valve Button PRI and SEC (after fuel flow starts) – Press simultaneously. Fuel flow stops within 10 seconds.

CAUTION

If fuel flow does stop when either PRI or SEC button of precheck valve is pressed, discontinue single-point fueling and go to manual fueling.

6. Precheck Valve Button – Release.

REFUELING

1. Battery Switch – BATT.
2. External Fuel Centerline Switch – CL (if installed).
3. Battery Switch – OFF.
4. Refueling – Accomplish. After fuel flow stops, close fuel nozzle shutoff valve.
5. Battery Switch – BATT.
6. External Fuel Switch(es) – OFF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

After Refueling Complete

7. Battery Switch – OFF.
8. Grounding Cables – Disconnected.

MANUAL

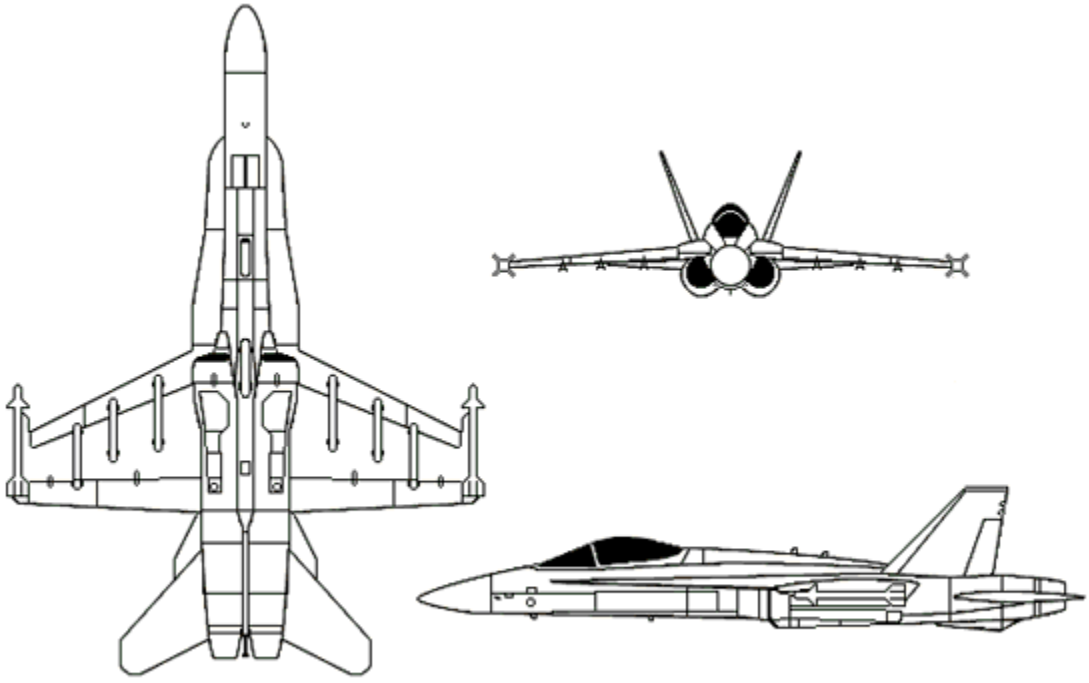
Service fuel tanks in the following sequence;

1. Left internal system.
2. Right internal system.
3. Centerline tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

F/A-18A/B/C/D

The F/A-18A/C Hornet is a single-place fighter/attack aircraft built by McDonnell Douglas Aerospace. It is powered by two General Electric F404-GE-400 or F404-GE-402 (enhanced performance) turbofan engines with afterburner.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: F/A-18A/B/C/D

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread Folded	 37 ft 6 in 27 ft 6 in	Maximum Gross Weight — 51,900 lbs
Length	56 ft	Maximum Footprint — 315 psi
Height	15 ft 3 in	

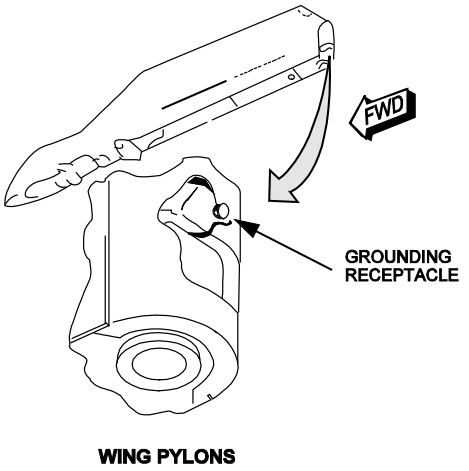
TABLE OF FUEL CAPACITIES

Tank			Gallons
Internal	Fuselage	Number 1	418
		Number 2	263
		Number 3	206
		Number 4	532
	Wing	Left	85
		Right	85
	Total Internal		1,589
External	Elliptical Wing or Centerline Tank		314
	Cylindrical Wing or Centerline Tank		330

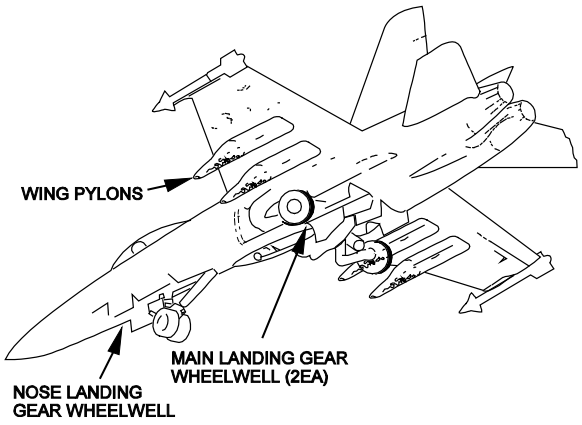
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

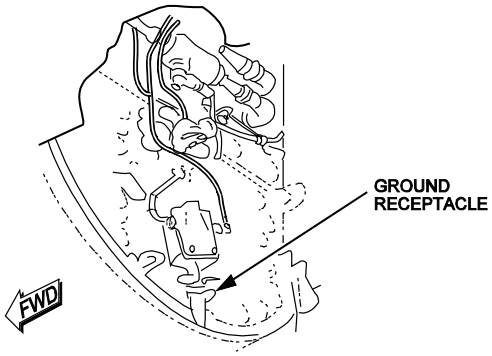
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points

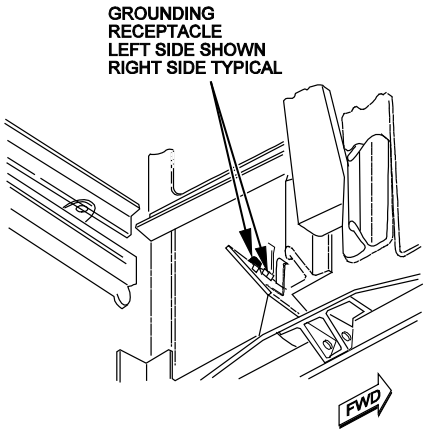


WING PYLONS





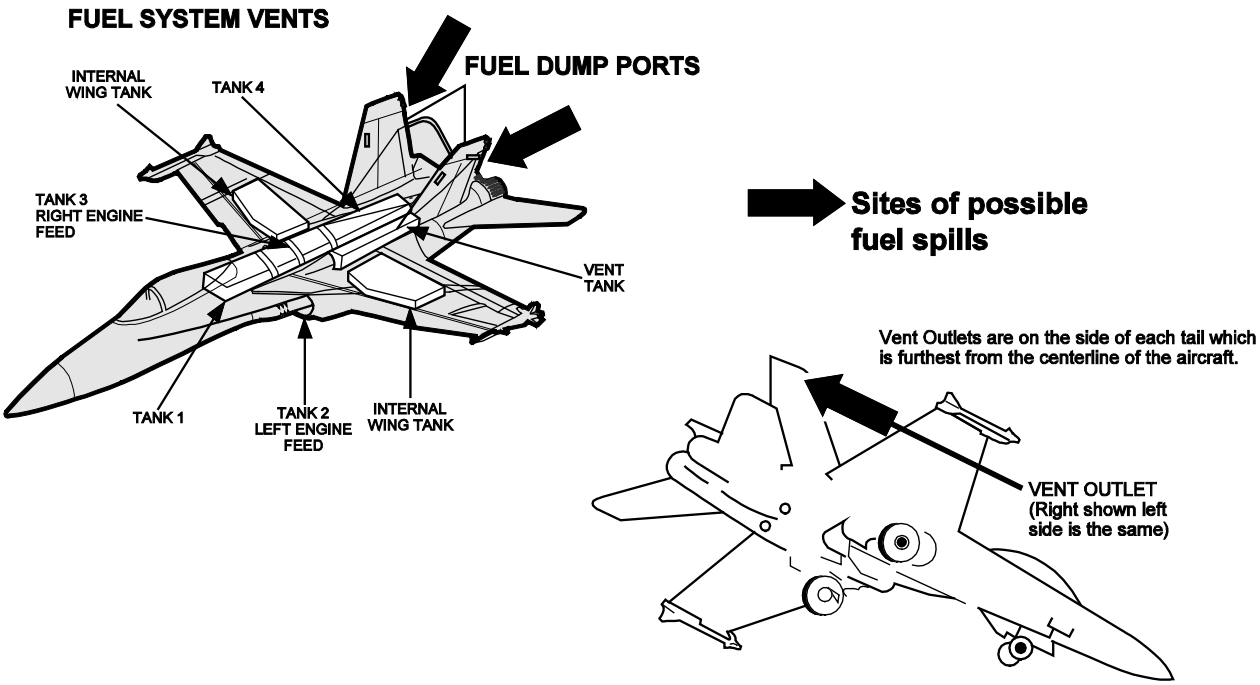
NOSE LANDING GEAR WHEELWELL



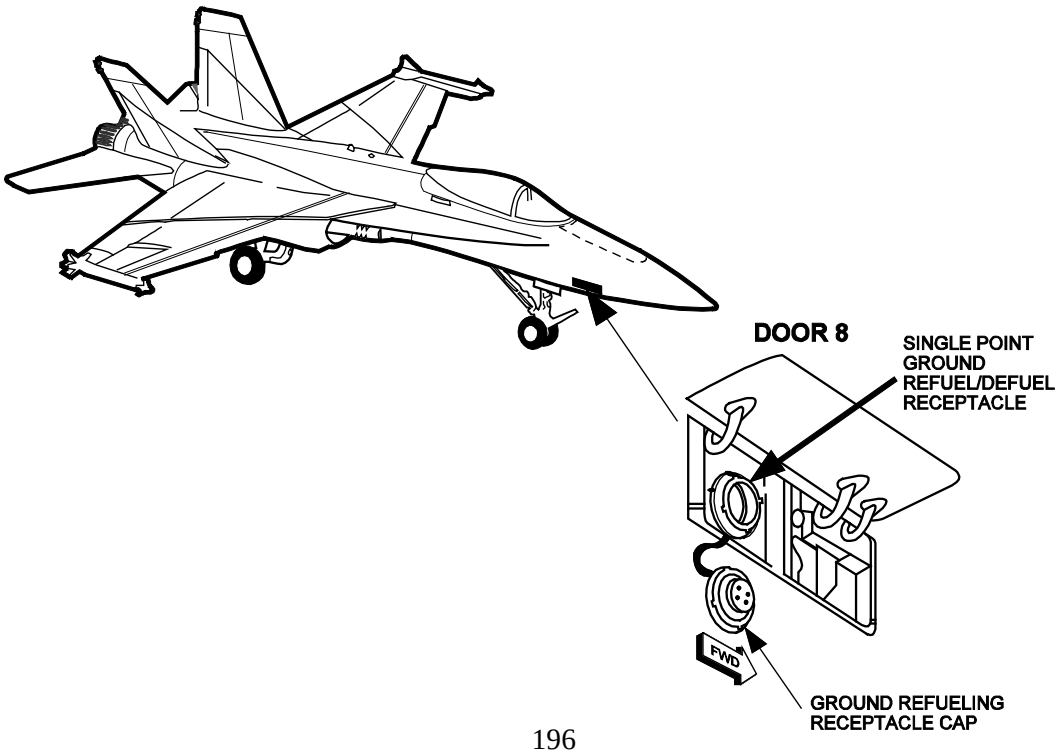
MAIN LANDING GEAR WHEELWELL

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Location of Fuel Vents, Dump Ports, and Low Point Drains
Ground Refueling Panel

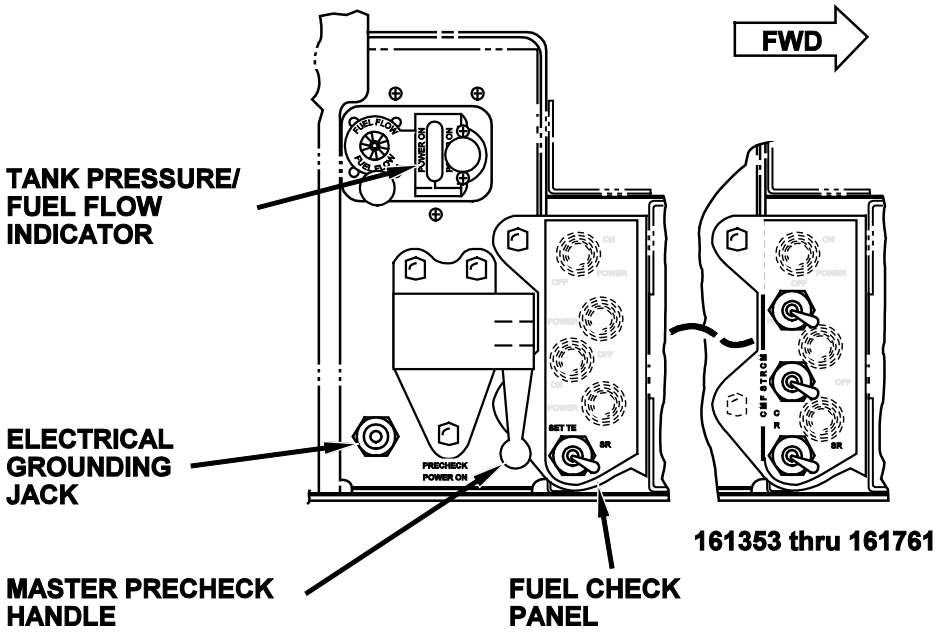


GROUND REFUELING PANEL AND RECEPTACLE



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

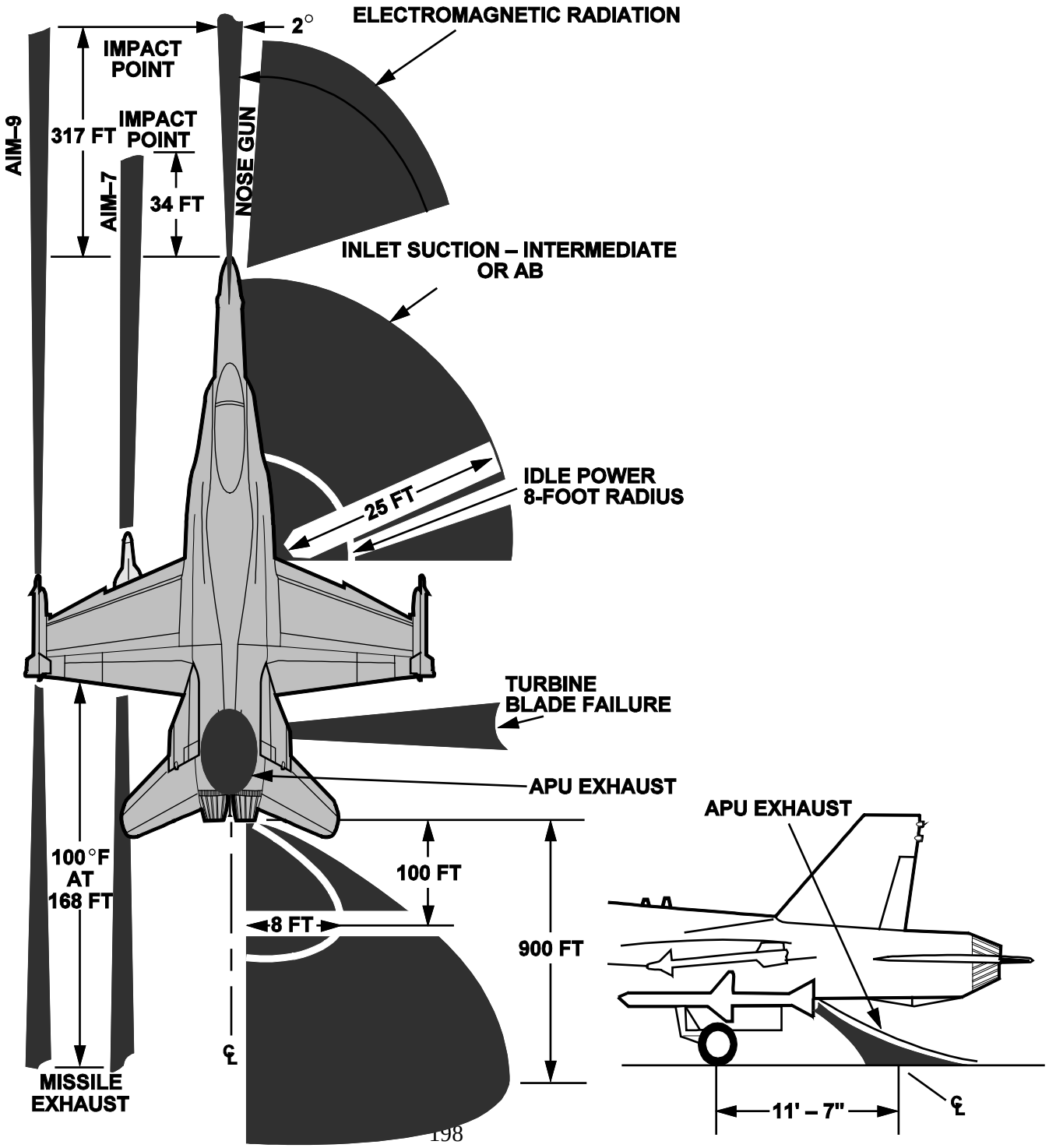
c) Location of Flow Indicator, Electrical Grounding Jack, and Fuel Check Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Engine/APU Inlet/Exhaust/Blast Area. Electromagnetic Radiation Area, and Turbine Blade Failure Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the F-18A/B/C/D aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

CAUTION

F-18 aircraft are normally hot refueled with both engines operating. DO NOT ASSUME RIGHT ENGINE TURNED OFF until confirmation obtained. Operation of the auxiliary power unit (APU) is prohibited during normal fueling.

1. Attach refueling nozzle to aircraft and place nozzle flow control handle in the open position.

2. Initiate fuel flow and observe tank pressure gauge on ground refueling panel.

CAUTION

Immediately stop fuel flow if tank pressure gauge indicates increasing pressure.

3. Exercise precheck system by moving the "Master Precheck Handle" into the up position. Observe "Fuel Flow Indicate" movement to verify that fuel flow into the aircraft stops.

   a. No external tanks installed. Fuel flow into the aircraft should stop in approximately 45 seconds.

   b. External tank(s) installed. Wait approximately 45 seconds, then press the "Ext Tk" button on the "Fuel Check Panel." Fuel flow into the aircraft should stop within approximately 10 seconds.

CAUTION

If fuel flow does not stop, discontinue refueling operation immediately. System failure **should** be investigated and resolved before hot refueling can be accomplished**.**

4. Return "Master Precheck Handle" to the down (off) position and continue fueling until aircraft tanks are filled and fuel flow automatically stops.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks, including external tanks, exits the aircraft through the common "Fuel System Vents."

2. If any high level shut-off valves fail to operate correctly, fuel may spill from the "Fuel System Vents."

3. A malfunction within an external fuel tank may cause fuel to spill from the bottom center of tank (pressure relief vent).

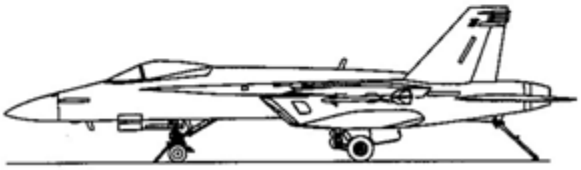
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

F/A-18E/F

The F/A-18E/F Super Hornet is a carrier based strike fighter aircraft built by Mcdonnell Douglas Corporation. The aircraft is powered by two General Electric F414-GE-400 turbofan engines utilizing Full Authority Digital Engine Control. The F/A-18F is the two seat model of the Super Hornet and is configured with tandem cockpits.







Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread Folded	 42 ft 10 in 32 ft 8 in	Maximum Gross Weight (E) — 31,550 lbs Maximum Gross Weight (F) — 32,000 lbs
Length	60 ft 2 in	Maximum Footprint — 315 psi
Height	16 ft	

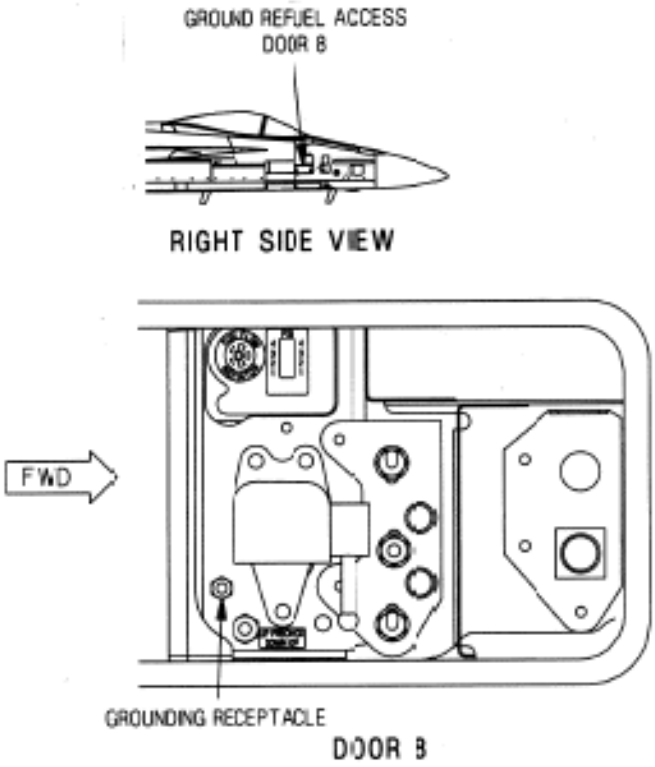
TABLE OF FUEL CAPACITIES

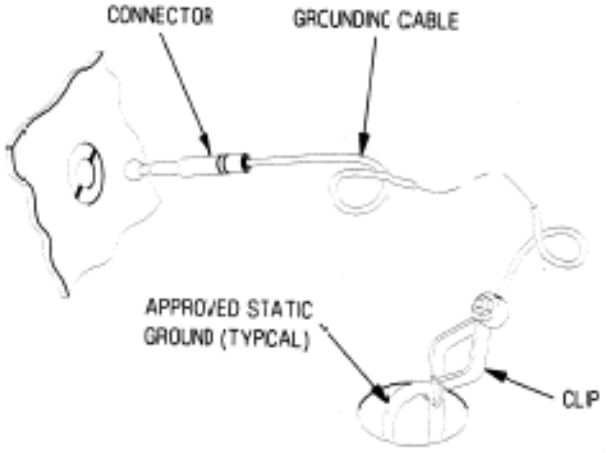
Tank			Gallons
Internal	Fuselage	Number 1 (E)	350
		Number 1 (F)	212
		Number 2	383
		Number 3	385
		Number 4	555
	Wing	Left	244
		Right	244
		Total Internal (E)	2162
		Total Internal (F)	2024
External		Drop Tank (s)	480 per tank

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

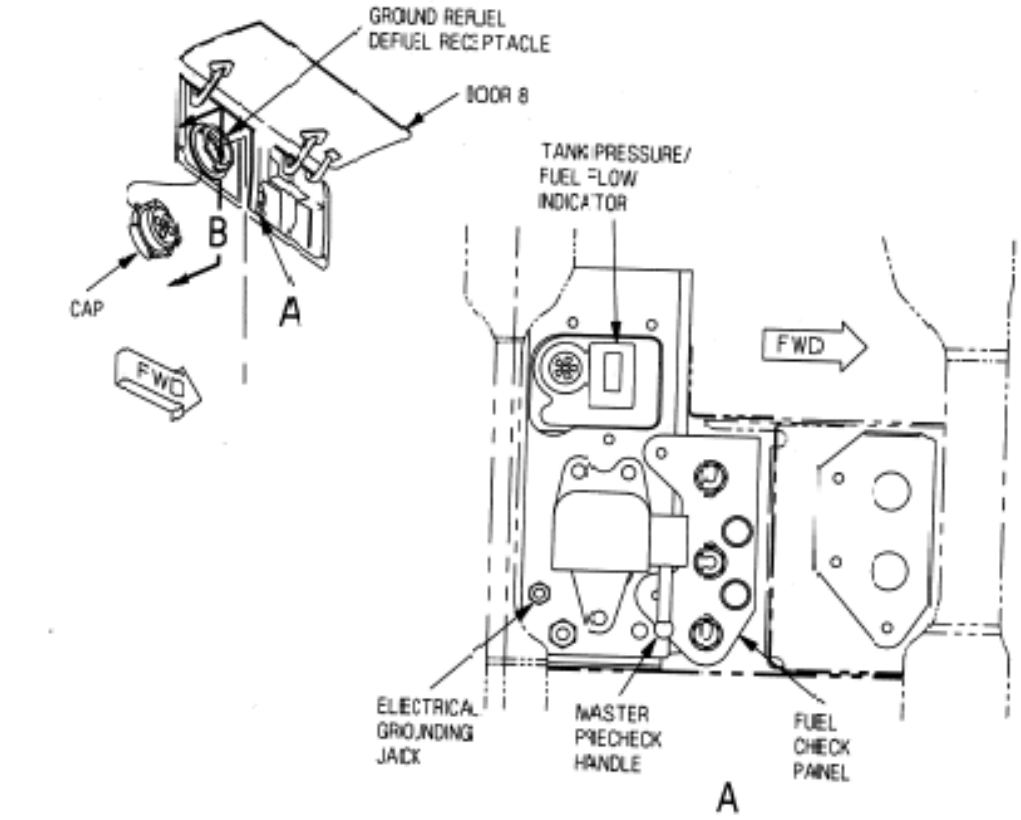
a) Electrical Grounding/Bonding Points

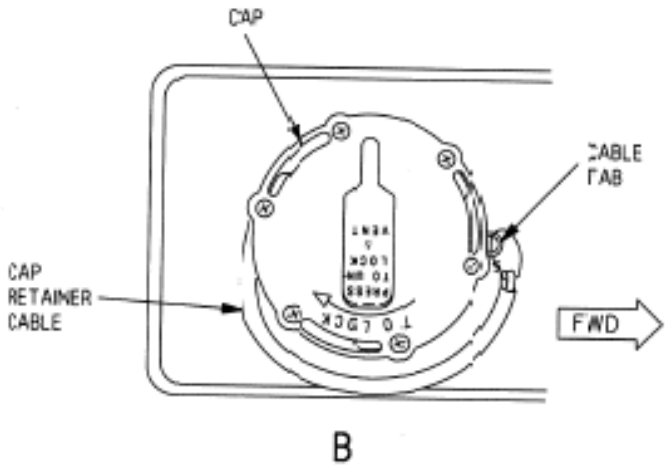




Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

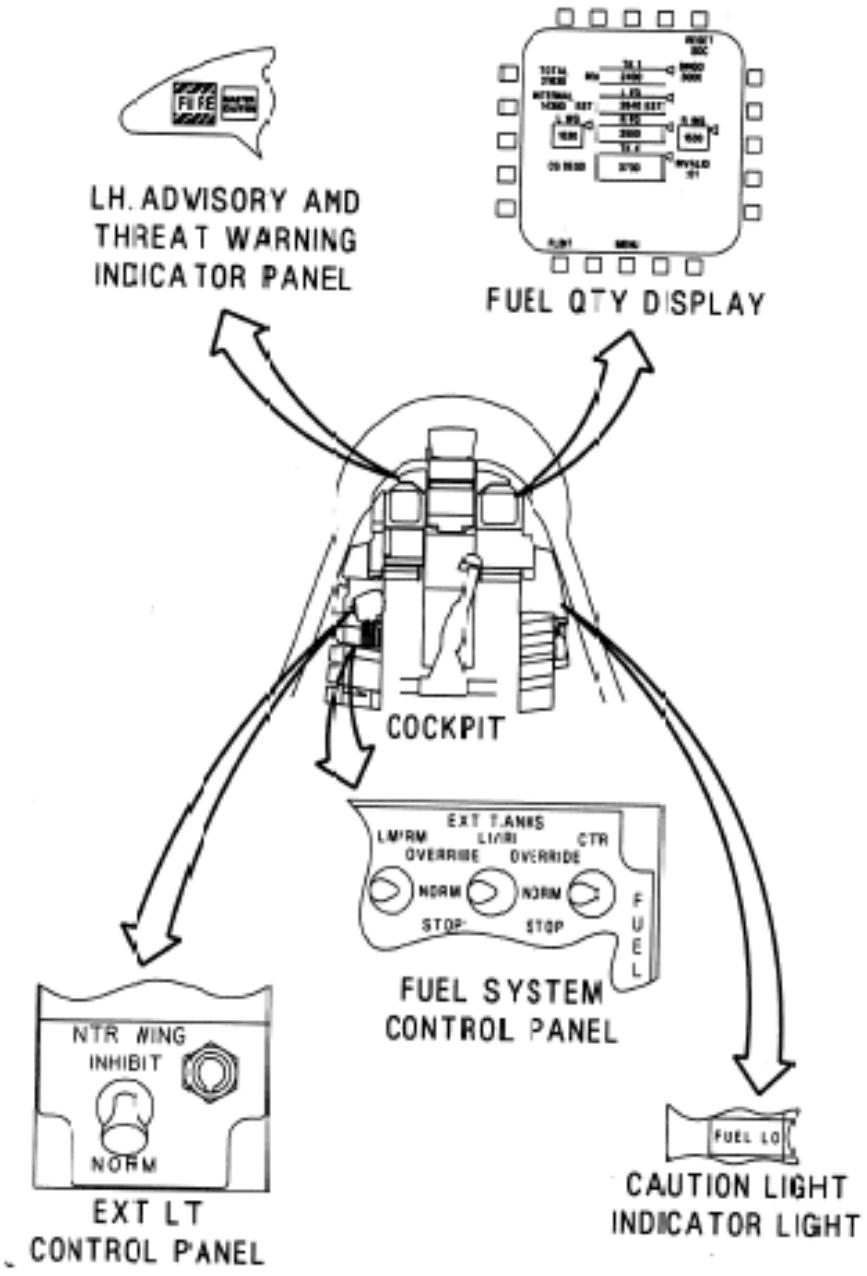
b) Ground Refueling Panel





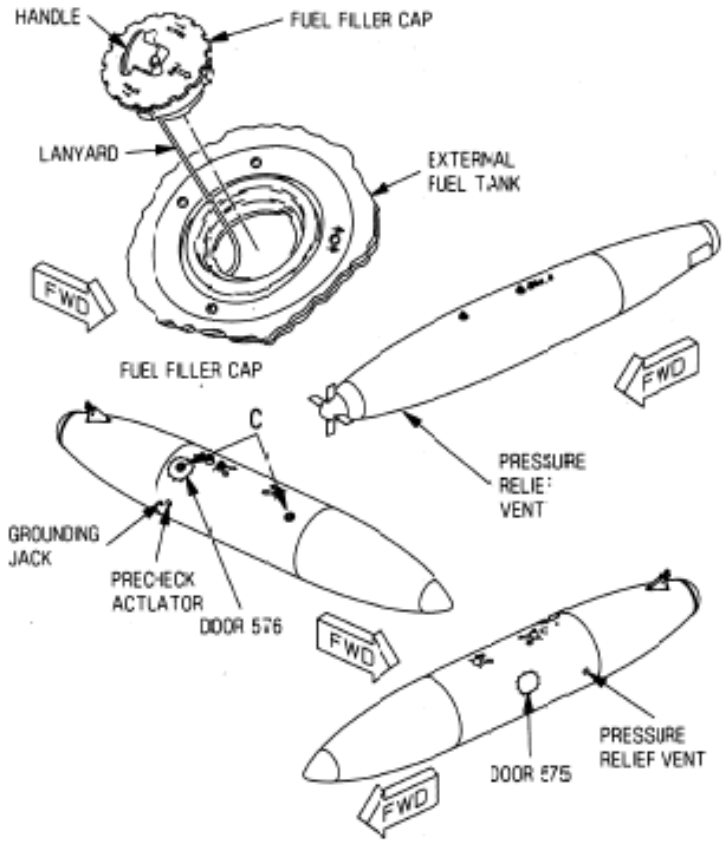
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Cockpit Controls for Refueling

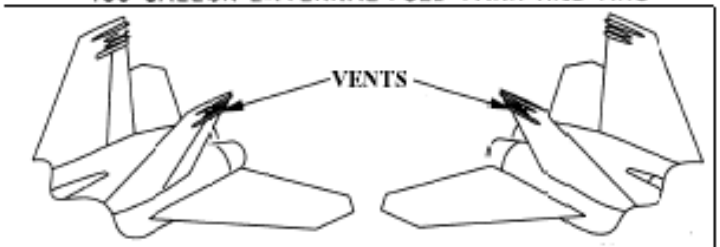


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Drop Tank Info and Fuel Vent Locations



480 GALLON EXTERNAL FUEL TANK AND ARS



LOCATION OF FUEL VENTS

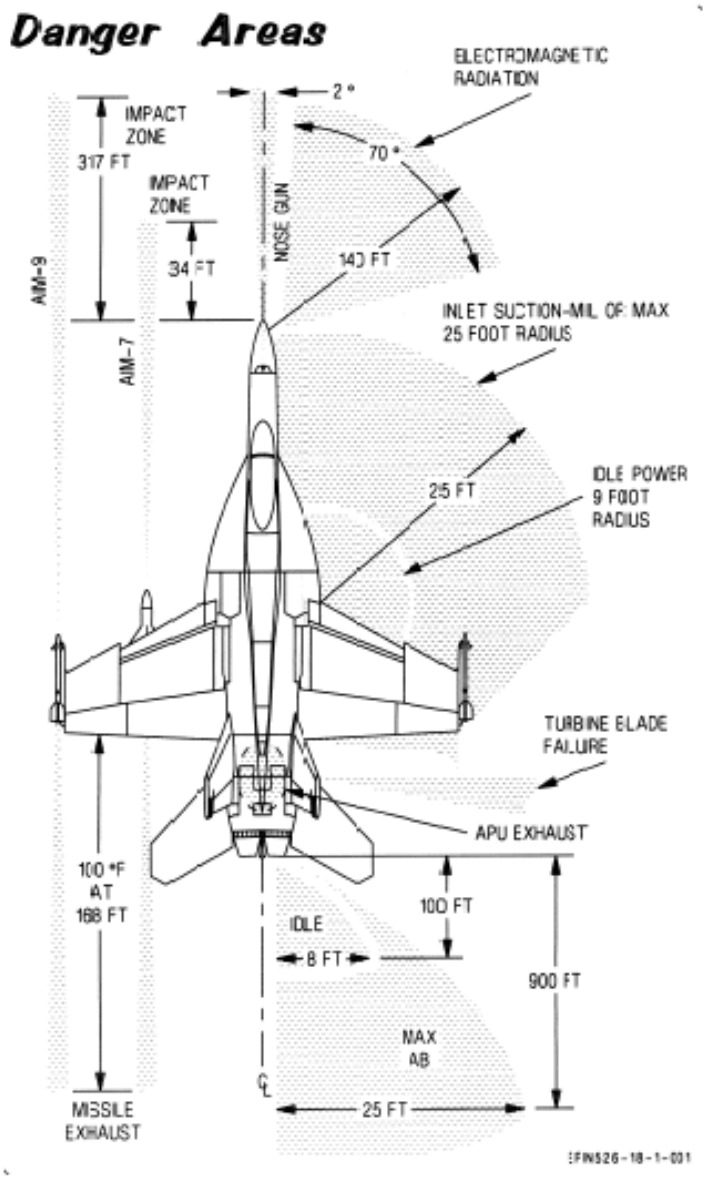


480 GALLON TANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Engine/APU Inlet/Exhaust/Blast Area. Electromagnetic Radiation Area, and Turbine Blade Failure Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

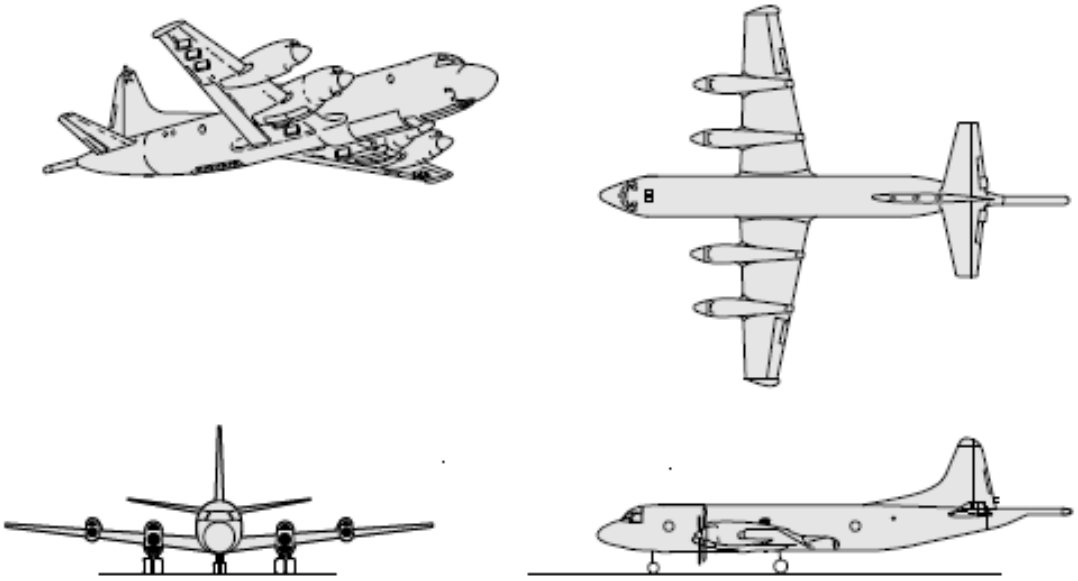
1. Refer to the F/A-18E/F NATOPS Servicing Checklist (A1-F18EA-NFM-600) for refueling procedures.

2. In addition to the procedures in the above referenced manual, the applicable basic refueling procedures contained in Chapters 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, **should** be followed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

P-3

The P-3 is a four-engine, low-wing aircraft designed for patrol, antisubmarine warfare, and fleet support. It is in the 135,000-pound gross weight class and powered by four T56-A-14 engines. Distinguishing features of the aircraft include surface and subsurface detection gear, including computer interfacing of the detection gear, the ordnance, and armament systems.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	99 ft 8in	Maximum Gross Weight — 139,760 lbs
Length	116 ft 10 in	Maximum Footprint — 250 psi
Height	34 ft 3 in	

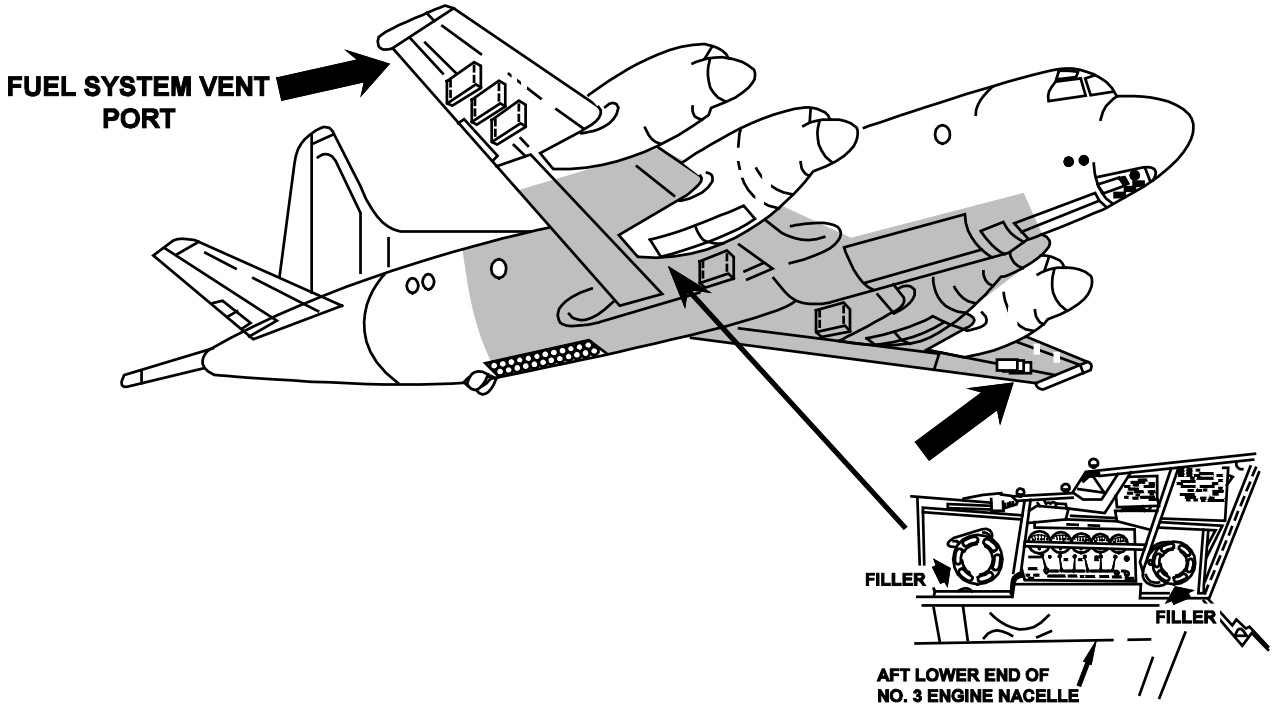
TABLE OF FUEL CAPACITIES

Tank		Gallons
Outboard	No. 1	1,606
	No. 4	1,606
Inboard	No. 2	1,671
	No. 3	1,671
	No. 5	2,646
	Total Internal	9,200

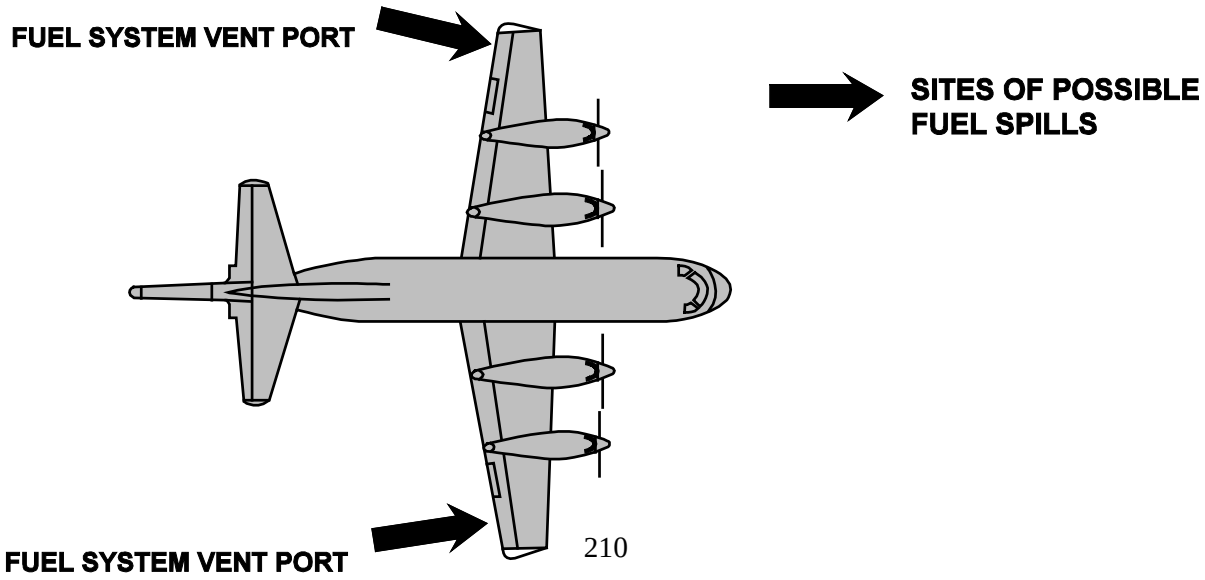
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Location of Fuel Vents, Dump Ports and Ground Refueling Panel

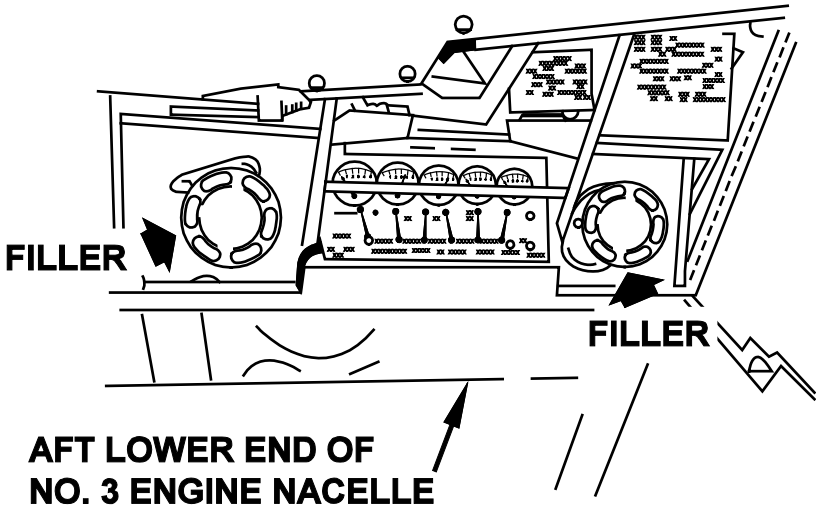


GROUND REFUELING PANEL AND RECEPTACLE



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

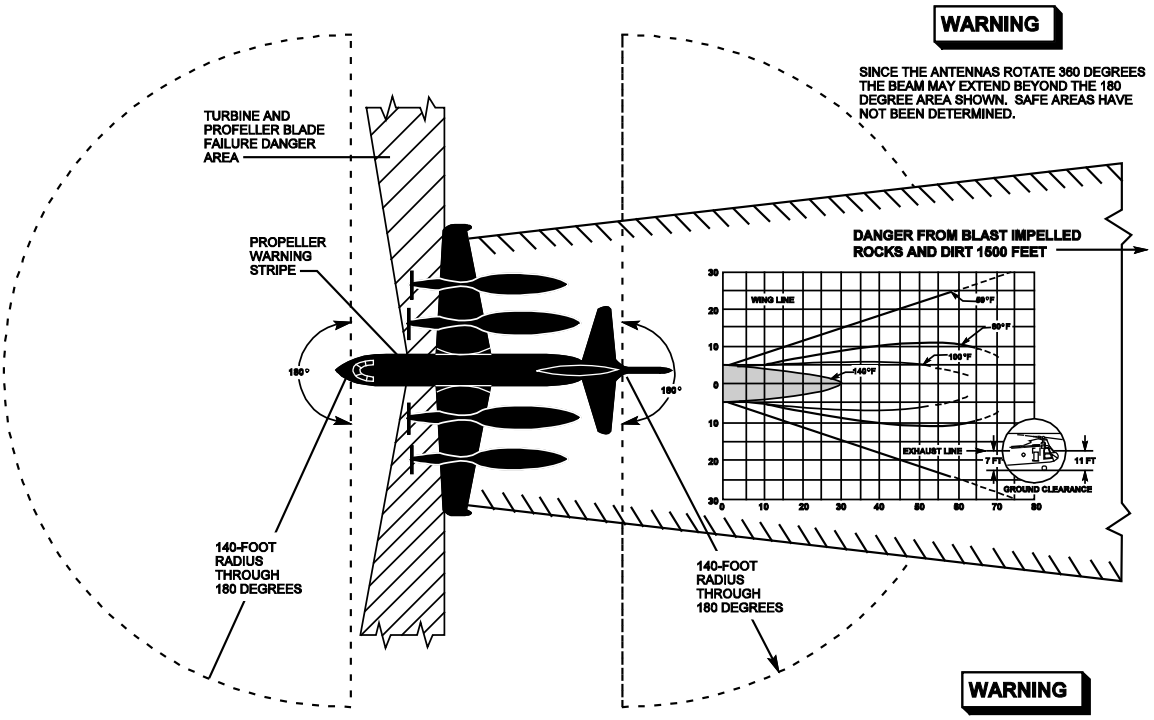
b) Ground Refueling Panel

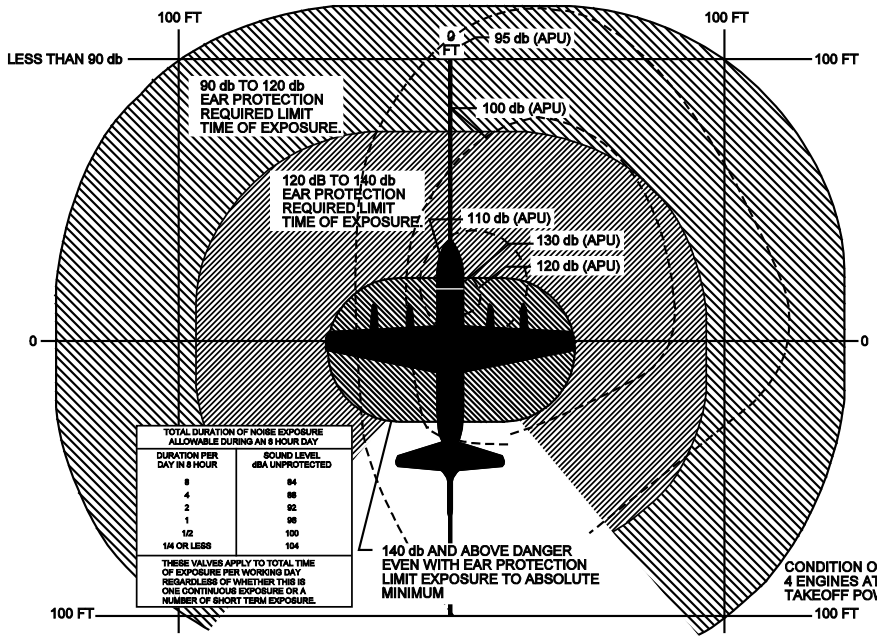


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Inlet/Exhaust/Blast Area, Prop Blade Failure Area





WARNING

- DO NOT APPLY HIGH VOLTAGE TO RADAR ANTENNA WITH ANT/DUMMY LOAD SWITCH IN ANT POSITION WHEN FUEL TRUCKS OR FUELING OPERATIONS ARE WITHIN A RADIUS OF 140 FEET FROM RADAR ANTENNA.
- PERSONNEL MUST BE WARNED TO STAY 75 FEET FROM A RADAR ANTENNA, WHEN HIGH VOLTAGE IS APPLIED AND ANTENNA IS SELECTED. (200 FEET FOR THE AN/APS 137 (V) 2)
- HERO ORDNANCE MUST REAMIN AT LEAST 140 FEET FROM THE ANTENNA WHEN HIGH VOLTAGE IS APPLIED AND ANTENNA IS SELECTED. (250 FEET FOR THE AN/APS 137 (V) 2)

NOTE

- TO ACCURATELY ASSESS THE EFFECTIVENESS OF HEARING CONSERVATION MEASURES ALL PERSONNEL EXPOSED TO NOISE HAZARDOUS AREA SHOULD RECEIVE ROUTINE MONITORING AUDIOMETRY IN ACCORDANCE WITH OPNAVINST 6260.6 SERIES.
- SOUND INTENSITY LEVELS OF 85 dBA AND HIGHER HAVE BEEN RECORDED INSIDE P-3 AIRCRAFT DURING FLIGHT. CONSIDERATION SHOULD BE GIVEN TO WEARING HEARING PROTECTION WHILE OFF HEADSET. AS UNPROTECTED EXPOSURE MAY RESULT IN TEMPORARY OR PERMANENT HEARING LOSS.
- A COMBINATION OF INSERT AND CIRCUMAURAL HEARING PROTECTION DEVICES SHOULD BE UTILIZED WHEN SOUND LEVELS EXCEED 100 dBA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the P-3 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Open Ground Refueling Panel access door.

2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.



Ensure that air is venting from "FUEL SYSTEM VENTS." If no venting is indicated, cease fueling operation immediately.

3. Locate the Precheck system valve test switches on the ground refueling panel. Hold left "VALV CONT" switch to "PRI CLOSE" and the right "VALV CONT" switch to "SEC CLOSE." Fuel flow into the aircraft should stop within a few seconds.

4. Release switches and fuel flow should resume.

5. Repeat the precheck process in step 3 above, this time holding the left "VALV CONT" switch to "PRI CLOSE."



If fuel flow does not stop step 3 or 5 above, discontinue hot refueling operation immediately. System failure **should** be investigated and resolved before hot refueling can be accomplished.

6. Release switches and refuel aircraft until fuel flow into the aircraft stops and the fuel quantity gauge on the ground refueling panel indicates the tanks are full.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

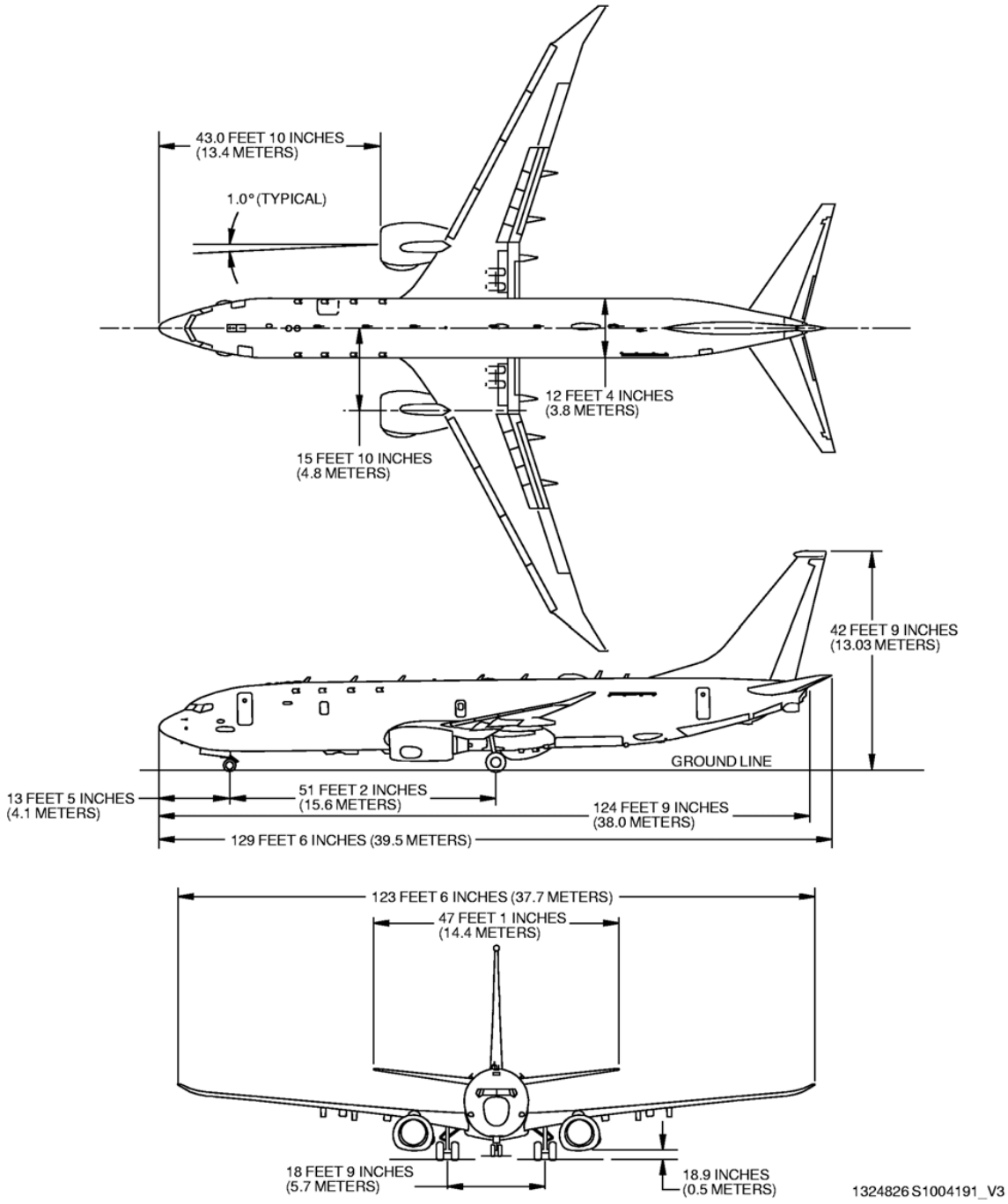
1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common "Fuel System. Vent Ports," which are located on each side of aircraft.

2. If any high level shut-off valves fail to operate correctly, fuel may spill from one of the "Fuel System Vent Ports." In addition, the fuel tanks, which are located in the Stubwings, may rupture and spill fuel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

P-8A

The P-8A Poseidon is a long-range anti-submarine warfare, anti-surface warfare, intelligence, surveillance and reconnaissance aircraft derived from a heavily modified Boeing 737NG airframe.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	42 ft, 9 in	Maximum Gross MTW 189,700 lbs	(lbs)
Length	129 ft, 6 in		
Wingspan Spread	123 ft, 6 in	Maximum Footprint -	(psi)

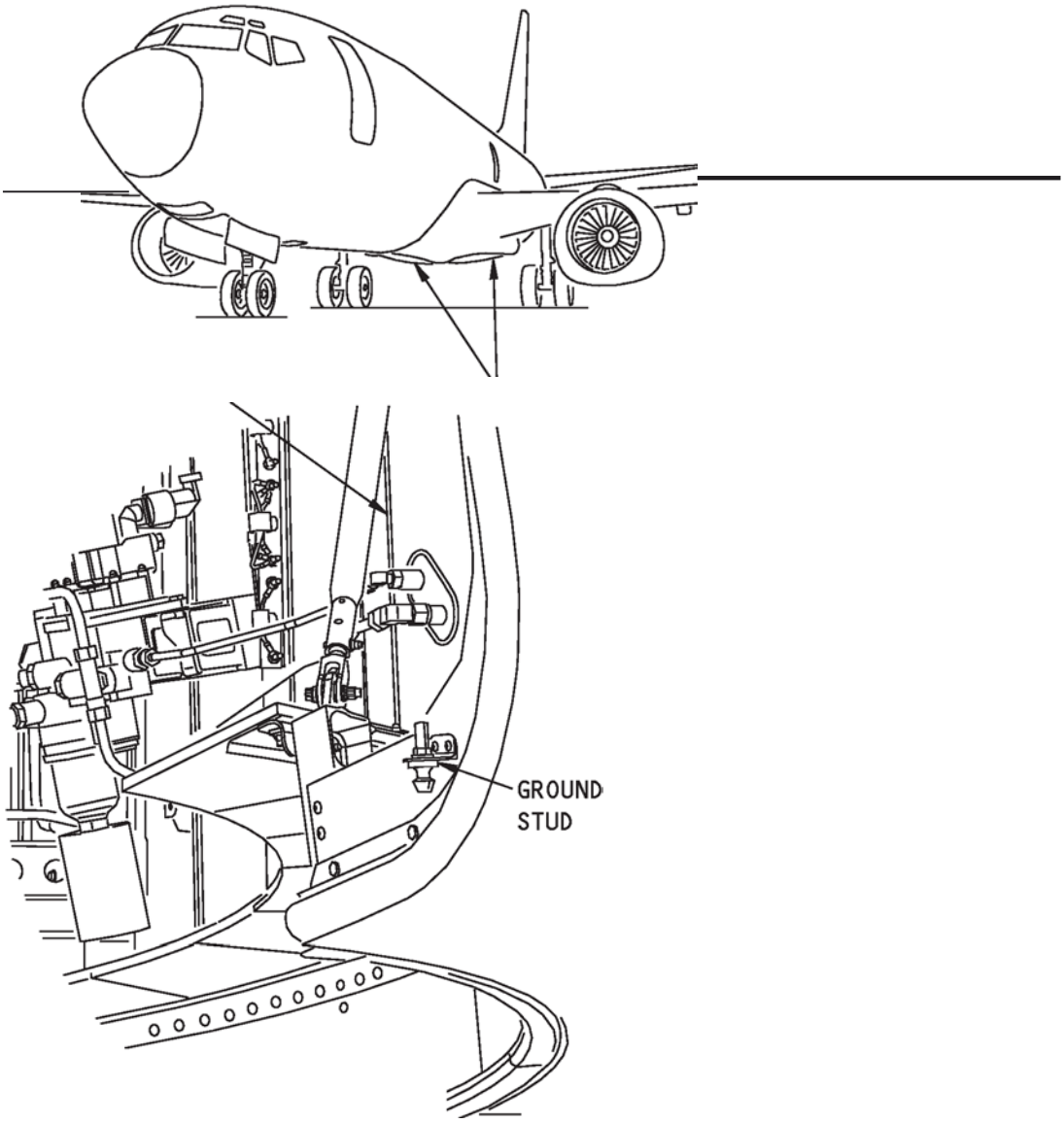
FUEL CAPACITIES

Tank		Gallons	Pounds (Approx) JP-5 @ 6.7lbs/gal
Main and Center wing tanks	M1 (Left)	1,226	8,214
	M2 (Right)	1,226	8,214
	Center	4,238	28,395
Total Main and Center		6,690	44,823
Auxiliary (Ground Pitch)	FWD AUX	1,499	10,0843
	AFT AUX	2,477	16,596
AUX Total		3,976	26,639
Aircraft Total	Wing + AUX	10,666	71,462

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

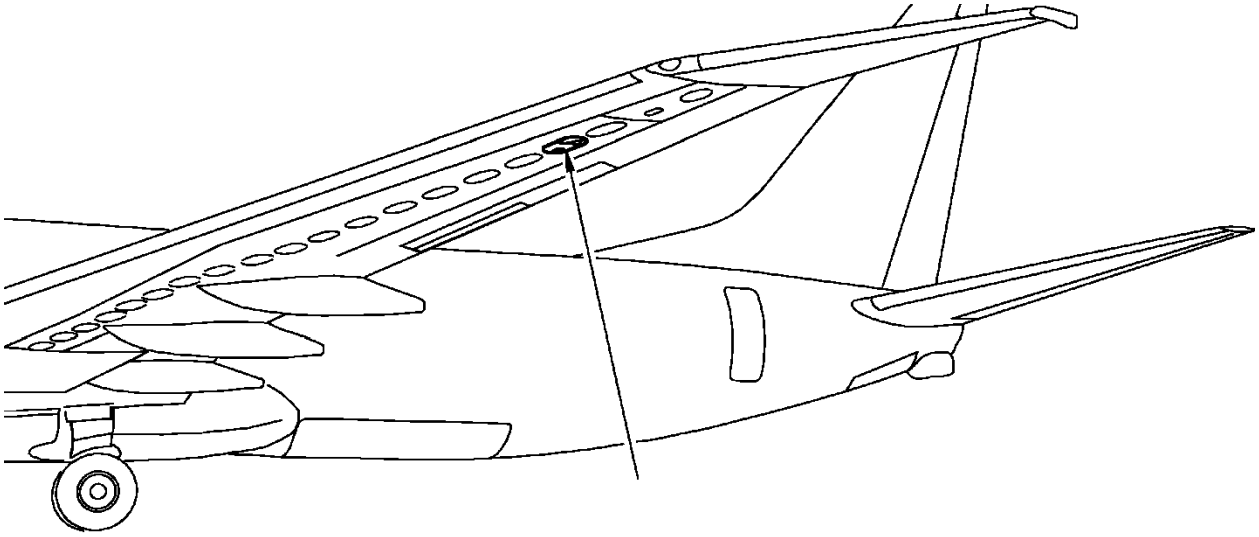
a) Electrical Grounding/Bonding Points



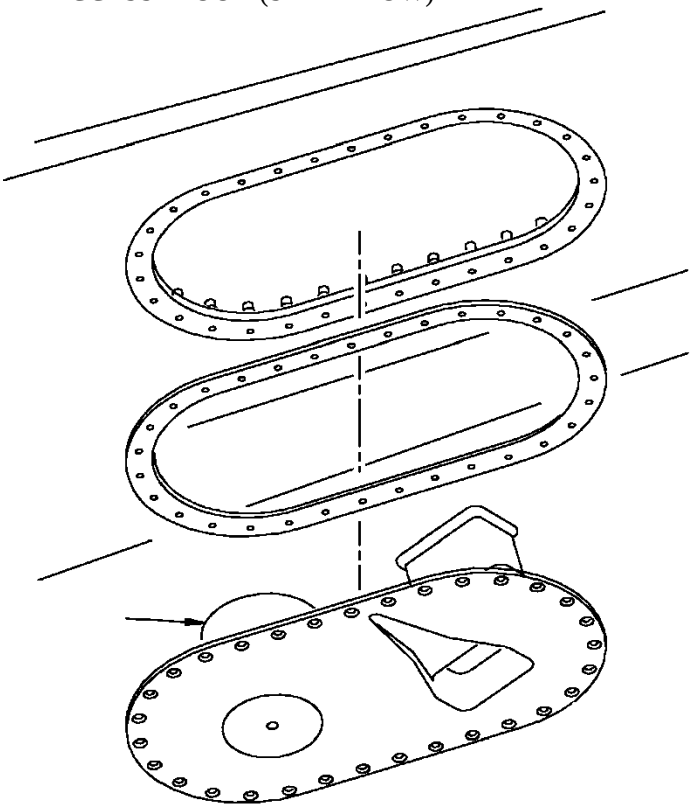
Main Wheel Well Ground Stud Location

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Location of Fuel Vents and Low Point Drains



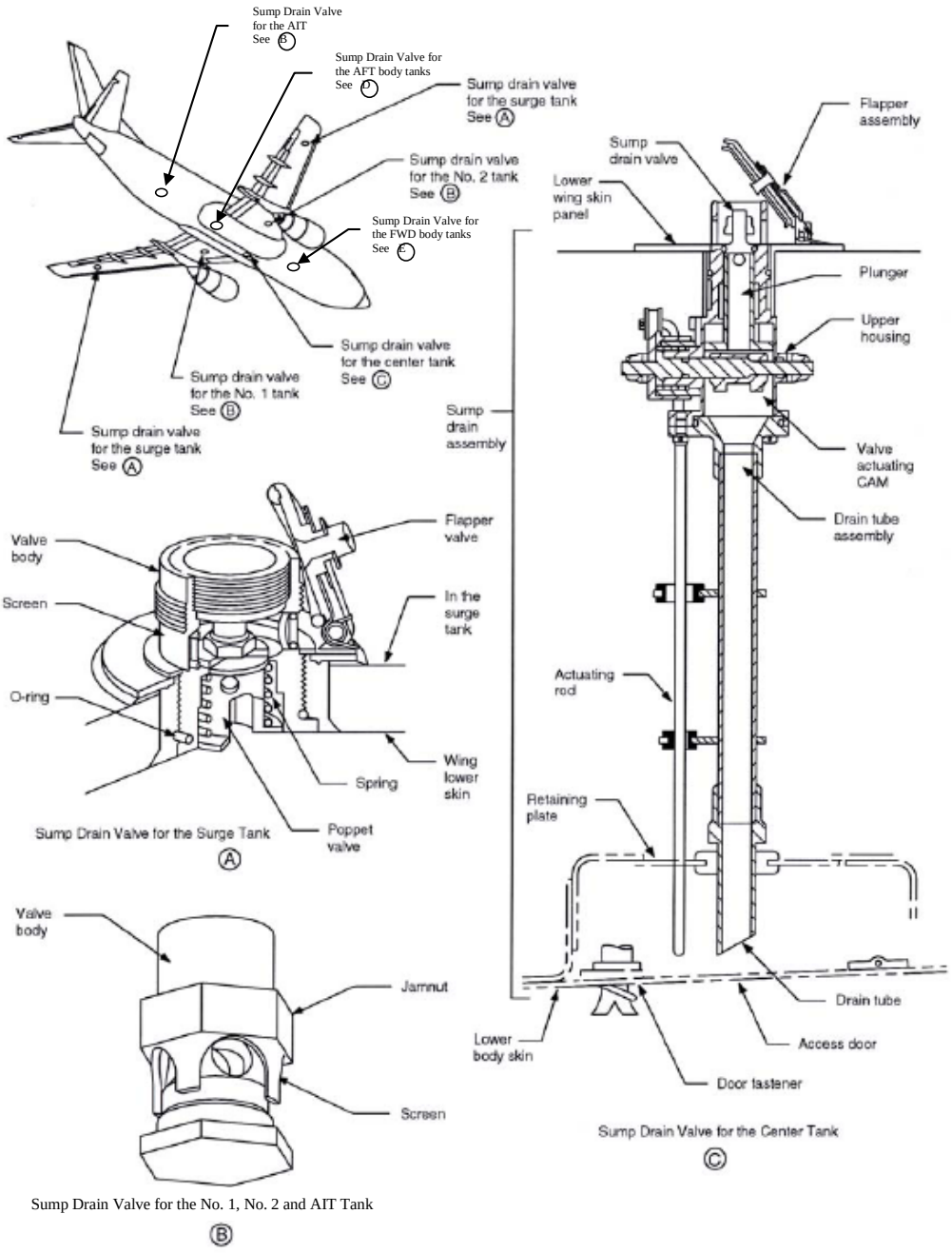
LEFT WING SURGE TANK ACCESS DOOR (SEE BELOW)



SURGE TANK ACCESS DOOR

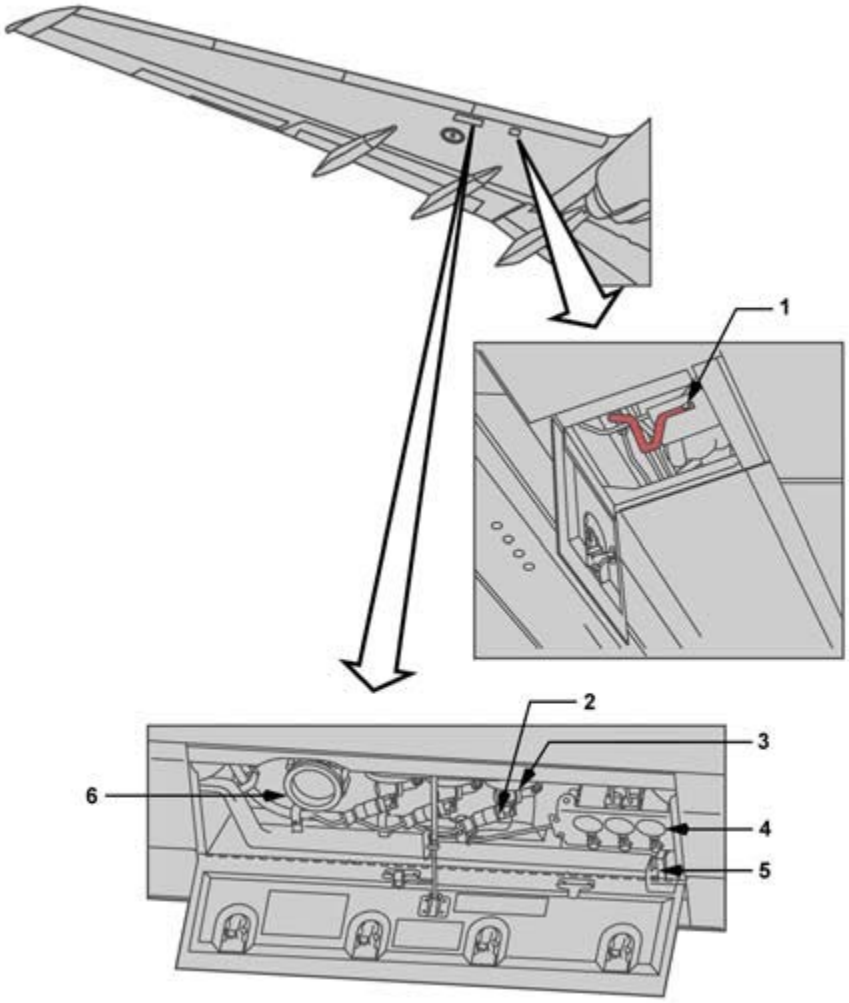
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Location of Sump Drains



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Ground Fueling Panel and Locations of Receptacles



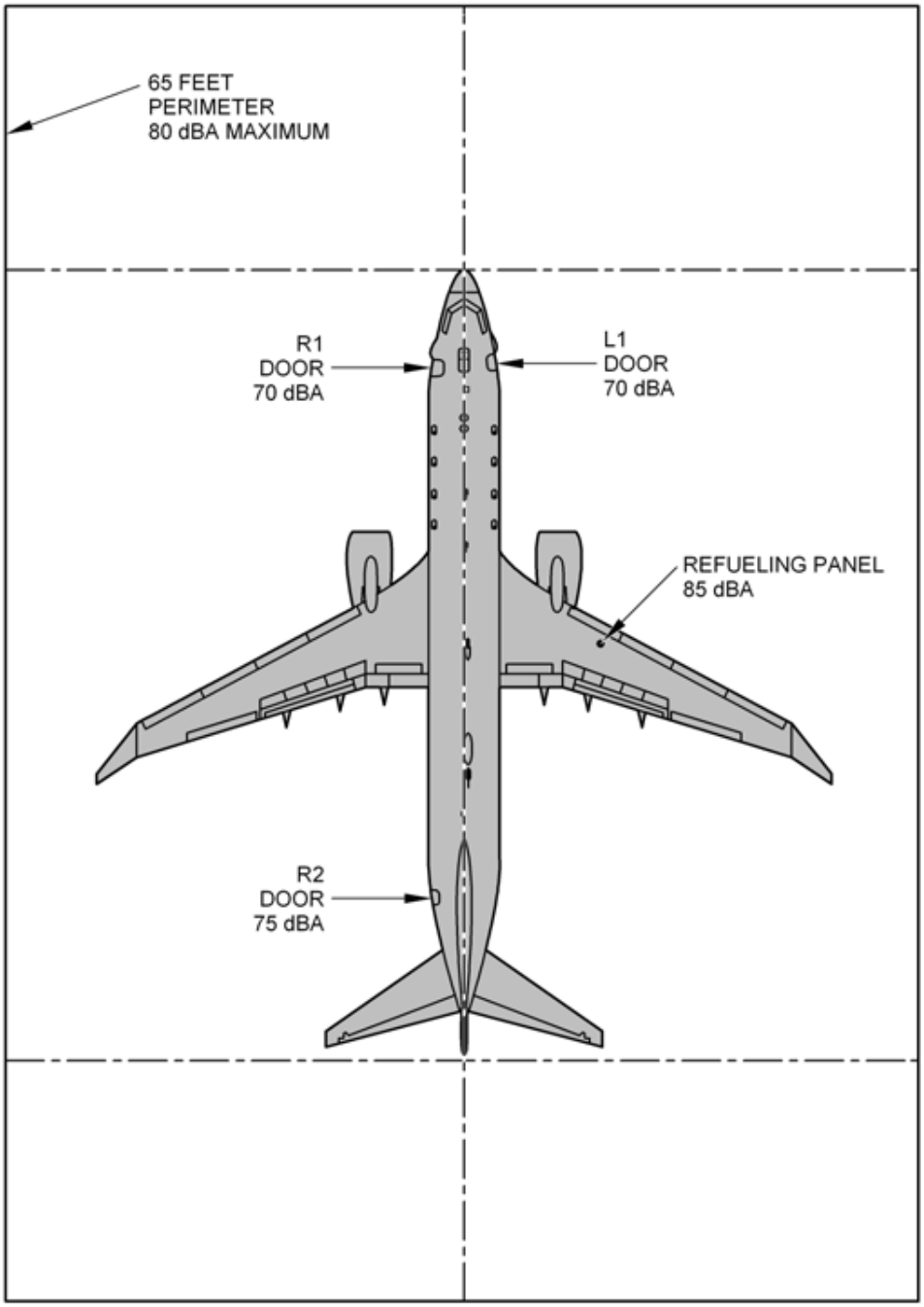
RIGHT WING LEADING EDGE

1 Manual Defueling Valve; 2 Manual Override Handle; 3 Fueling Valves; 4 Ground Fueling Panel; 5 Refueling Power Control Relay; 6 Fueling Receptacle
Note: The height of the Ground fueling Panel is 9.4 feet above the ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) APU Sound Levels

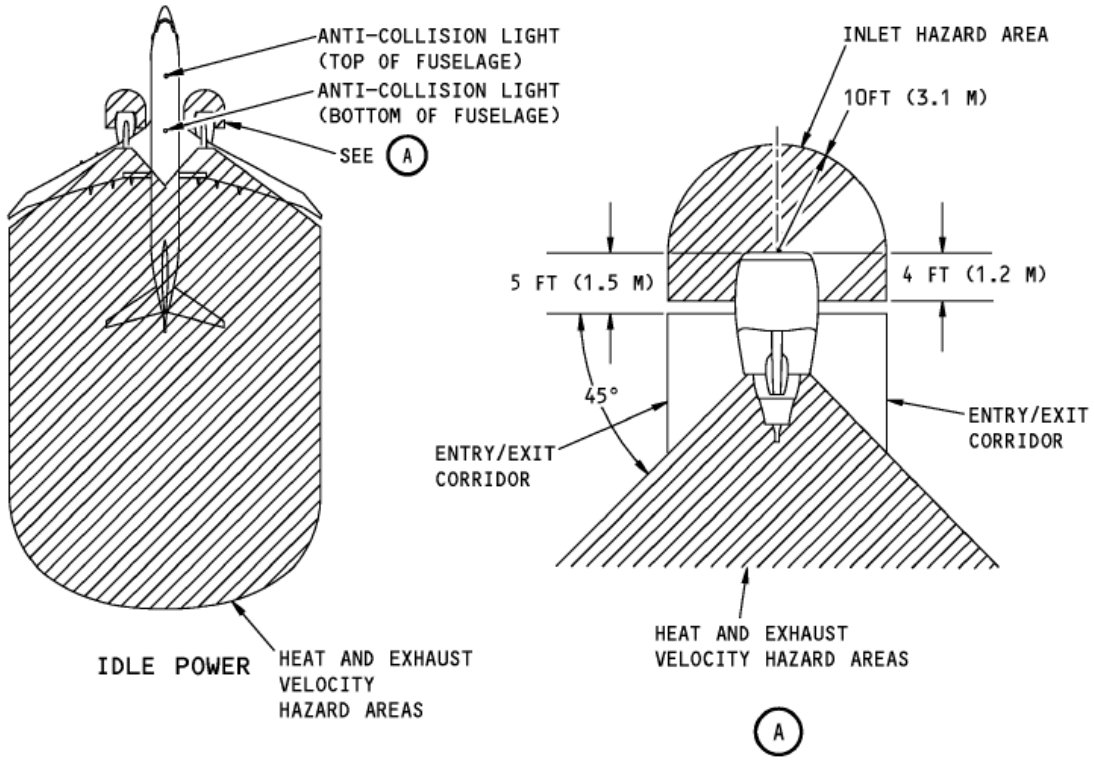


Note

APU, BOTH ECS PACKS, AND ALL COOLING FANS IN OPERATION.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Engine/APU Inlet/Exhaust/Blast Area



Engine Inlet Hazard Areas — Idle/Above Idle Power, Forward Thrust (Figure 3-13 from A1-P8AAA-NFM-000)

c) Separation Distances During Fueling Operations

Separation Distances During Fueling Operations	
EQUIPMENT OR IGNITION SOURCES	**REFUEL SEPARATION DISTANCE[1]**
Adjacent airplane engine or APU	50 feet
Fuel service equipment - measured from engine or exhaust system	10 feet from fuel vents
Ground power units	20 feet
Airplane servicing equipment - measured from the engine or exhaust system	10 feet
Electrical equipment that is likely to cause arcs or sparks	50 feet
Photographic equipment	10 feet
Battery powered equipment	10 feet from fuel servicing equipment or fuel

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Open flames, heat sources, lighted smoking material, and any other potential ignition sources	50 feet
Electrical transmitting equipment of more than 100 watts (radio or radar)	200 feet
Electrical transmitting equipment of 25 to 100 watts (radio or radar)	50 feet
Electrical transmitting equipment of less than 25 watts[3]	10 feet

NOTES:

1. The distance is measured from a point on the ground directly below the fuel vents or from fueling equipment.

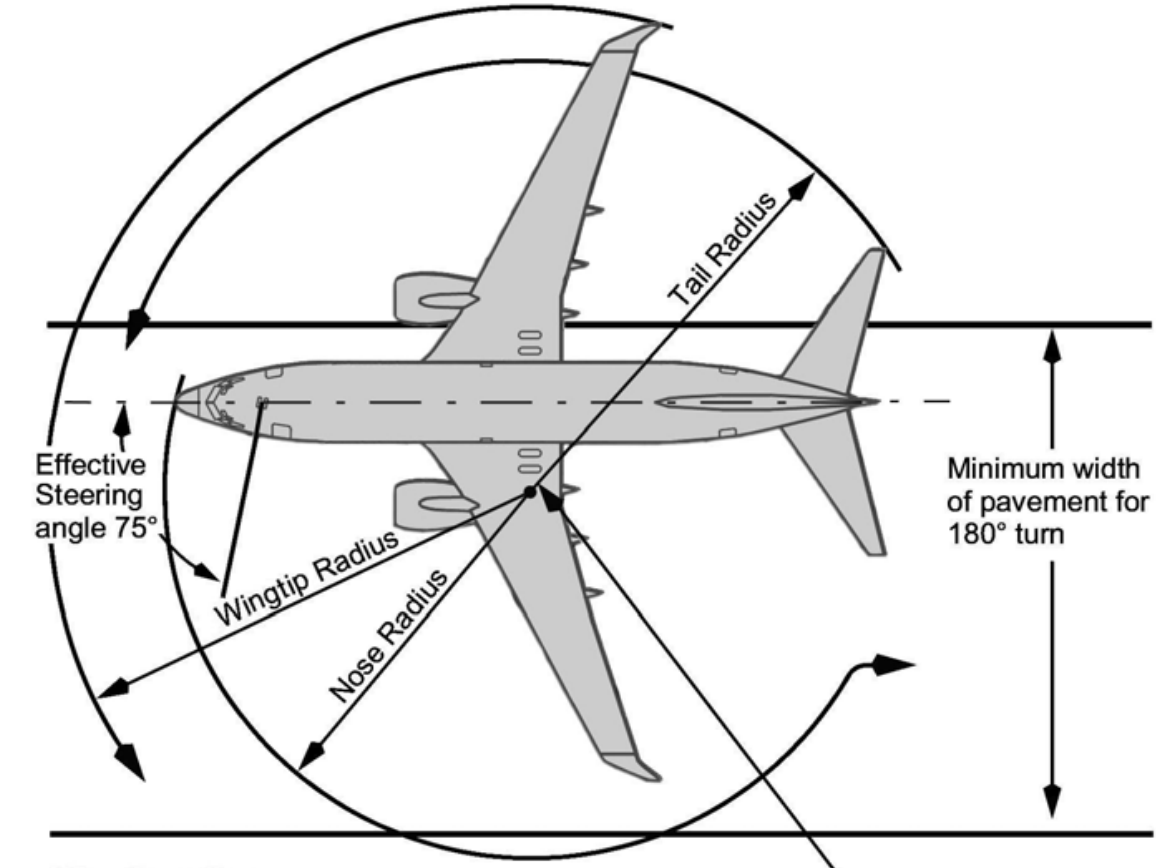
2. Does not apply to battery powered equipment approved for use in Class I Division 1 hazardous locations.

3. This category includes mobile phones, pagers, two-way radios, and similar wireless communication equipment. There are low power, intrinsically safe communication systems that are approved for use in hazardous locations. These devices can be used safely in areas that contain fuel vapor (UL 913 or equivalent standards).

(Figure 3-7 from A1-P8AAA-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Turning Radius



CAUTION

Do not attempt to make a turn away from an obstacle within 15 feet of the wingtip or within 24 feet of the nose.

Note

Minimum width of pavement for 180° turn: 79 feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

Stop the refuel operation if any conditions change that could cause an unsafe condition for personnel or equipment. During a refuel operation, continuously monitor the airplane for fuel leaks and fuel spills at the wingtip.

Do not put ground equipment below the fuel system vents at the wingtips. The fuel tanks are vented through the wingtips vents. An explosive mixture of fuel vapor can exist at these locations.

If a fuel spill occurs:
1. Stop the fueling operation.
2. APU switch OFF.
3. Notify fire department and maintenance of fuel spill.
4. Do not start the fueling operation or start the APU again until the fire department or the personnel in
charge have given approval.

• Stop fueling operations during atmospheric electrical activity. Disconnect and remove any external headsets. Do not touch any electrical connections. Lightning strikes may cause injuries to personnel. A fire or explosion may occur during fueling operations.

• Any Radio Frequency (RF) transmission is a potential source of ignition. Use of transmitting equipment during fueling operations should be avoided. Injury to personnel or fire may occur.
•
Do not key the HF radio while the airplane is being fueled. Injury to personnel or fire may occur.

• Do not apply or remove ground power while the airplane is being fueled. Injury to personnel or fire may occur.

• Do not fill or change oxygen bottles while the airplane is being fueled. Injury to personnel or fire may occur.

• Do not start fueling if a fire or engine overheat warning CMS message is shown or if any part of the landing gear is unusually hot. Injury to personnel or damage to equipment may occur.

• Obey separation distances during fueling operations shown in *c) Separation Distances During Fueling Operations*. Injury to personnel or damage to equipment may occur.

• Ensure the fuel source is an approved fuel grade per the platform NATOPS or maintenance manuals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

• Do not operate affected hydraulic pumps on the ground when main tanks No. 1 or No. 2 contain less than 1,675 pounds of fuel.

• Visually ensure the wingtip vents are not blocked before fueling. Fuel spills or damage to the airplane can occur.

Maintenance should check to make sure the wing pressure relief valves (adjacent to each wing tip vent outlet) are in the closed position.

• Ensure all ground equipment is clear of the airplane as fuel weight will compress the landing gear shock struts and lower the airplane.

The following is a list of APU operating procedures and limits during fueling operations:

• If there is a protective automatic shutdown of the APU or a failure to start condition, do not try to start the APU again during the fueling operation. Injury to personnel or damage to equipment may
occur.

• Ensure fueling vehicles are clear of APU exhaust stream. Injury to personnel or damage to equipment may occur.

1. The APU may be started during fueling if the start is an initial start or a restart after normal shutdown.

2. The APU may be shut down normally during the fueling operation.

3. If an APU fire occurs during fueling:
a. Stop the fueling operation.
b. Perform an emergency shutdown of the APU if it does not shut down automatically.
c. Notify personnel on board the airplane, airport fire services, and maintenance.
Ensure the landing gear wheel chocks do not touch the tires. The wheel chocks can wedge against the tires after fueling.

Minimum defuel pressure: -5psig

SPECIAL NOTES – AIRCRAFT FUEL SYSTEM

Main tanks No. 1 and No. 2 should normally be serviced equally until full. Main tanks No. 1 and No. 2 need to be scheduled to be full if the center tank contains more than 1,000 pounds of fuel. The main tanks and center tank need to be scheduled to be full if the auxiliary tanks are to be fueled to any level.

Hot refueling with a main engine operating is not authorized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

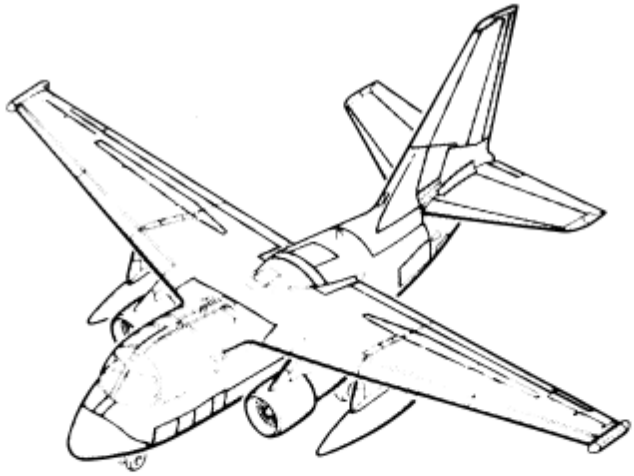
To help prevent tip up, AFT AUX tank quantity is limited to 6200 lbs unless the FWD AUX tank contains at least 3170 or the Center tank contains at least 7400 lbs.

A ground attitude of 1.148 nose down pitch and 0.08 roll permits you to put the maximum quantity of fuel in the tanks.

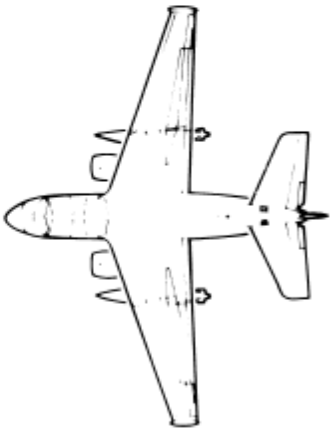
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

S-3

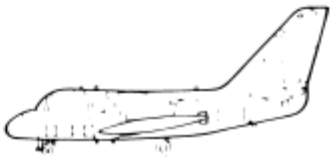
The S-3B is a high-wing, jet-powered, twin-engine, carrier-based sea control aircraft equipped with folding wings, folding vertical fin, a launch bar for catapult takeoffs, and a tailhook. It carries surface search equipment integrated target acquisition and sensor coordinating systems that can collect, process, interpret, and store surface warfare sensor data. It is powered by two General Electric TF34 high-bypass turbofan engines.



S-3B







Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread Folded	 68 ft 8 in 29 ft 6 in	Maximum Gross Weight — 50,000 lbs
Length	53 ft 4 in	Maximum Footprint — 288 psi
Height	22 ft 9 in	

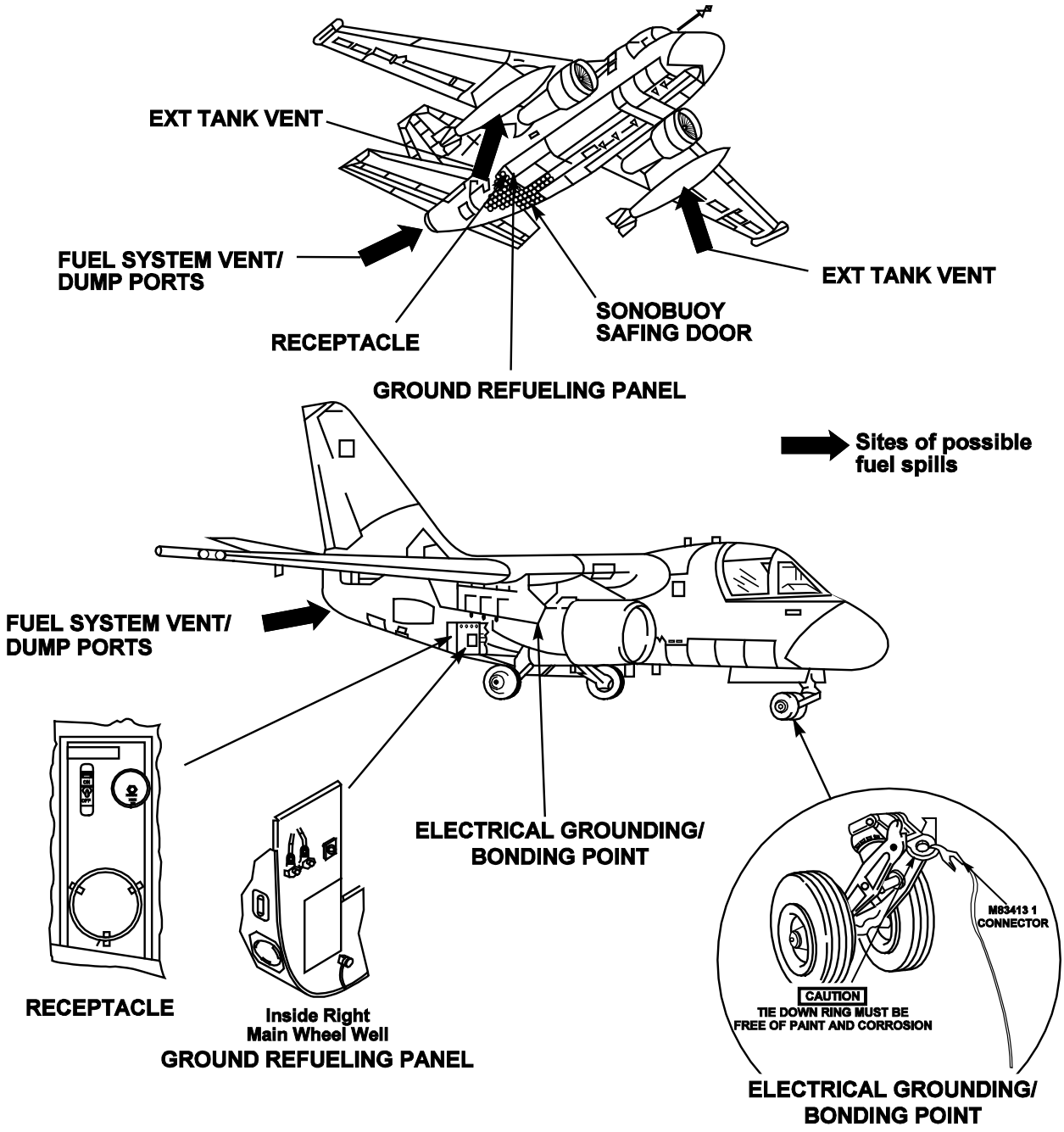
TABLE OF FUEL CAPACITIES

Tank		Gallons
Internal	Left Feed	176.5
	Right Feed	176.5
	Left Transfer	790.0
	Right Transfer	790.0
Total Internal		1,933.0
External	Left Pylon	265.0
	Right Pylon	265.0
Total	Internal + External	2,463.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

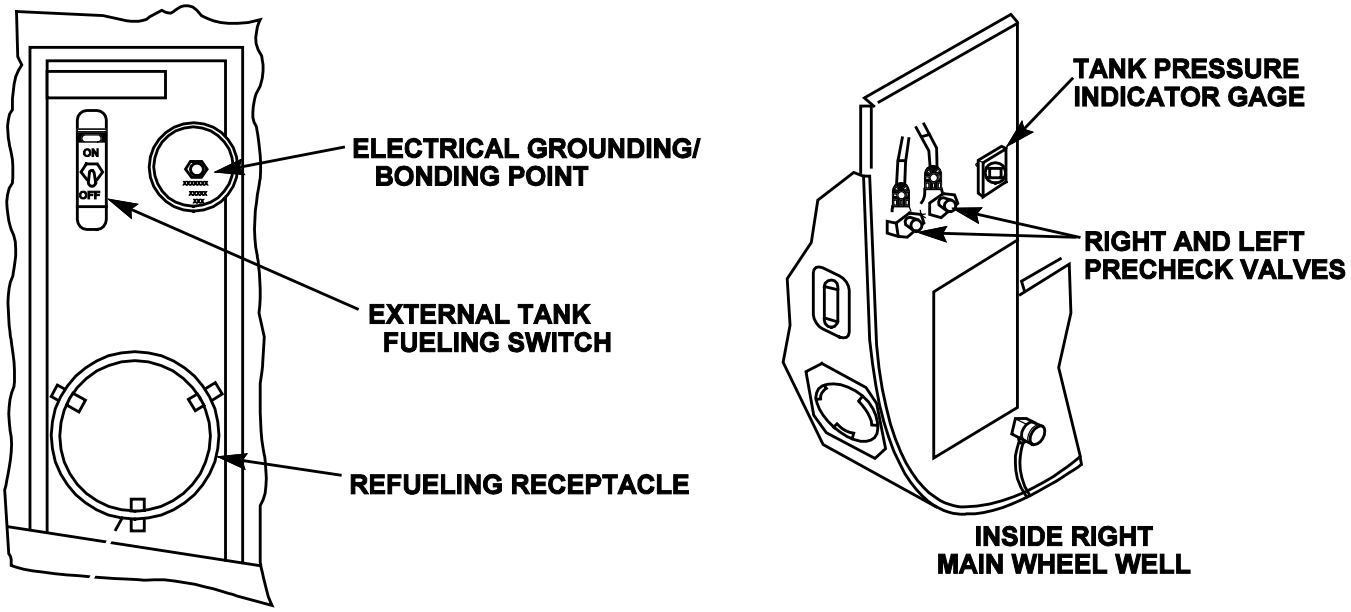
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Point, Location of Fuel Vents, Dump Ports, and Ground Refueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

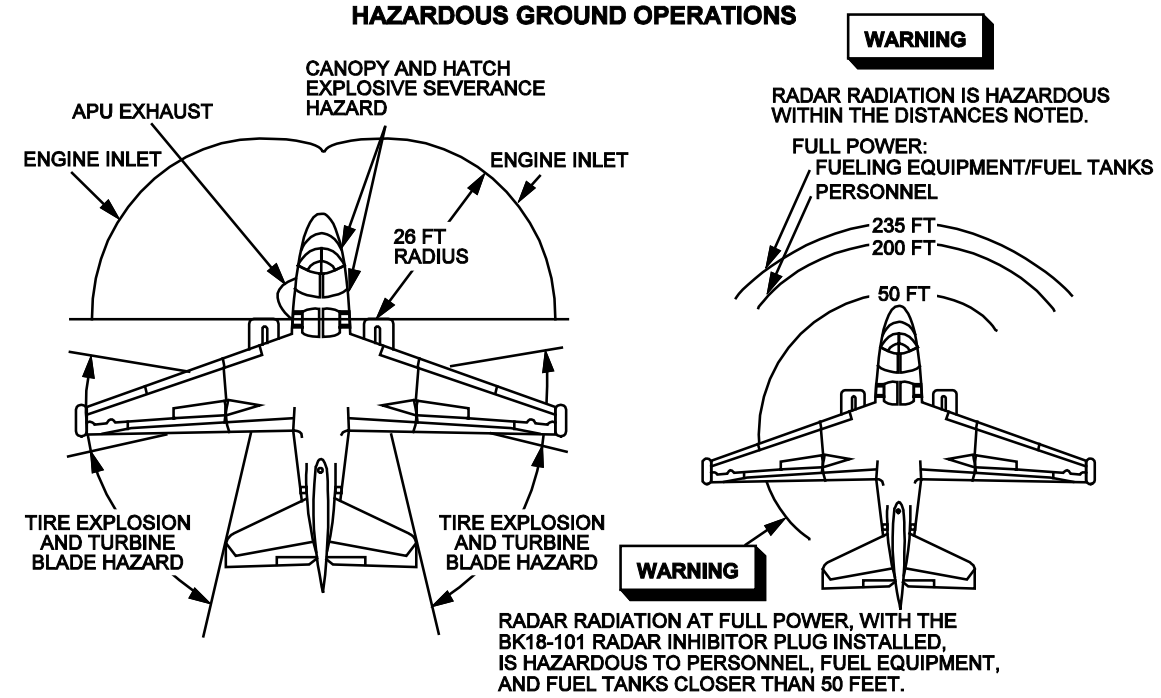
b) Ground Refueling Panel

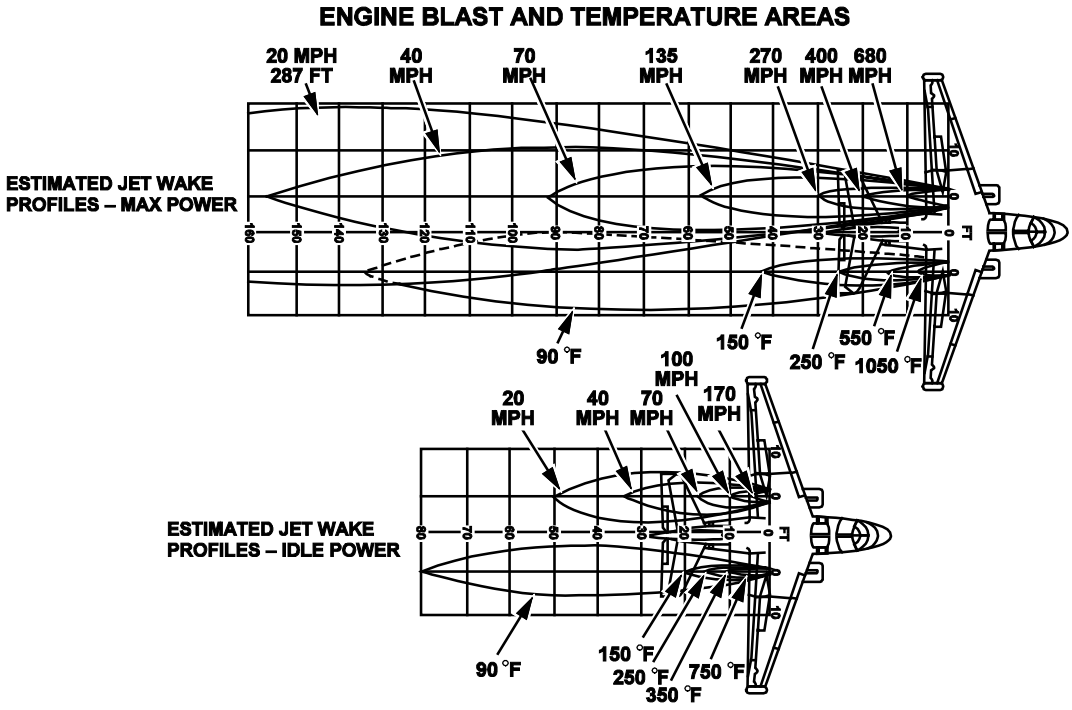


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine/APU Inlet/Exhaust Blast Area, Radar Radiation Area, and Turbine/Tire Failure Area





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the S-3 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.



Sonobouy Safing Door **should** be opened to ensure that the jettison system has been disarmed.

1. Attach refueling nozzle to aircraft and place nozzle flow control handle in the open position.

2. Initiate fuel flow and observe "TANK PRESSURE INDICATOR GAUGE" on ground refueling panel (inside the



right main landing gear wheel).

Immediately stop fuel flow if indicator moves into the red band labeled "STOP REFUELING."

3. Exercise precheck system by twisting and holding both "PRECHECK VALVES" on the ground refueling



panel in the "OPEN" position. Fuel flow into the aircraft should stop within 20 seconds.

If fuel flow does not stop, discontinue refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

4. "PRECHECK VALVES" are spring-loaded to the "CLOSED" position. Release them and continue fueling until aircraft tanks are filled and fuel flow automatically stops.

5. If external tanks are installed, they can be refueled by turning the "EXTERNAL TANK FUELING SWITCH" to the "ON" position. This switch should be placed in the "ON" position only after the high level shutoff valves in the internal tanks have been tested using the "PRECHECK VALVES" in steps 3 and 4 above. Return this switch to the "OFF" position at the conclusion of the refueling operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the internal tanks exits the aircraft through the common “Fuel System Vent Port.”

2. If any high level shutoff valves fail to operate correctly, fuel may spill from the “Fuel System Vent Port” and/or tank(s) may rupture.

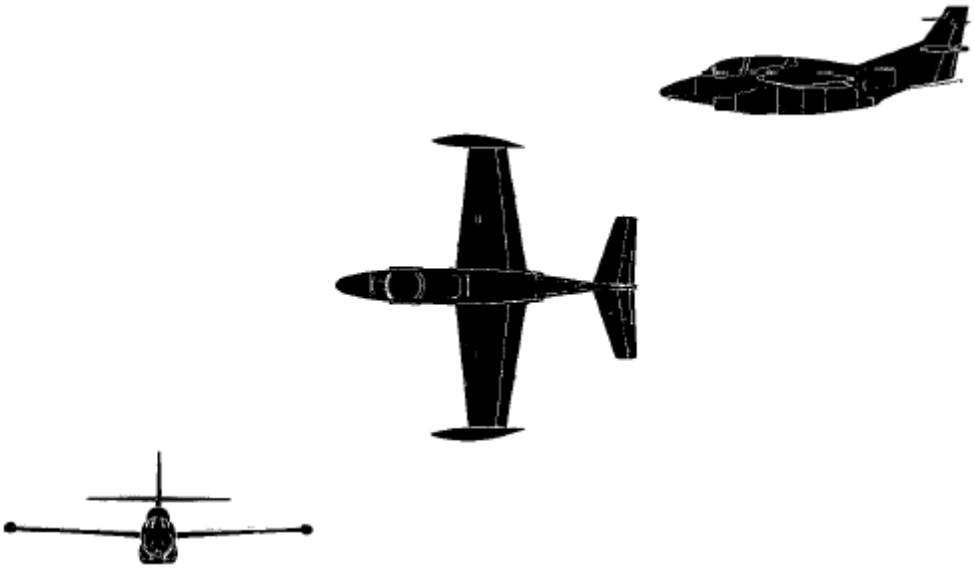
3. A malfunction within the external fuel tanks may cause fuel to spill from the bottom center of the external tank (pressure relief vent).

4. The Precheck system does not exercise the high level shutoff valves in the external tanks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-2

The T-2C is a two-place, subsonic trainer, powered by two axial flow turbojet engines. The T-2C uses the J85-GE-4 engine. The aircraft is designed for land-or-carrier based operations.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	38 ft 1 in	Maximum Gross Weight — 14,000 lbs
Length	38 ft 4 in	Maximum Footprint — 150 psi
Height	14 ft 10 in	

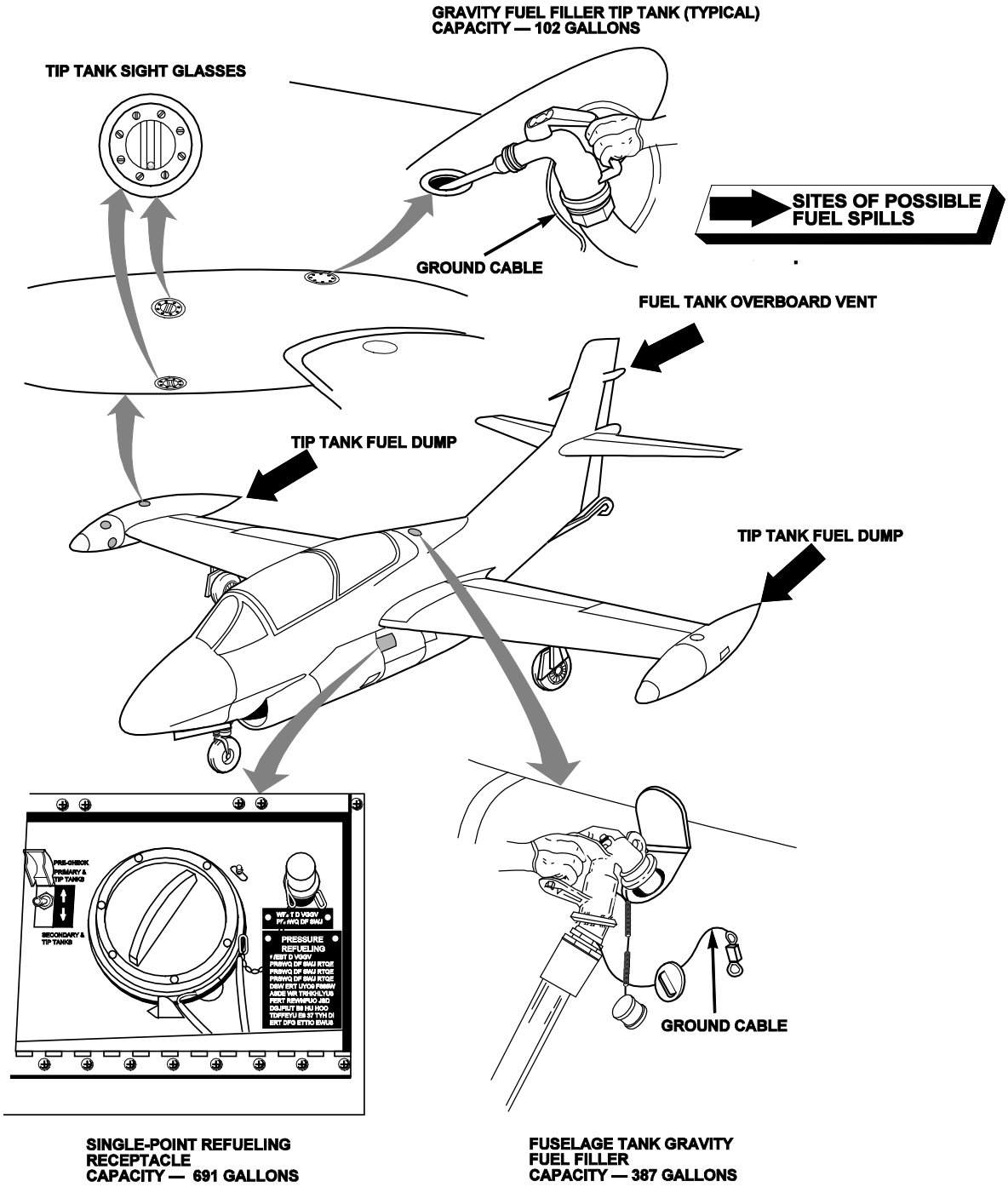
TABLE OF FUEL CAPACITIES

Tank	Gallons
Fuselage	387
Wing Leading Edge	50 (Each)
Tip	102 (Each)
Total	691

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

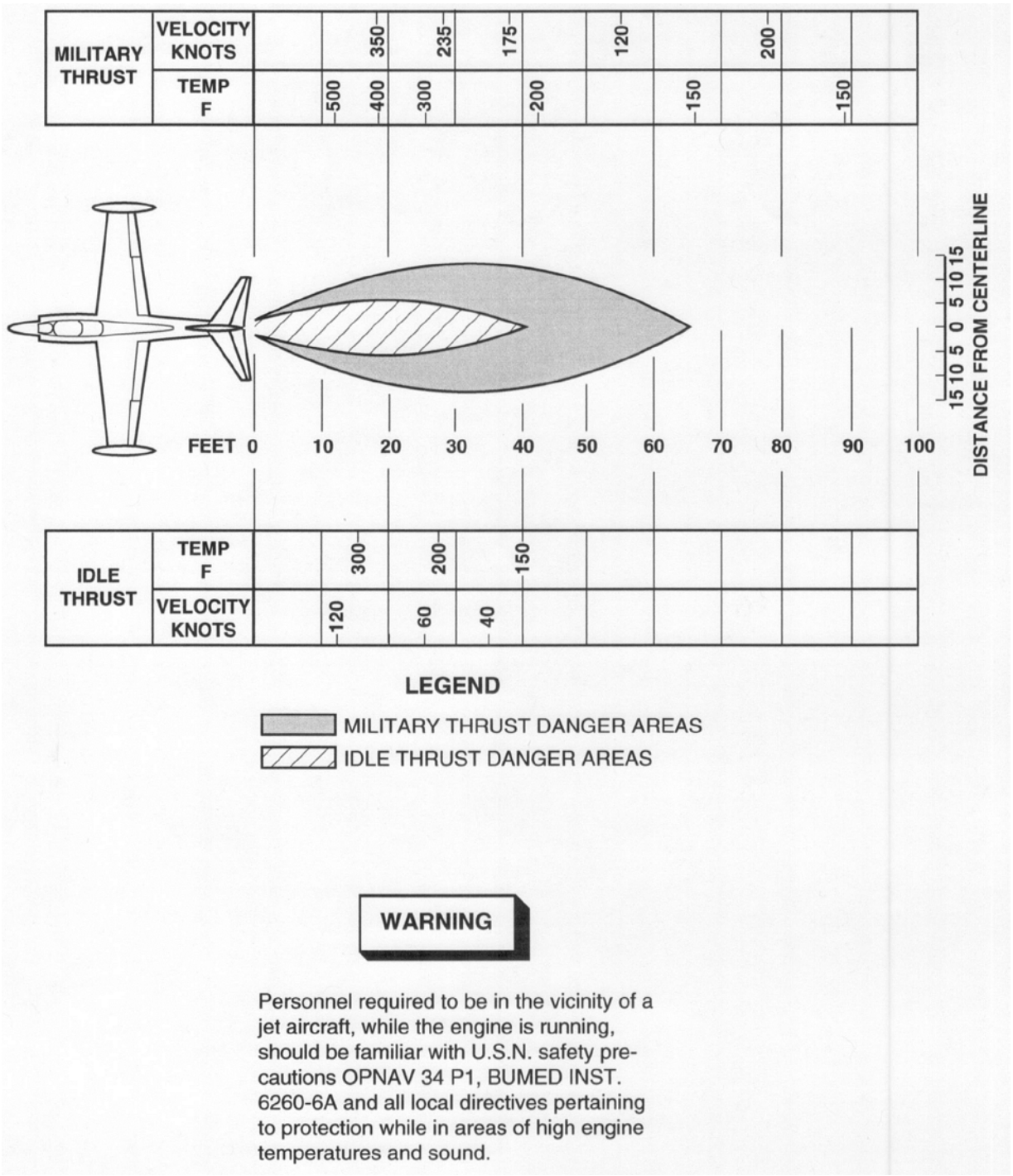
a) Electrical Grounding/Bonding Points, Location of Dump Ports, Ground Fueling Panel, and Fuel Vent



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

b) Engine Exhaust/Blast Area



WARNING

Personnel required to be in the vicinity of a jet aircraft, while the engine is running, should be familiar with U.S.N. safety precautions OPNAV 34 P1, BUMED INST. 6260-6A and all local directives pertaining to protection while in areas of high engine temperatures and sound.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures that are unique to the T-2 aircraft, primarily the operation of the “pre-check” system.

PRE-CHECK PROCEDURE

1. Prepare aircraft and refueling system in accordance with the applicable procedure in Chapter 6, 12, or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109.

2. Initiate refueling.

3. Move TIP TANKS switch to HOLD if not refueling tip tanks, or check switch at REFUEL if refueling tip tanks (aircraft 156722 and subsequent and aircraft with AFC-132 incorporated.)

4. Refuel the tanks with fuel. Test the level control valves during the first minute of refueling as follows:

   a. Position TEST switch to PRIMARY & TIP TANKS. Fuel flow should stop (hose jerk) within 5 seconds.

   b. Position TEST switch to SECONDARY & TIP TANKS. Fuel flow should stop (hose jerk) within 5 seconds.

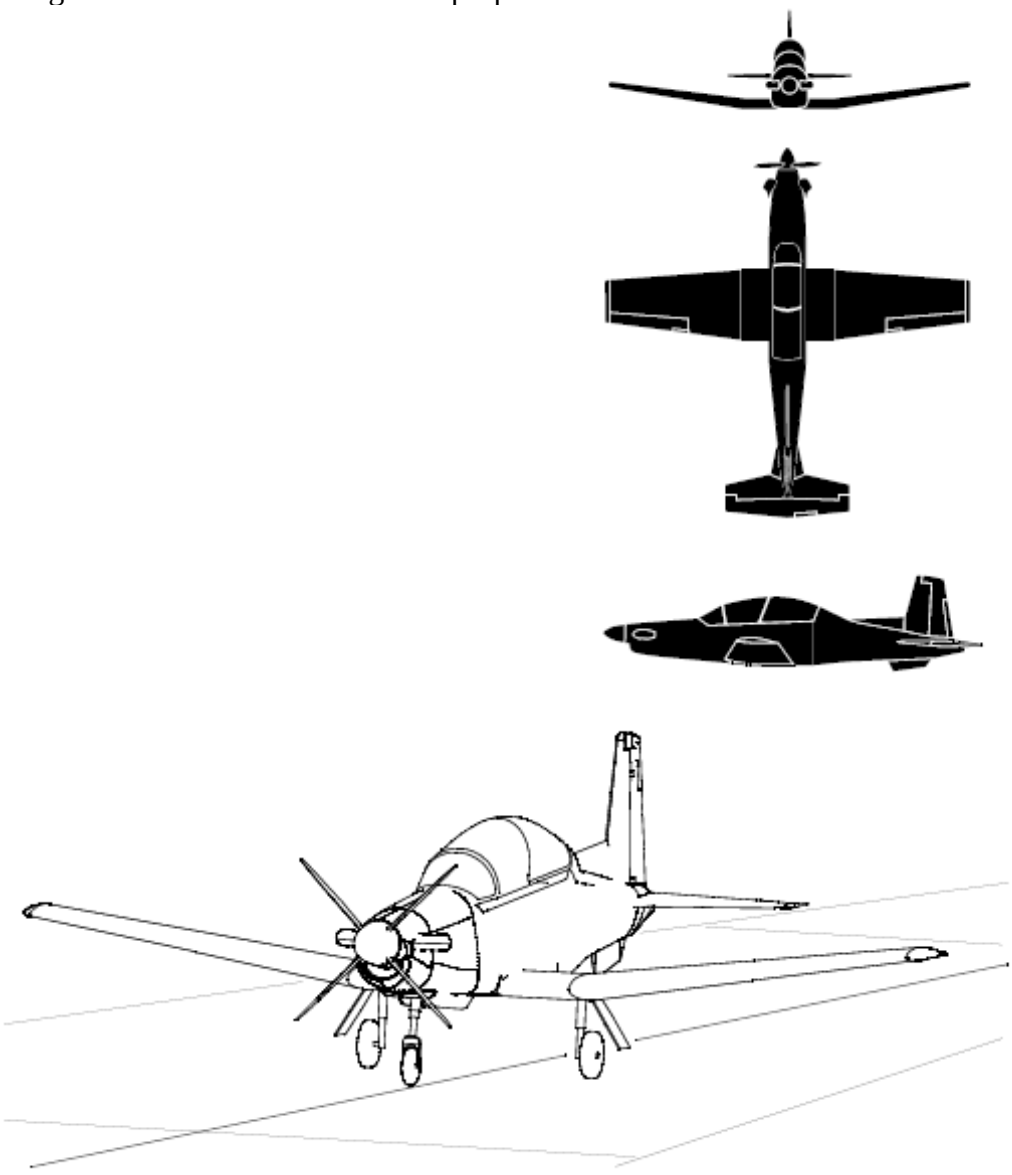


Do not hold TEST switch in either position more than 30 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-6

The T-6A is a single-engine, two-place (tandem seat), pressurized, low-wing training aircraft manufactured by Hawker Beechcraft Corporation, Wichita, KS. The aircraft is approved for day or night VFR and IFR flight. The aircraft is powered by a Pratt & Whitney PT6A-68 free turbine turboprop engine with a Hartzell four-blade propeller.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	33 ft 5 in	Empty Gross Weight — 4900 lbs
Length	33 ft 4 in	Maximum Gross Weight – 6500 lbs
Height	10 ft 8 in	

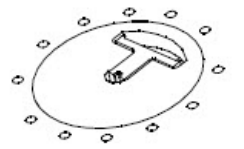
TABLE OF FUEL CAPACITIES

Tank	Gallons	Weight (JP-5)
Left Wing	79	527
Right Wing	79	527
Collector Tank	7	47
Total	164	1100

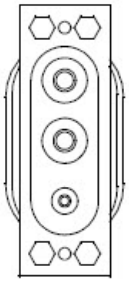
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

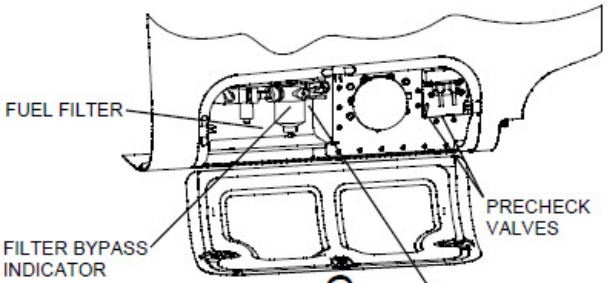
a) Servicing diagram



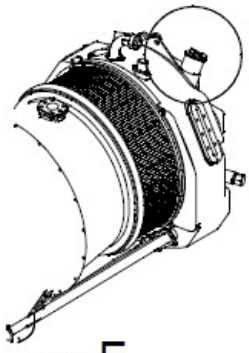
DETAIL A
WING TANK GRAVITY REFUELING (2)



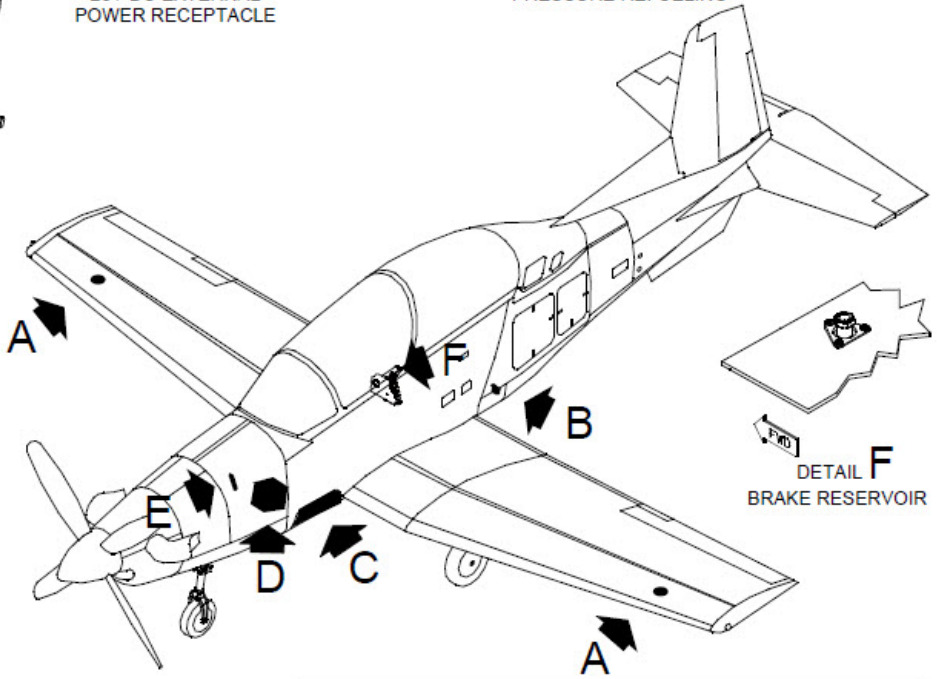
DETAIL B
28V DC EXTERNAL POWER RECEPTACLE



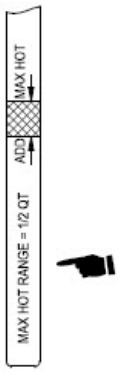
DETAIL C
SINGLE-POINT PRESSURE REFUELING



DETAIL E
ENGINE OIL SERVICING AND DIPSTICK



DETAIL F
BRAKE RESERVOIR



ENGINE OIL DIPSTICK MARKINGS



DETAIL D
BATTERY

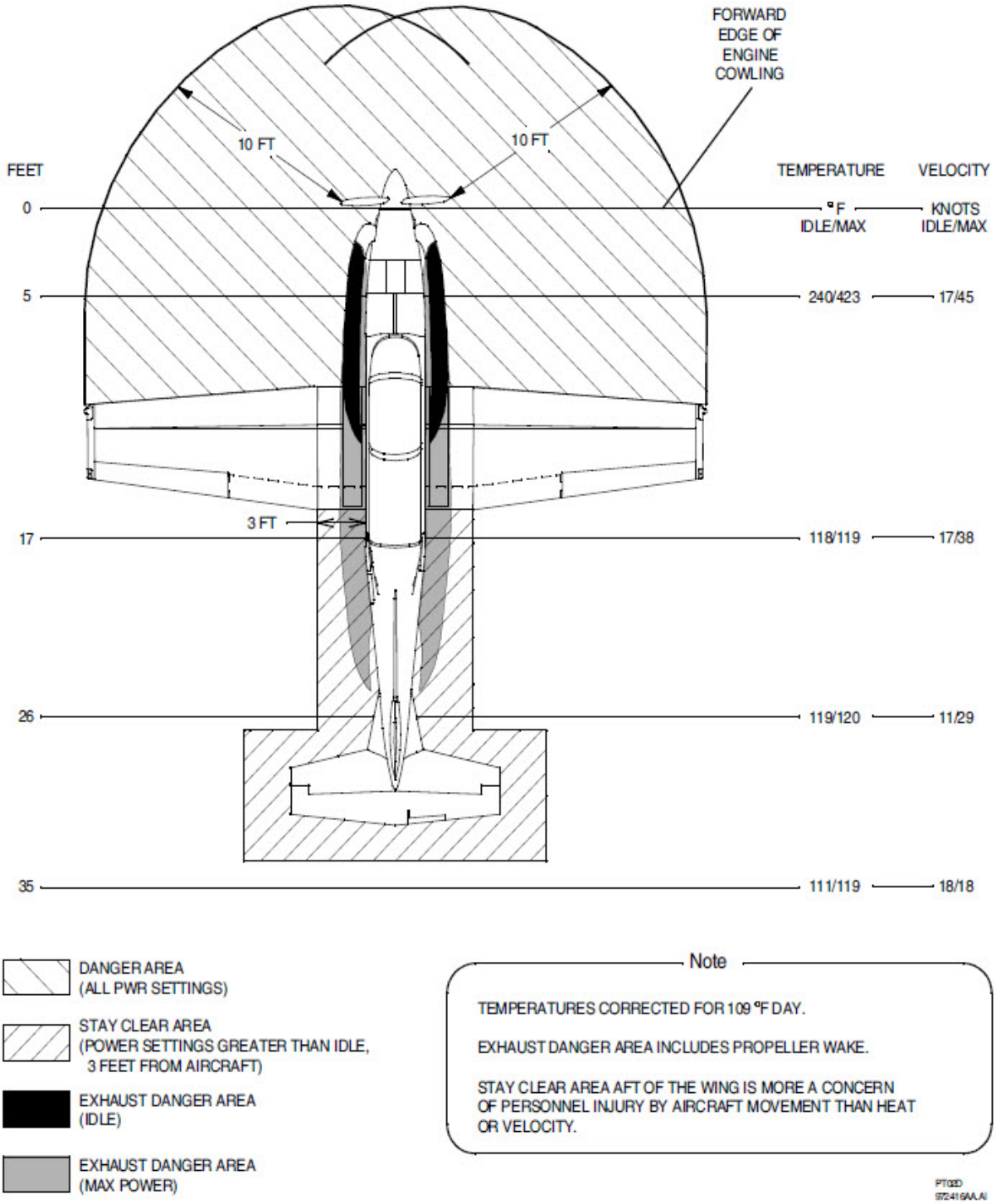
FUEL	FULLY SERVICED		USABLE	
	GALLONS	POUNDS	GALLONS	POUNDS
LEFT WING	79	527	79	527
RIGHT WING	79	527	79	527
COLLECTOR	11	74	7	47
TOTAL	170	1128	164	1100

NOTE:
- CHART IS BASED ON JP-8 FUEL (MIL-T-83133), SINGLE POINT REFUELING (SPR)
- ADD 15 GALLONS (100 POUNDS) TO TOTAL IF GRAVITY REFUELED TO TOP OF FILLER PORT
- ALL FUEL WEIGHTS ARE ROUNDED TO THE NEAREST POUND
- ALL GALLONS ARE ROUNDED TO THE NEAREST GALLON

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

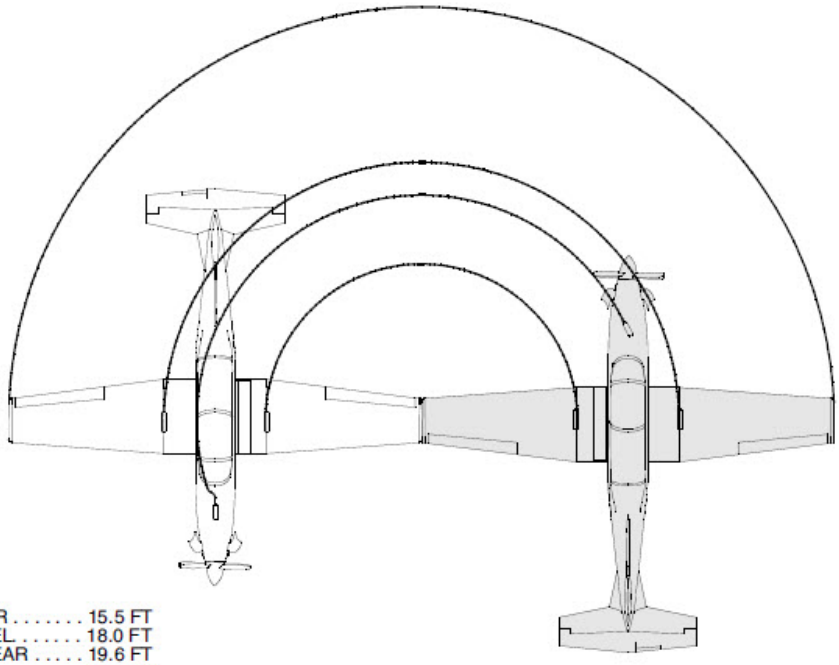
PERSONNEL DANGER ZONES

a) Engine/APU Exhaust/Blast Area



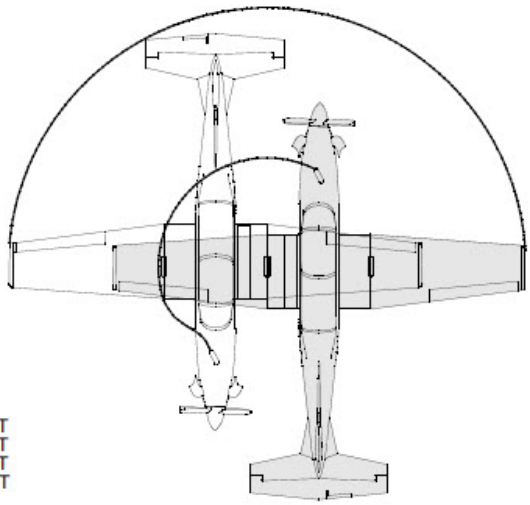
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Turning Radius



RADIUS FOR INSIDE GEAR . . . . . 15.5 FT
RADIUS FOR NOSE WHEEL . . . . . 18.0 FT
RADIUS FOR OUTSIDE GEAR . . . . . 19.6 FT
RADIUS FOR WINGTIP . . . . . 33.2 FT

NOSE WHEEL STEERING TURNING RADIUS



RADIUS FOR INSIDE GEAR . . . . . 0.0 FT
RADIUS FOR NOSE WHEEL . . . . . 8.3 FT
RADIUS FOR OUTSIDE GEAR . . . . . 8.6 FT
RADIUS FOR WINGTIP . . . . . 21.0 FT

DIFFERENTIAL BRAKING TURNING RADIUS (NWS OFF)

P T02 D
970511AA.AI

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The fuel system provides approximately 1100 pounds of usable fuel through the single point refueling system. Approximately 100 pounds additional fuel is available if manually filled to the base of the filler neck in each wing tank. Single point pressure refueling is the primary refueling method. Three integral tanks built into a single-piece wing provide fuel storage with usable fuel information listed above.

NOTE
When parking on ramps with greater than 1% slope, the fuel system may vent fuel overboard through a pressure relief valve.

The fuel system incorporates an auto balance system to keep the fuel level in the wing tanks within 20 pounds of each other. When a fuel imbalance of 20 pounds or more is detected for more than 30 seconds, the transfer valve will close the motive flow line to the light tank. This action stops fuel in the light tank from being transferred to the collector tank while fuel continues to be transferred from the heavy tank to the collector tank. If the fuel imbalance is not reduced to less than 30 pounds within 2 minutes, the FUEL BAL annunciator will illuminate and the auto balance system will shut off. The FUEL BAL annunciator will remain illuminated until the system is reset.

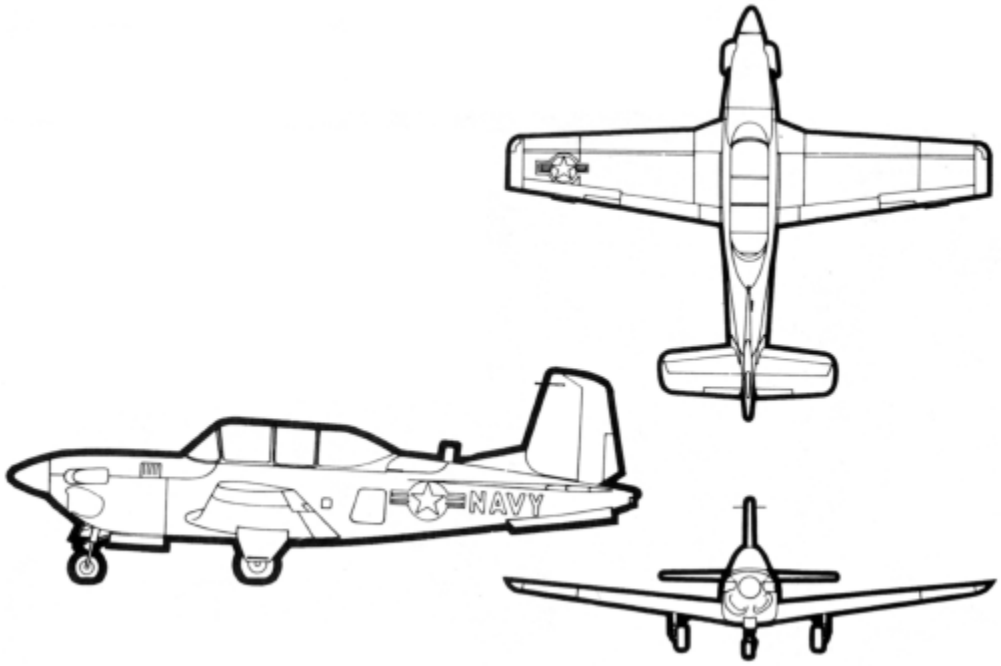
NOTE
If the auto balance system shuts off without reducing the fuel imbalance to 30 pounds or less, the auto balance system may be reset to provide an additional 2 minutes to balance the fuel load. The system may require multiple resets to balance the fuel load. The fuel load may also be manually balanced using the Fuel Imbalance procedure in Section III of A1-T6AAA-NFM-100. The fuel system includes six annunciators to indicate fuel system operations:

- The red FUEL PX annunciator is activated by the low pressure switch and indicates less than 10 psi fuel pressure in the motive flow/return flow supply line.
- The green BOOST PUMP annunciator is illuminated manually by selecting the BOOST PUMP switch ON, automatically by the low-pressure switch if the PCL is above the IDLE position, or whenever the starter is activated regardless of fuel pressure. The BOOST PUMP circuit breaker is located on the front cockpit battery bus.
- The amber L FUEL LO and R FUEL LO annunciators are activated by optical sensors and indicate fuel quantity below approximately 110 pounds in the respective wing tank. The optical sensors are independent of the fuel probes and fuel quantity gages. The FUEL QTY LO circuit breaker is located on the front cockpit battery bus prohibited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-34

The T-34C aircraft is an unpressurized two-place, tandem cockpit, low-wing, single-engine monoplane manufactured by Beech Aircraft Corporation, Wichita, Kansas.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: T-34

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	33 ft 4 in	Maximum Gross Weight — 4,425 lbs
Length	28 ft 6 in	Maximum Footprint — 90 psi
Height	9 ft 8 in	

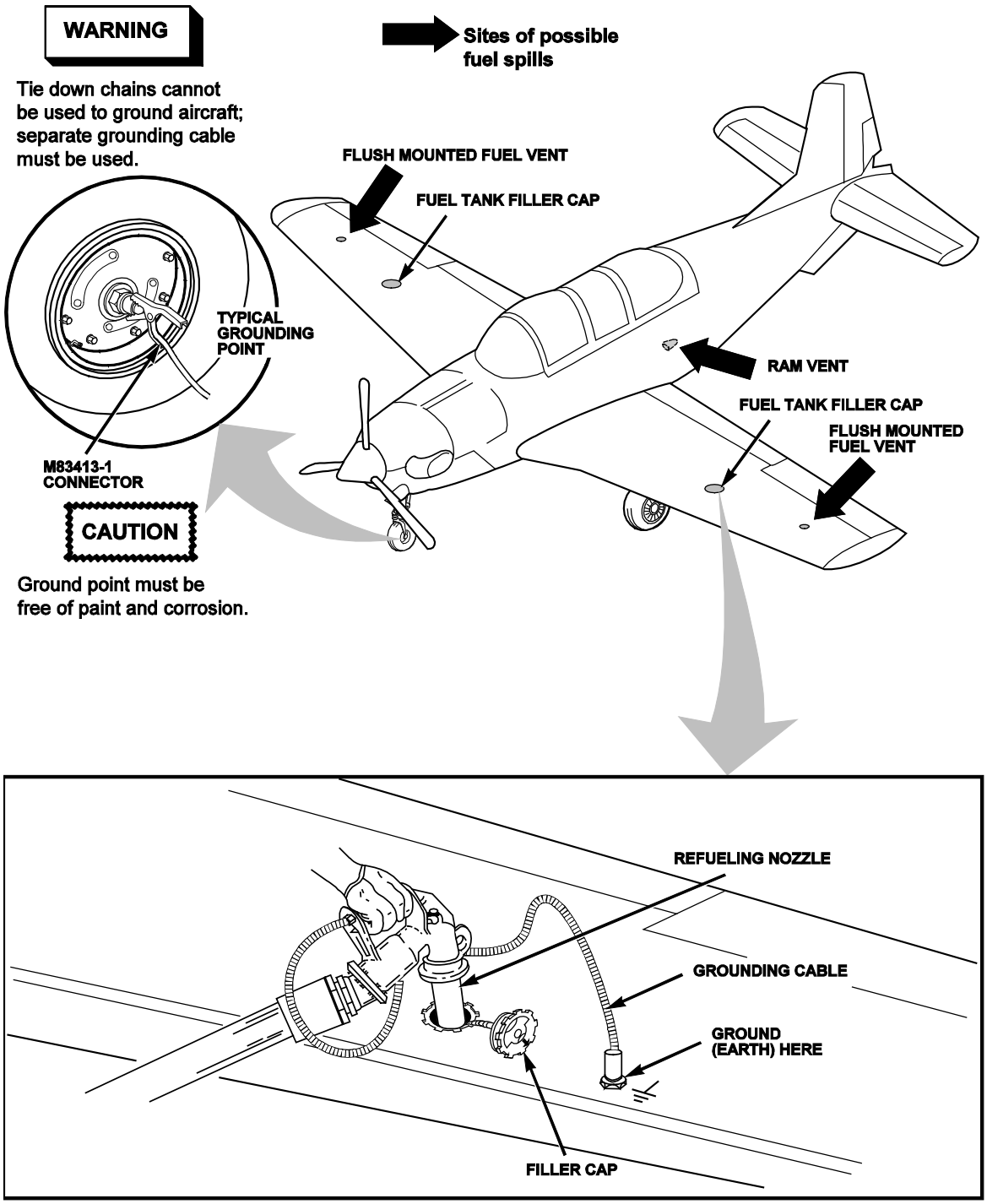
TABLE OF FUEL CAPACITIES

Tank	Gallons
Wing Leading Edge	40 Each
Wing Panel	25 Each
Total	130

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

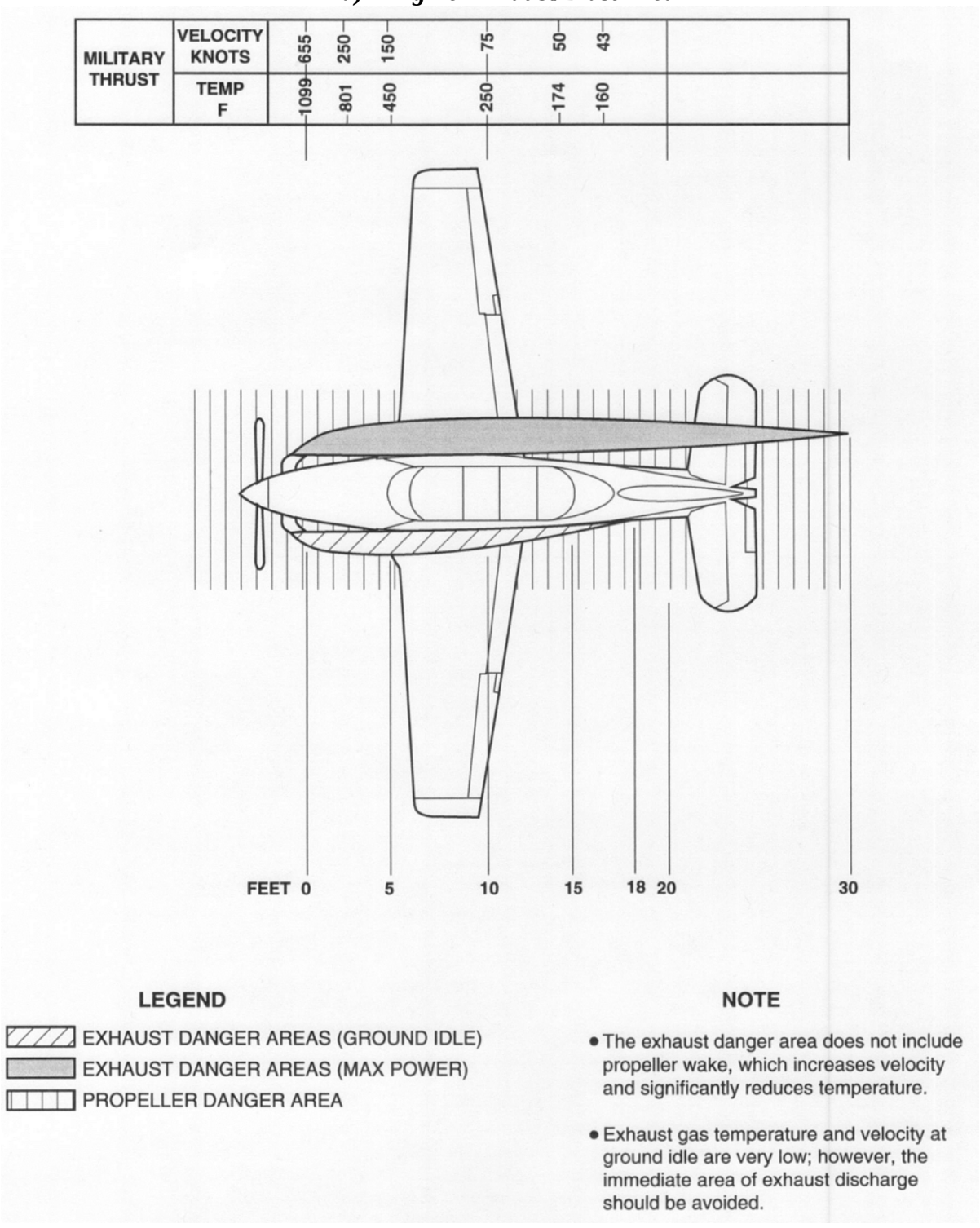
a) Electrical Ground/Bonding Point, Location of Fuel Vents, and Ground Fueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Exhaust/Blast Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The T-34 aircraft does not have a single-point refueling adapter and, therefore, can only be gravity refueled.

GRAVITY FUELING

1. Attach bonding cables to aircraft.
2. Attach bonding cable from hose nozzle to ground socket adjacent to fuel tank being filled.
3. Open applicable fuel tank filler cap.



Do not insert fuel nozzle completely
into fuel tank because of possible
damage to bottom of fuel cell.

4. Fill fuel tank with fuel.
5. Secure applicable fuel tank filler cap.



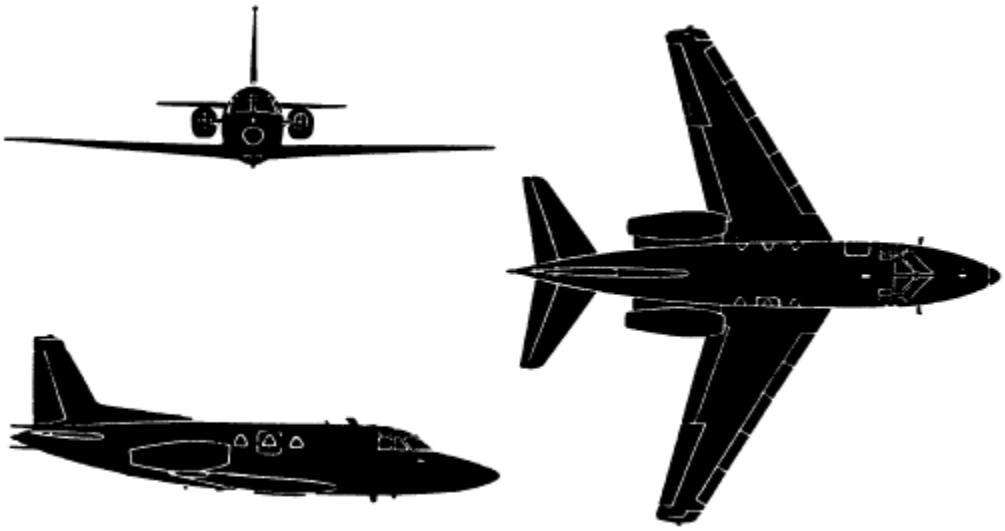
Make sure latch tab on cap is pointed aft.

6. Disconnect bonding cables from aircraft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-39

The T-39 aircraft is a low-wing, twin-jet monoplane with an axial-flow, pod-mounted engine on each side of the aft fuselage.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	44 ft 8 in	Maximum Gross Weight — 20,400 lbs
Length	46 ft 11 in	Maximum Footprint — psi
Height	16 ft	

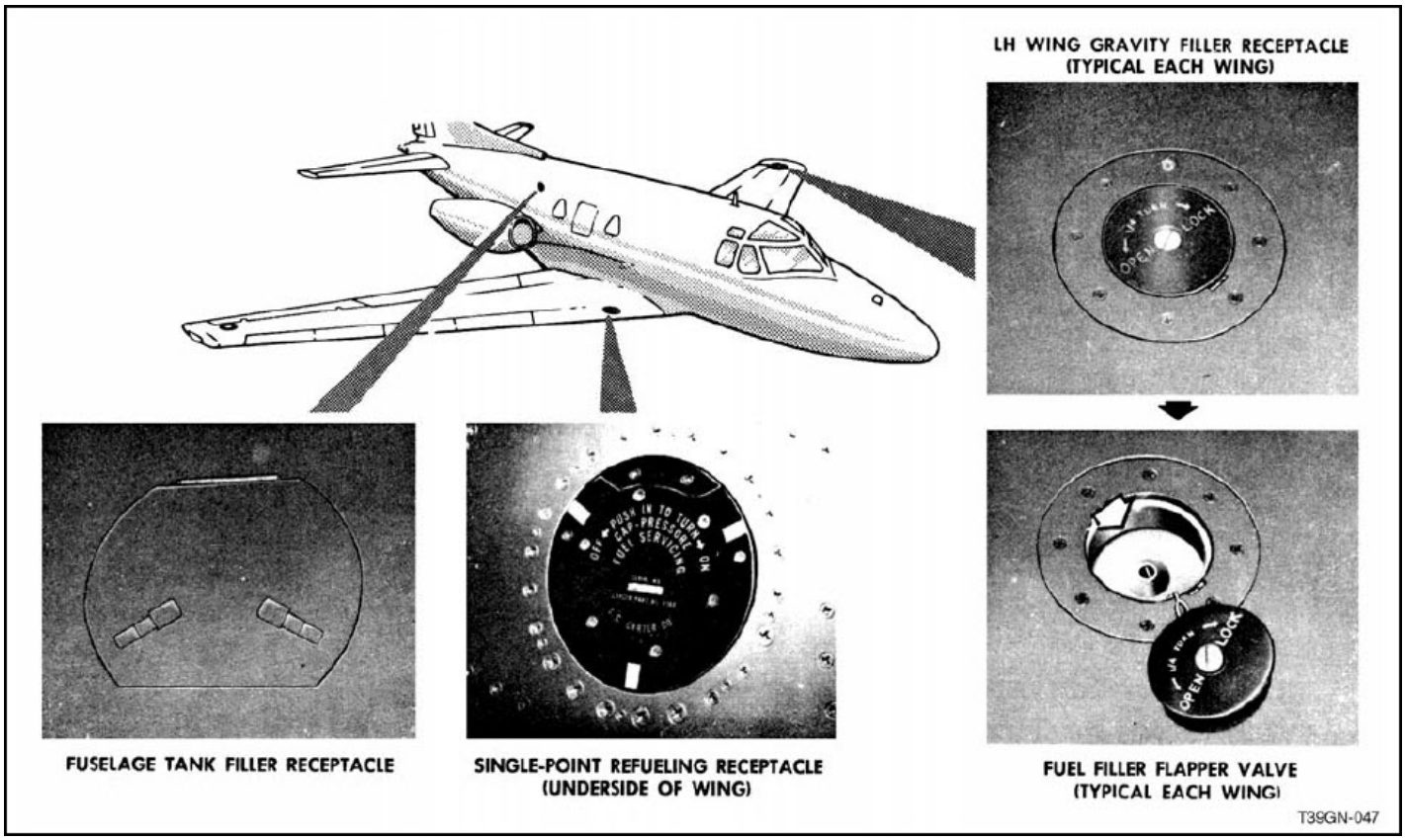
TABLE OF FUEL CAPACITIES

Tank	Gallons	Pounds (JP-5)
Left Wing	460	3150
Right Wing	460	3150
Fuselage	160	1100
Total	1080	7400

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

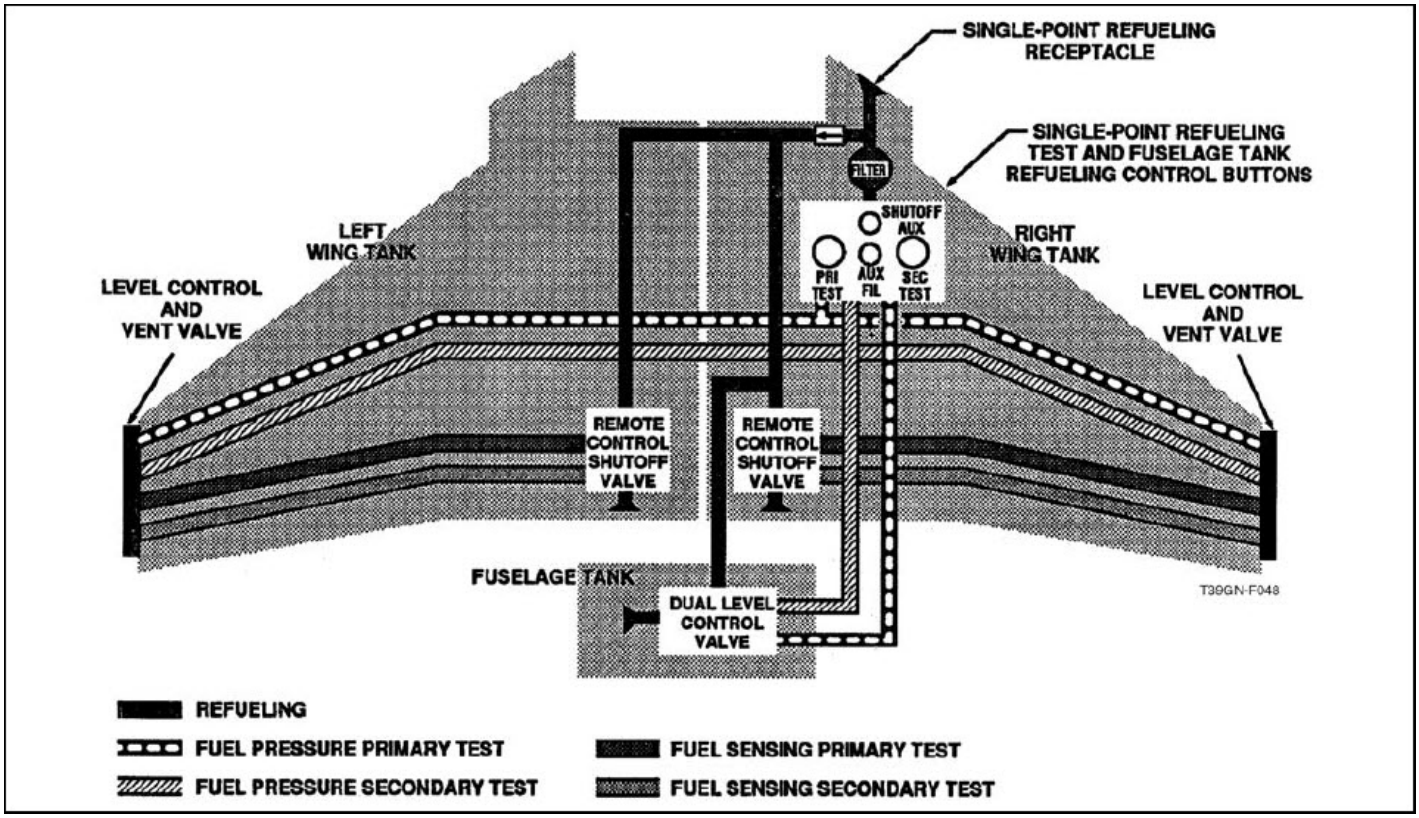
AIRCRAFT CONFIGURATION

a) *Location of Filler Receptacles and Valves*



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Fuel System Layout

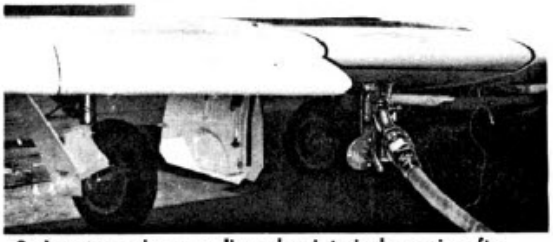


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

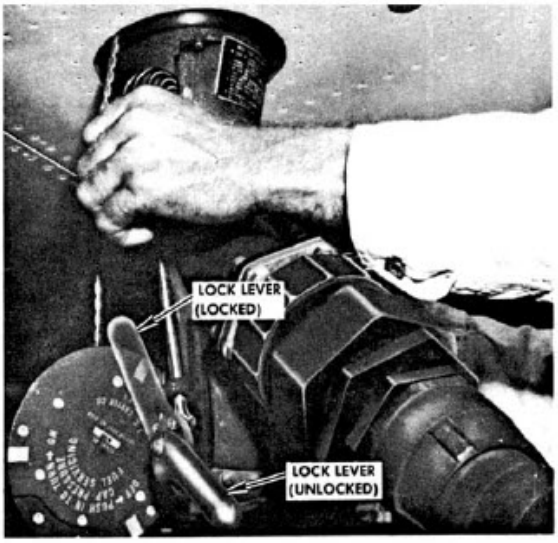
c) Refueling Instructions



1. Remove pressure refueling cap.



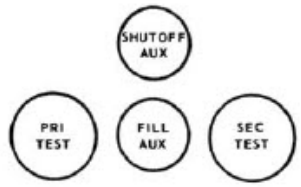
2. Insert nozzle grounding plug into jack on aircraft.



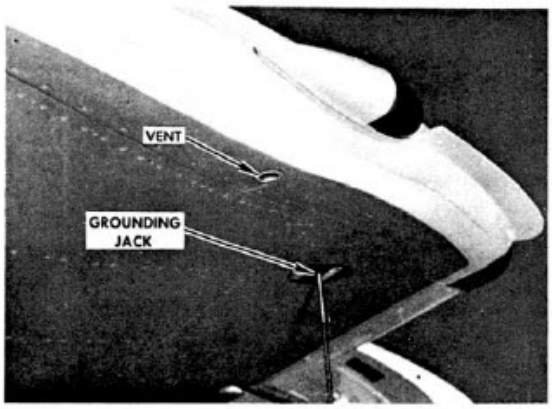
3. Connect and lock refueling nozzle. Start ground servicing unit. Refueling pressure should be 50 psi.

CAUTION

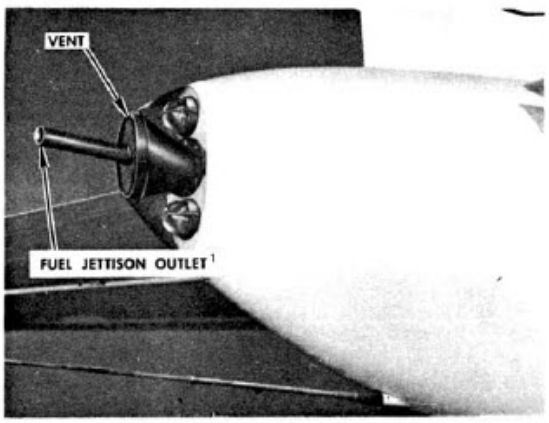
- Check to be sure that the single-point refueling receptacle is completely engaged prior to refueling. If it is not, damage may be done to the refueling receptacle.
- Before start of refueling, ensure that refueling hose is free from kinks.



4. Depress ***PRI TEST*** button. Fueling should stop (hose will stiffen).
5. Depress ***SEC TEST*** button. Fueling will stop (hose will stiffen).
6. Depress ***FILL AUX*** and ***SHUTOFF AUX*** buttons together. Release ***SHUTOFF AUX*** button first; ***FILL AUX*** button should remain in depressed position. Full servicing should be completed in approximately 12 minutes.
7. Check wing fuel tank vents at both wing tips for obstructions and passage of air to ensure proper wing tank venting. If no passage of air is noted, secure the fueling operation and have discrepancy investigated.



8. Fuel may come out wing tip vents. If this occurs, turn off refueler momentarily. Valves will reseat.

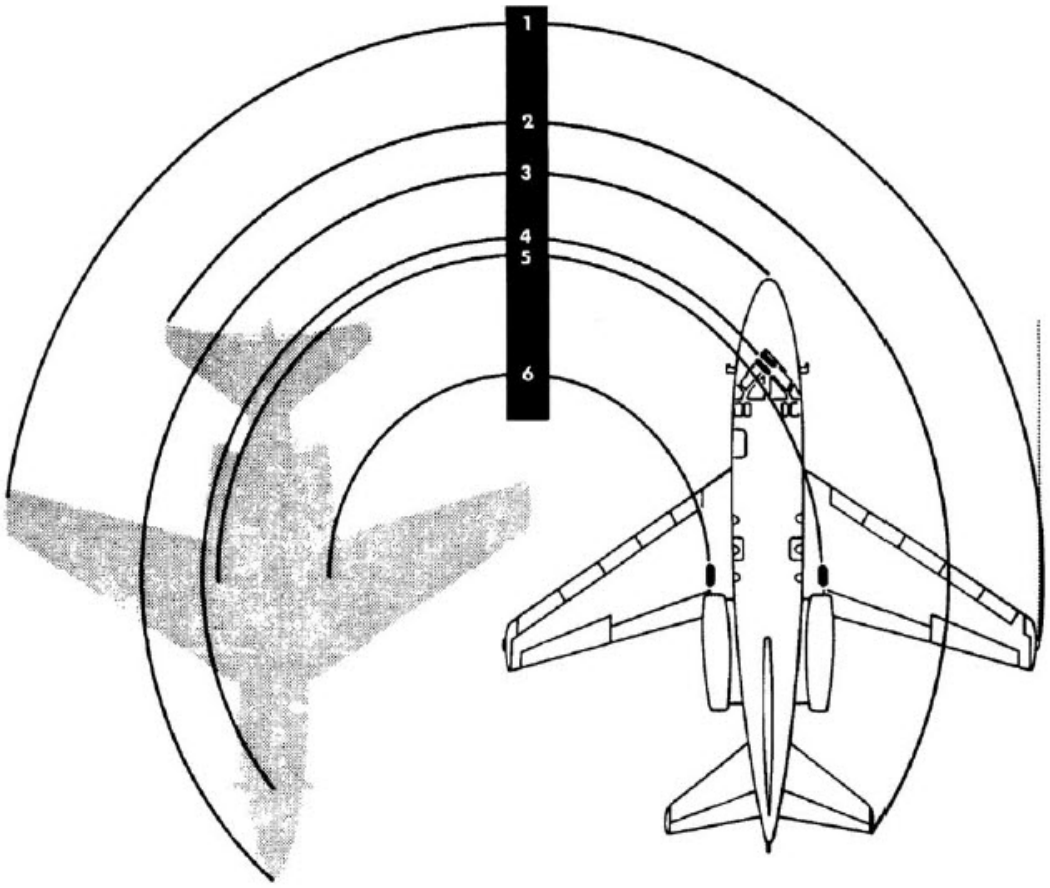


T39GN-F046

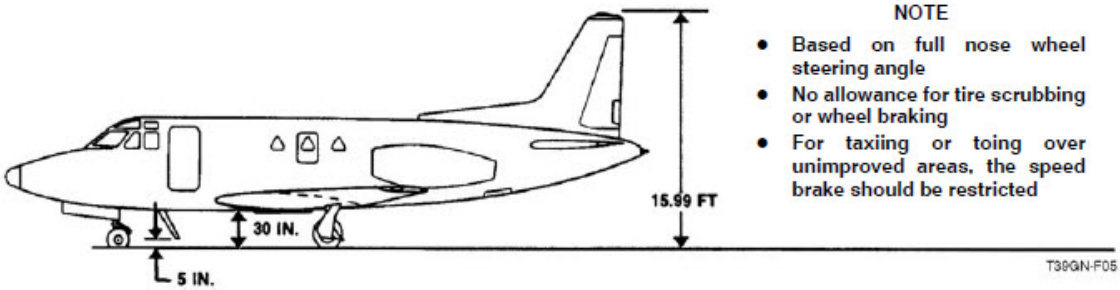
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONE

a) Turning Radius



	T-39N	T-39G
WING TIP	44.2 FT	44.64FT
HORIZONTAL STABILIZER	36.1 FT	35.4 FT
NOSE	31.3 FT	30.9 FT
NOSE GEAR	25.9 FT	25.3 FT
RIGHT MAIN GEAR	25.1 FT	24.3 FT
LEFT MAIN GEAR	17.9 FT	17.1 FT

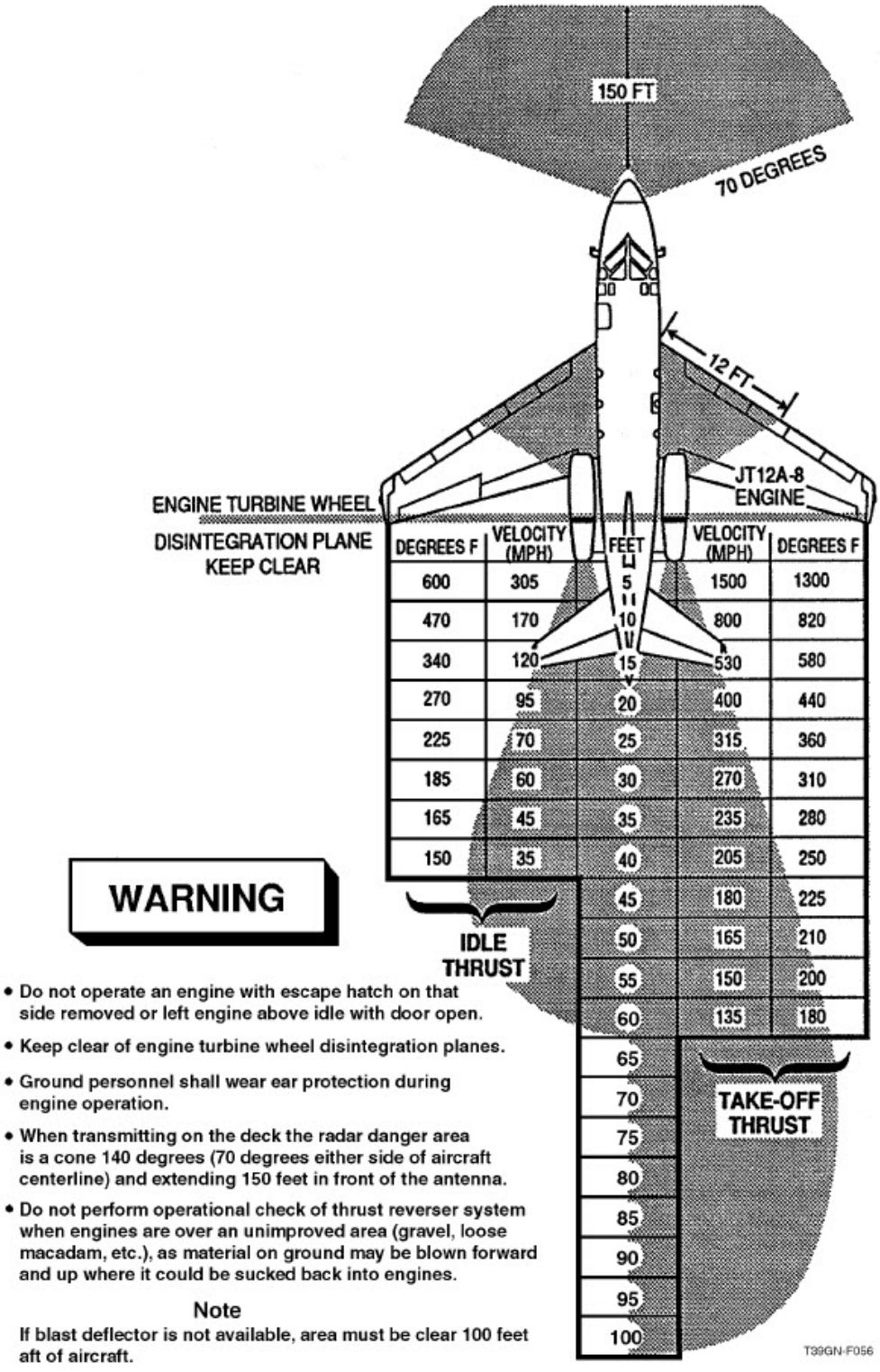


NOTE

- Based on full nose wheel steering angle
- No allowance for tire scrubbing or wheel braking
- For taxiing or toing over unimproved areas, the speed brake should be restricted

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Engine Inlet/Exhaust/Blast Area, and Turbine Failure Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

SINGLE POINT REFUELING

1. Attach bonding cables to aircraft.
2. Attach bonding cable from hose nozzle to ground.
3. Remove pressure refueling cap.
4. Insert nozzle grounding plug into jack on aircraft.
5. Connect and lock refueling nozzle. Start ground servicing unit. Refueling pressure should be 50 psi.



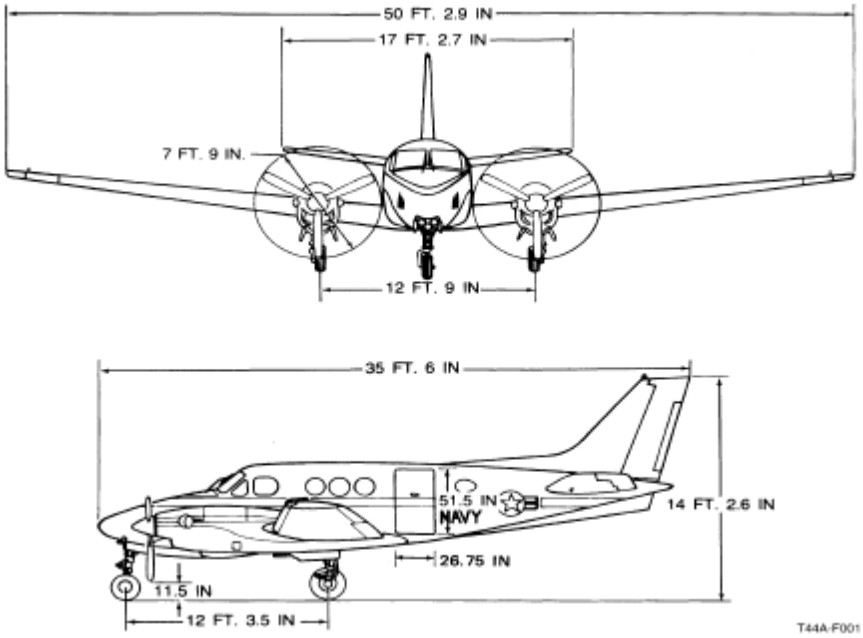
- Check to be sure that the single point refueling receptacle is completely engaged prior to refueling. If it is not, damage may be done to the refueling receptacle.
- Before start of refueling, ensure that the refueling hose is free of kinks.

6. Depress PRI TEST button. Fueling should stop (hose will stiffen).
7. Depress SEC TEST button. Fueling should stop (hose will stiffen).
8. Depress FILL AUX and SHUTOFF AUX buttons together. Release SHUTOFF AUX button first; FILL AUX button should remain in depressed position. Full servicing should be completed in approximately 12 minutes.
9. Check wing fuel tank vents at both wing tips for obstructions and passage of air to ensure proper wing tank venting. If no passage of air is noted, secure the fueling operation and discrepancy investigated.
10. Fuel may come out wing tip vents. If this occurs, turn off refueler momentarily. Valves will reseat.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-44

The T-44A aircraft is a twin-engine, pressurized, fixed-wing monoplane manufactured by Beech Aircraft Corporation, Wichita, Kansas. The T-44A is powered by two 550-shaft horsepower PT6A-34B turboprop engines manufactured by Pratt & Whitney of Canada Limited. The primary mission of the T-44A is to train student military aviators to fly multiengine turboprop aircraft.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	50 ft 2.9 in	Maximum Gross Weight — 9,710 lbs
Length	35 ft 6 in	Maximum Footprint — 57 psi
Height	14 ft 2.6 in	

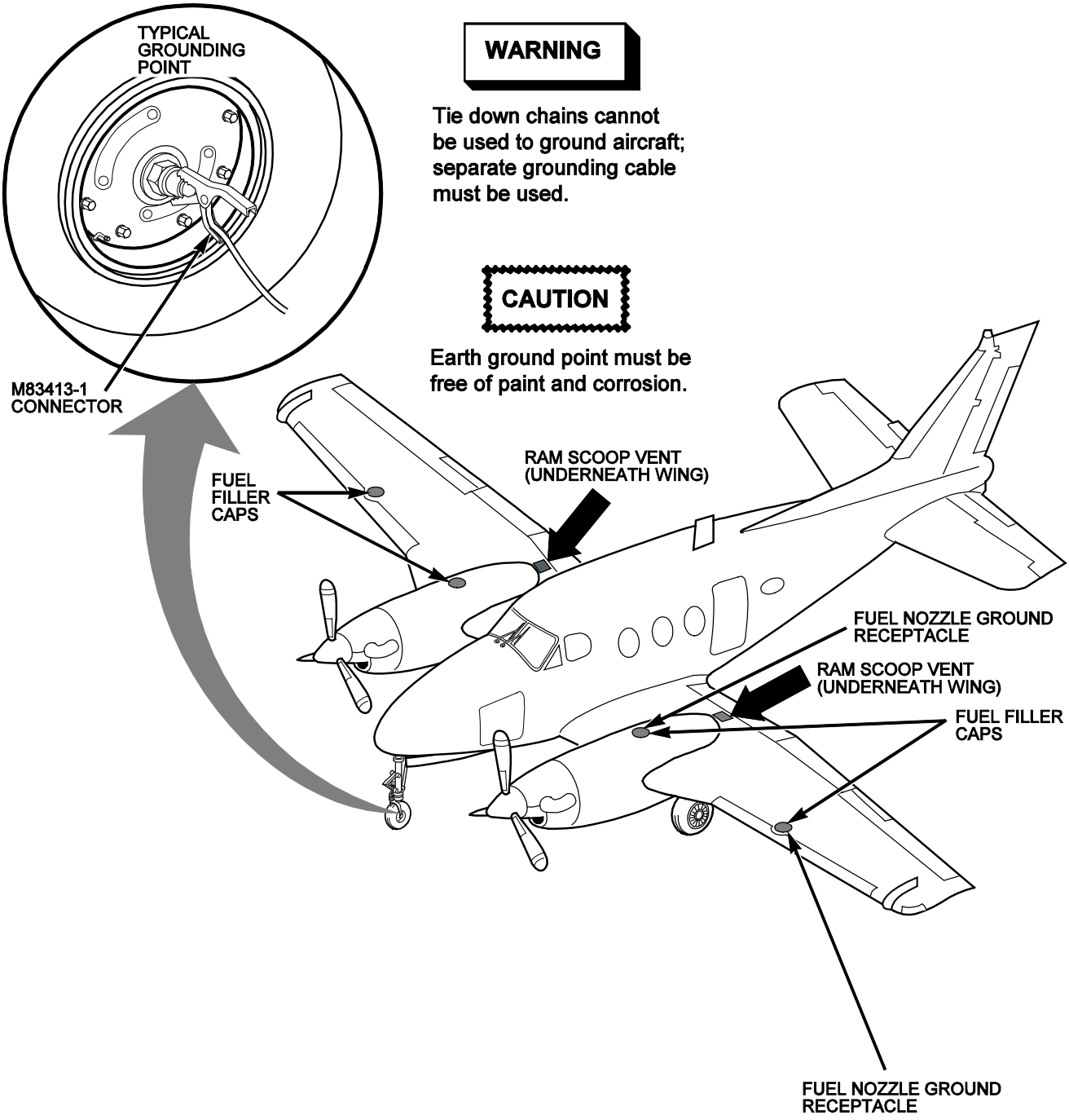
TABLE OF FUEL CAPACITIES

Tank	Gallons	Pounds (JP-5)
Nacelle	61 Each	415
Wing Tanks	132 Each	900
Total Useable	384	2610

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

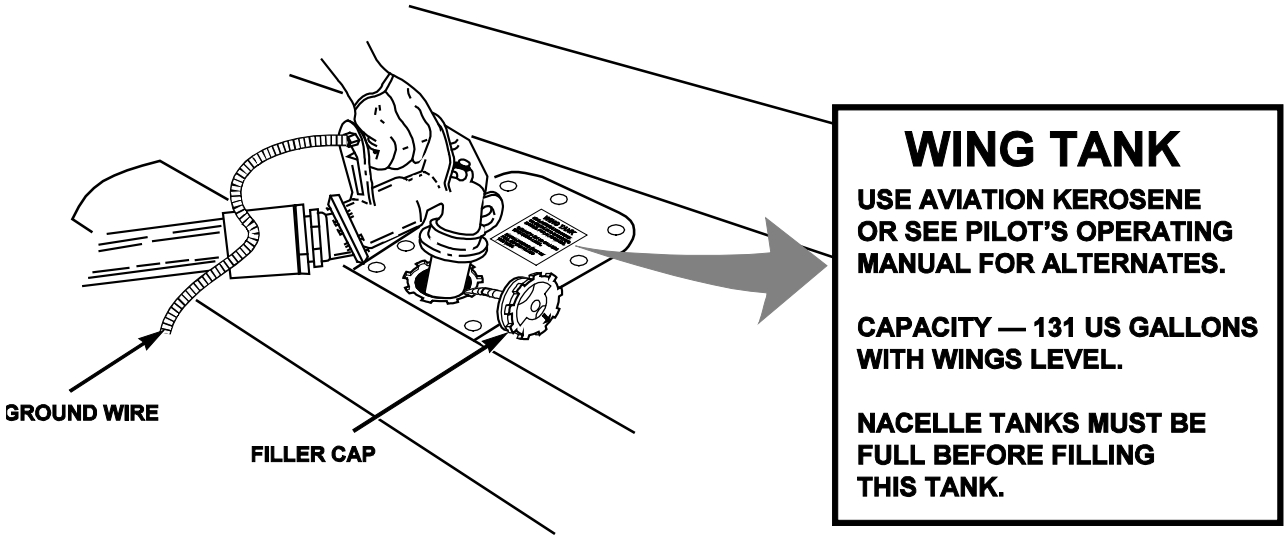
a) Electrical Grounding/Bonding Points, Location of Fuel Filler Caps



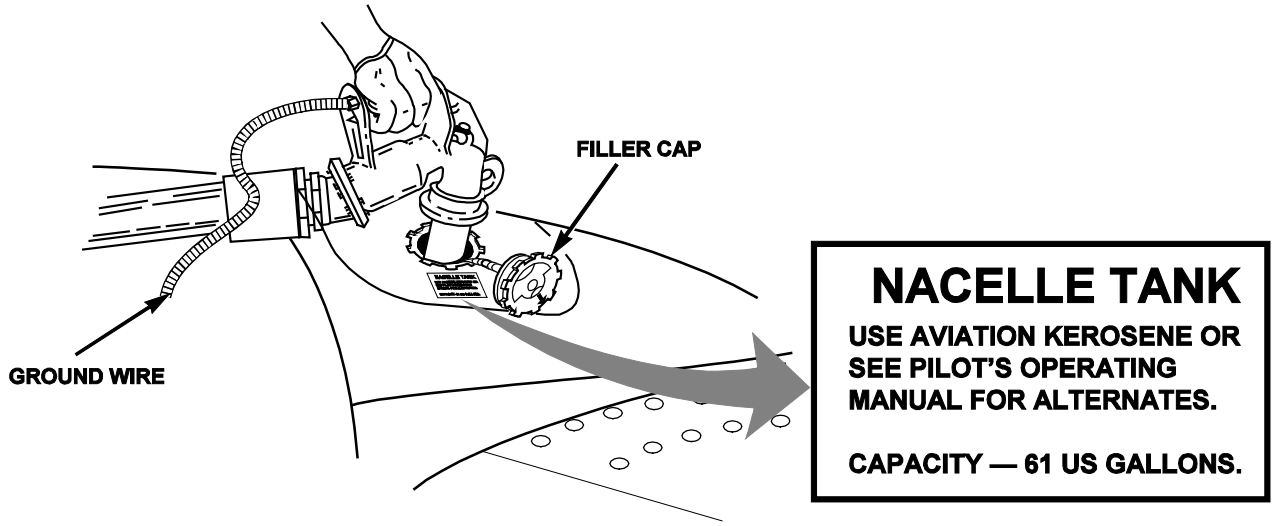
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Gravity Fill Ports

WING TANK GRAVITY FILLER



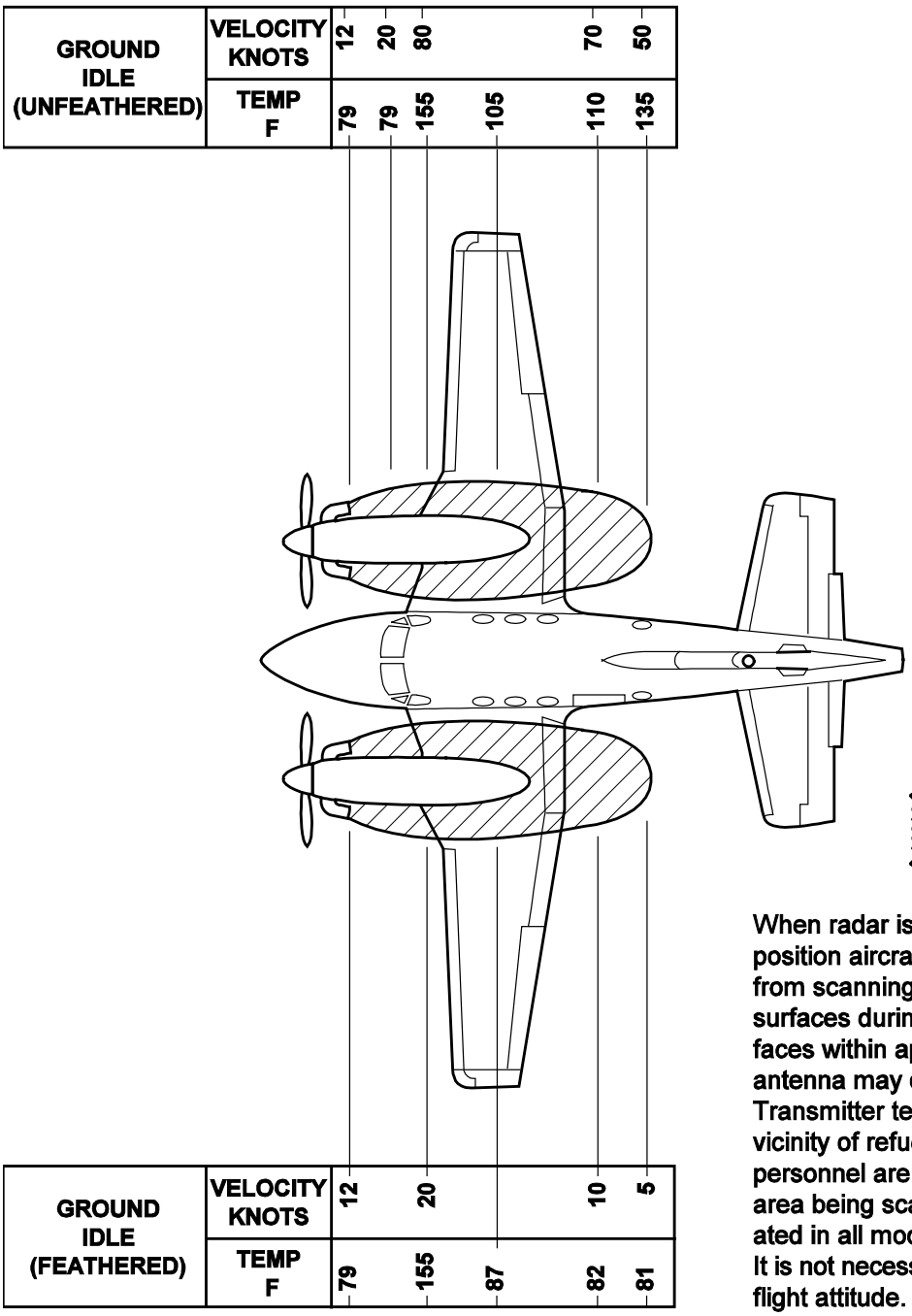
NACELLE FUEL CELL GRAVITY FILLER



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Exhaust/Blast Area



CAUTION

When radar is operating into antenna, position aircraft to prevent the antenna from scanning any large, close-by metallic surfaces during test. Scanning such surfaces within approximately 50 feet of the antenna may damage receiver crystals. Transmitter tests must not be made in the vicinity of refueling operations, or when personnel are standing within 18 feet of area being scanned. RF energy is generated in all modes except Standby and Test. It is not necessary to place aircraft in a level flight attitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The T-44 aircraft can only be gravity refueled.

GRAVITY FUELING

1. Attach bonding cables to aircraft.
2. Attach bonding cable from hose nozzle to ground.
3. Open applicable fuel tank filler cap.

WARNING

Do not insert fuel nozzle completely into fuel cell because of possible damage to bottom of fuel cell.

4. Fill fuel tank with fuel using slow fuel discharge rate.
5. Secure applicable fuel tank filler cap.

CAUTION

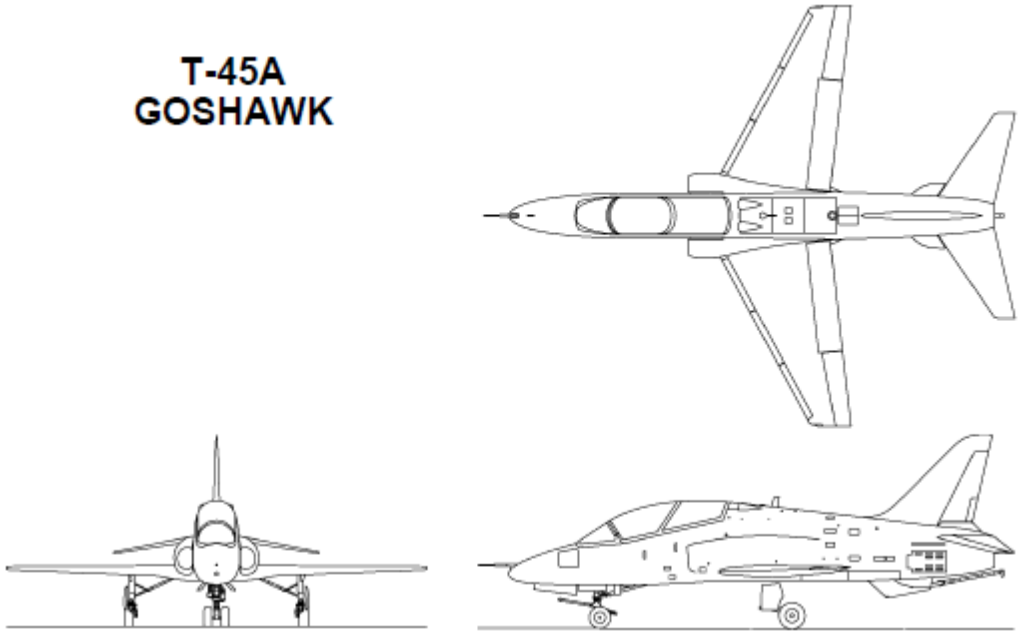
Ensure latch tab on cap is pointed aft.

7. Disconnect bonding cables from aircraft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

T-45

The Navy model T-45A Goshawk, manufactured by McDonnell Douglas Aerospace is a two place, lightweight, high performance, fully carrier capable, version of the British Aerospace Hawk. It is powered by a single Rolls Royce F405-RR401 turbofan engine, producing a sea level, installed, static thrust of 5,527 pounds.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	30 ft 10 in	Maximum Gross Weight — 14,500 lbs
Length	39 ft 4 in	Maximum Footprint — 145 psi
Height	13 ft 6 in	

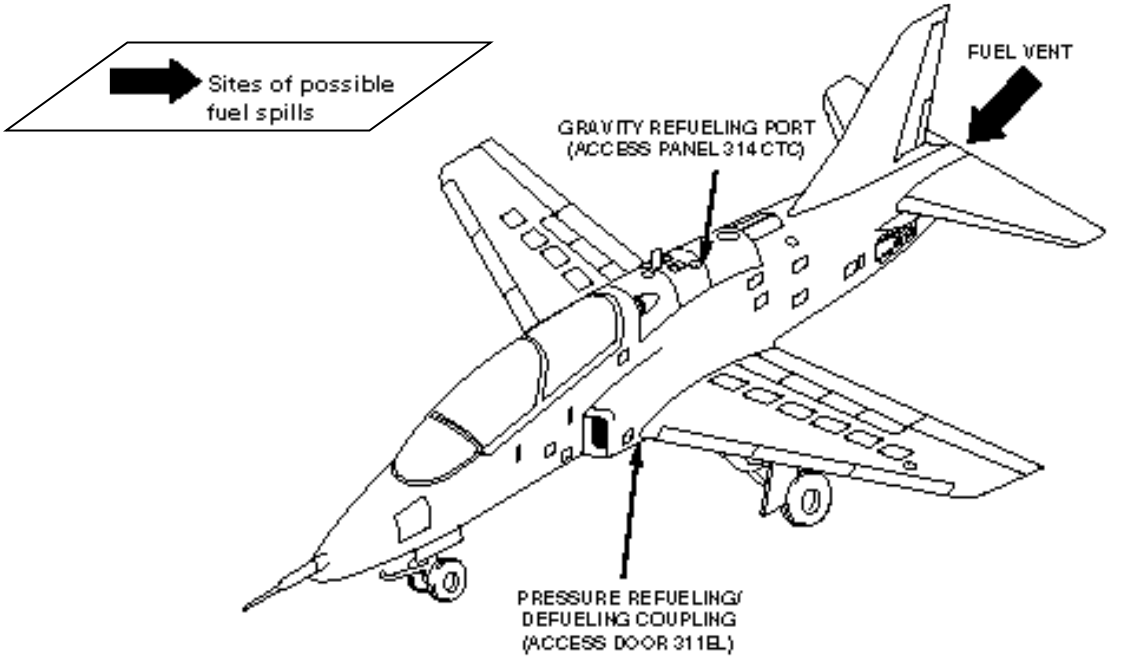
TABLE OF FUEL CAPACITIES

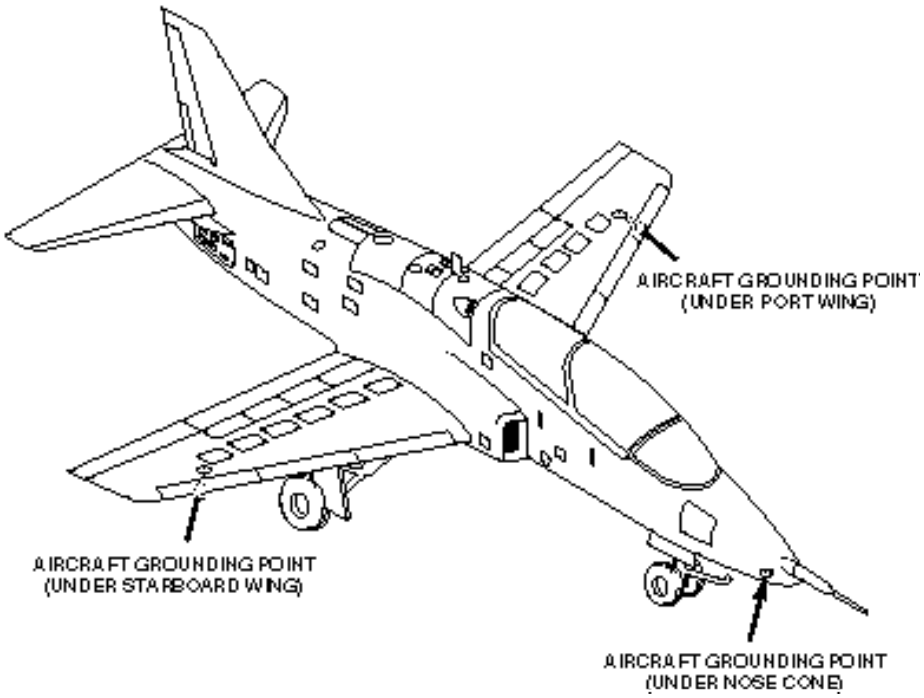
Tank	Gallons	Pounds (JP-5)
Integral Wing and Fuselage	432	2940
Useable	432	2940
Total Capacity	443	3015

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

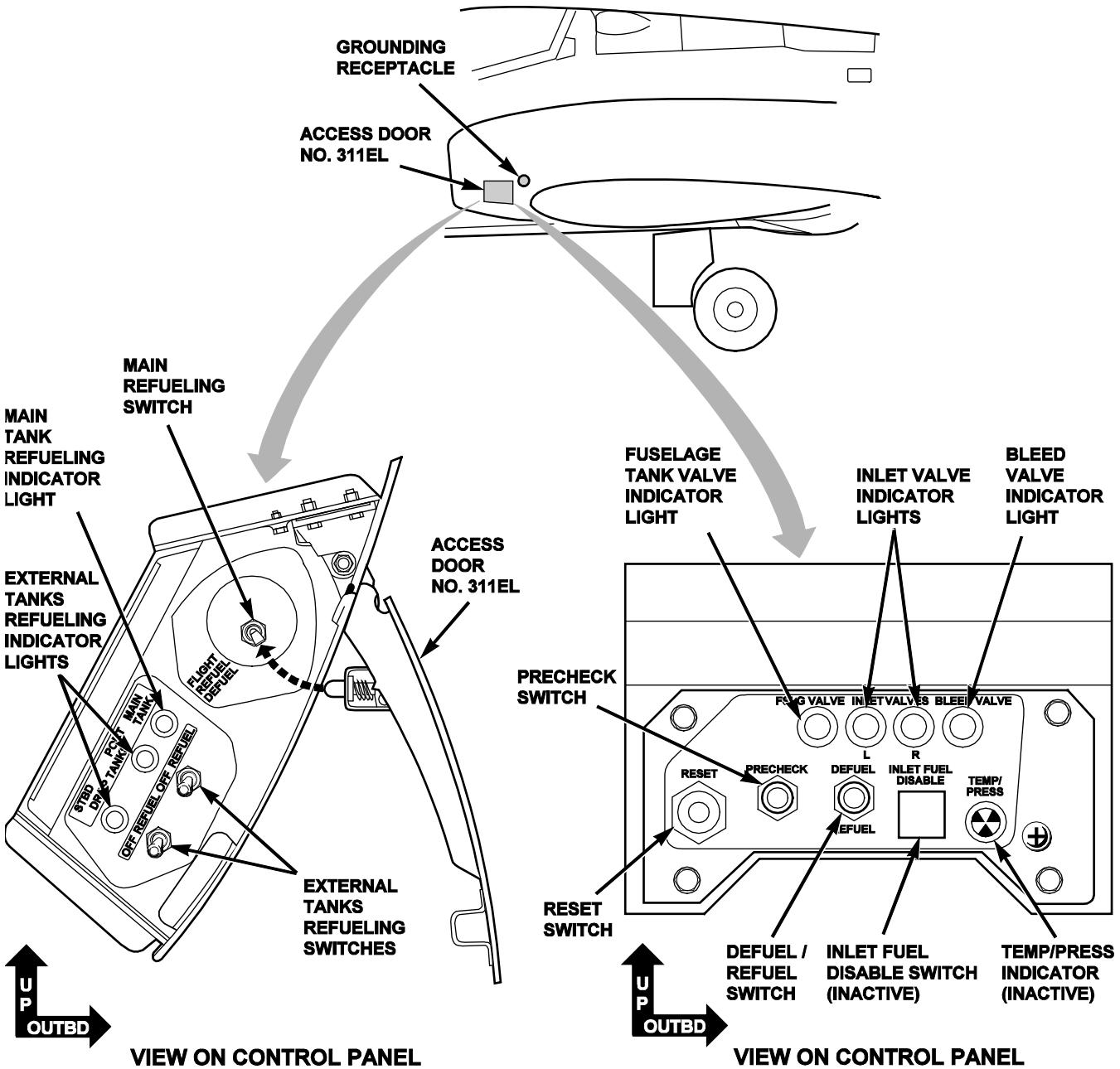
a) Electrical Grounding/Bonding Points, Location of Fuel Vent and Gravity Refueling Port





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Ground Refueling Panel



PRESSURE/HOT REFUEL
Aircraft 163600 thru 163646

PRESSURE/HOT REFUEL
Aircraft 163647 and up

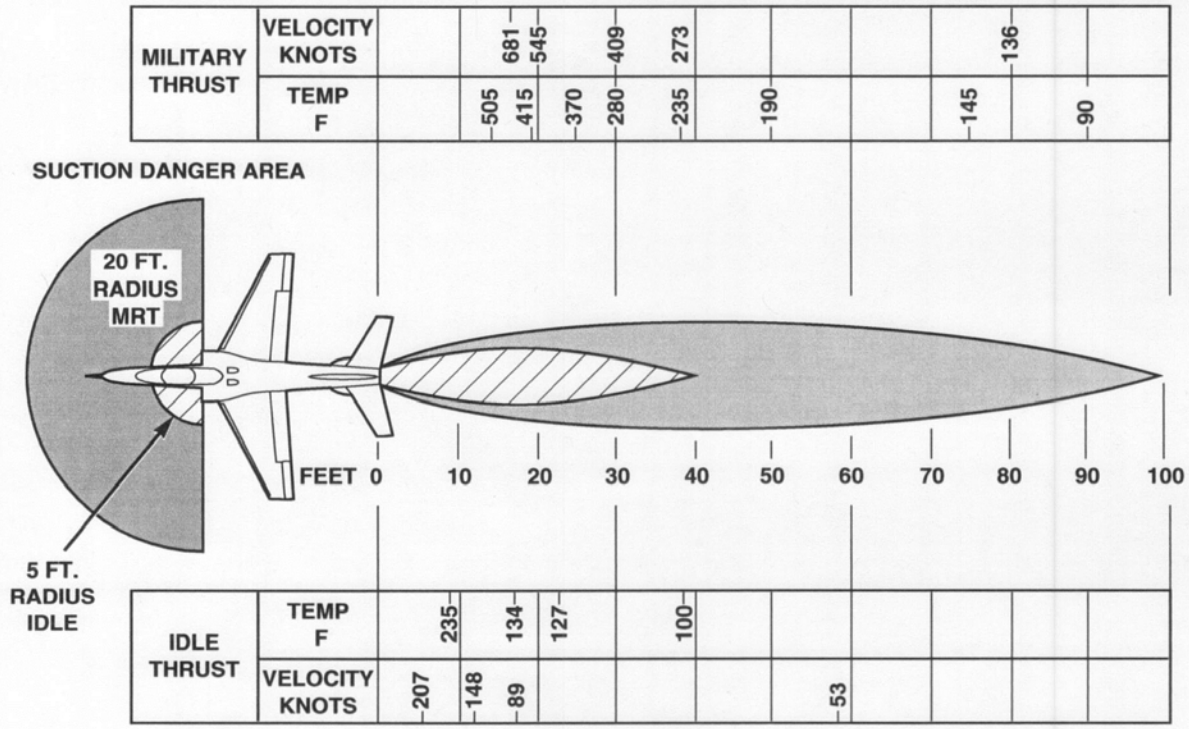
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Inlet/Exhaust/Blast Area

WARNING

- Remove all objects under aircraft. Failure to do so can result in damage to aircraft or injury to personnel.
- Stay clear of area within 100 feet directly behind the aircraft when the engine is operating at MRT.



LEGEND

MILITARY THRUST DANGER AREAS

IDLE THRUST DANGER AREAS

CAUTION

- Make certain that suction danger area is clear of debris.
- Jet blast zone varies according to prevailing wind.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELLER AND NOZZLE OPERATORS

The following procedures cover only the refueling procedures that are unique to the T-45 aircraft, primarily the operation of the "pre-check" system.

PRE-CHECK PROCEDURE

Note

Aircraft cannot be pressure fueled without electrical power available. If aircraft electrical power is not available, external power can be used.

1. Ensure REFUEL/DEFUEL switch is set to REFUEL.
2. Verify that INLET FUEL DISABLE indicator light and INLET VALVES L and R indicator lights are illuminated.

WARNING

If lights do not illuminate, discontinue refueling procedure; one or both transfer valves may be open resulting in a fuel spill.

3. Place tanker control in REFUEL position and start to pump fuel.
4. Press and hold PRECHECK switch and verify the following:
   a. FSLG VALVE light will illuminate within 15 seconds.
   b. Fuel flow will reduce to less than 5 GPM.
   c. Release PRECHECK switch.
   d. FSLG VALVE light extinguishes and fuel flow returns to normal.
5. Verify that air is being exhausted from fuel vent on the tail cone. If fuel is discharged, discontinue refueling immediately.
6. Continue refueling until fuel flow automatically stops.
7. Verify FSLG VALVE light is illuminated.

Note

The FSLG VALVE light will extinguish when refueling pressure is removed from the aircraft.

Since Aircraft 163600 through 163646 are not equipped with a "pre-check" system, the following special procedures for Hot Refueling apply to these aircraft:

HOT REFUELING (AIRCRAFT 163600 THROUGH 163646)

Perform hot refueling in the following sequence.

CAUTION

- The aircraft should not be hot refueled by the gravity fueling method.
- Due to the close proximity of the pressure refuel/defuel coupling to the port engine intake, ground crew should remain clear and aft of engine intake during hot refueling operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

1. Ensure that defueling valve is in the CLOSED position and defueling key removed.

2. Open access door 311EL.

3. Test and record refueling tanker quantity gauge and set flowmeter to zero.

4. Connect tanker grounding cable to static grounding point and aircraft grounding points.

5. Remove blanking caps from pressure fueling/ defueling coupling.

6. Connect nozzle grounding jack to aircraft grounding receptacle. Attach nozzle to refuel/ defuel coupling.

7. Ensure main refueling switch is set to REFUEL.

CAUTION

Failure to place the main refueling switch in the REFUEL position will energize the refuel/ transfer valves and result in a fuel spill.

8. Ensure external tanks refueling switches are in the OFF positions.

9. Place tanker control in REFUEL position, and start to pump fuel.

10. Verify that MAIN TANK refueling indicator light is illuminated.

11. If fuel discharges from the fuel vent, stop fueling immediately.

12. Pilot signals to stop refueling at 2,500 pounds of fuel.

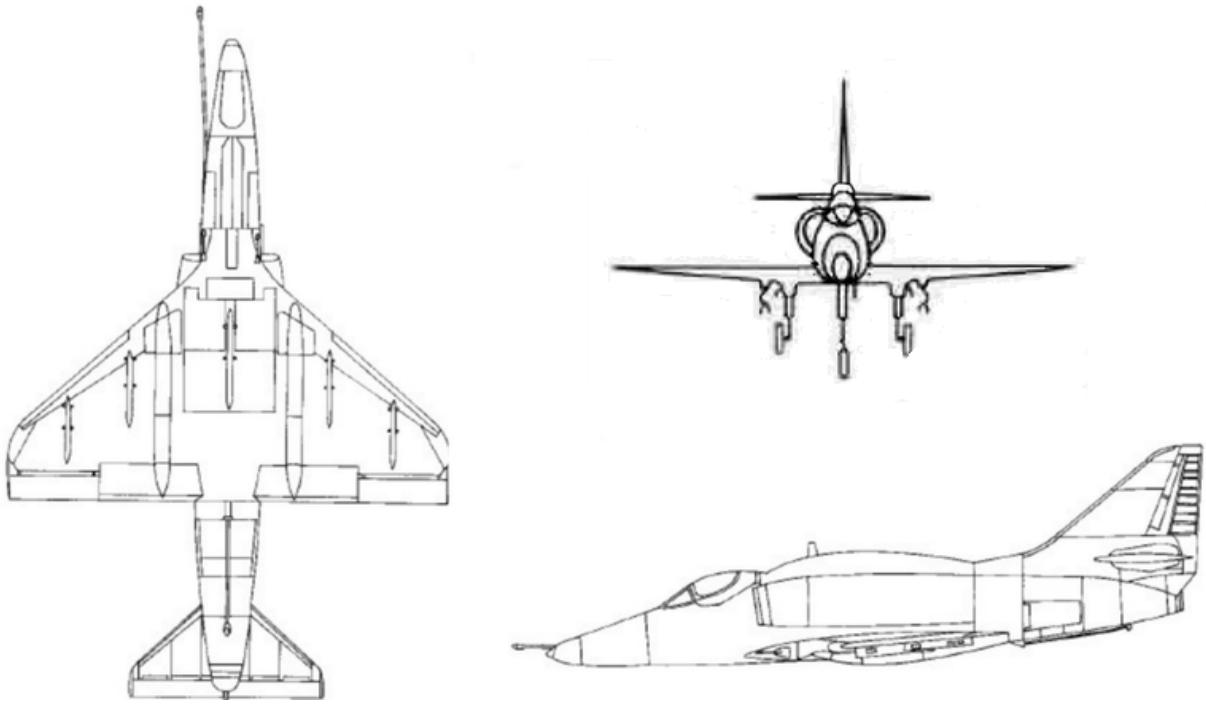
13. Verify that fuel flow stops when tanks are full and MAIN TANK light extinguishes. Stop tanker pump when refueling is complete.

14. Place tanker valve in OFF position. Remove nozzle and disconnect nozzle grounding jack from aircraft grounding receptacle. Install cap on fueling/defueling coupling and align yellow marks. Install sealing cover to nozzle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TA-4

The Douglas TA-4 Skyhawk is a carrier-capable ground-attack aircraft. It is powered by a single Pratt & Whitney J52 engine.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span	27 ft 6 in	Maximum Gross Weight — 24,500 lbs
Length	43 ft 7 in	Maximum Footprint — 350 psi
Height	15 ft 7 in	

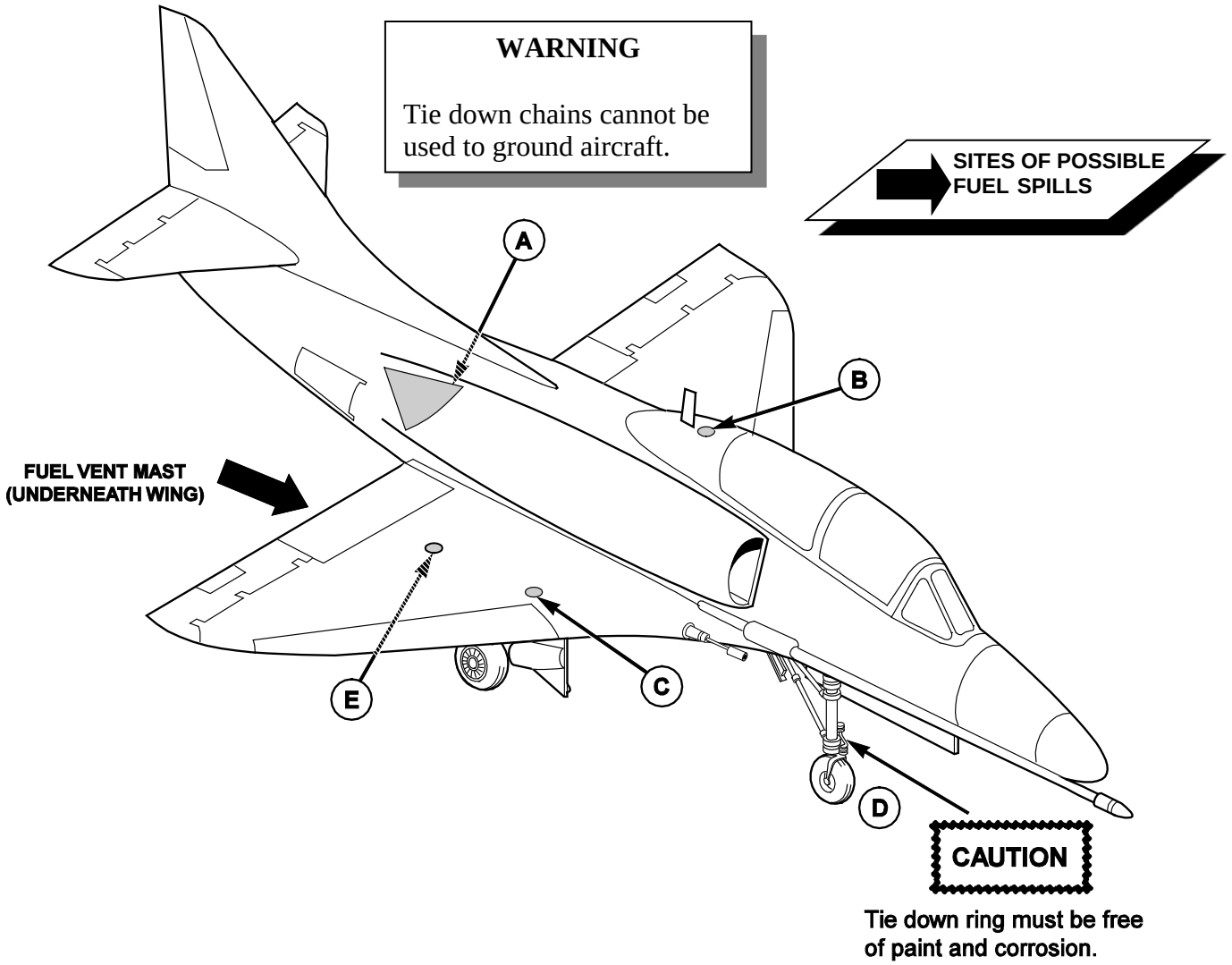
TABLE OF FUEL CAPACITIES

Tank		Gallons
Internal	Fuselage	104
	Wing	560
External	AERO 1C	147
	AERO 1D	295
	ATP-D1B	396
	Air Refueling Store	295

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

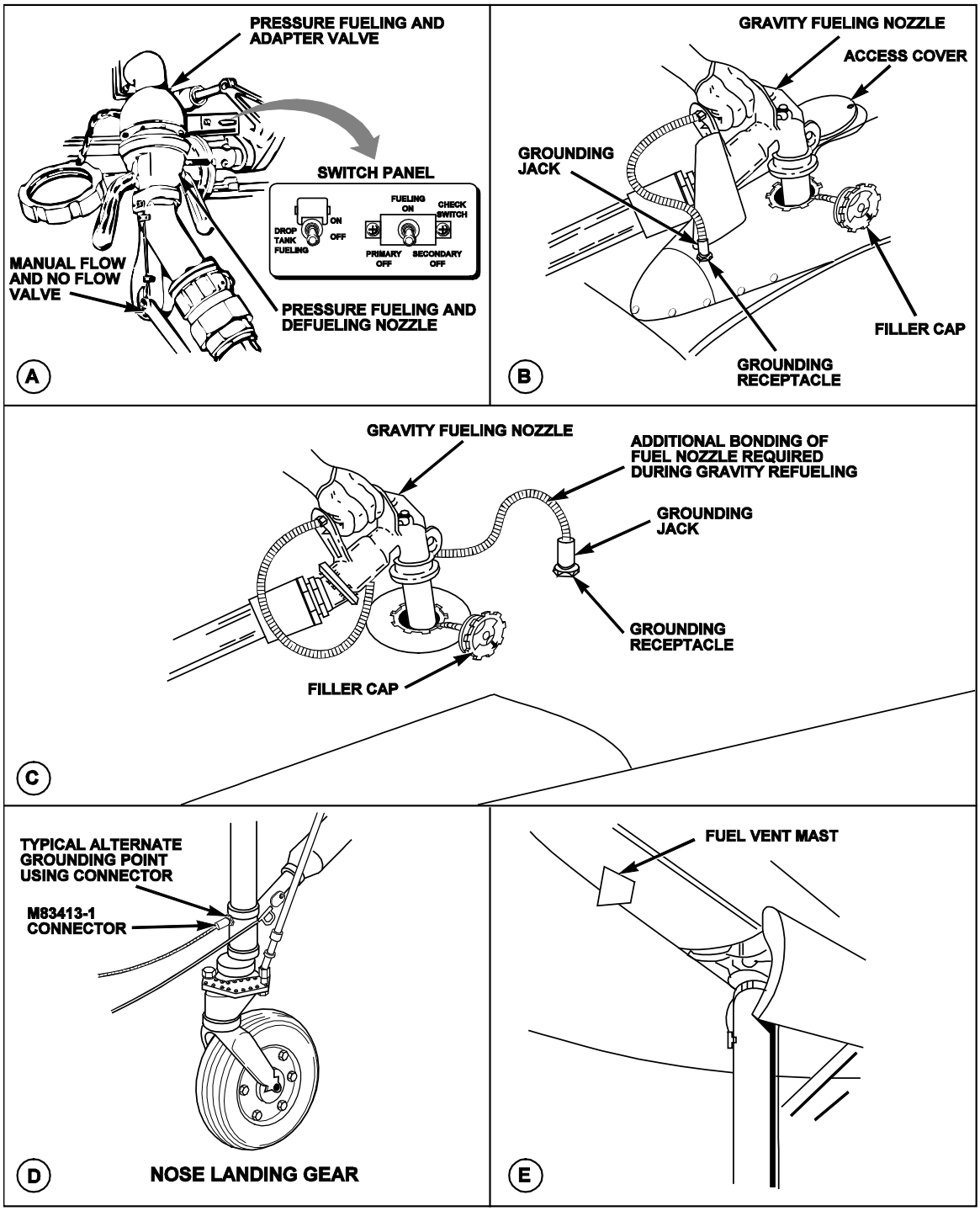
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points



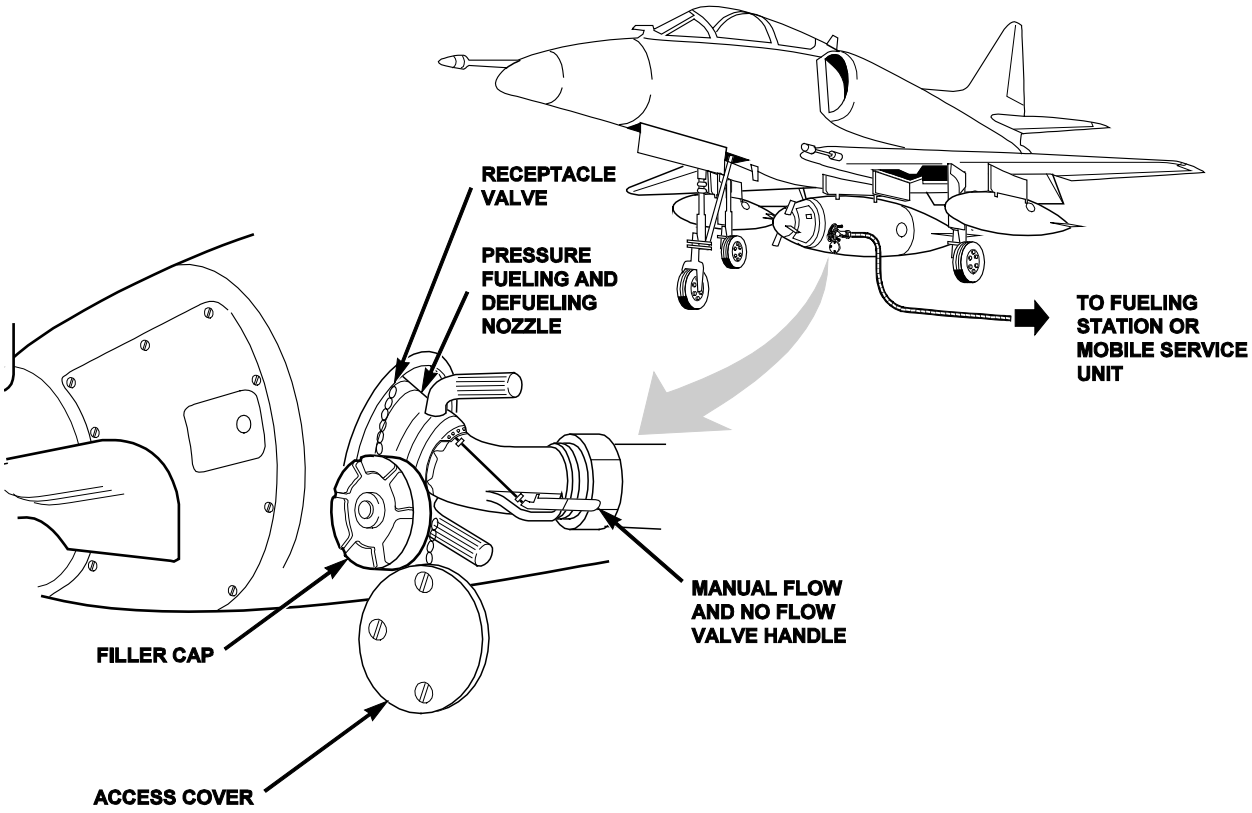
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) External Tank Refueling



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

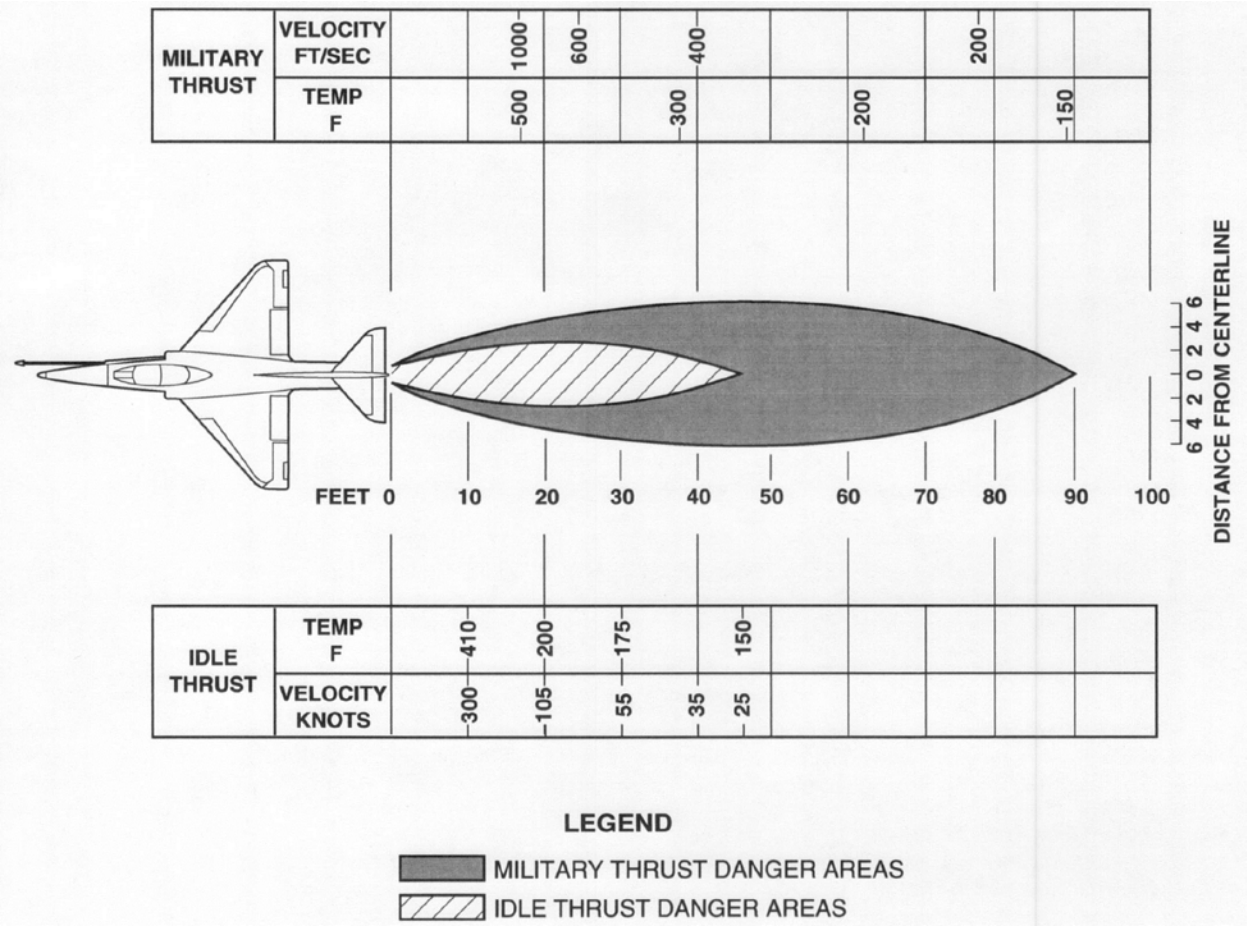
c) Electrical Grounding/Bonding Points



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Exhaust/Blast Area



CAUTION

- Approach aircraft from side, but not in line with turbine disintegration pattern.
- Stay clear of engine air inlet, exhaust, and gun and rocket patterns.
- Avoid wearing loose clothing or carrying loose articles near engine air inlet.

Do not foolishly experiment with safety margins noted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures that are unique to the TA-4 aircraft, primarily the operation of the “pre-check” system.

Pre-Check Procedures

1. Prepare aircraft and refueling system in accordance with the applicable procedure in Chapter 6, 12, or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109.

2. Ensure that CHECK SWITCH on fueling panel is in FUELING ON position 3. Connect external ac power to aircraft. (Refer to External Power Application.)

WARNING

Proper connection of ac external power cable plug to aircraft external power receptacle should be made. Failure to insert plug completely into receptacle can result in presence of high voltage on aircraft metal surfaces.

4. Place external ac power switch in EXTERNAL position.

5. Start pressure fueling equipment and open manual flow and no-flow valve on pressure fueling nozzle.

6. Immediately after pressure fueling has started, test the fuel vent system for proper functioning by holding the hand beneath fuel vent mast.

CAUTION

If air is not exhausting from fuel vent mast, stop pressure fueling immediately and investigate the fuel vent system. Failure to comply may result in damage to equipment and injury to personnel.

7. During the initial stage of pressure fueling, perform functional test of pressure fueling shutoff components in sequential order as shown in steps 8 through 12.

8. Place and hold CHECK SWITCH in PRIMARY OFF position. Fuel flow will stop in 1 to 3 seconds.

WARNING

If fuel flow does not stop in 1 to 3 seconds, stop fueling immediately and investigate cause.

Note

Because of fuel flow through pilot lines of float valves, a 2-gpm maximum flow (registered on pressure fuel meter) is permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

9. Return CHECK SWITCH to FUELING ON position. Fuel flow should start.

10. Place CHECK SWITCH in SECONDARY OFF position. Fuel flow will stop in 1 to 3 seconds.

11. Return CHECK SWITCH to FUELING ON position. Fuel flow should start.

12. Upon satisfactory completion of functional test, continue pressure fueling of aircraft.

13. During pressure fueling, inspect for evidence of fuel leakage. Correct if required.

CAUTION

If partial aircraft internal fuel load with full external tanks is desired, proceed as follows, disregarding steps 21, 22, and 23:

14. Place and hold CHECK SWITCH in the PRIMARY or SECONDARY OFF position, and place DROP TANK FUELING switch in the ON position. Fuel flow should start into external tanks only.

15. When fuel flow stops, return CHECK SWITCH to the FUELING ON position and return DROP TANK FUELING switch to tile OFF position.

CAUTION

If fuel flow does not shut off and overflows from an external fuel tank, stop fueling immediately and investigate the cause.

16. When internal quantity reaches desired amount, close manual flow and no-flow valve and shutdown pressure fueling equipment.

CAUTION

If a full fuel load is desired, disregard steps 14, 15, and 16, and proceed as follows.

17. Place DROP TANK FUELING switch in the ON position. Fuel flow should commence to the external tanks.

18. After fuel flow has stopped (cell and tanks full), check fuel-delivered meter for indication of pressure fueling system internal leakage. Maximum leakage is not to exceed 1 gpm.

19. Close manual flow and no-flow valve, and shut down pressure fueling equipment.

20. Place external ac power switch in INTERNAL position.

21. Disconnect external ac power from aircraft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

22. Disconnect pressure fueling nozzle from pressure fueling/defueling adapter valve and install cap.

23. Place DROP TANK FUELING switch in OFF position.

24. Close and secure aft fuselage lower access door, if applicable.

25. Replace external fuel tank caps, if removed.

HOT REFUELING

The TA-4 aircraft can be hot refueled only (engine running) through the aerial refueling probe because of the location of the pressure refueling adapter. The aircraft and shore-based refueling system should be prepared in accordance with paragraph 12.9 of Chapter 12 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109.

The following procedures should be strictly followed by pilots and ground personnel when refueling aircraft with the engine running.

Note

TA-4 aircraft will be hot refueled through the probe only.

PRIOR TO REFUELING

1. The aircraft will be chocked.

2. The plane captain should attach grounding wire to aircraft before any other connections are made.

3. The pilot will close and lock the canopy and select RAM air. The radio will be set on ground control or, if aboard ship, other appropriate frequency.

4. Drop tank pressurization switch should be placed in OFF position.

5. Ensure that FUEL TRANSFER BYPASS switch (A-4 AFC-317) is in the OFF position.

WARNING

Ensure that fuel dump switch is in OFF position prior to refueling.

6. The work stand should be positioned in front of inflight refueling probe and wheels should be locked.

7. The nozzle operator should attach nozzle adapter to the probe.

8. The pilot should signal to fuel pit coordinator when he is ready for commencement of fueling by thumbs-up signal in daytime, and by thumbs-up illuminated by flashlight at night.

AFTER COMMENCEMENT OF HOT REFUELING

The plane captain should visually check drop tanks upon completion of hot refueling to ensure that tanks are either completely full or completely empty. This procedure applies whether the drop tank fueling switch has been placed in the ON or OFF position during refueling. In all hot refueling operations the pilot should check his external fuel gauge to ensure the proper fuel load after completion of refueling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

1. Nozzle operator should slowly open valve and check for fuel leaks until the valve is fully open.

2. Immediately upon commencement of fueling, the plane captain should conduct the primary and secondary valve checks in accordance with the Maintenance Instruction Manual, NAVAIR 0140AVD-2-4.l. If this check is not satisfactory, the refueling operation should be secured immediately.

3. If the drop tanks are to be filled, the plane captain should either place the DROP TANK FUELING switch in the ON position, or the pilot should place the DROP TANK pressurization switch to the FLIGHT REFUEL position.

Note

When refueling ashore, gravity fueling methods should be used for partial drop tank loads.

4. Appropriately assigned personnel should monitor vent mast on wing to ensure that it is not obstructed.

WARNING

If pressure cannot be felt coming from the vent mast, the refueling operation should be secured immediately.

5. The pilot should signal the pit coordinator by a cut signal in the daytime and by flashlight at night when refueling is completed. The pit coordinator should signal the nozzle operator and pit operator. The nozzle operator should close the valve on the nozzle but should not remove the probe adapter until the pit operator has evacuated all fuel from the hose.

Note

The rotating beacon should be used as an emergency fuel cutoff signal at night.

6. The ground wire should be detached after all other refueling equipment is removed.

7. A qualified lane director should direct the aircraft out of the pits.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. The fuel grade selector on the engine fuel control should correspond to the grade of fuel being used. After JP-4 is introduced into the fuel system of the engine, the specific gravity switch on the fuel control should be set to JP-4. The setting should not be returned to JP-5 until after the first flight during which JP-5 has been used. When using a combined fuel load of JP-4 and JP-5 on the JP-4 setting, the pilot should monitor EGT to ensure that full throttle operation remains within the prescribed limits. Use the JP-5 setting on the control when using JP-8 fuel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

UC-35

The Navy UC-35C and UC-35D aircraft are derivatives of the Cessna Citation family of aircraft. These aircraft are turbofan-powered, small-to-medium sized business jets.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Wing Span Spread	52 ft 5 in	Maximum Gross Weight — 16,300 lbs
Length	48 ft 11 in	
Height	15 ft 0 in	

TABLE OF FUEL CAPACITIES

UC-35C (Aircraft –001 thru –0538)

Total Left Wing	433.2 USG
Total Right Wing	433.2 USG
Usable Left Wing	431.0 USG
Usable Right Wing	431.0 USG

(based on 6.75 pounds/USG)

UC-35D (Aircraft –0539 and on)

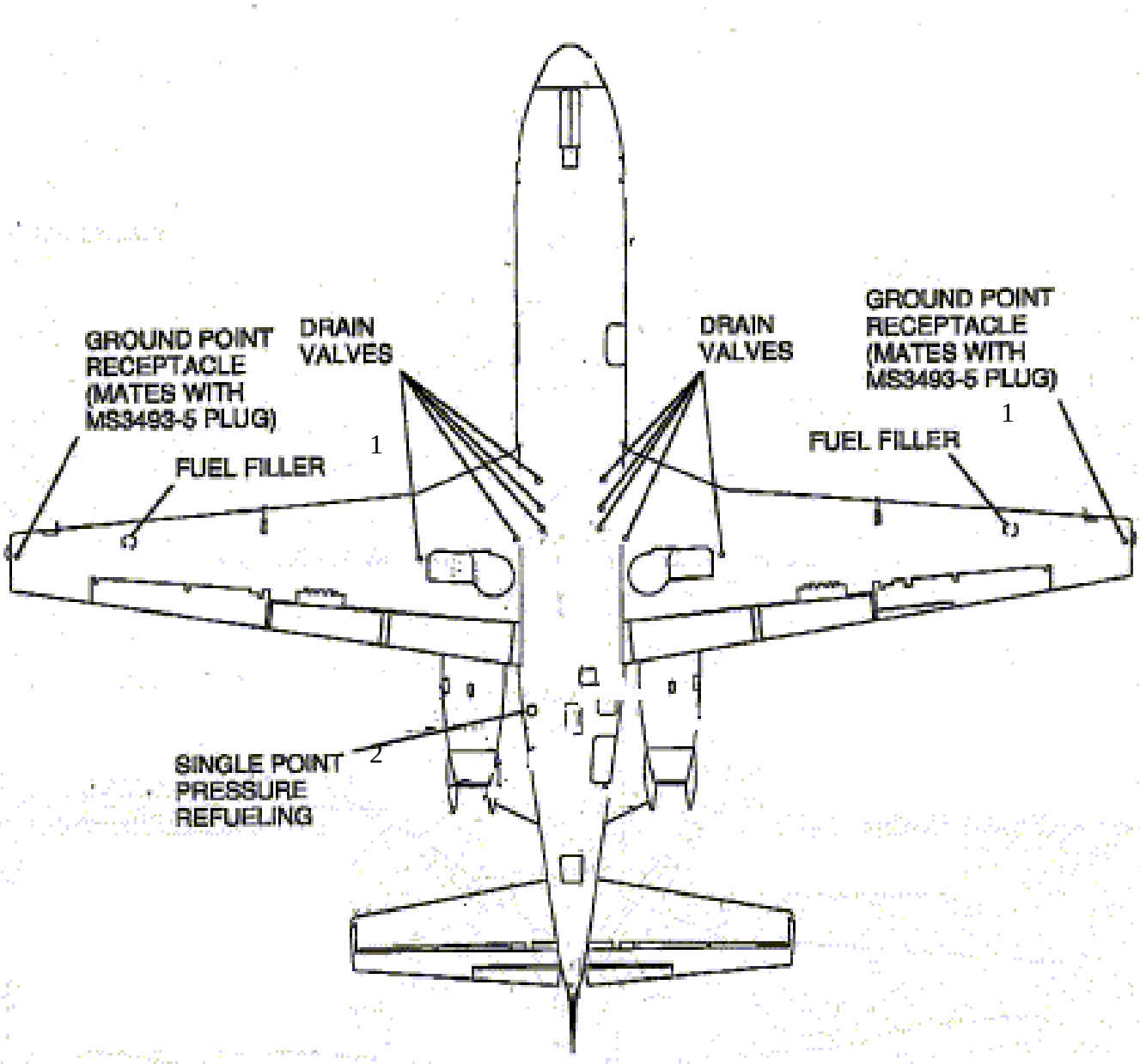
Total Left Wing	406.4 USG
Total Right Wing	406.4 USG
Usable Left Wing	403.0 USG
Usable Right Wing	403.0 USG

(based on 6.75 pounds/USG)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Drain Valves
(Underside, looking up)

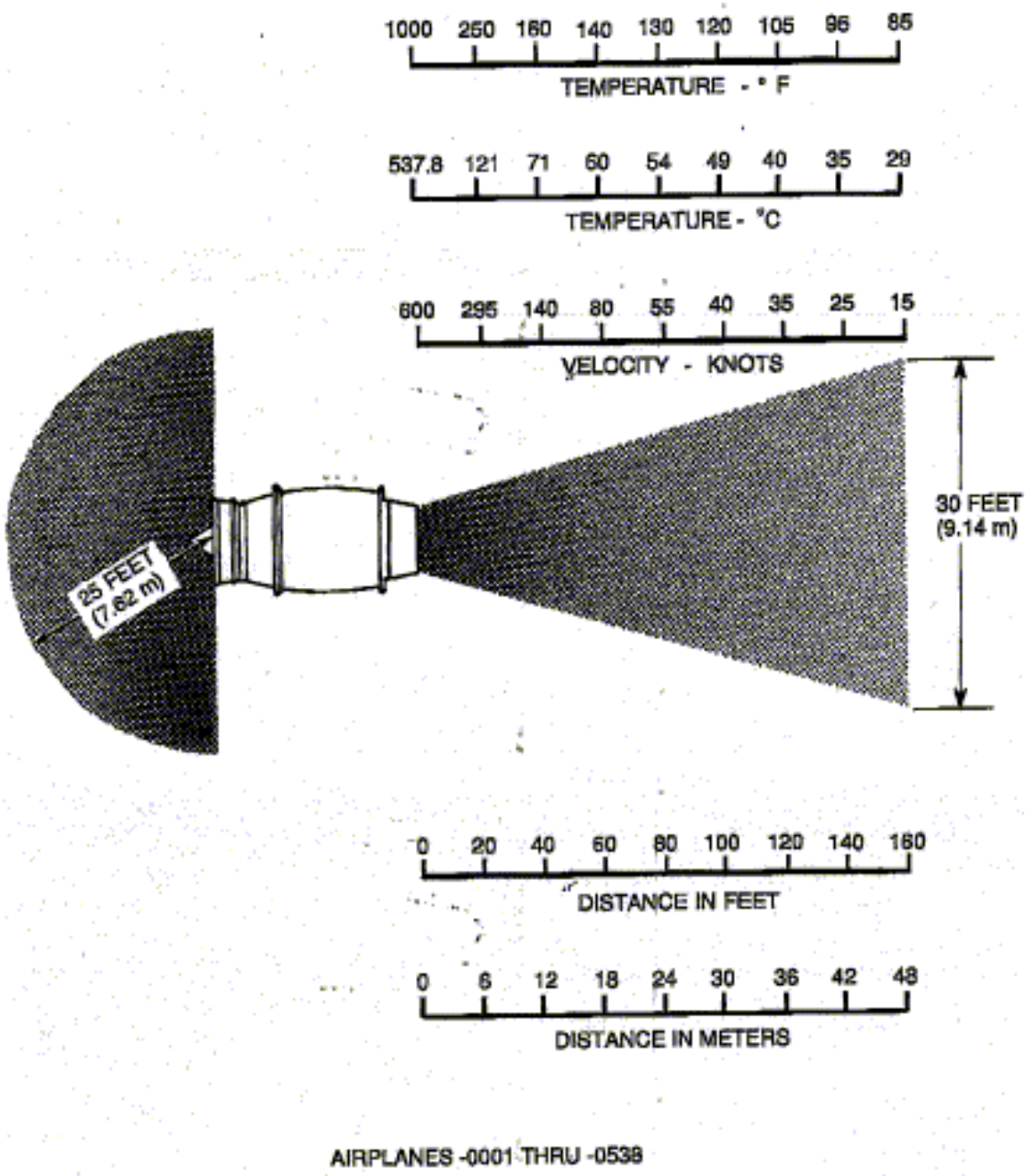


Notes: 1. Fuel filler caps are located on topside of wings.
2. SPR adapter is on aircraft -0293; -0307 thru -0538; and -0539 and on.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

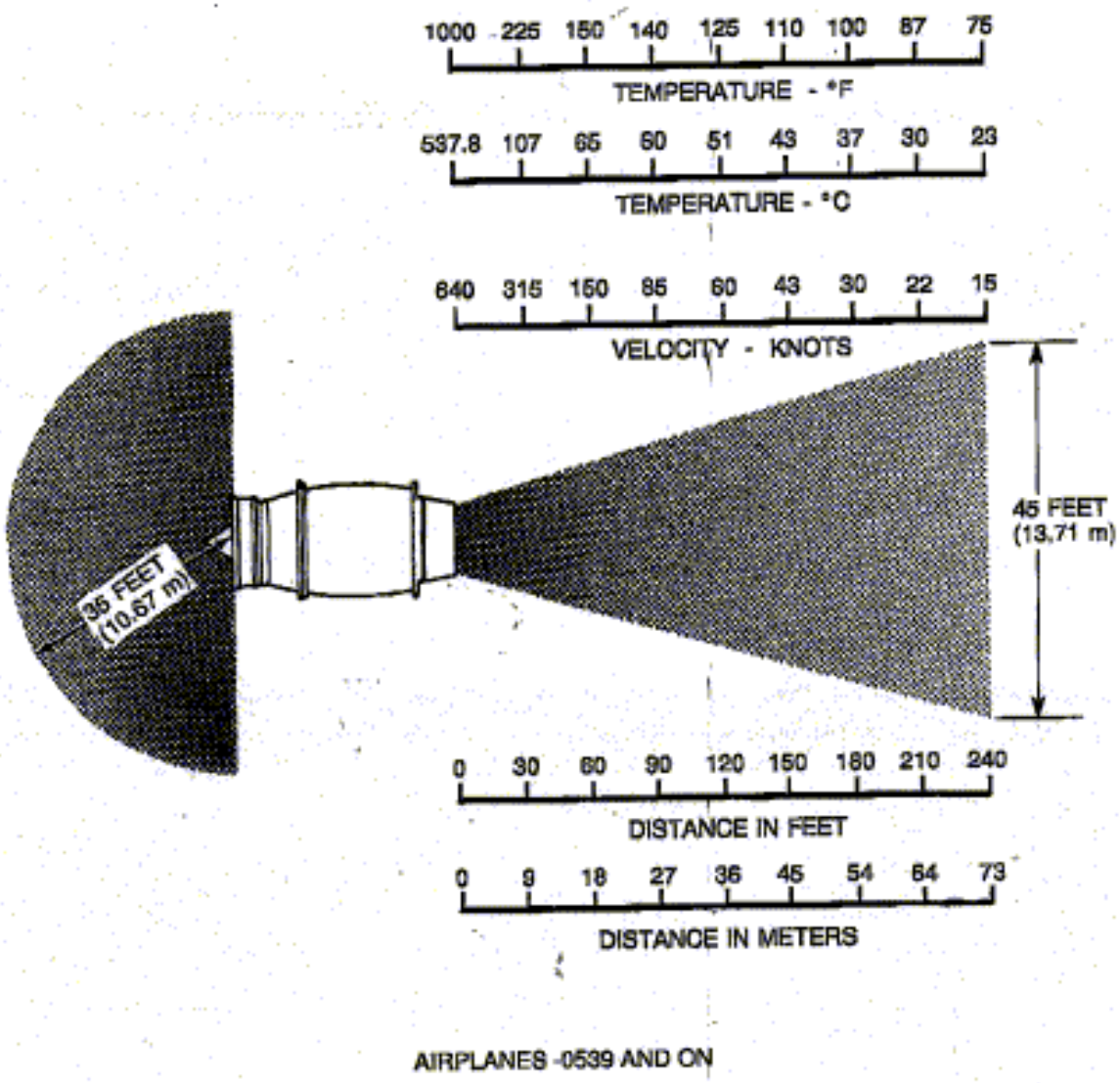
a) U-35C Engine Inlet/Exhaust/Blast Area



AIRPLANES -0001 THRU -0538

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) U-35D Engine Inlet/Exhaust/Blast Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

Overwing Tank Filling Procedures

WARNING: Perform fuel loading in area that permits free movement of fire equipment

WARNING: Ensure that fuel supply unit and airplane are grounded

1. Connect fueling nozzle ground to the airplane's grounding receptacle, located on the lower side of the wing, outboard of the filler cap.
2. Place a protective pad on the wing adjacent to the fuel filler and remove the filler cap.
    a. NOTE: Due to the position of the keyholes lock freezing may be encountered on airplanes with locking type filler caps. Heating the key prior to inserting it in the lock during inclement weather can reduce the freezing possibilities.
3. Service as follows
    a. If the turbine fuel has fuel system icing inhibitor added, fill wing tanks.
    b. If the turbine fuel does not have fuel system icing inhibitor added, select an inhibitor, refer to Tools and Equipment, and add as described by the inhibitor manufacturer or in accordance with Mixing Icing Inhibitor Procedures

    Caution: Ensure Filler Cap Is Secured
4. Remove fuel nozzle and protective pad, disconnect fueling nozzle ground and install fuel filler cap.

SINGLE-POINT PRESSURE REFUELING

CAUTION

Ensure the proper grade and type of fuel are used to service the airplane

The single-point refueling control panel, if installed, is located on the right side of the fuselage aft of the wing. The control panel consists of the refuel/defuel adapter (receptacle) and a refueling precheck panel. For access to the refueling control panel, open the control panel access door.

WARNING

Observe all technical and personal safety precautions when handling fuel.

Note

Single-point fuel pressure at the refueling nozzle should not exceed 60 psi maximum for airlines -0001 thru –D538 and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

should not exceed 50 psi maximum for airplanes –D539 and on.

Note

Fuel flow should not exceed 105 gpm maximum.

CAUTION

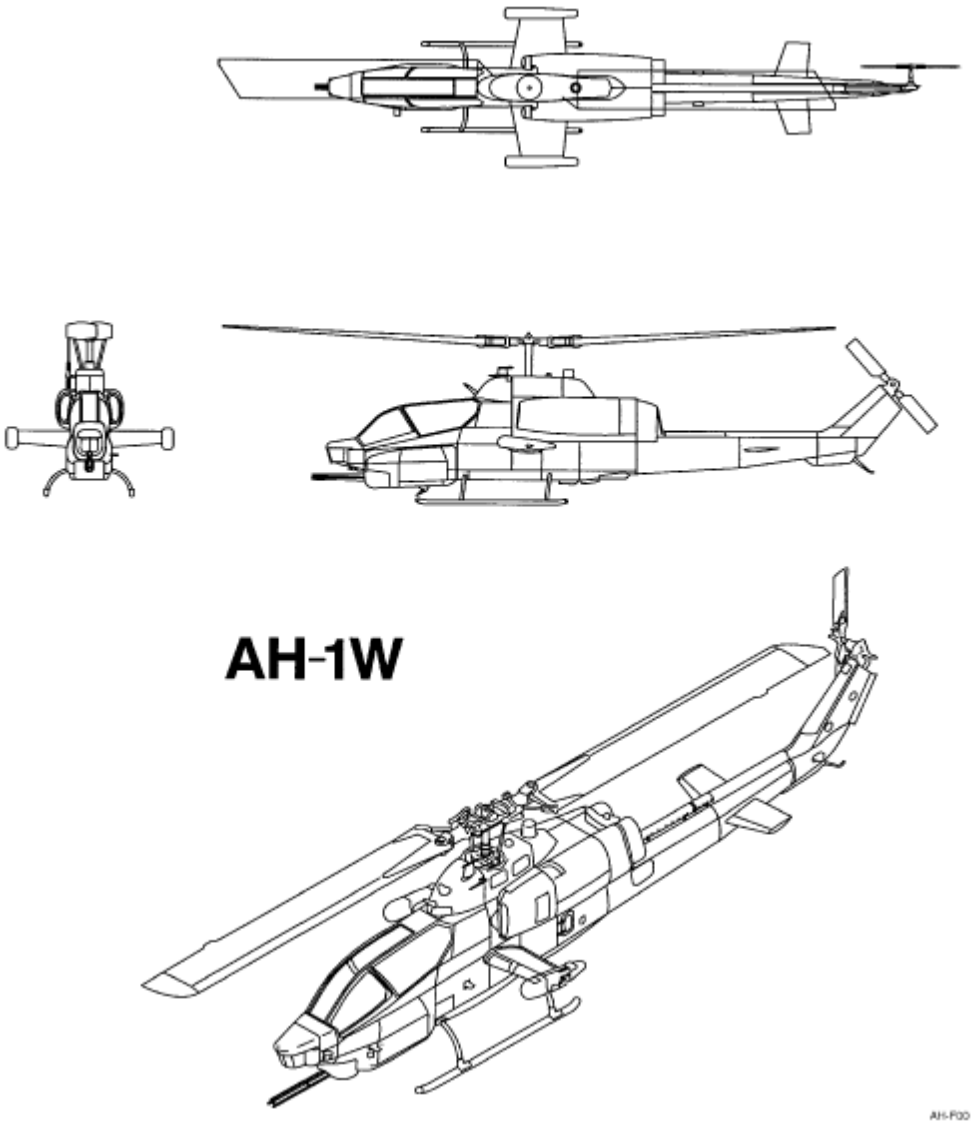
Perform a refueling precheck before each single-point refueling.

1. Verify fire-fighting equipment is readily available.
2. Open the refuel/defuel control panel access door.
3. Prepare the airplane for refueling by properly grounding the airplane and refueling vehicle/equipment together with an approved static grounding source. Refer to safety and maintenance precautions.
4. Ensure the airplane fuel vents are not obstructed.
5. Remove adapter cap.
6. Insert the refueling nozzle into the receptacle, turn clockwise and latch in place, open nozzle.
7. Start fuel flow and perform a system precheck to ensure the pilot valves and/or fuel shutoff valves are operating properly.
8. When the airplane fuel reservoirs become full, the high level pilot valves cause the fuel shutoff valves to close and fuel flow is discontinued automatically. Shut down pumping equipment (vehicle or hydrant equipment).
9. Verify the airplane fuel reservoirs are fully serviced by operating and checking the fuel quantity indicators.
10. Disconnect the refueling nozzle from the adapter (receptacle), install adapter cap.
11. Close and secure the refuel/defuel control panel access door. Remove all grounding cables and move the airplane or refueling vehicle from the area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AH-1

The AH-1 helicopter is an armed, tactical helicopter manufactured by Bell Helicopter Textron, and powered by two General Electric T700-GE-401 turboshaft engines. The primary mission is that of search and target acquisition, laser designation and range finding, low-altitude, high-speed flight, multiple weapon fire support, reconnaissance by fire, and assault support escort.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width	 10 ft 9 in	Maximum Gross Weight — 14,750 lbs
Length	58 ft	Maximum Footprint — 70 psi
Height	14 ft 2 in	

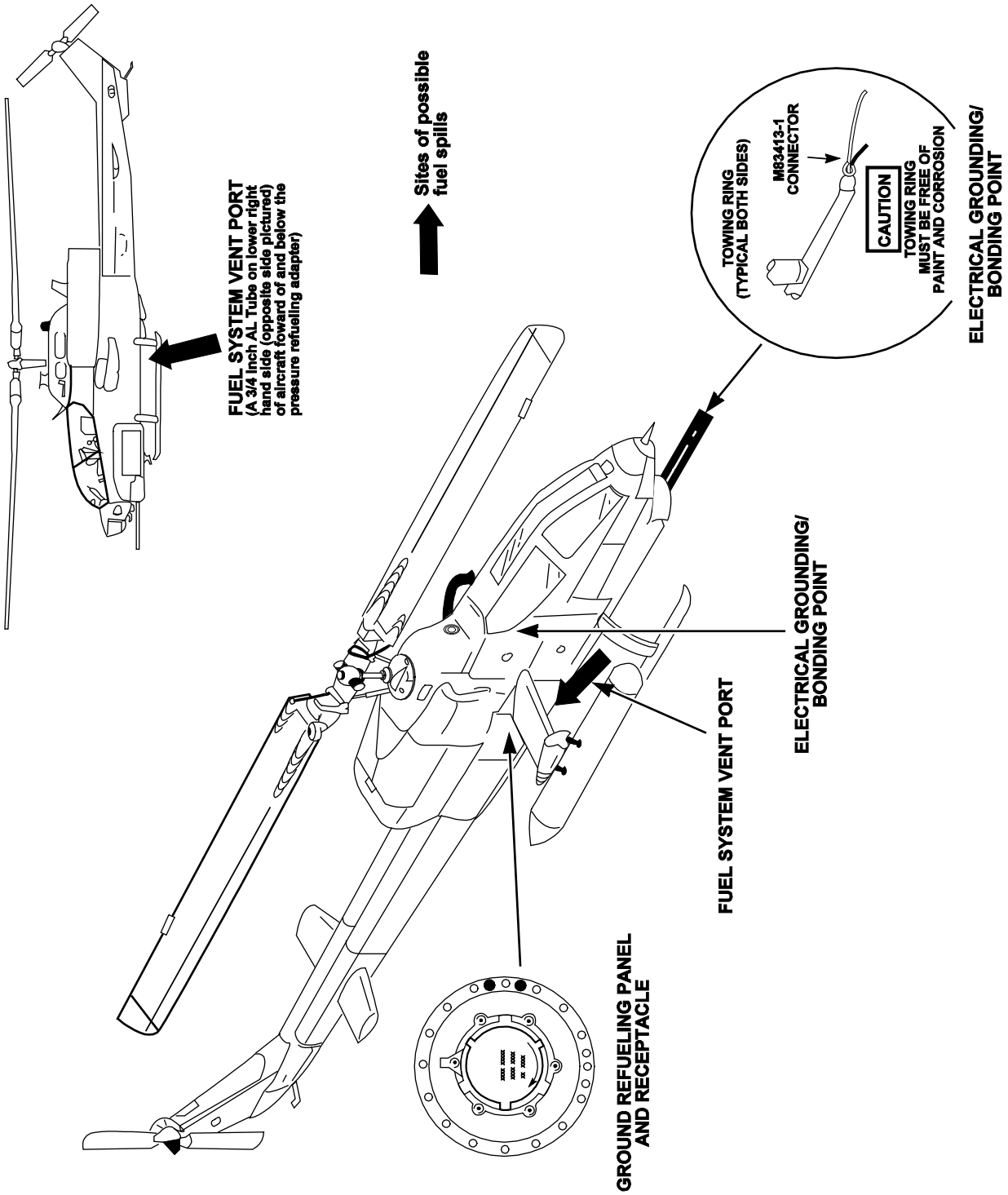
TABLE OF FUEL CAPACITIES

Tank		Gallons
Internal	Forward	190
	Aft	123
	Total	313
External Wing Tanks (each)	2 Maximum	100
	4 Maximum	77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

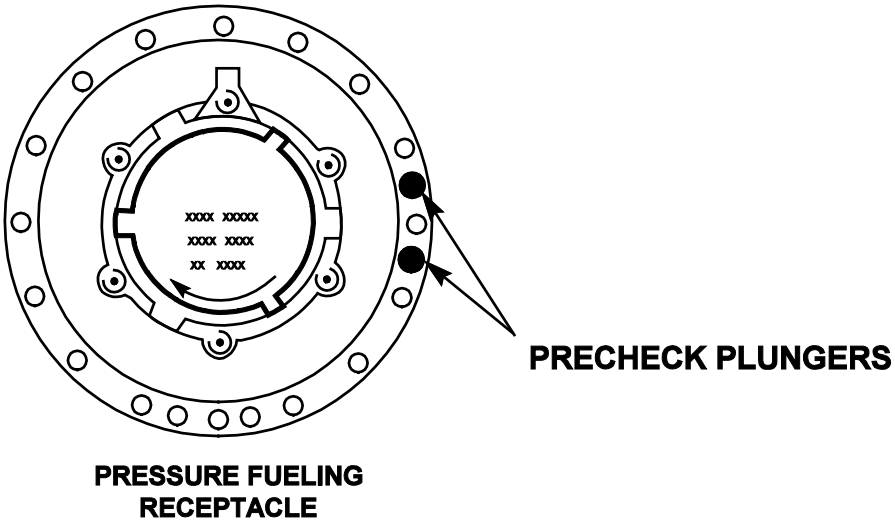
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Location of Fuel Vents and Ground Refueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

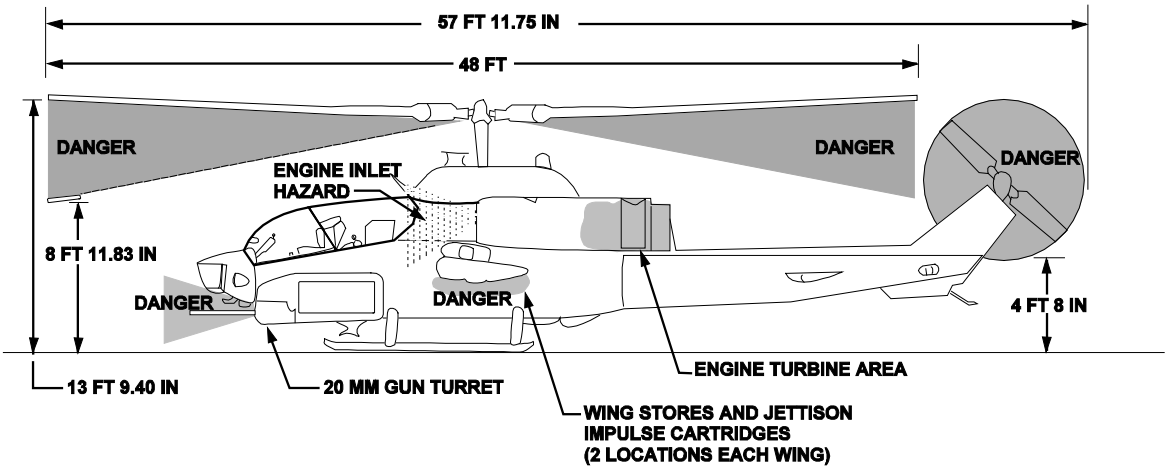
b) Ground Refueling Panel

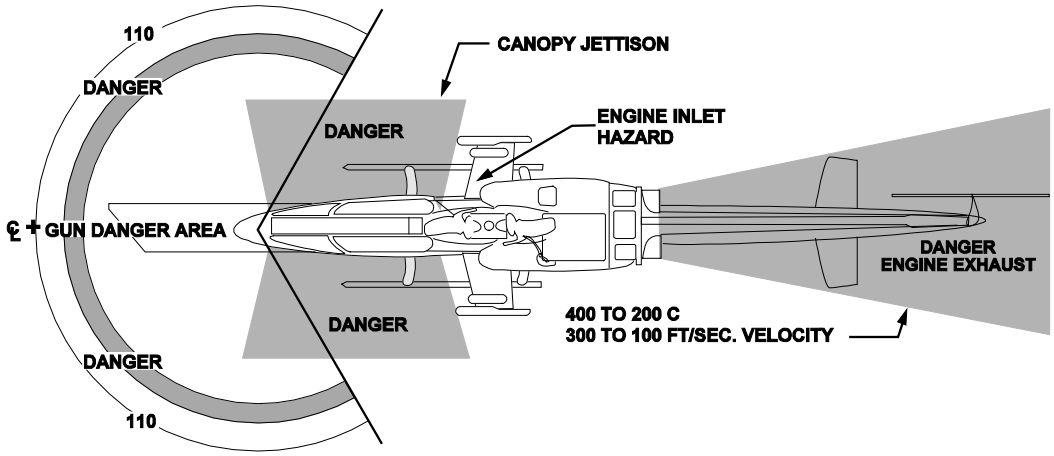


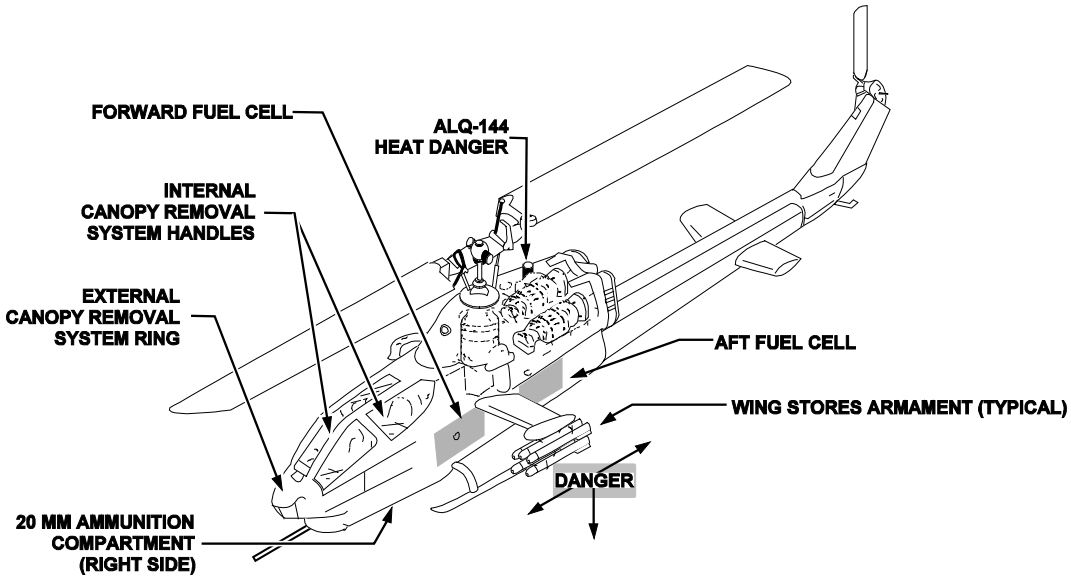
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

Engine Exhaust and Helo Roto Clearance







Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the AH-1 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.

WARNING

Ensure that air is venting from "FUEL SYSTEM VENTS." If no venting is indicated, cease fueling operation immediately.

2. Exercise the Precheck system. Press and hold one of the "PRECHECK" plungers on the rim of the fueling valve. Fuel flow into the aircraft should stop within 20 seconds. Release the "PRECHECK" plunger and fuel flow should resume into the aircraft.

3. Repeat the last step this time holding the other "PRECHECK" plunger down.

WARNING

Flow of fuel while either of the "PRECHECK" plungers is held down indicates a failed shutoff valve. Stop refueling immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

4. Release both "PRECHECK" plungers and refuel aircraft.

5. Continue fueling until fuel flow stops automatically.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common "Fuel System Vent Port," which is a 3/4-inch AL tube on the lower right-hand side of the aircraft forward of and below the pressure refueling adapter. It is almost on the bottom of the aircraft.

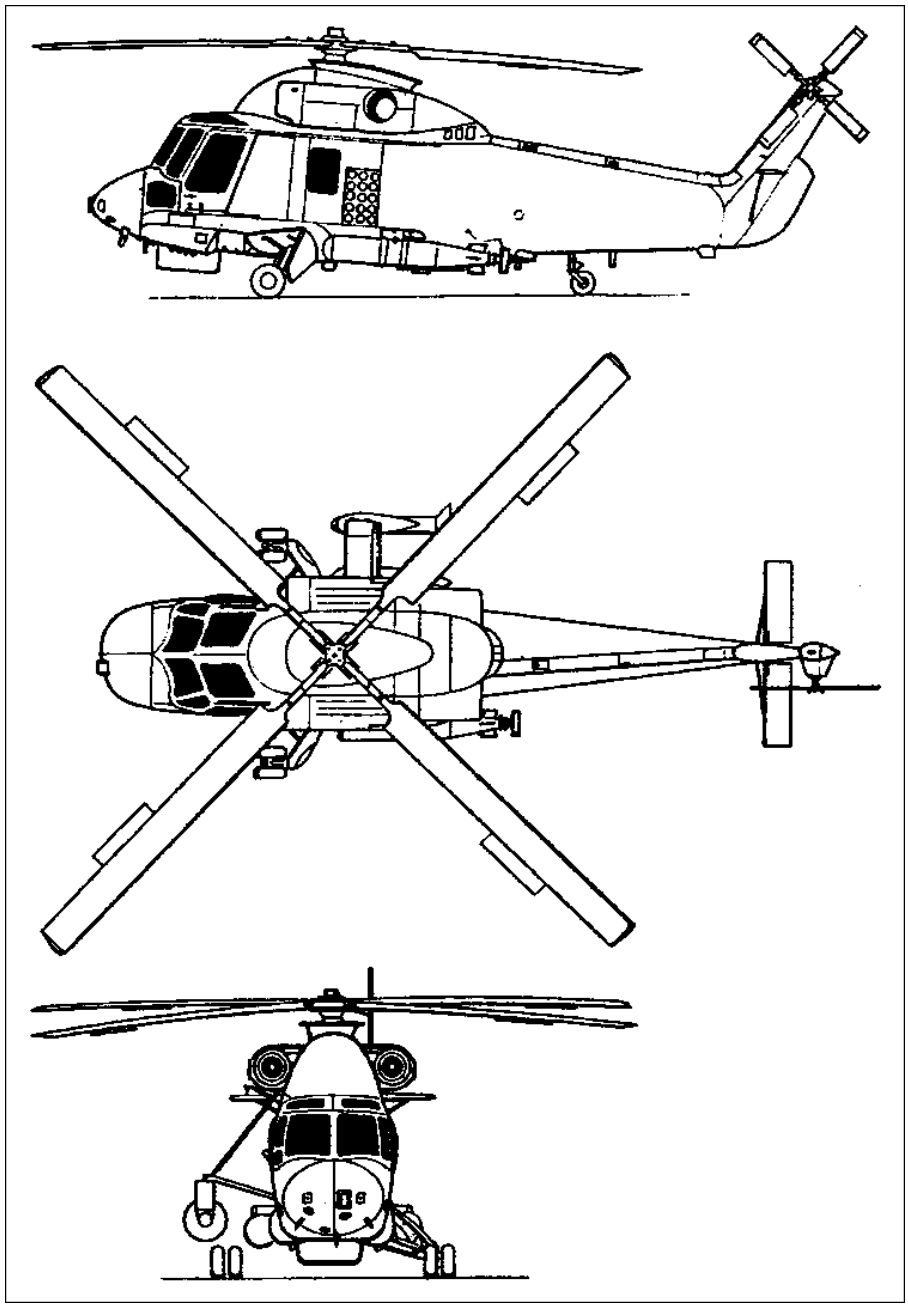
2. If any high level shut-off valves fail to operate correctly, fuel may spill from one of the "Fuel System Vent Port." In addition, the fuel tanks may rupture and spill fuel.

3. The external tanks cannot be pressure refueled. Each external tank should be gravity refueled separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-2

The Kaman H-2 is a ship-based helicopter, originally developed in the late 1950s as a fast utility helicopter. In the 1970s, anti-submarine, anti-surface threat capabilities were added to the design, including over-the-horizon targeting.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width	12 ft	Maximum Gross Weight — 13,500 lbs
Length	52 ft 0.3 in	Maximum Footprint — 225 psi
Height	14 ft 4.3 in	

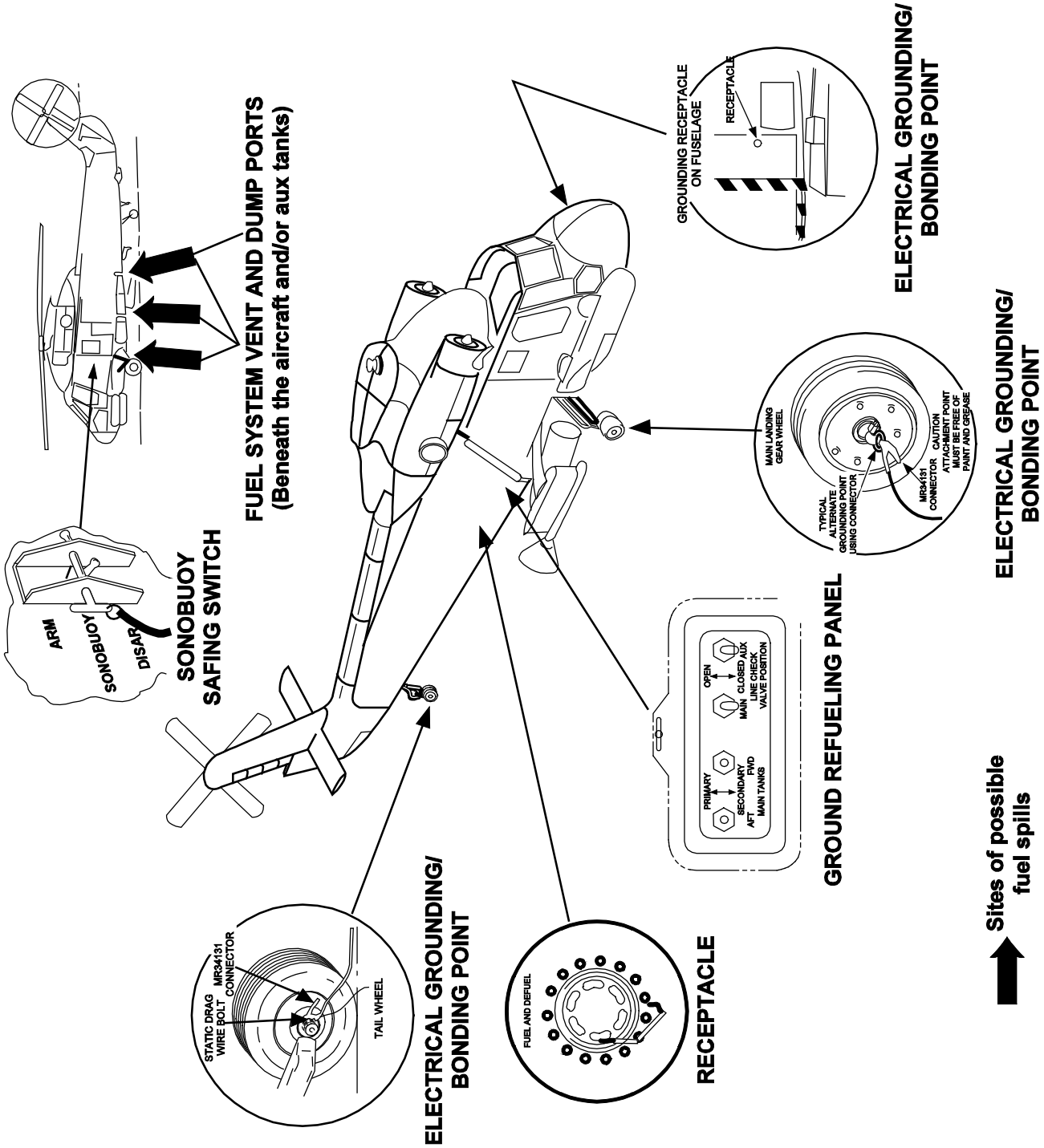
TABLE OF FUEL CAPACITIES

Tank			Gallons
	Internal	Sump	99
		Aft	176
		Total	275
Ext Aux Tanks (each)	Before AFC 293		59
	After AFC 293		99
Totals (With 2 Aux Tanks)	Before AFC 293		393
	After AFC 293		473

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

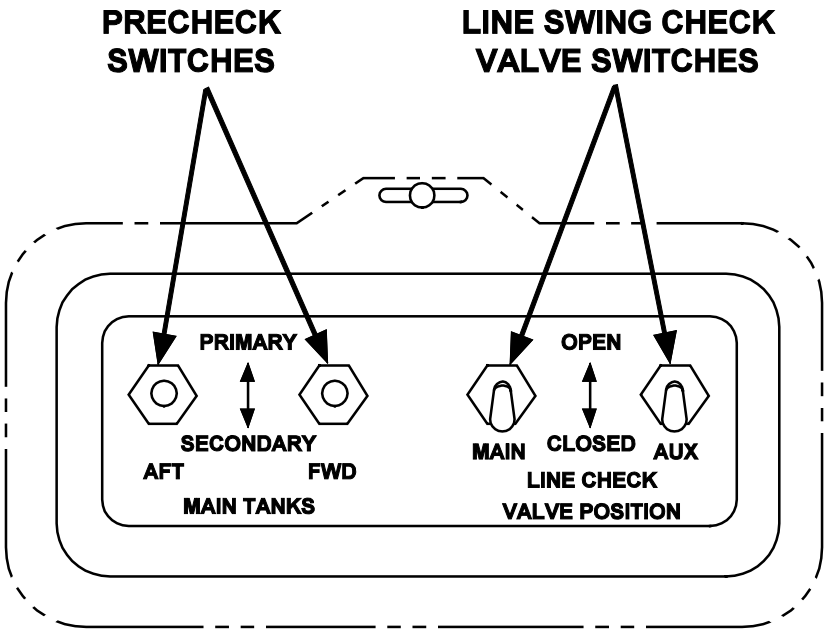
AIRCRAFT CONFIGURATION

a) Electrical Grounding/Bonding Points, Location of Fuel Vents, Dump Ports, and Low Point Drains



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

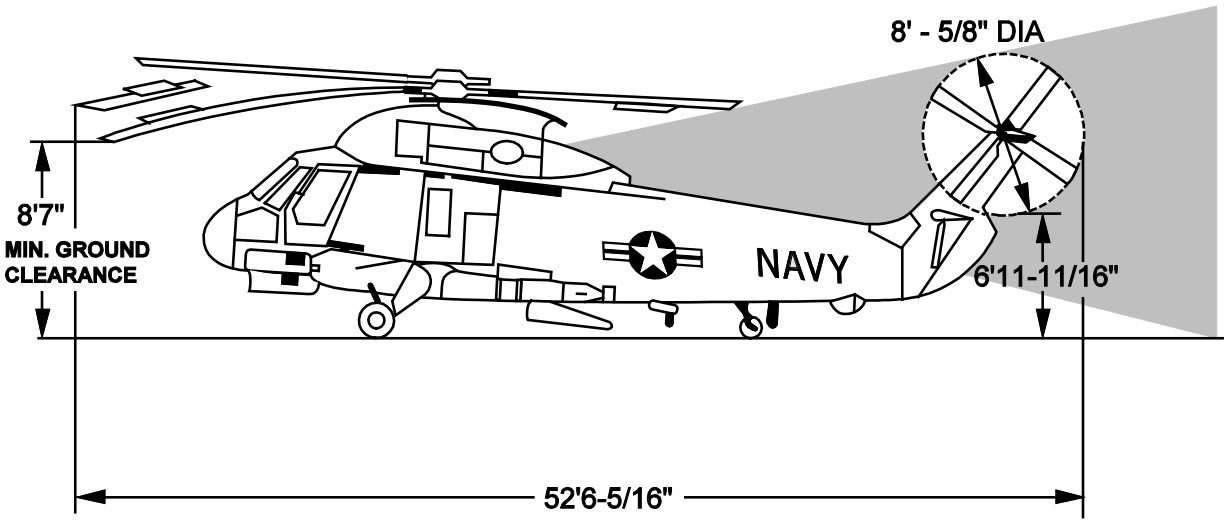
b) Ground Fueling Panel and Locations of Receptacles

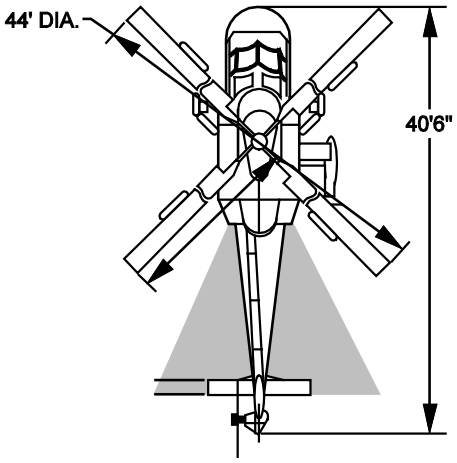


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Helo Rotor and Engine Exhaust Area





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the H-2 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

CAUTION

Sonobuoy Safing Switch on the SH-3 aircraft should be placed in the "DISARM" position prior to the initiation of the hot refueling operation.

1. Remove receptacle cap and attach refueling nozzle.
2. Open Ground Refueling Panel door. Place the "MAIN" and "AUX" "LINE WING CHECK VALVE SWITCHES," in the "CLOSED" position.
3. Initiate fuel flow into the aircraft.
4. Exercise the Precheck system. Press and hold both "PRECHECK SWITCHES" on the Ground Refueling Panel to "SECONDARY" position. Fuel flow into the aircraft should stop within 20 seconds. Repeat the last step this time holding both "PRECHECK SWITCHES" in the primary holding position.

WARNING

Flow of fuel while precheck switches are in the "SECONDARY" or "PRIMARY" positions indicates a failed shutoff valve. Stop refueling immediately. System failure needs to be investigated and resolved before hot refueling can be accomplished.

5. Return both "PRECHECK SWITCHES" on Ground Refueling Panel to their normal positions.
6. Stop the flow of fuel into the aircraft by closing the poppet valve on the aircraft refueling nozzle.
7. Place the "AUX" and "MAIN" "LINE WING CHECK VALVE SWITCHES" in the "OPEN" position.
8. Open refueling nozzle poppet valve and refuel aircraft.
9. Continue fueling until fuel flow stops automatically.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common “Fuel System Vent Ports,” which are located on the bottom of the aircraft and the auxiliary tanks (if installed).

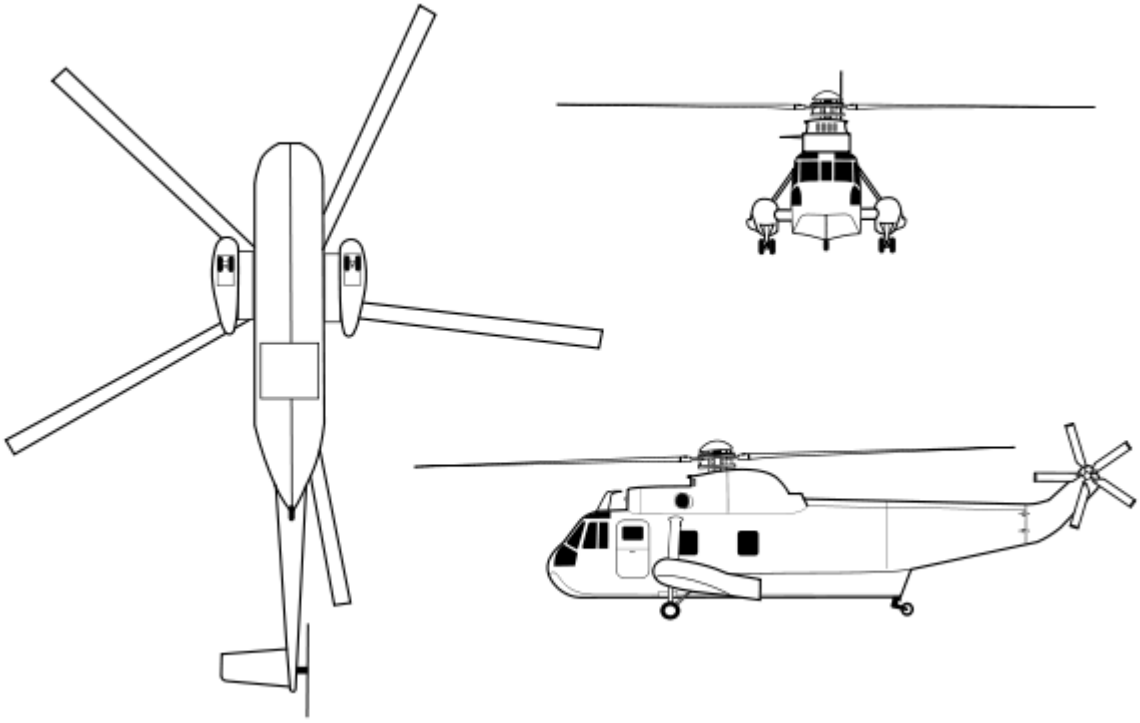
2. If any high-level shut-off valves fail to operate correctly, fuel may spill from one of the “Fuel System Vent Ports.” In addition, the fuel tanks may rupture and spill fuel.

3. The “LINE WING CHECK VALVE SWITCHES” can be used to selectively refuel only the main internal tanks on the aircraft. Place both of these switches in the “OPEN” position to obtain maximum fuel load (including auxiliary tanks).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-3

The Sikorsky SH-3 Sea King is a twin engine anti-submarine warfare, search and rescue, and utility helicopter.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width	14 ft	Maximum Gross Weight — 20,000 lbs
Length	72 ft 8 in	Maximum Footprint — 90 psi
Height	16 ft 10 in	

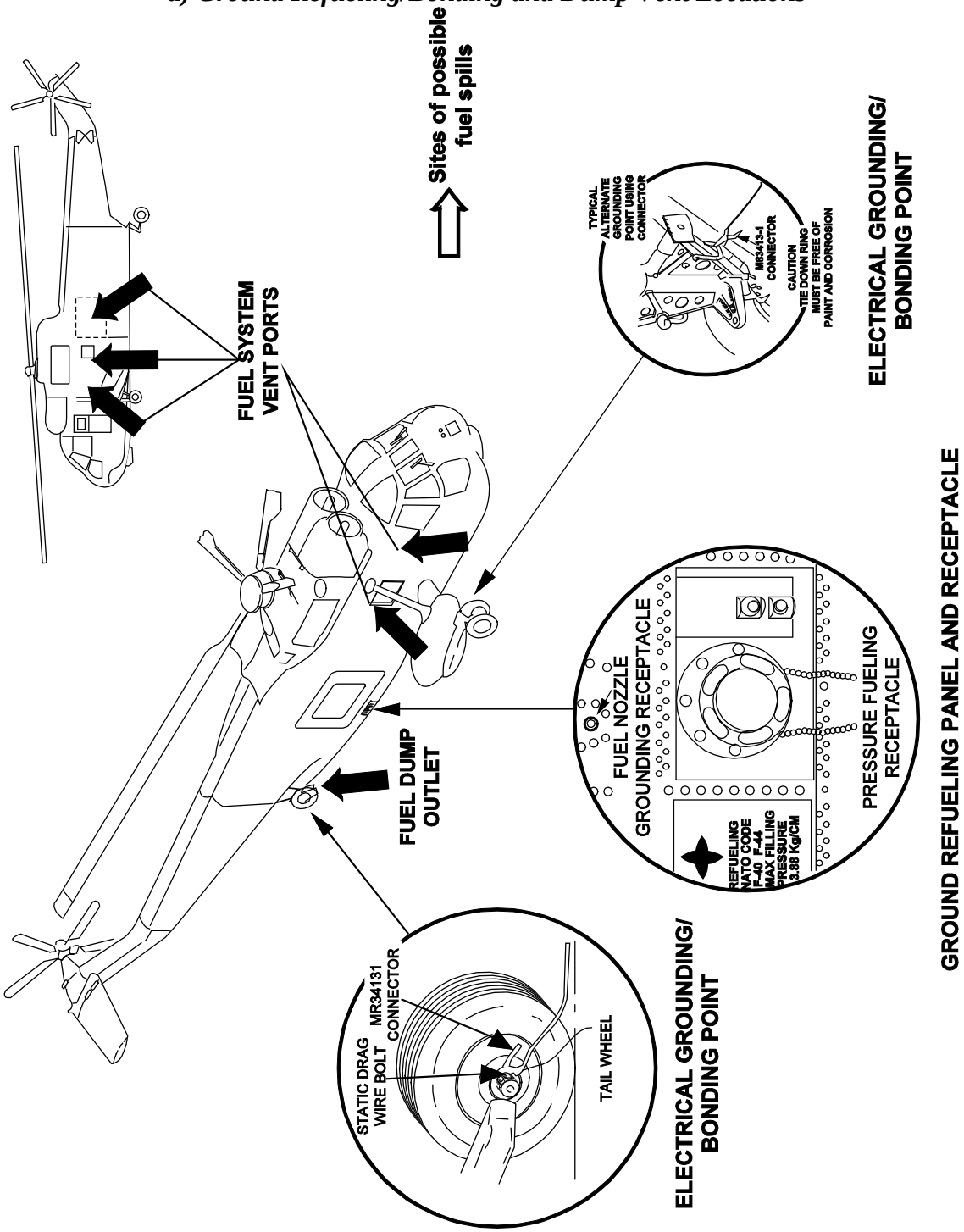
TABLE OF FUEL CAPACITIES

Tank		Gallons
Internal	Forward	341
	Aft	344
	Total	685

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

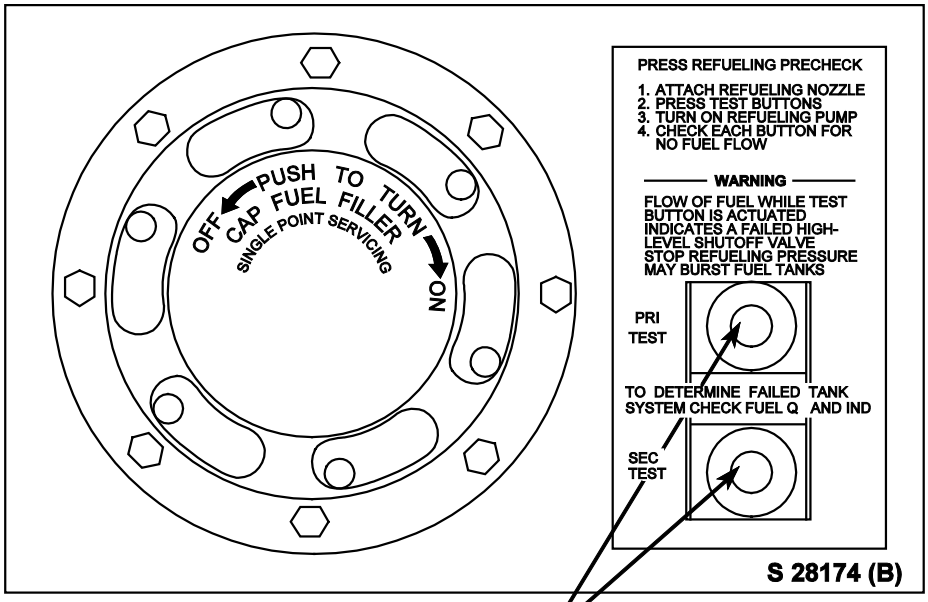
AIRCRAFT CONFIGURATION

a) Ground Refueling/Bonding and Dump Vent Locations



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

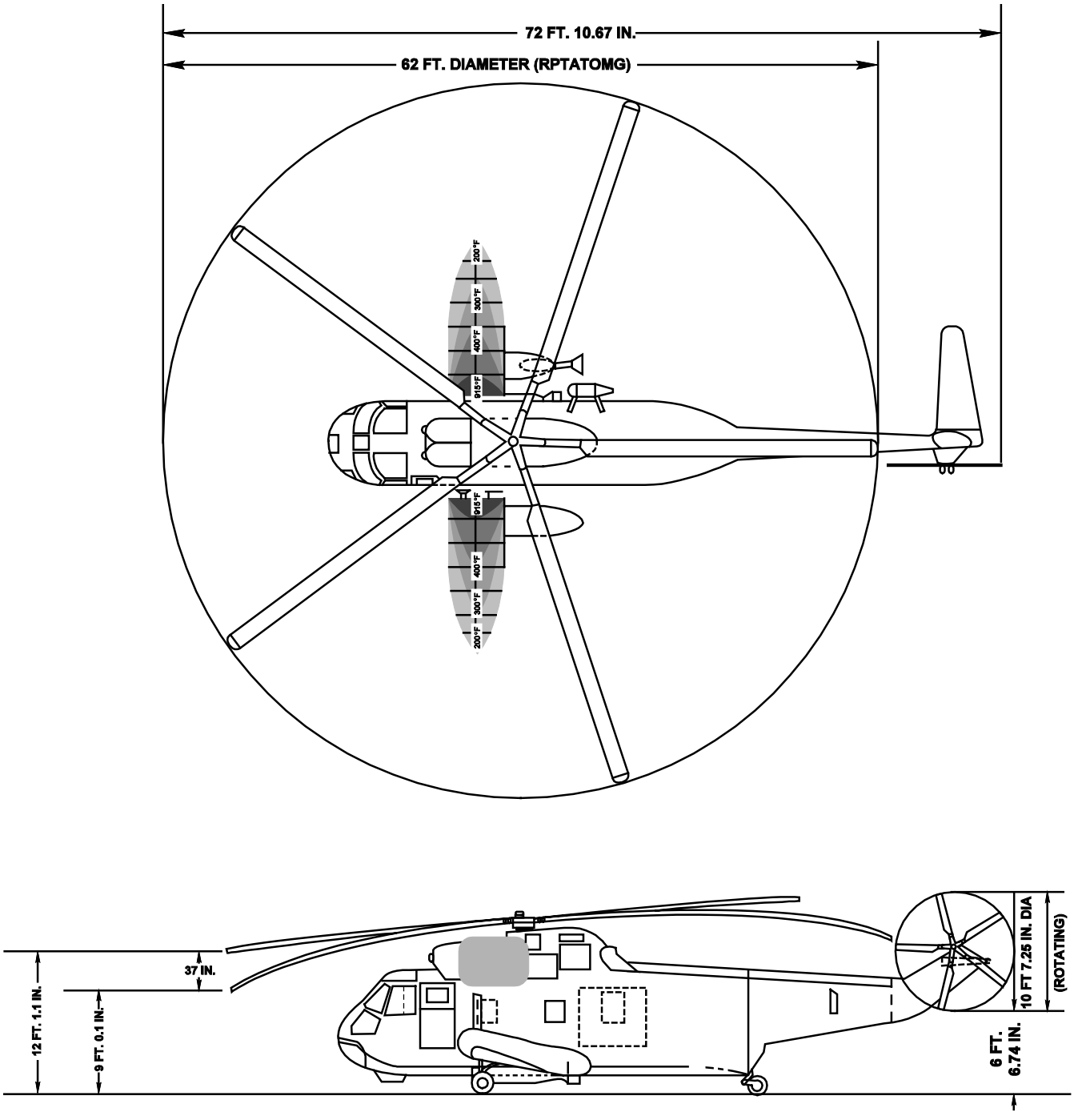
b) Ground Refueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Helo Rotor Clearance



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the H-3 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Open Ground Refueling Panel access door. (VH-3 only)
2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.
3. Press and hold the switch on the Ground Refueling Panel labeled "PRI TEST." Fuel flow should stop within 10 seconds.
4. Release the "PRI TEST" switch and resume fuel flow.
5. Repeat the precheck process in step 4 above, this time pressing and holding the "SEC TEST" switch. Again fuel flow should stop within 10 seconds.



If fuel flow does not stop in steps 3 or 5 above, discontinue hot refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

6. Release precheck switches and refuel aircraft until fuel flow into the aircraft automatically stops indicating the aircraft's tanks are full.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common "Fuel System Vent Ports," which are located on the cabin skin in the approximate places indicated on the aircraft configuration diagram.

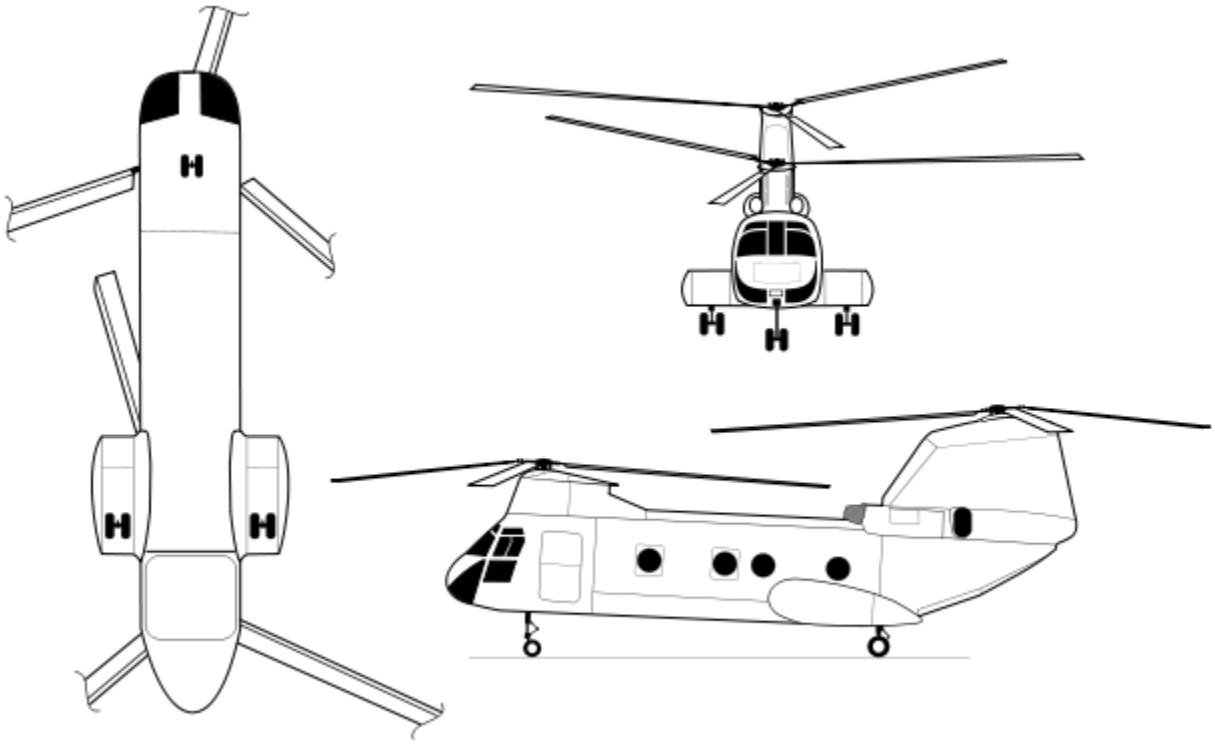
2. If any high-level shut-off valves fail to operate correctly, fuel may spill from one of the "Fuel System Vent Ports." In addition, the fuel tanks may rupture and spill fuel.

3. If externally mounted auxiliary tanks are installed, they can be refueled only by gravity. It is not recommended that these tanks be hot refueled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-46

The Boeing Vertol CH-46 Sea Knight is a medium-lift tandem rotor transport helicopter. It is used to provide all-weather, day-or-night assault transport of combat troops, supplies and equipment. Additional tasks include combat support, search and rescue, support for forward refueling and rearming points.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width	14 ft 9 in	Maximum Gross Weight — 23,000 lbs
Length	84 ft 4 in	Maximum Footprint — 170 psi
Height	18 ft 4 in	

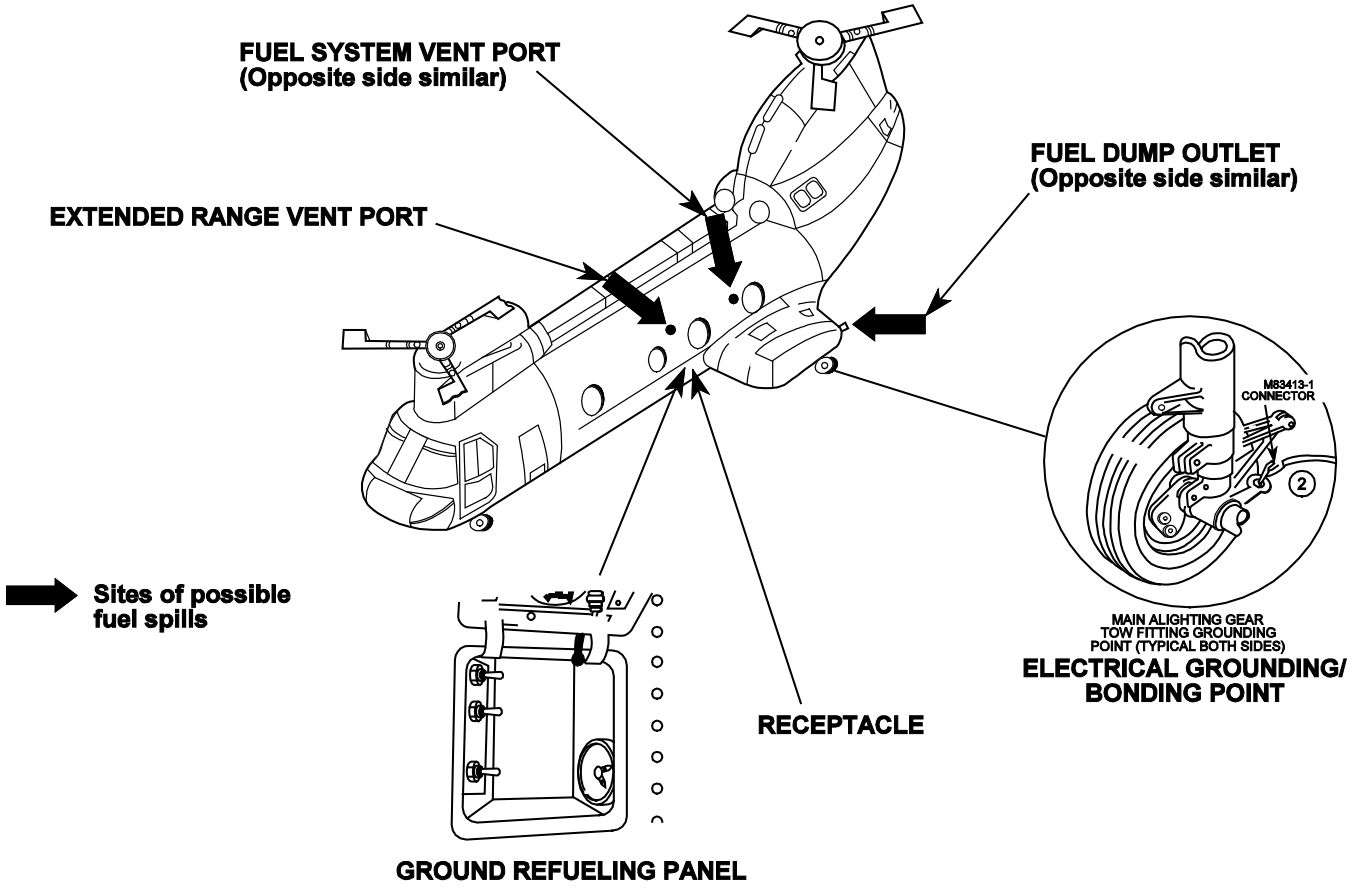
TABLE OF FUEL CAPACITIES

Tank		Gallons
Stubwing	Left	190*
	Right	190*
Total Internal		380
Extended Range Tanks (Each) Maximum of Three		250
*Certain aircraft have special extended capacity in their stubwing tanks — 330 gal each.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

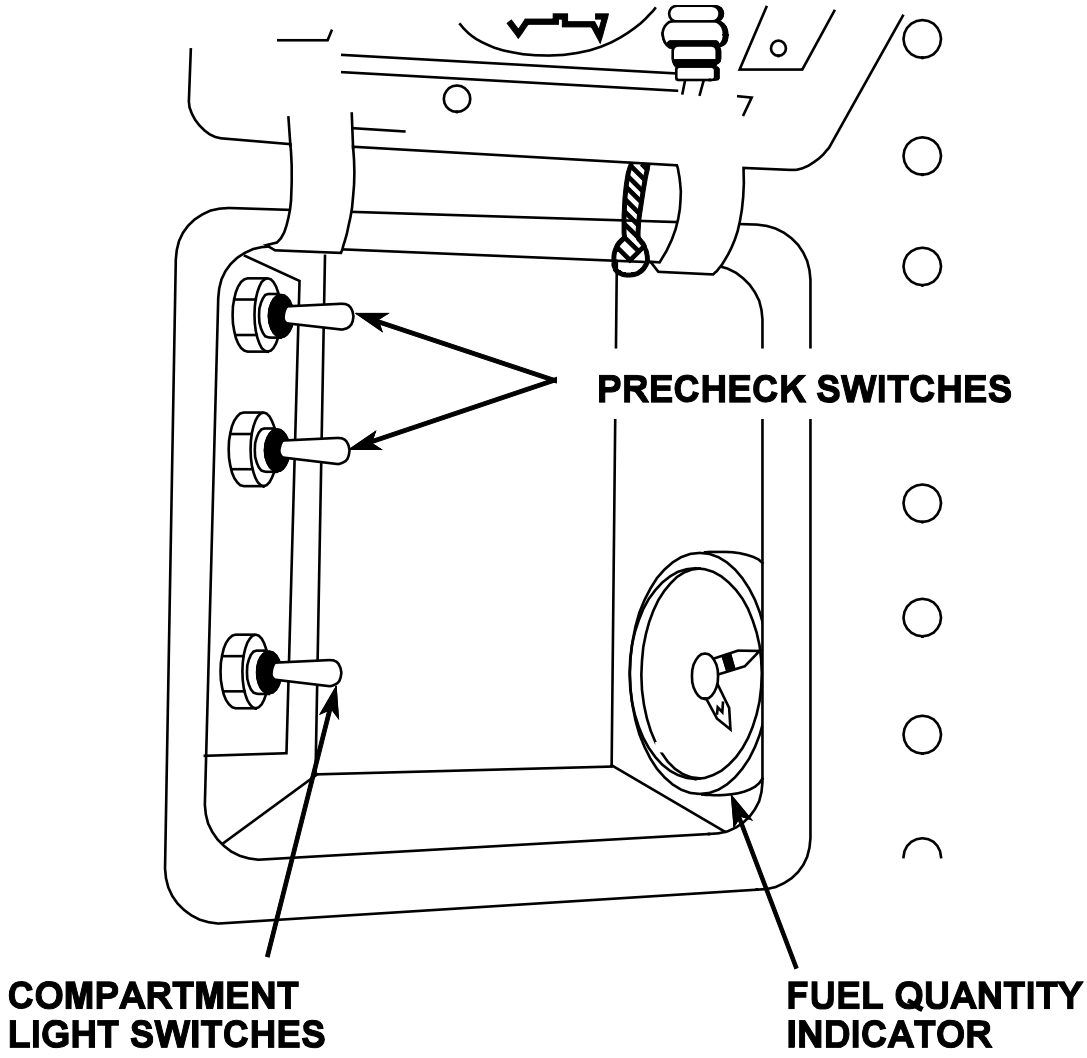
AIRCRAFT CONFIGURATION

a) Ground Fuel Panel Fuel Dump Outlet, Vent Ports, and Electrical Grounding/Bonding Locations



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

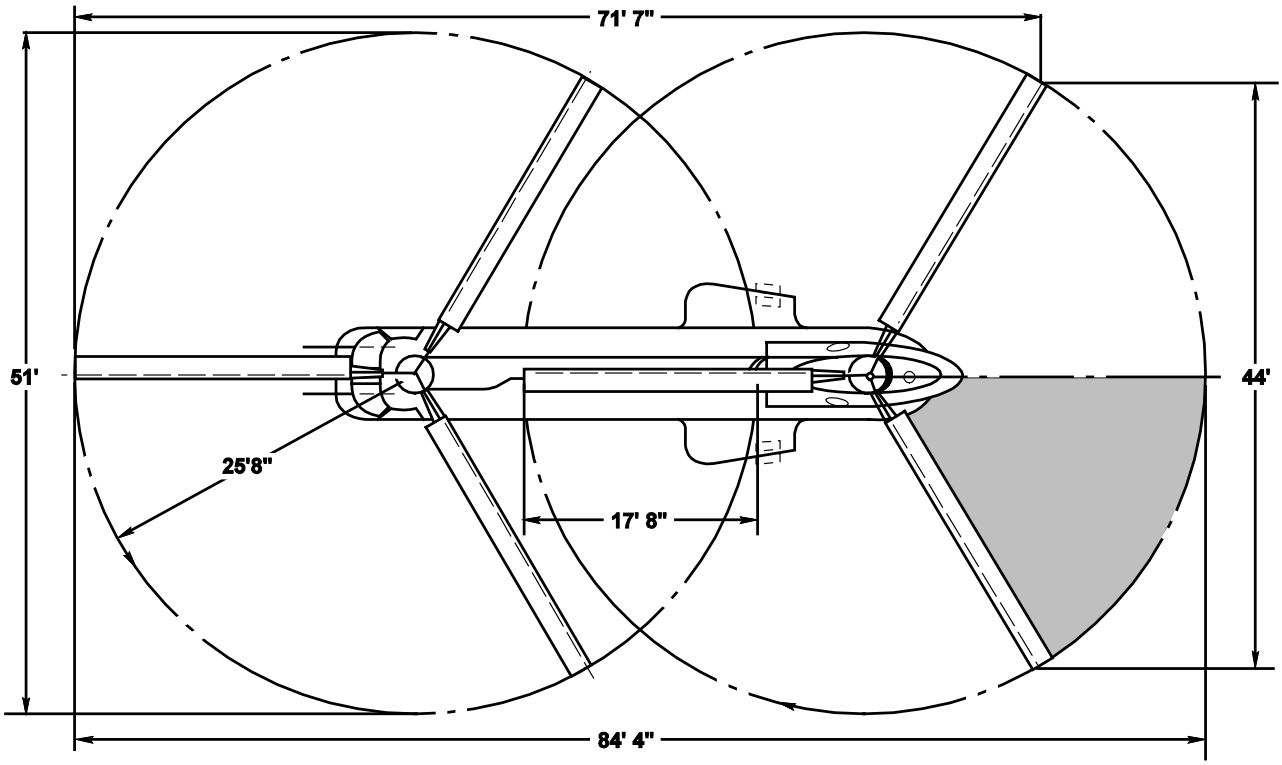
b) Ground Fueling Panel

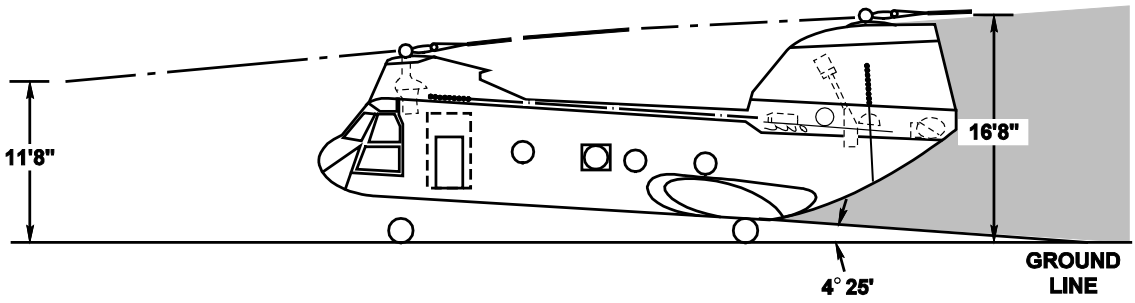


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Helo Rotor Clearance





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the H-46 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Open Ground Refueling Panel access door.

2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.



Ensure that air is venting from "FUEL SYSTEM VENTS." If no venting is indicated, cease fueling operation immediately.

3. Locate the Precheck system valve test switches on the ground refueling panel. Hold left "VALV CONT" switch to "PRI CLOSE" and the right "VALV CONT" switch to "SEC CLOSE." Fuel flow into the aircraft should stop within a few seconds.

4. Release switches and fuel flow should resume.

5. Repeat the precheck process in step three above, this time holding the left "VALV CONT" switch to "PRI CLOSE."



If fuel flow does not stop step 3 or 5 above, discontinue hot refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

6. Release switches and refuel aircraft until fuel flow into the aircraft stops and the fuel quantity gauge on the ground refueling panel indicates the tanks are full.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

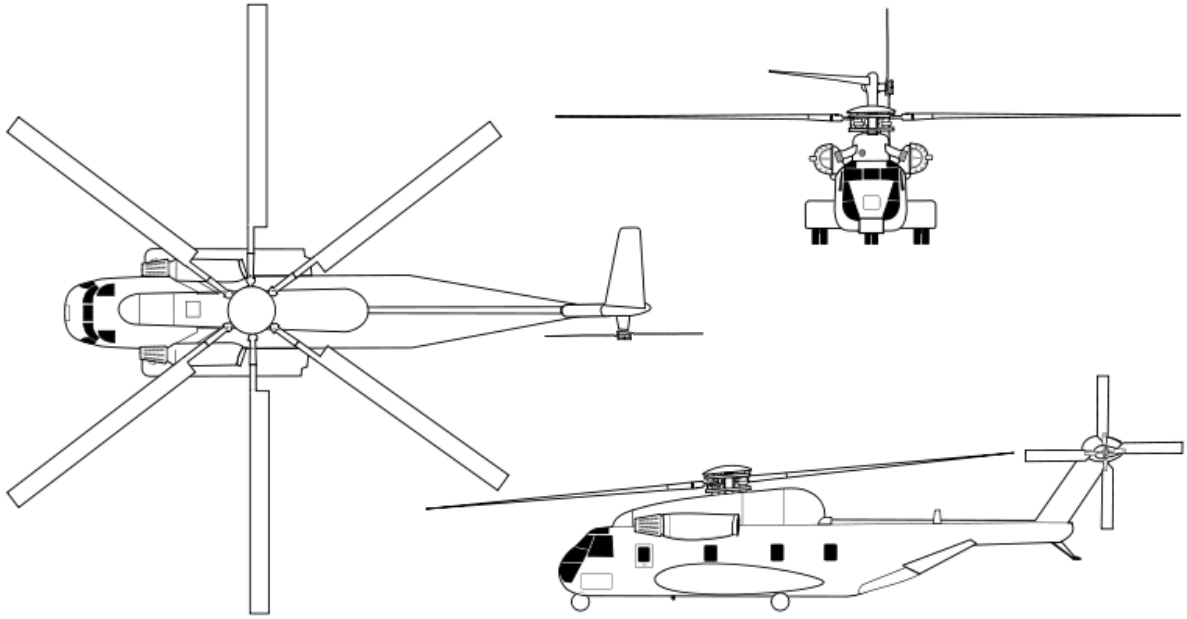
1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common “Fuel System Vent Ports,” which are located on each side of aircraft.

2. If any high-level shut-off valves fail to operate correctly, fuel may spill from one of the “Fuel System Vent Ports.” In addition, the fuel tanks, which are located in the stubwings, may rupture and spill fuel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-53

The H-53 or “Sea Stallion” is designed and manufactured by Sikorsky as a part of the Jolly Green Giant Series. The CH-53A features a six-bladed main rotor and four-bladed tail rotor. The twin General Electric turbashaft engines provide nearly 3000 shaft horsepower and are a variant of the T64 engine. The main mission of the H-53 is heavy lifting.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

CH-53E, MH-53E

Aircraft Dimensions		Aircraft Weight
Fuselage Width (With Aux Tanks)	14 ft	Maximum Gross Weight — 73,500 lbs
Length	99 ft 0.5 in	Maximum Footprint — 135 psi
Height	28 ft 0.5 in	

CH-53A/D

Aircraft Dimensions		Aircraft Weight
Fuselage Width (With Aux Tanks)	24 ft	Maximum Gross Weight — 42,000 lbs
Length	88 ft 6 in	Maximum Footprint — 135 psi
Height	24 ft 11 in	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TABLES OF FUEL CAPACITIES

MH-53E

Tank		Gallons
Internal	No. 1	914.6
	No. 2	1,383.2
	No. 3	914.6
	Total	3,212.4

CH-53E

Tank		Gallons
Internal	No. 1	300
	No. 2	377
	No. 3	300
	Total	977
External Aux Tanks (Each)		650
Total	With 2 Aux Tanks	2,277

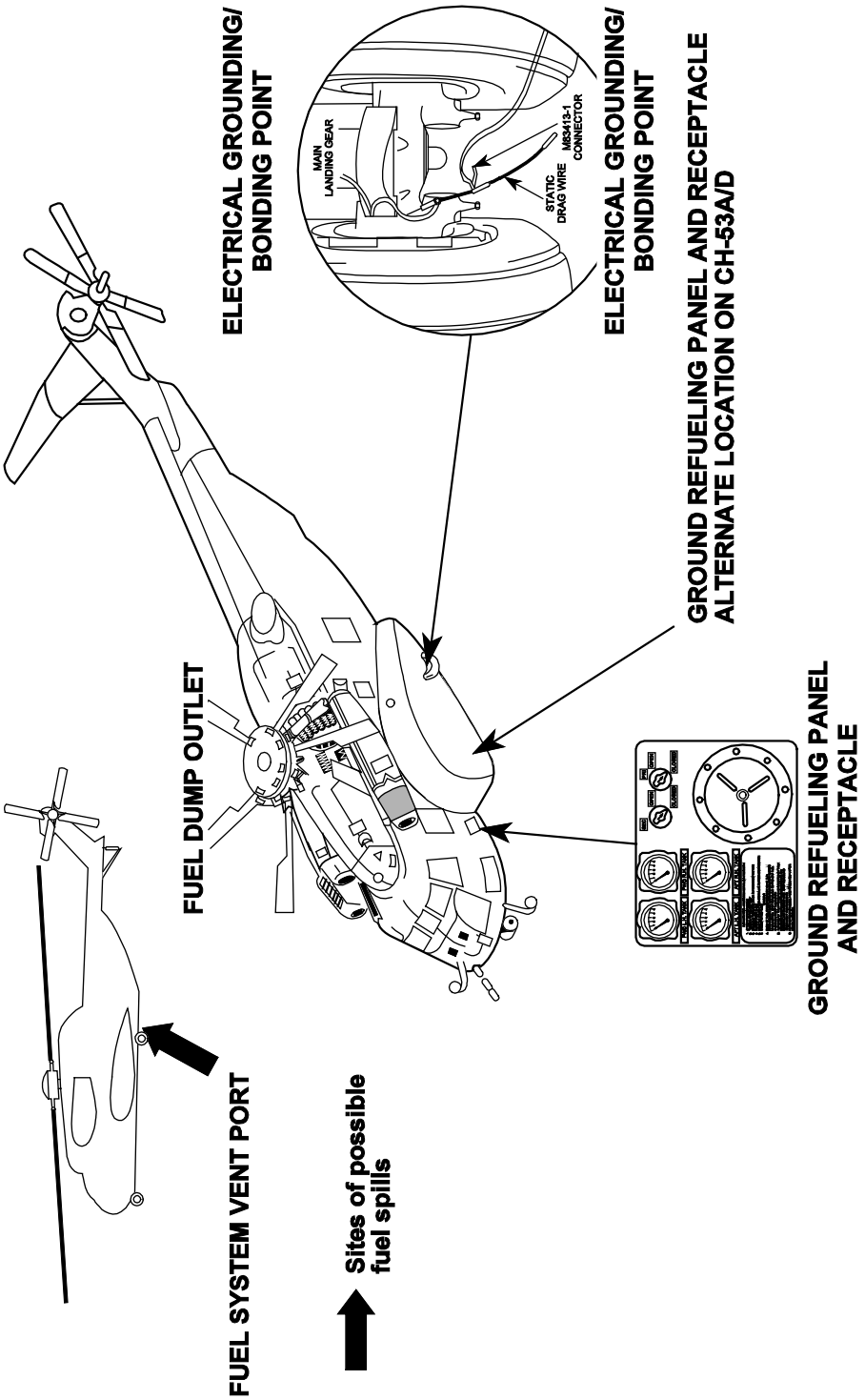
CH-53A/D

Tank		Gallons
Internal	Left	319
	Right	319
	Total	638
External Aux Tanks (Each)		650
Total	With 2 Aux Tanks	1,938

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

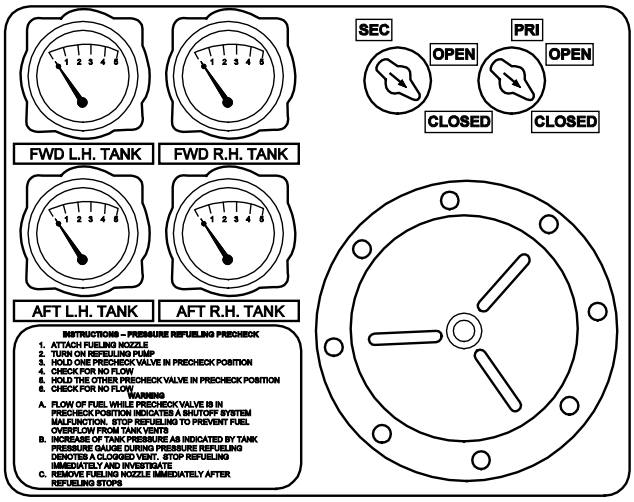
AIRCRAFT CONFIGURATION

a) Ground Fuel Panel Fuel Dump Outlet, Vent Ports, and Electrical Grounding/Bonding Locations

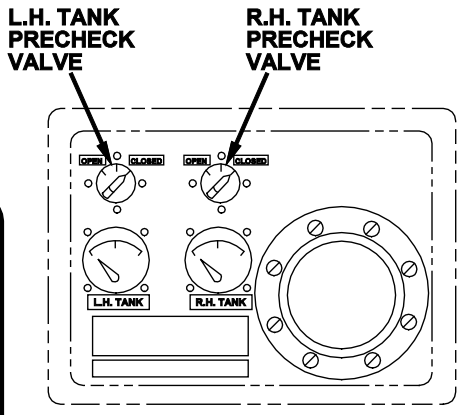


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

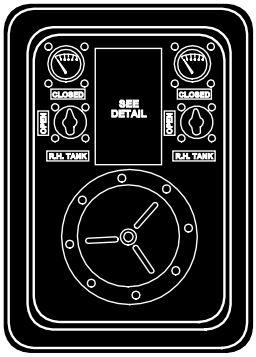
b) Ground Fueling Panel



GROUND REFUELING PANEL
CH-53E and MH-53-E



GROUND REFUELING PANEL
CH-53A/D

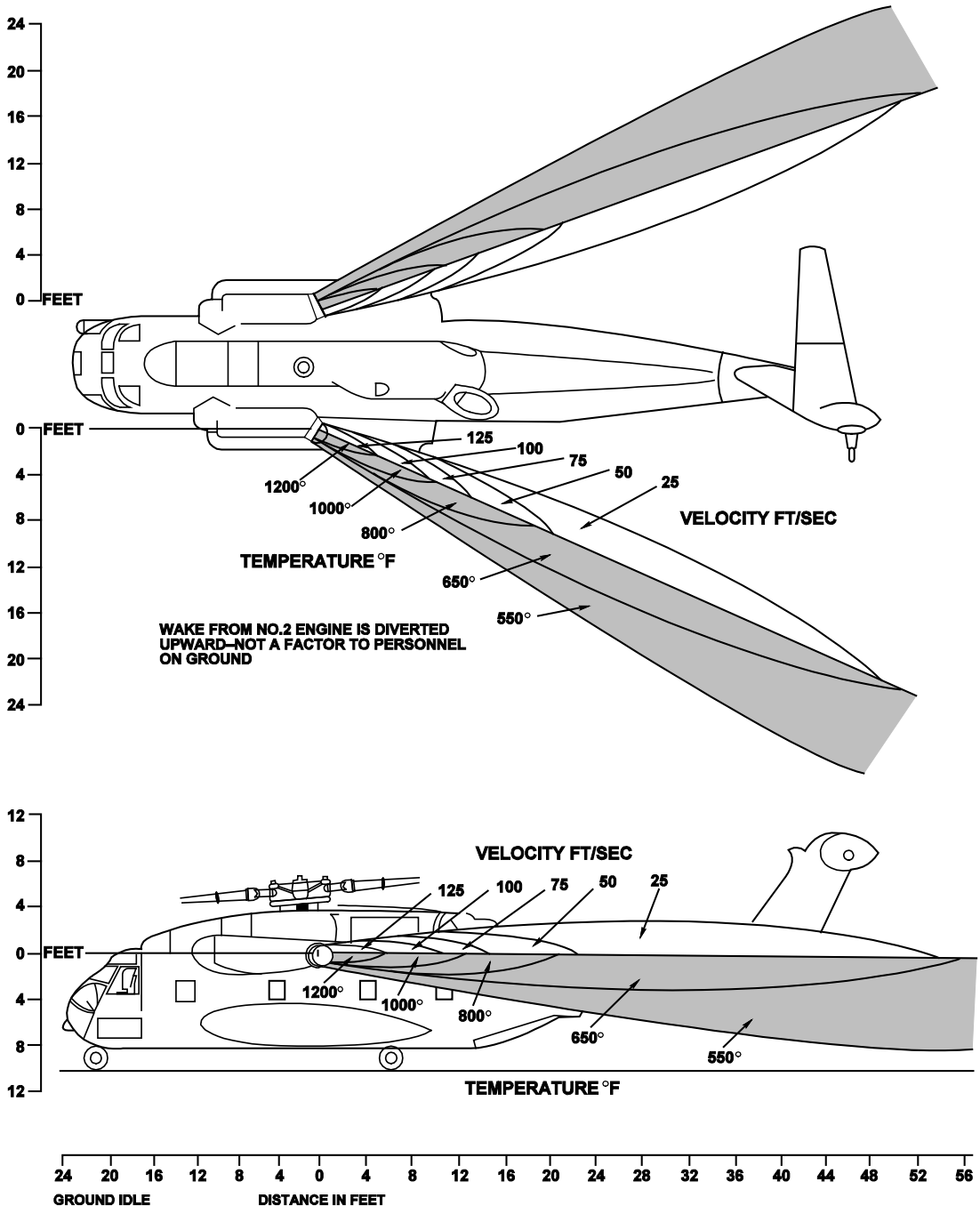


ALTERNATE GROUND REFUELING PANEL
CH-53A/D

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Engine Exhaust Area



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the H-53 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, is to be followed.

1. Open Ground Refueling Panel access door.

2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.

3. Exercise the Precheck system. On A and D models, simultaneously turn both precheck valves on the Ground Refueling Panel to "CLOSED" position. Fuel flow into the aircraft should stop within 10 seconds. On the CH-53E and MH-53E, turn the precheck valve labeled "PRI" to the "CLOSED" position and observe that fuel flow stops, then return the valve to the "OPEN" position. Repeat the last step, this time turning the precheck valve labeled SEC" to the "CLOSED" position.

CAUTION

Flow of fuel while precheck valve(s) are in closed position(s) indicates a failed shutoff valve. Stop refueling immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

4. Place precheck valves on Ground Refueling Panel in the "OPEN" positions and refuel aircraft.

5. Monitor the "Tank Pressure Indicators" on the Ground Refueling Panel.

WARNING

If any tank pressure indicator rises above 1.5 psi, stop refueling immediately. Blockage in vent system should be investigated and corrected prior to refueling.

6. Continue fueling until fuel flow stops automatically.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common “Fuel System Vent Ports,” which are located on the bottom of the aircraft's fuselage.

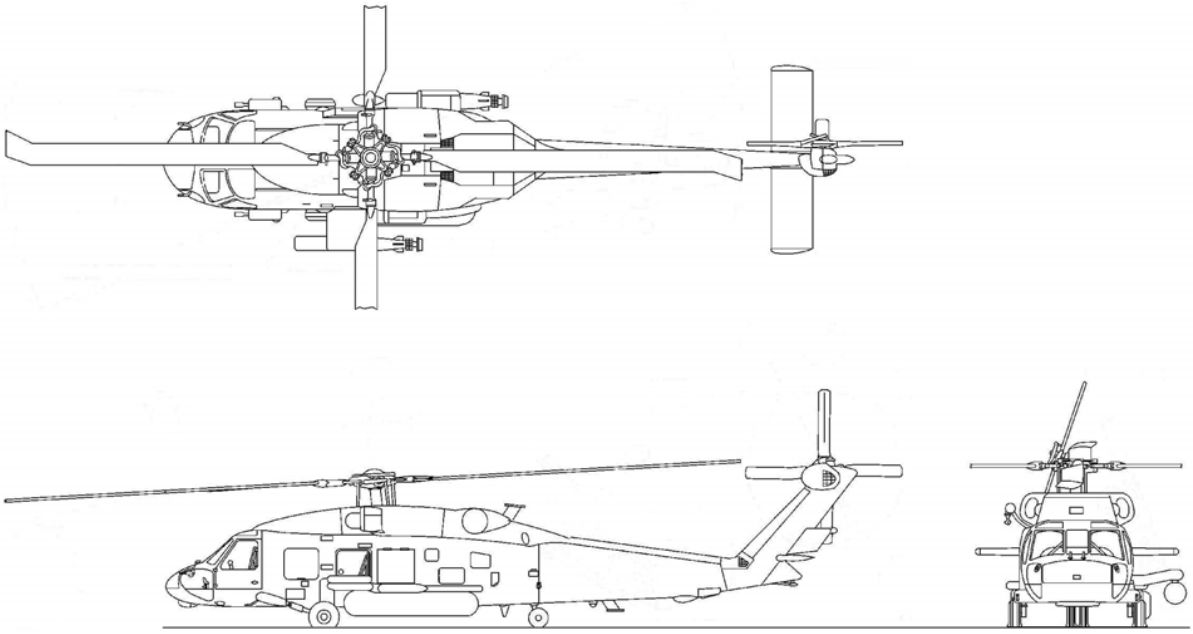
2. If any high-level shut-off valves fail to operate correctly, fuel may spill from one of the “Fuel System Vent Ports.” In addition, the fuel tanks may rupture and spill fuel.

3. The A and D model aircraft can be fitted with internal range extension tanks. Special procedures should be followed by the aircrew in order to refuel these tanks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H-60

The Sikorsky SH-60/MH-60 Sea Hawk is a twin turboshaft engine, multi-mission Navy helicopter based on the United States Army UH-60 Black Hawk and a member of the Sikorsky S-70 family. The most significant airframe modification is a hinged tail to reduce its footprint aboard ships. The Sea hawk can handle anti-submarine warfare, anti-surface warfare, naval special warfare insertion, search and rescue, combat search and rescue, vertical replenishment, and medical evacuation.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width (With Aux Tanks)	7 ft 9 in	Maximum Gross Weight — 22,000 lbs
Length	64 ft 10 in	Maximum Footprint — 90 psi
Height	17 ft	

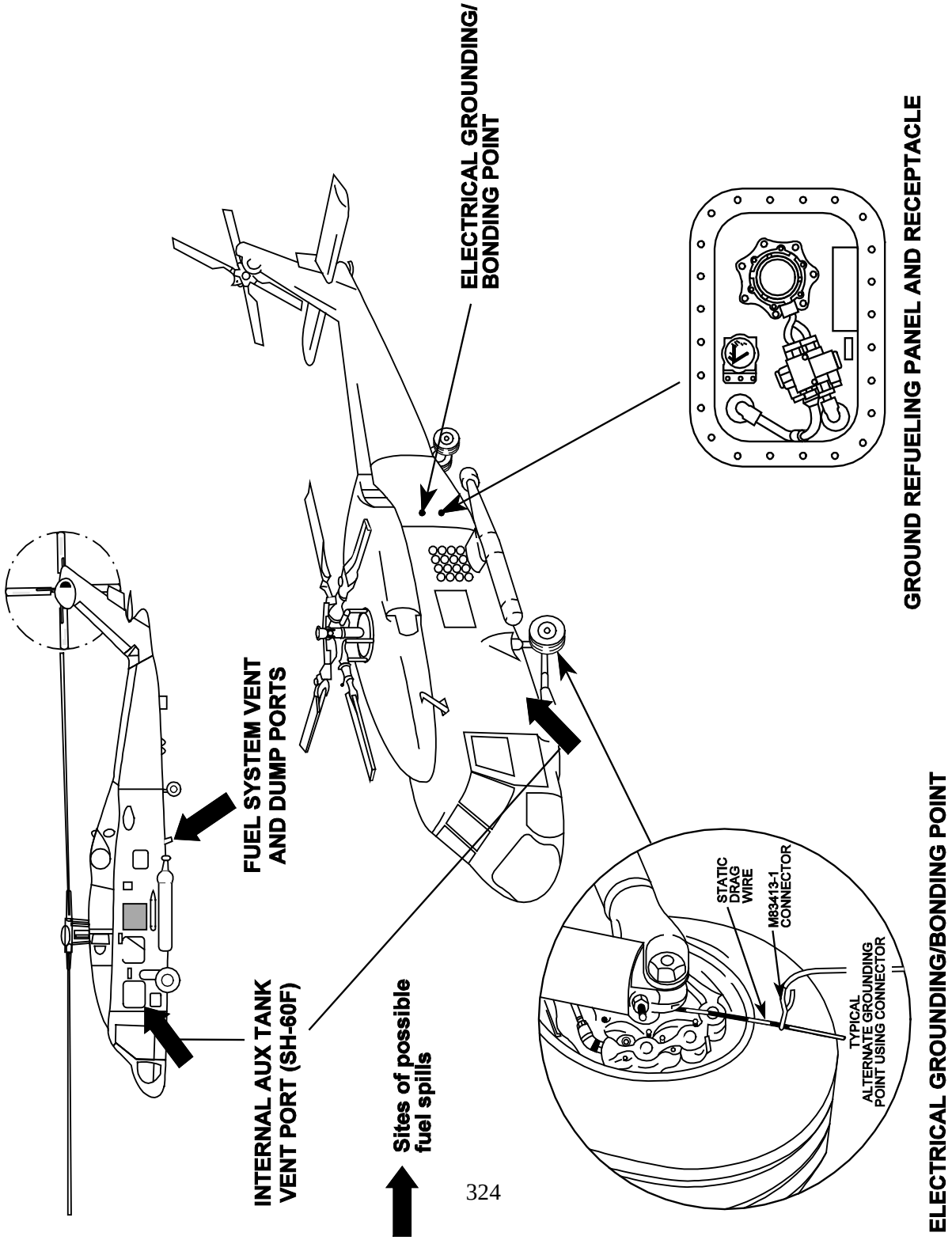
TABLE OF FUEL CAPACITIES

FuelTanks							
Variant	**Main Tanks**	**capacity lbs./tank**	**Internal Aux.**	**capacity lbs./tank**	**External Aux.**	**capacity lbs./tank**	**Notes**
SH-60B, HH-60H, MH-60RR	2	2040	0	0	2--L & R	816	The lower third of each main cell is self-sealing for 7.62 mm holes
MH-60S	2	1224	2	1360	N/A	0	Interchangeable, Crashworthy & Self-sealing main tanks-- crashworthy ballistically tolerant aux tanks
SH-60F	2	2040	1	721.5	2--L&R	816	The lower third of each main cell is self-sealing for 7.62 mm holes

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

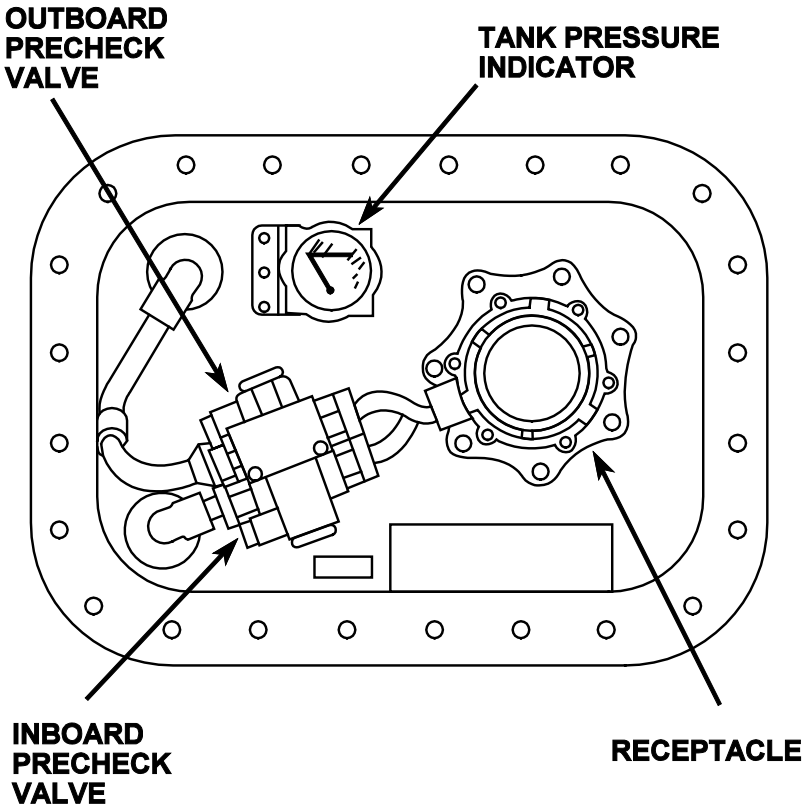
AIRCRAFT CONFIGURATION

a) Ground Fuel Panel Fuel Dump Outlet, Vent Ports, and Electrical Grounding/Bonding Locations



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

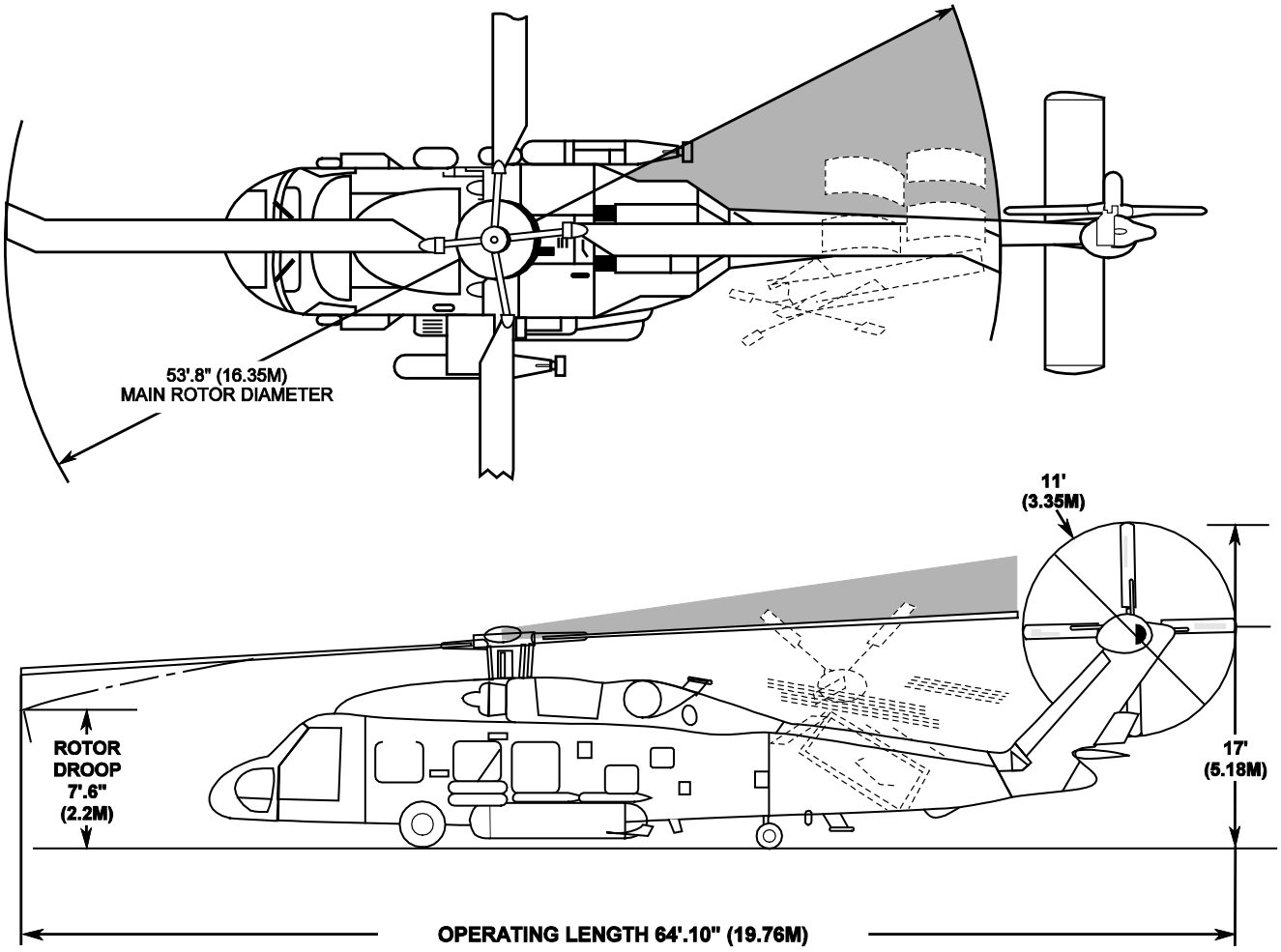
b) Ground Fueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Helo Rotor Clearance



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the H/SH-60 aircraft, primarily the operation of the “precheck” system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Open Ground Refueling Panel access door.

2. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.

3. Monitor the “Tank Pressure Indicator” on the Ground Refueling Panel.



If tank pressure indicator enters the red band, stop refueling immediately. Blockage in vent system should be investigated and corrected prior to refueling.

4. Hold the “OUTBOARD PRECHECK VALVE” in the “PRECHECK” position. Fuel flow should stop within 6 seconds.

5. Release the “OUTBOARD PRECHECK VALVE” and resume fuel flow.

6. Repeat the precheck process in step 4 above, this time holding the “INBOARD PRECHECK VALVE” in the “PRECHECK” position. Again fuel flow should stop within 6 seconds.



If fuel flow does not stop, discontinue hot refueling operation immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

7. Release precheck valves and refuel aircraft until fuel flow into the aircraft automatically stops indicating the aircraft's tanks are full.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common “Fuel System Vent Ports,” which are located on the bottom of the aircraft's fuselage. The H/SH-60F aircraft has an internal auxiliary tank that vents to an opening on the lower port side of the aircraft near the front. Each external auxiliary tank has a vent opening on its bottom approximately in the center.

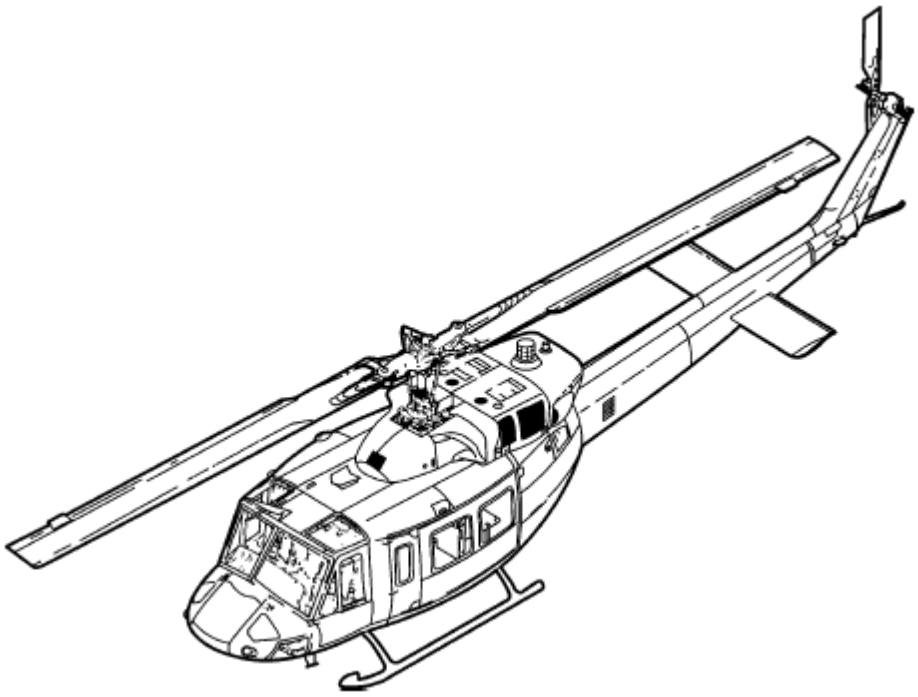
2. If any high-level shut-off valves fail to operate correctly, the tanks may rupture and/or fuel may spill from one of the Fuel System Vent Port.”

3. The H/SH-60F aircraft has a second Tank Pressure Indicator, which displays the pressure in the Internal Auxiliary Tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

H/UH-1

The U.S. Navy Series UH-1 utility helicopter is manufactured by Bell Helicopter Textron. It is capable of operating from prepared or unprepared takeoff or landing areas VFR or IFR, day or night. The helicopter is powered by a T400-CP-400 turbo shaft engine.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Aircraft Dimensions		Aircraft Weight
Fuselage Width	9 ft 1 in	Maximum Gross Weight — 10,500 lbs
Length	57 ft 3.3 in	
Height	13 ft 1 in	

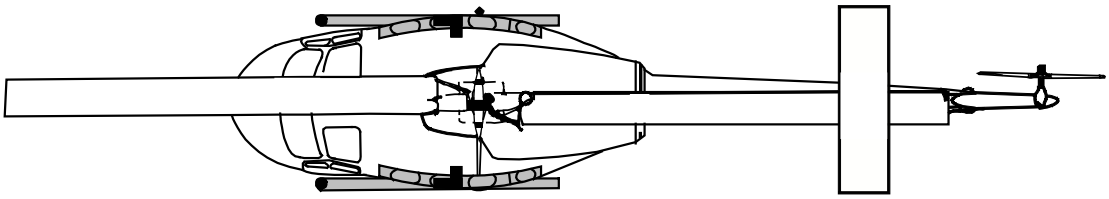
TABLE OF FUEL CAPACITIES

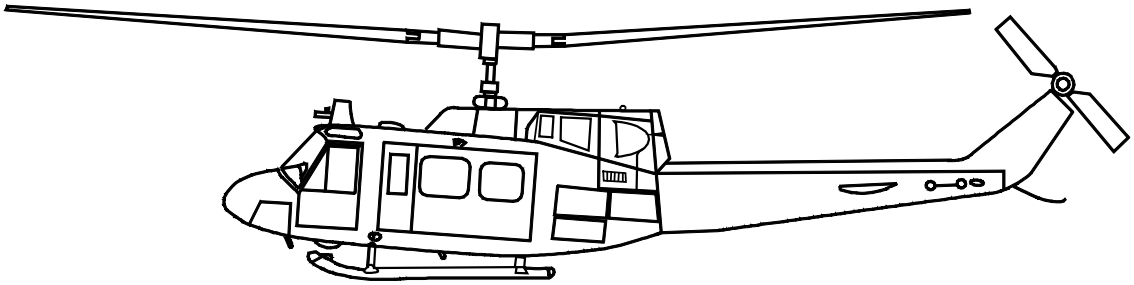
U.S. GALLONS AND POUNDS		
	MAX CAPACITY	**USABLE FUEL**
MAIN FUEL TANKS	195.5 Gallons 1,271 Pounds JP-4 1,329.5 Pounds JP-5 1,316 Pounds JP-8	193 Gallons 1,254.5 Pounds JP-4 1,312.5 Pounds JP-5 1,299 Pounds JP-8
AUXILIARY FUEL TANKS	(Both Tanks) 300 Gallons 1,950 Pounds JP-4 2,040 Pounds JP-5 2,019 Pounds JP-8	(Both Tanks) 300 Gallons 1,950 Pounds JP-4 2,040 Pounds JP-5 2,019 Pounds JP-8

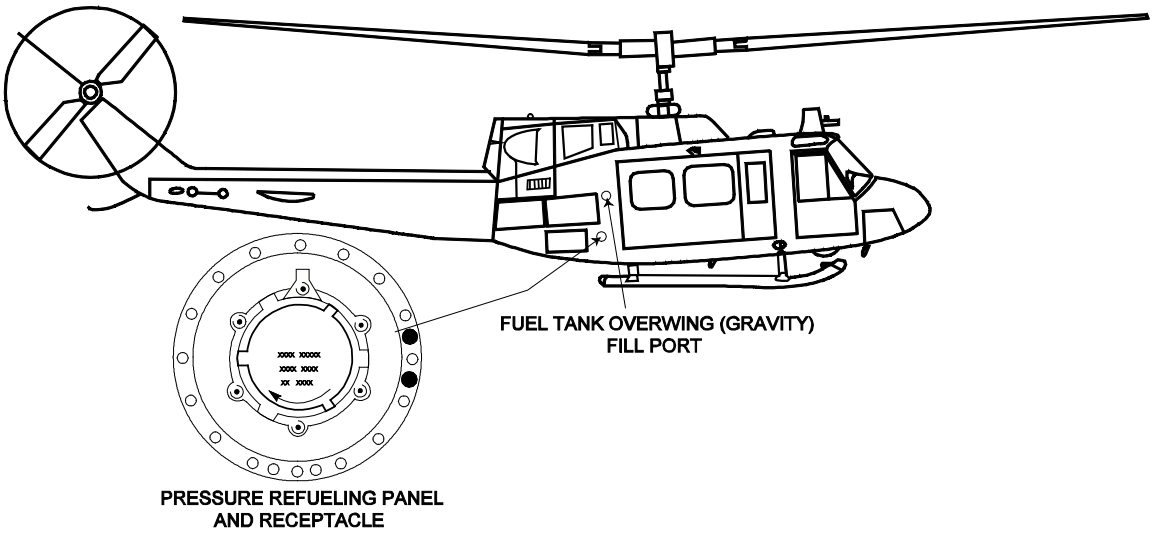
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Ground Fuel Panel Location

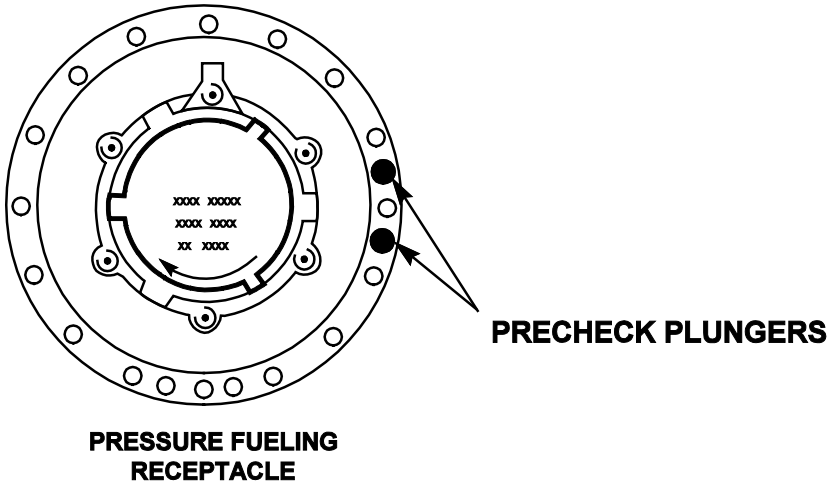






Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

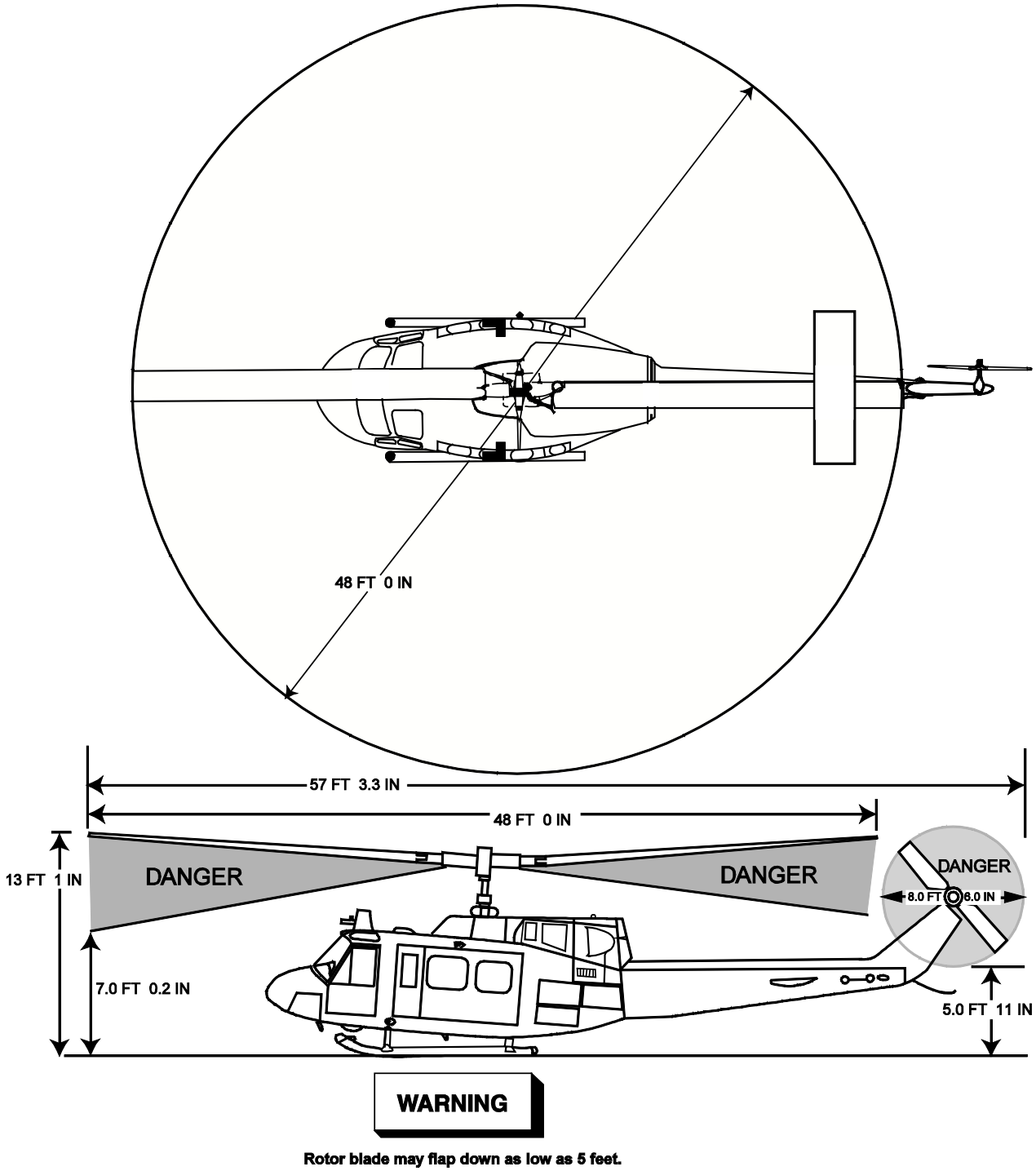
b) Ground Fueling Panel



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

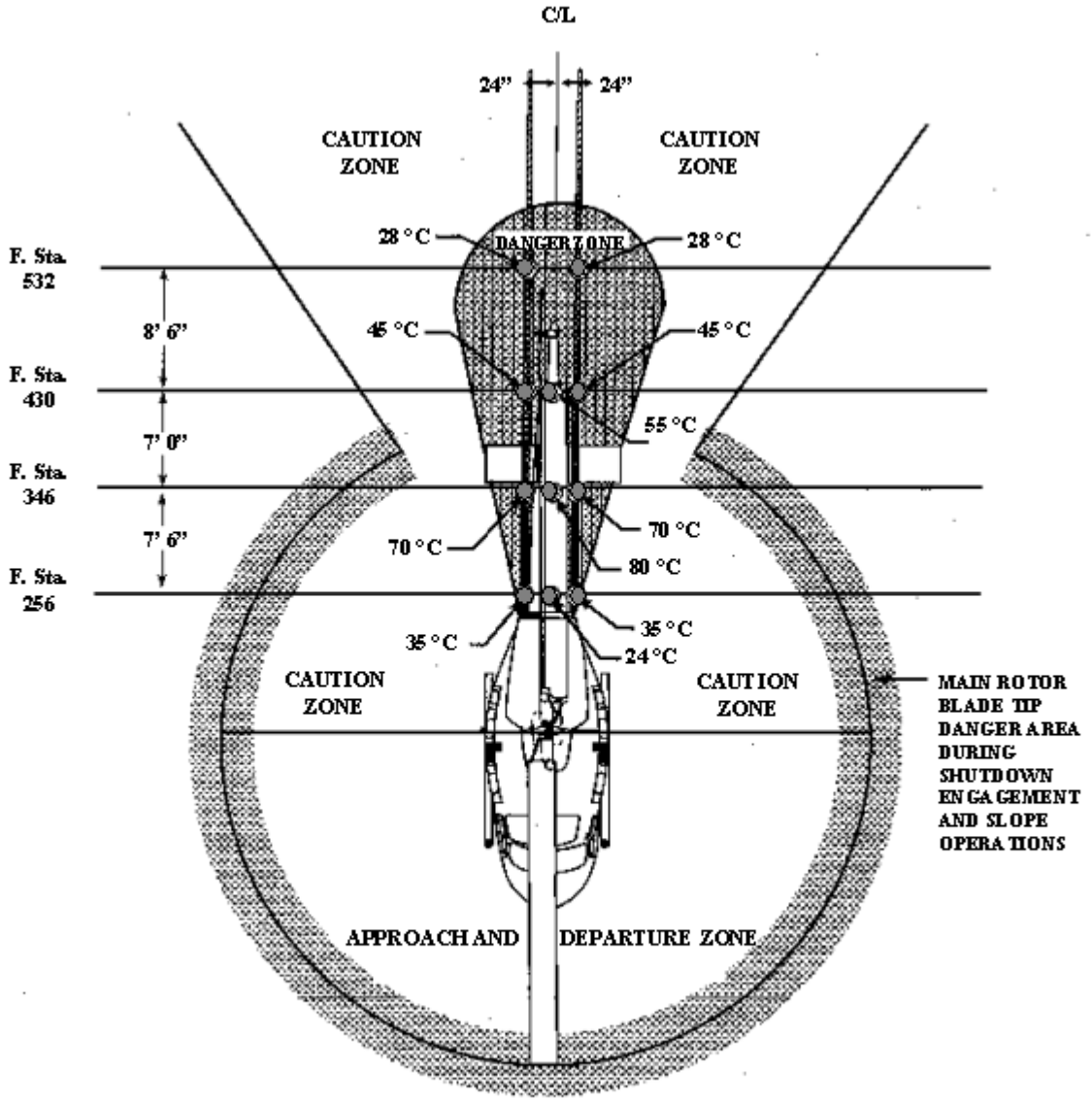
PERSONNEL DANGER ZONES

a) Helo Rotor Clearance



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Engine Exhaust Area



TEMPERATURES SHOWN ARE AT HEIGHT OF 6 FEET ABOVE GROUND LEVEL, WITH HELICOPTER OPERATING AT TAKE-OFF POWER ON GROUND RUN.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The following procedures cover only those refueling procedures unique to the UH-1 aircraft, primarily the operation of the "precheck" system. In addition to these specialized procedures, the applicable, basic refueling procedures contained in Chapter 6, 12 or 18 of the Aircraft Refueling NATOPS Manual, NAVAIR 00-80T-109, should be followed.

1. Remove receptacle cap, attach refueling nozzle, open nozzle to the fully open and locked position, and initiate fuel flow.

WARNING

Ensure that air is venting from "FUEL SYSTEM VENTS." If no venting is indicated, cease fueling operation immediately.

2. Exercise the Precheck system. Press and hold one of the "PRECHECK" plungers on the rim of the fueling valve. Fuel flow into the aircraft should stop within 20 seconds. Release the "PRECHECK" plunger and fuel flow should resume into the aircraft.

3. Repeat the last step this time holding the other "PRECHECK" plunger down.

WARNING

Flow of fuel while either of the "PRECHECK" plungers is held down indicates a failed shutoff valve. Stop refueling immediately. System failure should be investigated and resolved before hot refueling can be accomplished.

4. Release both "PRECHECK" plungers and refuel aircraft.

5. Continue fueling until fuel flow stops automatically.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES — AIRCRAFT FUEL SYSTEM

1. Under normal conditions, all air being displaced by fuel in the tanks exits the aircraft through the common “Fuel System Vent Port,” through the fuselage lower skin.

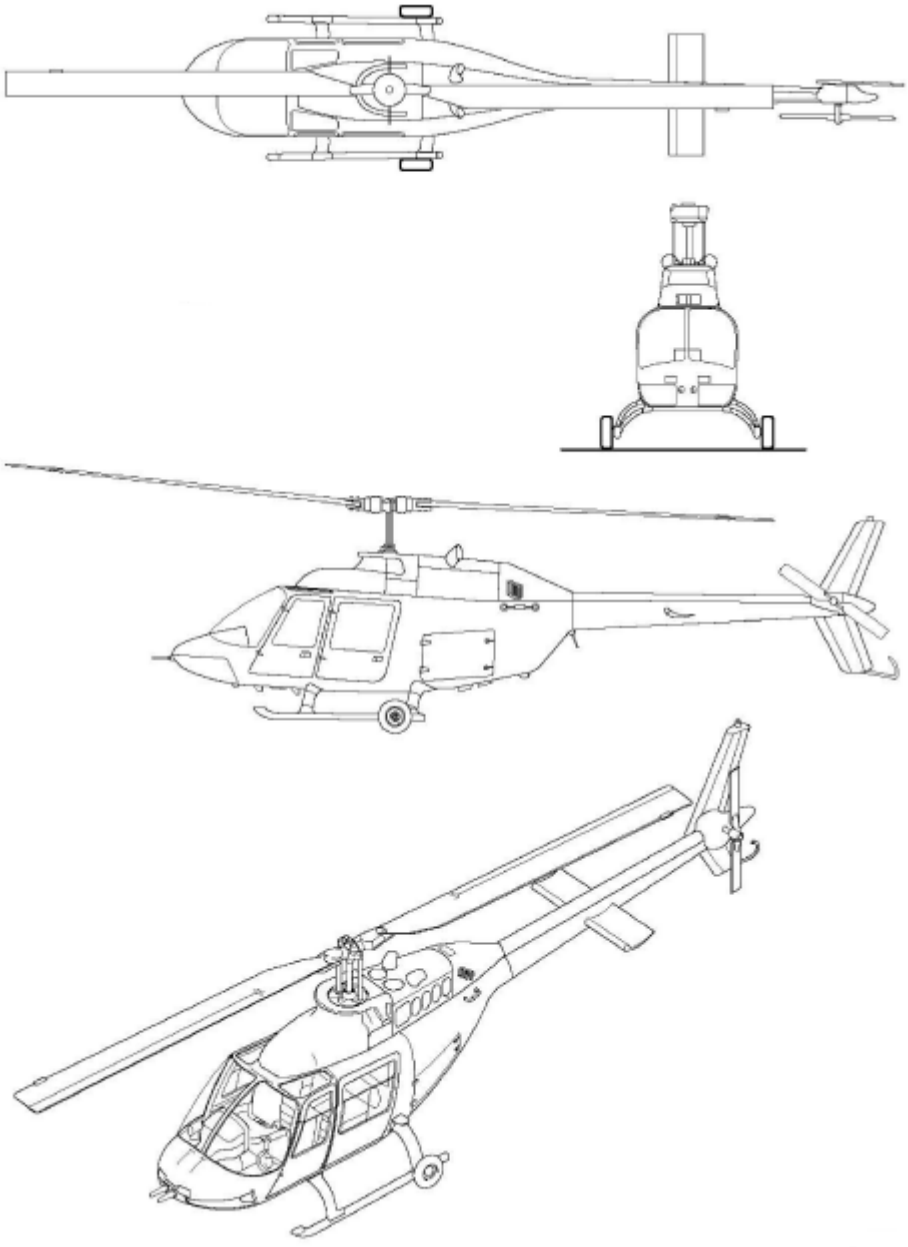
2. If any high level shut-off valves fail to operate correctly, fuel may spill from one of the “Fuel System Vent Port.” In addition, the fuel tanks may rupture and spill fuel.

3. The external tanks cannot be pressure refueled. Each external tank should be gravity refueled separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

TH-57

The TH-57 is a single-engine, land-based utility-type helicopter designed for reasonably level, firm terrain. The primary purpose of the TH-57 is training in both basic and advanced instrument (simulated or actual) conditions as well as introducing basic visual flight rules tactical maneuvering (to include shipboard operations).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	11 ft, 8 in	Maximum Gross MTW 189.7klbs	32000 lbs
Length	31 ft, 4 in		
Fuselage Width	6 ft, 4 in		

TABLE OF FUEL CAPACITIES

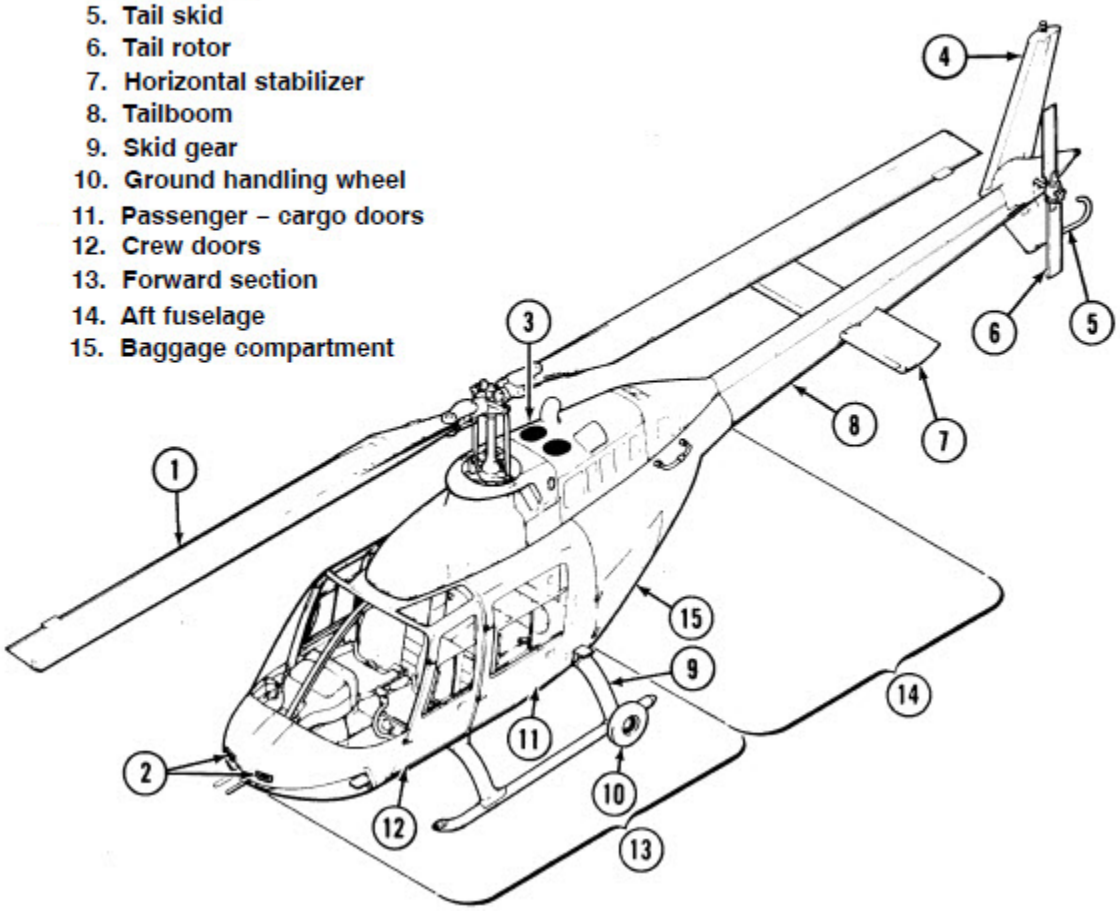
Gallons	Weight in Pounds(JP-5)	CG	Moment
30	204	110.8	22603
40	272	111.5	30328
50	340	113.8	38692
60	408	115.3	47042
70	476	116.4	55406
80	544	117.2	63757
90	612	117.9	72155

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

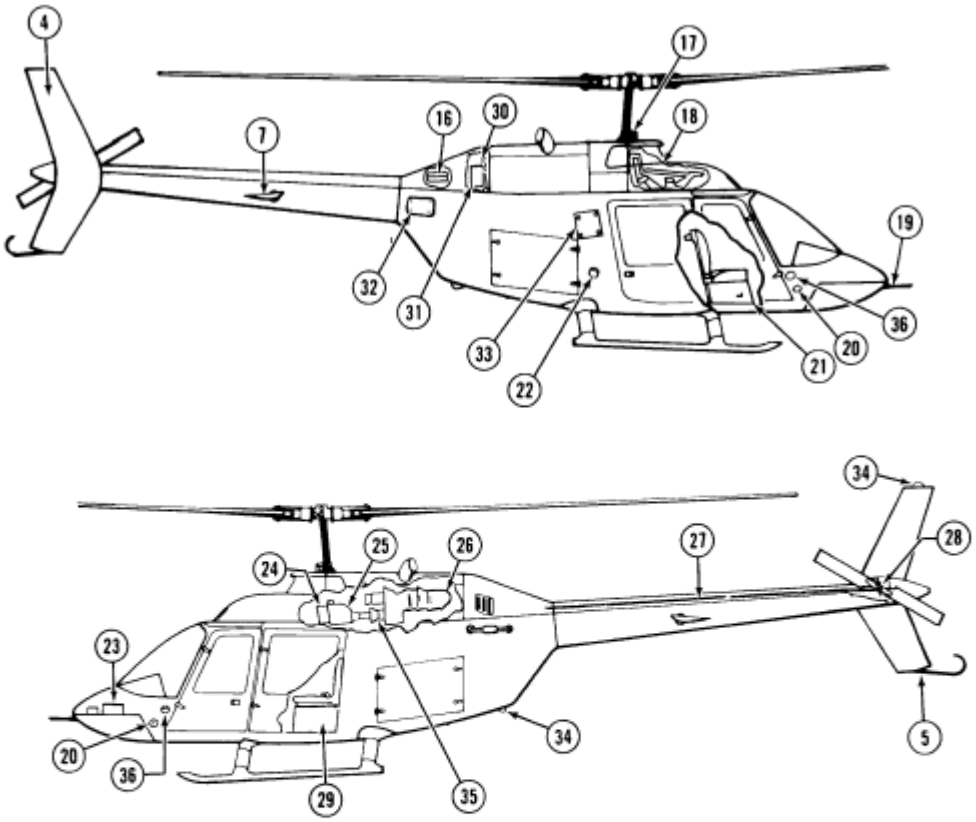
AIRCRAFT CONFIGURATION

a) Static Port, Fuel Shutoff Valve, and Fuel Tank Filter Locations

1. Main rotor
2. Air intake grill
3. Engine compartment
4. Vertical fin
5. Tail skid
6. Tail rotor
7. Horizontal stabilizer
8. Tailboom
9. Skid gear
10. Ground handling wheel
11. Passenger – cargo doors
12. Crew doors
13. Forward section
14. Aft fuselage
15. Baggage compartment



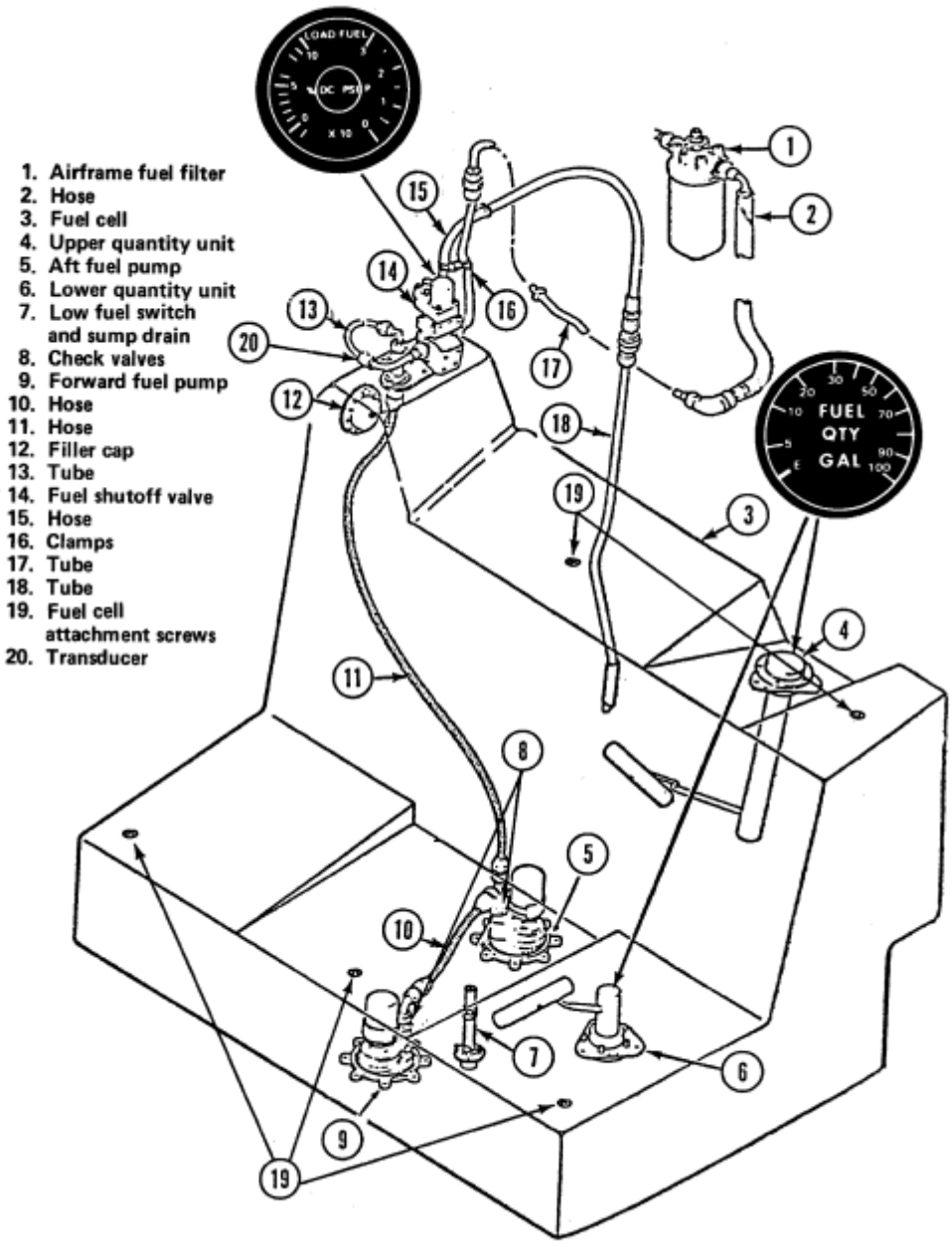
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.



16. Oil tank – engine
17. Swashplate drive link
18. Cyclic and collective servo actuators
19. Pitot tube
20. Static Port – pilot
21. Pilot's station
22. Fuel tank filter
23. Battery
24. Hydraulic pump and reservoir
25. Transmission assembly
26. Engine
27. Tail rotor driveshaft
28. Tail rotor gearbox
29. Passenger station
30. Oil cooler (engine)
31. Oil cooler blower
32. Tailboom attachment access panel
33. Fuel shutoff valve access panel
34. Anticollision beacon (2)
35. Freewheeling unit
36. Static port – co-pilot

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

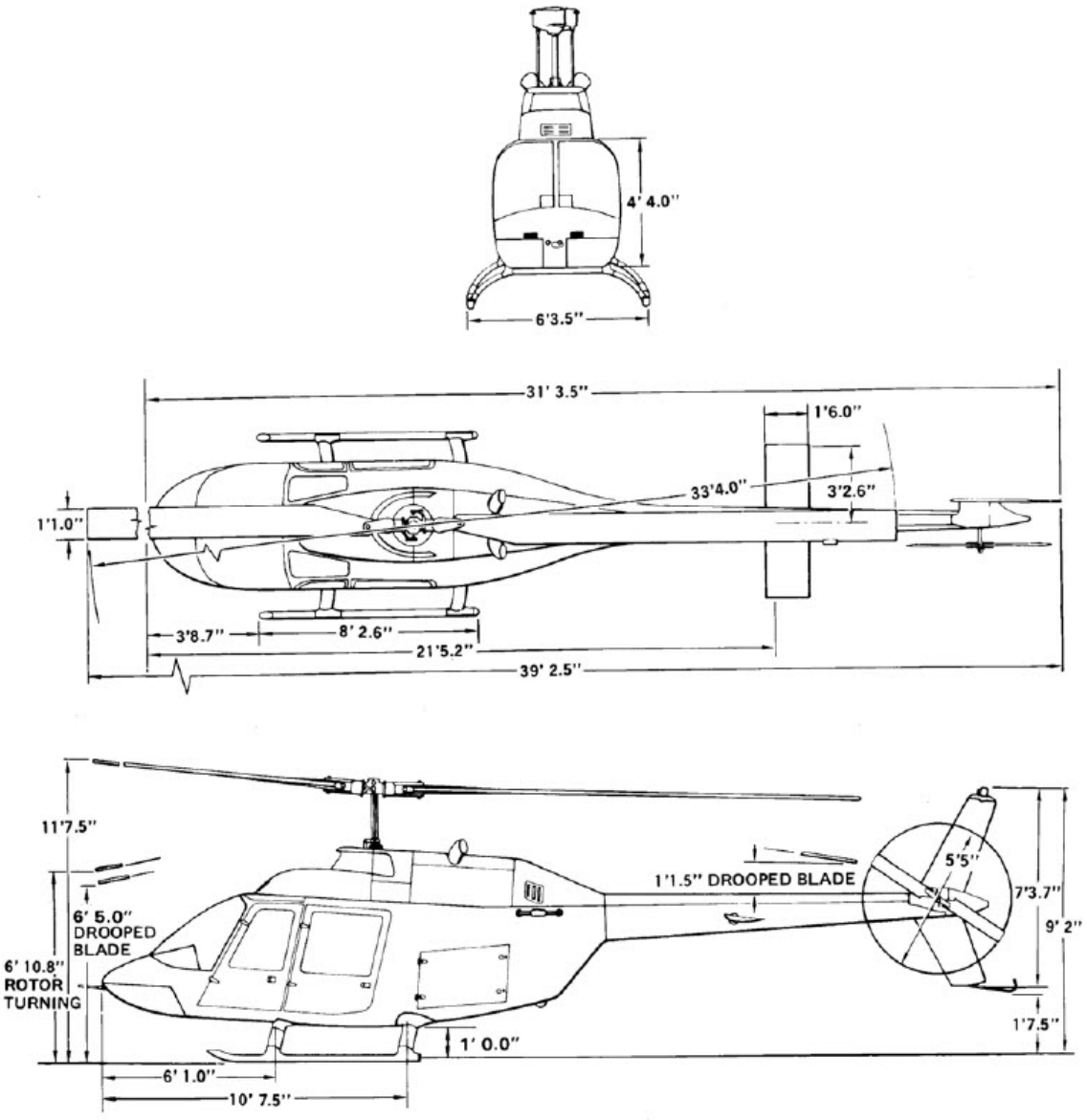
b) Fuel Cell System



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

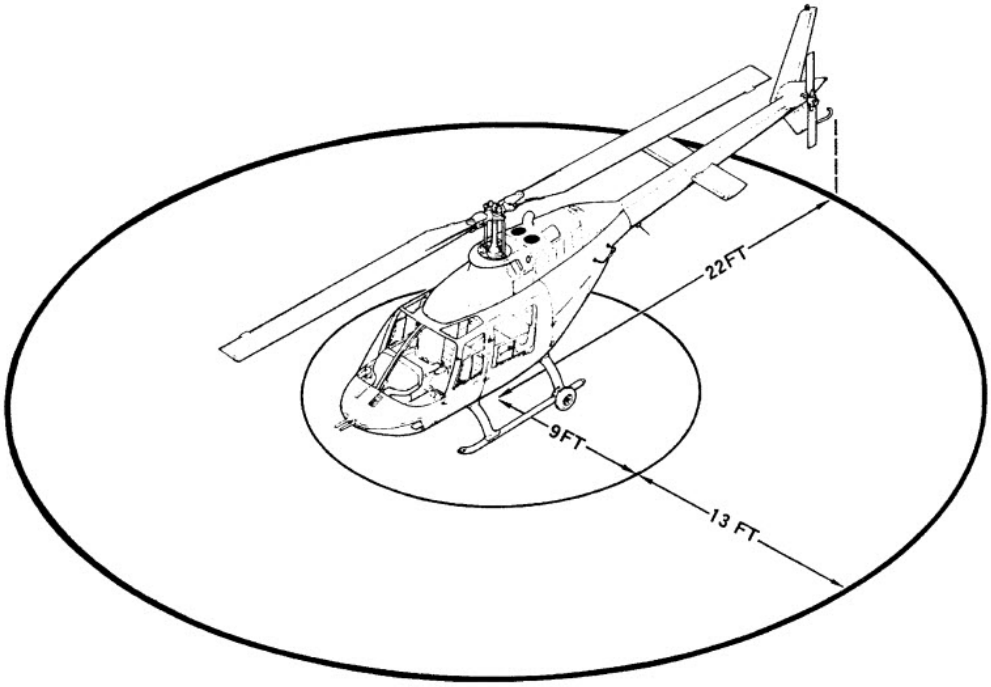
PERSONNEL DANGER ZONES

a) Principal dimensions



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Turning radius



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

The fuel system has a single bladder−type fuel cell located below and aft of the passenger seats. The fuel cell is gravity or pressure filled from the right side of the fuselage. Two electrically operated fuel boost pumps are located in the bottom of the fuel cell. The pumps are interconnected and furnish fuel through a single supply line to the engine−driven fuel pump. The boost pumps are equipped with a check valve, a pump drain port, a seal drain port, an intake screen, and a pump operating a pressure switch located in the pump discharge port. The pumps are controlled by two circuit breakers marked FUEL BOOST FWD and FUEL BOOST AFT located on the overhead console.

Note

Due to possible fuel sloshing in unusual attitudes or out of trim conditions with one or both fuel boost pumps inoperative, the unusable fuel is 10 gallons.

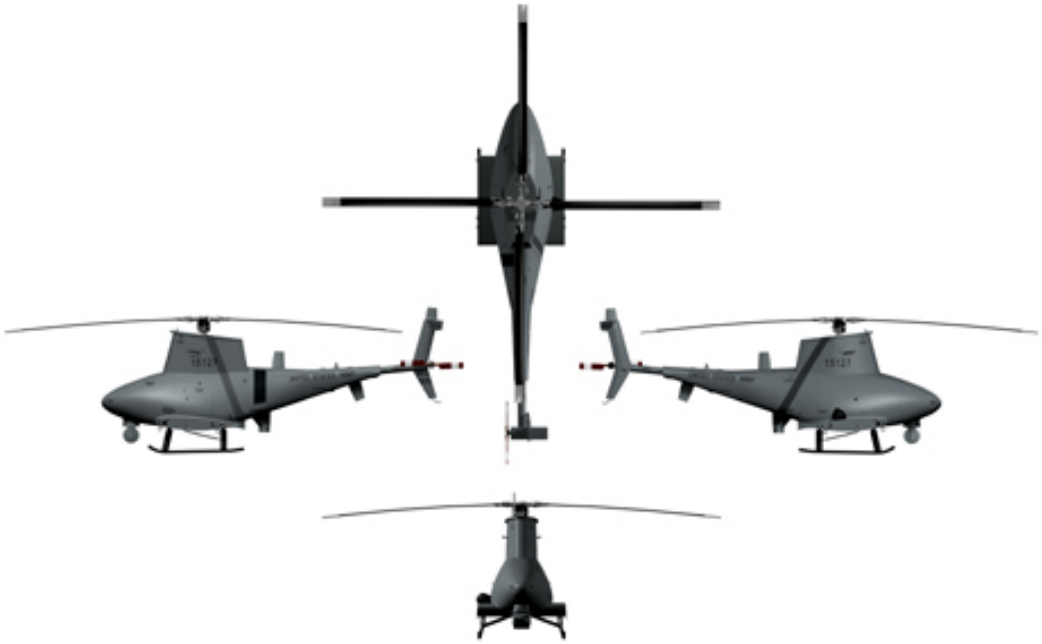
The engine is designed to operate without boost pump pressure under 6,000 feet PA and one boost pump will supply sufficient fuel for normal engine operations under all conditions of power and altitude.

Two float−type fuel level transmitting units are installed in the fuel cell. The lower unit is mounted in the bottom of the cell and measures fuel level up to the horizontal surface of the cell under the passenger seats. The upper unit is mounted on the top of the cell and measures the fuel level in the upper section of the fuel cell behind the passenger seats. Both transmitting units are connected to a common fuel quantity indicator. Unusable fuel in the aircraft is 1.03 gallons as indicated by “E” on the Fuel Quantity Indicator. The fuel filter assembly and single- or dual-element engine-driven fuel pump, which operates at 650 psi minimum and 750 psi maximum, are integral units mounted on the aft left end of the engine. Fuel enters the engine fuel system at the inlet port of the pump and passes through the filter before entering the gear elements of the pump. An electrically operated shutoff valve is installed min the main fuel supply line and is controlled by an ON−OFF switch located on the instrument panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

MQ-8B

The Vertical Takeoff and Landing Tactical Unmanned Air Vehicle (VTUAV) system provides reconnaissance and surveillance; target acquisition and tracking, laser designation; target damage assessment; and communication relay capability.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT: MQ-8B

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	9 ft, 9 in	Maximum Takeoff Gross	3,150 lbs
Length	23 ft, 3 in		
Width	6 ft, 2 in		

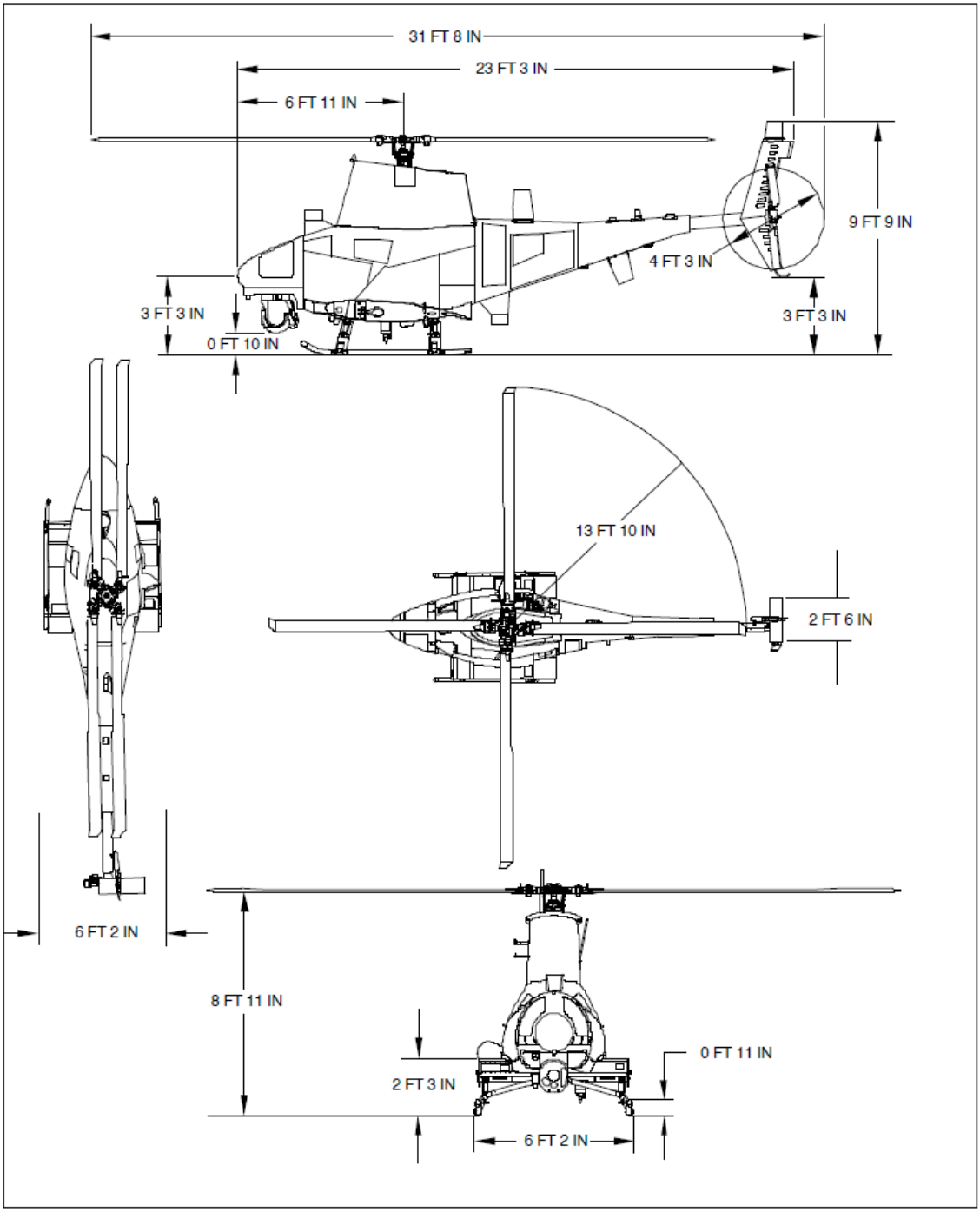
TABLE OF FUEL CAPACITIES

Tank	Gallons	Pounds (Appx.)
Total Internal	190	1295 lbs (JP-5) 1273 lbs (JP-8)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

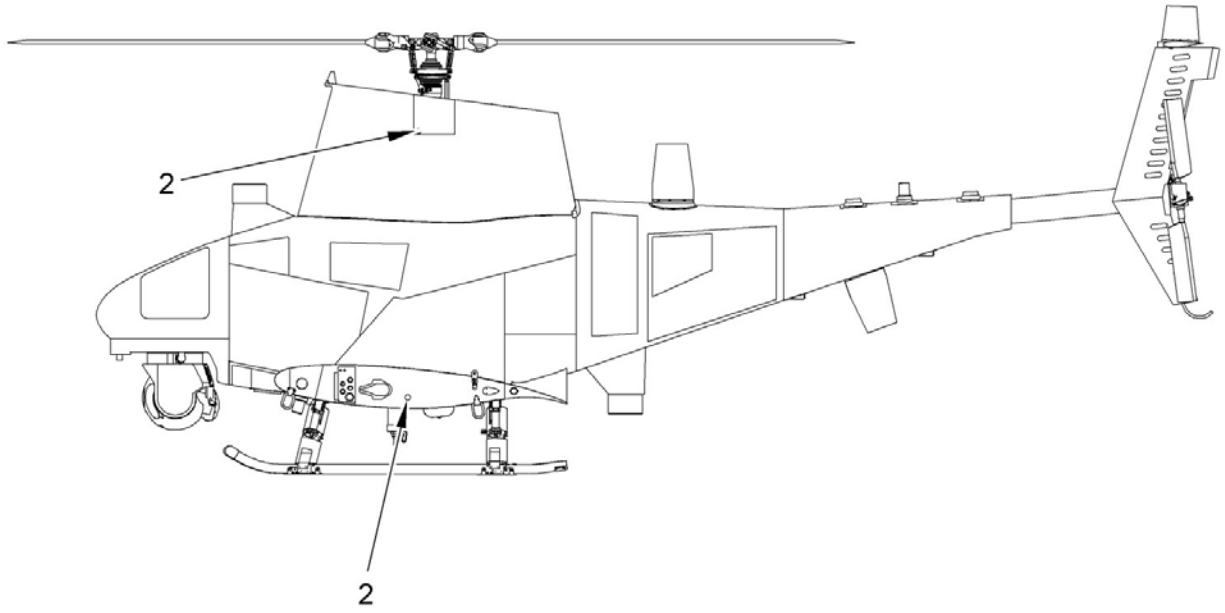
a) Aircraft Dimensions



(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

b) Aircraft Grounding Points

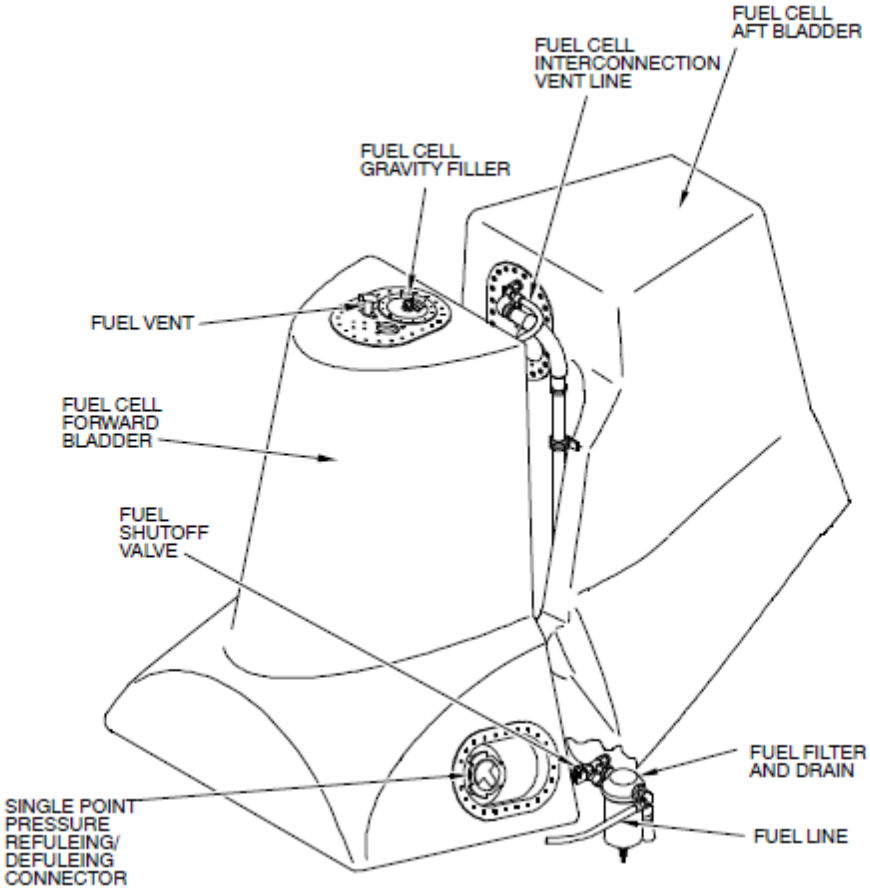


ICN-MQ8B-AA-12000000-78022-78022-21104-A-01-1

(Source: A1-MQ8BA-IETM)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

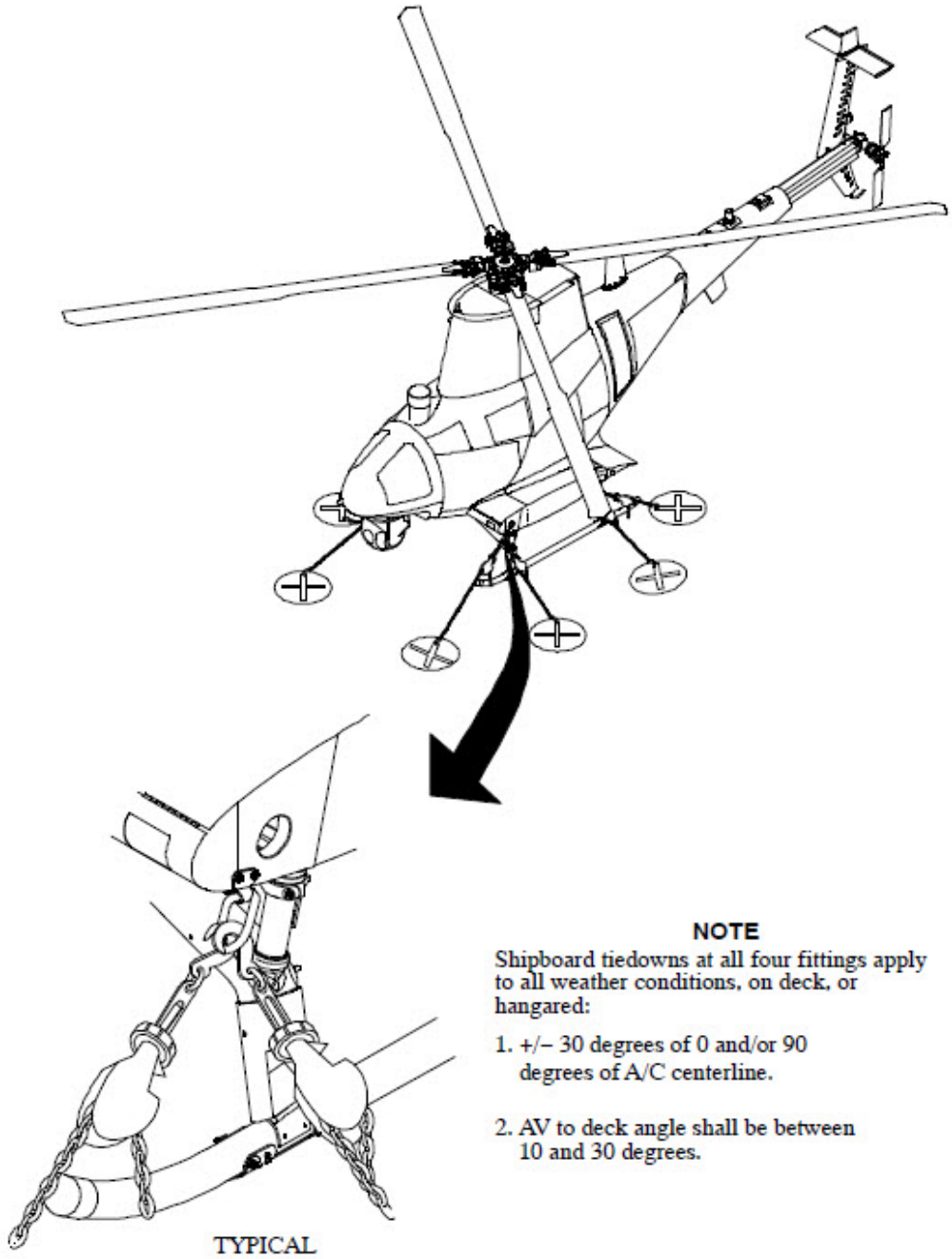
c) Location of Fuel Subsystem Components.



(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) Tie Down Locations



NOTE

Shipboard tiedowns at all four fittings apply to all weather conditions, on deck, or hangared:

1. +/– 30 degrees of 0 and/or 90 degrees of A/C centerline.

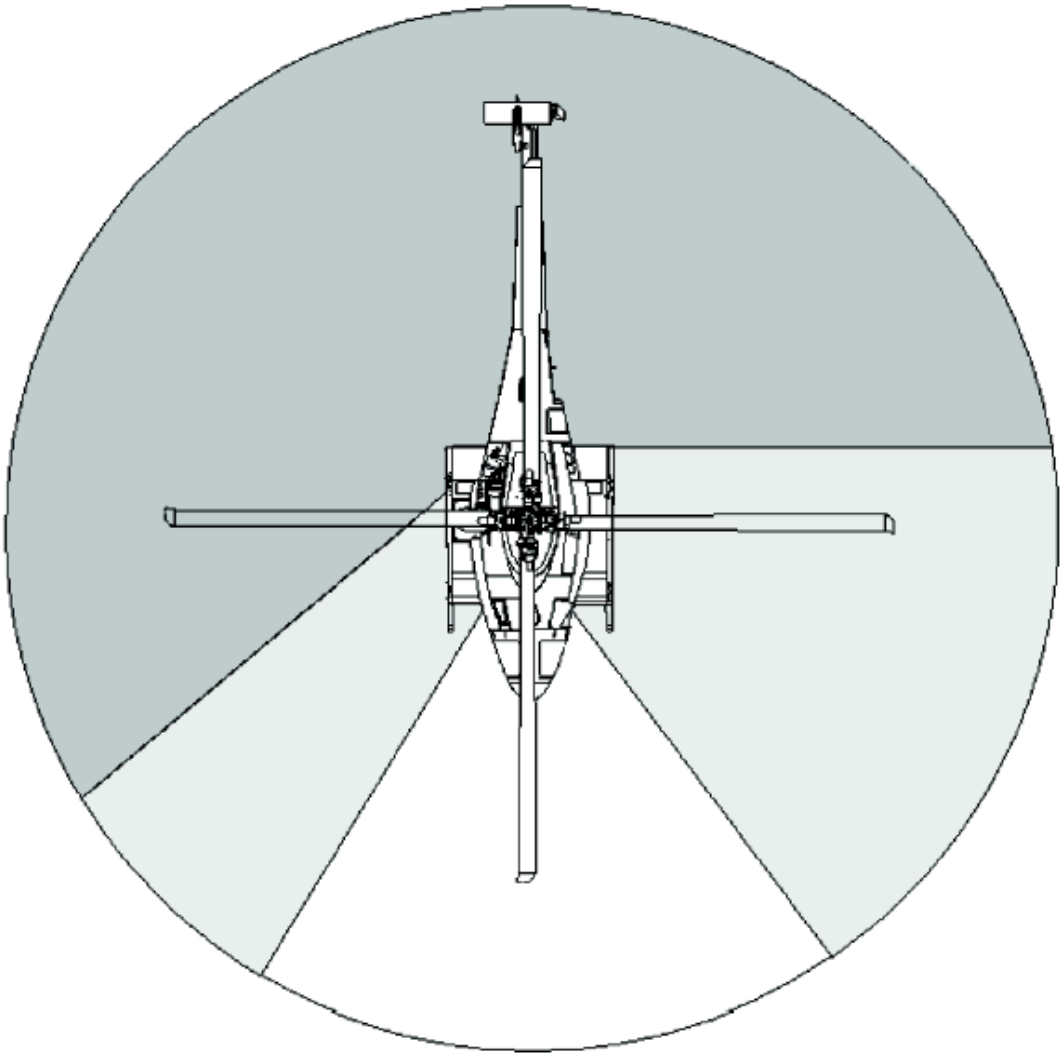
2. AV to deck angle shall be between 10 and 30 degrees.

(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Hazard Areas



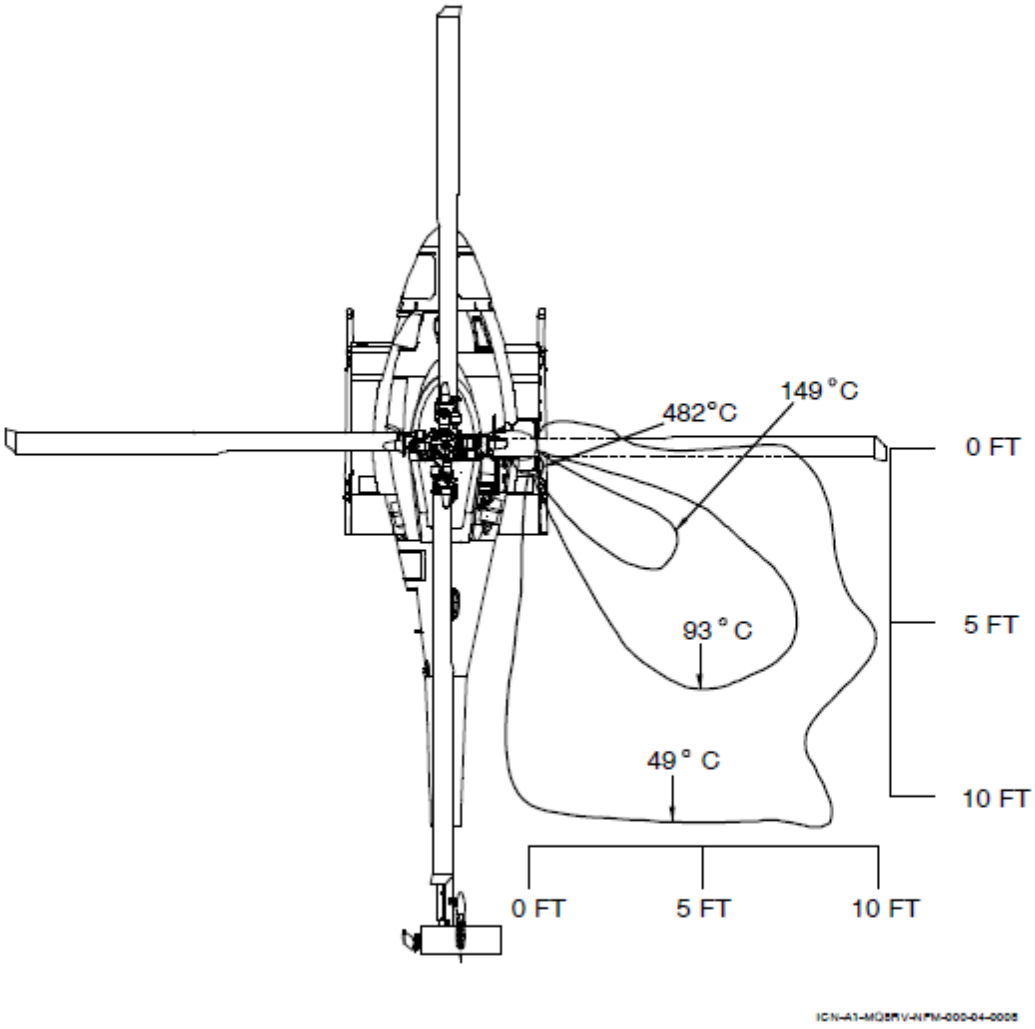
PERSONNEL APPROACH AREAS
(BLADES ROTATING)

	UNSAFE – DO NOT APPROACH
	APPROACH ONLY IF NECESSARY
	APPROACH WITH CAUTION
WARNING: COMPLY WITH ALL HELICOPTER SAFETY INSTRUCTIONS WHEN APPROACHING ROTATING BLADES	

(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

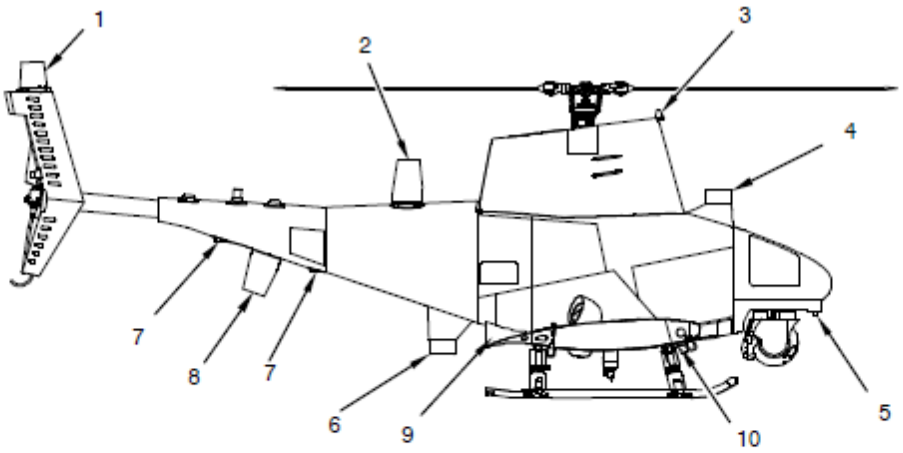
b) Engine Exhaust Areas



(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) Electromagnetic Radiation Hazard Distances



MINIMUM SAFE DISTANCE (FEET)

		HAZARDS OF ELECTROMAGNETIC RADIATION TO PERSONNEL (HERP)	HAZARDS OF ELECTROMAGNETIC RADIATIONTO ORDNANCE (HERO)		
OPERATING ANTENNA		(SIX–MINUTE EXPOSURE)	HERO UNSAFE/ UNRELIABLE	HERO SUSCEPTIBLE	HERO SAFE
1	UHF/VHF 3	5 FT	70 FT	17 FT	10 FT
2	UHF/VHF 1	3 FT	100 FT	25 FT	10 FT
3	IFF UPPER	1 FT	10 FT	10 FT	10 FT
4	TCDL FORWARD	2 FT	10 FT	10 FT	10 FT
5	UCARS	1 FT	10 FT	10 FT	5 FT
6	TCDL AFT	2 FT	10 FT	10 FT	10 FT
7	RADALT	1 FT	1 FT	1 FT	0 FT
8	UHF/VHF 2	3 FT	100 FT	25 FT	10 FT
9	IFF LOWER	1 FT	10 FT	10 FT	10 FT
10	TCDL OMNI	1 FT	10 FT	10 FT	10 FT

(Source: A1-MQ8B-NFM-000)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

When pressure or gravity fueling, it is recommended to calculate the exact fuel load desired prior to fueling to avoid over fill.

CAUTION: Only qualified and authorized personnel are to operate fueling equipment. Using loose pyrotechnics, smoking, striking matches, working on AV or using any device producing flame within 50 feet of the AV is prohibited during fueling. A minimum of 50 feet should be maintained from other
AVs or structures, and 75 feet should be maintained from any operating radar antenna.

NOTE: AV power and/or external power is allowable only during pressurized fueling.

Specific notes on Pressure Refueling:

Pressure refueling requires the AV to be powered up. The pressure fueling port is located behind the pressure refueling door. The pressure refueling door is located on the left side of the AV below the rotor mast pylon. The pressure fueling port is used to pressure fuel the AV. A Fuel Tank Fill button and Fuel S/O Valve switch are located on the ground control panel. The Fuel S/O Valve switch is a three way switch with a guard (cover). Guard up, switch UP is fuel flow ON; switch in center position (guard as required) is fuel flow OFF; lower switch position (MOM) is not used. The complete Pressure Refuel procedure for use by fueling personnel or maintainers is found in the MQ-8B Interactive Electronic Technical Manual (A1-MQ8BA-IETM) for Organizational Maintenance in the [12] Servicing Procedures section.

Steps for pressure fueling are:
1. AV and fuel equipment ground – Connect.
2. MPEDD – Connect, On. Select FUEL on screen to open the Fuel Status Page.
3. Fuel nozzle grounding wire to an AV ground receptacle – Connect.
4. Fuel source (truck/tank) ground wire to an AV grounding receptacle on either the LH or RH sponson – Connect.
5. High pressure fueling receptacle cap – Unlatch and remove.
6. Slots on nozzle faceplate to the lugs on the AV pressure refueling receptacle – Align.
7. Fueling nozzle – Press in firmly and rotate locking handles clockwise to a positive stop.
8. Nozzle flow control handle to full open position – Engage.
9. Fuel hose – Pressurize.

Note: Ensure there are no fuel leaks.

10. FUEL S/O VALVE switch on the GCP is in CLOSED position and switch guard is down – Verify.
11. FUEL TANK FILL circuit breaker on the GCP – Press to engage.

Note: The fuel system has a reducer that limits the fueling rate to 30 GPM. Prior

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

to refueling, calculate the desired fuel on-load in gallons and divide that number by 30 to get the total number of minutes needed to transfer the required fuel. The number can also be multiplied by 60 to get the total time in seconds to transfer the same amount of fuel. Example for a 50 gallon on-load: 50 / 30 = 1.667 minutes; 1.667 minutes X 60 = 100 seconds.

12. Fuel S/O VALVE switch – Position switch guard up and move to ON position.
13. Fuel shut-off pre-check – Perform after 10 seconds of fuel flow.
14. FUEL S/O VALVE switch – Move to CLOSED position and verify fuel flow stops.
15. FUEL S/O VALVE switch – Move to ON position and resume fueling
16. Stopwatch – Monitor the fuel on-load time as computed.
17. FUEL S/O VALVE switch – Move to CLOSED position when computed fuel on-load time has elapsed and verify fuel flow stops.
18. Fuel readings to stabilize for 120 seconds – Allow and verify fuel quantity on the MPEDD
19. Repeat Step 14 through Step 17 if more fuel is required.
20. FUEL TANK FILL circuit breaker – Pull.
21. Fuel hose – Depressurize.
22. Fueling nozzle – Press in firmly and rotate locking handles counterclockwise to disconnect.
23. Fuel nozzle ground wire – Disconnect from AV ground receptacle
24. Fuel source grounding wire – Disconnect from AV ground.
25. High pressure fueling receptacle cap – Install.
26. Pressure refueling door – Close and secure.
27. MPEDD – Disconnect and power down, as required.

Specific notes on Gravity Refueling:

Follow gravity refueling procedures per NATOPS 00-80T-109. However, note the following aircraft specific Warning, Cautions, and Notes:

The gravity fueling filler port is located inside the gravity fueling door on top of the rotor mast pylon. It is used to gravity fuel the AV.

WARNING: Prior to gravity fueling the AV power is to be turned off and external power disconnected. Gravity fueling with power on creates a spark potential. A static electrical charge could ignite a fire or an explosion resulting in injury or death and or damage of equipment.

CAUTION: To prevent fuel overflow caused by temperature change, fuel level should be approximately three inches from the top of the filler cap or damage to equipment may occur.

NOTE: When gravity fueling is used, the fuel tank can be overfilled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Specific notes on Pressure Defueling:

Perform pressure defuel in accordance with the following steps:

1. Confirm the Master switch (121) on the GCP (120) is in the OFF position and that external power is disconnected.
2. Verify that the AV grounding wire is securely connected to either the LH or RH sponson grounding receptacle (2) on one end and an approved aircraft ground on the other. Additionally, ensure that the fuel source (truck/tank) is securely connected to the same approved aircraft ground.
3. Connect the fuel nozzle grounding wire to AV grounding receptacle (2) on the left hand side of the upper pylon.
4. Connect the fuel source (truck/tank) grounding wire to the AV grounding receptacle (2) on either the LH or RH sponson.
5. Remove fuel cell gravity fill cap (3) to provide second redundant vent.
6. Unlatch and remove pressure fueling receptacle cap (1).
7. Align slots on nozzle faceplate to the lugs on the AV pressure refueling receptacle. Press the fueling nozzle in firmly and rotate locking handles clockwise to a positive stop.
8. Engage the nozzle flow control handle to the full open position.
9. Verify fuel hose is not kinked, routed on or near sharp objects or heat sources.
10. Turn on suction system from fuel source to begin defuel. Make sure there are no fuel leaks.
11. Visually observe fuel level through gravity fill opening.
12. Turn off suction system from fuel source when fuel level is below fuel pickup point or when fuel suction has stopped.
13. Press the fueling nozzle in firmly and rotate locking handles counterclockwise to disconnect the fueling hose nozzle. Remove fuel nozzle.
14. Install pressure fueling receptacle cap (1).
15. Install fuel cell gravity fill cap (3).
16. Remove residual fuel, if required for maintenance. MQ8B-AA-12-10-00-00A-221A-A. If not required, proceed to Step xviii.
17. Disconnect the fuel nozzle grounding wire from AV grounding receptacle (2).
18. Disconnect the fuel source (truck/tank) grounding wire from the AV grounding receptacle (2).
19. Disconnect fuel source grounding wire from the approved aircraft ground.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

SPECIAL NOTES – AIRCRAFT FUEL SYSTEM

- During refueling with engines/rotors turning, approach and depart the AV from the port side, abeam the centerline of the Main Rotor Mast to remain clear of the Tail Rotor and Engine Exhaust.

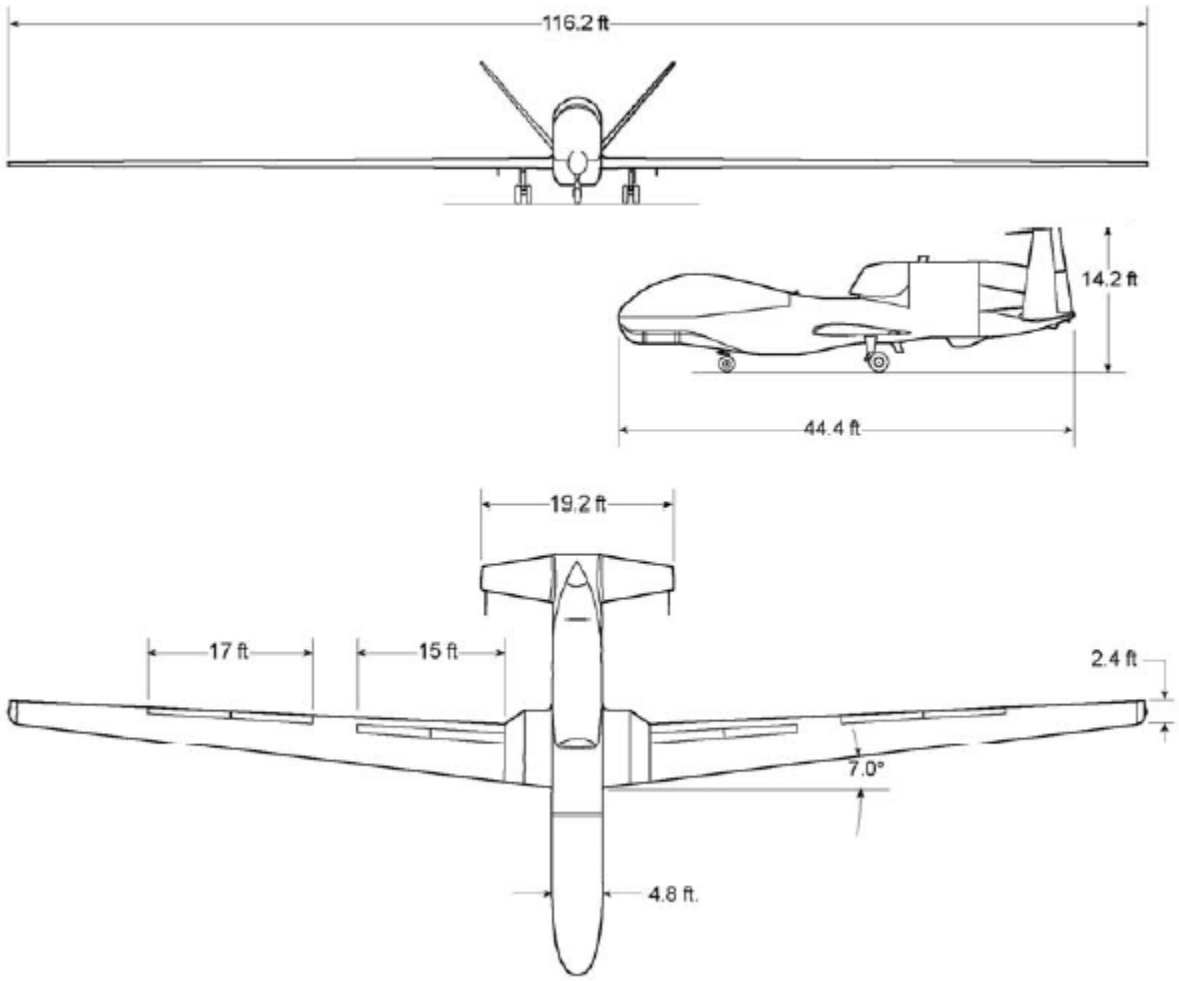
- During pressure refueling, verify fueling source supply is not greater than 55 psig. Damage to equipment may occur.

- Due to a delay in fuel quantity readings on the MPEDD, fuel onloads are to be monitored by use of a stopwatch to time the amount of fuel transferred. The fuel system has a reducer that limits the fueling rate to 30 GPM. Prior to refueling, calculate the desired fuel onload in gallons and divide that number by 30 to get the total number of minutes needed to transfer the required fuel. The number can also be multiplied by 60 to get the total time in seconds to transfer the same amount of fuel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

RQ-4

The BAMS-D, RQ-4 UNMANNED AIRCRAFT SYSTEM (UAS) is a high-altitude, unmanned, Multi-INT, persistent maritime patrol system which provides the Fleet with an experimentation and demonstration capability to support the Navy's transformational initiatives of Sea Power 21 and FORCEnet through the Fleet Readiness Training Program (FRTP).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	14.2 ft.	Maximum Gross	26,750 (lbs)
Length	44.4 ft.		
Wingspan	116.2 ft.		

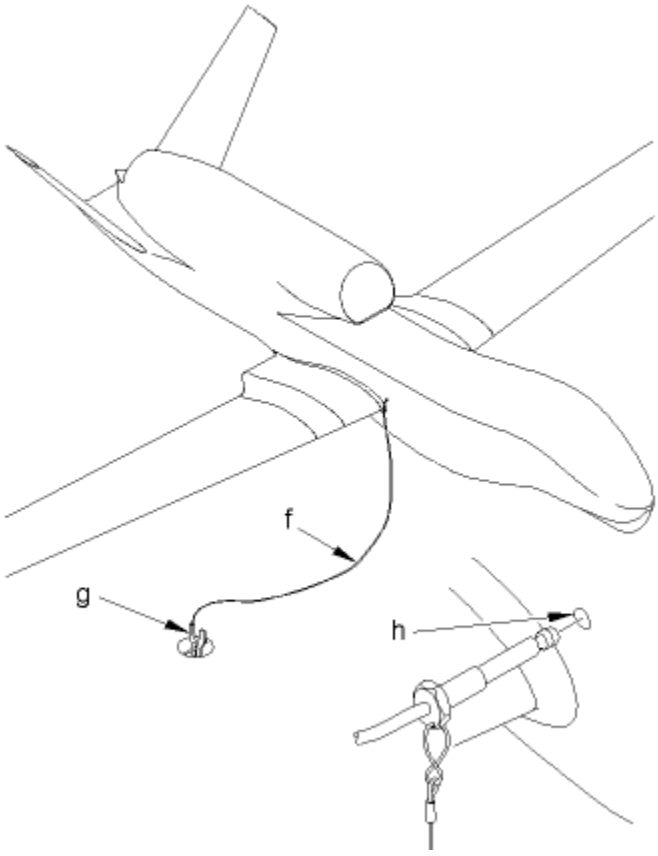
TABLE OF FUEL CAPACITIES

Tank		Gallons	Pounds (Optional)
Internal	Forward Fuselage	727.6	4948
	Aft Fuselage	262.4	1784
	Wing Tank, Left	325	2210
	Wing Tank, Right	325	2210
	Wing Root Tank	566	3847
	Sump Tank	64.4	438
Total Internal		2270.4	15,437

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

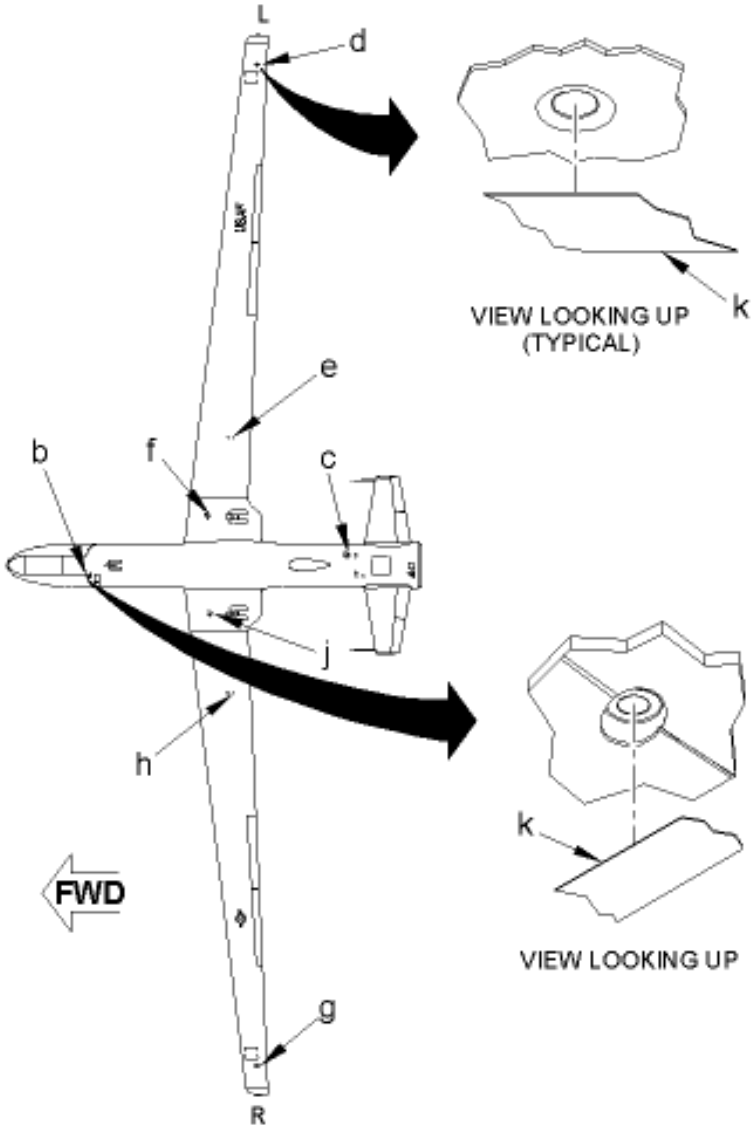
a) Aircraft Grounding Location



Refer to RQ4-A-00-20-00-00A-120A-A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

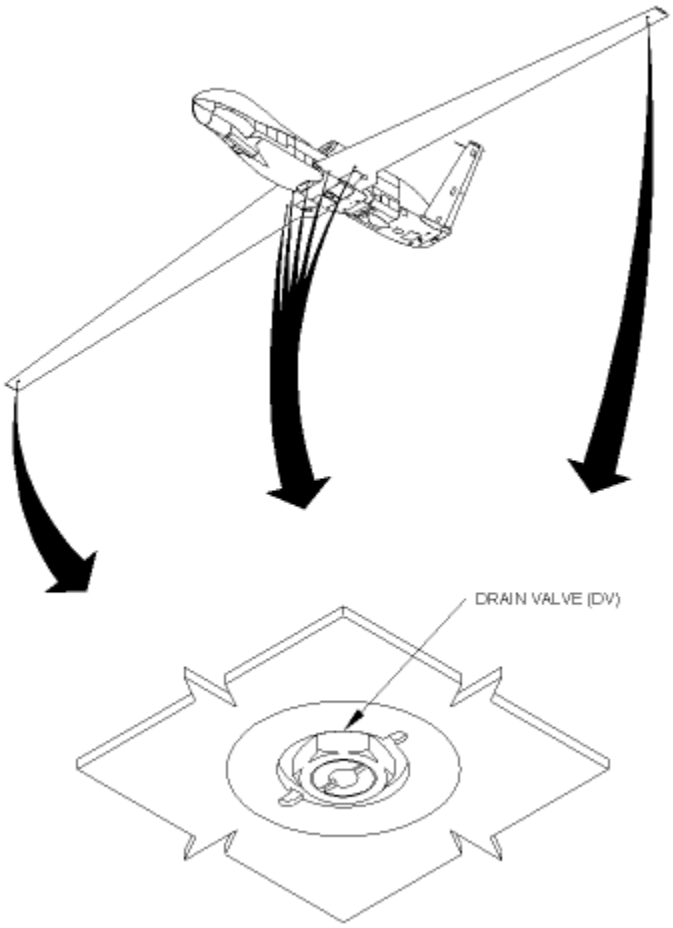
b) Aircraft Tie Down Locations



Refer to RQ4-A-10-20-01-00A-100A-A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

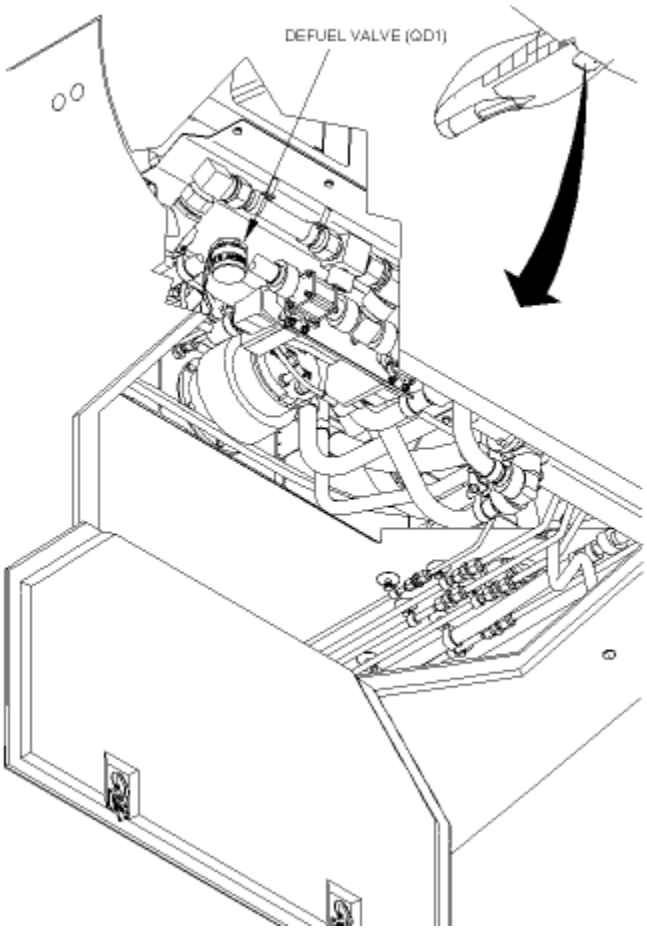
c) Fuel Tank Low Point Drain Location



Refer to RQ4-A-00-15-04-00A-100A-A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

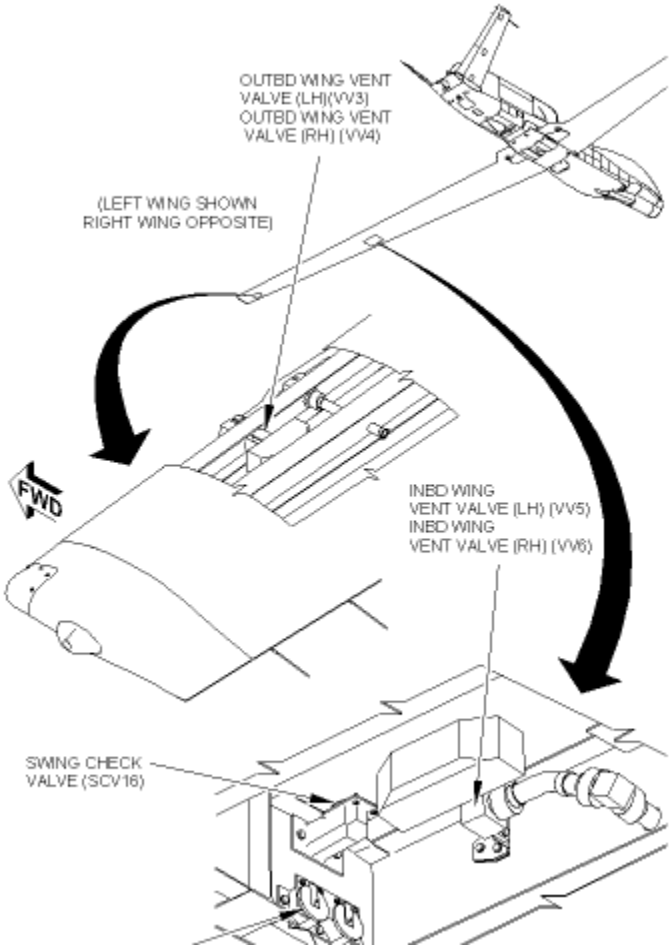
d) Defuel Valve



Refer to RQ4-A-12-11-03-00A-221A-A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

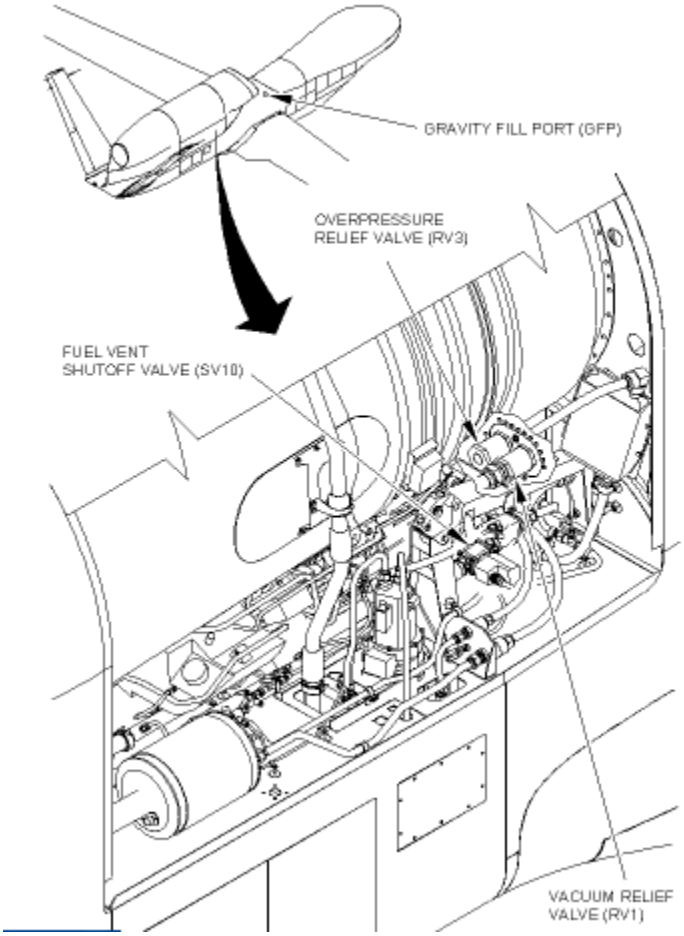
e) Wing Vent Locations



Wing Vent Locations

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

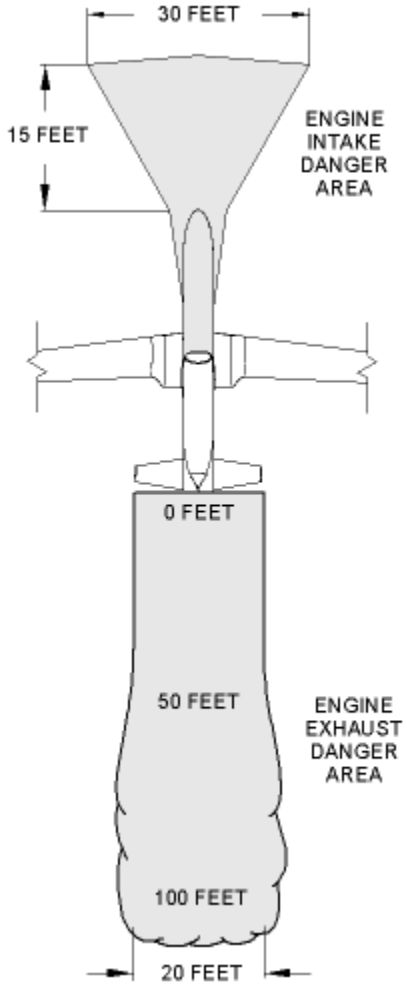
f) Gravity Fill and Relief Valve Locations



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

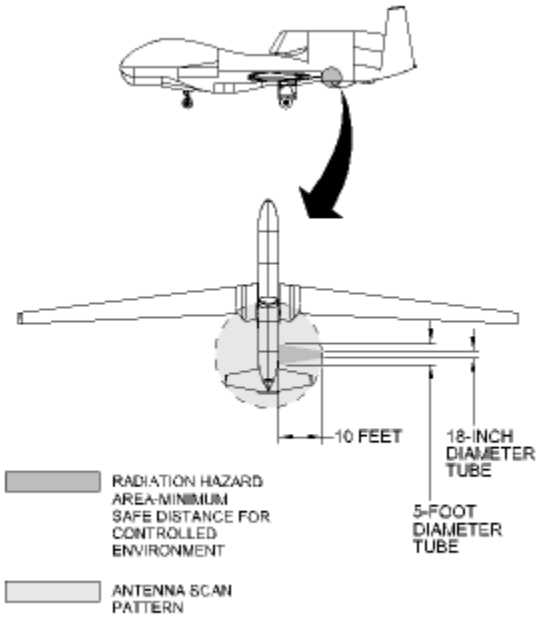
a) Engine Intake and Exhaust Danger Area



Refer to RQ4-A-06-30-00-00A-018A-A for details.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

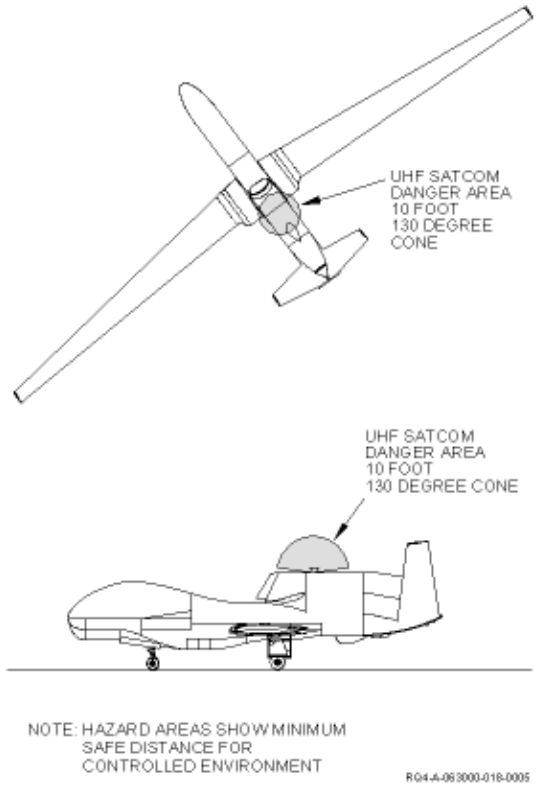
b) Common Data Link Danger Areas



COMMON DATA LINK

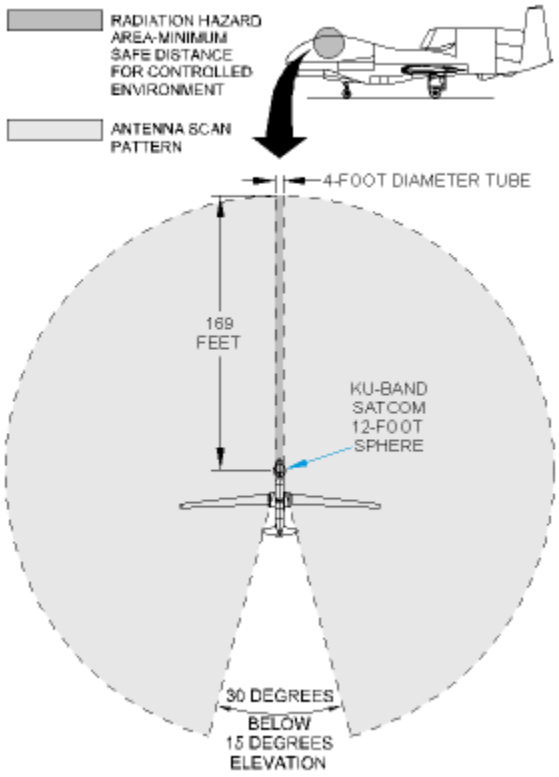
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

c) UHF SATCOM Danger Areas



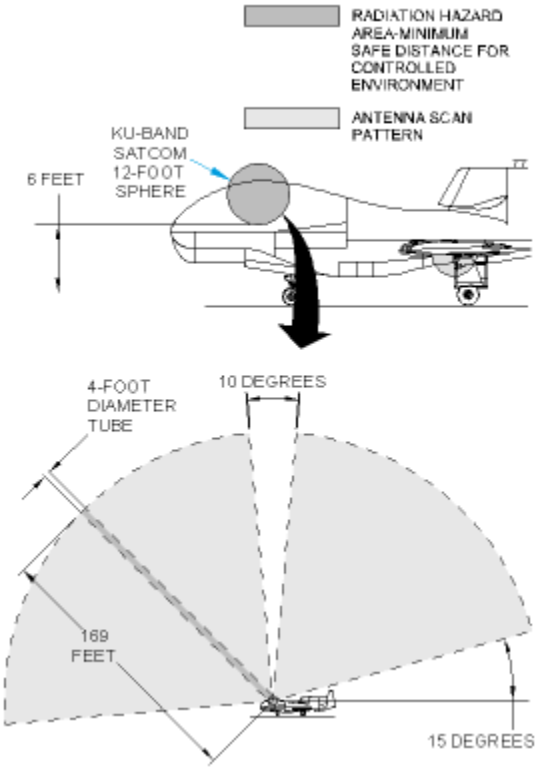
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

d) KU SATCOM Danger Areas



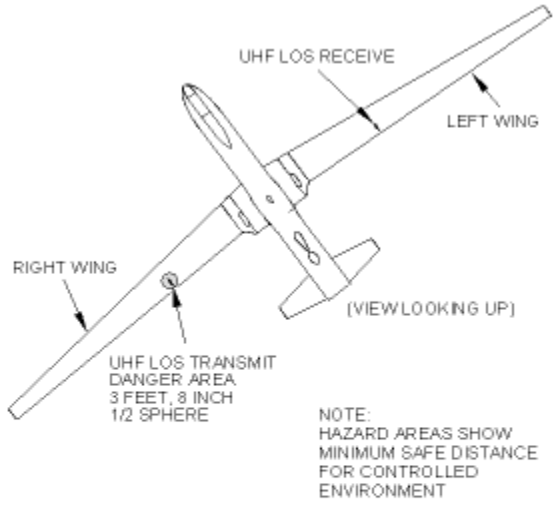
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

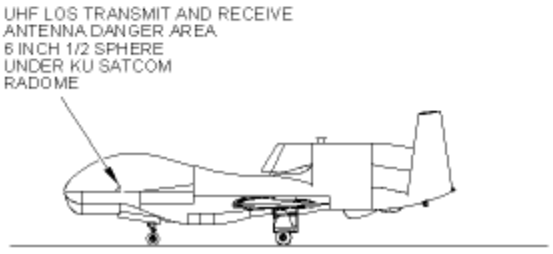
e) KU SATCOM Danger Areas



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

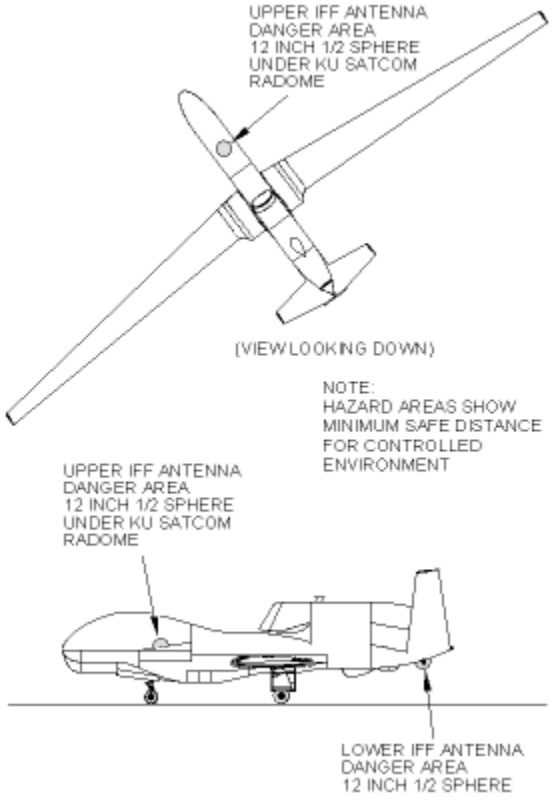
f) UHF LOS Danger Areas





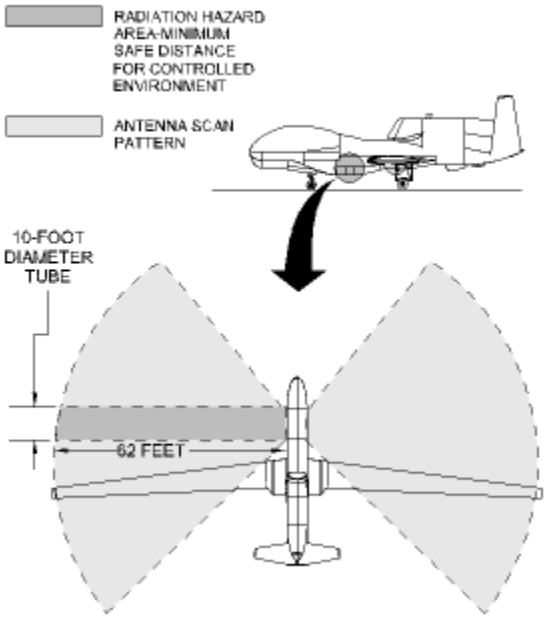
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

g) IFF Antenna Danger Areas



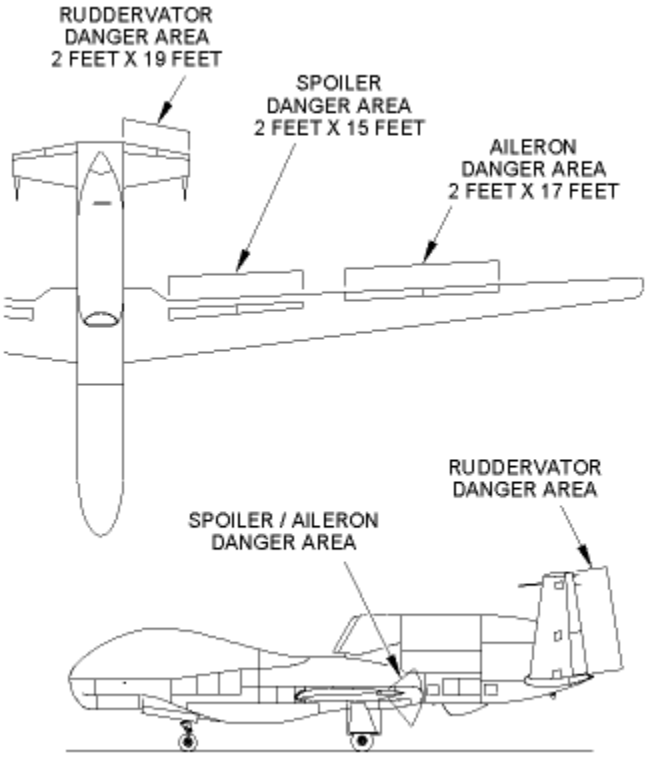
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

h) SAR Antenna Danger Areas



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

i) Flight Control Surface Danger Areas



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

Fueling RQ-4 is a complex procedure that requires special support equipment, fuel density determination, manual fuel load calculation, and weighing and balancing the AV. Refer to RQ4-A-12-11-01-00A-211A-A for details and required equipment.

DEFUELING PROCEDURES

Defueling RQ-4 is a complex procedure that requires special support equipment and procedures. Refer to RQ4-A-12-11-03-00A-221A-A.

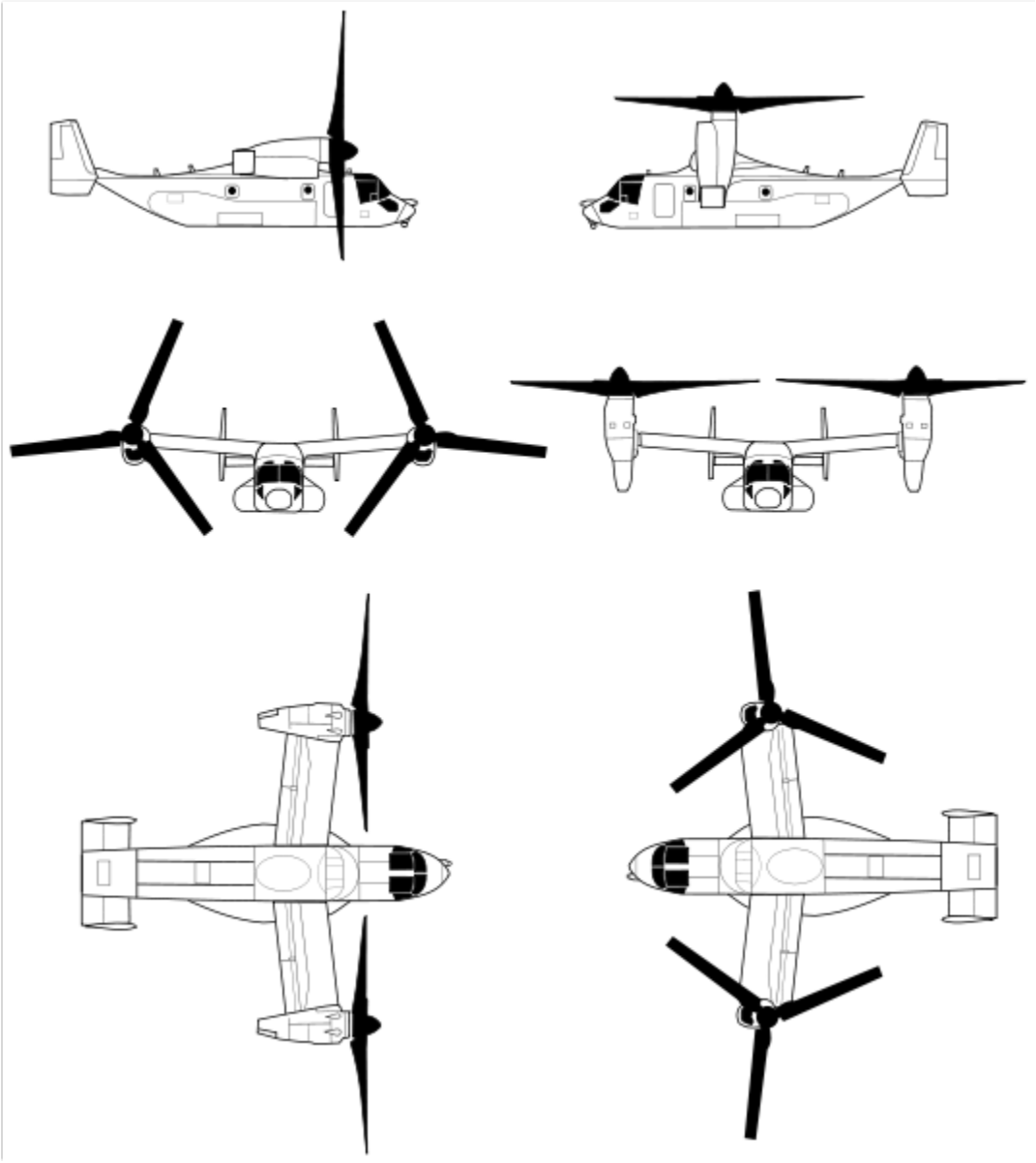
SPECIAL NOTES – AIRCRAFT FUEL SYSTEM

RQ-4 has a unique fuel system that requires special support equipment and procedures for refueling, defueling and trouble shooting. Only qualified personnel should service the RQ-4 fuel system. Refer to 1Q-4(R)A-2-WA-2 for procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

MV-22B

The Bell Boeing V-22 Osprey is a multi-mission, military, tiltrotor aircraft with both a vertical takeoff and landing, and short takeoff and landing capability. It is designed to combine the functionality of a conventional helicopter with the long-range, high-speed cruise performance of a turboprop aircraft.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CHARACTERISTICS

Dimensions		Weight	
Height	22 ft, 1 in	Maximum Gross	52,600 lbs
Length	57 ft,4 in	(VTOL)	57,000 lbs
Spread	63 ft	(STO)	60,500 lbs
Folded		(Max Alt GW)	
Wingspan			
Spread	84 ft, 7in		
Folded	including rotor diameters 18 ft, 5 in		

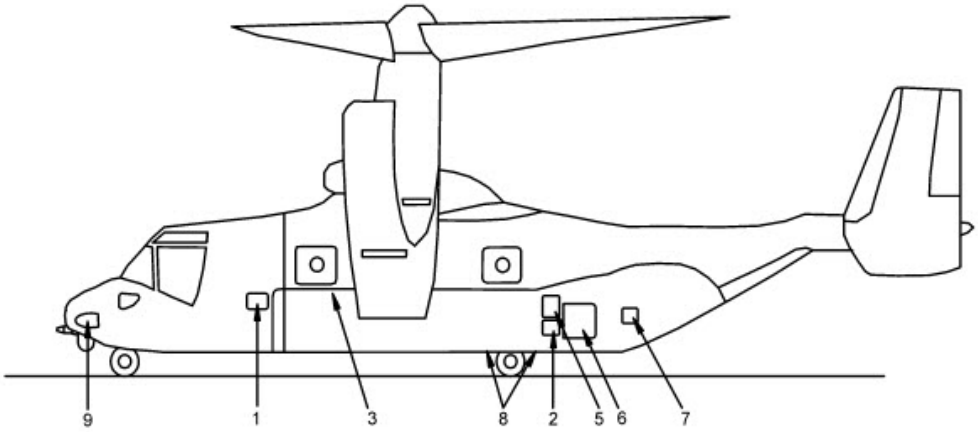
TABLE OF FUEL CAPACITIES

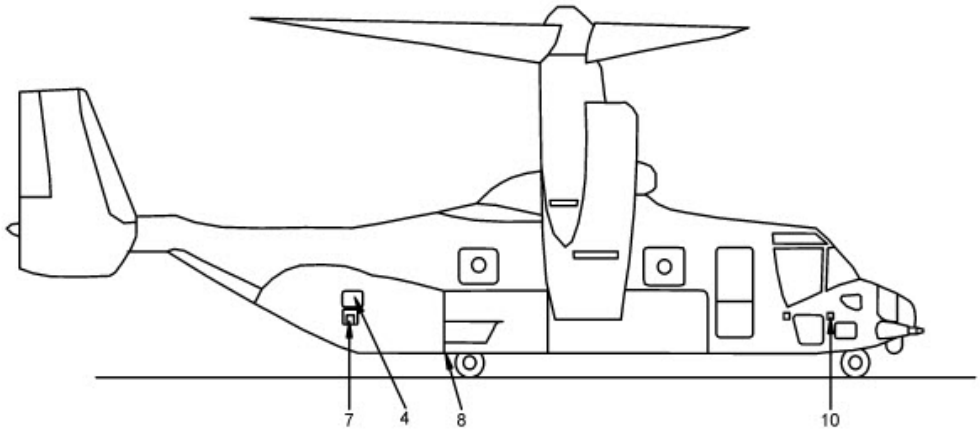
Tank		Gallons	Pounds (Optional)
Internal	Wing Auxiliary Tank (L/R)	294	2000
	Sponson Tanks (L/R)	378	3250
	Aft Sponson (optional)	301	2050
	Mission Auxiliary Tank System (FWD, MID, AFT)	430 Max 80% FWD (345) 75% MID (324) 75% AFT (324)	2853 (secondary shutoff) 2350 2200 2200
	Feed Tanks (L/R)	88	600
Total		Variable Depending on Configuration	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

AIRCRAFT CONFIGURATION

a) Various Connection and System Locations



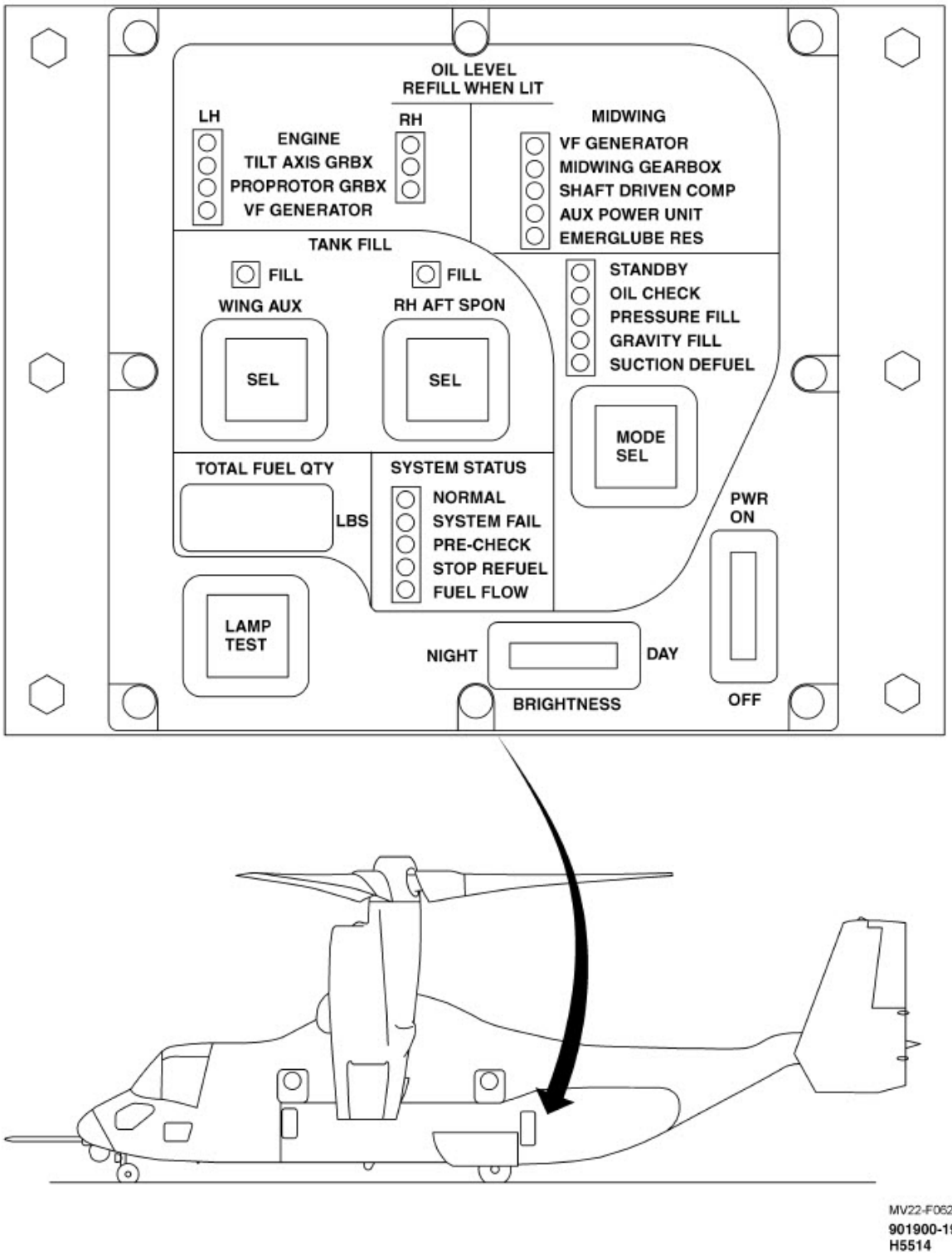


Item	Function	Location
1	External Electrical Power Connection	Forward Left Hand Fuselage
2	Pressure Fuel/Defuel Connection	Aft Left Hand Sponson
3	Gravity Fueling Access	Mid Left Hand Sponson
4	Emergency LNG Blow-Down Access	Aft Lower Right Hand Sponson
5	GTC-85 Air Connection	Aft Left Hand Sponson
6	Environmental Control Unit	Aft Left Hand Sponson
7	Hydraulic Ground Connection No. 1 Flight Control No. 2 Flight Control No. 3 Utility System	 Aft Left Hand Sponson Aft Right Hand Sponson Aft Right Hand Sponson
8	Fuel Drain Sump Access	Under Sponson (Left and Right Hand Side)
9	Nose Gear Hike Nitrogen Valve	Forward Left Hand Fuselage
10	Nose Gear Ground Lock Pin	Forward Right Hand Lower Fuselage

MV22-F127

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

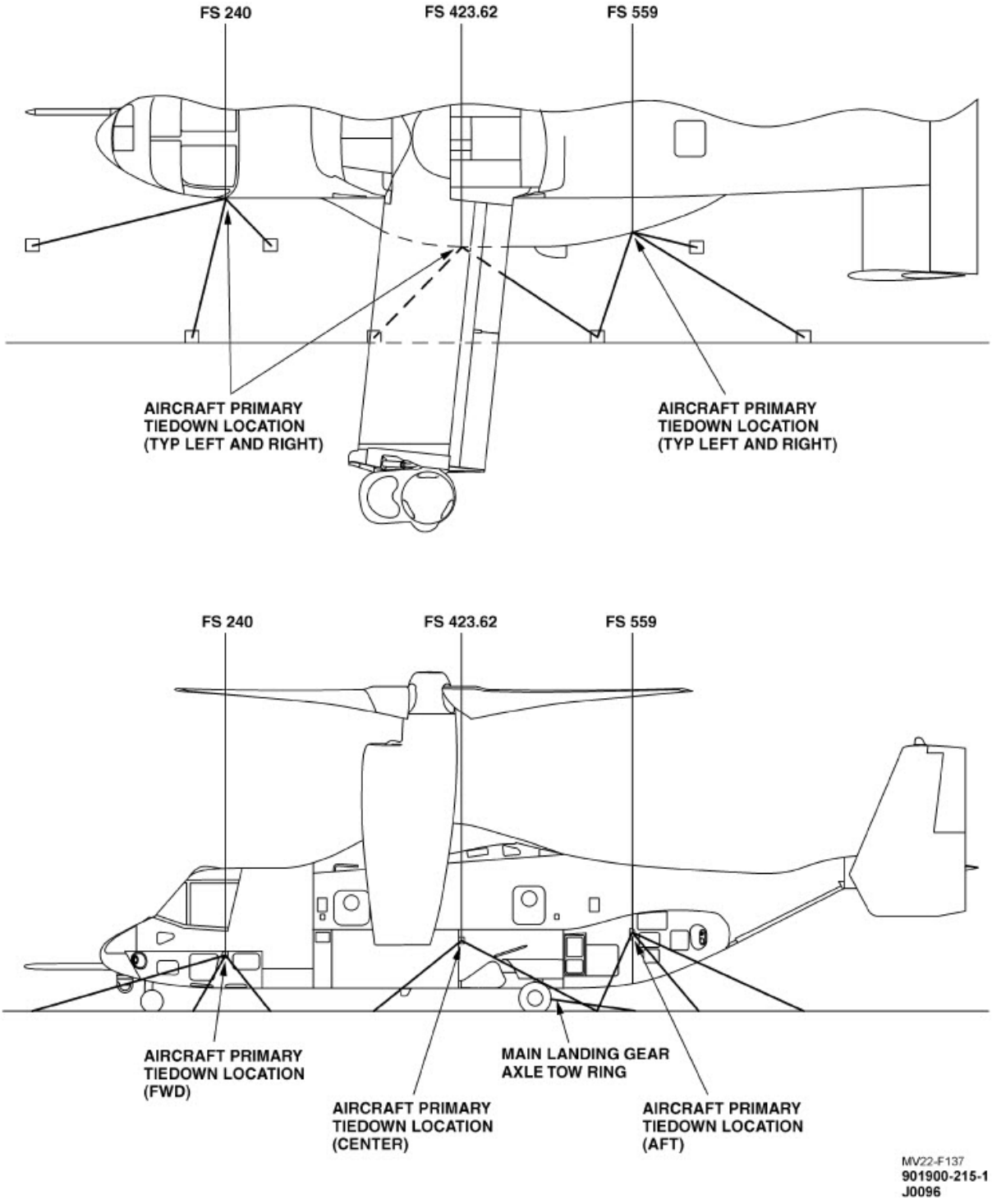
b) Ground Refuel/Defuel Panel



MV22-F062
901900-192
H5514

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

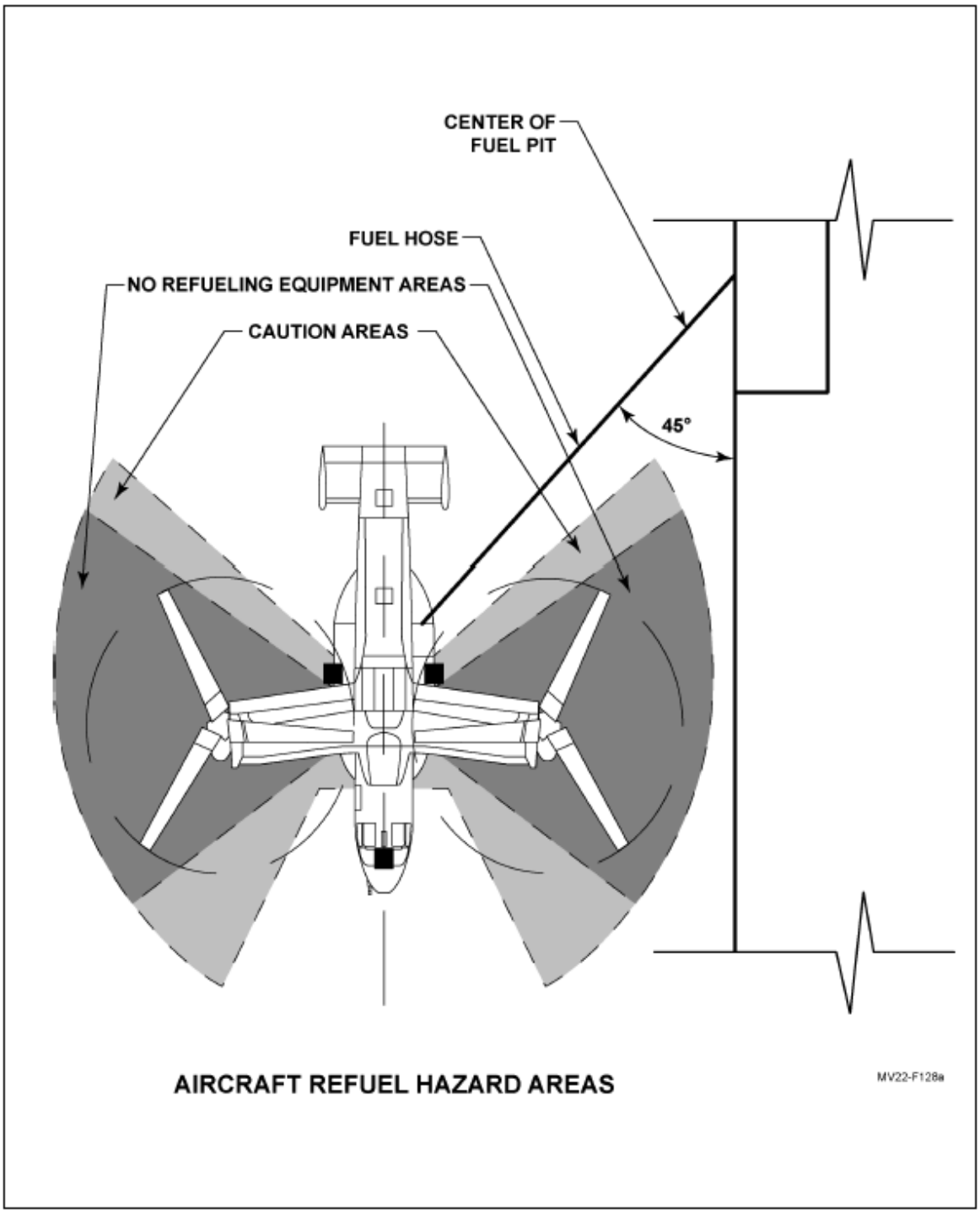
c) Tiedown Fitting Locations



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PERSONNEL DANGER ZONES

a) Aircraft Refuel Hazard Areas



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

PLATFORM SPECIFIC NOTES FOR REFUELER AND NOZZLE OPERATORS

(Cold) Pressure Refueling with Engines/APU Off

Single point pressure fueling (and suction defueling) of all airframe fuel tanks is provided by the GRDP through the single point refuel/defuel adapter. The GRDP is located aft of the left main landing gear behind panel 7LS1. Electrical power for the Fuel Management Units, fuel valves, and the GRDP is supplied from the essential/battery bus. No external power is required for pressure refueling. Refueling is controlled by a push-button switch marked MODE SEL, and two TANK FILL SEL push-button switches marked WING AUX and RH AFT SPON. System status is displayed by indicator lights and a digital total fuel quantity indicator. The maximum refueling rate is approximately 300 gallon/minute at 50 psi.

CAUTION

Pressure fueling operations are to be halted immediately upon indications of fuel spillage from any fuel tank vent, or loud/unusual noise accompanied by wing vibration. Inspect suspect fuel tank and immediate area, including the internal wing (As required), for structural damage.

CAUTION

- To prevent structural damage ensure that fueling pressure does not exceed 55 psi.
- Cold refueling aircraft with JP-4 or commercial JET B requires a dedicated fire extinguisher operator in addition to the nozzle operator and the station/ truck operator.
- MATS (if installed) should be limited to 2350 lbs JP5 in the forward tank and 2200 lbs JP5 in the mid and aft tank for normal operations. If these levels are exceeded, the restraint system for the MATS may not prevent tank movement during a crash. Visual inspection of the fuel quantity indicator (dipstick) is required to determine crashworthiness fuel limitations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

To pressure refuel the aircraft:

1. Disembark all non-essential personnel
2. Verify fire extinguisher is properly positioned
3. Fuel truck/ship ground cable — Connect to ground
4. Fuel truck/ship ground cable — Connect to aircraft
5. GRDP panel and SPR adapter panel — Open
6. SPR adapter — Remove fuel cap
7. GRDP power switch — ON (Total fuel quantity should display)
8. Panel lighting — Adjust as desired
9. LAMP TEST — Press (All lights should illuminate in sequence)
10. Pressure refueling nozzle — Connect to aircraft SPR adapter

WARNING

Nozzle needs to seat firmly on the adapter and not be cocked. A cocked nozzle indicates a malfunction of nozzle's safety interlock system, which may lead to fuel spray or spill.

11. Open fueling nozzle valve

CAUTION

Fuel flow is not to be regulated by flow control handle on fuel nozzle. The fuel nozzle handle is to be placed in either of two locked positions — fully open or fully closed. The handle is not to be used as a flag to indicate fuel flow. Excessive wear on the aircraft adapter and the fuel nozzle poppet will result if the handle is allowed to "float" in the unlocked position.

12. MODE SEL switch — Press and release until the PRESSURE FILL light illuminates (Wait two seconds for process to start)
13. Under TANK FILL — Press SEL FILL WING AUX and/or FILL RH AFT SPON (If desired)
14. Fueling truck/ship station — Start refueling (Depress and hold deadman switch if equipped)
15. Verify under SYSTEM STATUS on GRDP that PRECHECK light is illuminated

Note

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Once fuel flow to the aircraft has begun, the FUEL FLOW indicator will illuminate, and the fuel system automatic PRECHECK function will begin.

16. Verify that the SYSTEM FAIL light is not illuminated. If the SYSTEM FAIL light is illuminated, stop the refueling process. If the PRECHECK has successfully completed, the aircraft will begin to take on fuel again

Note

- Twenty seconds or less after the PRECHECK light is illuminated, fuel flow will stop. The PRECHECK light will extinguish and the NORMAL light will illuminate.
- If the SYSTEM FAIL light is illuminated, pressure refueling should not be continued unless individual tank quantities are monitored and if necessary, isolated to prevent a tank overfill, through the FUEL Status Layer MENU 3 or MENU 4 in the cockpit.

17. Observe SYSTEM STATUS display

CAUTION

- Monitor the SYSTEM STATUS lights during refueling. If the STOP REFUEL indicator illuminates, immediately cease refueling. Failure to do so may cause fuel spillage resulting in aircraft damage and possible fire hazard.
- Immediately cease fueling if the GRDP total fuel quantities exceed the following values:
  - [A] — 9750 lbs
  - [B] — 11700 lbs

18. Fuel aircraft until flow automatically stops or until desired fuel quantity is indicated on the GRDP
19. Fuel nozzle — Close, disconnect, and install fuel cap on SPR adapter
20. Grounding wires — Remove
21. GRDP PWR switch — OFF
22. GRDP access panel and SPR adapter access panel — Close.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Pressure Refueling with Engines Off/APU Operating

The APU may be used to supply electrical power to the aircraft during pressure refueling. All personnel (except necessary refueling personnel) are to disembark from the aircraft and move at least 50 ft away from aircraft during refueling operations with the APU operating. There are five crew personnel requirements: one person (qualified APU operator/ pilot) located in the cockpit monitoring APU operations and Fuel Status page during refueling operations, and four ground crew (one nozzle operator, one refueling coordinator (plane captain) who have to maintain constant communication with the APU operator/pilot and in hand signal communication with ground personnel, one station operator (deadman control), and one fire guard located outside the aircraft within 10 ft of the APU exhaust). Personnel are to wear proper hearing protection.

CAUTION

MATS (if installed) should be limited to 2350 lbs in the forward tank and 2200 lbs in the mid and aft tanks for normal operation. If these levels are exceeded, the restraint system for the MATS may not prevent tank movement during a crash.

1. Weapons/countermeasure systems and BFT transceiver — DISARM/OFF
2. BFT transceiver — OFF
3. WXR — OFF or STBY
4. Parking brake — Set
5. Searchlights/landing lights — OFF (Minimum two minutes prior to refueling)
6. All nonessential personnel — Disembark and move 50 ft from aircraft
7. APU operator/pilot — Maintain ICS contact with plane captain
8. APU operator/pilot — Monitor UHF (As required)
9. Cockpit side windows — Closed
10. Ramp door — Open
11. Ramp — Level
12. Landing gear — Chocked
13. Landing gear pins — All in
14. Aircraft — Grounded
15. EMCON — Select
16. Pitot heat — OFF
17. RADALT — OFF
18. IFF — Standby
19. TPUMP Boost — Off (Confirm)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

WARNING

Once fueling evolution has commenced, the aircraft's electrical power status and connections are to not be changed until fueling has been stopped. Engines or auxiliary power units are not to be started or stopped and external power is not to be connected, disconnected, or switched on or off. Changing the aircraft electrical power status can create significant ignition sources.

20. Refuel aircraft using standard pressure refueling procedures described previously ((Cold) Pressure Refueling with Engines/APU Off).

CAUTION

- Monitor the SYSTEM STATUS lights during refueling. If the STOP REFUEL indicator illuminates, immediately cease refueling. Failure to do so may cause fuel spillage resulting in aircraft damage and possible fire hazard.
- Immediately cease fueling if any of the following tank quantities are exceeded:

	JP-5 (lbs)	**JP-8/JET A (lbs)**
Feed tanks	650	640
Wing auxiliary tanks (if installed)	2050	2015
Forward sponson tanks	3300	3250
Aft sponson tank (if installed)	2110	2075
MATS (if installed)	2875	2830

Note

If the SYSTEM FAIL light is illuminated, pressure refueling should not be continued unless individual tank quantities are monitored and if necessary, isolated to prevent a tank overfill, through the FUEL Status Layer MENU 3 or MENU 4 in the cockpit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

(Hot) Pressure Refueling with Engines Operating/Proprotors Engaged

Pilots are to monitor Fuel Status page during refueling. The refueling coordinator is to maintain constant communication with the cockpit and the nozzle operator during refueling operations. There are three ground crew requirements: one station operator, one fireguard, and one plane captain/nozzle operator who directs all movements of the aircraft, maintains communication between the fuel crew and pilots, and operates the fueling nozzle.

CAUTION

- To prevent possible damage, ensure that all portions of the refueling hose remains aft of the leading edge of the GRDP.
- Monitor the SYSTEM STATUS lights during refueling. If the STOP REFUEL indicator illuminates, immediately cease refueling. Failure to do so may cause fuel spillage resulting in aircraft damage and possible fire hazard.
- Immediately cease fueling if any of the following tank quantities are exceeded:

	JP-5 (lbs)	**JP-8/JET A (lbs)**
Feed tanks	650	640
Wing auxiliary tanks (if installed)	2050	2015
Forward sponson tanks	3300	3250
Aft sponson tank (if installed)	2110	2075
MATS (if installed)	2875	2830

Note

If the SYSTEM FAIL light is illuminated, pressure refueling should not be continued unless individual tank quantities are monitored and if necessary, isolated to prevent a tank overfill, through the FUEL Status Layer MENU 3 or MENU 4 in the cockpit.

12. Nacelles — Modulate as required to minimize gearbox over temperatures and deck heating

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Note

Modulating nacelle settings between 97_ and 75_ spreads the thermal load below the nacelles over a greater area, decreases exhaust heat density and mitigates deck heating. Ensure personnel and equipment are clear of the nacelles prior to any nacelle movement.

If gearbox temperatures cannot be maintained in normal range, stop refueling and:
13. Extended ground loitering − Hot ambient conditions — Conduct upon completion of refueling:
14. Chocks/ground wire — REMOVE
15. EMCON — DESELECT
16. TPUMP BOOST — AUTO
17. ECLs — FLY.

CAUTION

The MATS (if installed) should be limited to 2350 lbs in the forward tank and 2200 lbs in the mid and aft tanks for normal operation. If these levels are exceeded, the restraint system may fail during a crash.

1. Weapons/countermeasure systems — DISARM/OFF

WARNING

Failure to ensure that the internal ALE-47 chaff/flare safety pin is installed and locked after landing and the ALE-47 CMDS CCU Mode switch is in the off position may result in inadvertent release of stores.

2. BFT transceiver — OFF
3. WXR — OFF or STBY
4. Parking brake — SET
5. No. 1 ECL or overdeck eng — AFT of FLY until Nr ~83%
6. Hot brake check — COMPLETE
7. Landing gear — CHOCKED
8. Aircraft — GROUNDED
9. EMCON — SELECT
10. TPUMP BOOST — OFF

WARNING

- Once fueling evolution has commenced, the aircraft’s electrical power status and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

connections are not to be changed until fueling has stopped. Engines or auxiliary power units are not to be started or stopped and external power is not to be connected, disconnected, or switched on or off. Changing the aircraft electrical power status can create significant ignition sources.

- SATCOM transmissions are not to be made during refueling operations as they can create a significant ignition source.

11. Aircraft — REFUEL

CAUTION

- To prevent possible damage, ensure that all portions of the refueling hose remains aft of the leading edge of the GRDP.
- Monitor the SYSTEM STATUS lights during refueling. If the STOP REFUEL indicator illuminates, immediately cease refueling. Failure to do so may cause fuel spillage resulting in aircraft damage and possible fire hazard.
- Immediately cease fueling if any of the following tank quantities are exceeded:

	JP-5 (lbs)	**JP-8/JET A (lbs)**
Feed tanks	650	640
Wing auxiliary tanks (if installed)	2050	2015
Forward sponson tanks	3300	3250
Aft sponson tank (if installed)	2110	2075
MATS (if installed)	2875	2830

Note

If the SYSTEM FAIL light is illuminated, pressure refueling should not be continued unless individual tank quantities are monitored and if necessary, isolated to prevent a tank overfill, through the FUEL Status Layer MENU 3 or MENU 4 in the cockpit.

12. Nacelles — Modulate as required to minimize gearbox over temperatures and deck heating

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Note

Modulating nacelle settings between 97_ and 75_ spreads the thermal load below the nacelles over a greater area, decreases exhaust heat density and mitigates deck heating. Ensure personnel and equipment are clear of the nacelles prior to any nacelle movement.

If gearbox temperatures cannot be maintained in normal range, stop refueling and:
13. Extended ground loitering − Hot ambient conditions — Conduct

Upon completion of refueling:
14. Chocks/ground wire — REMOVE
15. EMCON — DESELECT
16. TPUMP BOOST — AUTO
17. ECLs — FLY

Gravity Refueling with External Power or APU Operating

The aircraft is gravity refueled through a filler cap on the left forward sponson fuel tank. Fuel from the left forward sponson is transferred to the remaining tanks by the left and right sponson transfer pumps. External or APU electrical power is required for operation of the transfer pumps. There are four ground crew requirements: 1 cockpit, 1 nozzle, 1 fire guard/GRDP, and 1 deadman control operator. Personnel are to wear proper hearing protection.

CAUTION

MATS (if installed) should be limited to 2350 lbs in the forward tank and 2200 lbs in the mid and aft tanks for normal operation. If these levels are exceeded, the restraint system for the MATS may not prevent tank movement during a crash.

1. Verify fire extinguisher is properly positioned
2. External power unit or start APU — Apply electrical power to aircraft

CAUTION

- Ensure left and right avionics ram air exhaust covers, APU cover, and pitot static probe covers have been removed before applying electrical power.
- Precheck is not performed during gravity fueling operations. Gravity fueling

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

operations are to be halted immediately upon indications of fuel spillage from any fuel tank vent, or loud/unusual noise accompanied by wing vibration. Inspection of suspect fuel tank and its immediate area, including the internal wing for structural damage, is to be conducted.

3. Aircraft — Ground
4. Aircraft and all Fueling Equipment — Connect to ground

WARNING

Always bond nozzle to aircraft prior to removing fuel cap. This connection is to remain in place until entire fueling evolution is complete. Failure to ground nozzle and/or maintain contact can result in explosion.

5. Left sponson tank filler cap — Remove and insert refueling nozzle

WARNING

Once fueling evolution has commenced, the aircraft's electrical power status and connections are not to be changed until fueling has been stopped. Engines or auxiliary power units are not to be started or stopped and external power are not to be connected, disconnected, or switched on or off. Changing the aircraft electrical power status can create significant ignition sources.

6. GRDP PWR switch — ON (Total fuel quantity should display)
7. Panel lighting — Adjust as desired
8. LAMP TEST — Press (All lights should illuminate in sequence)
9. MODE SEL switch — Press and release until GRAVITY FILL light illuminates
10. TANK FILL SEL switches — Press and release until desired FILL lights illuminate
11. Fueling truck/ship station — Start refueling

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CAUTION

Observe SYSTEM STATUS lights constantly during refueling. If the STOP REFUEL indicator illuminates, immediately turn off the fuel supply pump (by releasing deadman switch if equipped) and release the handle on the overwing fuel nozzle to stop fuel flow. Failure to do so may cause fuel spillage resulting in aircraft damage and possible fire hazard.

Note

The FUEL FLOW indicator will illuminate once total fuel quantity increases by at least 100 lbs in any one-minute interval. The FUEL FLOW indicator will stop illuminating when the total fuel quantity does not increase by at least 100 lbs in any one minute interval.

12. Verify that the SYSTEM FAIL light on the GRDP is not illuminated. If the SYSTEM FAIL light illuminates, stop the refueling process
13. Tanks — Refuel slowly until all are full, or the desired fuel quantity displays on the GRDP

CAUTION

- Fuel flow from the refuel nozzle needs to be monitored during gravity refueling to ensure that the left forward sponson tank is not over filled.
- Immediately cease fueling if any of the following tank quantities are exceeded:

	JP-5 (lbs)	**JP-8/JET A (lbs)**
Feed tanks	650	640
Wing auxiliary tanks (if installed)	2050	2015
Forward sponson tanks	3300	3250
Aft sponson tank (if installed)	2110	2075
MATS (if installed)	2875	2830

14. Fueling truck/ship station — Stop
15. Refuel nozzle — Remove, secure fuel tank cap, remove ground wire
16. GRDP PWR switch — OFF
17. GRDP access panel — Close
18. APU or external power — OFF or disconnected
19. Battery — OFF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

DEFUELING PROCEDURES

CAUTION

- The APU or main engines cannot be running during defueling operations. The wing feed tanks will empty first causing APU or main engine flameout and subsequent loss of prime.
- During aircraft storage, fuel from the Wing Aux Tanks may migrate into the sponson tanks causing an overfill and disc rupture. If the aircraft is to be left static in excess of 10 days, consideration should be given to defueling the Wing Aux Tanks.

Note

If aircraft has a preexisting or verified vent system leak or burst rupture disc, perform maintenance and troubleshooting in accordance with A1-V22AB-TIS-000. Defueling any of the sponson tanks with fuel trapped in the vent system may damage the fuel cells and/or cell supports.

1. Verify that the aircraft has been properly grounded. If not, connect ground wire to deck and to one of the aircraft grounding points
2. Prior to initiating defueling, depress the climb dive valves on the L/R and AFT sponson fuel tanks

If more than one quarter (1/4) gallon is drained from vent:

a. Review A1-V22AB-TIS-000 for maintenance procedures
b. Maintenance procedures direct to vent the top of the sponson tank by loosening the gravity fill cap (left forward sponson only) or a fuel quantity probe (right sponson forward or aft tanks) to allow outside air to ventilate the tank.

3. If MATS are installed, select CBNTNK ISOL on the FUEL STAT layer on the MFD to isolate all MATS
4. Open GRDP access panel 7LS1 on the left aft sponson
5. Connect suction defueling equipment to pressure refueling port below the GRDP
6. Set GRDP PWR switch to ON

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

Note

If GRDP times out or STANDBY is selected, verify that the truck-defueling nozzle is closed and deactivate defueling equipment and restart at Step 6.

7. Press LAMP TEST to ensure all lights illuminate
8. Press and release MODE SEL switch until SUCTION DEFUEL light illuminates. Wait 10 seconds for process to start
9. Activate defueling equipment to apply suction to the nozzle flange and open nozzle valve
10. Verify total fuel quantity is decreasing and the FUEL FLOW indicator is lit on the GRDP
11. When Wing Auxiliary Tanks and Feed Tanks are empty, de-select CBNTNK ISOL on the MFD FUEL STAT layer to de-isolate all MATS for defueling
12. Defueling is complete when no flow is evident on the GRDP and FUEL QTY indication stops decreasing
13. Close the defueling nozzle valve
14. Deactivate defueling equipment and disconnect from aircraft
15. Press and release MODE SEL switch until STANDBY light illuminates
16. Turn OFF the GRDP by depressing the PWR switch
17. Close the GRDP access panel
18. If a fuel tank is to be opened, the appropriate tank drain valve needs to be opened to completely empty the tank
19. Remove ground wire if aircraft is to be moved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

GLOSSARY

D.1 Definitions.

D.1.1 AFFF. The acronym for Aqueous Film Forming Foam, the Department of Defense standard firefighting agent for flammable and combustible liquid fires.

D.1.2 Adsorption. A separation method where one component is concentrated on the surface of a porous solid. Surfactants (surface active agents) are separated from jet fuel by adsorption on clay.

D.1.3 Ambient temperature. The air temperature surrounding a specific area.

D.1.4 API gravity. The petroleum industry's scale and method of measuring density of petroleum products.

D.1.5 Aviation gasoline (AVGAS). Specially blended gasolines used to power reciprocating piston aircraft engines.

D.1.6 Clay treater. A treating unit that utilizes special clay (Fuller's earth) to remove surfactants from turbine fuel.

D.1.7 CMC. Commandant of the Marine Corps.

D.1.8 CNATRA. Chief of Naval Air Training.

D.1.9 CNO. Chief of Naval Operations.

D.1.10 Coalescence. The property of a filter cartridge to bring together very fine droplets of free and entrained water to form large droplets that are heavy enough to fall to the bottom of the filter/separator vessel.

D.1.11 Contaminants. Either foreign or native substances that may be present in fuel and which detract from fuel performance.

D.1.12 Cowling. Removable covering around engine sections.

D.1.13 Cyclone separator. A device that used the principle of centrifugal force to cause the contaminate in a fuel to settle to the bottom of the vessel without the use of filter media.

D.1.14 Defense Working Capital Fund. Fuel is owned by the Defense Logistics Agency.

D.1.15 Density. The amount of mass (weight) in a unit volume of a material.

D.1.16 Differential Pressure (Delta P). The measured difference in pressure between any two points, generally at the inlet and outlet of a filter or a filter/separator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

D.1.17 Disarming action. As applied to filter/separators, the rendering of the elements incapable of performing their designed functions; e.g., coalescer elements incapable of coalescing water and separator elements incapable of separating water from fuel.

D.1.18 Dissolved water. Water that is in solution in the fuel. This water is not free water and cannot be removed by conventional means.

D.1.19 Eductor. A device for inducing flow of a fluid from a chamber or vessel by using the pressure of a jet of air, water, steam, etc., to create a partial vacuum in such a way as to entrain the fluid being removed

D.1.19.1 A device placed in a hose line to proportion liquid foam or wetting agents into the fire stream.

D.1.19.2 An ejector that siphons water by creating a vacuum from the velocity of water passing through it.

D.1.20 Effluent. Stream of fluid at the outlet of a filter or filter/separator. This is the opposite of influent.

D.1.21 Empennage. Aircraft tail assembly including stabilizers, elevators, rudders, etc.

D.1.22 Emulsion. A dispersion of two dissimilar immiscible droplets in the continuous phase.

D.1.23 Entrained Water. Small droplets of free water in suspension that may make fuel appear hazy.

D.1.24 EOD. Explosive ordnance disposal.

D.1.25 ETA. Estimated time of arrival.

D.1.26 Exhaust. The part of the engine through which the exhaust gases are ejected.

D.1.27 Filter. A device to remove solid contaminants from fuel.

D.1.28 Filter membrane (Millipore) test. A standard test in which fuel is passed through a small filter membrane housed in a plastic holder. The cleanliness of the fuel can be determined by examining the membrane.

D.1.29 Filter/Separator. A mechanical device used to remove entrained particulate contaminants and free water from a fuel.

D.1.30 Fin. A fixed or adjustable airfoil for directional stability, such as a tail fin or a skid fin. A common name given the vertical stabilizer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

D.1.31 Fixed Base Operator (FBO). Common title for aviation fuel dealer at the airport.

D.1.32 Flaps. A movable airfoil attached to the trailing edge of the wing that improves the aerodynamic performance of the aircraft during takeoffs and landings.

D.1.33 Flash point. The lowest fuel temperature at which the vapor above the fuel will ignite.

D.1.34 Floating suction. A floating device used in a tank for drawing product from the upper level of the fuel.

D.1.35 Free water. Water in the fuel other than dissolved water. Free water may be in the form of droplets or haze suspended in the fuel (entrained water) and/or a water layer at the bottom of the container holding the fuel. Free water may also exist in the form of an emulsion which may be so finely dispersed as to be invisible to the naked eye.

D.1.36 Freezing point (fuel). The lowest fuel temperature at which there are no crystals.

D.1.37 Halogenated agent. An extinguishing agent composed of hydrocarbons in which one or more hydrogen atoms have been replaced by halogen atoms; the common halogen elements used are fluorine, chlorine, bromine, and iodine.

D.1.38 Hazardous Material. Any substance that, by reason of being explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful, is likely to cause death or injury.

D.1.39 Hydrophilic. Water accepting or water wettable.

D.1.40 Immiscible. Liquids that are mutually insoluble. This is the opposite of miscible.

D.1.41 Influent. Stream of fluid at the inlet of a filter or filter/separator. This is the opposite of effluent.

D.1.42 Landing gear. The understructure that supports the weight of an aircraft when in contract with land. It usually contains a mechanism for reducing the shock of landing (also called undercarriage).

D.1.43 Longerons. The principal longitudinal structural members of the fuselage.

D.1.44 MarForLant/Pac/Res. Marine Forces Atlantic, Pacific, or Reserve.

D.1.45 Micron (Micrometer). A unit of linear measurement. One micron is equal to 0.000029 in. and approximately 25,400 microns equals 1 in. For example, the average human hair is about 100 microns in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

D.1.46 Miscible. Liquids that are mutually soluble. This is the opposite of immiscible.

D.1.47 Monitor. A device that shows or gives warning of improper performance (noun); or to test or check performance on a continuing basis (verb).

D.1.48 Nacelle. The enclosed streamlined housing for a powerplant. A nacelle is usually shorter than a fuselage and does not carry the tail unit.

D.1.49 NATSF. Naval Air Technical Services Facility.

D.1.50 NAVAIR. Naval Air Systems Command.

D.1.51 NAVFAC. Naval Facilities Engineering Command.

D.1.52 NAVSEA. Naval Sea Systems Command.

D.1.53 NAVSUP. Naval Supply Systems Command.

D.1.54 NFPA. National Fire Protection Association.

D.1.55 OPNAVINST. Chief of Naval Operations Instruction.

D.1.56 Oxidizer.

D.1.56.1 A substance that readily gives up oxygen without requiring an equivalent of another element in return.

D.1.56.2 A substance that contains an atom or atomic group that gains electrons, such as oxygen, ozone, chlorine, hydrogen peroxide, nitric acid, metal oxides, chlorates, and permanganates (also called oxidizing agent).

D.1.57 Particulate matter. Solid contaminants (e.g., dirt, rust, scale, sand, etc.) sometimes found in fuel.

D.1.58 Pod. The enclosed streamlined housing around the jet engine.

D.1.59 POL. Petroleum, Oil, Lubricants.

D.1.60 Prefilter. A filter that has a high dirt-holding capacity that is installed up-stream of other filtration equipment.

D.1.61 Pressure drop. See Differential Pressure.

D.1.62 Pylon, nacelle strut. The structure that attaches a jet engine to the wing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

D.1.63 Relative density (specific gravity). In fuel, this is the ratio of weight of any volume of fuel to the weight of an equal volume of water.

D.1.64 Rudder. The upright movable part of the tail assembly that controls the direction (yaw) of the aircraft.

D.1.65. SDA. Static Dissipater Additive.

D.1.66 Settling time. The time allowed for water or dirt entrained in the fuel to drop to the bottom of the storage tank.

D.1.67 Slat. A movable auxiliary airfoil, the primary function of which is to increase the stability of the aircraft. Slats are found on the leading edge of the wing.

D.1.68 Specific gravity. See Relative Density above.

D.1.69 Stabilizer. Any airfoil the primary function of which is to increase the stability of the aircraft. The term stabilizer is most commonly used in reference to the fixed horizontal tail surface of the aircraft.

D.1.70 Sump. A low point in a system for collection and removal of water and solid contaminants.

D.1.71 Surfactant. Any wetting agent.

D.1.72 Surfactants (surface active agents). Chemical substances that make it difficult to separate fuel and water and that disarm filter/separators.

D.1.73 T.A.U. Twin agent fire extinguisher.

D.1.74 T.O. Technical order.

D.1.75 Thief (sump) pump. A small pump having a suction which extends to the low point of a tank for the purpose of drawing off water that may have accumulated.

D.1.76 Turbine fuel. A group of various kerosene or wide-cut types of fuels used to power aircraft turbine engines.

D.1.77 TYCOM. Type Commander.

D.1.78 Water Slug. A large amount of free water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Navy – AS

Preparing activity:
Navy –AS
(Project 9130-2014-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:52Z
Check the source to verify that this is the current version before use.